COURT FILE NO.: 8842/95
KINGSTON

ONTARIO

SUPERIOR COURT OF JUSTICE

B E T W E E N:	)	
REED and LAWLESS	)	Warren H.O. Mueller Q.C., for the plaintiffs
Plaintiffs	)	
- and -	)	
DONALD GARBUTT and EMMONS & MITCHELL CONSTRUCTION INC.	)	Robert M. Nelson and Ritu Gambhir, for the defendant Emmons & Mitchell Construction Inc.
	)	Norman Donald Garbutt appearing on his own behalf
Defendants	)	
	)	**HEARD:** May 13-17, 21-24, 27-31; June 3-5; November 12-15; 18-22, 25-28, 2002 and January 6-9, 2003

MORIN J.

REASONS FOR JUDGMENT

[1] The plaintiffs seek to recover damages from the defendants for breach of contract and/or negligence in respect of the design and construction of their commercial building in the City of Kingston, Ontario.

[2] Both the plaintiffs are dentists licensed to practise dentistry in the Province of Ontario. The defendant Donald Garbutt ("Mr. Garbutt") is and at all material times was a professional

engineer, licensed to practise professional engineering in the Province of Ontario. The defendant Emmons & Mitchell Construction Limited ("Emmons & Mitchell") is and at all material times was a corporation carrying on business as a general contractor in the Province of Ontario.

[3] In 1983, the plaintiffs purchased a parcel of land at 275 Queen Street in the City of Kingston on which they intended to construct a commercial building for the purpose of housing offices for their own dental practices and for housing other commercial tenants. Sometime in 1984 the plaintiffs were referred to Mr. Garbutt as an engineer who was involved in commercial construction. They subsequently met with Mr. Garbutt who agreed to design the building and to act as a consultant during the course of its construction. The plaintiffs had no written contract with Mr. Garbutt. From their conversations with him, they understood that he would design the building and thereafter make sure that the building was properly constructed by carrying out inspections during the course of construction. I find that that was indeed the agreement between the parties. For his services, Mr. Garbutt was to receive a fee of five per cent of the cost of construction. Mr. Garbutt in turn, on his own behalf, retained the architectural services of Roberts Engineering Inc. to assist in the preparation of some of the drawings.

[4] The construction of the building eventually went to tender and Emmons & Mitchell submitted the lowest bid.

[5] In due course a stipulated price contract was entered into between the plaintiffs and Emmons & Mitchell in a form authorized by the Canadian Construction Documents Committee. This formal contract is dated August 12, 1985 and names the plaintiffs as "Owner", Emmons & Mitchell as the "Contractor" and Mr. Garbutt as the "Consultant", all of which terms are defined in the contract.

[6] Article A-2 of the contract identifies certain contract documents including specifications and details prepared by Mr. Garbutt, drawings prepared by Mr. Garbutt, and drawings prepared by Roberts Engineering Inc. The contract also has appended to it the usual general conditions of a stipulated price contract.

[7] Construction was commenced by Emmons & Mitchell in August 1985 and completed in or about July 1986 with the final billing from Emmons & Mitchell dated as of December 1986 for a total contract price of $838,000.

[8] Dr. Reed has occupied the fourth floor of the building from the beginning and throughout, and so was able to chronicle at least the obvious problems relating to water penetration into the building which occurred during heavy or prolonged rainstorms from almost the very beginning of the building's history.

[9] Dr. Lawless occupied the third floor of the building from 1986 to 1991 and gave evidence with respect to his observations during that period of time with respect to leaking problems.

[10] As matters eventually developed, the plaintiffs' main complaints to date are with respect to the roof, the parapet above the roof around the perimeter of the walls of the building, and most importantly the walls of the building themselves.

[11] In this action, evidence was heard over a period of some 35 trial days from May 13, 2002 to January 7, 2003. Over that period of time some 142 exhibits were entered into evidence including contract documents, reports, drawings, and physical evidence from the building site. Written submissions of the parties were concluded in May 2003.

[12] During the course of the trial *viva voce* evidence was received from some 19 witnesses, many of whom were experts retained by the parties for the purpose of conducting on site examinations and analyses, and providing opinions and recommendations. As one would expect in a case of this kind, there are many conflicts in the evidence.

[13] During the course of this trial, I took fairly detailed notes of the evidence of each of the witnesses and then prepared summaries of the evidence with appropriate references to exhibits that had been entered during the *viva voce* evidence. I have attached to these reasons for judgments as Addendum 'A' my summary of the evidence. Needless to say, in arriving at my conclusions with respect to the issues in this case I have taken into consideration all of the evidence put before the court.

[14] In some instances during the course of this trial, when one of the parties opposed the tendering of certain evidence, I received that evidence and reserved my decision as to its admissibility. In all such instances, after due consideration I was satisfied that the evidence was relevant and admissible. Of particular note was the evidence of Emmons & Mitchell's site supervisor, James Sinclair. It had been determined that during the course of construction shelf plates had been cut from the beams at each of the floor levels on two sides of the building. These plates were intended to support the brick cladding at each floor level. Mr. Sinclair proposed to testify that Mr. Garbutt had advised him to cut the shelf plates and that Mr. Sinclair in turn had asked the steel sub-contractor, Christmas Steel, to do the cutting. The proposed evidence was objected to by Mr. Mueller, counsel for the plaintiffs and by Mr. Garbutt who was self-represented during this trial. The objection was made on the basis that Emmons & Mitchell had not fulfilled its undertaking given on examination for discovery where Mr. Mitchell testified that he could not confirm that Mr. Garbutt had been asked for permission to remove the shelf plates, and then counsel for Emmons & Mitchell gave an undertaking to provide particulars of any such discussion. Shortly before trial, Mr. Nelson, counsel for Emmons & Mitchell, through his associate Ms. Gambhir advised the parties that "it is our position that we were instructed by Mr. Don Garbutt to cut the shelf angles." In the circumstances I received Mr. Sinclair's evidence and reserved on the issue of admissibility.

[15] Rule 31.07(2) provides that where a party fails to answer an undertaking given on examination for discovery, that party may not introduce the information at trial except with leave of the trial judge. On a review of the evidence as it went in prior to the calling of Mr. Sinclair, it would have been obvious to the remaining parties that Mr. Sinclair would be testifying that he had permission from Mr. Garbutt to cut the shelf plates. In the circumstances I see no prejudice to the parties in admitting this evidence. See Rule 53.08(1). In the circumstances the evidence will be admitted.

[16] It would be worth while at this point to identify the witnesses who were heard during the course of this trial and to indicate very generally the nature of the evidence that each gave.

The Evidence of the Plaintiffs

1. Drs. Reed and Lawless testified as to the problems that they have had with this leaking building, the attempts that they made to determine the causes of the problems, the reasons why they did not institute action prior to May 17, 1995, and the reasons why they have not yet undertaken the repairs to the building envelope which their experts are telling them are required.

2. Tony Garofalo is a general contractor who testified that it would cost $75 per affected window to make cosmetic repairs to the surrounding drywall and paint. The total cost of this repair would be $2,525.

3. Robert Kendall is a roofing contractor and consultant to the roofing industry. He has, over some 50 years, been involved in major roofing projects including new roof installations and re-roofing. He was asked by the plaintiffs' architect, Paul Johnston, to assist with respect to resolving the roof problems. He attended the site with Mr. Johnston in October of 1994 and observed an "irma" type roof that had been installed. He described the construction of such a roof style and explained that the waterproof membrane is laid directly on to the concrete slab roof. Insulation is then laid on top of the membrane and ballast is placed on top of the insulation. He explained the need in such a roof system for slope to drain and the lack of adherence of the insulation to the top layer of felt. He described certain repairs that were effected to the roof in 1994 and the observations that were made with respect to the roof at that time. He gave evidence with respect to the ultimate replacement of the roof in 1998.

4. Gordon Robb is a general contractor with 20 years experience in the construction industry including cost analysis of projects. He is the president of Detra Builders Inc., a company which bids approximately $30,000,000 in projects every year. Mr. Robb has priced remedial work and wall construction called for by architect Johnston with respect to the subject building. Mr. Robb's costing sets out two alternatives as did Mr. Johnston's scope of work. Both alternates include the removal and replacement of the existing brick face. Mr. Robb costed alternate A which is essentially a veneer wall at $401,245.85, and alternate B which is a cavity wall at

$433,313.84. These are 1995 prices and Mr. Robb testified that there should be an inflation adjustment from that year to the present because of an increase in the cost of building materials and labour at the rate of five per cent per year.

5. Dr. Robert Drysdale holds his M.A.Sc. in civil engineering from the University of Toronto and obtained his Ph.D. in civil engineering from the same University in 1967. He is eminently qualified to provide opinion evidence as an expert on masonry construction and building envelopes. Dr. Drysdale was retained by architect Paul Johnston to comment on the subject building under the categories of design and construction and to comment on the reports of Mr. Chris Roney, a structural engineer retained by Emmons & Mitchell. After reviewing the three Folios prepared by Mr. Johnston, Exhibits 54, 55 and 56, wherein Mr. Johnston addressed General Commentary, Roof and Parapet Design, and Wall Assembly and Design, and after reviewing Mr. Roney's reports, Dr. Drysdale came to some very disturbing conclusions. In a nutshell, he determined that the masonry walls are not structurally adequate, that Mr. Roney's suggested remedial work of the addition of Helifix ties and the introduction of vertical control joints in the brick is inadequate and that the replacement of the brick veneer cannot be avoided in order to properly address the structural and building envelope issues.

6. Paul Johnston is an architect. After graduating from the University of Toronto with an Honours Degree in architecture in 1971 he worked in Helsinki, Finland and Toronto before beginning his own practice in 1979. He remains active in the field today. He provided the court with opinion evidence as to the original design and construction of the building's roof and walls, his concerns with respect thereto, his views with respect to building problems arising out of the design and construction, and the resolution of those problems both with respect to work already done and to be done. He has been involved with the plaintiffs since the fall of 1994, providing them with advice as to the problems in the building, directing them to other appropriate consultant experts, and providing them with advice with respect to appropriate remedial measures. He has been involved as a consultant and supervisor of work

with respect to local repairs to the roof in 1994 and with respect to its replacement in 1998, and provided the court with his observations and opinions arising out of his involvement with that work. He has done extensive investigations with respect to the walls of the building and provided the court with evidence of his observations and with his opinions relating to the problems. It was because of the nature of his involvement with the plaintiffs and their building since 1994 that defence took the position that Mr. Johnston was not independent and that his opinion evidence should not be admitted. For reasons delivered during the course of the trial, I denied the defendants' motion. Indeed at all times throughout the course of his evidence Mr. Johnston testified in a straightforward and impartial manner. He was in many ways the "treating physician" to the plaintiffs, retained by them in 1994 to finally find out what was wrong with this building and to propose reasonable remedial measures. I found him to be a most impressive witness both with respect to his observations and his opinions.

7. Stephen Saxe is a real estate appraiser and provided the court with his opinion as to the present fair market value of the building, both without reference to the building deficiencies as he understood them, and with reference to those deficiencies. Leaving aside the issue of deficiencies, he appraised the building's fair market value based on comparative sales at $745,000 and based on capitalized value at $672,084. As well, he expressed the opinion that based on his understanding that the remedial works required to the building are estimated to cost approximately $613,000, then that figure in his view is considered to reflect the minimum discount from market value that would be necessary to attract a purchaser to the subject building. He also expressed the view that it may be that the building in its present condition is simply not saleable at all.

The Evidence of the Defendant Mr. Garbutt

1. Mr. Garbutt is a professional engineer and has been practising since 1953. He testified that he had been approached by the plaintiffs in 1984. The building was initially

designed by him as a two-story building and then as a four-story building with the contemplation that another two stories might be added to it at some time in the future. Under the contract between the plaintiffs and Emmons & Mitchell, Mr. Garbutt was the consultant and by his own admission was thus responsible for general review of the project as it proceeded. He testified that the intention of his design with respect to the walls was for a solid masonry wall consisting of the two wythes of block and brick with infilling between the two wythes (as it turned out the specifications prepared by him called for an 18 millimetre gap between the block wythe and the brick wythe). He testified that he intended that the building have an inverted roof, that is an "irma" style roof which was in fact built (the specifications and literature required a two per cent slope to drains for an irma roof and in fact the roof was built without any slope at all, indeed with a reverse slope to the parapets. As well the manufacturer's materials called for no adhesion of the insulation to the top felt and the court heard evidence that in fact there was significant adhesion). Mr. Garbutt who had required in his drawings shelf plates on all four sides of the building at every floor level to support the brick as required by the Code, testified that he never gave permission to Emmons & Mitchell or anybody else to have the shelf plates cut off on the east and west sides and indeed did not know that the cutting had been done until long after the building was completed. His specifications called for a "soft" joint between the underside of the shelf plate and the top of the brick course below, and in fact the contractor Emmons & Mitchell had installed a hard mortared joint, a fact which Mr. Garbutt had missed during the course of construction. Mr. Garbutt testified in general that the difficulties with the building were matters of construction as opposed to matters of design.

2. John Wilson is an architect retained by Mr. Garbutt. He has been a practising architect since 1980 and has significant experience in renovation work. Mr. Wilson attended at the building for the first time on March 20, 2002 to gain an understanding of the amount of water penetrating into the interior of the building. He gave his evidence with respect to his observations at that time. He does not agree with Mr. Johnston that the masonry design is deficient. He does not agree with Mr. Johnston that the roof had originally been

poorly designed and specified. In Mr. Wilson's view, Mr. Garbutt prepared original drawings and specifications for the walls which met the requirements of the National Standard Council of Canada and thus the masonry requirements of the National Building Code. Mr. Wilson sees the wall as built to be a hybrid between a mass wall and a classic rain screen wall. He thinks that the walls are good walls that meet the Code requirements. He completely agrees with the evidence of Antonio Pascoal who is a professional engineer retained by Emmons & Mitchell and who costed out certain remedial recommendations made by structural engineer Chris Roney, also retained by Emmons & Mitchell. Those recommendations do not include the complete removal of the brick wythe and replacement thereof, but rather the installation of flashings, weepers, support angles on the east and west elevations, soft and expansion joints and new Helifix stainless steel skewer type brick ties into the existing brick wythe at a significantly lesser expense than Mr. Johnston's recommendations.

The Evidence of the Defendant Emmons & Mitchell Construction Inc.

1. Eric Gault is an employee of the Bank of Nova Scotia and gave evidence with respect to certain business records of the Bank relating to the mortgage financing put in place by the plaintiffs with respect to this building. The Bank documents show the financial situation of each of the plaintiffs as disclosed by them to the Bank at any given point in time from 1995 on.

2. Stephen Murphy is the Supervisor of the Building Section for the City of Kingston. He testified that he had been served with a subpoena requiring him to bring from the City any documents relating to this project, such as drawings, orders, complaints and inspection reports, that he had conducted a search of the City archives and current files, that no such documentation has been found, and that he is not aware of any complaint made by Dr. Drysdale, Mr. Johnston or Drs. Reed and Lawless with respect to the subject property.

3. Chris Roney has been a licensed engineer since 1993. His expertise is in the structural field. Mr. Roney was retained by Emmons & Mitchell in 1999 to inspect

the property and to prepare certain reports. His first report was dated July 2, 1999 and was subsequently incorporated into his second report dated April 24, 2002. The scope of his report in 2002 was to examine the exterior and interior of the building for the purpose of addressing structural problems as raised by Mr. Johnston and to determine the validity of the problems as seen by Mr. Johnston. Mr. Roney gave the court his evidence with respect to his observations, calculations and opinions. Contrary to the views of Dr. Drysdale and Mr. Johnston, he is of the opinion that the walls are structurally sound and that while he concedes that remedial work is required, he is of the view that it is not required to the extent suggested by Mr. Johnston and as supported by Dr. Drysdale.

4. Robert Vinkle is presently one of the three principals of the new company, Emmons & Mitchell Construction (2000). It was his evidence that the original company, the named defendant in this action, is still active although all contract work is now done by Emmons & Mitchell Construction (2000). Mr. Vinkle was involved in this project in April of 1985 as a junior estimator when Mr. Garbutt brought drawings to him and asked if his company would be interested in bidding on the job. Initially, Mr. Garbutt presented only the drawings and not the specifications. Mr. Vinkle quoted on the job on May 8, 1985 at a time when he did not yet have the specifications. Eventually a draft contract was prepared dated June 27, 1985 which refers to specifications and details. Mr.Vinkle testified that by this date Mr. Garbutt had brought in the specifications and three smaller drawings which were received in evidence during the course of the trial and which had been prepared by Mr. Roberts of Roberts Engineering Inc. retained by Mr. Garbutt as an architectural sub-consultant for the masonry and building envelope, as required by the then existing Building Code. The final contract was eventually entered into on August 12, 1985. Construction was commenced and was completed by the summer of 1986. Mr. Vinkle personally had no discussions with the plaintiffs between 1986 and 1992. He was aware that the plaintiffs' lawyer sent a letter to his company on September 16, 1993. As far as he knows, his company was not advised of the roof repair in 1994, nor the installation of

a new roof in 1998. In cross-examination by Mr. Mueller, Mr. Vinkle conceded that during the job he was never advised by the masonry contractor Abe Dick or the roofing contractor that they lacked any contract documents nor did either entity ever enquire of Mr. Vinkle with respect to the interpretation of the contract documents. As well, neither Mr. Sinclair, the site supervisor, or Mr. Emmons who was ultimately in charge of the project ever advised Mr. Vinkle that there had been such inquiries made.

5. Primo Santin has been working with the masonry company Abe Dick since 1956. He was employed by Abe Dick on this particular job as a field supervisor. At any given time a field supervisor may have eight or nine jobs on the go. Mr. Santin testified that with respect to this particular job he recalls getting a drawing which he sent on to head office to price out. He testified that with respect to this particular job he checked out the site every day and would check with both his own foreman and with Mr. Sinclair, the site supervisor for Emmons & Mitchell. In his evidence, he complained about the John Price brick chosen by the plaintiffs. It was his evidence that the only drawing he ever received and saw was Drawing No. 4 from Exhibit #9 which is a section drawing. He described the wall that he was building as a “shell wall” or “cool wall” and not a cavity wall. He explained that in this particular instance the mason who was laying the brick had to grab it at the top and the sides and therefore he needed room for his fingers, thus a void was created between the brick wythe and the block wythe. He discussed the type of ties which were laid between the two wythes along the length of the wall, and the fact that the brick and block courses did not match.

6. Peter Harding is the owner of Amherst Roofing and installed the new roof for the plaintiffs in 1998. In his evidence he gave his views with respect to the alleged quality of the roof which he observed in 1998. The cost of the new roof was $62,000 plus $32,000 for the mechanical work and masonry, and a $4,500 extra for the mechanical work for upgraded aluminium facer waterproofing (which amount does not form part of the plaintiffs’ claim).

7. James Sinclair is presently retired. He was the construction supervisor on the plaintiffs' building, employed by Emmons & Mitchell. As construction supervisor he reported to Mr. Emmons, Mr. Vinkle and Mr. Garbutt. He was on the site every day and was not responsible for any other jobs during this project. He too complained about the use of John Price bricks on the project. He explained that on the east and west sides of the building there was a problem with the coursing of the brick wythe. The brick course would not fit under the plates that were riveted to the steel beams to support the weight of the brick above. There was no such problem on the north and south sides where the brick coursing came to the underside of the plate, however the beams on the east and west sides had been placed higher in order to accommodate the height of the steel deck which was sitting on floor joints which in turn were sitting on four inch shoes. It was Mr. Sinclair's evidence that with respect to this problem he spoke to Mr. Garbutt who came over and gave permission to him to have Christmas Steel cut the shelf angles off the beams on the east and west sides. With respect to the shelf plates, Mr. Sinclair testified that they were required for compliance with the Building Code, and he understood as well that Mr. Garbutt would have filed with the Municipality, drawings showing the plates in place. In cross-examination Mr. Sinclair really had no answer to the question how he thought that Mr. Garbutt would permit the cutting off of the plates if they were required by the Code and shown on his drawings filed with the Municipality. He gave evidence with respect to the control joints which were called for by the specifications but not shown on the drawings, and none of which were installed apart from one in the centre of the south side wall. He also testified with respect to the joint between the top of the brick and the underside of the beam at each floor, which was specified to be a soft joint but was installed as a hard mortar joint. According to Mr. Sinclair, what he was building was a solid wall with a finger width of void between the brick wythe and the block wythe. When he retired in 1992 he threw out his diaries in which he had kept a complete day to day record of the job during its course. He professed not to ever know that Mr. Garbutt had intended that the collar joint between the brick wythe and the block wythe be filled. Mr. Sinclair admitted that the paper work on the job was a disgrace.

He added that he was so sick of the job at one time that he wanted to quit. He described it as a “haphazard way of doing business”.

8. Richard Quirouette, although not a licensed architect, graduated from Carleton University with his Bachelor of Architecture in 1973. After graduation, he worked from 1973 to 1987 with the National Research Council in the field of building envelopes. In 1987 he joined a private consulting firm and in 1993 started his own consulting firm. He has conducted over 250 cases relating to envelope failure. About one-third of his work at the present time is work that he does for lawyers. I found Mr. Quirouette qualified to give expert opinion evidence to the court in the field of building envelope technology. He was retained by Mr. Nelson for Emmons & Mitchell in early 2002. His first site visit to the building was on February 5, 2002 and thereafter he attended at the site on March 16, 2002 and assisted Mr. Roney with respect to test openings numbers one and two. Mr. Quirouette gave evidence with respect to his observations on both occasions and provided the court with his opinion that the wall as constructed was neither a mass wall with the collar joint filled with mortar nor a cavity wall. He saw it to be a hybrid wall with perhaps more features of a mass wall than a cavity wall. He noted that the design drawings do not clearly indicate a void between the block wythe and the brick wythe, but the specifications on the other hand do provide for a 19 millimetre void between the two wythes. He noted that with a cavity wall there is flashing material provided to divert cavity water to the outside, and also includes openings in the masonry veneer (weep holes). All in all, Mr. Quirouette arrived at several conclusions, most of which are generally favourable to Emmons & Mitchell. He concludes that it is not necessary to replace the masonry cladding to restore the rain penetration control of the exterior wall system.

9. Antonio Pascoal is a professional engineer and president of J.V. Masonry Limited. He has prepared two reports for Emmons & Mitchell, one dated June 28, 1999 and the other, more a quote than a report, dated April 26, 2002. Mr. Pascoal was initially retained by Emmons & Mitchell to review the project and assist Mr. Roney in making

some internal openings. Eventually he was asked to assist Mr. Roney in opening some external holes in the building. Mr. Pascoal gave evidence to the court as to his observations in both cases. He proposes remedial work which does not involve removing and replacing the brick wythe but does involve the installation of certain waterproof membrane flashings, weepers, support angles, soft joints, expansion joints and new Helifix stainless steel skewer type brick ties at a cost in all of $90,000.

[17] In order to better understand the difficulties that arose in the design and construction of the subject building and the responsibilities of the parties in that respect, it would be useful to review the General Conditions and Specifications of the contract entered into by the parties on August 12, 1985 with some general comments with respect to the evidence as it applied to the General Conditions. The contract follows the form of the 1982 standard Canadian Construction Documents Committee's form of Stipulated Price Contract. The following General Conditions ("GC's") are relevant.

1. GC 1.2 provided as follows:

> The Contract Documents are complementary and what is required by any one shall be as binding as if required by all.

As the evidence eventually disclosed, the specifications attached to the contract contained requirements not found on the drawings.

2. GC 1.6 provided as follows:

In the event of conflicts between Contract Documents, the following shall apply:

> a) figured dimensions shown on a drawing shall govern even though they may differ from dimensions scaled on the same drawing,
> b) drawings of larger scale shall govern over those of smaller scale of the same date,
> c) specifications shall govern over drawings,
> d) the General Conditions shall govern over specifications,
> e) supplementary conditions shall govern over the General Conditions, and
> f) the executed agreement between the owner and contractor shall govern over all documents.

> Notwithstanding the foregoing documents of later date shall always govern.

Again, as the evidence disclosed, the specifications prepared by Mr. Garbutt and dated June 27, 1985 contained requirements not shown on and inconsistent with the drawings and also post-dated the drawings prepared by Mr. Garbutt and the drawings prepared by Mr. Roberts on Mr. Garbutt's behalf.

3. GC 2.1 provided as follows:

> During the progress of the Work, the Consultant will furnish to the Contractor such additional instructions to supplement the Contract Documents as may be necessary for the performance of the Work. Such instructions shall be consistent with the intent of the Contract Documents.

The evidence consistently demonstrated that no efforts were made by those involved in the design and construction to resolve the inconsistencies between the drawings and specifications, nor to provide required detail where it was lacking.

4. GC 3.3 provided as follows:

> The Consultant will not be responsible for and will not have control or charge of constructions means, methods, techniques, sequences or procedures, or for safety precautions and programs required for the Work in accordance with the applicable construction safety legislation, other regulations or general construction practice. The Consultant will not be responsible for or have control or charge over the acts or omissions of the Contractor, his Subcontractors or their agents, employees or other persons performing any of the Work.

The significance of this particular GC is that where there may be any construction deficiencies, Emmons & Mitchell cannot blame those deficiencies on Mr. Garbutt.

5. GC 3.4 provided as follows:

> The Consultant will visit the site at intervals appropriate to the progress of construction to familiarize himself with the progress and quality of the Work and to determine in general if the Work is proceeding in accordance with the Contract Documents. However the Consultant will not make exhaustive or continuous on-site inspections to check the quality or quantity of the Work.

Without defining at this point the scope of Mr. Garbutt's obligation as consultant, it became clear from the evidence that Mr. Garbutt failed on occasion to identify construction errors and deviances from the specifications that ought to have been obvious to him. For example Mr. Garbutt in his evidence very candidly admitted that he "simply missed" the fact that the masons had installed a solid mortar joint between the top brick course and the underside of the angle iron at each floor on the north and south sides, whereas the specifications called for a soft joint.

6. GC 3.8 provided as follows:

> The Consultant will have authority to reject Work which in his opinion does not conform to the requirements of the Contract Documents. Whenever he considers it necessary or advisable he will have authority to require special inspection or testing of Work whether or not such Work be then fabricated, installed or completed. However, neither the Consultant's authority to act nor any decision made by him either to exercise or not to exercise such authority shall give rise to any duty or responsibility of the Consultant to the Contractor, his subcontractors, or their agents, employees or other persons performing any of the Work.

Thus, for example, should the installation of the solid mortar joint as described above be adjudged defective work under the terms of the contract, then Mr. Garbutt's failure to reject it does not impact on the liability of Emmons & Mitchell.

7. GC 10.1 provided as follows:

> The Contractor agrees to preserve and protect the rights of the parties under the contract with respect to work to be performed under subcontract and to:
>
> a) enter into contracts or written agreements with the Subcontractors to require them to perform their work in accordance with and subject to the terms and conditions of the Contract Documents, and

> b) be as fully responsible to the owner for acts and omissions of his Subcontractors and of persons directly or indirectly employed by them as for acts and omissions of persons directly employed by him.

The Contractor therefore agrees that he will incorporate the terms and conditions of the Contract Documents into all subcontract agreements he enters into with his Subcontractors.

In practical terms, this means that Emmons & Mitchell will be responsible for any deficiencies of their masonry or roofing subcontractor.

8. GC 11.1 provided as follows:

> Except as provided in GC 12 – Valuation and Certification of Changes in the Work, paragraph 12.4:
>
> a) the Owner, through the Consultant, without invalidating the contract, may make Changes in the Work with the Contract Price and Contract Time being adjusted accordingly by written order, and
> b) no changes in the work shall be proceeded with, without a written order signed by the Owner and no claim for a change in the Contract Price or change in the Contract Time shall be valid unless so ordered and at the same time valued or agreed to be valued as provided in GC 12 – Valuation and Certification of Changes in the Work.

As matters turned out, and as disclosed by the evidence, no change order exists for the removal by Emmons & Mitchell of the shelf plates at all floor levels on the east and west sides, which plates were designed to support the brick wall. The site supervisor Mr. Sinclair, testified that he sought and obtained Mr. Garbutt’s verbal authority to have the shelf plates removed. Mr. Garbutt in his evidence denies that to be the case. Obviously, this is one of many conflicts in the evidence that would have been avoided if the General Conditions of the contract had been complied with.

9. GC 14.6 provided as follows:

> Notwithstanding the provisions of paragraph 14.5 and notwithstanding the wording of such Certificates, the Contractor shall ensure that such Work is protected pending the Total Performance of the Work and be responsible for the correction of defects in it regardless of whether or not they were apparent when such Certificates were issued.

GC 14.5 provided that the Contractor may seek from the Consultant certification that the work of a subcontractor has been totally performed to the Consultant's satisfaction. The thrust of GC 14.6 is that Emmons & Mitchell cannot avoid responsibility for a subcontractor's defective work by reliance on Mr. Garbutt having certified it for payment.

10. GC 14.10 provided as follows:

> No payment made by the Owner under this Contract or partial or entire use or occupancy of the Work by the Owner shall constitute an acceptance of work or products which are not in accordance with the requirements of the Contract Documents.

11. GC 24.1 provided as follows:

> The Contractor shall be responsible for the proper performance of the Work only to the extent that the design and specifications permit such performance.

12. GC 25.1 provided as follows:

> The Contractor shall have complete control of the Work and shall effectively direct and supervise the Work so as to ensure performance with the Contract Documents. He shall be solely responsible for construction means, methods, techniques, sequences and procedures and for co-ordinating the various parts of the Work under the Contract.

13. GC 25.5 provided as follows:

> The Contractor shall review the Contract Documents and shall promptly report to the Consultant any error, inconsistency or omission he may discover. Such review by the Contractor shall be to the best of his knowledge, information and belief and in making such review the Contractor does not assume any responsibility to the Owner or to the Consultant for the accuracy of the review. The Contractor shall not be liable for damage or costs resulting from such errors, inconsistencies or omissions in the Contract

> Documents which he did not discover. If the Contractor does discover any error, inconsistency or omission in the Contract Documents he shall not proceed with the Work affected until he has received corrected or missing information from the Consultant.

As the evidence eventually disclosed, there were many instances and situations encountered during the course of construction that required consultation between the Contractor and the Consultant for the purpose of clarification, and no consultation was had.

14. The Contract Definition section which immediately precedes the GC's provides that the Contract Documents form the Contract and stipulates that the Contract is the undertaking by the parties to perform their respective duties, responsibilities and obligations as prescribed in the Contract Documents and represents the entire agreement between the parties. The Definition section also provides that the Contract Documents consist of the executed agreement between the Owner and Contractor, the General Conditions of the Contract, Supplementary Conditions, Definitions, specifications, drawings and such other documents as are listed in Article A.2 – Contract Documents. As well, the Definition section provides that the Work means the total construction and related services required by the Contract Documents.

[18] The Construction Contract had annexed to it specifications dated June 27, 1985, prepared by Mr. Garbutt's office and signed off by Emmons & Mitchell. The specifications are divided into various divisions covering such headings as GENERAL REQUIREMENTS, SITE WORK, MASONRY, METALS, etc. and each division has several sections. The following specifications are worth noting at this time.

1. Section 7(1) of the GENERAL REQUIREMENTS (DIVISION 1) provided as follows:

> All work shall be carried out in accordance with the best standard practice, by mechanics skilled in the type of work concerned.

2. Section 11(1) of the GENERAL REQUIREMENTS provided as follows:

> As-built drawings shall indicate all changes, deviations, omissions, additions, etc., made to the project during the construction period on a complete set of drawings, in the form of permanent reproducibles.

One of the difficulties encountered in this case is that no as-built drawings are available although the court heard evidence from the contractor's site supervisor Mr. Sinclair, that he religiously kept an up-to-date set of as-built drawings and when the work was completed gave those as-built drawings to Emmons & Mitchell.

Section 1(1) of DIVISION 4 – MASONRY, provided under the heading Standards as follows:

> Brick masonry to comply with NBC (National Building Code) 1975 sup.4 subsection 4.4.2 materials and to CSA (Canadian Standards Association) A-224 'A Guide for the Design and Construction of Unit Masonry' (subsection 4)

Section 2.1 of DIVISION 4 provided as follows:

Brickwork: (based on an average unit cost of $300/1000)

> 1. Bricks shall be laid in running or stretcher bond with control joints as noted on drawings. Units shall be tied every other course as per drawing. Maintain continuous 19 millimetre air space behind brick throughout project.
>
> Provide 6 millimetre continuous expansion joint caulked continuously between steel support angles and brick panels each floor.
>
> 2. Tool joints to smooth concave profile.

As it turned out, control joints were not shown on the drawings and as well the masonry ties were not identified on the drawings. The drawings were confusing as to whether an air space was called for between the brick wythe and the block wythe. I heard evidence from Mr. Garbutt that his design intent was for a solid wall with the collar joint between the two wythes to be filled with mortar. In fact, the collar joint was not filled with mortar. I also heard evidence that the number and placement and type of ties was deficient. It was clear on the evidence that the “soft” joint called for in this section between the steel support angles and the brick was not installed but rather a solid mortar joint was installed. As previously mentioned the steel support angles on the east and west sides of the building were completely cut off.

3. Section 2(2) provided as follows under the heading <u>Reinforcement and Anchorage</u>:

> 1. Reinforced masonry walls indicated with continuous masonry reinforcement in every second block course.
>
> 2. Provide control joints as dictated by structural design.
>
> 3. All block walls shall be bonded at intersections.
>
> 4. All beams and lintels to have 200 millimetre minimum bearing on masonry and seats to have two courses filled solid with concrete unless noted otherwise.
>
> 5. Anchors: hot dipped galvanized steel of sizes and types to suit conditions.
>
> …
>
> 8. Anchor and bond masonry to conform with NBC (National Building Code).

I eventually heard evidence that control joints were not shown on the drawings and were not in fact placed as required during the course of construction.

4. Section 4 of the Masonry Specifications provided in part as follows:

> 6. Bond abutting masonry work. Use metal anchors where natural bonding is impractical.
>
> 7. Anchor masonry abutting concrete with metal dovetail anchors.

I heard evidence, at least from the plaintiff’s experts that the anchoring procedure on this wall was deficient as to number, type and location of metal ties between the two wythes and anchorage of the block wall to the steel structure.

5. Section 6 of the Masonry Specifications provided under the heading Wall Flashing as follows: "Provide 56.7 gram (2 oz.) copper fibreen." The evidence disclosed that in fact no flashing was installed in the walls.

6. Section 8 of the Masonry Specifications under the heading Wall Drainage provided as follows: "Provide brick weepers at 600 millimetres o.c. (24 inches) at all bearings and over horizontal obstructions." Again the evidence disclosed that in fact no wall drainage at all was provided.

7. Section 9 of the Masonry Specifications under the heading Job Mock-Up provided as follows:

> Construct sample brick wall of mortar and brick. Units to show colour range, texture, bond, jointing, for approval. Remove wall if not approved, and re-construct until approval is received. Retain approved wall until project completion. Remove when directed. Only masonry matching approved sample will be acceptable on this project.

It is clear on the evidence that a separate sample wall was not constructed as such. There is evidence that the initial portion of the brick wall was treated as a sample wall and eventually removed and reconstructed when it was determined that the first soldier course bricks were badly out of plumb. However the evidence in my view did not disclose that that initial piece of wall was viewed by anybody for the purpose of determining whether or not the nature of the cavity between the brick wythe and the block wythe was acceptable, and whether or not sufficient ties of an acceptable quality were being installed in the right places.

8. Section 1 of Division 5, the METALS Specification, provided in part as follows:

> 3. Submit shop drawings clearly indicate framing plans, bearing and anchorage details, framed openings, accessories, schedule of materials, camber and loadings, welded and fasteners.
>
> …
>
> 10. Obtain written permission of engineer prior to field cutting or altering of structural members.

The evidence disclosed that shop drawings relating to the steel frame were never submitted and that eventually the shelf plates were cut off the steel framing on the east and west sides without Mr. Garbutt's written permission.

9. Section 4 of the METALS Specifications provided under the heading Plates and Anchors "Provide bearing plates, anchor plates and anchors indicated." The drawings showed the location of the anchor plates as required by the Specifications.

10. Section 7 of the METALS Specifications provided as follows under the heading Field Requirements:

    1. Before beginning fabrication, this Contractor shall field check all dimensions, and shall report any major variance from the plans to the Contractor's Consultants.

    Arguably, if this specification had been followed it would have been determined prior to fabrication of the steel framing that on the east and west sides having regard to the existing dimensions the brick course at the top of each floor was not going to fit under the shelf plate.

11. Section 2 of Division 7, the THERMAL AND MOISTURE PROTECTION Specifications, provided as follows under the heading Roof Drainage:

> Provide all roofs with roof drains. Provide complete underground storm sewer drainage system. For inverted roofs, slope structures to drains at least 5m/100mm gradient.

Although as it turned out the drawings showed no such slope, clearly the Specifications called for what amounted to a two per cent slope of the roof to the roof drains. The evidence disclosed that in fact the roof was constructed without such a slope and indeed had a reverse slope away from the drains and toward the parapet. I heard evidence with respect to the water pooling at the edge of the parapet and thus adversely affecting the underlying membrane at the cant, that is the area where the horizontal roof meets the vertical parapet.

12. Section 6 of the THERMAL AND MOISTURE PROTECTION Specifications provided in part as follows under the heading Membrane Flashings:

    1. Construct membrane flashings at all junctions of roofs with vertical surfaces with the same bitumen and roofing felts as used for the roof application. Carry four ply felt and asphalt membrane flashings up the surface of the wall or curb to a point not less than 200 millimetres (eight inches) nor more than 370 millimetres (15 inches) above the surface of the roof. Lay strips of felt so as to break joints with the underlying ply. End laps of all strips shall not be less than 100 millimetres (four inches) and be firmly imbedded in bitumen.

    The evidence disclosed that this specification in fact had not been followed by the roofing subcontractor.

[19] The evidence discloses that there were problems with this building from almost the very beginning of its construction, although the scope of the problems and the cause of the problems and how to best remedy the problems were often in dispute and indeed remained in dispute until the trial of this action. It would be worthwhile at this point to review some of the documentation created during and after construction, to better understand how the disputes among the parties were brought to a head.

[20] For example, a review of the Site Meeting Minutes discloses the following:

1. On November 6, 1985 it was noted that a section of the brick wall laid by the mason Abe Dick had been rejected and was to be rectified. (The evidence here at trial showed the bottom soldier course of bricks to be very much out of plumb to the point that one must wonder how they could have been laid in that fashion to begin with.)

2. The Minute of February 12, 1986 notes that the masonry walls were to be cleaned and pointed as necessary.

3. On March 13, 1986 the Minute notes that Emmons & Mitchell were instructed by Mr. Garbutt to re-locate the fourth floor HVAC unit with a new hole for the duct work to be cut and the existing hole to be filled in, and repairs to the roofing to be effected as required.

4. In the Minute of April 9, 1986 it was noted that the roof had to be made water tight in the area of the changed ducts and heating units as quickly as possible in order to finish topping on the concrete floors.

5. The Minute of April 25, 1986 required Emmons & Mitchell to call Abe Dick Masonry to set up a meeting once all their work had been completed (the evidence discloses that by this time it had been recognized that the masonry workmanship was aesthetically unsatisfactory).

[21] The preliminary lists of deficiencies sent by Mr. Garbutt to Emmons & Mitchell on June 5, 1986 noted the following under the heading MASONRY:

- Some portions not cleaned
- Fitting around fixtures in brick walls not acceptable
- Finished brickwork on lobby walls scored and not cleaned
- Pointing at stone surrounds not complete
- Openings from basement to stairwell not filled
- In general, <u>Poorly done</u> [underlining mine]

[22] The Minute of the July 3, 1986 Site Meeting, at which were present the plaintiffs, Abe Dick of Abe Dick Masonry Limited, and Don Emmons of Emmons & Mitchell, noted that a site inspection had been carried out by the parties present to determine the method and procedure to follow to remove mortar, stains and discolouration in order to obtain a more uniform appearance to the overall job. The Minute notes as well that Mr. Dick agreed to an overall clean down of the entire building, using a solution of his choice with Mr. Dick pointing out that there was a

possibility that this cleaning could bring out further efflorescence that might not recede for some length of time.

[23] On September 4, 1986 Emmons & Mitchell sent to the roofing subcontractor, Quintal and England Limited a deficiency list relating to the roof and which noted a break in the flashing at one of the corners of the roof, a hole in the flashing near the roof hatch, and poor caulking on the flashing.

[24] On September 11, 1986 Emmons & Mitchell corresponded with the plaintiffs advising them that Mr. Emmons was continuing to review the masonry problems and would arrange a meeting with them.

[25] On September 4, 1986 Emmons & Mitchell sent a list of deficiencies to its window subcontractor, Fort Glass Incorporated which list contained the following items:

- General all caulking should be checked, as general poor workmanship.
- Fourth floor window leaks onto sills.
- Third floor window leaks.

The list also refers to poor caulking, a leak around the entrance skylight, and water leaks around the doors.

[26] Emmons & Mitchell's letter to Abe Dick Masonry Ltd. dated September 15, 1986 confirms that the mason has tried a complete cleaning of the brick work in an attempt to remove mortar splatters and improve the overall appearance of the brick, and that the owners still felt that the overall appearance was unsatisfactory and were requesting a further meeting.

[27] In the fall of 1986 the plaintiffs retained the services of Mill & Ross Architects to perform a series of visual exterior inspections with respect to the walls. Mr. Mill reported to the plaintiffs under date November 14, 1986 and that report was received in evidence not for the truth of its contents but rather as part of the narrative and to indicate what information the plaintiffs were receiving with respect to the building. The report deals only with the external aesthetics of the building and notes that the workmanship is of substandard quality.

[28] By letter to them dated November 20, 1986 Emmons & Mitchell acknowledged that the plaintiffs were seeking a $12,000 credit to cover the masonry deficiencies. (Eventually by letter dated January 15, 1987 Emmons & Mitchell confirmed a $9,000 masonry credit to the plaintiffs.) It is clear on the evidence that this settlement related only to the external appearance of the walls.

[29] By invoice dated November 20, 1986 Emmons & Mitchell charged the plaintiffs $1,500 for “caulking of windows throughout” (it would appear that the Specifications required this caulking to be done as part of the Contract Price).

[30] On November 26, 1986 the plaintiffs sent to Emmons & Mitchell a list of items to be completed which included the following:

- ensure water problems in brick around middle window on fourth floor (north side) and brick around rear exit are resolved completely to owners’ satisfaction (regardless of what must be done to rectify the problem).

- correct water leaks around all exterior doorways (Queen St., main entrance and rear exits) -- in particular at the junction of the side glass panels and the brick walls at floor level and around the door frame in the rear exit.

[31] On December 31, 1986 Emmons & Mitchell sent to the plaintiffs an adjusting invoice showing the $9,000 masonry credit. On that invoice it is noted that there is a water leak in the rear entrance, watermarks on the bricks around the window where flashing drips and some persistent leaks.

[32] Throughout the course of the work, Emmons & Mitchell submitted progress billings which were approved by Mr. Garbutt. As of July 31, 1986 Mr. Garbutt approved the work as 100% complete at a value of $838,285.53 with a holdback of $50,128.26. As well, from time to time, Mr. Garbutt issued inspection reports with respect to various aspects of the construction wherein he gave the following certification, “I hereby certify that all work indicated in the above report is in accordance with the plans and specifications, relevant municipal and provincial requirements, the Ontario Building Code, and good construction practice.” As well, during the

course of the construction, Mr. Garbutt would invoice Drs. Lawless and Reed based on the value of the work done to that date. His invoices indicated his total fee to be based on five per cent of the construction cost with 75% of the total fee attributable to DESIGN and 25% of the total fee attributable to SUPERVISION.

[33] At some point early in 1987 Mr. Garbutt became involved in a dispute with his sub-consultant architect Tom Roberts. The fact of that dispute is evidenced by a letter sent by Mr. Roberts to Mr. Garbutt dated February 25, 1987 relating to non-payment of Mr. Roberts's fees which were to be one per cent of the total cost of construction to be paid out of Mr. Garbutt's five per cent.

[34] The evidence discloses that although the window caulking performed in the fall of 1986 appeared to correct to some extent the leakage around the windows, periodic leakage into the building continued to occur, particularly during severe rain storms. For example, in the fall of 1991 some of the building's windows developed drops of water at the top extremity which drops then proceeded to fall onto the window sills. Dr. Reed at the time contacted the glass company which showed no interest in his observations and complaint.

[35] It appears from the evidence that in the late summer of 1992 Dr. Reed met with Mr. Emmons to discuss his complaints of water leakage and Mr. Emmons advised that the windows once again be caulked. The windows were in fact caulked once again by the plaintiffs in November of 1992 but the leakage problems were not resolved. From the fall of 1992 to the spring of 1993 there was, according to Dr. Reed's evidence which I accept, a severe progression of water leakage with increasing interior damages. There occurred during this period of time an overflow through the roof hatch on two occasions. The roof hatch is elevated above the roof level by approximately 10 inches and gives access to the roof. It is located over the stairwell. At the time of these occurrences, Dr. Reed assumed that water had flowed over the top of the roof hatch but on later occasions, when water was coming from the roof hatch, he observed that there was no water flowing down the inside wall of the hatch. When the hatch area was repaired in 1994 Dr. Reed observed that there were tears in the membrane in the area where the roof met the hatch wall.

[36] In the spring of 1993 Dr. Reed met with Mr. Emmons and Mr. Primo Santin, a representative of the Abe Dick Masonry company. They discussed the fact that water was still coming in through the windows, notwithstanding that the windows had been caulked a second time in the fall of 1992. Messrs. Santin and Emmons suggested that the water was coming in through the porous bricks that had been used and they suggested the possibility of applying a silicone sealant over the entire walls.

[37] It is clear from the documentation that, at least at this point in time, discussions were being had between Mr. Emmons and Mr. Garbutt as to the possible causes of the leakage problem (these were serious problems – Dr. Reed's video shows the trough and drain system that he had himself devised to collect water at the top of his windows on the fourth floor and the video shows the water literally pouring out of the drain tube into a container during a rainstorm). Mr. Garbutt sent a letter to Mr. Emmons on May 10, 1993 referencing their telephone conversation of May 7, 1993 with respect to the masonry work. The letter reads in part as follows:

> 2) See if the following can be determined:
>
> a) that the collar joint (between brick and block) is filled – there is no such thing as a one-half inch air space
>
> …
>
> b) type of masonry ties used
>
> 3) If the collar joint has not been filled and the wall subsequently not constructed as a cavity or veneer wall (flashing, weep holes etc.) serious problems result.

Mr. Garbutt's latter statement proved to be prophetic. In fact the collar joint had not been filled and the wall not constructed as a cavity or veneer wall with flashings and weep holes, and it is an understatement to say that serious problems resulted.

[38] During the course of this litigation there was produced from the Emmons & Mitchell file a note that reads as follows:

> Primo – J.P. Price – antique
>
> Built as a solid masonry wall collar joints filled. Walls tied together with adjustable ties.

Although the author of the note is unidentified, one can infer that in an attempt to answer the questions raised by Mr. Garbutt in his letter to Mr. Emmons May 10, 1993 Mr. Emmons spoke to Mr. Santin to get the answers and Mr. Santin provided the information as to the type of brick used and the fact that the collar joint had been filled and that the walls were tied together with adjustable ties. Again, the fact of the matter is that the collar joint was not filled, there were no flashings or weep holes, and adjustable ties were <u>not</u> used. Rather, strap ties were used that were on occasions bent down from the brick wythe to the block wythe in order to accommodate off-coursing.

[39] At this point in time, the Summer of 1993, the plaintiffs were confused and dissatisfied with the position and suggestions of Emmons & Mitchell who seemed to be laying the blame on the porosity of the bricks chosen and were suggesting that the walls be sealed with silicone. This in the face of information that they had received from Mr. Thomson in the fall of 1992 that while caulking the windows he had noticed gaps between the mortar and brick through which water could enter. Mr. Thomson had been retained by the plaintiffs to do the caulking on the recommendation of Emmons & Mitchell.

[40] As a result of their dissatisfaction at this time and their concern about the leakage problems, the plaintiffs began to consult various professionals for independent advice on the cause of the leakage and appropriate remedies. They retained Inspec-Sol (Ontario) Limited to look at the roof system, the parapets, and the brick veneer. By a report dated September 28, 1993 they were advised by Inspec-Sol that internal leaking problems did not conclude that the roof membrane was at fault and that while the bitumen flashings appeared to be in good condition it was noted that the membrane flashings were not installed over the entire parapet but rather only three-quarters of the way across the parapet. With respect to the brick veneer, Inspec-Sol advised the plaintiffs that “the relatively porous properties of the brick veneer create a distinct probability of water penetration through to the wall cavity, and into the interior of the

building at window openings." Inspec-Sol recommended to the plaintiffs that additional testing be performed to "conclude the direct source of water infiltration".

[41] Again, because they were unhappy with the building and felt that they needed outside consultation, the plaintiffs retained the services of a professional engineer, C.H. MacAdam. Mr. MacAdam is today an elderly man who has suffered two strokes and could not be called to testify. The reports that he eventually provided to the plaintiffs between August 1, 1993 and November 30, 1993 were received in evidence, not as proof of the truth of their contents but solely to determine what the plaintiffs knew through these reports at any particular time.

[42] In his report to the plaintiffs dated August 1, 1993, Mr. MacAdam notes that it was agreed between himself and the plaintiffs that he and S.D.F. Reszetnik, an architect, would meet with the plaintiffs to go over the problems, inspect the building and the drawings and if possible suggest a course of action to remedy the problems. In his report, Mr. MacAdam points out that he expressed concern at the absence of any flashings or expansion joints in the walls and at the removal in part or completely of the masonry support plates from the perimeter beams. As well, Mr. MacAdam confirms that he advised the plaintiffs to engage a roofing consultant to give an opinion on the adequacy of the roof flashings and the roofing ballast. As well, Mr. MacAdam suggested to the plaintiffs as confirmed in the letter that they consult lawyers immediately to determine their legal position on the matter and that Mr. Garbutt be consulted to suggest remedial measures.

[43] In due course the first "lawyer's letter" was sent out to the defendant company and to Mr. Garbutt. It is from the plaintiff's then lawyer, Martin J. Szcepaniak and is dated September 16, 1993. The letter was put in evidence not for the proof of its contents but rather as part of the narrative and to establish what was being told to the defendants on behalf of the plaintiffs. The letter refers to water leaks at the building that started prior to the completion of construction and which had continued to date, and which had gotten progressively worse. The letter alleges that the water problems seemed to be the result of improper design and construction of the building. The lawyer asks for written reports setting out the parties' opinions as to the causes of the water problems and their proposed solutions.

[44] By covering letter dated September 20, 1993 Mr. Garbutt sent to Mr. Emmons a draft of his answer to the lawyer's letter asking for his comments. That draft has never been produced but by letter dated September 21, 1993 Mr. Garbutt responded to the plaintiffs' lawyer saying in part the following:

> I have been contacted by Mr. Emmons regarding this matter, have had conversations with him and have inspected the building. I am also in receipt of two reports prepared by Rigney Building (suppliers of the brick) which were forwarded to me by Mr. Emmons and which I believe were forwarded to your clients, and in general I agree with their contents. Since your clients were solely responsible for the selection of the brick and hence ramifications as to scheduling and installation, it would be helpful toward the preparation of further reports as requested if your clients would provide maintenance records as they relate to the building envelope. The brick selected (no longer in production) was probably the most porous one available and while this in itself is not necessarily detrimental to imperviousness of the walls, must be accompanied by a comprehensive regular compensating maintenance program to prevent penetration of moisture into the building components with subsequent adverse effects. The most effective way, and the most economic procedure, of combating damage to buildings from moisture is to control its penetration of envelope components in the first place. The primary information required is the frequency of caulking and what material was used to seal the surfaces of the brick with application dates so that various effects can either be eliminated or evaluated. Your letter refers to your preliminary investigations and these reports would also be of assistance so please forward these as well.

[45] Dr. Reed in his evidence, which I accept, testified that the letter of Mr. MacAdam dated August 1, 1993 was the first time that he and Dr. Lawless were advised that the leakage problem might relate to the absence of flashings or expansion joints and the removal of masonry support plates. As well, in his evidence, Dr. Reed testified that prior to the receipt of Mr. Garbutt's letter dated September 21, 1993 Mr. Garbutt had never advised the plaintiffs that there might be a problem with the porosity of the bricks used or that any kind of maintenance was required for the brick. I accept Dr. Reed's evidence in that respect.

[46] Having gotten the above noted information from Mr. Garbutt, the plaintiffs sought advice from Mr. MacAdam. Mr. MacAdam in his report to the plaintiffs dated September 25, 1993 commented that Mr. Garbutt appeared to have forgotten or not be aware of the basic design

principles of masonry cavity walls. Mr. MacAdam advised that all masonry structures are porous and that the accepted design principle known as the 'rain screen principle' was to allow water to penetrate into the cavity, run down therein and to be diverted outside by wall flashings and weep holes. Mr. MacAdam further reported that sealing the exterior surface of the brick with silicon should be regarded as a last resort and is rarely very satisfactory, requiring renewal usually about every three years.

[47] The next report received by the plaintiffs providing them with information relating to the problems was that of Inspec-Sol (Ontario) Limited dated September 28, 1993, the contents of which have already been reviewed. It should be noted that Inspec-Sol was retained by the plaintiffs on the recommendation of Mr. MacAdam.

[48] Mr. MacAdam provided another report to the plaintiffs dated October 11, 1993 wherein he gave the opinion that the water was penetrating into the building interior due to the absence of wall flashings, vents or weep holes, as well as the fact that the roof perimeter flashings were neither complete nor as shown on the drawings. Mr. MacAdam recommended drilling weep holes and closing masonry cracks by caulking or pointing and explained that a full and permanent solution would be extensive and expensive. Mr. MacAdam in his report pointed out that while flashings and weepers are specified, none were shown in the drawings nor were any installed and advised the plaintiffs that this would indicate deficiency in both design and in supervision and inspection of the work. Mr. MacAdam was of the view that the cracks in the masonry were probably due to the absence of expansion and control joints, both of which were specified but omitted from the drawings and the work. Mr. MacAdam expressed the view to the plaintiffs that a full and proper and permanent solution would involve extensive demolition and reconstruction and that it would be prohibitively expensive and that he would not suggest it seriously. (It should be noted that in his report dated November 30, 1993 Mr. MacAdam changed his mind and stated that a full, proper and permanent solution involves the complete removal of the outer wythe of bricks and its replacement with new bricks including expansion joints, control joints, flashings and weep and vent holes which should have been incorporated initially.) Dr. Reed took from the report of October 11, 1993 that the roof was probably not involved in the leaking problems. It was Dr. Reed's evidence that when he and Dr. Lawless

received the report of Mr. MacAdam dated October 11, 1993 referring to a full, proper and permanent solution involving extensive demolition and reconstruction, they felt that this is what they were looking for as a remedy, but Mr. MacAdam had provided no details and as a result they asked Mr. MacAdam for details as to what the proper and permanent solution would involve.

[49] In his October 11, 1993 report Mr. MacAdam addresses the removal of the shelf plates and opines that it was a mistake to remove them. I accept Dr. Reed's evidence that at that point in time this was contrary to what the plaintiffs believed and on balance with respect to the entire report from Mr. MacAdam dated October 11, 1993, the plaintiffs were not convinced that what Mr. MacAdam was saying was correct.

[50] Mr. MacAdam's report dated November 30, 1993 was forwarded to the plaintiffs in response to their request for details as to what the proper, permanent solution would involve. In that report of November 30, 1993 Mr. MacAdam provided an alternative to replacement of the exterior masonry which would include removing the windows and cutting expansion and control joints into the existing masonry to limit further cracking, and then applying over the masonry an entirely new and separate cladding of glass and metal.

[51] As it turned out, following Mr. MacAdam's report of October 11, 1993 a severe rainstorm occurred in Kingston on October 21, 1993. Dr. Reed testified that during the storm water came in the windows and through the back and front entrances. It was necessary for him to call his wife to obtain buckets to collect the water. It was his evidence, which I accept, that he later that day went to the roof to examine it and found no pluggage in the drains and no standing water which confirmed his belief and that of Dr. Lawless that the roof was not involved.

[52] By covering letter dated December 23, 1993 the plaintiffs' lawyer provided Mr. Garbutt and Emmons & Mitchell with copies of all of Mr. MacAdam's reports and with information relating to the maintenance of the building. In his covering letter the lawyer asked Mr. Garbutt and Emmons & Mitchell for their reports.

[53] Mr. Garbutt in his letter dated January 10, 1994 acknowledged receipt of Mr. MacAdam's reports and critiqued those reports. It was his position that Mr. MacAdam was in error because the masonry wall was not designed as a cavity or veneered rain screen type, but rather was designed as a solid masonry wall in respect of which all joints were required to be filled solid. Mr. Garbutt in his report said, "The most important aspect is that the collar joint (longitudinal vertical joints between wythes, i.e. brick and concrete block) be filled at all locations." In his report to the plaintiffs' lawyer, Mr. Garbutt took the position that it was important to determine by way of field investigation whether the collar joint had been filled in its entirety or had been filled except for localized areas and if so to ascertain the extent and location of those areas. Mr. Garbutt took the position that "until factual data is obtained from field investigation all reports from this point are speculative and non-contributive." He said as well,

> Any remedial work suggested to essentially convert this wall into a veneer type (weep holes, vents, flashings etc.) are ill advised because of the minimal width of the collar joint (approximately 12 millimetres) facilitating bridging and blockages by mortar droppings in an erratic unforeseeable pattern. The width of the collar joint is well under that which is generally accepted for veneer construction (40-50 millimetres,…).

[54] Mr. Garbutt suggested in his report that should collar joint filling prove to be required the procedure consisted of anchoring the brick to the back-up block wall and pressure grouting the joint. Mr. Garbutt closed his report by saying that it was most important at this stage to either perform the field investigation work or to proceed with external wall coating as suggested in his letter of September 21, 1993.

[55] It was Dr. Reed's evidence, which I accept, that at this point in time, upon receipt of Mr. Garbutt's letter dated January 10, 1994 that Dr. Reed and Dr. Lawless considered that their own consultant Mr. MacAdam could be wrong and could be talking about a different wall system than was actually installed. They realized that Mr. MacAdam had never done more than actually look at the contract documents. They were at this point in time getting conflicting advice. Mr. MacAdam on the one hand was saying the proper procedure would be to turn this into a veneer type wall with weep holes, vents, flashings, etc., and Mr. Garbutt was saying that this would be ill advised. At this point in time having received Mr. Garbutt's report, the plaintiffs did not feel

much further ahead in terms of resolving the problems. As a result, in March of 1994 architect Ernest Cromarty who happened to be a patient of Dr. Reed and who happened to be in Dr. Reed's office at the time, was asked if he would look at the problem, which he agreed to do. Mr. Cromarty convened a meeting with the builder, consultant, masonry and structural steel subtrades in Mr. Cromarty's office on April 20, 1994. As he said in his letter to the plaintiffs dated March 31, 1994 advising of the meeting, "We expect at this meeting which will be held without prejudice to better ascertain how the building was actually constructed and the reasons what and why changes were made to the contract documents."

[56] In due course, on May 17, 1994 Mr. Cromarty reported to the plaintiffs the results of his meetings with the various players, including the general contractor Emmons & Mitchell, the architect Mr. Roberts whom Mr. Garbutt sent to the meeting in his place, and a representative of the masonry subtrade. The report advises in part that with respect to the wall section, the parties did not agree nor seem to know or acknowledge who wrote the specification and why the specification did not relate to the drawings and/or as built conditions. It was Dr. Reed's evidence, which I accept, that this came as a complete surprise to him. Again, it should be noted that the report was received in evidence not for the truth of its contents but simply as part of the narrative and to aid in establishing what the plaintiffs knew with respect to the problem at any given time. The report goes on to advise that although the specifications called for a cavity type wall with flashing and weep holes, the structural consultant claims that the building was designed for an exterior wall of solid masonry. Mr. Cromarty in his report advises that the drawings tend to indicate solid wall construction without cavity flashing, cavity weepers or flashing at lintels.

[57] Mr. Cromarty in his report recommended that before any remedial work is carried out, tests should be conducted to determine the following:

(a) the porosity and water absorption qualities of the existing face brick;

(b) if both the existing vertical and horizontal collar joints in total are fully grouted;

(c) whether there is a brick veneer only at the steel beams at each floor with no masonry back-up beam infill;

(d) if the eight inch by ¼ inch continuous steel plate is in fact continuous in part or total;

(e) if and where an existing vapour barrier occurs; and

(f) the type and location of anchorage provided between interior and exterior wythes and whether the wythes are anchored to each other and to the building structure proper.

At or about the same time, Mr. Cromarty provided advice and rough costings for several types of remedial methods.

[58] To further confuse matters for the plaintiffs, one Paul Desroches, President of Paul Desroches Construction Limited appeared at Dr. Reed's office on May 19, 1994 and provided certain views which are confirmed in his letter to Dr. Reed dated May 20, 1994. Again, the letter was received into evidence not for the truth of its contents but as part of the narrative and to ascertain the nature of the information that the plaintiffs were receiving. Mr. Desroches advised Dr. Reed that the roof was not tied into the wall at the perimeter of the roof/wall interface, and that the roofing was deficient at various locations at the base of the parapet curb. Dr. Reed noted that this information was contrary to Inspec-Sol's report. As well, Mr. Desroches advised that the brick was excessively porous and that coupled with the sub-standard wall assembly facilitated the infiltration of moisture into the finished areas of the building. To Dr. Reed, this information was somewhat consistent with the information coming from Mr. Garbutt. In due course, Mr. Desroches provided Dr. Reed with estimates for various alternative remedial works.

[59] On June 20, 1994 Mr. MacAdam provided to the plaintiffs a detailed response to Mr. Garbutt's letter dated January 10, 1994, taking great and detailed issue with Mr. Garbutt's various positions.

[60] Mr. Garbutt in his report dated January 10, 1994 stressed the importance of field investigation to determine whether the collar joint had been filled in its entirety and stressed that until factual data was obtained from field investigation, all reports were "speculative and non-

contributive". By the same token, architect Ernest Cromarty in his report dated May 17, 1994 recommended extensive testing before any remedial work be carried out.

[61] In August of 1994 the plaintiffs retained the services of architect Paul Johnston to inspect, evaluate, and provide proposals for remedying the roof and wall systems. At about the same time, on August 19, 1994 the plaintiffs' lawyer wrote to Emmons & Mitchell and Mr. Garbutt. The letter reads in part as follows:

> Further to the letter dated January 10, 1994 from Mr. Garbutt, my clients have now reached a stage where they feel that they have sufficient information regarding the cause or causes of the water problems at their building at 275 Queen Street, the alternatives for remedial work and cost estimates for such work, and they would like to proceed with remedial work as soon as possible. My clients would like an opportunity to meet with Mr. Garbutt and Emmons & Mitchell with or without the parties' respective lawyers and their technical advisors present, in order to present the information which they have gathered, and to determine if a settlement can be reached. …

[62] Mr. Johnston thereafter, with the assistance of roofing expert Robert Kendall undertook an onsite investigation of the roof, parapets and walls, and in due course reported the results of his investigations and recommended certain remedial steps. In due course his various reports were combined into three folios titled FOLIO 1, Building Envelope Review and Analysis, General Commentary; FOLIO 2, Building Envelope Review and Analysis, Roof and Parapet Design; and FOLIO 3, Building Envelope Review and Analysis, Wall Assembly and Design.

[63] In due course the plaintiffs instituted action against Mr. Garbutt and Emmons & Mitchell by statement of claim issued May 17, 1995.

Findings of Fact

[64] I turn now to my findings of fact in this case. I have carefully reviewed all of the evidence which I dictated from my notes on a daily basis throughout the course of the trial and which evidence is attached as Addendum 'A' to these Reasons. The case has been a difficult one in many ways but primarily because of the conflicting opinions from the various experts as to the nature of the problems in the building as constructed, and the nature of the appropriate steps that

should be taken to remedy the problems. In arriving at my conclusions I have relied heavily on the observations and opinions of the plaintiffs' experts, architect Paul Johnston, roofing expert Robert Kendall, and structural engineer Robert Drysdale. It was not simply a matter of deciding to choose the evidence of the plaintiffs' witnesses over that of the defendants' witnesses. Rather, my review of the evidence as a whole and a comparison of the competing views led me comfortably to the conclusion that the evidence of the plaintiffs' experts to the extent that it consisted of site observations, was accurate and credible, and to the extent that it consisted of opinion evidence had a logical base and was technically correct. For example, Dr. Drysdale's reply evidence addressing certain calculations made by the defendant Emmons & Mitchell's structural engineer Mr. Roney, in my view demonstrated clearly, for the reasons given, that Mr. Roney's analysis of the structural integrity of the exterior masonry walls was incorrect and could not be used as the basis for remedial measures. As well, it must be remembered that Mr. Johnston and Mr. Kendall were retained in 1994 by the plaintiffs to investigate the problems that had hounded this building from its completion in 1986 and to determine appropriate remedial measures to once and for all solve those problems. They were, as I have mentioned earlier, akin to the "treating physicians".

[65] With respect to the evidence of Dr. Reed and Dr. Lawless, they both gave their evidence in a credible fashion and I have no reason to disbelieve anything that they testified to during the trial of this action.

The Roof and Parapet

[66] The Specifications, Division 7 – THERMAL AND MOISTURE PROTECTION, section 2 under the heading Roof Drainage clearly required that for an inverted roof a two per cent slope to the drains was required. Ours was an inverted roof and no such slope was provided. This was contrary to the Specification, section 2 and also contrary to section 1 of the Specification which required as the standard for the job adherence to CRCA Roofing Specifications. This constituted a breach of contract.

[67] Not only was the two per cent slope not in existence but the evidence disclosed that the roofing subcontractor Quintal and England installed a negative “back slope” running to the parapet where water collected and pooled at the cant, resulting in deterioration of the membrane.

[68] Because of faulty workmanship a repair to the northwest corner of the roof was required and affected in late 1994. As Mr. Johnston and Mr. Kendall described in their evidence, between the horizontal surface of the roof slab and the vertical inside surface of the parapet, a cant strip is placed which from a side view is a triangular shaped strip that “softens” the angle between the two surfaces. As Mr. Kendall explained, the split or tear observed in the photographs in Mr. Johnston’s report, Folio 2 page 15 is at the top of the cant. The problem was caused by the cant not being placed so as to abut the vertical inside wall of the parapet, thus allowing a space between the vertical surface of the cant and the vertical inside surface of the parapet. This provides a space between the cant and the vertical inside surface of the parapet where there is no support for the membranes that are being run up the side of the parapet. As it turned out, contrary to what was called for in the specifications, only a single felt was brought down along the side of the parapet to overlap the horizontal membranes. Whereas this should have been a very strong joint, in the circumstances, and in the absence of proper support the joint was very much weakened and as a result the bitumen broke away in a “potato chip fashion” when very slight pressure was applied to the rupture. Another problem with the northwest corner of the roof was that it was negligently built with a ridge that served to isolate the corner from the closest drain, so that the northwest area remained totally saturated and subject to break-down of the membranes.

[69] As well, Emmons & Mitchell through its roofing subcontractor was negligent in permitting widespread adherence between the hot asphalt of the roofing membrane and the insulation above it. Both Mr. Kendall and Mr. Johnston were very clear in their evidence, which I accept, that such adherence was contrary to the recommendations by Dow Chemical which were in place by early 1980. Dow Chemical was the supplier of the Roofmate system used by Emmons & Mitchell on this job. The difficulty caused by adherence between the roofing membrane and the insulation is that with the collection of rainwater on the roof, the insulation

has a propensity to float and as it does so it tears the underlying membrane that it is attached to. In due course, it was determined as a fact that this had happened.

[70] In my view, both Emmons & Mitchell's job superintendent James Sinclair and Mr. Garbutt failed in their contractual responsibilities to clarify the roofing requirements and to supervise and review the work of the roofing subcontractor in order to ensure that roofing deficiencies as did occur, did not occur. Mr. Garbutt had purposely designed a flat concrete roof to allow for the possible addition of two additional stories to the building, and knew or should have known that the irma roof that he specified required a two per cent slope and that that slope could be built into the insulation. He was negligent in not ensuring that such a slope was built into the roofing system.

[71] The parapet above the roof surface around the perimeter of the building walls was badly constructed due to breaches of the requirements of the plans and specifications and negligent construction practices. As a result the parapet had to be completely rebuilt when the roof was replaced in 1998.

[72] The third "small" drawing prepared by Mr. Roberts on Mr. Garbutt's behalf, which drawing constituted part of the Contract Documents, called for the vertical distance between the top of the roof ballast and the top of the parapet to be eight inches. As built, it was only about two inches which was insufficient to allow for a two per cent slope of the roof to drains.

[73] Division 7 of the Specifications – THERMAL AND MOISTURE PROTECTION, section 3.6.1 required in part as follows, "Construct membrane flashings at all junctions of the roof with vertical surfaces with the same bitumen and roofing felts as used for the roof application. Carry four ply felt and asphalt membrane flashings up the surface of the wall or curb to a point not less than 200 millimetres (eight inches)…" As built, there were only one or two roofing felts, depending on the location, running up the inside of the parapet from the roof membrane. As earlier noted, in many places the cant strip was not placed so as to abut the vertical inside wall of the parapet, thus allowing a space between the vertical surface of the cant and the vertical inside surface of the parapet and thus resulting in even more weakening of the deficient membrane at this point.

[74] As a result of poor construction practice and contrary to the CRCA Roofing Specifications which set the standards for the roofing work, the horizontal bituminous membrane generally ran only from one-half to two-thirds of the distance across the horizontal face of the top of the parapet so as to leave the plywood beneath the metal coping exposed to any water entering into the parapet, resulting in deterioration of the plywood which is shown in Mr. Johnston's photographs.

[75] I accept Mr. Kendall's evidence that the felt membrane should have been extended completely across the top of the parapet and a distance down the outside of the brick wall, and the failure to do so was a fundamental error on the part of the roofing subcontractor. As Mr. Kendall pointed out, this was particularly so having regard to a large gap which existed between the brick wall and block wall, which gap should not have been there in the first place and constituted negligent building practice in its own right. As Mr. Kendall explained, the staining and rottenness of the plywood shown in photographs 2, 3 and 4 of Mr. Kendall's report of October 11, 1994, was caused by water being pushed through the joints of the top metal cladding and from the plywood topping the water would then go down the space between the brick and the blocks.

[76] As a result of negligent construction, the top metal cladding of the parapet extended only one-half inch below the top of the brick, allowing for water during a rainstorm to be blown up between the lower edge of the metal cladding and the outside wall of the brick. As Mr. Kendall explained in his evidence, proper construction practice would be the installation of a two-inch starter strip extending along the top of the brick wall to receive the lower edge of the top cladding. Again contrary to good building practice, on both sides of the parapet the two-part flashing was not interlocked in a water shedding arrangement, again allowing wind driven water to be blown into the assembly.

[77] Contrary to good building practice and contrary to CRCA Roofing Specifications, the top coping was constructed flat rather than sloped towards the roof in order to shed rain water into the roof basin.

[78] Contrary to Code requirements, there were no anchors or ties installed to attach the parapet wall to the interior block system, and this for a considerable distance below the bottom of the parapet.

[79] The emergency roof work carried out in late 1994 was clearly required as a result of design and construction deficiencies. That work was confined primarily to the northwest corner of the roof. The work was done by Oliver Brothers who invoiced the plaintiffs a total of $20,986.30.

[80] In my view it was completely reasonable for the plaintiffs to rebuild the roof and parapet system in 1998 at a net cost of $101,561.64 (after deducting from the total re-roofing contract price of $106,092.64, the sum of $4,531 which constituted an upgrade to the mechanical units). It was clearly evident in 1994 through the investigations of Mr. Johnston and Mr. Kendall and through the emergency repair work that had been conducted primarily in the northwest corner, that the roof was defective and would have to be totally replaced. Mr. Johnston, whose evidence I accept, concluded that the lack of positive drainage and the insulation adherence to the top membrane had impaired the life expectancy of the entire roofing system. Mr. Johnston in his report dated April 20, 2002 in response to Mr. Wilson's observation that the design and construction of the main roof membrane was never a contributing factor in water infiltration, pointed out that Mr. Kendall, a very experienced roofing inspector, had carefully inspected and found the parapet flashings deficient throughout the roof area, and that they were indicative of flashings that were in his opinion inappropriate, and deficient elements of practice that in other applications had led to failure. Mr. Johnston pointed out that substantial justification to suspect the entire roof application, substrate and standard of construction was evident from the visible and total failure of flashings in the northwest corner. As well, Mr. Johnston pointed out that deterioration of the main roof membrane was identified and confirmed during its removal in 1998. In all, Mr. Johnston was of the opinion, which I accept, that the re-roofing of the main roof was overdue considering the number of faults in detail, slope, standards compliance, specification, competent substrate and field quality that he found on investigation. He was of the view that deterioration of the built-up layers of felt and bitumen had already begun in 1994

adjacent to the more exposed areas at the parapet perimeter, and again that this was corroborated in the subsequent removal.

[81] Mr. Kendall's, whose evidence I accept, clearly found the roof deficient and constructed contrary to CRCA Standards. Mr. Kendall, in his report and in his *viva voce* evidence commented that this inverted roof had been assembled flat and the retention of water over long periods of time had the effect of weakening the tensile strength of the felts, and that condition may well shorten the service life of the roof. At the time of the repairs in 1994 the roof was some eight years old. It was Mr. Kendall's evidence that based on his observations it was difficult to say how much further life it had beyond eight years and bearing in mind that construction of a roof is usually consistent through its entirety, when it begins to leak it is a very bad sign and it is usually the beginning of the end. Here, two places had been breached, the roof hatch and the west end, by 1994. As a result, according to Mr. Kendall's evidence, which I accept, as of 1994 one would have to think seriously about replacing the entire roof. Mr. Kendall in his evidence, explained that when he viewed the roof in 1994 his immediate reaction was to repair the west end and rework the parapet detail, but his gut feeling as expressed to Mr. Johnston was that the roof was reaching the end of its life and that if it was his building, he would start replacing it within the next two or three years. His main concern would be a major failure of the membrane which would let significant amounts of water into the building.

[82] I accept Mr. Kendall's evidence that properly designed and constructed, an irma roof should have a life span of approximately 30 years. By the time of its replacement, the roof had been serving the plaintiffs for some 12 years. It was however not trouble free service, nor was it the roof that the plaintiffs had bargained and paid for.

The Wall

[83] The wall system as built represented a "bastardized" or hybrid system that incorporated a narrow cavity between the brick wythe and the block wythe associated with a typical veneer wall, but without the flashings and weepers needed to drain such a cavity. Flashings and

weepers were in fact called for by the specifications which specifications came from Mr. Garbutt's office, were written by his specification writer, and were approved by Mr. Garbutt.

[84] Mr. Garbutt throughout this litigation has maintained that it was his intention that a solid wall be built with the collar joint between the two wythes filled with mortar.

[85] I agree with plaintiffs' counsel's submission that the bastardized wall system resulted from a negligent lack of communication and ineffective site supervision and review among the mason Abe Dick Limited through its manager Primo Santin, Emmons & Mitchell's site superintendent James Sinclair, and Mr. Garbutt. On his own evidence, Mr. Santin had in his possession only one schematic plan of the wall system that clearly was at too small a scale to show the intended mode of construction. Again on his own evidence, Mr. Santin did not have in his possession during the course of construction either the specifications or the three small drawings of Mr. Roberts which were detail of the wall section drawings. One wonders how the masonry subcontractor could have undertaken to build these walls without seeing those drawings and specifications. The masonry subcontract executed in September of 1985 clearly notes that the obligation of Abe Dick Limited was to build "in accordance with the plans and specifications… prepared by Garbutt P.Eng.". Even at that it is interesting to note that Mr. Roney for Emmons & Mitchell testified that the Roberts's wall section drawings were not clear and that one would have to look at the specifications for further guidance. He was of the view that consultation between the contractor and the consultant would have been reasonable in the circumstances. In any event it is clear that the mason Santin assumed that a narrow cavity or veneer wall system was intended and as a result constructed the walls with a gap between the brick and the block wythes.

[86] Mr. Sinclair testified that he assumed that what was supposed to be constructed was a "solid wall with a finger width of void between the brick wythe and the block wythe", which at first glance appears to be a contradiction. Mr. Sinclair explained in his evidence that the finger width of void between the two wythes resulted from the fact that the bricklayers had to hold on to the top of the brick in order to lay it. Interestingly, Mr. Sinclair in his evidence testified that he was certain that at no time did he see the three small Roberts's drawings which are the only

drawings that might be interpreted to suggest a gap between the two wythes. In any event, Mr. Sinclair never did discuss with Mr. Garbutt what Mr. Garbutt wanted. It was Mr. Sinclair's evidence that it never occurred to him to actually fill the void between the block and brick wythes of the subject wall and acknowledged that he never realized that Mr. Garbutt intended the collar joint to be filled.

[87] It is clear on the evidence however that Mr. Garbutt intended a solid wall with the three-quarter inch collar joints shown on the three small drawings and as required by the specifications being filled with mortar. In my view, if that is what Mr. Garbutt intended, then he was negligent in failing to clearly indicate on the plans or in the specifications that this was necessary. As well, in all of the circumstances Mr. Santin and Mr. Sinclair were negligent in failing to seek clarification from Mr. Garbutt as to his design intent and in failing to seek the clarification that was required, contrary to General Condition 25.5 which required the contractor to review the Contract Documents and promptly report to the consultant any error, inconsistency or omission that he might discover.

[88] During the course of his evidence, Mr. Garbutt attempted to justify the wall as built as qualifying as a panel wall under the Code. In my view, there is no merit in that position. Dr. Drysdale in his evidence, which I accept, pointed out that the wall thickness of 210 millimetres inclusive of the brick and block wythes and the specified cavity would be insufficient for the bottom story where 290 millimetres was required for panel walls under the Code. As well Dr. Drysdale pointed out that the lack of horizontal movement joints at the shelf plates and beams resulted in the walls being load bearing, whereas panel walls under the Code must be "non-load bearing exterior masonry walls". As well, the 600 millimetres of additional brick opposite the joists and steel beams resulted in the wall not being "solid masonry" over the distance, again contrary to the requirement for a panel wall under the Code. Finally, Dr. Drysdale pointed out that in any event even if this were to be defined as a panel wall, then flashings and weep holes would still have been required over the windows in accordance with s.3.16.1 of the Code M-78 and in fact none were installed.

[89] The evidence with respect to the type of ties that were actually used to bond together the brick and block wythes is extremely confusing. From Mr. Sinclair's evidence it would appear that he believed that "ladder" truss type reinforcing was to be used only within the block wall itself and understood that the masons were using only flexible strap ties to tie together the inner block wythe and outer brick wythe. It would appear as well from Mr. Sinclair's evidence that he knew that the use of such flexible strap ties was inappropriate for the bonding of the two wythes in a true cavity or true veneer wall. According to Mr. Santin's evidence, he understood that Mr. Sinclair had told him to use "ladder" reinforcing to bond together both wythes where the coursing of the brick and block permitted, and elsewhere to use flexible strap ties. It would appear that Mr. Garbutt was content with the use of strap ties because he believed that the cavity was being filled with concrete to create a solid wall. This much is clear from the evidence, and that is that in fact where ties were used to bond the two wythes, flexible strap ties were used almost exclusively and were, where necessary, bent between the offset courses of brick and block, often with the tie sloping from the brick wythe <u>down</u> to the block wythe. This type of tie system was structurally inadequate.

[90] Much was said during the course of this trial with respect to Mr. Garbutt's role as the consultant on this building project. The definition section of the construction contract provides that the Consultant is the person, firm or corporation identified as such in the Agreement, and is an Architect or Engineer licensed to practice in the province or territory of the Place of the Work, and is referred to throughout the Contract Documents as if singular in number and masculine in gender. The Agreement identifies Mr. Garbutt as the Consultant.

[91] General Condition 3.4 provides as follows.

> The Consultant will visit the site at intervals appropriate to the progress of construction to familiarize himself with the progress and quality of the Work and to determine in general if the Work is proceeding in accordance with the Contract Documents. However, the Consultant will not make exhaustive or continuous on-site inspections to check the quality or quantity of the Work.

[92] Although Mr. Garbutt when billing the plaintiffs for his services stipulated in his invoice that 25 per cent of his fee was earned for what he termed as "supervision", Mr. Garbutt in his

evidence made it clear that from his point of view it was Mr. Sinclair's responsibility as the General Contractor's Superintendent to supervise the project, and that Mr. Garbutt did not supervise the Work as such. Mr. Garbutt in his evidence appeared to feel that the guideline which he included in his materials, titled Professional Engineers Providing General Review of Construction as Required by the Ontario Building Code fairly set out the appropriate performance standards of the professional engineer who has undertaken a general review of construction. Performance Standard No. 1 set out in that document provides as follows:

> 1. The Professional Engineer with respect to the matters that are governed by the Building Code, shall,
>
> i. make periodic visits to the site to determine, on a rational sampling basis, whether the Work is in general conformity with the Plans and Specifications for the building.

[93] In terms of Mr. Garbutt's obligations to the plaintiffs as their consultant under the contract, perhaps Dr. Reed put it at its simplest when he testified that although the plaintiffs had no written contract with Mr. Garbutt, they understood that he would design the building and thereafter make sure that the building was properly constructed by carrying out inspections during the course of construction. I accept that that was their understanding of the agreement and Mr. Garbutt did not take a contrary position.

[94] While the evidence discloses that during the course of construction Mr. Garbutt did visit the site once or twice a week, it is clear that he breached his obligations as consultant to the plaintiffs by negligently failing to observe and recognize as such the several construction deficiencies that were being incorporated into the building, including the fact that contrary to his design intention a "non-solid" wall was being constructed and constructed without weepers or flashings, the steel support plates were cut off on the east and west sides of the building at each floor level, at each floor level on the north and south sides of the building a hard joint was installed between the top of the brick course and the underside of the shelf plate, vertical control joints were not installed in the wall as required, the appropriate type and number of ties were not being installed at the appropriate locations, the parapet was being constructed contrary to the Contract Documents and contrary to good building practices, and the roof itself was being

installed contrary to Mr. Garbutt's own specifications and contrary to the requirements of the manufacturer Dow. However one defines Mr. Garbutt's role as Consultant, he was negligent and in breach of his contract with the plaintiffs in failing to spot and have remedied these defects. It goes without saying that Emmons & Mitchell which was responsible for the work of its sub-trades negligently breached its contract with the plaintiffs by permitting their building to be constructed with these deficiencies which deficiencies rendered the building structurally unsound, causing it to leak during rain storms when a properly constructed building would not have. It is to be noted that many of the problems with this building were contributed to by Contract Documents that were less than clear and in some case ambiguous, and by the failure of Emmons & Mitchell on the one hand and Mr. Garbutt on the other to in any way consult with one another for the purpose of resolving the ambiguity and lack of clarity when it was obvious that such consultation was required and would have been reasonable in the circumstances.

[95] It should be pointed out that throughout the trial it was suggested by the defendants when the opportunity arose that somehow Drs. Reed and Lawless were significantly involved in and responsible for the design and construction of this building, and that, as such, to a large extent were authors of their own misfortune. I reject that position. The fact of the matter is that they had no construction knowledge or experience. They had hired a professional engineer of good repute and in good standing to design their building and to act as Consultant during its construction. They had entered into a contract with a reputable construction company that had been in business in the City for many years to build their building. It was entirely normal for them to take a keen interest in the building as it went up. It was the first time they had been involved in such a project and they were investing a great deal of money in the project. While they did in fact supervise some of the changes to the project the changes were not structural in nature but rather were of a type that any interested owner would be capable of supervising. While they knew that shelf plates were being cut from some of the beams, they had no idea of the structural significance of that cutting. I find as a fact that they did not veto the installation of more control joints as was suggested by a defence witness. I accept Dr. Reed's evidence in that respect and find that again he and his partner had no real understanding of the significance of such control joints. Much was said during the course of the trial about the type of brick that had

been chosen by the plaintiffs and the choice of stone surrounds at the windows. I find as a fact that the plaintiffs were never told by either Mr. Garbutt or any representative of Emmons & Mitchell that the choice of brick and/or stone surrounds might contribute to problems with the building. It was the responsibility of Mr. Garbutt and Emmons & Mitchell to consider the choices being made by the plaintiffs and to advise the plaintiffs whether those choices could be incorporated into the building without problems. They were never cautioned in that respect and I find that if they had been so cautioned they would have bowed to the recommendations of Mr. Garbutt and Emmons & Mitchell. In all, there was no conduct on the part of the plaintiffs that in any way caused or contributed to the construction of a defective building.

[96] When one begins to list the various wall system deficiencies, one wonders how this building could have been designed and consulted on by an otherwise competent professional engineer and constructed by an otherwise competent building contractor. With Mr. Sinclair's diary gone, the as-built drawings gone, and the City of Kingston building file having disappeared, we have very little written record of what went on during the course of construction apart from the Site Meeting Minutes where the plaintiffs sat in. We have however some hint of site supervisor Mr. Sinclair's frustration during the course of construction from his discussion with Mr. Garbutt during the course of cross-examination, wherein Mr. Sinclair agreed with Mr. Garbutt's comment that on site Mr. Sinclair had not been given a full deck of cards to deal with. Mr. Sinclair admitted to Mr. Garbutt that he was so sick of the job at one time during the course of construction that he wanted to quit the job but had never done such a thing before. He described the overall construction project as a haphazard way of doing business and that "a lot of things happened right from the beginning". He testified that everything was very difficult in the field.

[97] Whatever the details of what was happening on site during the course of construction, we do know that the wall system was ultimately constructed with the following deficiencies.

[98] The gap between the brick wythe and the block wythe existed from the bottom to the top of the walls on all four sides, and ranged from about three-quarters of an inch (19 millimetres) to one and a half inches (38 millimetres) and more in some locations. Indeed, Mr. Johnston found

the gap to be close to four and a half inches at one location at the parapet level. The gap existed notwithstanding Mr. Garbutt's intention that the wall be designed as a solid wall and varied in dimension notwithstanding that the Specification required the mason to "maintain continuous 19 millimetre air space behind brick throughout project". As previously noted, notwithstanding the gap between the two wythes and the fact that rainwater was entering into the wall cavity through visible and invisible cracks in the brick mortar joints, and through locations at the parapet, there were no flashings and weep holes installed in the wall to direct such moisture to the outside of the building.

[99] While it is not possible to be certain on this issue without removing the entire brick wythe, I am satisfied on the basis of the "sample" evidence that is available as a result of various test openings into the wall that it is a reasonable likelihood that there are an insufficient number of proper ties installed in this wall to bond the brick wythe to the block wythe, and that there was a <u>total lack of ties</u> of any type for an extended distance of about six feet in height near the top of the walls where one would expect structural stability to be particularly important. The area opened by Mr. Johnston in December of 1994 on the west wall below the third floor disclosed ties at every other block course which varied between some ladder ties (which are intended to be installed <u>within</u> the block wythe) and simple corrugated strap ties which were bent to accommodate the "off coursing" between the brick wythe and the block wythe. I accept Mr. Johnston's evidence that the ties, being bent, would not provide a direct horizontal transfer of any wind pressure between the brick wythe and the block wythe. It is worthwhile to quote from Mr. Johnston's Folio 3 (Exhibit 56), page 23 where he concluded the following:

> In general we found large areas of wall that had neither consistent anchorage nor suitable masonry ties and, as discovered in several areas, there were no ties installed at all. This is conclusively a veneer wall construction, and with only a 90 millimetre (four inch) concrete block inner wythe, the wall must interact through 'cross ties' to effectively share the wind load and transfer it on to the structure without excessive and damaging flexure. This movement of the entire wall in membrane fashion exposed to natural wind forces is of course more a concern on the east and west walls, where the shelf plates were removed during the construction.

[100] It would appear on the evidence that because of the coursing discrepancy between the brick wythe and the block wythe, the appropriate tie to use for the purpose of bonding the two wythes would have been an adjustable tie which is designed for that very situation, and which were available as a standard construction device in 1985-1986. The Specifications however did not call for adjustable ties and even Mr. Roney comments in his evidence that the contractor would not have known the designer's intent but could conclude that the strap ties were satisfactory. One wonders why the contractor would make such an assumption, rather than simply asking the engineer what he actually intended.

[101] Many of the strap ties were observed to be sloping down from the brick wythe to the block wythe, acting as bridges from the outer brick through and into the inner wall and inside finishes, all contrary to good construction practice. As well, many of the ties were found to be significantly corroded and rusted by the rainwater that had entered the cavity. Again, it is a reasonable likelihood that this condition seen by Mr. Johnston in 1994 and in the tie that was actually taken from the wall and entered as an Exhibit is fairly representative of the ties as they exist throughout the entire wall. It is likely that the situation with respect to rusting ties is even worse today than it was in 1994.

[102] Investigations by both Mr. Johnston and Mr. Roney disclosed that the masons when taking the two wythes up had negligently allowed fins of mortar to squeeze out from the mortar joints of the brick wythe, which fins sometimes crossed the gap between the two wythes. These fins were not purposeful but rather were as a result of sloppy work on the part of the masons. In my view the fins served only as another transfer mechanism for water coming into the cavity to be transmitted to the block wall and then to the interior of the building. I am not satisfied that the fins served any structural function in terms of stabilizing the wall as suggested by the defence, nor am I satisfied that mortar droppings which in some cases fell to the top of the strap ties added any structural integrity to those ties which were deficient in and of themselves and were made even more deficient by being bent up or down between the brick coursing and the block coursing.

[103] As previously noted, although the wall was constructed with a gap between the brick wythe and block wythe varying from three quarters of an inch to one and a half inches and even greater in localized places, there was a complete absence of flashings and weepers as called for by the specifications, and as required by the Design Code M-78. As Mr. Johnston noted in his Folio 3, Exhibit 56, page 24, "On its own, the lack of flashing anywhere in the building envelope was without doubt the single most convincing reason that any incursive water cannot be deflected back out to the exterior." Mr. Johnston is critical, and I believe justifiably so, of Mr. Quirouette's approach that totally disregards the provision of flashings and weep vent holes as originally specified, and that fails to recommend such flashings and weep holes in the remedial work suggested by him. Further, I accept the evidence of Dr. Drysdale who disagreed with Mr. Garbutt's suggestion that the steel shelf angles on the north and south sides could be treated as a "flashing" because the angles do not extend to the outside of the brick as would proper flashings. I also accept Dr. Drysdale's evidence that the Building Code would not have allowed one to eliminate weep holes and flashings, even if this had been a solid wall as apparently Mr. Garbutt intended it to be. Nor did the many mortar fins squeezed between the two wythes at the brick mortar joints constitute this to be a solid wall.

[104] It seems clear on the evidence, because of the observed variance in the width of the gap between the brick wythe and the block wythe over the height of the building, that one of the elements, either the brick wythe, the block wythe, or the structural steel frame was constructed out of plumb. While it is not necessary for the purpose of these reasons to make a finding of fact as to which of the elements is in fact out of plumb, it is interesting to note the conflicting evidence on the issue. Mr. Santin testified that his foreman had told him during the course of construction that the structural steel was out of plumb. Mr. Santin went to Mr. Sinclair and pointed this out to him and Mr. Sinclair said that it was too late to do anything. Mr. Sinclair denied that such a conversation took place, and testified that he was personally responsible for seeing that the steel was installed plumb and that in fact it was.

[105] There is no doubt that the shelf plates which were called for by Mr. Garbutt's drawings and which were in fact installed by the structural steel supplier, were required by the

Ontario Building Code at each of the four floor levels, in order to support the bricks above. Section 6.2.3 of Code M-78 (Exhibit 47) provides as follows:

> 6.2.3 Unit masonry veneer more than 11 metres above the top of the foundation wall shall bear on masonry, concrete or other non-combustible bearing supports spaced not more than 3.6 metres vertically.

As it turns out the brick wythe resting on the foundation is more than 12 per cent over the maximum height of 11 metres (36 feet) allowed by the Ontario Building Code. I reject Mr. Roney's interpretation of s.6.2.3 which would require shelf plates only at the fourth floor level, rather than all floor levels as was intended by Mr. Garbutt, as testified to by Mr. Johnston and Dr. Drysdale, and as a fair reading of s.6.2.3 would lead one to conclude. In fact the shelf plates were, during the course of building construction, cut off from the east and west sides of the building in order to resolve a coursing problem. As Mr. Sinclair explained in his evidence, when the brick came up to the plates on the east and west sides of the building, the top course would not fit under the plates. It was Mr. Sinclair's evidence that he spoke to Mr. Garbutt about the problem, and Mr. Garbutt attended at the site and gave him verbal permission to have the steel supplier cut the shelf angles off the beams on the east and west sides. Mr. Garbutt testified that no such conversation took place and that at no time did he ever authorize the removal of the shelf plates. It was his evidence that to do so would have meant that he was authorizing a breach of the Ontario Building Code requirements as implemented in his stamped drawings on file with the Kingston Building Department, and to do so could have "cost him his ticket". The removal of the shelf plates at each of the floor levels on the east and west sides of the building was contrary to the Ontario Building Code and contrary to good building practice, and had an adverse structural impact on the east and west sides of the building. The removal of the plates would have been a very significant piece of work. It would have taken a substantial effort of time and cost to cut the plates from the beams with an acetylene cutting torch.

[106] There is no written record of the alleged conversation between Mr. Sinclair and Mr. Garbutt. Again, Mr. Sinclair's diary has been disposed of, and there are no as-built drawings. Having regard to all of the evidence, I have concluded that it is not likely that Mr. Sinclair asked for and obtained from Mr. Garbutt permission to remove the shelf plates. As early

as July 8, 1999 during the course of his examination for discovery, Mr. Mitchell was unable to confirm that anyone from his company went to Mr. Garbutt to ask for permission to remove the shelf plates. On the same date, July 8, 1999, Mr. Mitchell was asked, if he later did take that position, to provide full particulars as to any such conversation, which he undertook to do. In the face of that undertaking, on May 9, 2002 a few days before the start of the trial, defence counsel wrote to plaintiffs' counsel answering the undertaking with the words, "…it is our position that we were instructed by Mr. Don Garbutt to cut the shelf angles." This was an important issue for the defendant Emmons & Mitchell, and one would have thought that if there had been such a conversation between Mr. Sinclair and Mr. Garbutt on the point, then full details of that conversation could have been and should have been given to plaintiffs' counsel in the three years that elapsed between the time of the undertaking and the terse note to plaintiffs' counsel. As well, it would appear on the evidence that the problem presented by the shelf plates could have been resolved on site without a significant money investment, and in a fashion that would have maintained the Code requirement. I find as a matter of probability that there was no discussion between Mr. Sinclair and Mr. Garbutt as alleged by Mr. Sinclair and that as a matter of probability Mr. Garbutt never gave permission to have the shelf plates removed on the east and west sides of the building.

[107] It should be noted that had the contractor followed the requirements of the Specifications and the General Conditions, it is likely that the problem with the interfering shelf plates on the east and west sides of the building would have been detected and rectified before the steel was manufactured and installed. Division 5 of the Specifications – METALS, s. 7 thereof, provides as follows, under the heading <u>Field Requirements</u>:

> 1. Before beginning fabrication, this contractor shall field check all dimensions, and shall report any major variance from the plans to the Contractor's Consultants.

General Condition 34.3 provides as follows:

> Prior to submission to the Consultant the Contractor shall review all shop drawings. By this review the Contractor represents that he has determined and verified all field measurements, field construction criteria, materials, catalogue numbers and similar data or will do so and that he has checked and co-ordinated

> each shop drawing with the requirements of the Work and of the Contract Documents. The Contractor's review of each shop drawing shall be indicated by a stamp, date and signature of a responsible person.

[108] As well and in any event with respect to the removal of the shelf plates, the Contractor is in breach of General Conditions 11 and 12 of the Contract in failing to obtain a written change order relating to their removal.

[109] While Drs. Lawless and Reed were generally aware at the time that some shelf plates were being cut off the spandrel beams, they had no idea of the structural significance of the cutting, nor in the circumstances would they be reasonably required to make any enquiries with respect to the removal of the shelf plates. On the other hand, accepting Mr. Garbutt's evidence that he did not know that the shelf plates were being or had been removed, and did not learn of their removal until well after construction was completed, it goes without saying that he was in breach of his duties as Consultant and was negligent in failing to determine during his attendances at the site either that the shelf plates were in the process of being removed or had in fact been removed, thus rendering the building structurally deficient.

[110] It is of interest to note that there is a simple enough requirement in the job specifications which if followed likely would have avoided the problems with the wall system, including the question of the gap, the type and spacing of the ties and the absence of weepers and flashings. The Division 4 – MASONRY specification, s. 9 under the heading <u>Job Mock-Up</u> provides as follows:

> Construct sample brick wall of mortar and brick. Units to show colour range, texture, bond, jointing, for approval. Remove walls not approved and reconstruct until approval is received. Retain approved wall until project completion. Remove when directed. Only masonry matching approved sample will be acceptable on the project.

[111] I find as a matter of fact notwithstanding Mr. Sinclair's evidence that he was aware of the specification requirement for a mock-up and that he had been told by Mr. Emmons that in fact a separate mock-up wall had been built, perhaps in the Abe Dick yard by Mr. Santin, that no such separate mock-up wall was ever constructed. Arguably, the first portion of the

lower wall that was rejected because of the grossly out of plumb soldier course bricks, could constitute a "job mock-up", but clearly no specific approval was sought or given as required by the specification after the original wall was removed and replaced with a new wall.

[112] As previously pointed out, Specification Division 4 – MASONRY s.2.2 required the Contractor to "provide control joints as dictated by structural design." It seems not in issue that this particular Specification referred to vertical control joints which were required by the Specifications to be placed for the purpose of reducing unnecessary stressing of the wall which would result in cracking such as did in fact occur. One of the problems was that the drawings did not show any control joint locations and that in fact only one vertical expansion joint was built into the south face of the building. While it was suggested during the course of the defence evidence that Dr. Reed had refused to accept more than one vertical expansion joint, I do not accept that evidence but rather accept Dr. Reed's evidence to the effect that he saw the one expansion joint only once, and that was after the brick work was complete and the tarpaulins removed. I find that Dr. Reed did not by his actions or words contribute to the fact that as it turned out the building was deficient in that it was absent an appropriate number of vertical control joints in appropriate locations. It is of interest to note that the majority of the experts who testified were in agreement with author T. Ritchie's comments in his article, <u>Rain Penetration of Walls of Unit Masonry</u> in the National Research Council Building Digest (page 21-3 of Exhibit 53), where he said the following, under the heading Control Joints:

> The outer part of a cavity wall forms a relatively thin skin around a building and may be subjected to appreciable changes in temperature and moisture content, producing stresses which lead to cracking. In addition, the outer part of the wall may be affected by movements taking place in other components of the building. Experience in the design of cavity walls has indicated the value of providing vertical control joints in the outer part of the wall to accommodate these movements. It has been found that the corners of cavity walls are particularly susceptible to cracking when a structural frame has been used. Accordingly, a vertical control joint is usually provided in the outer part of the wall about three or four feet from the corner…

There was negligence on the part of both Mr. Garbutt and Emmons & Mitchell that contributed to the fact that there were an inadequate number of vertical control joints at appropriate locations in the brick wall. Mr. Garbutt was negligent in failing to identify their locations in the specifications and in failing to show them on his drawings, and Emmons & Mitchell in the face of those inadequacies in the specifications and drawings, was negligent in failing to seek appropriate instructions from Mr. Garbutt as to their number and location. As well, Mr. Garbutt was negligent in discharging his duties as Consultant during the course of site attendances, in failing to observe that vertical control joints were not being properly installed as he intended. It is of interest to note that Mr. Roney's suggested remedial work includes the cutting of vertical control joints at each corner of each of the four walls. It is clear on the evidence that the absence of appropriate vertical control joints has resulted in cracking of the brick mortar joints in various locations, one of which cracks progressed from approximately eight feet in length to 20 feet in length over the course of less than 10 years.

[113] Section 2.1.1 of the Specifications, Division 4 – MASONRY required the contractor to "provide 6 millimetre continuous expansion joint caulked continuously between steel support angles and brick panels each floor." As was explained in the evidence, the soft joint was required between the top row of bricks and the underside of the spandrel beam in order to avoid the cracking of the mortar joints when the spandrel beam rotated as it would. As it turned out, what was installed between the top brick course and the underside of the spandrel beam, at least on the north and south sides where the support plates had not been cut, was a solid mortar joint. It was Mr. Santin's evidence that it was Mr. Sinclair who told him to use mortar rather than caulking to solidly fill in the space below the shelf angle plates on the north and south sides. Mr. Sinclair denied the conversation. In any event, Emmons & Mitchell were clearly negligent in allowing such a solid mortar joint to be installed, contrary to the specifications and contrary to good building practice. Mr. Garbutt on the other hand was negligent in failing to detect the error and failing to have it remedied. In fact, Mr. Garbutt testified that during the course of site inspections, he did notice that the joint between the top brick course and the underside of the shelf plate was in fact solid mortar, but it simply did not register with him at the time.

[114] The inner block walls in this building sit on the concrete floors between the steel columns and beneath the steel spandrel beams. Section 3.6.1 of the National Standard of Canada Design Code M-78 provides as follows, under the heading Anchorage:

> 3.6.1 Masonry walls and partitions shall be anchored to their lateral supports where these supports are other than masonry by:
>
> a) metal anchors spaced vertically at not more than four times the nominal thickness of the wall or partition where lateral supports are vertical, or not more than two metres o.c. (on centre) horizontally where the lateral supports are horizontal; or
>
> b) other approved construction systems.

As well, the Specifications in Division 5 – METALS, s.2.9 require the contractor to "supply all fastenings, anchors, accessories required for fabrication and erection of work of this section."

[115] As it turns out, the inner block walls in the subject building were constructed without any form of anchorage to the steel columns and steel spandrel beams. Mr. Johnston during the course of his investigation observed that the space between the top of the inner block wall and the underside of the perimeter steel beam was variously filled with mortar, bricks on edge, half blocks or at some locations was left completely unfilled. As well, when Mr. Johnston did find mortar laid between the top of the block walls and the bottom of the spandrel beams at certain locations, he observed that it was not laid uniformly from front to back of the joint so as to avoid eccentric loading. Mr. Johnston discovered as well that the inner block wall simply butted up to the vertical column and did not engage the column flange. I accept Mr. Johnston's evidence with respect to his observations and find as a matter of probability that his observations in this respect were representative of the actual construction of the inner block wall throughout the building, and that throughout the building the inner block wall was not anchored to the steel frame as required by s.3.6.1 of M-78. I find as well that the various types of "infill" used in some locations between the top of the block wall and the underside of the spandrel beam did not constitute "other approved construction systems" as referred to in s.3.6.1.

[116] Mr. Johnston on his inspection also found that there were gaps or holes between the block wall and the steel frame which prevented a proper air barrier from existing in the walls, and which precluded any remedial solution that would involve only the introduction of weepers and flashings.

[117] As Dr. Drysdale pointed out in his reports and evidence which I accept, the lack of positive connection of the inner block wall to the steel frame renders the wall unstable. This situation which in my view exists throughout the entire building will need to be corrected to ensure even distribution of the load throughout the panels and to avoid successive sheer failure of individual panels. As Dr. Drysdale pointed out, this very significant problem had not been addressed by Mr. Roney in his analysis.

[118] In my view, the problems with the inner block wall as outlined above which resulted in the wall being left without the required lateral stability, occurred as a result of negligent design and construction and deficient inspection/review on the part of both Mr. Sinclair and Mr. Garbutt in their respective capacities.

Remedial Work

[119] Mr. Roney, supported by the other defence witnesses, in his reports and *viva voce* evidence recommended remedial work on the building consisting of the following:

a) Six millimetre Helifix ties to be installed at the rate of one per four square feet of wall surface area over the entire north and south walls, from an elevation approximately four feet above the level four floor up to the top of the perimeter steel beam (at the roof). 400 ties would be required for this exercise.

b) At the east and west walls the same ties to be installed at a rate of one per square foot of wall surface area over the entire surface area, from a point 16 inches above the level four floor up to the top of the perimeter steel roof beam. Approximately 600 ties would be required.

c) A horizontal row of ties is to be installed approximately two inches below each floor level and at the roof level, spaced at 24 inches on centre. Approximately 300 ties would be required.

d) All of the above ties will be installed from outside the building by drilling holes in the brick wall. The ties are self-tapping into the block wall and adhered to the outer brick wall by epoxy.

e) Vertical control joints are to be cut into the wall at each corner, always on the north or south wall.

[120] Mr. Pascoal for the defence recommended the following additional remedial work:

a) Waterproof membrane flashings to be introduced at all shelf angle locations and all window and door locations, and foundation perimeters.

b) PVC grade weep vents to be introduced over new flashings.

c) New support angles at acceptable intervals to be installed along the east and west elevations.

d) New soft joints to be saw cut and caulked horizontally at shelf angle locations.

e) New expansion joints to be saw cut vertically as recommended by Mr. Roney.

f) Installation of new Helifix stainless steel brick ties as recommended by Mr. Roney.

g) Detergent cleaning of the brick wall.

Mr. Pascoal anticipates a cost of some $90,000 for all of the above remedial work.

[121] Mr. Johnston, supported by Dr. Drysdale is of the view that the only reasonable and appropriate manner in which to remedy the deficiencies in the building is to remove and discard the face brick, repair the back-up block wall, ties and structure, introduce appropriate

flashings and weep holes and re-install a new brick veneer with appropriate control joints. Contractor Gord Robb who gave evidence on behalf of the plaintiffs costed the work out in 1995 at between $401,000 and $433,000, depending on whether alternate a) or alternate b) as set out in his quote was chosen.

[122] A review of the competing positions taken by the experts discloses the following.

[123] One of the significant problems with the "Roney approach" if I may call it that, is that it does nothing to provide lateral stability to the inner block wall as previously discussed in these reasons. Mr. Roney, through his calculations, is satisfied that the walls as designed and constructed can safely resist the vertical and lateral wind loads specified in the Ontario Building Code in effect at the time of construction, with the exception of a portion of the east and west walls above the level four floor, which deficiency can be remedied by way of reinforcement. In due course Mr. Roney produced his calculations upon which he based his opinion and Dr. Drysdale had an opportunity to critique those calculations. In my view, Dr. Drysdale presented his critique both by way of report and *viva voce* evidence in a logical and technically correct fashion. Dr. Drysdale pointed out that Mr. Roney in arriving at his conclusion that the walls as constructed could safely resist the out of plane lateral and anticipated vertical loads specified in the OBC, used an analysis that was fundamentally flawed. Dr. Drysdale testified, and I accept his evidence, that the analysis presented by Mr. Roney proceeded on a number of implicit assumptions, all of which assumptions were in fact violated. As pointed out by Dr. Drysdale, the analysis made by Mr. Roney was valid only if,

a) neither wythe cracks (cracking of the brick wythe has been established as a fact);

b) the wythes have identical support conditions and are of the same height (neither of these conditions is met);

c) ties connecting the wythes are sufficiently stiff to force the wythes to deflect equally (in fact, as we know, flexible ties were used and even then were bent between the two wythes);

d) additional bending due to eccentric axial loads is not applied to one or both wythes (eccentric axial load is applied in indeterminate and inconsistent ways to both wythes); and

e) additional stresses are not introduced by restraining deformations due to moisture and temperature changes (expansion or contraction), and in plane forces caused by the infill resisting lateral loads (in fact all of these stresses do exist).

[124] Contrary to the conclusion reached by Mr. Roney and the other defence witnesses, I am satisfied on all of the evidence that there has in fact (in addition to observable cracking) been significant de-bonding between the brick and the mortar at the bed joints which lack of bonding cannot be repaired by pointing. This de-bonding has occurred primarily on the east and west walls where the shelf plates were removed. One will recall that Mr. Johnston when he opened the wall from the inside, observed "archeological" evidence of water entering at the bed joints between the top of the brick course and the bottom of the mortar, where no crack was visible. He then during his spray test and during an actual rain storm observed water entering at the bed joint and actually following the "archeological" moisture markings on the inside face of the brick. The "Roney approach" does nothing to address or solve this critical problem. It seems clear from the evidence that the composite action of the two wythes required for the walls to withstand Code winds is simply not possible because of the de-bonding.

[125] I have considered carefully all of the expert evidence, both by way of reports and *vive voce* evidence and I am satisfied that Mr. Johnston and Dr. Drysdale have reached correct conclusions with respect to the inadequacy of the building's walls. In particular, Dr. Drysdale as a result of his analysis reached the following conclusions which I accept:

a) The brick veneer wall is reported to have very low bond values and there is not a reliable source of flexural resistance to resist lateral wind pressure. Even if it had tensile strength comparable to Code design values, cracking would be expected and in the event of cracking the brick veneer is no longer able to share in resisting the load.

b) Although the bond conditions for the block wall are not known, Code values are an optimistic prediction of this characteristic. Even such values are exceeded by a wide margin by the predicted stresses from wind loading.

c) The calculated tie forces are very large and likely well beyond the ability of helical types of ties embedded in 26 millimetres thick block face shelves or cracked brick mortar joints.

d) Dr. Drysdale observed from his calculations that the tensile strength of the masonry will be exceeded and that there is as a result potential for cracking in the brick veneer and subsequent cracking in the block back-up wall.

e) There is already evidence of cracking in the brick veneer and so there is now reliance on the block back-up wall to withstand wind pressure.

f) There will be a progression of cracking as a matter of high probability.

[126] In these reasons, I have touched on some aspects of Dr. Drysdale's evidence which I accepted and which led me to conclude that Mr. Roney's analysis is incorrect and that his recommendations for remedial work are not acceptable, having regard to the actual state of this building. In his evidence, Dr. Drysdale identified many areas in which Mr. Roney's analysis is incorrect and inappropriate and in that respect I would direct the reader to the summary of Dr. Drysdale's evidence which is contained in Addendum 'A'.

[127] In my view, having regard to all of the evidence, it would be reasonable for the plaintiffs to follow the recommendations of Mr. Johnston, Dr. Drysdale and their other experts to adopt a remedial course of action in repairing the deficiencies in their building that would see the brick cladding removed in order to permit necessary repairs to the block wall before re-installing the brick wythe with proper ties, shelf angle plates, flashings, weepers and control joints. In Mr. Johnston's Folio 3, Exhibit 56 at Tab L is Mr. Robb's 1995 costing of the remedial work recommended by Mr. Johnston and Dr. Drysdale, based on alternate a) and alternate b). As Mr. Robb explained in his evidence alternate a) essentially results in a brick veneer wall, while

alternate b) results in a cavity wall. The alternate a) cost of remediation as of 1995 was $401,245.85 and the cost of alternate b) was $433,313.84. Both of those numbers include the cost of parapet repairs which were actually undertaken and paid for in 1998. The difference in the costing between alternate a) and alternate b) is accounted for by the increase to cost of alternate b) resulting from increasing the width of the cavity in order to permit the installation of steel posts for the purpose of stabilizing the inner block wall. In alternate b), not only is the cost of the remedial work increased by some $32,000, but it is further increased by an additional cost of some $62,000 attributable to the addition of the steel posts. Plaintiffs' counsel in his submissions takes the position that the reasonable costs of the remedial program for the walls should include alternate b) plus the additional $62,000 for some 116 steel posts which is the system preferred by Dr. Drysdale. I note however that at the time that Mr. Johnston gave his evidence, he admitted to Mr. Garbutt on cross-examination that he had not yet determined the best manner of reinforcing the back-up block wall.

[128] In all of the circumstances I find the reasonable cost of remedial work to be as calculated by Mr. Johnston in his Folio 3, Exhibit 56 at pages 30 and 31, as follows:

1994 cost of alternate a) as calculated by Mr. Robb	$401,245.00
Inflation value added at 5% per annum for 8 years from 1995 to 2002 for a total of 40%	160,000.00
	$561,245.00
Less: Value of parapet repairs achieved in 1998	24,000.00
	$537,245.00
Hoarding permits, temporary services (excludes winter allowances	16,500.00
	$553,745.00
Professional fees, inspections, testing, certifications, 11% of $537,000	59,070.00
TOTAL COSTS to remedy	$612,815.00

[129] To bring the inflation factor up to date, there should be added to the 1994 costs a further five per cent which amounts to an additional approximately $20,000 for a total cost of repair then of $632,815.

[130] It should be noted that Mr. Robb gave evidence that in his view there should be an inflation adjustment from 1995 to present at the rate of five per cent per year because of an increase in the cost of building materials and labour. He was of the view that five per cent would be an average inflation factor and explained that some years it has been higher and in other years lower. As well, he advised that the inflation factor has varied with the item under consideration. I am prepared to rely on Mr. Robb's evidence with respect to the inflation factor.

[131] Notwithstanding the position taken by the defendants, I am of the view that the plaintiffs are entitled to have the quantum of the necessary wall remediation work assessed in 2003 dollars rather than 1994 dollars. In my view, the plaintiffs have acted quite reasonably in deferring undertaking these repairs pending the completion of this law suit. Those repairs will cost a very substantial amount of money. It is money that the plaintiffs do not have in the sense that the evidence discloses that they have very little cash reserves and without the proceeds from this law suit, the cost of the repairs would have to be financed which financing would obviously involve significant payments of interest. The plaintiffs have done nothing wrong in this case. They hired a reputable engineer and construction firm in 1985 and have been plagued with a leaking building since completion of construction. They have diligently pursued answers to the causes of the problems and the correct remedial approach. Throughout they have gotten conflicting advice with respect to those questions. As well, they have been told by both defendants that they did not start their action in time and that their action is statute barred. The plaintiffs have been faced with real and worrisome uncertainties as to the collectability of any judgment that they might obtain against the defendants. Emmons & Mitchell has had its ongoing construction business conducted through a new corporation (Emmons & Mitchell 2000 Limited) for the last three years. Mr. Garbutt has no professional liability insurance covering these claims against him notwithstanding that such insurance was required to be in place at the time of this design and construction. In all of the circumstances, the plaintiffs acted reasonably in effecting roof repairs in 1994 and eventually replacing the roof and parapets in 1998. Equally, the

plaintiffs have acted reasonably in deferring the remedial wall work pending the outcome of this action.

[132] In the circumstances of this case, the plaintiffs should be reimbursed by Emmons & Mitchell the window caulking charge of $1,500.00 as paid by them to the company in October 1986. That charge was included in the Contract Price. It is not my intention to make any adjustment for the $2,035.14 caulking charge paid to Alan Thompson & Sons in December 1992. There is some issue as to whether the second piece of caulking work around the windows, entrance and some brick joints was normal maintenance.

Joint and Several Liability

[133] The plaintiffs in their statement of claim have framed their action in negligence and breach of contract. They have alleged defects in the design of the building as set out in paragraph 18 of the statement of claim. They have alleged defects in the construction of the building as set out in paragraph 19 of the statement of claim. And in paragraph 21 of the statement of claim they have pleaded that the defects in the design and construction of the building were caused by the negligence or breach of contract or both of the defendants and thereafter set out particulars of the negligence or breaches of contract with respect to each defendant. The plaintiffs have made out their case as pleaded. As a result of the negligence and breaches of contract of the defendants as disclosed by the evidence and as set out in these reasons, the plaintiffs have been left with a defective leaking unsound building which they have already repaired to some extent at significant cost and which they will be required now to repair further at even more substantial cost.

[134] Defence in their submissions take the position that Emmons & Mitchell and Mr. Garbutt are not jointly and severally liable under s.1 of the *Negligence Act* because:

a) they are consecutive tort feasors; and

b) the damages are divisible.

Section 1 of the *Negligence Act* provides that where damages have been caused or contributed to by the fault or neglect of two or more persons, the court shall determine the degree in which each of such persons is at fault or negligent, and where two or more persons are found at fault or negligent they are jointly and severally liable to the person suffering loss or damage for such fault or negligence, but as between themselves in the absence of any contract express or implied each is liable to make contribution and indemnify each other in the degree in which they are respectively found to be at fault or negligent. Defence argues that the jurisprudence has restricted the application of s.1 of the *Negligence Act* to concurrent wrongdoers. See *Martin v. Listowel Memorial Hospital* (2000), 51 O.R. (3d) 384 (C.A.).

[135] With due respect to the submissions of defence, one need only read the evidence of how this building went wrong to appreciate that as plaintiffs' counsel has submitted, there is simply no reasonable or practical basis on which the responsibility for the damages connected with each of the three elements of the remedial work being roof repair, parapet rebuilding and wall rebuilding can be separately attributed to, or segregated on a causal basis as between, Emmons & Mitchell and Mr. Garbutt. I agree with all of plaintiffs' counsel's submissions in that respect as follows.

a) The need for the roof work due to poor slope arose not only from the roofing sub contractor's failure to follow the specification sections on both slope and CRCA standards for irma roofs, but also from some degree of confusion in Mr. Garbutt's specification between a conventional built-up roof and the irma roof where the requisite slope because of the anticipated addition of floors had to be achieved within the roofing materials rather than by sloping the concrete base, but no specific direction was given by Mr. Garbutt in the Contract Documents in this respect.

b) The need for rebuilding the parapets was the result of not only Emmons & Mitchell's poor workmanship and departures from the specifications, but also was the result of Mr. Garbutt's negligent failure to have recognized any of the seven separate parapet deficiencies hereinbefore referred to during his on-site review. Clearly on the evidence, he should have detected those problems during construction and rectified them at the time.

c) The need for the extensive wall remediation as previously described is similarly the result of a composite pattern of conduct on the part of both Mr. Garbutt and Emmons & Mitchell. Concurrently, Mr. Garbutt negligently failed to adequately review the site to prevent the continuance of Emmons & Mitchell's construction deficiencies and Emmons & Mitchell did in fact continue to act negligently, contrary to the Specifications, and contrary to good building practices including failure to seek clarifications as to Mr. Garbutt's intent of filling the collar joint to create a solid wall, deficient tie practice both with respect to the type of ties and spacing thereof, sloppy mortar work that produced fins, the elimination of weepers and flashings, and the use of a hard joint rather than a soft joint at the north and south shelf plates.

d) The need to remove the outer brick wall so as to remedy the inner block wall deficiencies again is the result of the same combination of poor construction practice and poor site review with respect to:

i) the sporadic and non-uniform manner of infilling between the top of the block walls and the bottom of the spandrel beams;

ii) the leaving of air gaps between the block walls and the steel frame; and

iii) the creation of a structurally inadequate wall system composed of two wythes of brick and block separated by an air space and insufficiently tied together to be able to support reasonably anticipated design wind loads.

e) The shelf plates on the east and west walls were removed by Emmons & Mitchell negligently, contrary to the Contract Documents, and contrary to the Code requirements. I have already found that Mr. Garbutt did not give permission for the cutting of the shelf plates, but surely he was negligent in failing to detect during the course of his site inspections that they were in fact being removed or had in fact been removed.

[136] In the result, I agree with the submission of plaintiffs' counsel, and that is that on the evidence it is simply not possible to segregate the causal impact of the respective fault of Emmons & Mitchell and Mr. Garbutt so as to be able to say, for example, that 'x' dollars of the damages associated with remedying the wall deficiencies flows solely from identifiable fault of Emmons & Mitchell and 'y' dollars flows solely from identifiable fault of Mr. Garbutt. In the result, responsibility for the damages sustained by the plaintiffs in this case should rest jointly and severally on Emmons & Mitchell and Mr. Garbutt.

Is the Plaintiffs' Action Prescribed?

[137] Emmons & Mitchell takes the position that the plaintiffs' action is prescribed both in contract and tort. In terms of the Contract, Mr. Nelson for the defendant company points to Division 7 of the Specifications, s.7 of which requires the Contractor to provide a written guarantee to the owner, guaranteeing that built-up roofing and membrane flashings will be maintained free from leakage for a period of two years from the date of Substantial Completion. As well, again in terms of the Contract, the corporate defendant relies on s.14.1(2) of the General Conditions which provides that as of the date of Total Performance of the Work as set out in the Certificate of Total Performance of the Work, the owner expressly waives and releases the Contractor from all claims against the Contractor including without limitation those that might arise from the negligence or breach of contract by the Contractor except those made in writing within a period of six years from the date of Substantial Performance of the Work.

[138] In terms of tort, the defendant Emmons & Mitchell says that the plaintiffs' action is statute barred by reason of s.45(1)(g) of the *Limitations Act*, R.S.O. 1990 c.L-15.

[139] Emmons & Mitchell takes the position that its roofing subcontractor provided the plaintiffs with a written guarantee for the roof dated September 5, 1986 and the roof leaks occurred in 1992 or 1993, well after the guarantee had expired. As well, Emmons & Mitchell argues that the plaintiffs discovered the material facts that support their cause of action in 1985 and 1986.

[140] Mr. Garbutt submits that the plaintiffs' action against him is statute barred by reason of s.46 of the *Professional Engineers Act*, R.S.O. 1990 c.P-28, which provides in the first instance that proceedings shall not be commenced against an engineer for damages arising from the provision of a service that is within the practice of professional engineering, after twelve months after the date on which the service was or ought to have been performed, but goes further to provide that a court may extend the limitation period if the court is satisfied that there are apparent grounds for the proceedings and that there are reasonable grounds for applying for the extension.

[141] The plaintiffs' action was instituted by way of statement of claim issued May 17, 1995. The limitation periods relied upon by the defendants are subject to the "discoverability rule". In *Consumers Glass Co. Ltd. v. Foundation Co. of Canada Ltd.* (1985), 51 O.R. (2nd) 385 (C.A.), Dubin J.A. speaking for the court, said the following beginning at page 398:

> In my opinion, in cases which are based on a breach of duty to take care, a cause of action does not arise and the time does not begin to run for the purposes of the *Limitations Act* until such time as the plaintiff discovers or ought reasonably to have discovered the facts with respect to which the remedy is being sought, whether the issue arises in contract or in tort.
>
> As I read the judgment of the Supreme Court of Canada in *Kamloops, supra*, the underlying policy consideration was 'the injustice of a law which statute-bars a claim before the plaintiff is even aware of its existence'. That principle, in my opinion, is equally applicable where the issue arises in cases sounding in contract or in tort. That is not to say that the plaintiff would have to know the extent of the damage complained of before the time begins to run, but the cause of action does not arise, in my opinion, until the plaintiff could first have brought an action and proved sufficient facts to sustain it, or ought reasonably to have discovered the facts upon which the cause of action is premised.

[142] And in *Peixeiro v. Haderman* (1995), 25 O.R. (3rd) 1 (C.A.), Carthy, J.A. speaking for the court at page 4 said:

> …the rule appears to be established that a limitation statute commences to run when the material facts upon which the action is based have been discovered or ought to have been discovered by the exercise of reasonable diligence.

> In *Consumers Glass Co. v. Foundation Co. of Canada* (1985), 51 O.R. (2nd) 385, this court held that the 'cause of action arose' against a contractor when the plaintiff had sufficient knowledge to institute the proceeding.

[143] And further, in discussing the quandary in which an accident victim may find herself when in doubt as to whether to institute the action, Carthy, J.A. said the following at page 7 of *Piexeiro, supra*:

> In summary the injured person who was advised by a doctor that the injury is not serious and permanent is in no different a position than a homeowner who sees a crack in the wall and does not realize that the building is going to fall down because of the contractor's negligence. He or she is incapable of identifying the cause of action, and worse, in the case of an accident victim, if the action is asserted speculatively, it could be struck down on the motion of the defendant. On the other hand, the insurer will be faced with an increased burden of investigation and of potential claims beyond the two year period. The balance between these interests remains in favour of the discoverability rule.

[144] In my view, in all of the circumstances disclosed by the evidence in this case, the time period for limitation purposes should not be considered to have started to run against the plaintiffs with respect to both defendants, until at the earliest the date that the plaintiffs received Mr. Cromarty's comprehensive report addressed to them and dated May 17, 1994 (see Tab 8 Exhibit 2) as prepared by Mr. Cromarty at the conclusion of his meetings with the defendants. At the latest, the limitation period should start to run at the conclusion of Mr. Johnston's investigations and reports to the plaintiffs in December of 1994. Each of these start times is within one year from the date of the institution of the action.

[145] I am satisfied on the evidence that the plaintiffs made all reasonable efforts and used due diligence in attempting to ascertain the cause of their leakage problem and the appropriate remedial work required to rectify the problem. It would appear that in the 1992-1993 period, the plaintiffs came to the realization that they had a serious leakage problem, but the advice that they received from Emmons & Mitchell and Mr. Garbutt at that time was that theirs was a maintenance problem that required either or both window caulking and silicone sealant, having regard to the porosity of the brick that they had chosen.

[146] The plaintiffs had never been made aware that in the spring of 1993 Mr. Garbutt wrote to Emmons & Mitchell enquiring as to whether or not the collar joint had been filled and what kind of ties had been used. Mr. Garbutt in that communication with Emmons & Mitchell expressed the concern that "serious problems result" if the collar joint is not filled. Nor were the plaintiffs ever aware that in response to his enquiries Mr. Garbutt had been falsely advised that the collar joint had in fact been filled and that adjustable ties had been used.

[147] On January 10, 1994 after seeing Mr. MacAdam's report, Mr. Garbutt wrote to advise the plaintiffs that MacAdam was "in fundamental error" because the wall was a solid wall, making MacAdam's "conclusions and recommendations… neither applicable nor appropriate." It was Dr. Reed's evidence, which I accept, that this left him and Dr. Lawless confused and in a state of doubt that perhaps Mr. MacAdam's comments did not really reflect the way in which the building was actually constructed. As well, Mr. Garbutt was insisting that Mr. MacAdam's concern about absent flashings and weep holes was misguided because again as the wall was designed and built as a solid wall, flashings and weepers were not required.

[148] It is fair to say then, on the evidence, that until the latter half of 1994 the plaintiffs notwithstanding reasonable efforts on their part to get to the bottom of the problem, were confused and had no real understanding as to the cause or appropriate cure of their leakage problem. It is fair, as well, to say that until the latter half of 1994 the plaintiffs had no idea that someone was responsible for that problem, never mind who that someone might be.

[149] It would appear that it was only in June of 1994 as a result of the meetings arranged by Mr. Cromarty that it was learned that the walls had been built as cavity walls. Even at that time Mr. Garbutt was taking the position that all of the opinions expressed by the plaintiffs' experts were speculative until such time as destructive wall investigation was undertaken. As a matter of interest, Mr. Cromarty agreed with that position.

[150] That destructive testing was undertaken by Mr. Johnston in late 1994 and reported on in December of 1994. The testing, among other things, disclosed to Mr. Johnston that rainwater was entering into the cavity through mortar bed joints in the brick wythe that had

become unbonded as a result of stresses applied to the wall because of design and construction deficiencies, and which debonding rendered the wall structurally unsound.

[151] While it makes no difference (because each of the alternate start dates is within one year from the date of institution of action), on all of the evidence I prefer the start up date of late 1994 when Mr. Johnston finally concluded and reported the results of his destructive investigation. Interestingly, Mr. Garbutt in his evidence has admitted that in his own view it would have been irresponsible for the plaintiffs to have commenced action against him before verifying the true condition of the building and its actual mode of construction by destructive testing.

[152] With respect to the contractual limitation period in the Contract, it is dependant in law on the same discoverability principles as already discussed. As well, in this particular case it is dependant on the delivery of a formal Certificate of Total Completion of the Work, a step that was never taken. Indeed, it appears that no attempt was made by Emmons & Mitchell to obtain a Certificate of Total Performance.

[153] With respect to the express two-year guarantee on the roof, the law seems clear that an express time limited warranty does not exclude the owner's right to claim for breach of the original contract requirements, beyond the time limited warranty.

[154] I should point out that had I found that the limitation period under the *Professional Engineers Act* had expired, I would have been inclined in all of the circumstances of this case to extend that limitation period on the basis that there were obviously apparent grounds for the proceedings and that there were reasonable grounds for applying for the extension. When the leaking problems began to increase in about 1993, both Emmons & Mitchell and Mr. Garbutt advised the plaintiffs that it was simply a maintenance problem that could be cured by either caulking or applying a silicone sealant to the bricks. Mr. MacAdam's reports in late 1993 were answered by Mr. Garbutt who took the position that Mr. MacAdam was in fundamental error because the wall was a solid wall resulting in Mr. MacAdam's conclusions and recommendations being neither applicable nor appropriate. As late as January 10, 1994 Mr. Garbutt was taking the position that no conclusion was more than speculative without the benefit

of a field investigation which the plaintiffs eventually arranged through Mr. Johnston. Again, Mr. Garbutt in his own testimony said it would have been irresponsible for the plaintiffs to commence action against him before undertaking the destructive investigation that he was recommending.

[155] Mr. Garbutt in his submissions argues that he has been prejudiced by reason of the plaintiffs not having given him adequate notice of their claim, he having been given no reasonable opportunity to inspect and remedy the problems. In my view, there is no merit to that submission. Mr. Garbutt was provided with Mr. MacAdam's reports for comment and was invited to the meetings with Mr. Cromarty.

[156] Mr. Garbutt suggests that he has been prejudiced by the absence of the as-built drawings, but it is to be noted that Mr. Garbutt himself was responsible under the Contract to have obtained delivery of the as-built drawings by the Contractor, and he failed to do so. With respect to Mr. Sinclair's missing job journal and the removal of the shelf plates, it is to be noted that there was a duty on both the Contractor and Mr. Garbutt to have seen to a proper change order, and it appears that neither did.

[157] In all of the circumstances I am satisfied that the plaintiffs have brought their action in time and that they have neither breached any statutory limitation period or any limitation period under the Contract.

The Result

[158] In the circumstances, there will be judgment for the plaintiffs against the defendants jointly and severally for the following amounts:

(a) the cost of the roof repairs carried out in late 1994 in the amount of $20,986.30;

(b) the cost of rebuilding the roof and the parapet system in 1998 after proper reductions for upgrades and betterment. The total cost of the work in 1998 was $106,092.64. From this figure must be deducted the sum of $4,531.00 for an upgrade to the mechanical units for a net cost of $101,561.65. Properly designed and constructed the roof should have had a life span of approximately thirty years. By the time of its replacement the roof had been serving the plaintiffs for some twelve years. However, it was not trouble free service nor was it the roof that the plaintiffs had bargained and paid for. At the discovery stage in this action the plaintiffs advised the defendants that they were claiming only a portion of the net cost of replacement. In all of the circumstance, it would be fair and equitable that the plaintiffs recover two thirds of the net cost of replacement. As a result, judgment on account of this item will be for the sum of $67,707.76;

(c) the cost of rebuiling the wall system in the amount of $632,815.00.

[159] There will be judgment against the defendants Emmons & Mitchell only in the amount of $1,500.00 on account of monies paid by the plaintiff to the company on account of caulking which was part of the contract price.

[160] Mr. Garbutt’s counterclaim for approximately $6,500 for additional compensation for certain extras connected with the original work in 1986 is statute barred and must be dismissed. It appears from the evidence that the plaintiffs did not pay Mr. Garbutt’s last invoice of $6,500 because they were of the view that he was entitled to compensation for a mark-up on only those change orders in which he had been involved. It is to be noted that Mr. Garbutt did not pursue or even raise the matter again until he was sued in the within action. In the circumstances, his counterclaim is dismissed.

[161] I will receive written submissions with respect to prejudgment interest and costs.

The Honourable Mr. Justice G.R. Morin

Released: October 29, 2003

ADDENDUM "A"

SUMMARY OF THE EVIDENCE

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ADDENDUM "A"

SUMMARY OF THE EVIDENCE

PLAINTIFFS' EVIDENCE

Witness – Brian Reed

Examination-in-Chief

Dr. Reed is one of the plaintiffs in this action. He is 54 years old and a licensed dentist. He moved his practice to Kingston in 1983 and eventually joined with his co-plaintiff Kenneth Lawless who is also a dentist, to locate a property on which they could build a four-story office building in which they could locate their practices. The building was eventually constructed at 275 Queen Street in the City of Kingston.

Dr. Reed's practice has occupied the fourth floor of the building since October of 1986 to present. Dr. Lawless occupied the third floor of the building until he moved his practice to Belleville in December of 1991. Presently another dentist, Dr. Tomlak is the tenant of the third floor.

The second floor has been unoccupied since February of 1997. The main floor is presently occupied by commercial tenants.

Dr. Reed has no experience in the construction business and no particular knowledge of construction in general.

Subsequent to the acquisition of the property in December of 1983 for a purchase price of approximately $175,000, the contractor who was doing the renovations to existing row housing on the property recommended the defendant Donald Garbutt, as someone who was involved in high-rise construction. The plaintiffs subsequently met with Mr. Garbutt who suggested the defendant Emmons & Mitchell Construction Limited (the defendant company) as a possible builder. The construction of the building eventually went to tender and the defendant company submitted the lowest bid. The plaintiffs discussed the bids with Mr. Garbutt and he assisted them in choosing the contractor.

The plaintiffs had no written contract with Mr. Garbutt. They understood that he would design the building and thereafter make sure that the building was properly constructed by carrying out inspections during the course of construction. For these services he was to receive a fee of 5% of the cost of construction which finally came in at $833,000. Mortgage financing was arranged for the acquisition of the property and the construction of the building. There is presently some $618,000 owing on what was an original $1.1 million mortgage.

During the course of discussions with Mr. Garbutt the plaintiffs indicated to him that they would like to have the option of adding two more floors to the four-story building at some time in the future, and they understood that Mr. Garbutt had designed the building accordingly.

Prior to construction, Dr. Reed had attempted to read the specifications. It was a substantial book which he had difficulty with and retained very little of. Dr. Reed also went through and tried to understand the drawings. He discussed them with Mr. Garbutt but there was no intensive review of the drawings. He made no attempt to correlate the drawings to the specifications. He could understand a few things from the drawings but not very much from the specifications.

With respect to the brick cladding, the defendant company referred the plaintiffs to Rigney Building Supplies to choose the bricks. They selected a brick known as 'John Price' and when they advised the defendant company of their choice, the defendant company indicated that those bricks would be more costly and that they would have to pay more than was previously allowed in the contract. With respect to the porosity of the bricks chosen, neither defendant prior to 1993 indicated a possible adverse implication with respect to absorption of water.

Plaintiffs' counsel then reviewed with Dr. Reed the various documents contained in Exhibit #1. There are two further books of documents entered as Exhibit #2. Although there is no admission with respect to the truth of the contents of the documents, the parties are in agreement that the documents are authentic. There are some documents especially letters from experts retained by the plaintiffs which constitute hearsay in that the experts will not be called to testify. Mr. Mueller explained that the letters were not being produced for the truth of their contents but rather to address the discoverability issue in connection with the limitations defence, that is to show what the plaintiffs knew of the building problems at any given point in time.

At Tab 1A of Exhibit #1 is a letter from the defendant company to the plaintiffs which references in part a change to exterior walls. Dr. Reed explained that this item relates to the interior walls where studs and insulation were added as an extra.

In the minutes of the office meeting dated May 27, 1985 there is reference under Item 10 to "add for cavity wall insulation at exterior walls of elevator". The doctor testified that he had no idea at the time what a cavity wall was.

Item 14 of the same minutes refers to the addition of two roof drains. The doctor testified that from his understanding at the time there were none.

Further on in Tab 1A is hand-written list of sub-trades. Neither of the plaintiffs were involved with the selection of any of these 'subs'.

Following the list of sub-trades is a document entitled 'Stipulated Price Contract'. It is dated June 27, 1985 and is signed only by Mr. Emmons of the defendant company. The witness explained that this was the contract as initially presented to the plaintiffs by the defendant company. The final formal contract is contained at Tab 2 of Exhibit #1.

Contract Documents

Both contracts stipulate that they include specifications and details dated June 27, 1985 as prepared by Mr. Garbutt, drawings prepared by Roberts Engineering Inc. dated June 19, 1985 and drawings prepared by Mr. Garbutt dated April 15, 1985.

Attached as Appendix A to the Contract is a Contract Price Summary which *inter alia* sets out certain allowances carried in the base bid. Brick is not included in those items. Appendix A also shows a change to the roof insulation to 6 inches. The witness explained that the plaintiffs had requested this change in order to ensure adequate insulation on top of the fourth floor where Dr. Reed's offices would be located.

Attached to the Contract are several hand-written sheets headed up "Cost Analysis". Item 11 of those sheets relates to roofing and indicates that the roof will be an "irma" system. The doctor understood that an irma system involved laying the insulation on top of the waterproof membrane and then laying the ballast gravel on top of the insulation. The doctor also indicated that the company listed in the list of sub-trades as previously discussed (Amherst Roofing) was not in fact the roofing company that installed the original roof although it did do some subsequent repair.

The formal Contract at Tab 2 was executed August 12, 1985. It shows that the consultant on the contract may be served at 42 Lower Union Street which is Mr. Garbutt's address. Attached to the Contract are some 24 pages of standard general conditions. Although Dr. Reed reviewed these conditions they did not make a lot of sense to him.

At Tab 3 is a summary of events prepared in Dr. Reed's handwriting. It was prepared by him to give to the consultants that were eventually being retained to assist the plaintiffs with respect to the problems being encountered in the building. The summary indicates that in September of 1986 the third and fourth floor windows were still leaking as were the rear and front entrances. It is recorded that the windows were caulked for the sum of $1,500 which was billed on

December 31, 1986. The summary also records a masonry credit of $9,000 being given to the plaintiffs for masonry deficiencies. On page two of the summary there is further references to periodic episodes of window leaking. In the fall of 1991, some windows developed drops of water at the top and the drops then fell onto the windowsill. The witness contacted the glass company which showed no interest.

The summary shows that in the period from the fall of 1992 to the spring of 1993 there was a severe progression of water leakage with increasing interior damages. There was an overflow through the roof hatch on two occasions. The roof hatch is elevated above the roof level by approximately 10 inches and gives access to the roof. It is located over the stairwell. At the time the witness assumed that water had flowed over the top of the roof hatch. On later occasions when water was coming from the roof hatch he observed that there was no water flowing down the wall of the hatch. When the hatch area was repaired in 1994, he observed that there were tears in the membrane at the roof level. On the particular occasion that he went to the roof, he had been warned by the first floor tenant that water was pouring in the roof hatch. On that particular day there was no pool of water unable to get through the drain.

The summary indicates the following. In the late summer of 1992 the witness met with Mr. Emmons who advised that the windows be caulked by Alan Thompson. The windows were caulked by Thompson in November of 1992. Mr. Thompson advised that he could see at least two horizontal and two vertical cracks in the mortar, some two to four feet in length which he caulked. The witness noted in his summary that there was one vertical expansion joint that ran from the top of the building to the bottom on the Queen Street wall. He did not observe any other expansion joints. There were no weepers and no flashings over the windows. In the spring of 1993 he met with Mr. Emmons and Mr. Primeau from the masonry company that did the initial work. They discussed the fact that water was still coming in through the windows notwithstanding that the windows had been caulked. Primeau and Emmons suggested that the water was coming in through porous bricks and they suggested at that time the possibility of applying a silicon sealant. The last entry on the summary is dated August 27, 1993. Dr. Reed feels this was an add-on and that the rest of the document was prepared some time before that.

At Tab 6 there are two inspection reports from Mr. Garbutt, one dated August 20, 1985 indicating the project was approximately 11% complete, and the other dated September 30, 1985 indicating that the project was 23% complete. The witness had never seen these inspection reports before litigation was commenced, and in fact never received any reports at all from Mr. Garbutt during the course of construction. There were in fact weekly site meetings which the plaintiffs attended and for which the defendant company prepared the minutes. Mr. Garbutt was present for about 50% of the meetings.

At Tab 10 is an excerpt from the Professional Engineers Act which was in force at the time. Section 88 required Mr. Garbutt to have professional liability insurance of not less than $250,000. Dr. Reed pointed out that he never had any discussions with Mr. Garbutt with respect to insurance, and eventually it was disclosed that Mr. Garbutt had no such insurance. One of the reasons that the plaintiffs had not repaired the wall to this point in time is concern about recovery of the costs of any necessary repairs.

At Tab 11 is a sample general review commitment certificate issued by Mr. Garbutt with respect to another project. The certificate points out that the Professional Engineer shall make periodic visits to the site to determine on a rational sampling basis whether the work is in genera conformity with the plans and specifications for the building, and record deficiencies found during site visits and provide the client, the contractor and the owner with written reports of the deficiencies and the actions that must be taken to rectify the deficiencies. Dr. Reed testified that the plaintiffs had no specific discussions with Mr. Garbutt as to how many site visits he might make or what in particular he might do. They understood generally that he would oversee the building so that there would be no problems.

At Tab 49 is a site meeting minutes dated November 6, 1985 indicating that a section of the brick wall that had been laid the day before had been rejected. The brick wall had just started to be laid at this point.

The specifications in Division 4 (Tab 1) require by paragraph 9 a 'job mock-up' where a sample brick wall of mortar and brick is constructed for approval. Dr. Reed testified there never was such a mock wall built and he wasn't aware of the requirement for one. Again referring to Tab 49, although not referred to in the site meeting minutes, Dr. Reed explained that in fact Dr. Lawless had observed some of the shelf plate on the east and west walls being removed because the bricks were not matching up and this issue had been raised at the site meeting. No-one suggested any adverse implications on the structure by the removal of the shelf plates, nor did anyone suggest a formal change notice.

Running from Tab 26 to Tab 41 are monthly accounts rendered by the defendant company to the plaintiffs, showing where the building was at any particular stage and how much they owed. At Tab 28 is a sample statement of account which has noted in typewriting, 'Examined and Approved' and then Mr. Garbutt's signature.

The site meeting minute dated January 17, 1986 at Tab 51 indicates that the major portion of the masonry had been completed.

At Tab 54 a site meeting minute dated February 12, 1986, there is reference to the fact that the masonry walls will be cleaned and pointed as necessary as soon as weather permitted. Dr. Reed explained that there had been a lot of mortar dripped on to the bricks and a significant amount of efflorescence, that is white colouring, on the bricks.

At Tab 56 is a minute dated March 13, 1986 indicating the re-location of the heating and air conditioning units on the roof.

At Tab 48 is a document in Dr. Reed's handwriting setting out a number of change orders in which Mr. Garbutt had been involved. It is to be noted that Item 36 indicates that Mr. Garbutt was involved in required roofing and flashing due to the re-location of the heating and air conditioning units.

At Tab 58 is the minute dated April 9, 1986. One of the items refers to the defendant company checking to see if there were any windows missing the 'window water drip' and to correct same. The window water drip was a cut along the length of the stone either at the top of or at the bottom of the window to permit drainage of water.

At Tab 59 is a minute dated April 25, 1986 indicating that Mr. Emmons was to call Abe Dick Masonry to set up a meeting. This meeting was required because the plaintiffs were unhappy with the appearance of the brick.

Tab 13 is a minute of the meeting on September 4, 1986 referring to the fact there is a break in the flashing on the corner near the rust check, a hole in the flashing near the roof hatch, and that the caulking on the flashing is poor. The witness explained that the flashing being referred to was the metal flashing around the perimeter of the roof on the parapet. The parapet extended above the roof level by a few inches. He believes these particular problems were rectified.

At Tab 14 is the minute for September 4, 1986 indicating general poor workmanship with respect to the caulking, that the fourth floor windows were leaking on to the sills, and that a third floor window was also leaking. The witness explained that some efforts were made to rectify these problems.

Tab 17 is a letter from the defendant company to Dr. Reed confirming that Mr. Emmons would review the masonry problem and set up a date for a meeting. Dr. Reed explained that the plaintiffs were having continuing concerns about the appearance of the brickwork. At Tab 18 is a letter from the defendant company to Abe Dick Masonry Ltd. confirming that there has been a complete cleaning of the brickwork and an attempt to remove mortar splatters and improve the over-all appearance of the brick but that the plaintiffs are still not satisfied.

At Tab 19 is a letter from Mill & Ross Architects, to Dr. Lawless dated November 14, 1986. Dr. Reed explained that they were still not satisfied with the building and did not know if they were being reasonable or unreasonable with their dissatisfaction. They felt they needed a consultation.

Mr. Mill conducted a visual inspection only and in his report made no comment with respect to flashing or weepers. In the report Mr. Mill incorrectly advises Dr. Lawless that as the progress payments were authorized during the course of the project, such authorization implies that the work delivered to date was acceptable and therefore could continue. That advice is in fact contrary to the general conditions of the contract at Tab 7 and in particular general conditions 14.10 and 14.11. Dr. Reed confirmed that at no time did the plaintiffs seek from architect Mill any advice with respect to the structure of the building as such.

At Tab 20 is a letter from the defendant company to the plaintiffs discussing the possibility of resolving the aesthetic deficiencies of the masonry by giving the plaintiffs a credit of $12,000 which was the amount still owing to the masonry contractor.

At Tab 21 is a document dated November 26, 1986. Its authorship is unknown but it addressed concerns that the plaintiffs had at the time. It reads in part:

> Ensure water problem in brick around middle window on fourth floor (north side) and brick around rear exit are resolved completely to owners' satisfaction (regardless of what must be done to rectify the problem), correct water leaks around all exterior doorways (Queen Street, main entrance and rear exit) in particular at the junction of the side glass panels and the brick walls at floor level and around the door frame in the rear exit.

The latter deficiency was fixed but to this date there is still water coming in on occasions.

At Tabs 24 and 25 are documents that came with a small maintenance manual that in fact contained no maintenance directions.

At Tab 39 is an account from the defendant company to the plaintiffs charging *inter alia* for caulking of the windows in the amount of $1,500. This same item appears in the statement of account at Tab 41, although the caulking was only done once. Dr. Reed was not aware that in accordance with the specifications at Tab 1, Division 7, paragraph 6.1, the caulking was required as part of the Contract.

Tabs 40 and 41 are statements of account which show that there was a holdback owing to the masonry contractor of $12,688.75 and eventually the plaintiffs were given a credit of $9,000 of that amount. This was arrived at by way of agreement (see Tab 45). The writing on the statement of account at Tab 41 is Dr. Reed's and refers to the roof flashing. It was Dr. Reed's evidence that at that time, December 31, 1986, there was still water leaking at the top of the inside of windows, notwithstanding the caulking that had occurred. At that particular time, it was occurring infrequently but now the leaking is very persistent and affecting many windows, particularly when there is driving rain from the south.

Tabs 42 and 43 are invoices from Mr. Garbutt which confirm the agreement that his fee would be 5% of tendered costs, and 75% of that would be allocated to design while 25% would be allocated to supervision.

At Tab 44 is the last invoice received by the plaintiffs from Mr. Garbutt in the amount of $6,506.50. Dr. Reed did some calculations which are reflected in his notes at Tab 48, which show that by his calculation the plaintiffs had overpaid Mr. Garbutt by some $2,186.95 which he in fact owed them. The billing date was November 1, 1986. The plaintiffs attempted to arrange a meeting to discuss the invoice with Mr. Garbutt but were unable to arrange such a meeting. Dr. Reed explained that Mr. Garbutt never attempted to collect that amount until they brought action against him in 1995.

At Tab 46 is a letter from Roberts Engineering Inc. (Tom Roberts who is an architect) dated February 25, 1987. In the letter Mr. Roberts is looking for his fee of 1% of the total costs of construction in connection with this project. The plaintiffs were never aware of this arrangement. In fact they had had discussions with Mr. Roberts before their initial discussions with Mr. Garbutt and had decided not to use Mr. Roberts. In the letter, Mr. Roberts refers to Mr. Garbutt's allegations that drawing dimensions that Mr. Roberts had prepared were not correct and states that the contractor Mr. Emmons has confirmed that they were in fact correct. Dr. Reed confirmed that at the site meetings there was never any complaint with respect to the drawings.

At Tab 47 is a letter from Mr. Garbutt to Mr. Roberts dated February 27, 1987 acknowledging the letter of February 25.

That brought us into the two volumes that comprise Exhibit #2.

At Tab 62 of Exhibit #2 is a note from Dr. Reed dated November 23, 1993 relating to the maintenance of the building. This note was prepared at the request of Mr. Garbutt. The note points out that in November of 1992 the windows were caulked on the recommendation of Mr. Emmons. The person hired at that time on the recommendation of Mr. Emmons was a Mr. Thompson who, during the course of his work, made no statement to the plaintiffs with respect to flashing or weepers. The meeting with Mr. Emmons in the fall of 1993 arose as a result of water continuing to come in around the windows. The November 23, 1993 note also advises that in October of 1992 the flashings on the roof were re-caulked.

At Tab 64 is a note from Mr. Garbutt to the defendant company dated May 10, 1993 which reads in part as follows:

> See if the following can be determined:
>
> a) that the collar joint (between brick and block) is filled – there is no such thing as a one-half inch air space

and further:

> If the collar joint has not been filled and the wall subsequently not constructed as a cavity or veneer wall (flashing, weep holes, etc.) serious problems result.

Dr. Reed noted that at the time of this letter from Mr. Garbutt to the defendant company he had no knowledge that Mr. Garbutt was still involved, and that he did not see this document until the litigation had been commenced.

In the spring of 1993 Mr. Emmons with a representative of the masonry sub-contractor attended at the site and advised the plaintiffs that water was going through the brick itself. They suggested application of silicon. The plaintiffs had previously been advised by Mr. Thompson that there were gaps between the mortar and the brick through which water was entering. They were not happy with the suggestion to apply silicon.

At Tab 65 is a document produced by the defendant company which refers to the wall being built as a solid masonry wall. The plaintiffs had not seen this document before the litigation.

At Tab 67 is a report from a professional engineer, C.H. MacAdam dated August 1, 1993. Mr. MacAdam is elderly and has suffered two strokes and cannot be called to testify. I remind myself that I can receive this report solely on the issue of discoverability to determine what the plaintiffs knew through this report and others at any particular time. Mr. MacAdam's reports cannot be considered for the truth of their contents. Dr. Reed explained that they retained the services of Mr. MacAdam because they were unhappy with the building and felt that they needed outside consultation. Mr. MacAdam was recommended to them. In this report, August 1, 1993 Mr. MacAdam points out that he expressed concern at the absence of any flashings or expansion joints in the walls, and at the removal in part or completely of the masonry support plates from the perimeter beams. As well, MacAdam confirms that he advised the plaintiffs to engage a roofing consultant to give an opinion on the adequacy of the roof flashings and the roofing ballast. Mr. MacAdam also suggested to them as confirmed in the letter that they consult lawyers immediately, and Mr. Garbutt be consulted to suggest remedial measures. Dr. Reed confirmed in his evidence that this was the first time that they were advised that the problem might relate to the absence of flashings or expansion joints, and the removal of the masonry support plates.

At Tab 69 is the first letter from the plaintiffs' then lawyer Martin J. Szcepaniak, dated September 16, 1993 addressed to the defendant company and Mr. Garbutt. The letter refers to water leaks at the building that started prior to the completion of construction and which have continued to date, and which have gotten progressively worse. The letter alleges that the water problems seem to be the result of improper design and construction of the building. The lawyer is asking for written reports setting out the parties' opinions as to the causes of the water problems and their proposed solutions.

At Tab 70 is a note from Mr. Garbutt to the defendant company dated September 20, 1993 which refers to a draft of answers to the lawyer's letter being enclosed. The draft answer has never been produced.

At Tab 71 is a letter from Mr. Garbutt to the plaintiffs' lawyer dated September 21, 1993 which reads in part as follows:

> I have been contacted by Mr. Emmons regarding this matter, have had conversations with him and have inspected the building. I am also in receipt of two reports prepared for Rigney Building (suppliers of the brick) which were forwarded to me by Mr. Emmons and which I believe were forwarded to your clients and in general I agree with their contents. Since your clients were solely responsible for the selection of the brick and hence ramifications as to scheduling and installation it would be helpful toward the preparation of further reports as requested if your clients would provide maintenance records as they relate to the building envelope. The brick selected (no longer in production) was probably the most porous one available and while this in itself is not necessarily detrimental to imperviousness of the walls must be accompanied by a comprehensive regular compensating maintenance program to prevent penetration of moisture into the building components with subsequent adverse effects. The most effective way and the most economic procedure of combating damages to buildings from moisture is to control its penetration of envelope components in the first place. The primary information required is the frequency of caulking and what material was used to seal the surfaces of the brick with application dates so that various effects can either be eliminated or evaluated. Your letter refers to your preliminary investigations and these reports would also be of assistance so please forward these as well.

As Dr. Reed testified, prior to this letter the defendant had never advised the plaintiffs that there might be a problem with the porosity of the bricks used or that any kind of maintenance was required for the brick. It was in response to this letter that Tab 62 was forwarded by the plaintiffs to Mr. Garbutt.

Having gotten this information from Mr. Garbutt, the plaintiffs sought advice from Mr. MacAdam. At Tab 72 is a letter to them from Mr. MacAdam dated September 25, 1993, commenting that Mr. Garbutt has obviously forgotten that this was a cavity wall and has forgotten or is not aware of the “rain screen principle” which allows the rain to penetrate through the exterior masonry width into the cavity between the exterior masonry and interior width where it then runs down the cavity until it encounters a wall flashing and runs out through weep holes. Mr. MacAdam advises against the sealing of the exterior surface as a last resort. It should be noted that the plaintiffs at this point in time, in the fall of 1993, were receiving conflicting views as to what the problem was with the wall and how it could be remedied.

At Tab 73 is a roofing report by Inspec-sol (Ontario) Limited dated September 28, 1993. This was obtained by the plaintiffs on the recommendation of Mr. MacAdam. The report indicates that the roof system appeared to be in good working condition and that the leaking problems were not as a result of a faulty roof membrane. As well, the report concludes that the flashings on the parapets were in generally good condition and were not the main source of the leakage problem. It is to be noted that all this was consistent with what the defendants were telling the plaintiffs. The report states that the relatively porous properties of the brick veneer created a distinct probability of water penetration through to the wall cavity and into the interior of the building and window openings. The report recommended that additional testing be performed to include the direct source of water infiltration and suggested a possible solution would be to flood the roof on a clear day to eliminate the possibility of a roof leak.

At Tab 74 is another report from Mr. MacAdam dated October 11, 1993. MacAdam expresses the opinion that water is entering the building through the masonry by penetrating the exterior brickface through cracks and through the mortar joints, and goes on to say that due to the absence of wall flashings, vents or weep holes, the water then penetrates the inner masonry wall into the interior of the building. MacAdam says that if properly designed and constructed the degree of absorption of the bricks should not be relevant. MacAdam says that it is also possible but not certain that water also finds its way into the wall cavity from the roof as the flashings at the perimeter of the roof are neither complete nor as shown on the drawings. Dr. Reed took from this that the roof was probably not involved. MacAdam points out that while wall flashings and weepers are specified, none are shown in the drawings nor were any installed. This he says would indicate deficiency in both design as well as in supervision and inspection of the work.

He points out as well that vents are neither specified nor shown on the drawings, and opines that cracks in the masonry are probably due to the absence of expansion and control joints, both of which are specified but omitted from the drawings and from the work. He states that a full, proper and permanent solution would involve extensive demolition and reconstruction and that it would be prohibitively expensive and that he would not suggest it seriously. However, it is to be noted that at Tab 75 in a report dated November 30, 1993 MacAdam changes his mind and states that a full, proper and permanent solution involves the complete removal of the outer width of bricks and its replacement with new bricks including expansion joints, control joints, flashings and weep and vent holes which should have been incorporated initially. With respect to the report of November 30, 1993, Dr. Reed explained that when he and Dr. Lawless received the report of October 11 referring to a full, proper and permanent solution involving extensive demolition and reconstruction, they felt that was what they were looking for but MacAdam had provided no details and as a result they asked MacAdam for details as to what the proper and permanent solution would involve. As a result he sent them the report dated November 30, 1993. In his report of October 11, 1993 MacAdam addresses the removal of the brick supports and opines that it was a mistake to remove them. At this point in time this was contrary to what the plaintiffs believed and on balance with respect to the entire report, the plaintiffs were not convinced that what MacAdam was saying was correct.

In passing, Dr. Reed testified that he recalled a severe rainstorm occurring on October 21, 1993 when water came in the windows and through the back and front entrances. The doctor had patients at the time and called his wife to obtain buckets to collect the water. He later that day went to the roof to examine it and found no pluggage in the drains and no standing water which confirmed his belief that the roof was not involved.

In 1998 there was work done on the parapets. The entire roof was replaced and the parapets redeveloped with the membrane extending out completely over the raised parapets. The Inspec-sol report pointed out that the membrane had extended only ¾ of the way across the parapet.

In 1994 there were repairs at the roof hatch and at the northwest corner of the roof because of problems in those areas in the event of heavy rains. The northwest corner involves the extension that houses the elevator. It is an outcropping from the main building.

The problem with water entering at the windows on the south side continues to occur, usually when there is a driving rain. On the south face there are eight windows on each floor and there have been some modifications of those windows on the first floor to accommodate a tenant's needs. Two windows were made into a door and on either side of the door two sets of windows were joined together. With respect to this renovation, the doctor knows of one incident when the tenant placed a bucket in the window but as far as he knows there have been no other complaints.

Recounting the history of the water problems, the doctor pointed out that initially in 1986 there were a few windows on the fourth floor and one or two on the third floor where drops of water collected at the top of the window and then fell to the sill. Efforts were made to solve this problem by caulking which corrected the situation for a time, but then the problem recurred. By 1992 there were increased water problems which were more frequent and involved more windows. The windows were caulked which seemed to solve the problem but Dr. Reed pointed out that this was in late October when the weather was turning cold and there was less rain.

In the period from 1992 to 1994 there were problems again at the roof hatch and increased window problems in 1993. The roof repairs that were conducted in late 1994 at the hatch and northwest corner of the roof seemed to correct the leaking into the front stairwell but then the leakage came back and the roofing contractor came back and appeared to repair the problem, but then the walls began to get wet. The plaintiffs were not sure if water was still coming down from the roof and staying inside the wall.

In 1998 the full roof was replaced. There have been no further problems with the hatch or with the northwest corner, but water is still coming in at the windows on the north and south sides of the building, and sometimes on the east and west sides. The south side is particularly bad.

Counsel took up with Dr. Reed portions of the report of Mr. Wilson that was prepared for the defendant Mr. Garbutt. The report is dated March 20, 2000. Dr. Reed pointed out that Wilson was at the site for about 2 hours and did not use any instrument to take any measurements. Wilson in his report says at the time there was light drizzle and the temperature was five degrees. Dr. Reed says at the time there was mixed snow and rain and the temperature was hovering around zero degrees. Wilson says that there is light damages to the gypsum board at the edge of the windows. Dr. Reed points out that on the fourth floor one can observe extensive damage to the gypsum board once the eavestroughs that he designed to collect the water into a pail are taken off. On the second floor, the paint has peeled off the gypsum board which has deteriorated and expanded. On the third floor, wallpaper hangs down over the windows. Mr. Wilson in his report comments on slow dripping water at the windows. Dr. Reed points out that there was not in fact a lot of water coming in that day, he thinks because of the low temperature. In any event, there was significant water flow and it got worse after Wilson left. That being the case, the doctor decided to video the water flow. Wilson in his report comments that by his calculation there would be one-half litre of water collected per day at one window. The doctor did his own measurements a few days later, and calculated that at the rate at which he collected 30 millilitres of water, a litre container would be filled in five hours and that over a 24-hour period approximately five litres would be collected from one window. Mr. Wilson in his report refers to Dr. Tomlak's third floor walls being painted. Dr. Reed points out that it is on this floor where the wallpaper has pulled back and falls down into the window space. As well, while the arborite window sills do not soak up water, they are pulling away from the metal surround. Wilson in his report says that the north and east side windows do not leak. Dr. Reed points out that on his floor, the fourth floor, the windows leak on both the north and the east sides. Wilson in his report says the situation is not unhygienic, but Dr. Reed points out that a smell has developed in the hygienist's room and there is concern that mould is developing in the walls.

With respect to the hallways outside the offices on the fourth, third and second floors, the wooden sills on the windows at the ends of the hallways have had to be replaced.

Referring again to Tab 75, MacAdam's report of November 30, 1993 the doctor in his evidence notes that MacAdam suggests that an alternative remedy would be to remove the windows and cut expansion and control joints into the existing masonry to limit further cracking and then applying over the masonry an entirely new and separate cladding of glass and metal. The doctor explained that he and his partner purposely constructed the building of brick in order that it would blend in with the neighbourhood and were not interested in having a glass façade.

At Tab 76 is a letter from the plaintiffs' lawyer, forwarding to Mr. Garbutt copies of MacAdam's reports as requested by Mr. Garbutt. The lawyer also provided Mr. Garbutt with maintenance details as provided by Dr. Reed, that is that the windows were caulked in 1986, the roof flashings in October, 1992, and then Alan Thompson caulked the windows, entrances and some brick joints in November of 1992. In conclusion, the lawyer asked Mr. Garbutt for the report that had first been requested on September 16, 1993 (see Tab 69).

At Tab 77 is Mr. Garbutt's response to the lawyer dated January 10, 1994. Garbutt takes the position that the basic premise of the MacAdam reports is in fundamental error in that the masonry wall was not designed as a cavity wall, so that the comments and conclusions of MacAdam contained in the reports based on this assumption are "neither applicable nor appropriate". Again at page two of his report, Garbutt points out that the wall was designed as a solid masonry wall, the most important aspect being that the collar joint between the masonry and the bricks be filled at all locations. Garbutt points out that it is important to determine whether the collar joint has in fact been filled in its entirety, and says, "until factual data is obtained from field investigation, all reports from this point are speculative and non-contributive." Garbutt in his report comments that any remedial work suggested to essentially convert this wall into a veneer type with weep holes, vents, flashings are ill advised because of the minimal width of the collar joint which is approximately 12 millimetres, facilitating bridging and blockages by mortar droppings. He further comments that should collar joint fillings prove to be required the procedure consists of pressure grouting.

What Dr. Reed and his partner took from Mr. Garbutt's report was that their own consultant, Mr. MacAdam could be wrong and could be talking about a different wall system than was actually installed. MacAdam had never done more than actually look at documents. The plaintiffs thought Garbutt's suggestion of a full field investigation was a good one and as a result they retained Mr. Johnson in 1994 to do a physical examination. They were getting conflicting advice; MacAdam on the one hand was saying the proper procedure would be to turn this into a veneer type wall with weep holes, vents, flashings etc., and Garbutt was saying that this would be ill-advised. At this point in time having received Mr. Garbutt's report, the plaintiffs didn't feel much further ahead in terms of resolving their problems.

At Tab 78 is a document dated March 22, 1994 prepared by Mr. MacAdam as a comment on Mr. Garbett's letter of January 10, 1994 to Mr. Szcepaniak. At Tab 92 is a report from MacAdam dated June 20, 1994 to the plaintiffs enclosing two copies of comments in response to Mr. Garbutt's letter of January 10, 1994. Dr. Reed testified that the report of March 22, 1994 (Tab 78) was not received until June 20, and that the plaintiffs had not seen Tab 78 until it was received on June 20. In the report, MacAdam states that the solid composite wall as described by Mr. Garbutt is incomplete and that to deal with water penetration, flashings, damp-proof course and weep holes are all required, and that the inner masonry wall must be fully parged. This was confusing to the plaintiffs. Garbutt was saying that MacAdam was wrong in thinking that this was a cavity wall. Then it appeared that MacAdam in his report of March 22 was saying that even if it was a solid wall, weepers and flashings are still required. MacAdam expresses doubt that what he calls a "hybrid wall" could be converted to a solid composite masonry wall as Mr. Garbutt was suggesting by grouting the void spaces, and MacAdam recommended against it.

At Tab 79 is a note from Mr. Garbutt to Ernest A. Cromarty Architect Inc. dated March 29, 1994. Dr. Reed explained that by this point in time he and his partner were receiving a lot of conflicting information with respect to the construction of the wall. They were confused and no further ahead and as a result architect Cromarty who happened to be a patient of Dr. Reed and in his office at the time was asked if he would look at the problem, which he agreed to do. At Tab 81 is Mr. Cromarty's letter to the plaintiffs dated March 31, 1994 advising that a meeting has been arranged with the builder, the consultant, the masonry and structural steel sub-trades on April 20, 1994 to better ascertain how the building was actually constructed.

At Tab 82 is Cromarty's letter to GeoTech Contracting dated March 31, 1994 explaining that it was apparently intended that the building be constructed with a solid masonry wall but it appears it was actually built as a bastard type cavity wall as the collar joint was not filled and the wall was built without weep holes, flashings etc. Cromarty is asking GeoTech for a quote to pressure grout the collar joints.

At Tab 87 is the agenda for the April 20 meeting.

At Tab 88 is Mr. Cromarty's report to the plaintiffs dated May 17, 1994. The report advises that with respect to the wall section, the parties did not agree nor seem to know or acknowledge who wrote the specification and why the specification does not relate to the drawings and/or as built conditions. Dr. Reed testified that this came as a complete surprise to him. The report goes on to advise that although the specifications calls for a cavity type wall, for flashing over windows, etc., the structural consultant claims that the building was designed for an exterior wall of solid masonry. Cromarty advises that the drawings tend to indicate solid wall construction without cavity flashing, cavity weepers or flashing at lintels. Dr. Reed was of the impression that this was what Mr. Garbutt was saying.

Under the heading 'Structure', Cromarty states that there appears to be some doubt whether the continuous steel plate indicated to be welded to the structural beam at each floor was actually installed or modified during the construction. In this respect, Dr. Reed knew that his partner Dr. Lawless actually had pieces of the steel plate in his possession.

A number of recommendations were made by Mr. Cromarty and the plaintiffs agreed with all of them. It appeared to them that whatever might be done it was going to be very

expensive. Cromarty's report addresses the question of pressure grouting and lists the various disadvantages of it. At this point, the plaintiffs did not think that pressure grouting was viable.

At Tab 90 is a letter from Paul Desroches Construction Limited dated May 20, 1994. Paul Desroches was a young man referred by Mr. MacAdam. He simply appeared at Dr. Reed's office once day. He further confused matters by reporting that the roof was not tied into the wall at the perimeter of the roof-wall interface and that the roofing was deficient at various locations at the base of the parapet curve. This information was contrary to Inspec-sol's report. As well, Desroches advised that the brick was excessively porous and that this coupled with the substandard wall assembly facilitates the infiltration of moisture into the finished areas of the building. This information was somewhat consistent with the information coming from Mr. Garbutt.

At Tabs 91 and 93 are further communications from Mr. Desroches providing estimates for various alternative remedial works. The brick system estimated by Desroches seems similar to the system recommended by MacAdam; see Tab 92, a report from Mr. MacAdam dated June 20, 1994 where on the last page thereof he suggests the following: "remove face brick and replace with new brick and cavity on new foundation. We regard this as the best of the solutions proposed. If properly carried out it should solve the water entry problems entirely." As Dr. Reed testified when dealing with Mr. Johnson, they decided to go with this remedy. They have not actually had the work done because it is very costly.

At Tab 94 is a letter from the plaintiffs' lawyer to the defendants, dated August 19, 1994 advising that "my clients have now reached a stage where they feel that they have sufficient information regarding the cause or causes of the water problems at their building, the alternatives for remedial work and the cost estimates for such work and they would like to proceed with remedial work as soon as possible." The lawyer then seeks a meeting with all involved for the purpose of settling the matter. Dr. Reed testified that the advice being given by the lawyer to the parties was correct. The plaintiffs by this time believed that the problem was that water was going in through cracks in the mortar and not going back out because there were no flashings and weepers, and was therefore coming in to the interior through the windows.

At Tab 95 is a report to the plaintiffs by architect Paul Johnston dated August 29, 1994. He had been referred by Mr. Desroches. Dr. Reed had met with him and provided him with all of the information. Mr. Johnston in his report suggested a schedule of steps to be taken in sequence and Dr. Reed agreed.

At Tabs 96, 97 and 98 are pieces of correspondence between Mr. Szcepaniak and Mr. Garbutt, discussing the proposed meeting.

At Tab 101 is a comprehensive report from architect Johnston dated October 5, 1994.

At Tab 115 are several photographs taken by Dr. Reed. These photographs were provided to Mr. Johnston with handwritten comments on the back thereof made by Dr. Reed. At Tab 115 the handwritten comments are typed out. Dr. Reed explained that the information on the backs of the photographs was obtained by him at site meetings. See particularly photo 25 which discusses the shelf angles being removed; photo 33 which again discusses the shelf angles; photo 36 which again refers to the shelf angles; and photo 40, again referencing the removal of the shelf angles. The original colour photographs which comprise Tab 115 were entered as Exhibit #4.

At Tab 102 is a report dated October 11, 1994 from Kendall Consulting Inc., a roofing expert who was working with Mr. Johnston.

At Tab 103 is a further report from Mr. Johnston dated October 12, 1994 with conclusions set out at page 7 thereof. The conclusions firstly address roof repairs and the plaintiffs accepted the conclusion and the recommendation and carried out the roof repairs in 1998.

At Tab 113 is a summary of costs incurred to date which Dr. Reed prepared and which he testified was accurate.

Dr. Reed again reviewed in some detail the summary of costs set out at Tab 113.

Entered as Exhibit #5 was a series of colour photographs taken by Dr. Reed in approximately mid-April of 2002.

On page one, the top right photograph and bottom right photograph show the north face of the building toward the east end with bricks removed at the top of the wall. The top left photo shows the east face of the building with bricks removed at the right hand corner; the corner brick removal can also be seen in the top right photograph.

Page two of the photo's shows the corner of the building where the west and south faces meet. The dark areas shown in the brick at the corner are separate holes that had been made by extracting the brick and that had been filled in.

Page three of the Exhibit shows the same area as shown on page two but close up. It was the doctor's evidence that he did not observe any ties in the exposed area.

Page four shows another small exposed area on the brick wall with a metal tie protruding from the inside masonry wall. It was the doctor's evidence that the metal tie was rusty and eaten through.

Page five of the Exhibit contains photo's of the same area as seen in other photo's and in particular an area where bricks had been removed from the south wall. It was the doctor's evidence that neither in the top two photo's which show the south wall nor the bottom two photo's which show the west wall, were any ties observed.

Page six of the Exhibit shows the same hole in the brick on the south face as shown on page five.

Page seven of the Exhibit on the top left is the same corner at the meeting of the west and south faces as seen in other photo's, and the bottom photo's show the same hole in the brick on the corner as is shown on page one of the photo's.

It should be noted that all of these holes in the brick were intentionally made in order to observe the inner wall.

Filed as Exhibit #6 was a plan prepared by Roberts Engineering showing the renovations executed in 1989 in connection with the south side, first floor and on the east face, first floor. As previously explained by Dr. Reed, the two centre windows were made into an entrance way on the south side and each set of two windows on either side of the entrance way were made into one large window as was the case on the east face. In connection with these renovations, the doctor knows of only one incident where the tenant had a bucket in the window to collect water.

The doctor in his evidence explained that the plaintiffs waited until 1998 to replace the roof because of the lack of funds and for the same reason the repairs to the walls have not been executed. The doctor expressed concerns as to the collectability of any judgment that is obtained in this case. He has been advised that Mr. Garbutt has no insurance and the name of the defendant company has been changed as recently as 2000.

Counsel asked Dr. Reed as to why the plaintiffs did not institute an action either in late 1993 or early 1994 as opposed to May of 1995. The doctor explained that in 1993-1994 the plaintiff did not know what the problem was. Many people were looking into the problem and the plaintiffs were receiving confusing and conflicting responses from them. There had been no thorough investigation but rather just walk-arounds. MacAdam was telling them that the walls were cavity walls; Garbutt was telling them that they were solid walls. Garbutt was suggesting that there be an investigation to find out what kind of walls they were. The plaintiffs thought that Garbutt should know. In the final analysis no one could tell them why water was coming in through the windows and that was their basic concern. By 1993-1994 they were being told that there were at least four options with respect to the manner to repair. Simply put, it was not clear to the plaintiffs at that time what the situation was.

Counsel requested Dr. Reed to advise with respect to his financial situation in 1994, and at present. His evidence was as follows. In 1994, operating expenses were approximately $18,000 per month and he had loans outstanding for the operation of his medical practice. He had RSP's of $40,000 and cash in the bank of approximately $10,000, and no other liquid assets. His disposable income was between $100,000 and $110,000 per annum, out of which he paid the mortgage, life insurance, health care, disability insurance etc., which left approximately $40,000 for living expenses. Today the carrying costs remain about the same and he still has practice loans. His liquid assets today are approximately the same. His disposable income today is about the same but he has the added expense of two children in university at approximately $16,000 for each child. He is presently borrowing money to live.

The mortgage on the building today stands at $618,000. The building is presently appraised at $100,000 in the condition in which it is, so that there is no possibility of raising further money on the building.

The $170,000 that has been expended by the plaintiffs to date on account of the items shown at Tab 113 has been paid by way of cash flow and borrowing.

Before Mr. Mueller became involved, legal costs incurred by the plaintiffs amounted to some $75,000. Since Mr. Mueller has been involved additional legal costs of $125,000 have been incurred.

At this point in his evidence, Dr. Reed showed a video that he took at various times. The opening scenes are of the inside of the fourth floor window with the trough in place as designed by the doctor. At one end of the trough is a tube and one can see the water pouring out of the tube into a container. The leaking occurs at the top of the window well where the horizontal wallboard meets the metal frame of the window. The troughs designed by Dr. Reed consist of a plastic trough with a tube located at one end. The moisture collects in the trough and goes out the tube and into a container. In the video, Dr. Reed removed the trough and the water simply leaked onto the window sill. He replaced the trough and within a very short period of time the water began to drip out of the tube again. He indicated that depending on the nature of the rain outside the water dripping from the tube can vary from drops to a full stream. That particular video of the fourth floor window opening was taken some time after 1994.

A video taken on April 8, 2000 shows leaking in the window wells on the north face. The doctor explained that usually it is the south face that gets problems. The video shows a window well on the north side with water on the sill running to the edge of the sill and showing the damaged drywall on the side. There was no trough in place at the time this water accumulated. The video on this date also shows damaged wallboard on the south side.

Further video taping was done on March 20, 2002, the day that Wilson attended for Mr. Garbutt. The video shows a bucket in place in a window well in the waiting room behind decorations acting as a camouflage. The time is 12:43 and there was an extensive accumulation of water in the bucket. The video goes on to show severely damaged drywall around windows on the south side of the fourth floor. Dr. Reed was leaving his office at about 10:00 p.m. on March 20, 2002 when he went to the second floor and took further videos. One can observe in the video significant deterioration of the drywall adjacent to the window and all around the window on the west wall. Similar deterioration is shown in another window well. Similar severe damage to the drywall is shown in a south wall window on the second floor. The video shows three windows in the open area where water has collected on the sill and there is bad damage to the drywall. Similar damage is shown in window wells on the north side, still on the second floor, which according to the evidence is vacant at this time and was at that time. The video shows a window well on the south side that had been taken apart for investigatory reasons and then put back together, and repainted in 1999. The video shows that by March 20, 2002 it is damaged again at the window surround. Both paint and the wallboard is flaking off. The video shows a window well on the east face of the building, second floor, where there is water collecting on the sill. Further video taping was done on April 8, 2002 at 10:00 p.m. In the waiting room a window on the south face is dripping water. The doctor is shown collecting the water out of the trough in a 30 millilitre cup. It was the doctor's testimony that it took eight minutes to fill the 30 millilitre cup which translates into one litre in five hours or approximately five litres of water collected in 24 hours. The video was entered as Exhibit #7.

Cross-Examination by Mr. Nelson

Dr. Reed explained that when he and Dr. Lawless bought the property there was located on it a garage, a parking lot and one end house of a four-unit row house. That end unit was eventually renovated and the next unit was eventually purchased and renovated as well so that at the present time the property contains the subject building, a garage, a parking lot and the two row houses.

Dr. Reed confirmed that in this action there is no claim for lost income with respect to the building, or any personal claim for lost income or damages for lost financing. He confirmed that at the time of the acquisition and building, the plaintiffs put on a mortgage for $1.1 million, $175,000 of which was used to purchase the property and $838,000 of which was used to erect the building. The balance of the funds were used to renovate the end-unit row house and the garage to some extent. Dr. Reed noted that the plaintiffs actually took over the responsibility for the interior finishing of the building. The mortgage was initially for a five-year term which has been renewed from time to time. Although there is a right to make capital payments upon renewal it was the doctor's evidence that he and Dr. Lawless had never made any capital payments.

The doctor explained that he and Dr. Lawless at first planned on a two-story building. Mr. Garbutt was retained to do the drawings. Dr. Reed did not know until long after the building was completed that architect Roberts was involved with some of the drawings.

Entered as Exhibit #8 were eight sheets of what were referred to as "Tender Drawings". The first two drawings were prepared by Garbutt in 1984 and depict a two-story building. The remaining series of drawings are dated March of 1985 and are of the four-story building. Dr.

Reed explained that between July of 1984 and March of 1985 the plaintiffs decided to go for a four-story building and so advised Mr. Garbutt. Drawing #4 of the Tender Drawings shows an irma system roof with three inches of roof insulation. Dr. Reed conceded that he vaguely realized that an irma roof at the time meant the membrane was on the bottom. He does remember that he wanted more insulation on the top. He conceded that he reviewed with Mr. Garbutt all of the drawings to some extent because they were eventually going to be used for the purpose of tendering.

Dr. Reed conceded that he understood Mr. Garbutt to be an engineer and that he and the co-plaintiff did not use an architect nor did they ask Garbutt to involve an architect. They simply asked Mr. Garbutt to design and supervise the construction of the building for a fee of 5% of the cost of construction. Mr. Nelson put to the witness that that appeared to be a rather light fee, and Dr. Reed responded that he had no knowledge of whether the fee was light or heavy.

At Tab 2 is a letter from the defendant company to Dr. Lawless dated May 8, 1985 advising that the company has compiled a price to complete all work as outlined on the drawings prepared by Mr. Garbutt. Dr. Reed conceded that the drawings referred to in the letter are probably the drawings that went in as Exhibit #8 but he could not say for sure.

At Tab 1A is a letter dated May 22, 1985 from the defendant company to the plaintiffs referring to several change items. The various items from one to 11 are headed up “Items as per your letter”. That letter has never been produced by the defendant company and the plaintiffs do not have a copy of it. In any event, Dr. Reed explained that he and his partner did receive the May 8 tender at Tab 2 and reviewed it, and suggested changes to the defendant company.

Four pages in at Tab 1A is an office meeting minute of May 27, 1985 prepared by the defendant company. Dr. Reed confirmed that at the meeting various electrical components of the building were discussed.

There is then a letter from the defendant company addressed to the plaintiffs dated May 30, 1985 wherein various items are addressed including an extra of $5,400 for six inches of roof insulation as opposed to three. The letter represents final decisions made by the plaintiffs and the defendant company as a result of discussions back and forth. The doctor explained that by this time it was understood that the defendant company would be doing the building, and as well it was always understood to this point in time that the plaintiffs wanted the option of putting on two more floors in the future.

Entered as Exhibit #9 was a set of drawings which were characterized as "Contract Drawings". These are drawings dated June 27, 1985, for the most part by Garbutt with one or two prepared by architect Roberts. Again, Dr. Reed explained that he never really noticed that Roberts was involved with some of the drawings and he never took Roberts' presence up with Mr. Garbutt. As far as he was concerned, Garbutt was doing the drawings.

Entered as Exhibit #10 were a group of drawings designated as "Planning Drawings". These were drawings that eventually were approved by the Planning Department. They were submitted for the purpose of getting a building permit. To begin with in this grouping there are three small sketches to which Dr. Reed contributed after discussions with the defendant company. He explained in his evidence that he did not prepare these sketches for submission to the Planning Department but it appears that his sketches were added to by some person and eventually submitted for approval. The doctor marked on these three small sketches with yellow magic marker the portions that were prepared by him. Included among these drawings is a site plan which illustrates the building having been moved a few feet south in order to increase the parking behind the building. These drawings are prepared by Garbutt and some by architect Roberts. Again, Dr. Reed testified that he did not know of Roberts' involvement until long after the building was constructed.

Filed as Exhibit #11 was a group of drawings referred to as "Miscellaneous Drawings".

At Tab 2 of Exhibit #1 is the Contract entered into by the parties for a contract price of \$729,857.86. The Contract purports to be made on the 12th day of August, 1985, and lists Mr. Garbutt as the consultant. Article A-2 of the Contract lists the contract documents. Dr. Reed conceded that he would have gone through the Contract before signing it but pointed out that the project was well under way by the time the Contract was signed.

Under Article A-2 the Contract lists part of the contract documents as “specifications and details dated June 27, 1985 as prepared by N.D. Garbutt, P.Eng. and signed by Emmons & Mitchell Construction Limited”. Dr. Reed agreed that the specifications are those at Tab 1. He did read the specifications but did not completely understand them, and he had no specific discussions with Mr. Garbutt with respect to the specifications. He assumed that Mr. Garbutt had drafted them. Division 7 of the specifications addresses thermal and moisture protection, and under Item 3 refers to a “standard built-up roof”. The doctor agreed that at the time he understood that with respect to the roof the insulation would be on top of the membrane. He was not aware at the time that in fact the specifications provided that the insulation would be under the membrane.

Back to Article A-2 of the Contract, the doctor was unable to say what the “details” were that were referred to in the first item of contract documents.

The second item under contract documents is “Drawings prepared by Roberts Engineering Inc. dated June 19/85 number 2b.”

The third item under contract documents is “Drawings prepared by N.D. Garbutt, P.Eng. number 1 dated 15/04/85, number 3, number 4, number 5, ME1, ME4 dated 01/03/85, ME2, ME3, ME5, ME5, ME6 revised 13/06/85”. It was agreed that these drawings plus three small drawings comprise Exhibit #9 as filed.

Mr. Nelson reviewed with Dr. Reed general condition 14.12. At Tab 3 is Dr. Reed’s short history of the project which indicates in part that it was billed 100% complete July 31, 1986 in the amount of \$838,285. The doctor explained that he took that information form the defendant company's invoice. He confirmed that the masonry settlement for $9,000 took place probably around December of 1986 and that the caulking was done some time in late 1986. General condition 14.12 provides for a limitation period of six years from the date of substantial performance of the work. Mr. Nelson attempted unsuccessfully to obtain from Dr. Reed some acknowledgement of the expiry of the limitation period. It should be noted that the general condition refers to substantial performance of the work "as set out in the certificate of substantial performance of the work". As we now know there was no such certificate issued by Mr. Garbutt.

Dr. Reed acknowledged that in the summer of 1986 there was a further application made to the City to approve a re-positioning of the building and to convert it to four stories. Initially the City approval was for a two story building only. The plaintiffs were actively involved in the process and actually took the application in to City Hall. The doctor has no recollection of having called City Hall with respect to any specific point. He does acknowledge having received from the City a letter dated August 1, 1985 addressed to the plaintiffs with respect to the site plan control application. Attached to the letter is a report of Rupert Dobbin, Director of Planning and Urban Renewal. The letter was entered as Exhibit #12. The plans referred to in the report are the planning set of drawings already filed as an exhibit at this trial. In the report the designer is designated as Garbutt and Roberts Engineering Inc. On the second last page is a memo from the Deputy City Engineer to Mr. Dobbin referencing a call that he had gotten from the doctors a few days before, relating to a possible encroachment of the steps onto City property. In his evidence the doctor recalled the issue but did not specifically recall a conversation.

At Tab 7 is the Building Permit issued August 28, 1985. The construction started some time in late August.

Filed as Exhibit #13 was the defendant company's Exhibit Book, Volume 1.

The persons from the defendant company with whom the plaintiffs had contact during construction were President D. Emmons, Secretary-Treasurer R. Vinkle, and J. Sinclair who was the site foreman. The plaintiffs' offices were not a great distance from the construction site and both attended frequently. As well, they attended virtually all of the weekly site meetings. Mr. Garbutt came to some of the site meetings as well.

Mr. Nelson then reviewed with Dr. Reed the various site meeting minutes contained at Tab 1 of Exhibit #13. During the course of that review, the doctor confirmed that during the course of construction he took some 110 photographs. He explained that the construction of this building was a significant point in his history, the history of his practice and the history of his family. He considered it to be a great adventure and as a result took photographs as the construction went along. Mr. Nelson attempted to make out that the photographs were being taken in order to provide the plaintiffs with control of the construction but Mr. Nelson failed to make his point. [In my view it would be perfectly normal for the owners to have taken photo's as construction went along and indeed videotaping the construction as apparently Dr. Lawless did.]

With respect to the performance of masonry work in the winter time, it is noted in the minute of January 8, 1986 that it was agreed that the work would proceed with winter protection and heating. The doctor explained that while the plaintiffs were aware that the masonry work would go on in the winter months, they did not actually want it to do so. The doctor recalls a discussion as to whether it would continue through the winter and the plaintiffs agreed that it do so with the proper protection. Mr. Nelson put to Dr. Reed that he and Dr. Lawless were very insistent that the masonry work continue through the winter months, and Dr. Reed disagreed with that proposition. The plaintiffs were anxious about the masonry work being done in cold weather.

The minute of January 17, 1986 notes that the major portion of masonry has been completed. The doctor confirmed that the shelf angles had been cut by this time. It was his recollection that they were cut in the fall while the beams were in the air.

The minute of March 26, 1986 indicates that Dr. Reed will be moving from the fourth floor to the third and that Dr. Lawless would move from the second floor to the fourth. As it turned out, Dr. Reed ended up on the fourth floor and Dr. Lawless on the third.

Mr. Nelson then reviewed with Dr. Reed a number of issues.

With respect to the shelf angles, the doctor conceded that the steel beams were ordered on the basis of drawings prepared by Mr. Garbutt. He conceded that design was not the obligation of the defendant company. He conceded as well that the windows and the roof came from drawings.

The doctor conceded that he was aware in the fall of 1985 when the beams were put up that the shelf angles would not fit. He appreciated at that time that they had to be cut although he did not observe the cutting. He knew that they had to be cut on the east and west sides of the building and knew that on the north and south sides the shelf angles remained and the bricks sat on the angles at each floor. He does not know whose decision it was to cut the shelf angles. He is certain that he knew at the time that the shelf angles had to be cut from the first floor beams on the east and west side, and by the time he was dealing with consultants in the '90's he knew that all of the shelf angles had been cut on the east and west side. Filed as Exhibit #14 was a piece of shelf angle that had been cut from the beams.

Another issue during the course of construction was expansion joints. The doctor recalls observing some time toward the end of the masonry work one straight line of mortar down the centre of the Queen Street wall from top to bottom. The line was straight and did not follow the bricks, and as such stood out. He asked what it was and he was told that it was there to accommodate the expansion and retraction of the bricks. The doctor explained that the masonry work was being done under tarps and that he did not observe the straight line until the work was finished. Mr. Nelson pointed out to the doctor that the evidence of the defendants would be that they proposed expansion joints at each corner and that Dr. Reed said 'no' and said he would only accept the one expansion joint that was presently placed. Dr. Reed has no recollection of that, in fact he does not remember expansion joints being at issue at all until he observed the straight line of mortar on the front of the building. He had no idea what it was when he first saw it.

With respect to the windows, the doctor conceded that the plaintiffs were not holding the defendant company liable for the design of the windows. There is a stone surround that goes all the way around each window. The surround was suggested by Mr. Garbutt when the plaintiffs commented that the building looked rather plain. Garbutt added this as a architectural feature. The doctor indicated that he was not aware of any design drawings with respect to the windows and Mr. Nelson pointed out that there were in fact no design drawings for the window. Mr. Nelson pointed out that the evidence will be from Primeau Santin, the mason, that when it came time to put in the windows, he was not happy with the stone surround. he was concerned that the surround and particularly the top stone, that is the stone over the top of the window, would be a trap for water and the windows would leak. It will be his evidence that he spoke to the site super, Sinclair, about his concern and Sinclair in turn talked to Mr. Garbutt who dismissed the concern and directed that the stone surround be installed. It was the doctor's evidence that Garbutt never discussed that matter with him, and that this simply was not an issue and indeed it was Garbutt's suggestion for the stone surround. Nor did Dr. Reed have any discussion about the discussion with Primeau.

With respect to the 110 photographs taken by the doctor in 1985 and 1986, the doctor conceded that he did not show the photographs to Tom Mill, the architect that he retained in the fall of 1986 with respect to the appearance of the brickwork, nor did Dr. Lawless show him the video. The doctor explained that there was no need to do so as his only concern at that time was with the appearance of the brick.

The writing on the back of the photo's was done in 1994 when Mr. Johnston was retained by the plaintiffs as a consultant. When Johnston asked the plaintiffs to provide to him whatever they had relating to the building, Dr. Reed for his part went through the photographs and wrote on them what he thought might be pertinent information for Johnston.

Mr. Nelson went through some of the photo's with Dr. Reed. In photo #21, the shelf angle is depicted and it is referred to in the note on the back of the photo. Photo #25 shows the steel work and it is apparent that the end beams and the side beams are of different heights and don't line up. The doctor has made a note of that fact on the back of the photo but explained that he at the time that the photo was taken was not aware that the side and the end beams did not line up. Photo #53 is a shot from the fourth floor looking down inside the building and shows a portion of the inner wall.

With respect to the renovations performed in 1989, Dr. Reed confirmed that architect Robert Culvert was retained to prepare the drawings. Two drawings relating to the renovations were filed as Exhibit #15. The doctor explained that on the particular side where the renovations were performed, the two sets of end windows were taken out and the walls between the two windows were taken out and larger windows inserted. In the centre of the wall, the two windows were again taken out, the wall between the two windows was taken out, the wall underneath the windows was taken out and double doors were installed. A window was similarly enlarged around the corner. All this was done to accommodate a retail tenant.

In response to a question that I put to the doctor, he testified that when he was taking the photo's in 1985 and 1986 he never anticipated the problems that the plaintiffs eventually had with the building.

Dr. Reed was referred to paragraph 12 of his statement of claim wherein it is alleged that in or about the month of September, 1986 the plaintiffs became aware that water penetration was occurring at the building in a number of locations, particularly at the windows, entrances and roof. The doctor testified that he recalls observing some drops of water on the window sills in the fall of 1986 and also at times water in the entrances, but was not aware of any water penetration occurring on the roof.

With respect to the caulking that was done in the fall of 1986, the doctor testified that he thought it was successful at the time but as things turned out of course it was not. Mr. Nelson confronted the doctor with an affidavit sworn August 14, 1995, one paragraph of which he suggested contradicted his evidence here at trial. Suffice it to say the affidavit which is filed as Exhibit #16 does not in any way contradict what the doctor said here at trial.

At Tab 41 is an account from the defendant company to the plaintiffs dated December 31, 1986 at the bottom of which the doctor has noted in his own handwriting, “some persistent window leaks”. The doctor conceded that this was written by him after the caulking was completed.

The doctor was then shown portions of Dr. Lawless’s video which was filed as Exhibit #17. The doctor has never seen the video in its entirety. He believes that in the early years, perhaps 1986, 1987 he saw the beginning of the video and he attempted to view it a month ago but was able to view it for only one hour. The video is of some interest. Shown was the soldier course at the bottom of the brick wall, just after bricking began. The soldier course is out of whack by, according to what Lawless says on the video, some 3/8 of an inch. One can actually observe the fact that it is out of true. The doctor believes that it was taken down and rebuilt. [As a matter of editorial comment it is amazing to think that bricklayers would do a job like that in the first place.] The bricklayers on the video are shown laying the bricks. At least on the lower courses one is not able to see on the video any significant spacing between the bricks and the inside block layer.

On the video, Dr. Lawless expresses concerns about corrosion on the joists. Dr. Reed testified that Dr. Lawless had no discussions with him about that. As well, Dr. Reed conceded that he saw the roof during construction and had no concerns about it.

The video shows a bad job on the fourth floor slab which apparently was poured in cold weather. It shows as well defects in corners of the roof and Lawless comments on the video that these defects are potential leak sources.

Dr. Reed explained that it was not his financial situation that stopped him proceeding with remedial work in 1993. He explained that in 1993 the plaintiffs did not proceed with remedial work because they did not know what was wrong. They eventually retained architect Johnston in the late summer of 1994. He supervised the initial roof repair and drew plans for it, and eventually advised them to replace the roof in 1998. Johnston felt that there were multiple problems causing leaking around the windows and the roof was one, but Johnston told them that replacing the roof would not stop the water leakage around the windows. Dr. Reed personally did not think that the replacing of the roof would solve the leaking problems, although he understands that Dr. Lawless believed that the problem around the windows would be cured by replacing the roof.

Dr. Reed disclosed that he had a video that he took of the roof replacement, and he undertook to produce it to counsel and Mr. Garbutt.

Dr. Reed conceded that the defendant company was not given notice of the roof replacement. The plaintiffs are presently claiming the full cost of the roof replacement against the defendant company notwithstanding that Dr. Lawless on discovery said he was claiming only a pro rated portion.

That completed Mr. Nelson's cross-examination of Dr. Reed, except for any questions that may arise out of his viewing the roof video.

Cross-Examination by Mr. Garbutt

Dr. Reed had no discussions with Mr. Roberts after receiving a copy of Mr. Roberts's letter to Mr. Garbutt dated February 25, 1987 (Tab 46). The doctor is presently aware that Mr. Roberts is both an engineer and an architect. Dr. Reed in his evidence indicated that he and his partner had spoken to Mr. Roberts before having spoken to Mr. Garbutt but decided not to go with Mr. Roberts. The doctor could not recall whether he had advised Mr. Garbutt at the time that they had spoken to Mr. Roberts before speaking to him.

At Tab 19 is a report from Mill & Ross Architects dated November 14, 1986 which refers to a series of visual exterior inspections performed by them to assess the quality of brickwork on the building. The doctor did not know how many times they had attended for visual inspections. He did know that he met on site with Mr. Mill on one occasion, pointing out his dissatisfaction with the appearance of the brick. This would have been two to four weeks before November 14,

1986. He has no recollection of meeting with Mr. Mill after the letter of November 14. He never provided Mr. Garbutt or Mr. Roberts with a copy of Mr. Mill's report, nor did he ever provide a copy of that report to the mason, Mr. Dick.

The doctor conceded that by the end of 1986 there were only two amounts being withheld on the contract, that is the $9,000 to the mason and Mr. Garbutt's last invoice.

Referring to Tab 3, the doctor's chronological notes, the doctor conceded that there was really no contact with Mr. Garbutt with respect to the water problems between 1986 and 1993. The doctor explained that the plaintiffs were dealing with the defendant company during that period of time. They viewed it as a problem with the building with Mr. Emmons being their primary contact.

The doctor conceded that during his 16 years of occupancy, the air ducts of the building have not been cleaned.

The doctor conceded that in his report dated December 25, 1993 (Tab 72) Mr. MacAdam pointed out that caulking around windows, etc. is normal to prevent entry of excess water and that flashings and weep holes are usually also installed at these locations. MacAdam goes on to point out that caulking is not permanent, should be inspected at intervals and maintained.

The plaintiffs manage their own building. They have an elevator maintenance contract, and an air conditioning and heating service contract together with a fire alarm contract. They do not perform any regular inspection of the outside wall for caulking and repointing.

The doctor was referred to Tab 113, his summary of costs, and was of the view that nothing on that list referred to routine maintenance. There has been no caulking on the walls since the last entry on that list, September 1999.

The doctor conceded that he does not know how many site inspections were performed by Mr. Garbutt.

With respect to the 1989 alterations on the Queen Street side (Exhibit #6) the doctor had no idea how the lintels were joined after the bricks in between were removed. In any event, the architect did not at the time of that work report any deficiencies with respect to the wall.

The gross rents for the building are as follows: basement - $1,250 per month; first floor flower shop - $2,100 per month (the store is not doing well and the tenants are not paying on a regular basis); second floor – vacant; third floor - $28,000 per annum; fourth floor - $28,000 per annum.

Their mortgage is with a commercial lender. None of the deficiencies have been reported to the lender.

The doctor was asked whether Mr. Johnston and Mr. Drysdale have any concerns about public safety arising out of the wall. The doctor does not know what Drysdale's reports say but does know that Mr. Johnston says that the bricks are not well attached to the block and therefore there is a safety concern. He has no idea whether Mr. Johnston has reported the safety issue to the City.

The roof video was filed as Exhibit #18.

Re-examination of Dr. Reed

Dr. Reed confirmed for Mr. Mueller that it was in 1993 when the plaintiffs were dealing with the consultants that Dr. Reed first found out that all of the shelf angles had been removed on the east and west sides.

He confirmed as well that he would not have accepted the design for the window surrounds had he known of Primeau’s concern about their constituting water traps.

At no time did Dr. Reed or Dr. Lawless consider themselves to be site controllers or supervisors, and that it was Mr. Garbutt who had responsibility for site review beyond the contractor.

Prior to their discussion with Mr. MacAdam with respect to the removal of the shelf angles, Dr. Reed had no concern with respect to any structural deficiencies arising out of the removal of the shelves.

Dr. Reed did not start to look at the drawings in detail until after Mr. MacAdam was retained.

Between the opening of the walls and the closing of the walls for the purpose of permitting the experts to view the construction, at least a couple of weeks passed.

With respect to Dr. Reed's dispute with Mr. Wilson's report, Dr. Reed prepared a written response and it has been filed as Exhibit #19. Dr. Reed testified that all of his comments contained in Exhibit #19 are true and accurate.

Witness – Kenneth Lawless

Examination-in-Chief

Dr. Lawless is 61 years of age and has been practising dentistry since 1969. He started his practice as a specialist oral surgeon in 1976. He began to occupy the third floor of the building in the summer of 1986, running his practice with an associate. Dr. Lawless operated a second dental office in Belleville until 1991 when he began to practice full time in Belleville, and his associate took over the Kingston practice.

By way of a short history, Dr. Lawless explained that because of the absence of a pension plan and the expense of rent, he was looking to either buy or build an office building to house his practice.

During the course of construction, Dr. Lawless videotaped the construction because he considered it to be a major event in his life and he wanted a record of the event. The taking of the video had nothing to do with enabling the plaintiffs to somehow critique the construction. The video was never to be viewed by anyone outside their families.

With respect to Dr. Lawless's comment on the videotape, "all is not well", he explained that he visited the site the first day that the masons started laying block and brick. He readily observed that the bottom soldier course of bricks was decidedly out of plumb. He was upset. He told the mason that he was not satisfied and thought that the work should stop. It was the doctor's evidence that was the one and only time that he ever gave any instructions on site. He confirmed that the incident is referred to in the minutes of the site meeting held November 6, 1985 under point 10.1, which notes that a section of the brick wall that had been laid the day before was rejected and the mason would rectify.

With respect to his comment on the video that "it showed how bricks should not be laid", the doctor in his evidence explained that he was observing the same kind of problem at a different point in time, that is that a row of bricks somewhere above the soldier course was not plumb.

Heard on the video is a comment by Dr. Lawless with respect to measles. He explained that he made the comment over his concern about white patches on the wall. He described the wall as having multiple, various sized white patches. The plaintiffs had it cleaned and washed down on two occasions to get concrete and efflorescence off the walls. It was with respect to this appearance that they were given a $9,000 credit from the holdback owing to the mason, but even after that they observed efflorescence again from time to time.

With respect to the water problems that Dr. Lawless observed for the period that he occupied the building, 1986 to 1991, the doctor testified that water dripping around the windows began in the fall of 1986. It was relatively minor at that time with small droplets falling onto the window sill. Mr. Emmons of the defendant company was called. He came to the building and advised the plaintiffs to caulk around the windows which they did in 1986. There were no problems in the winter of 1986-87 which the doctor puts down to the fact that there was no rain during that period. Then inconvenient light drops of water occurred from time to time. As time went on it got progressively worse. The doctor explained that the weather was a large determinant. During long hot periods he did not observe much water leakage on his floor, but if a steady period of rain occurred then he got dripping on his floor. He normally used a sponge to mop up the sills when leakage occurred. He believes there were only one or two occasions when he had to place buckets in his windows in order to catch the water.

He really had nothing to add to Dr. Reed’s evidence as to the problem up until 1991, other than to say that he happened to be in the building in 1992 when the bad leakage occurred involving water running down the fourth floor wall into the entrance way.

Dr. Lawless has absolutely no experience with construction. As a hobby he did take a course in welding one year. It was his habit to take a different course each year after he came to Kingston. He has access to a welding machine and can do such things as weld a trailer hitch for himself.

Dr. Lawless conceded that he read the specifications which he found to be very technical and much of which he did not understand. He did not compare the specs to the drawings.

With respect to the removal of the shelf angle, Dr. Lawless happened to be passing by the building site at the end of November, 1985. He observed someone on one of the beams cutting with a cutting torch. Dr. Lawless stopped and spoke to the steel company representative and was told that a shelf plate was being cut off the beam. At the time Dr. Lawless did not know the function of the shelf plate. Either then or at a subsequent site meeting it was explained that the brick was not lining up properly and therefore the shelf plate had to be cut. There were pieces of cut shelf plate on the ground that day, and Dr. Lawless eventually took eight or 10 home after they had been laying there several days.

At the time that Dr. Lawless made his observation of the cutting of the shelf angle, the cutter was cutting from the beam on the second or third floor on the east end of the building. He did not know if all of the shelf plates were being cut off and he did not ask. At a subsequent site meeting, he asked what the purpose of the shelves was, and he was told that they were to support the bricks. He asked what would hold the bricks in the absence of the shelf plate, and he was told that the concrete foundation was strong enough to support the bricks. He does not know who told him this and he does not know whether Mr. Garbutt was at that particular site meeting. Certainly no one suggested that there would be any structural problem as a result of the removal of the shelf plates, and no one suggested that there should be a change order with respect to the cutting of the shelf plates. It was after Mr. MacAdam was retained and the plaintiffs were being interviewed that Dr. Lawless told Mr. MacAdam of the cutting of the shelf angles, and that as a result of the advice of MacAdam, Dr. Lawless realized for the first time that this could be a problem.

With respect to the leakage problems at the windows, the first person that gave Dr. Lawless a technical explanation of the problem was Mr. Johnston in 1994.

With respect to Mr. Garbutt's final invoice, Dr. Lawless arranged with Mr. Garbutt a meeting to discuss the invoice but Mr. Garbutt did not appear.

In 1994 there were two local roof repairs at the northwest corner and at the roof hatch, and finally in 1998 a new roof was installed. Dr. Lawless explained that the four year delay was caused by the fact that the new roof was an expensive proposition and he was reluctant to put out that kind of money. In 1994 his disposable income was $130,000, out of which he paid $5,000 per month on a bank note, insurance of $1,500 per annum, alimony of $81,000 per annum, plus his own living expenses. In that respect he explained that after the video was completed, it fell into the possession of his former wife and it was not until 1994-1995 that his lawyer was able to obtain it from her, at which time it was given to Mr. Johnston.

Again with respect to his finances, the doctor explained that his income decreased markedly in 1995 and 1996, and the began to recover to the point where last year he reached his 1994 level of income. He presently has land on the St. Lawrence valued at $400,000 and mortgaged to the extent of $200,000. He and his new wife are building a home there. In 1994 he owned two student housing units which were valued at $285,000 the pair, and which were mortgaged to the extent of $170,000. He still owns those properties. As well he is presently part owner of the building in which he practises in Belleville. He has a 25% interest in the building. In 1994 his share was worth $150,000 but the building was mortgaged to its full value. Today the mortgage is approximately $95,000.

In 1994 he also owned a condominium property valued at $130,000 and mortgaged to the extent of $30,000 then. He still has the property and it is now mortgaged to the extent of $75,000.

In 1994 he had a business loan outstanding of $200,000 which today has been reduced to $150,000. In 1994 he had RRSP's valued in the amount of $225,000 which are today valued at $295,000. He has never had any significant cash then or now.

The doctor explained that has not been prepared to encumber his assets in order to finance the repairs, apart from the extent to which the repairs have presently been completed. He does hope to be successful in this action but he has a concern with respect to the collectability of any judgment that the plaintiffs may be awarded. He knows that Mr. Garbutt is not insured and he knows that the defendant company changed its corporate name in the year 2000.

The doctor explained that the plaintiffs did not institute court action prior to May of 1995 because he had not received conclusive information to convince him as to what the definitive problem was. As he put it, they "had no diagnosis and no treatment plan". He explained that in 1993-94, Mr. Johnston had disassembled portions of the wall and looked into the walls. Prior to that time they were receiving conflicting reports. He and his partner could make no rational sense of it until Mr. Johnston disassembled the walls and explained the problem to them. The doctor went on to explain that prior to Mr. Johnston coming on the scene in the late summer/fall of 1994, the plaintiffs had been receiving various suggestions as to what should be done including pressure grouting and the installation of a glass skin, but nothing really addressed the problems. They did not understand what needed to be done to resolve the problems.

Cross-Examination by Nelson

The doctor explained that the property was purchased in the late fall of 1983 and the plaintiffs started planning the building in 1984. They first had discussions with architect Roberts but did not retain him or any other architect, but rather retained Garbutt whom they knew to be an engineer. Dr. Lawless was not aware that Mr. Roberts prepared any plans until this action was started. Although he reviewed all of the drawings with a view to office space use, he did not look at or see the bottom right hand corner where the author of the drawings was identified. He explained further that they did not retain Roberts because they did not feel they had a good personality fit with him.

The doctor conceded that he went to the site almost every day during the course of construction.

With respect to the shelf angle that was being cut, he knew it was on the second or third floor and he knew that it was on the east wall. He does not now know if he knew at the time that the shelf angles were going to be cut from the west wall as well.

Reviewing with Mr. Nelson the water leakage, the doctor confirmed that it began in 1986, and that it was worse with extended periods of wet weather and with south winds. In 1986 there was precipitation between the window frame and the edge of the drywall which dropped to the sill, and this leakage was worse on the south side. As well, Dr. Reed's fourth floor was always much worse than Dr. Lawless's third floor. While Dr. Lawless had problems on the third floor, again he confirmed that there were only two instances when he had to place containers in the window to catch the leakage. Otherwise he used a sponge to mop the sill.

After the caulking by Mitchell in 1986, the leakage problem never totally ceased. It reduced to a minor nuisance from time to time but eventually went on to get much worse.

The doctor confirmed that he and his partner did not start to talk to the experts until 1993. Architect Mills was retained in 1986, only for the aesthetic problem of the wall faces, and the water problem was relatively minor at that time.

Dr. Lawless was separated from his wife in 1991 and he confirmed to Mr. Nelson that he had the video in his possession from 1986 until 1991.

With respect to the local roof repairs done in 1994, the doctor explained that Mr. Johnston in 1994 gave advice with respect to deficiencies, but never said that putting a new roof on would stop the penetration of water through the wall and leaking at the windows. Dr. Lawless believes that at the time Mr. Johnston said that a new roof would not reduce the water problem at the windows. Dr. Lawless went on to explain that there were deficiencies in the roof, at the roof hatch and the northwest corner in 1994 and repairs were affected there. However, concerns continued to exist with respect to the failure of the roof membrane and the damage that could be caused to the fourth floor, and so it was decided to replace the roof in 1998. Mr. Nelson asked whether the roof was upgraded in any way in 1998, and the doctor explained that the only difference was that a slope was put in, a slope which was actually called for in 1986 and not installed. The doctor confirmed that the drawings show an irma roof, that is an inverted roof with the membrane on the bottom then the insulation on top of the membrane and then the gravel on top of the insulation. The doctor recalled at the time a mention of an irma roof with a 2% slope. Mr. Nelson asked Dr. Lawless whether the drawings showed an irma roof which apparently they do, and Dr. Lawless testified that he believed at the time they were getting an irma roof. He never had any conversation with Mr. Garbutt with respect to a conflict between the specifications and the drawings with respect to the roof.

At Tab 67 is a letter from Mr. MacAdam to the plaintiffs dated August 1, 1993, referring to a meeting with them on July 29. Mr. MacAdam in his letter states that during his visit with them he expressed concern at *inter alia*, the removal, in part or completely, of the masonry support plates from the perimeter beams. It was Dr. Lawless's evidence that he does not know whether it was at that time that he mentioned the cutting of the support plates from the beams. The letter also confirms Mr. MacAdam's expression of concern at the absence of any flashings or expansion joints in the walls, and it was the doctor's evidence that to that point in time he was not aware of the significance of flashings or expansion joints in the walls.

Dr. Lawless is aware that Don Emmons has now passed away, Mr. O'Donnell has passed away, the Kingston files relating to the construction cannot be found and the shop drawings no longer exist.

Filed as Exhibit #20 were photographs of Dr. Lawless's new home on the St. Lawrence River. It looks to be a beautiful house, some 6,000 square feet.

With respect to the cost of the new roof in 1998, while his then lawyer on discovery said that the plaintiffs were not claiming the full cost but rather a *pro rata* share of it, the plaintiffs now take the position that they are claiming the full cost.

At Tab 48 is a document prepared by Dr. Reed which *inter alia* lists the change orders in which Mr. Garbutt was involved. In the middle of the page the following words are written: "additional costs (plus or minus $200,000) over tendered costs are itemized by change orders. The majority of these were ordered and supervised by the owners, not the engineer." It was the doctor's evidence that notwithstanding that notation, while the plaintiffs asked for various changes they never felt that they were supervising the changes. Dr. Lawless went on to say that he was very naïve. He put great faith in the defendant company that was constructing the building. When he spoke to the company with respect to changes he and Dr. Reed had ultimate faith that the changes would be implemented properly.

With respect to his regular attendance at the site meeting, the doctor explained that it was not at all unusual that the person or persons that were spending the money would attend the site meetings.

Cross-examination by Mr. Garbutt

Dr. Lawless confirmed that no photo's were taken of the 1989 alteration to the south face first floor for the flower shop, but the doctor recalled that there may be some part of the video that shows this. He confirmed that no samples were kept of the walls and the bricks were used for a planter.

At Tab 2, the General Conditions, Item 34 discusses shop drawings. The term 'shop drawings' means drawings, illustrations, schedules etc. and other data which are to be provided by the contractor to illustrate details of a portion of the work. In this regard, the doctor recalled that when they discussed changes, someone would draw a sketch and inquire as to whether what was depicted would suit the purposes.

Re-examination by Mr. Mueller

At some point in the cross-examination, Mr. Nelson and Dr. Lawless were talking about precipitation. By this, the doctor meant water droplets.

With respect to the alterations in 1989, Mr. Mueller inquired of Dr. Lawless whether at that time he observed any ties, and the doctor responded that he would not have known what to look for.

Witness – Tony Garofalo

Examination-in-Chief

Mr. Garofalo has been a general contractor since 1960. Approximately two weeks ago he was escorted to floors two and four of the subject building by Dr. Lawless for the purpose of preparing an estimate to repair the drywall damage and painting around the windows on those floors. He provided a quote at $75 per window to repair drywall and paint. In his quote, Exhibit #21, he notes that any wallpaper required would be additional. In his evidence, he notes that the $75 per window is an average, with some windows requiring $100 worth of work and some $150, etc.

Cross-Examination by Ms. Gambhir (junior counsel to Mr. Nelson)

Mr. Garofalo is the general contractor with respect to the construction of Dr. Lawless's new home on the St. Lawrence River. He confirmed that no architect has been engaged by Dr. Lawless on that project.

Referring to a covering letter from Mr. Mueller which accompanied Mr. Garofalo's quote, Ms. Gambhir suggested to the witness that there were some 35 windows in all involved at $75 per window, for a total of $2,525. Ms. Gambhir illustrated her deductions on a sheet of paper which together with the letter from Mr. Mueller were marked as Exhibit #22.

Re-examination by Mr. Mueller

Mr. Garofalo confirmed that with respect to the construction of his home, Dr. Lawless is giving no instructions on matters of trade or construction; his role is confined to choosing colours and floor materials, etc.

Witness – Robert Kendall

Examination-in-Chief

Mr. Kendall's c.v. was filed as Exhibit #23. He has been involved in the roofing industry as both a contractor and a consultant, providing specialized inspection and consulting services to the construction and architectural community since 1953. Major projects which have been carried out by him include the SkyDome, the SunLife Towers, and St. Michael's Cathedral re-roofing in the year 2000. He is an extremely impressive and well-spoken witness. He has been involved with three to four roofs per day for some 50 years, half of them being new roof installations and half re-roofing. He explained that the inverted irma roof installed in the plaintiffs' building came into existence in the early '70's and he has been involved with approximately 100 such roofs since that time.

In all of the circumstances I found Mr. Kendall well-qualified to provide the court with, in addition to direct evidence, his opinion evidence with respect to the roofing problems.

At Tab 102 is Mr. Kendall's report to architect Paul Johnston dated October 11, 1994. Marked as Exhibit A was Mr. Johnston's report Folio 2 for the purpose of permitting Mr. Kendall to refer to Folio 2 when appropriate. It was later marked as Exhibit 55.

After being asked by Mr. Johnston to assist with respect to resolving the roof problems on the building, Kendall and Johnston attended the building on October 5, 1994. Kendall's report notes that he observed a roof which was an inverted assembly, employing two layers of 3" Roofmate, and built-up felt waterproofing membrane on the concrete slab. He notes that the assembly also included a polyolifin filter basket and approximately 2 ½" of crushed limestone ballast. The flashings were pre-painted sheet metal.

In his evidence Mr. Kendall indicated that he was familiar with this type of roof system that came into existence in or about 1970. The roof is essentially inverted or turned upside down compared to the standard roofing. Mr. Kendall explained that the waterproof membrane is laid directly on to the concrete slab roof. Insulation is then laid on top of the membrane and ballast is then placed on top of the insulation to avoid flotation and to keep the sunlight off it. Mr. Kendall explained that generally this type of roof, the "irma" roof has done fairly well in Canada. Where there have been failures, often they were caused by saturation of the felt membrane, often as a result of lack of slope.

Mr. Kendall explained that when this roofing system came out in 1970 or thereabouts, inventor Dow Chemical stipulated that the insulation should be glued to the top plies of felt. This stipulation was changed in 1977 when Dow Chemical recommended that the insulation could be either loose or adhering to the top layer of felt. In the early 1980's, Dow Chemical stipulated that adherence of the insulation to the top layer of felt was no longer acceptable and that the bitumen should be cool before the first layer of insulation is applied, to avoid adherence. The problem with adherence was that in prolonged rain storms the storm sewers could not take the load, the drains would fill up and water would collect on the roof, perhaps to a depth of several inches. The insulation has a strong tendency to float and when it did, if it was adhered to the top layer of felt, then it had the effect of tearing off the top layer of felt. Roof felts are essentially scrap paper bonded together with bitumen, a waste oil product that is put on the felt to saturate it. When membranes split off, the result can be disastrous.

Mr. Kendall did point out however that if the irma roof has proper slope and there is no adherence between the insulation and the top layer of felt, the roof system does very well. Many today are 28 to 30 years old and still doing well. Mr. Kendall speculates that this type of roof, if properly constructed, could be a 50-year roof but it is not possible to say at this time because that period of time has not expired, the system being put in place in 1970 for the first time.

Mr. Kendall explained that one great advantage of the irma roof system is that the membrane avoids the fluctuations in temperature from minus 20 degrees to 150 degrees Fahrenheit as is experienced by a conventional roof. The membranes of an inverted roof on the other hand are always at a relatively consistent temperature equal to the interior temperature of the building which fluctuates only 15 degrees Fahrenheit from winter to summer.

When problems are experienced in an irma roof, usually saturation is the root of the problem. The structure of the roof is such that water percolates through the ballast down to the insulation and then drains through the joints of the 4’ x 2’ squares of insulation to the membrane where it moves to the drains. If the water does not move to the drains, the felts become saturated from the standing water inside the joints, and algae, slime and moss begin to grow. This further blocks the joints between the insulation blocks. One must bear in mind that the drains are at membrane level, not insulation level.

In order to get the proper slopes in an irma roof, the slope could be incorporated in the concrete roof slab but here the roof slab was left flat in anticipation of another story being added. Where the slope cannot be incorporated in the roof slab, then insulation is cut in sloped wedges and placed beneath the membrane to achieve the proper drainage. This technique has been around since the ‘70’s and is widely used.

At Exhibit A, Tab K, is the Dow Chemical specification for the irma roof system and more particularly for Roofmate insulation which is the Dow Chemical product used in the system. This particular insulation is of itself waterproof by virtue of a skin that surrounds the insulation. At page 3 of the document under the heading <u>Design Considerations</u>, it is recommended that membrane flashings be extended well above the expected high water level, and there is reference to the Canadian Roofing Contractor’s Association (CRCA) manual for details. Mr. Kendall explained that this is often referred to as the flood rim. The membrane must go a minimum height above the main field which is the membrane that sits on the roof slab. The aim is to construct a basin of membrane up against the parapet. Mr. Kendall explained that in standard low parapets which have a height of up to 18” or two feet, the appropriate construction practice is to come up the inside of the parapet with the membrane, then completely over the top of the parapet and down the outside of the parapet approximately two inches.

The Dow Chemical specification at page 3 further points out that “all roof decks should have proper drainage of the membrane surface. The CRCA recommends the deck have a minimum slope of 1:50 (1/4 inch in 12 inches) toward the roof drains for bituminous membranes." Mr. Kendall explained that this specification is universally accepted as a roof construction standard. It is recommended by both the NRC and the CRCA. It represents a 2% slope. Mr. Kendall further explained that on occasion, depending on practical considerations, construction may go to a one per cent slope which is 1/8 inch in 12 inches, however he pointed out that most institutional buildings such as we have here would stay at a 2% slope for the roof.

At page 4 of the Dow Chemical specification it is pointed out that "if a bituminous membrane is used, the final flood coat must be allowed to cool before applying the insulation loose. Some single ply membranes and coat tar pitch require a slip sheet. It is the roofer's responsibility to ensure that <u>bonding of the insulation does not occur</u>." Again Mr. Kendall pointed out that this change in specification occurred in the late '70's, early '80's when it became generally accepted in the industry that the insulation should not adhere to the membrane.

At Tab B of Exhibit A is the standard specification from the CRCA which strongly recommends a minimum 2% slope for proper drainage for irma roofs. At page 3 of the specification is reference to a requirement in an irma roof for positive drainage. Mr. Kendall explained this means that the roof must have appropriate slope.

At Tab C of Exhibit A, the CRCA specifications with respect to irma roof systems again stipulate a minimum slope range of 1:50 which is 2% and stipulate that the flood coat be allowed to cool before laying the insulation.

At Tab E is the general condition for Division 7 – Thermal and Moisture Protection. It again provides that for inverted roofs, there should be slope structures to drains at at least 100 millimetres per 5 metres which is again 2%. This general condition is found as well at Tab 1 of Exhibit #1.

By paragraph 1 of Division 7, the contractor is also referred to CRCA roofing specifications.

Referring to paragraph 3 of Division 7, under the heading Standard Built Up Roof, Mr. Kendall explained that this paragraph is describing a conventional roof as opposed to an inverted roof. For example, it calls for a vapour barrier which an inverted roof does not have. It also calls for the insulation to go on before the membrane which again is opposite to an inverted roof.

Paragraph 6 of Division 7 requires the construction of membrane flashings at all junctions of the roof with vertical surfaces, with the same bitumen and roofing felts as used for the roof application. The specification further requires that 4-ply felt and asphalt membrane flashings be carried up the surface of the wall to a point not less than 8" above the surface of the roof. Mr. Kendall explained that this was the "basin" he referred to earlier. He explained that the four plies are built up with bitumen as the roof is constructed. That is, there are four individual felt sheets that are used to eventually construct one membrane. The membrane obtains its strength as a result of the lamination. In this case when there is reference to the membrane being brought up a minimum of 8" above the surface, the surface referred to is the top of the ballast.

At page 15 of Folio 2, Exhibit A, there is reference to the fact that only one ply of the bitumen and felt was carried up to the top surface of the parapet, and even it did not extend to the exterior face of the brick wall but rather was discontinued approximately 2/3 of the way across the width of the parapet. Mr. Kendall explained that that observation and what is demonstrated in the photographs does not comply with the specifications, and it permits water to get through the joints of and under the metal cladding to the underportion of the parapet, which water would in this case end up flowing into the space between the bricks and the concrete blocks. It was Mr. Kendall's evidence that the membrane should have gone completely across the parapet top and over onto the face of the brick wall. Mr. Kendall on October 5, 1994 observed this defect and indeed took a photograph of it which is at his report, Tab H, photograph #3. He explained that the only thing on top of this assembly as shown in photograph #3 was the tin cladding which allowed water to come into the top of the parapet and thence to run down between the bricks and the blocks.

Referring again to Johnston's report, Folio 2, page 15 and the photographs at the bottom of the page which serve to record the repair to the northwest corner in 1994, Mr. Kendall testified that based on what he observed he saw the problem to be this. Between the horizontal surface of the roof slab and the vertical inside surface of the parapet, a cant strip is placed which from a side view is a triangular-shaped strip that "softens" the angle between the two surfaces. He explained that the split or tear observed in the photographs is at the top of the cant. In his view the problem was caused by the cant not being placed so as to abut the vertical inside wall of the parapet, thus allowing a space between the vertical surface of the cant and the vertical inside surface of the parapet. This provides a space between the cant and the vertical inside surface of the parapet where there is no support for the membranes that are being run up the side of the parapet. In this respect, he explained that generally one runs four plies of felt up to the top of the cant, and then four new ones down the side of the parapet for a total of eight plies at the juncture of the roof slab and the parapet. This should be a very strong joint but in the absence of proper support it is weakened, particularly if there is only one layer coming down to overlap the horizontal membranes. Exhibit #24 is a sheet of paper on which Mr. Kendall illustrated the problem. What Mr. Kendall described and illustrated in his evidence is referred to at page 16 of Folio 2, where Mr. Johnston notes that at the area of the cant, only a single felt was brought down along the side of the parapet and as a result the bitumen broke away in a "characteristic potato chip fashion" when very slight pressure was applied to the rupture. Mr. Kendall testified that what was being described he would characterize as a workmanship deficiency.

Filed as Exhibit #25 are the original colour photographs attached to Mr. Kendall's report of October 11, 1994 at Tab 102 of Exhibit #2, and also contained at Tab H of Exhibit A. For the purpose of this evidence it is preferable to refer to the colour photographs which are attached to Mr. Kendall's report at Tab H of Exhibit A. Photograph #1 shows the east end of the outside south wall. Photograph #2 shows the top of the parapet with the metal removed, exposing the plywood top. At the joinder of one of the plywood sheets there is a rotten corner and there are water stains on both ends of the plywood. These photographs by the way were taken on October 5, 1994.

Photograph #3 shows the top of the parapet with the portion of the plywood topping removed in order to observe what is underneath. Photograph #4 again shows the rotten corner and severe staining on the top plywood and shows the membrane coming about ¾ of the way across the wooden board that lays on top of the blocks. It also shows the cavity between the inside of the brick wall and the blocks. In this regard in Photograph #5, the opening of this cavity is even clearer. It was Mr. Kendall's evidence that the failure to extend the felt completely across the top of the parapet and a distance down the outside of the brick wall was a fundamental error, particularly having regard to the gap between the brick wall and the block wall which should not have been there in the first place. Mr. Kendall explained that the staining and rottenness on the plywood shown in Photographs #2, #3 and #4 was caused by water being pushed through the joints of the top metal cladding and from the plywood topping the water would then go down the space between the brick and the blocks. Mr. Kendall explained that the top metal cladding is constructed such that one end floats freely in a lap joint of the next piece and one anticipates expansion, depending on the weather such as to permit water to enter at the so-called lap joints. The protection with respect to the water extending further is to extend the membranes to the outside wall and turn them down against the outside wall. In this case the membrane ended up a couple of inches short of the outside of the block, when it should have gone right across the block, across the cavity, across the brick and down the outside of the brick wall.

Photograph #6 shows that the top metal cladding of the parapet extends only ½" below the top of the brick, allowing for water during a rainstorm to be blown up between the lower edge of the metal cladding and the outside wall of the brick. Mr. Kendall explained that proper practice in this circumstance would be the installation of a 2" starter strip, extending along the top of the brick wall to receive the lower edge of the top cladding. In this case the membrane did not extend over the top of the parapet, there was no starter strip and there was no extension down the outside face of the brick wall, all of which allowed wind to blow water up and over the brick wall. Photograph #7 shows water spraying upwards in the wind. The source of the water is a hose that was left hanging over the side of the building. According to Mr. Kendall's evidence, on the day in question he observed the wind blowing the water some 4' over the top of the building. With respect to the use of this hose, Mr. Kendall strongly denied the anticipated evidence of Mr. Quirouette to the effect that Mr. Kendall and Mr. Johnston had directed water directly down the gap between the brick and the block.

Photograph #8 shows a piece of insulation that was taken up from the top of the membrane during the course of the inspection. The photo depicts part of the top membrane adhering to the insulation and part of the insulation adhering to the membrane.

Photograph #9 shows the filter blanket pulled back showing previous cuts in the insulation which Mr. Kendall could not explain.

Photographs #10 - #16 illustrate some of the testing that was done on October 5. In photograph #10 one can see the channel that was cut leading to a drain. The channel was filled with water by a hose at the top of the photograph and it was allowed then to drain away. Photograph #11 illustrates that after drainage had completed the depth of water left in the channel was 5/8" with the result that after drainage was complete 5/8" water was left at the parapet, indicating negative drainage and allowing water to accumulate. Photographs #12 - #16 illustrate the testing that was conducted on the roof in the location of the west stairway and west parapet. It was known that there was a leak into the upper corridor and the testing was designed to determine whether the wall or the roof was the source of the leakage. In order words, water would be introduced onto the roof. If there was no leak in the corridor then the cause would be attributed to the wall. If there was leakage into the corridor then the cause would be attributed to the roof. It was Mr. Kendall's evidence that once water was applied to the roof, there was a darker colour observed on the outside of the brick wall below the top of the wall as depicted in photograph #14. That told Mr. Kendall that the water was coming from the roof down the interior of the brick wall, and this was occurring when the water on the roof was still 3" – 4" below the parapet. His conclusion was that there was "a hole in the boat" at the cant as illustrated by the photographs at page 15 of Exhibit A.

Photographs #15 and #16 show the same wall as depicted in photo #14, and according to the evidence of Mr. Kendall show signs of leaking over a long period of time. His evidence was that in order to get so much efflorescence as depicted in photograph #15, leakage would have had to occur over a long period of time. It could take months of leakage to get the kind of efflorescence depicted. His conclusion was that this was evidence that water was getting in behind the brick wall for some time.

Referring to Tab E of Exhibit A, the Division 7 specifications, on the second last page, Item #1 specifies that flashing shall be 26 gauge, pre-finished steel, and Mr. Kendall is able to testify that this is what he saw on his inspection. Item #7 of the specifications on flashings provides that details must conform to the principles in the guide specifications contained in CRCA Roof Specifications, "Bituminous Built-Up Flashing". It was Mr. Kendall's evidence that he was familiar with the details of those principles and that what he observed was not in conformity with those principles for the following reasons:

1. The cant strip was not sealed to the inside of the parapet wall;
2. The felts did not extend over the top of the parapet and down the outside of the brick wall;
3. There was no starter strip for the sheet metal top, the outside of which did not extend down the outside of the brick wall the required 2".

It was Mr. Kendall's evidence that these would have been minimum requirements for a competent roofer.

Referring to his report dated October 11, 1994 at Tab 102, paragraph 3, Mr. Kendall confirmed that he was advised that the building had been experiencing water incursion noted by the occupants to be associated with rain storms, and having a tendency not to be a problem during the winter. Mr. Kendall in his evidence confirmed paragraph 4 of his report in that upon inspection he could see that there was water damage consistently on the top floor in the window reveals, with an emphasis at the window heads.

Mr. Kendall in his evidence confirmed the contents of paragraph 3 on page 2 of his report to the effect that the use of 15-lb. organic felts, which were used here, would further increase the need for gradients in that this type of material must be permitted to dry out on occasion otherwise it can become water-logged which promotes rot and it can lose a lot of its tensile strength.

Mr. Kendall noted in his report and in his evidence that the cap metal flashing was poorly joined with the side metal flashing on the inside of the parapet. A proper design in accordance with CRCA standards would have the two pieces interlocking, which was not the case here.

Mr. Kendall in his evidence confirmed that when the hose was left running on the roof, water appeared in the interior of the building in the corridor a few minutes before it appeared on the outside of the brick wall.

Having effectively completed a review of his report by way of *viva voce* evidence, Mr. Kendall turned to the conclusions that he had reached as set out in the report, which were as follows.

1. This inverted roof had been assembled essentially flat and the retention of water over long periods of time can have the effect of weakening the tensile strength of the felts and may shorten the service life of the roof. At the time of the repairs in 1994 the roof was some 8 years old. It was Mr. Kendall’s evidence that based on his observations it was hard to say how much further life it had beyond 8 years, but bearing in mind that usually construction of a roof is consistent throughout its entirety, when it begins to leak it is a very bad sign and it is usually the beginning of the end. Here, two places had been breached; the roof hatch and the west end by 1994. As a result, according to his evidence, one would have as of 1994 to think about replacing the entire roof. Mr. Kendall went on to explain that when he viewed the roof in 1994 his immediate reaction was to repair the west end and rework the parapet detail, but his gut feeling as expressed to Mr. Johnston was that the roof was reaching the end of its life and that if it was his building he would start replacing it within the next two or three years. His main concern would be a major failure of the membrane which would let significant amounts of water into the building.

At page 23 of Folio 2 there are two photographs depicting the state of the roof prior to the 1998 replacement. It was Mr. Kendall's evidence that the fissures depicted in the top photograph are typical of water getting into the top ply of the membrane as are the decomposed felts also depicted in the photo, indicating that the felts had not been adequately covered with bitumen. As well, the bottom photo shows that the top of the membrane has lost a portion of the last pour of bitumen.

2. Adhering the first layer of insulation to the felts is contrary to the recommendations of Dow Chemical – the manufacturers of the insulation. In the event of flotation it is possible there will be some damage to the upper plies of the felts.

3. The perimeter flashing has to be reworked in the manner described in his evidence. In this respect, we looked at the roof video, Exhibit #18 which depicts the removal and replacing of the roof, beginning November 4, 1998 which work included the reworking of the parapets. The video depicts much inconsistent adhesion between the bottom layer of insulation and the top of the membrane. Mr. Kendall commented that in replacing the roof membrane, the sound old felts were left in place and two new bituminous felts which are tougher were applied over the old felts. The video shows that not only the roof parapets were increased by 7 or 8 inches, but the roof hatch perimeter was as well.

4. Finally, Mr. Kendall recommended that the west wing should be repaired as soon as possible, after removal of the ballast, the pulling back of the filter blanket and removing the insulation with care to examine the built-up felts. Defective areas should then be cleaned, dried and patched with 3 plies of 15-lb felt. The components could then be replaced.

Cross-examination by Mr. Nelson

Filed as Exhibit B for identification were two volumes of materials received from Amherst Roofing, which was the roofer that replaced the roof in 1998. It was Mr. Kendall's evidence that he did not recall assisting Mr. Johnston with the specifications for that replacement, nor with the invitation to bid prepared by Mr. Johnston. It was also his evidence that with respect to Tab 76 of Exhibit B, a letter from Johnston to Amherst advising that the owners were conditionally prepared to contract with Amherst for the replacement of the roof, Mr. Kendall was not involved in that process. Mr. Kendall testified that he had not seen the actual contract for the replacement of the roof at Tab 139 of Exhibit B. He agreed that Tab 139, the contract, lists Mr. Johnston as consultant, and that general condition 11.1 dealing with insurance provides that general liability insurance should be in the joint names of the contractor, the owner and the consultant. Presented to Mr. Kendall was a letter at Tab 70 of Exhibit B, from Johnston to Amherst dated October 2, 1998, requiring the certificate of insurance be amended to include the owners and the consultants Johnston and Kendall. Mr. Kendall was not prepared to concede that he was constituted a consultant by reason of the fact that architect Johnston retained him for special advice.

Mr. Kendall conceded that in preparing his report at Tab 102, he reviewed the original specifications but not all of the drawings. He recalls seeing a couple of drawings involving parapet sections. He confirmed paragraph 5 of his report at Tab 102 to the effect that the specification was confusing in so much as it called for a conventional assembly which had clearly been changed during construction. He elaborated that the specification in Division 7 and in particular s.3 had been written for a conventional roof that was not in fact installed. He indicated however that it was not unusual for things to change by either change orders or site meetings.

Mr. Kendall was shown the roof drawings for the first time during his cross-examination. While he conceded that the three small drawings of Exhibit #9 show an irma roof, it was his evidence that drawings 4 in Exhibits #8, #9, #10 and #11, while they purport to call for an irma roof, are scaled too small to determine if in fact they are reflective of an irma roof.

Mr. Kendall confirmed that in paragraph 7 of his report dated October 11, 1994 he made reference to there being an indication in the drawings and specifications that gradients of 2% were to be included in the roof assembly. It was his evidence here at trial that that paragraph should not include the word "drawings" but should refer only to the specifications calling for gradients. This was a mistake on his part in that the drawings do not show any gradients.

Mr. Kendall conceded that in addition to the CRCA there is also an association known as the Ontario Industrial Roofing Contractors Association (OIRCA).

The witness reviewed the CRCA specifications set out at Tab B of Exhibit A. That particular Tab was put in as a separate exhibit, Exhibit #27. Mr. Kendall agreed that specification 60.10 of the CRCA effective November, 1978 provided for a strong recommendation of minimum slope for proper drainage as being 1:50 or 2%. Exhibit #27 also contains specification 60.09 which appears to be effective August, 1986 and which provides for a requirement of positive drainage for the protective membrane roofing. The suggestion on cross-examination was that the requirement for positive drainage as per the CRCA specifications did not come into existence until shortly after the roof was completed. Mr. Kendall's effective reply was that it made little difference in that the 1978 specification required a 2% slope.

In his report dated October 11, 1994, Tab 102 at page 2, paragraphs 5 and 6, Mr. Kendall states that it was noted that the bottom layer of insulation was partially secured in hot bitumen to the flood coat at the top of the asphalt plies and that while in the early days of inverted roofs this had been Dow Chemicals recommendation, they departed from this recommendation in the early 1980's. It was pointed out to Mr. Kendall that in Exhibit A, page 7, paragraph 2 Mr. Johnston states that "while the practice of adhering the insulation to the membrane was not discouraged in the early '80's, depending on the height of the building, its location and prevailing winds, on buildings of this type and in a standing application, Dow Chemical currently recommends an unadhered application of the insulation." It was Mr. Kendall's testimony with respect to this apparent contradiction that he, Mr. Kendall, was correct.

Referring to his flood testing of the roof on October 5, 1994 it was Mr. Kendall's evidence that it was a very effective test whereby a rainstorm was simulated. When faced with the defendant's expert, Mr. Quirouette's anticipated evidence that this test was five times more powerful than that recommended by American standards, Mr. Kendall pointed out that if Quirouette is saying that then he is saying it with reference to the wall test and not the roof test. With respect to Mr. Quirouette's criticism that the test was tantamount to placing the hose directly into the cavity between the brick wall and the block wall, Mr. Kendall again expressed the view that Mr. Quirouette was purporting to criticize the wall test and that in fact that did not occur. As far as the roof test was concerned there was no pressure involved at all. The water was simply turned on and left to accumulate to see whether or not the roof was graded. It was Mr. Kendall's evidence that he has done hundreds of such tests and that's how you do it, that's the only way that you do it.

At Tab 3 of Exhibit #2 is the report of Inspec-sol dated September 28, 1993. I pointed out to the witness that this document had been admitted not for the truth of its contents but rather to ascertain the plaintiffs' knowledge at any given point in time on the issue of discoverability. In any event, it was Mr. Kendall's evidence that it made no sense to determine that the bitumen flashings were installed only ¾ of the way across the horizontal surface of the parapet, and then to express the belief that the perimeter flashings were not the main surface of the leakage problem.

In Exhibit B, the Amherst file at Tab 1 is a letter from Peter Amherst dated January 17, 2000 addressed to the defendant company, expressing the fact that "we found the existing membrane to be in good condition and the felt flashings also to be in good condition, though the felt flashing only extended to the outside of the inside masonry." On page 2 of the report, Amherst goes on to say that he cannot comment on how much of a problem the lack of felt flashing over the parapet caused, as he has no maintenance history of the building and that he felt that the existing roofing was in good shape, apart from some nuisance leaks at roof projections on the old roof which were maintenance problems. Mr. Kendall, faced with these portions of the report, said simply that it appeared that Amherst had a different view from him, although he, Mr. Kendall, does not recall seeing Mr. Amherst on site during the roof replacement work.

Counsel on cross put the following to Mr. Kendall. In 1993 Inspec-sol thought the roof was in good condition. In 1994 Kendall and Johnston recommended repairs which were completed. The problems continued through 1994 to 1998. In 1998 the roof was replaced and the water leaks continued. In the face of these facts, would Mr. Kendall agree that it was not necessary to replace the roof. Mr. Kendall disagreed. It was his evidence that the roof was never properly put together. The roof was improperly constructed with a lack of gradients and slopes and deficient perimeter details. It was his evidence that roofs that hold water have a minimal life as evidence by leaks at the hatch and the west end. When the insulation was finally removed in the 1998 replacement, there was obvious evidence of deterioration of the felt membrane. Mr. Kendall stood fast in his opinion that it was reasonable to replace the roof in 1998.

Referring to photograph #15 attached to Mr. Kendall’s report of October 11, 1994, Mr. Kendall was asked exactly what he meant when he said that the efflorescence shown in the photographs was indicative of leaking that had gone on for months and months. Mr. Kendall in reply said it was difficult to say. It could be the result of 7 or 8 months, or a longer period of leakage. What he meant to convey in his evidence was that the efflorescence shown was not that week’s problem or that the water test conducted on that day had caused the appearance. It was his evidence that efflorescence would start on a small scale and would be a progressive thing that could extend over a period of years.

Cross-examination by Mr. Garbutt

Mr. Kendall was referred to the comment contained on page 3 of the Dow Chemical specifications, which first calls for a minimum slope of 2% toward the roof drains but then goes on to say, “If, however, the roof is designed to allow ‘ponding’, the insulation should be applied loose with a permeable fabric over the insulation.” Mr. Kendall conceded that the construction of a roof designed to allow ponding would be consistent with a option of adding additional floors but pointed out that such a roof would not be designed with the particular felt that was used here. In the case of the construction of a roof designed to allow ponding, one would have to use a much stronger membrane, perhaps an artificial rubber membrane, and that it would be extremely risky to use the felt that was used here in a roof designed to allow ponding.

Witness – Gordon Robb

Examination-in-Chief

Mr. Robb has 20 years' experience in the construction industry including cost analysis of projects. He is the president of Detra Builders Inc. His c.v. is contained at Tab 114 of Exhibit #2. His is a general contracting firm and has been involved in the construction of major construction projects such as a 21-story residential tower in Toronto and a sewage treatment plant in Long Sault. He reviews with his estimators every tender that is made with respect to a project. His company bids approximately $30,000,000 in projects every year.

At Tab 108 of Exhibit #2 is Mr. Robb's report to architect Paul Johnston dated January 9, 1995, setting out budget costs for two alternative scopes of work, namely Alternate A and Alternate B at $395,000 and $435,000 respectively. The report notes that these budget cost quotations are subject to review of design drawings and details when available.

The report acknowledges Mr. Johnston's fax of December 20, 1994 regarding preliminary scope of work and that document is set out at Tab 107 in Mr. Johnston's handwriting. It is titled "Proposed Scope of Remedial Work and Wall Construction, 275 Queen Street, Kingston, Ontario."

The scope of work breakdown attached to Mr. Robb's letter of January 9, 1995 sets out the following alternatives, as did Mr. Johnston's scope of work.

<u>Alternate A</u>

"Apply Tyvec-type prophylactic membrane to existing back-up block. Provide new brick masonry face ($450 per thousand supply allowance). Reinstall lintels, caulking and control joints."

Alternate B

"Weld new shelf plates to allow brick to move out 25 to 40 millimetres. New shelf angles to stairwell. Move out existing stone window surrounds. New 25 millimetre AF fiberglass cavity wall insulation. Provide new brick masonry face ($450 per thousand supply allowance). Reinstall lintels, caulking and control joints."

Alternate A is essentially a brick veneer wall. Alternate B is a cavity wall.

At Tab 109 is a report from Mr. Johnston dated January 31, 1995 to the plaintiffs, setting out in the last three pages thereof a more formal typewritten scope of remedial work which is essentially the same as the handwritten scope of work set out at Tab 107.

At Tab 114 is a report from Mr. Robb to Mr. Mueller dated January 23, 2002 which sets out the details used by Mr. Robb in arriving at his cost analysis. In this detailed cost analysis the total budget cost for Alternate A is $401,245.85 and for Alternate B, $433,313.84. Mr. Robb explained that for the purpose of presentation to Mr. Johnston in his report dated January 9, 1995 he rounded those numbers to $395,000 and $435,000 respectively. As well, in his letter to Mueller dated January 23, 2002 Mr. Robb presented budget figures for the items that had originally been excluded from his pricing of January 9, 1995.

Further in his report of January 23, 2002 he presented his view that there should be an inflation adjustment from 1995 to present because of increase in the cost of building materials and labour at the rate of 5% per year. This in Mr. Robb's view would be an average inflation factor. In some years it has been higher, in others lower, and the inflation factor has varied with the item under consideration, eg masonry costs have gone up fairly significantly now because of a limited supply of masons.

Finally, Mr. Robb testified that in his costing there is no allowance for winter work.

Cross-Examination by Mr. Nelson

The details of the pricing contained in Tab 114 were the details prepared in 1995.

Alternate B includes new fiberglass wall insulation. Mr. Robb was asked whether the original building had wall insulation. He responded that he was not privy to that information but was fairly sure that it did. The new insulation in Alternate B comes from the scope of the work at Tab 107 and more specifically at the bottom of page 3. Alternate A did not include new wall insulation but called for the replacing of the existing insulation. Mr. Robb conceded that with respect to his opinion with respect to inflation, he did not do an analysis of costs in the Kingston area. His opinion is simply based on his experience.

At Tab 114, the details of the costing include a number of items under "General Requirements". According to Mr. Robb's evidence these were specific job costs and not overhead. Overhead is calculated in a separate item at page 4 at 5% of costs, and is meant to cover such things as office expenses.

Although for the purpose of providing an opinion with respect to inflation, he did not specifically review the costs in the Kingston area. Mr. Robb was familiar with work in Kingston generally.

Cross-examination by Mr. Garbutt

Mr. Robb conceded that there are certain indices available by year and location for the purpose of calculating inflation. He did have a look at Mean's Construction Index but did not spend a lot of time on it. He looked specifically at brick and noted that there was a 16% increase in the price of brick from 1995 to 2002. He defended his reluctance to depend on the indices because he does pricing every day and knows the extent to which costs have risen. He is bidding two to three projects every week and has a good sense of the increases in prices over time.

Witness – Robert Drysdale

Examination-in-Chief

Dr. Drysdale's c.v. was filed as Exhibit #28. He graduated with his BSc in civil engineering from the University of Manitoba in 1962. He received his M.A.Sc. in civil engineering from the University of Toronto in 1965 and his Ph.D. in civil engineering from the University of Toronto in 1967. His c.v. leaves no doubt whatsoever that he is eminently qualified to provide opinion evidence as an expert on masonry construction and building envelopes. He is presently the Director of the Centre for Effective Design of Structures at McMaster University and has been the recipient of many honours and awards since 1961 to present. Last year he was made a Fellow of the Canadian Academy of Engineering. He has co-authored several books on masonry structures and has published some 88 research papers since 1982, many of which relate to the design and construction of masonry walls.

Dr. Drysdale's report to architect Paul Johnston dated May 6, 2002 was filed as Exhibit #29. His back-up analysis of forces due to lateral wind pressure on the brick veneer, concrete block back-up walls for the subject building was filed as Exhibit #30. It should be noted that both Dr. Drysdale and the defendant's expert Mr. Roney were ordered by the court to provide their back-up calculations. Although Mr. Roney was to provide his first, as it turned out Dr. Drysdale because of time constraints prepared his analysis prior to receiving Mr. Roney's and delivered it to the defence. Dr. Drysdale eventually received Mr. Roney's calculations and saw that his calculations had been revised to take into consideration that the floors were not three metres in height but rather 3.2 metres in height.

Mr. Roney's original report of April 26, 2002 to which Dr. Drysdale was responding by way of his report dated May 6, originally assumed three metre stories. The problem for both Mr. Roney and Dr. Drysdale has been that the drawings showed a three metre floor height which was changed to 3.2 metres without any change order. Dr. Drysdale has now amended his calculations to take into consideration the fact that the stories are 3.2 metres in height and filed as Exhibit #31 is Appendix G to his report which shows a height of 3.2 metres per floor, which results in bending and stresses being extended by 16% to 17%. Filed as Exhibit #32 is a short response by Dr. Drysdale to Mr. Roney's engineering calculations dated May 14, 2002.

Filed as Exhibit #33 are certain Code provisions which are relevant with respect to these issues.

Mr. Garbutt, prior to the commencement of this trial, was served with a notice to admit and in response thereto, Code M78 was brought into the picture. Dr. Drysdale's brief comments with respect to Mr. Garbutt's position was filed as Exhibit #34.

Referring to Exhibit #29, his report dated May 6, 2002, Dr. Drysdale explained that the report deals with the conditions of the subject building under the categories of design and construction.

Under the heading of Design, Dr. Drysdale explained the following.

As noted in his report, he has concluded that the steel framing of this building was not designed as a moment-resistant frame, that is a frame resistant to stress and load. As a result the stability of the building against lateral loading depends on the infill concrete block masonry, and to properly do its job the masonry wall should be built tight to adjacent columns and to the underside of the beams. However in our case the block wall was not tight to the columns and therefore the strut action which would have resulted from the block wall being tight to the columns was not developed to ward off lateral load. As well, it appeared that the mortar joint between the top of the block wall and the bottom of the beam was not consistent in its application in that it appeared that mortar between the top of the block wall and the bottom of the beam was missing in some places. The Dr. commented that the plans were not clear with respect to the type of the joint required, however the specification did call for a soft joint under the brick shelving, that is a joint where the mortar has been left out. Notwithstanding the specification, in all cases the joint between the brick wall and the shelf was mortared solid, in other words was a hard joint which would cause the brick wall to be load bearing upon expansion. The brick wall was never meant to be load bearing.

In his report and in his evidence, Dr. Drysdale pointed out that in this building the open web steel joists bear on the flanges of the beams along the north and south sides of the building. As the floor is put in place there is a dish created in the middle of the steel joists from the weight of gravity. This causes the beam to twist and can result in a crack in the masonry. Once the wall is cracked it cannot resist bending imposed on it by wind sheer force from the side, and we are left with lateral instability. Tests show that the existence of cracking in the veneer brick wall will increase water penetration by a factor of five.

Neither under the Empirical Rules nor the Engineering Analysis method is the capacity of these masonry walls to resist out of plane bending resulting from wind pressure or seismic forces sufficient. As a result by Dr. Drysdale's calculations and in his opinion, the masonry walls are not structurally adequate.

In Dr. Drysdale's view it was necessary in this case to connect the inside block wall to the steel frame by way of mechanical connectors. This is required by the M78 version of CSA S304 (the masonry standard).

It was the Dr.'s evidence that structural movements of the building requires that movement joints or expansion joints be inserted, both horizontally and vertically. In the first instance this prevents transfer of gravity loads from the structure into the veneer. Vertical movement joints are required to relieve stresses caused by differential expansion and contraction. In this building, there was but one vertical movement joint on the south face. There were no horizontal movement joints. As a result there were no mechanics to relieve stress and tension and compression on the brick wall, resulting in the possibility of breaks in the wall with perhaps entire panels of brick falling off. In this respect the Dr. did observe some very visible cracks on the east end of the south face and the south end of the west face. The Dr. noted that the specifications did call for expansion joints, see the specifications at Tab 1, Division 4 – Masonry, section 2.2, "provide control joints as dictated by structural design." The Dr. explained that those control joints should be shown on the drawing and if not shown on the drawing then in such circumstances the contractor should go to the designer (Mr. Garbutt in this instance). The

Dr. noted as well that the designer should have put the control joints in the drawings in the first place. While specifications normally dominate over drawings, section 2.2 is simply not sufficient by itself to tell the contractor what to do with respect to expansion joints.

Finally under the heading of Design, Dr. Drysdale in his report and in his *viva voce* evidence gave his opinion that the building envelope was not adequately designed or specified. As a veneer wall with an airspace between the brick veneer and the concrete block back-up, full development of the rain screen approach was required, and this should have included rain resistant veneer and open vertical airspace between the blocks and the bricks, properly installed flashing at the base of the wall, open weep holes, perhaps but not necessarily vents at the top of the veneer at each floor, and some water resistance of the back-up wall. The above were not shown and were inadequately specified. Dr. Drysdale in his *viva voce* evidence went on to explain that flashings and weep holes at the top of every window is an absolute requirement whether or not it is a veneer wall or an open cavity wall.

As well, the Dr. explained that the above inadequacies, while they begin as a design problem, become a construction problem.

That brought us into the construction portion of Dr. Drysdale's report.

From photographs and descriptions of visual surveys, wall openings and instrument surveys, Dr. Drysdale concluded that there were many construction faults, some of which are related to or have aspects in common with design omissions. Specifically he listed the following:

a) lack of vertical movement joints which would address horizontal expansion and contraction. It was his evidence that this can cause failure by way of cracking of the mortar joints.

b) lack of positive connection of masonry to the steel frame. This is with reference to the block wall.

c) inadequate roof drainage.

d) improperly placed ties and use of improper ties for large cavity sizes. Dr. Drysdale pointed out that it has been documented that some of the ties were bent during installation and therefore lost strength. He noted also that the corrugated ties actually used on the project were under the Code limited to a void of 25 millimetres of less and the void between the block wall and the brick wall at the top of the building was more than that maximum.

e) lack of corrosion protection on the ¼ inch shelf plate designed to support the brick.

f) lack of construction to provide an effective rain screen design. Dr. Drysdale explained that effective rain screen design requires that the brick wall be made as impervious to rain as possible, then there should be provided a proper cavity. It was his evidence that none of these were provided other than a cavity that had some mortar in it and that would provide some drainage.

g) there was an ineffective vapour barrier provided which was not attached to the construction members.

h) there was a lack of effective air barrier. This was not called for on the drawings or specifications. This could have been constituted by the drywall if properly attached and painted. As well, it could have been constituted by the block wall if it had been parged and sealed. That was not the case here. As a result, warm moisture-laden air transfers into the wall cavity in the winter time where it condenses.

i) lack of veneer support and effective horizontal movement joints. Dr. Drysdale explained that he was referring to the lack of ¼ inch support plates on the east and west wall. He explained that the specifications called for open joints between the top of the brick and the bottom of the support plate on the north and south wall and in fact they were mortared solid.

It was the Dr.'s evidence that all of these construction faults caused him serious concerns relating to structural performances such as the build-up of stress in the masonry resulting in cracks and lack of connection of the block wall to the steel which could result in the wall being pushed out under stress.

Dr. Drysdale's report then goes on to comment on Mr. Roney's report dated April 24, 2002. Some of the report he finds helpful and agrees with. He notes that the report confirms that the wall system is a brick veneer system. He explained that a cavity wall, as opposed to a veneer wall, must have a minimum cavity of 50 millimetres according to the Code, and that does not exist in our wall.

Dr. Drysdale notes that the Roney report limits remedial measures related to "structural issues identified in this report. Other remedial work associated with building science issues and general items such as localized repointing shall be addressed by others." In Dr. Drysdale's opinion the building science issues are closely related to many of the same factors that have to be addressed structurally. He notes that a structural solution must, rationally, accommodate and take into account solutions developed for the building envelope. The structural solution recommended in the Mr. Roney report, he notes, is limited to the addition of Helifix ties and introduction of vertical control joints in the brick at the four corners of the building.

With respect to the suggestion of replacement ties, Dr. Drysdale makes the following points:

1) He disagrees with Mr. Roney's analysis that the walls as constructed can safely resist the out of plane lateral and anticipated vertical loads specified in the Ontario Building Code, with certain exceptions. He notes that the variable nature of axial load on the block and the even more uncertain self-weight effects of the brick veneer make accounting for the benefits of axial load very problematic. In his view the walls are under-designed for out of plane wind load for both the 1985 period and now.

2) The Helix ties suggested by Mr. Roney have high axial stiffness and therefore cannot be used to supplement the much more flexible metal strip ties presently in place. As the stiffer ties will attract most of the load, they would have to installed as a complete replacement system, not as a supplement to the present system. Dr. Drysdale noted that he did a computer model of the wall whereas Mr. Roney did not.

   Entered into evidence as Exhibits #36, #37 and #38 was the Helifix tie suggested by Mr. Roney, a larger Helix tie and the ordinary tie used in this construction.

3) The regions with wide air space in many locations and incorrect installation of the metal strip ties will require replacement ties. The Dr. noted again that the Code provides that the maximum width of cavity for the ties used is 25 millimetres or one inch, and that distance is exceeded in various place.

4) The introduction of Helix ties connecting the brick and block will not provide a composite action in the sense that the inside and the outside wall will work together in an effective way. In his view, the low flexural stiffness of the ties and only partial end fixity in the masonry including cracked mortar joints, makes this an unlikely solution.

5) The Dr. notes that the top two feet of the brick veneer is effectively not attached to anything because there is no block back-up wall at the top.

6) Mr. Roney makes no provision for attaching the block wall to the steel frame. Relying on friction is a very doubtful practice according to Dr. Drysdale, as axial load will be highly variable, including the potential for no axial load.

7) Mr. Roney makes no mention of the fact that the ends of the block panels do not solidly abut the steel columns. To be effective in plane sheer resistance, they must fit tightly to the columns and to the underside of the beam. The infill panels need to be mechanically attached to the columns and the top bearing must be ensured. None of this had been addressed in the Mr. Roney report.

8) Mr. Roney suggests the introduction of vertical control joints at the corners of the building to relieve horizontal stresses. Dr. Drysdale notes that the removal of support by the perpendicular wall will also place greater need on tie capacity in this high wind pressure region. In addition, Dr. Drysdale is of the view that because of the local weakness introduced by the wall openings, closer spacing of vertical control joints would be required over the lengths of the walls.

9) With respect to horizontal control joints, Dr. Drysdale notes that in the building there are cases where the ¼ inch plates exist and do extend into the brick veneer, whereas in other locations they have been cut off. In the regions where the brick support exists, the space between the upper and lower course of bricks that exists beyond the toe of the steel plate has been filled with mortar, with the result that there will be eccentric loading on the

brick wall due to brick expansion which can lead to veneer buckling or spalling. In his view, horizontal movement joints need to be installed.

10) Mr. Roney has not addressed the fact that corrosion of the structural steel has been observed in many places.

Dr. Drysdale concludes by providing his opinion that the remedial measures suggested by Mr. Roney which avoid replacement of the veneer will not effectively address the documented structural and building's envelope issues. The remedial work suggested by Mr. Roney will not ensure endurance and in many ways does not even address some of the major structural issues.

Exhibit #30 is Dr. Drysdale's report on the analysis of forces due to lateral wind pressure on the brick veneer/concrete block wall for the subject building. For this analysis, Dr. Drysdale used the structural analysis computer program known as SAP 2000, which is a popular two-dimensional program. The purpose of using this program was to simulate in the wall the Helix ties that were being recommended by Mr. Roney as a remedial measure. It seemed clear on Dr. Drysdale's evidence that in all cases he fed into the program assumptions, values that were more favourable to the remedies suggested by Mr. Roney than to the position taken by the plaintiffs.

The actual brick used in construction had a frogged bedding plan (indent) on one side of the brick, but the assumption fed into the program was a solid brick. The result of this approach would be to underestimate the flexural stresses on the wall as the indent does not become effectively filled with mortar. By way of further example, the block used in the inner wall was in fact only 60% solid, but Dr. Drysdale fed into the computer program the assumption of a solid block.

Dr. Drysdale described in his report the various analyses which were done with respect to a variety of support conditions, the results of which form the basis for Exhibit #29.

After discussing the loading assumptions that were used in the computer program, Dr. Drysdale in his report provided comments on the analytical results as follows:

1) The computer runs show that the ties provided negligible coupling of the brick wall to the block wall.
2) Regardless of the modeling used for the various cases, the results in terms of maximum tensile strength in the masonry and maximum axial tie force were very similar.
3) The maximum tensile stresses in the brick wall and block wall need to be increased to account for higher stresses resulting from openings in the wall and for higher pressures near corners.
4) Maximum tie forces were exceptionally high.

Dr. Drysdale as a result of his analysis reached the following conclusions:

1) The brick veneer wall is reported to have very low bond values and there is not a reliable source of flexural resistance, to resist lateral wind pressure. Even if it had tensile strength comparable to Code design values, cracking would be expected and in the event of cracking the brick veneer is no longer able to share in resisting the load.
2) Although the bond conditions for the block wall are not known, Code values are an optimistic prediction of this characteristic. Even such values are exceeded by a wide margin by the predicted stresses from wind loading.
3) The calculated tie forces are very large and likely well beyond the ability of helical types of ties embedded in 26 millimetres thick block face shells or cracked brick mortar joints.

It was the Dr.'s evidence that because he observed from his calculations that the tensile strength of the masonry will be exceeded, there is as a result potential for cracking in the brick veneer and subsequent cracking in the block back-up wall. There is already evidence of cracking in the brick veneer and so there is now reliance on the block back-up wall to withstand wind pressure. In the Dr.'s view there will be a progression of cracking, and he sees this as a high probability.

Filed as Exhibit #39 was a sheet on which Dr. Drysdale had drawn.

Referring to Exhibit #31 which is now Appendix G to Exhibit #30, Dr. Drysdale explained that Exhibit #30, his report on the analysis of forces due to lateral wind pressure on the walls for the subject building was prepared on the basis of the assumption that the floors were three metres in height. It was eventually disclosed that in fact the floors are 3.2 metres in height. Exhibit #31 (Appendix G) was then prepared by Dr. Drysdale to account for the taller wall. Page G-3 shows that the taller wall is subject to increased bending and stresses are increased some 17% with the increased floor height.

Again by way of explanation, Dr. Drysdale explained that he had done a computer analysis while Mr. Roney did an approximate type analysis which as far as Dr. Drysdale was concerned could only be useful if the two wythes are equal in height, have identical support conditions, and are connected by infinitely rigid ties. In this case, at least the first two pre-conditions are not met, although one can approximate infinitely rigid ties.

Dr. Drysdale commented on the fact that in his analysis Mr. Roney first used the present Code provisions and then went back and used the 1985 Code provisions, which is quite contrary to specifications for upgrade. If a building is under capacity, then it must be brought up to present Code provisions.

Referring to Exhibit #32, Dr. Drysdale's critique of Mr. Roney's engineering calculations appended to Mr. Roney's April 24 report, Dr. Drysdale explained as an overview that as a result of necessary adjustments to Mr. Roney's calculations, there are significant increases in stress for the various faces of the building. For example, once the windows are taken into consideration on the south wall which results in a smaller area of wall, the increase in stress on the remaining wall is some 56% greater. The east wall is even worse with a 75% increase in stress. He explained further that even assuming Mr. Roney was correct in using the old Code provisions, which he was not, then with the adjustments set out in Exhibit #32, the level of stress in the masonry, that is the tensile stress, far exceeds Code provisions.

Referring to Exhibit #32 point by point, Dr. Drysdale explained the following:

1) Mr. Roney reverted to the use of M78. This would be appropriate if the only reason was to check Mr. Garbutt's original calculations. However as clearly shown in Commentary

K, the 1995 National Building Code adopted in the 1997 Ontario Building Code, must be used for design of upgrade.

2) As well, because OBC '97 references CSA S304.1 (1994), it must be used in the design of the masonry.

3) Mr. Roney used a Cpi (co-efficient of interior pressure) of .3. The use of this co-efficient would require a category 1 building. See sentence 37 from Commentary B from 1995 NBC, Exhibit #33. This category includes building without large or significant openings but having small, uniformly distributed openings accumulating to less than .1% of the total surface area. Our building does not have uniformly distributed openings and the percent of area is approximately 9%. It was Dr. Drysdale's evidence that the proper design category of our building is category 2 and the proper Cpi (co-efficient of internal pressure) is therefore .7.

4) In Commentary B, it is stipulated in paragraph 38 that "figures B-7 to B-10 are most appropriate for buildings with widths greater than twice their heights, and a reference height that does not exceed 10 metres. Beyond these limits, figure B-14 should be used with a pressure co-efficient of .8." As a result:

5) According to Dr. Drysdale's calculations with an east-west wind the pressure on the wall is .86kPa;

6) With a north-south wind the pressure on the wall is .84kPa.

In other words the design pressure should be .86kPa. Mr. Roney used a design pressure of .7kPa with a resulting –23% impact on wind load.

7) Mr. Roney's calculations do not make any adjustments for the effect of windows which reduce the section resisting wind pressure. Properly, stresses due to bending caused by wind pressure need to be increased.

8) Because of the indented brick, the full brick cross-section does not resist bending. Accounting for this, increases tensile stress in the brick by about 10%.

9) Mr. Roney used 1978 Code EI values representing section stiffness. This would be a valid approach, according to Dr. Drysdale, only if,

a) neither wythe cracks (cracking of the brick was reported);

b) the wythes have identical support conditions and are of the same height (neither of these conditions is met);

c) ties connecting the wythes are sufficiently stiff to force the wythes to deflect equally (Mr. Roney relies on stiff ties only on the fourth floor. The remaining ties are very flexible;

d) additional bending due to eccentric axial loads is not applied to one or both wythes (eccentric axial load is applied in indeterminate and inconsistent ways to both wythes); and

e) additional stresses are not introduced by restraining deformations due to moisture and temperature changes, and in-plane forces caused by the infill resisting lateral loads (all of these stresses do in fact exist).

10) The analysis presented by Mr. Roney has all of the above implicit assumptions. Violation of any one of these assumptions renders the analysis incorrect. In fact, all of the assumptions are violated. As a result the analysis cannot be used as the basis for remedial measures. Alternatively, if it is assumed that the brick is not cracked (contrary to reports) over the story height and that the mortar bond is not weak (contrary to reports), the analysis in Appendix E of Dr. Drysdale's report shows brick stress of .42MPa which should be modified by multiplying by .86 for wind pressure, 1.56 to account for window openings and 1.10 to account for frogged brick. Then, tensile stress due to bending is .62MPa compared to the Code value of .275MPa. The difference of .345MPa would have to be balanced by concentric compression of 34,500N/m which would require 20.7 metres height of brick above the point in question, which is not the case in our building.

11) Both Mr. Roney's and Dr. Drysdale's analyses have been done assuming cavity wall behaviour, even though the brick is clearly a veneer. To be considered to share in resisting the wind pressure as a cavity wall, there is a greater requirement for a mortar bond than for a veneer.

12) Another major problem with the proposed solution is that individual tie forces are extremely high.

13) Dr. Drysdale explained that the brick veneer will crack first and stop sharing the load, leaving the blocks alone to resist wind pressure. The cracking capacity in Mr. Roney's calculations are about the same as Dr. Drysdale's. However the CSA Masonry Standard M78 in clause 3.7.3.2 thereof (Exhibit #33) requires the metal ties to be at least 5 millimetres diameter, which is not satisfied by either the existing joint reinforcing, the existing corrugated strip ties, nor the Helifix ties proposed by Mr. Roney.

14) The masonry is so rigid that tying the brick to the bottom flange of a beam will produce cracking upon a very small lateral deflection of .3 millimetres.

Dr. Drysdale then turned to Exhibit #34, which is a document authored by Dr. Drysdale to respond to Mr. Garbutt's claim that he designed the masonry as solid masonry panel walls. Again going through Exhibit #34 point by point, Dr. Drysdale explained the following:

1) Assuming the collar joint (the cavity between the brick and block wythes) had been filled with mortar or grout, the total thickness of the wall would be 90 + 20 + 100 (block, cavity, brick) for a total of 210 millimetres.

2) By definition, a panel wall is a "non-load bearing exterior masonry wall having bearing support at each story."

3) The thickness of 210 millimetres is satisfactory for the top three stories but must be 290 millimetres for the bottom story because the building height exceeds 11 metres (see para. 5.3.3.2 of M78, Exhibit #33).

4) The maximum spacing of supports is 18 (210) or 3,780 millimetres so that 3.2 metre floor to floor spacing was satisfactory.

5) The following points are used to indicate that the walls were not correctly designed as panel walls:

    a) lack of horizontal movement joints makes the masonry walls part of the load-bearing system, and load-bearing walls are not panel walls;

    b) use of shelf angles is a detail usually associated with support of brick veneer. In our case, continuation of the brick past the concrete floor, the ends of the open web steel joists and the steel beams, means that the wall is not solid masonry over this 600 millimetre height. See the small drawings in Exhibit #9 which show the absence of blocks in this area resulting in the wall not being solid masonry.

c) panel walls must be anchored to their lateral supports. This is not called for by either the specifications or the drawings.

6) The claim by Mr. Garbutt that solid masonry did not require flashings and weep holes is incorrect. See para. 3.16.1, Exhibit #33, which quite clearly calls for flashing to be installed in solid walls in a number of places including over the heads of window openings in exterior walls. The provisions go on to call for weep holes as well.

7) Part IV of the Ontario Building Code provided general guidance to ensure that wall design included provisions to avoid moisture problems due to condensation and rain penetration. This was written mainly in performance terms rather than prescriptive terms, therefore there is more responsibility on the designer to be up to date and follow good practice. It appears that Mr. Garbutt was aware of CMHC Advisory Document (1981) by Gord Plewes, and CSA A371. He likely was also aware of the very helpful Canadian Building Digest published by NRC. All of this should have helped Mr. Garbutt reach a different conclusion regarding the need for drainage, flashing and weep holes. According to Dr. Drysdale, the design of brick veneer with block back-up was the proper decision and by the early 1980's was also the common decision. Solid composite walls of clay and concrete were very unusual and difficult to achieve. This is especially true for the lower part of the wall where the brick is supported at the bottom of the steel support beam and the block is supported at the floor level. Again this leaves 600 millimetres that cannot be solid masonry.

Cross-examination by Mr. Nelson

Dr. Drysdale's time is divided among teaching, research and consulting with his consulting work amounting to approximately 15%. His consulting work in the last 10 years has been largely confined to the retrofitting of buildings.

Dr. Drysdale was first retained by architect Johnston in mid-March 2002, and received three folios from Mr. Johnston in mid-April. Filed as Exhibit #43 was a letter of retainer from Mr. Johnston to Dr. Drysdale dated March 15, 2002.

Dr. Drysdale received on May 6, 2002 Mr. Roney's report dated April 24, 2002, and at the time was just in the process of completing his own report dated May 6, 2002.

In preparing his report of May 6 (Exhibit #29), Dr. Drysdale reviewed a few elevations drawings provided to him by Mr. Johnston.

Dr. Drysdale explained that by order of Justice MacLeod on May 7, 2002 Mr. Roney was ordered to provide his calculations by May 14, 2002 and Dr. Drysdale by May 16, 2002. In the result, Dr. Drysdale's report on the analysis of forces, dated May 9, 2002 (Exhibit #30), was delivered to the defendants on May 13, 2002. In this respect the Dr. explained that what he was involved in was a very standard structural analysis program, that is computer program SAP 2000. What is put into the computer is the information set out in Tab 1, Part A of Exhibit #30, that is the properties of the brick, block, ties and connections between those elements. What was needed to be inputted was the moment of inertia (I), area and modules of elasticity (E). With that, the computer runs the program.

Dr. Drysdale went on to explain that once having received Mr. Roney's calculations, he noted that Mr. Roney had prepared his calculations based on a floor height of 3.2 metres. Accordingly Dr. Drysdale did another run with his computer at this height, and that resulted in Appendix G, Exhibit #31. In making his run, Dr. Drysdale eliminated the cavity between the brick wall and the block wall as a variable, and simply entered it as 30 millimetres.

Dr. Drysdale was familiar with the article prepared by D.E. Allan from the National Research Council on Limit States Design, entered as Exhibit #44. The article describes the gradual replacement of allowable stress design by Limit States Design as a basic calculation tool for designing and evaluating civil engineering structures. Dr. Drysdale conceded that the January 1982 paper fairly did that.

The article explains that in allowable stress design, the adequacy of a structure is checked by calculating the elastic stresses in it due to the maximum expected loads, and comparing them with allowable stresses. The allowable stress is equal to the failure stress of the material divided by a safety factor determined by applying allowable stress design methods to successful structures existing at that time. The article points out that allowable stress design has formed the basis of structural codes and standards for most of this century. Dr. Drysdale agreed with all of that. As well, Dr. Drysdale agreed with the article's proposition that the allowable stress design method has a number of limitations.

By way of overview, Dr. Drysdale explained that in the allowable stress design approach, one first determines maximum expected load on the structure and then calculates the elastic stresses in the structure due to the maximum expected load. The stresses so calculated are then compared to the allowable stress or stresses and if any of those allowable values are exceeded then the structure must be redesigned.

Under the heading, Limit States Design the article explains that all structures have two basic requirements in common: safety from collapse; and satisfactory performance of the structure for its intended use. The limit states define the various ways in which a structure fails to satisfy these basic requirements. Ultimate limit states relate to safety and correspond to strength, stability and very large deformation. Serviceability limit states relate to satisfactory performance and correspond to excessive deflection, vibration and local deformation. Dr. Drysdale agreed with all of this and agreed that Limit States Design, over time, has replaced the allowable stress design.

Filed as Exhibit #46 was the CSA Masonry Design for Buildings (Limit States Design), published in December of 1994. Dr. Drysdale was familiar with the directive and indeed was on the technical committee that produced the recommendation, after working some four years. This document was incorporated into the Ontario Building Code in 1997. The Dr. explained that M78 (Exhibit #47) and the 1983 Building Code (Exhibit #48) were based on allowable stress design as opposed to Limit States Design. See page 172 of Exhibit #48, section 4.4.2.1 that provides that buildings and their structural members made of plain and reinforced masonry shall conform to CAN 3-S304-M78.

The Dr. conceded that by the mid-1980's the standard for masonry was allowable stress design and that when he prepared his analysis of forces (Exhibit #30) dated May 9, 2002 he used the 1994 masonry standard, but explained that he was simply using that device to come up with the properties of the materials and that he could have referred to those properties based on experience rather than based on any reference to the 1994 masonry standard. He explained further that his model is independent of Limit State Design or allowable stress design, and in any event when looking at a retro-fit, you look at the standards of the day, not the standards of the period of original construction, as is pointed out in point 2 of Exhibit #32.

The Dr. pointed out that Mr. Roney in his calculations, misinterpreted the pressure co-efficients and gust factors. See point 3 of Exhibit #32. In attempting to find the internal air pressure, Mr. Roney chose a category 1 building when he should have chosen a category 2 building. The Dr. further pointed out that this critique would apply regardless whether one was approaching from a Limit State Design or an allowable stress design. In the result, Mr. Roney used an internal air pressure of .3 whereas it should have been .7 by Code, and this would have a 20% impact on the result.

The Dr. explained that Exhibit #36 is the Helifix tie (brand name) referred to by Mr. Roney in his report. Dr. Drysdale was not sure if Mr. Roney in referencing the Helifix tie meant that specific brand or in general a twisted tie. Exhibit #37 is another twisted tie, but not of the Helifix brand. Dr. Drysdale calculated the properties of both Exhibit #36 and #37 and used primarily the properties of Exhibit #36, however from his analysis of both he knew that there was no significant difference in the properties and eventually went with the properties of Exhibit #37 because they were easier to determine.

Dr. Drysdale explained that in theory when constructing a brick and block wall, the height of three bricks and three mortar joints (the third mortar joint being on top of the third brick) equals eight inches, and the height of one block and one mortar joint on the top of that block also equals eight inches, and that a brick tie is placed across the top of the top mortar joints every six bricks in order to connect the bricks to the blocks. The Dr. went on to explain that in our particular construction, the base soldier course had the bricks standing on end and as a result in parts of the building, the bricks and the blocks did not match up. As well in many cases the mortar joints exceeded the standard 10 millimetres, again resulting in the bricks and the blocks not matching up. In our construction this mis-match was accommodated by bending the strip tie from the top of the block to the top of the third brick. It was the Dr.'s evidence that the strip tie is not meant to be bent and it is not proper practice to do so.

Referencing point 10 of Exhibit #32, his comments relating to Mr. Roney's May 14, 2002 calculations, the Dr. explained that the term 'MPa' is a measure of stress and explained once again that from the analysis in his Appendix E, the brick stress of .42 MPa must be modified by multiplying by: .86 for wind pressure, 1.56 for window openings and 1.10 for frogged brick for a total tensile strength due to bending of .62 MPa. He explained that that will be the stress on the wall after Mr. Roney's retro-fit and in particular on the south wall between the windows where the windows are set 2.5 metres apart. On other parts of the building where the windows are three metres apart the stress on the pier between the windows will be .57 MPa. In both cases, the MPa is in excess of the Code value of .275 MPa. In other words in both cases the stress on the walls is above Code limits after the 'Roney' retro-fit, and stress over Code leads to cracking.

Referring to point 14 on Exhibit #32, Dr. Drysdale confirmed that masonry is so rigid that tying the brick to the bottom flange of a beam and to the top will produce cracking simply because of the very small movement of the beam in the order of .3 millimetres. The fact that the beam will move is definite and accordingly there will be a crack in the brick wall in each floor level along the level of the beams. These would likely be very small cracks, not visible to the naked eye, and are different from the cracks described in Item 10 of Exhibit #32 which would be cracks between the windows. Dr. Drysdale explained that engineers design on a problematic basis so that even where a building is under-designed, it may not necessarily crack. What the design is determining is an acceptable risk of cracking, in other words the building may be designed for a risk of cracking in the order of one out of a hundred. Then in 99 cases there will be no cracking, but that risk is still unacceptable.

Cross-examination by Mr. Garbutt

Dr. Drysdale explained that his only visible observations of the building before May 21, 2002 came from photographs provided by Mr. Johnston. He did see the building on May 21 and observed cracks in the mortar at that time, but did not have time for an informed opinion then as to what was causing the cracks.

Mr. Garbutt took Dr. Drysdale through an analysis of the weight stresses on the building, based on various categories and various pounds per square foot coming to bear on the building, and asked Dr. Drysdale if he was aware that the building was designed for 145 pounds per square foot which he was not.

Mr. Garbutt reviewed with Dr. Drysdale some provisions of the 1997 Ontario Building Code. He agreed that the reference velocity pressure 'Q' for the design of cladding shall be based on a probability of being exceeded in any one year, of one in ten. The chart at page 2 of the 1997 Code which was entered as Exhibit #49 provides that in the Kingston area, the designed hourly wind pressure for one in ten is .35 KPa which equals approximately 51 mph. Part 4.1.1.4 of the same Code, under the heading <u>Structural Design</u> provides that buildings and their structural members shall be designed by one of the following methods:

a) standard design procedures and practices provided by this Part; or

b) one of the following three bases of design,

    (i) analysis based on generally established theory;

    (ii) evaluation of a given full scale structure or a prototype by loading test; or

    (iii) studies of model analogues.

Mr. Garbutt asked Dr. Drysdale whether the fact that this building has stood for 17 years would qualify it as a properly designed building under section b) (ii), and his response was a flat ‘no’.

The Dr. conceded that the filling of the cavity would best increase the bending strength of the wall, more so than the retro-fit ties suggested by Mr. Roney. If the cavity was filled it would make for a solid masonry wall <u>except</u> for the 600 millimetre strip of airspace at each floor. The Dr. conceded that while filling the cavity is a time consuming and expensive proposition, it is not considered to be complicated.

Dr. Drysdale explained that a wall becomes a load-bearing wall, that is a wall that is holding up part of a structure, even if the load is a lateral one.

With respect to flashings, the Dr. explained that there are a number of examples of acceptable flashing in the Code and that steel angles over the windows do not constitute a flashing. Indeed the steel angle over the window would be protected by a separate flashing.

With respect to weep holes, the Dr. explained that they have two purposes: one, to drain water from the cavity; and two, to pressurize the cavity. In hollow wall construction they are used to remove water from the top of the flashing and the Code provides that wherever there is a flashing, there should be corresponding weep holes.

With respect to joints and anchorage, Dr. Drysdale explained that if you put veneer on a shelf plate to support it over its height and don’t attach the veneer at the top or at the sides, then you aren’t faced with a cracking problem. It is only when you attach at the top and the bottom that you get cracking.

With respect to the possibility of sealing the brick façade against water penetration, the Dr. explained that a commercial sealant might be relied upon for a period of approximately five years, but that this was a very dangerous way to attempt to resolve the problem. While the sealant may preclude water entering through the bricks themselves, water will eventually get in at other places and once in, because of the sealant, it cannot evaporate. As a result, the Dr. does not recommend sealing.

Re-examination by Mr. Mueller

Exhibit #46 CSA S304.1-94, Masonry Design for Buildings (Limit States Design) was incorporated into the National Building Code in 1995, and into the Ontario Building Code in 1997. The Dr. confirmed once again, not only was it imprudent in 1995 to retro-fit on the basis of M78, M78 effectively did not exist in 1995 so that the choice in 1995 would have been between the 1984 or the 1994 version.

With respect to the ties, it was the Dr.'s evidence that adjustable ties were available in 1985-86, and in fact had been designed at that point in time for the very problem facing the masons, that is that the brick top and block top did not line up.

Referring to Mr. Garbutt's cross-examination, Dr. Drysdale was asked by Mr. Mueller about the possibility of 'pointing' large cracks. The Dr. explained that 'pointing' involves routing out some of the mortar and replacing it with new mortar. He explained that re-pointing does not re-establish bond, and also the crack will inevitably re-appear because of the building flexing.

With respect to Mr. Garbutt's representation that the building was designed to 145 pounds per square foot and that Dr. Drysdale had expressed the thought that that was a high load, Mr. Mueller referred Dr. Drysdale to Exhibit #9, drawing S1, which shows a TL of 725 pounds per linear foot for the joist design. The Dr. explained that in order to get the weight per square foot, one would have to look at the joint spacing, which he did, and in the final analysis the joist design was approximately 140 pounds per square foot for both live and dead weight. He explained further that normally one would design for a 40 pound per square foot live load, and for whatever the dead load would actually be. Dr. Drysdale acknowledged that in his cross-examination Mr. Garbutt had suggested that a number of loads could balance each other out. Dr. Drysdale explained that he had admitted to that possibility in general terms, but balancing out of loads would certainly not solve the problems in our case.

With respect to the suggestion of power grouting or pressure grouting of the cavity, Dr. Drysdale explained the difficulties encountered with pressure grouting:

1) without adequate ties the wall may be blown apart;
2) there is potential for defacing the building when the grout leaches out through cracks and other openings;
3) there is still the likelihood after pressure grouting that the cavity would not be completely filled, water would enter, and thus there would be a need for flashings and weep holes. Pressure grouting, if successful, would solve the structural problem between the floors but would do nothing to solve the rain problem.

Witness – Paul Johnston

Examination-in-Chief

Mr. Johnston is an architect. His c.v. was filed as Exhibit #51. After an appropriate *voir dire* I found Mr. Johnston's qualifications as an architect impeccable and that he should be permitted to provide the court with opinion evidence as to the original design and construction of the subject building's roof and walls, his concerns with respect thereto, his views with respect to building problems arising out of the design and construction, and the resolution of those problems both with respect to work already done and to be done.

Mr. Johnston has been involved since 1994 with the plaintiffs, providing them with advice as to the problems they have experienced and are experiencing, directing them to other appropriate consultant experts, and providing them with advice with respect to appropriate remedial measures. He has been involved as a consultant and supervisor of work with respect to local repairs to the roof in 1994 and with respect to its replacement in 1998.

It was because of the nature of his involvement with the plaintiffs and their building since 1994 that defence took the position that Mr. Johnston was not independent and that his evidence should not be admitted. I disposed of defence's position in that regard in favour of Mr. Johnston for reasons that were delivered at the conclusion of the *voir dire*.

Mr. Johnston opened his evidence in chief by reviewing a series of Canadian building digests released by the NRC and various technical notes released by the Brick Institute of America. The NRC building digests date from June of 1960 and the technical notes from 1977. The purpose of the review was to provide the court with a better understanding of the wall structure and the design and construction thereof. The various digests and technical notes were, with defence counsel's consent, highlighted by plaintiffs' counsel to assist me in a review of the documents prior to Mr. Johnston giving his evidence, which I did do. Mr. Johnston in his evidence referred to and adopted many of the highlighted areas.

During a rain storm the air pressure at the outside surface of the rain wetted wall is usually higher than that at the inside surface. The rain falling on a wall is forced through the wall by this pressure difference, provided that there are pathways within the units and mortar, or between them, through which the water can follow. The first digest refers to structural cracks in masonry walls, caused by differential movements of parts of the building as being a means of entry of rain into the wall. Mr. Johnston explained that the cracks being referred to there are visible cracks. He explained further that for water to enter through the brick wall it is not necessary that there be visible cracks.

In most instances of rain penetration of brick walls, leakage takes place between the brick and the mortar. In some cases, units, or bricks, are used which are sufficiently permeable to moisture that when exposed in a wall to heavy rain, leakage takes place through the brick units.

Unbonded areas between brick and mortar which usually are the cause of leakage in brick walls may result from faulty construction techniques in which insufficient mortar is used to form the joint. It is the bond that holds the wall together, explained Mr. Johnston.

Failure in the design of a building to provide for the accommodation of differential movements between its parts may lead to cracking of the masonry and thereafter entry of rain. Expansion or contraction of walls may also be a cause of cracking. Cracking may be caused by contraction when the brick units shrink in the drying process after installation.

The digest points out that parapet walls are frequently a source of entry of rain when they have not been isolated from the wall below by proper flashings.

In response to a question put by me, Mr. Johnston expressed the view that our subject wall was a veneer wall. By definition a cavity wall is a wall with a two-inch deep cavity, both walls of which provide a composite reaction for support. This is a veneer wall which has only one inch separation between the walls and in which there is no indication that it was designed as a composite wall. Tab D of Mr. Johnston’s Folio 1 is illustrative of the wall’s design. As a matter of interest, Mr. Johnston pointed out that from his examination the width of the cavity in our wall in fact begins at something less than an inch at the bottom and as it approaches the top of the wall it is expanded to some two inches.

As well, in explaining the difference between a cavity wall and a veneer wall, Mr. Johnston pointed out that typically a cavity wall has very strong anchors from the brick to the block. He explained that if it is not required that the brick act as part of the loaded wall, then lighter ties can be used which is typically the case in a two to three story building. Here, the building exceeds three stories and Part 3 of the Code applies, which in his view would require more robust ties. Mr. Johnston explained that at a height of four stories, the building will experience winds that would get a kite up, as opposed to winds that one might experience at lower levels. By way of example, on a visit to the building on December 5, 1994, Mr. Johnston observed rain water that was being held in place against the fourth story by the wind, as opposed to water flowing down the side of the building as it was on the lower floors. He explained that the wind is unable to blow by the face of the building and therefore starts to move up the face of the building. The same phenomenon was experienced by Mr. Johnston and Mr. Kendall when they performed a water test which has already been referred to by Mr. Kendall.

Building digest CBD 21, issued September 1961, discusses cavity walls and notes that cavity walls do not permit rain penetration. Rather, by their design, water cannot reach the inside surface of the wall. When rain falls on a cavity wall it may penetrate the outer wall, but the water then trickles down the inner surface of the outer wall and cannot traverse the cavity. The base of the wall is provided with metal flashings that direct any water that has entered the cavity outward through weep holes. It is noted as well that it is essential that the airspace be kept continuous and not bridged by mortar, or other material that would allow water to pass across the cavity.

It is noted as well that it is the function of the ties to anchor the two parts of the cavity wall together, so that adequate strength may be obtained from the wall assembly. Shown in Figure 2 at page 2 of this digest are various styles of sturdy ties that can be used for this purpose.

It is interesting to note from Figure 1, a cross section of a cavity wall that the typical flashing angles down from the inner wall to the outer wall and weep holes, so that water is unable to come in and get to the inside wall. Shown in Figure 3 at page 3 of the digest is another cross section of a cavity wall showing a shelf angle attached to the beam supporting the outer part of the cavity wall. Referring to Folio 1, Tab D, the construction cross section of our wall, Mr. Johnston noted that while the shelf angle is not shown in the drawing, it is referred to as PL, 8” x ¼”.

The digest points out with respect to control joints that the outer part of a cavity wall forms a relatively thin skin around a building and may be subjected to appreciable changes in temperature and moisture content, producing stresses which lead to cracking. Experience in the design of cavity walls has indicated the value of providing vertical control joints in the outer part of the wall to accommodate these movements. It has been found that the corners of cavity walls are particularly susceptible to cracking when a structural frame has been used and accordingly a vertical control joint is usually provided in the outer part of the wall, about three or four feet from the corner. As Mr. Johnston explained, the wall gets its strength from its corners and so it is not prudent to place the control joints too close to the corner.

It is pointed out in this digest released September 1961 that the National Building Code of Canada, 1960 by way of example states that the maximum height to which a cavity wall may be built above its bearing support is 36 feet and that for buildings taller than this, it is necessary to provide intermediate bearing support so that the allowable height above the support is not exceeded. The digest also points out that there are three essential requirements for cavity walls;

1) the cavity wall must have a gutter at its base to collect leakage water and drains to direct water out of it;
2) the two parts of the wall must be anchored together with metal ties that are corrosion resistant and adequately strong; and
3) the wall must have a cavity free of mortar or other material that may form a water bridge across the two walls.

It was Mr. Johnston's evidence that these requirements were fundamental.

Mr. Johnston pointed out that veneer walls should be constructed functionally as cavity walls but not structurally. In other words, veneer walls which we have here should be designed and constructed with proper flashing and weep holes for drainage, weep holes for equalizing the pressure of the cavity with the outside pressure, shelf angle supports, anchorage and proper bonding or tying between the walls.

Technical note 21 revised, shows how the mortar bed of the bottom brick can be properly beveled toward the inside so that when the top brick is placed on, the mortar is not squeezed into the cavity. Again, Mr. Johnston explained that the problem with allowing mortar fins extending from the inside of the brick wall is that they may form a bridge between the brick wall and the block wall. In fact, his inspection of the wall showed that at least one-third of the joints had mortar fins and 75% of those bridged all the way across between the brick wall and the block wall. The impact of connecting fins on drainage depends on how the cavity wall is performing. If the pressure inside the cavity is the same as the pressure on the outside of the wall, then the water will not flow into the cavity. The danger with a veneer wall with the air space as small as it is, and no weep holes, is that the full effect of the wind is taken on by the brick wall and there is no chance to equalize pressure. The water is thus driven through the brick wall onto the fins and across to the block wall.

Technical note 21 revised, at page 5 notes that when the wall height or the number and location of openings necessitates supporting the outer width of the cavity wall on shelf angles attached to the structural frame, it is recommended that horizontal pressure relieving joints be placed immediately beneath each angle. Pressure relieving joints may be constructed by either leaving an air space or placing a fully compressible material under the shelf angle and sealing the joint with a permanent elastic sealant. Mr. Johnston explained the concern here is the rotation of the shelf angle or the lintel along its length, which in turn applies stress to the brick wall, resulting in cracking. With respect to our building, Mr. Johnston testified that as opposed to soft joints, he observed hard joints of mortar between the top of the underbrick and the bottom of the shelf angle, notwithstanding that the specifications called for soft joints. As well, his examination of the wall made it apparent to him that the masons had attempted to place the block immediately below the beam with a full mortar joint which often was not achieved but often was. Mr. Johnston explained that as in the case of the bricks, the specifications called for a soft joint between the top of the block and the bottom of the flange of the beam. Filed as Exhibit #52 was Mr. Johnston's drawing which he referred to in the course of his evidence.

Technical note 21, in addressing flashing and weep holes, notes that good flashing details are an absolute necessity in cavity wall construction. In order to divert moisture out of the cavity through the weep holes, continuous flashing should be installed at the bottom of the cavity and wherever the cavity is interrupted by elements such as shelf angles or lintels (the steel plate that goes across the top of a window opening or across the top of a door opening).

Technical note 21 C notes that weep holes can be easily created or installed by various methods such as:

1) inserting oiled rods or pins in the head joint and removing before final set of mortar;
2) placing sash cord or other suitable wicking material in the head joint;
3) placing metal or plastic tubing in the head joint; or
4) by far the simplest method, just eliminating each second or third head joint.

Mr. Johnston explained that 99% of the time weep holes go with flashing and that in his view, by present standards, only number 3 above would serve as a satisfactory weep hole. It was his view that even in 1985-86, only number 3 would serve as a satisfactory weep hole. He explained that if such weep holes existed as they should in our building, then they should be visible to the eye from an outside inspection. They are not visible. It was Mr. Johnston's evidence that when he first was retained in 1994 he looked and saw no weep holes.

The same technical note points out that with respect to wall ties, the most important factors are:

1) corrosion resistance;
2) proper spacing;

3) full bedding of the bed joint and placing of the wall tie in the mortar to assure full bond;

4) crimping of the metal ties in the middle to form a drip is not necessary and could decrease the strength of the tie.

Mr. Johnston explained that all this was known in 1985-86.

Technical note 28 revised, notes that there are six requirements for satisfactory performance of brick veneer:

1) an adequate foundation;

2) a sufficiently strong, rigid, well braced back-up system;

3) proper attachment of the veneer to the back-up system;

4) proper detailing;

5) the use of proper materials; and

6) good workmanship in construction.

Mr. Johnston adopted these requirements and noted the following.

a) Re 1, the foundation was adequate.

b) Re 2, Mr. Johnston disassembled an area of the wall from inside. The area chosen was by chance and was located over the top of a window, and within four to six feet of a column. He found no anchorage between the block wall and the steel structure that he was able to observe, that is the beam, the shelf plate and the plate over the window. He expected to find some measure of attachment between the block wall and the beam because on its own, the block wall was of such a dimension that it could not support itself as back-up to the brick wall. It was his evidence that there should have been at least one strap anchor from the block wall to the structure and there was none in the area exposed.

At some later point in time Mr. Johnston was able to view the situation with respect to the ends of the block wall through disassembly from the outside. He anticipated that the end of the block wall would extend into the indentation of the column, or would otherwise be attached to the column. The columns exist only on the north side of the building. Contrary to expectations, Mr. Johnston observed the block wall butted up against the edge of the column but not either extending into the column nor attached to the column. This in his view was simply not sufficient.

Referencing 28 revised in Exhibit #53, Mr. Johnston pointed out that brick veneer wall assemblies are drainage type walls, and walls of this type which include cavity walls are recommended where maximum resistance to rain penetration is desired. The technical note goes on to indicate that it is essential to maintain the one inch (25 millimetre) clear space between the brick veneer and the back-up to ensure proper drainage.

In technical note 44 B in Exhibit #53 at Figure 1, there are drawn three types of ties, one of which is a corrugated tie similar to the type that Mr. Johnston noted in the lower floors. Generally, he observed a different type of tie in the middle floors.

In Figure 3 of the same technical notes are shown a number of continuous joint reinforcement ties. Mr. Johnston explained that typically these reinforcements are used at the back-up masonry, in this case in the concrete block wall. Again typically the reinforcement would stay within the block wall. In our case the specifications did call for masonry reinforcement in the form of truss type reinforcements which are illustrated in Figure 3. In fact, the three wire ladder type also illustrated in Figure 3 were used. While both types can be appropriate, the truss type which was actually specified is stronger. At Exhibit #56, Tab F is the masonry specifications. At section 2.6 under the heading Reinforcement, Dur-O-Wall Truss or Blok Truss are called for to go into the masonry block wall. At Tab K of Exhibit #56, photo's 31 and 32, one can see the ladder type joint reinforcements which are being used improperly as a tie between the brick wall and the block wall.

Also at Tab F, the masonry specifications, see 2.5 which specifies anchors: hot dip galvanized steel with sizes and types to suit conditions. Mr. Johnston explained that anchors are used to connect the block wall to the steel. At 2.8 it is specified that anchor and bond masonry is to conform with the National Building Code. Mr. Johnston explained that this related back to 2.1.1 which in part calls for units to be tied every other course as per drawings. He further explained that 2.8 relates back to 2.1.1 and to that extent it was a mis-specification to refer to an anchor in 2.8. Mr. Johnston noted that the specification itself does not really say what kind of a tie should be used.

Figures 5 and 6 in technical note 44 B show various types of adjustable unit ties to be used between the block and the brick where the courses do not line up, rather than bending the tie.

At this point in Mr. Johnston's evidence, we viewed a portion of Dr. Lawless's construction video, Exhibit #17.

We saw again that at the very beginning of the masonry work, the soldier course was out of plumb. This time Dr. Lawless is heard to say that the soldier course had been leaned forward in order that succeeding courses could clear the flange of the beam above.

With respect to another portion of the video, Mr. Johnston explained that as the masons rounded a corner with the bricks, they used a technique known as toothing, which generally was not recommended.

The video goes on to show that the soldier course has been replaced, and now appears to be vertical. There is no indication as to how they managed eventually to clear the flange of the beam above with succeeding courses.

Mr. Johnston explained that a portion of the video shows a mason laying a brick, then picking it up and putting it down. This was done twice and this is contrary to good masonry as the bond is broken when the brick is lifted.

The video shows that as the brick is laid, no mortar is being purposely put at the rear of the brick, between the brick and the block wall. Mr. Johnston inferred from this that at this point in time the mason was starting an air space, although as Mr. Johnston noted it was less than the 19 millimetres specified by Mr. Garbutt. There is a fair expanse of wall shown in the video as the bricks are being laid where no ties are visible at all.

As the bricks go up, Mr. Johnston explained that the air space appears to be getting wider, perhaps three-quarters to one inch.

A portion of the video shows the fins coming out between the blocks into the air space between the brick wall and the block wall. They are being squeezed out as the block is put in place and it does not appear that the mason is making any attempt to clear that mortar in the air space.

As well, the video shows the block butting against the flange of the column but not being inserted in the indent of the column. It appears that alignment is such that the block could not be inserted into the indent in the column, and as Mr. Johnston explained, it therefore should have had an alternate arrangement to connect the block wall to the column with some kind of anchor.

The video then shows a mason trying to clear mortar from the air space, but not doing a very good job of it. We then see from the outside a portion of the block wall which is higher than the adjacent brick wall with ties lying on top of the blocks well spaced but bent down from the block toward the brick. As Mr. Johnston explained, this breaks the galvanizing to permit oxidization to start on the tie. As well, the bending of the tie negatively affects lateral stiffness. A portion of the video shows brickwork in the winter time with no protection afforded it, and it appears they are laying bricks in -10^{o} weather as commented upon by Dr. Lawless.

A portion of the north wall is then shown with severe staining and efflorescence. The appearance was totally unacceptable. Mr. Johnston explained that the bricks as manufactured contain calcium products that are left in the brick after manufacture. If the brick is exposed to water, drying through to the outside of the brick wall, it takes the various calcium products and salts with it to the outside face. As well, condensation can have the same effect.

Mr. Johnston then went to Exhibit #54 which is his Folio 1 entitled Building Envelope Review and Analysis, General Commentary. Mr. Johnston explained that he first met with Dr. Reed in mid-August of 1994 at which time Dr. Reed expressed to him the following concerns:

a) deterioration and damage to interior finishes from water leakage around the windows;
b) water damage at the interior brick and drywall finishes of the stair/lobby tower;
c) residue and staining from water entry on interior finishes, floors and window surrounds; and
d) severe efflorescence, particularly at the north and west walls of the building where white chalky residue and dark gray staining of the brick was particularly evident.

At that time, Mr. Johnston had a walk-about and went to the roof and had a look at some of the available drawings, and as a result quickly became suspicious that limited detail on the original drawings, obvious faults with the execution of certain edge and flashing details, and certain installations or features which he would regard as deficient given the nature of the construction, could potentially account for all of the observed faults. At that time, Mr. Johnston advised Dr. Reed that further detailed investigation would need to be done. Mr. Johnston suspected that there was damaging and systemic water incursion through the exterior envelope of the building, rather than some localized problem. As well, he inspected the caulking around the windows at that time, and it seemed to be satisfactory.

Mr. Johnston sent his initial report to the plaintiffs on August 29, 1994. It is contained at Tab A of Exhibit #54 (Tab 95 of Exhibit #1). In this report Mr. Johnston gave his initial opinion as follows:

a) there appeared to be deficiencies in the roof and parapet construction, contributing substantially to water entry and penetration;
b) elements of and configuration of the parapets, flashings and the continuity and extent thereof appeared insufficient to perform adequately, and were inconsistent with details appropriate to a protected membrane roofing system as employed;
c) with no observed evidence of through wall or projecting flashings and none detailed, the staining and water markings were consistent with an inability of the wall design to shed

absorbed or penetrating water, let alone any roof collected precipitation retained on the roof until the roof drains accommodated the accumulation; and

d) although not corroborated by removal of flashings, roofing or masonry finishes, the limited details and Dr. Reed's description of what he recalled of the construction led Mr. Johnston to be concerned with a number of items relating to design and construction.

In due course, Mr. Johnston reviewed the copious materials provided to him by Dr. Reed and listed on four pages at Tab B of Exhibit #54.

Mr. Johnston explained that discussion of the investigation procedures and reports of his findings along with several recommendations were done separately as Folio 2 (Exhibit #55) for the roof analysis portion and Folio 3 (Exhibit #56) for the wall analysis. As the direct result of his testing and advice to minimize severe deterioration of the northwest corner, a repair was made to the roof locally at the stair/lobby tower in 1994 during the course of which the revealed area of fault demonstrated a substantially understandard application of membrane flashings. This for him raised real concerns that the balance of the roof was suspect. As a result, he encouraged the plaintiffs to consider replacement of the entire roof system to the specification that he subsequently recommended for a retro-fitted assembly. The replacement was completed in the latter months of 1998. At the time of the total re-roofing, the parapet construction was extended vertically and was strengthened in order to secure and isolate the wall construction from any roofing interface. The new protective flashings were well overlapped onto the brick masonry and a newly sealed drip edge was installed at the top cornice of the brick façade throughout. As Mr. Johnston pointed out, any continued wall problems (which have in fact been experienced) could not then have originated at the roof.

At Tab C of Exhibit #54 is the Inspec-sol report relating to the roof and dated September 28, 1993. That report said that the roof system appeared to be in good working condition and that internal leaking problems did not conclude that the roof membrane was at fault. Mr. Johnston of course took exception to the report, saying that in 1994 he found a fault (a membrane rupture) right at the very spot that Inspec-sol had inspected.

At Tab 1 D are two drawings. The first is from the contract set of documents detailing the exterior walls and particularly the parapet. The second drawing is the one that was established by Mr. Johnston for the 1998 retro-fit of the roof.

Compared to the retro-fit drawing, the contract drawing is almost devoid of any written detail or instruction. The parapet in the contract drawing is drawn as a solid parapet, construction of what appears to be two bricks with insulation in between, wood members over the bricks and coping over the wood members. The contract drawing appears to show the water proof membrane coming along the surface of the roof, up the side of the parapet, over the parapet and down the outside of the brick wall. Although the contract drawing refers to a ‘plate’which Mr. Johnston took to be the shelf angle required to support the brick wall, the location of the plate is not located on the drawing as it should have been.

Located at Tab I of Folio 3, Exhibit #56 is a sketch by Mr. Johnston of the wall and parapet as actually found during the course of retro-fitting at least to the extent that the wall could be opened up. That extent is shown by the heavy dotted line. Nothing to the right of that line as it is shown on the drawing was opened up. As opposed to the contract drawing, it was found on opening up that the bricks laying under the wood members were not bricks but indeed blocks and that the voids between the blocks had not been filled in as required. It was found that the space between the blocks and the bricks was up to four inches on two sides of the building and one inch to one and three quarter inches on the other two sides. In Mr. Garbutt’s design, there was to be eight inches from the top of the parapet to the top of the insulation, but it turned out to be not much more than two inches. The membrane in fact did not go right across the top of the parapet. The contract drawing showed some 10 to 12 inches of surface on the side of the parapet to which the membrane could adhere. In actual fact, there was only five to six inches of side surface. The lintels in the brick wall and block wall over the window opening were offset with a small space between them. Eventually Mr. Johnston concluded that water was going through the brick wall and running down the inside face of the brick wall to the lintels where it was directed to the top of the windows. Mr. Johnston believes that the water was coming in to the cavity through areas where the bricks had de-bonded from the mortar, and this was true for the whole wall.

With respect to the involvement of Mr. Thomas Mill of Mill & Ross Architects in 1986, Mr. Johnston was satisfied from the Mill report that Mr. Mill had been addressing only the aesthetic nature of the workmanship, and was not aware of and did not address at that time any particular failing in the wall performance.

It was Mr. Johnston's evidence that if a general contractor site supervisor had observed building practices or construction deficiencies, then these would be the subject matter of subsequent site meetings, and subsequent site inspections by the supervisor. As well, it would obviously trigger discussions between the site supervisor and the consultant, if good construction practice was being followed. As well, if the consultant had observed the same things, he in the scheme of things would be expected to change the nature of his inspection practices, and would be expected to be more vigilant.

With respect to the Cromarty report which addressed the possibility of pressure grouting the air space between the block wall and brick wall, it was Mr. Johnston's evidence that in view of his diagnosis that the bed joint between the bricks had cracked or de-bonded, and that that was the major cause of the water entry problem. The pressure grouting could create stresses on the brick wall that could cause serious damage. As well, pressure grouting would not address the lack of flashings and weep holds. Often with pressure grouting, there is a problem with the grouting not bonding with the back of the brick, and the problems remain. Additional problems can occur when the entire airspace is not filled (and entire filling cannot be guaranteed). As well the fins of mortar between the block wall and the brick wall, can obstruct proper grouting.

Mr. Johnston was aware that when problems persisted with the building leaking after the windows had been caulked on two separate occasions, the contractor recommended the application of a sealant to be applied to the outside of the brick wall. In Mr. Johnston's view that was an inappropriate recommendation. To begin with it is not a permanent solution. If the cracks between the bricks are of sufficient width, the sealant will not bridge the crack. The sealant, once applied, disallows the brick to breathe properly, resulting in water being contained in the air cavity, subject to a freeze-thaw cycle which could cause even more damage. It was Mr. Johnston's evidence that the suggestion of caulking or sealing did not address the basic design and construction problems that were inherent in the building.

Mr. Johnston pointed out that subsequent to the caulking of the windows in 1986 and 1992, the subsequent disassembly of the walls revealed that there was no flashings and no weepholes at the windows. The caulking itself around the windows was in reasonably good condition when this inspection was carried out in 1994. There is no indication that a deficiency in the caulking made any major contribution to the leaking problems.

It was Mr. Johnston's evidence that the drawings were not consistent with a solid masonry wall and indeed the specs called for a 19 millimetre air space between the brick wall and the block wall. As a result what was designed in this case, both in accordance with the drawings and with the specs, was a veneer masonry wall.

Mr. Johnston then went on to give what may be especially important evidence. It was his evidence that the specifications as drawn for a veneer wall were quite good and could have resulted in a good wall if they had been followed and if appropriate flashings and weepers had been installed and the air space maintained at 19 millimetres with proper ties and anchors. It was his view that a generally competent contractor could have constructed an acceptable wall from these specifications, with appropriate consultations with the consultant.

There were provisions in sub-section 9 of Division 7, Masonry, of the contract specifications for a mock-up or sample panel of the wall to be provided by the mason to ensure that the intended masonry was being installed with appropriate anchorages, flashings, ties and generally in a manner as was intended by Mr. Garbutt's drawings. This mock-up wall requirement should have addressed all of the issues that we have seen in this case concerning the wall.

By the same token, referring to the parapet, Mr. Johnston was of the view that a generally competent contractor should have been able to achieve an adequate parapet following Mr. Garbutt's drawing and specifications with the caveat that the contractor would have had to require more detail with respect to the parapet than was being provided by Mr. Garbutt.

Again by the same token, Mr. Johnston was of the view that Mr. Garbutt provided quite a good specification and drawings for the roof such that they would provide adequate direction for a contractor to build an adequate roof.

At Exhibit #55, Tab E is Specification Division 7 which covers the roof in this case. It was Mr. Johnston's evidence, apart from two aspects of this specification, that the provisions are appropriate for an irma roof. One of the exceptions is that section 3 has a heading '<u>Standard Built-up Roof</u>". Mr. Johnston pointed out that an irma roof is not typically known as a "Standard Built-up Roof". However, he pointed out that with three exceptions all of the specifications under that heading are appropriate for an irma roof. The other exceptions consist of three items under the heading, <u>Insulation</u> relating to vapour barrier, coating the first layer of insulation with hot bitumen, and securing the top layer of insulation to the wood nailers on the concrete decking. Mr. Johnston's evidence was that the remainder of the specifications are appropriate for an irma roof and in fact, in the actual building the contractor avoided the three items which were inappropriate.

Mr. Johnston noted in the last paragraph on page 4 of Folio 1 (Exhibit 54), that at one time Mr. Garbutt took the position that lack of maintenance of the wall hand-in-hand with ineffective sealant application could possibly be considered as the reason for water incursion at the exterior wall and attempted to identify that a "compensating maintenance program" would prevent moisture from entering. It was Mr. Johnston's evidence that in this case no rationale existed for a maintenance program nor was any called for in the specification, and if there was such a need for a program it should have been in the specification.

With respect to Mr. Garbutt's position, that he intended to design, by way of specs and drawings, a solid wall, it was Mr. Johnston's evidence that nothing in the specs or drawings shows that intention. He conceded that a solid wall could work with a porous brick, but the proper changes would have to be incorporated. He explained that detailing and specifications are substantially and completely different as between a solid masonry wall and a veneer wall. It was his evidence that if a client that chooses a porous brick in circumstances such as ours, then appropriate practice for the consultant would be to address measures to deal with that choice or to simply tell the owner that that particular brick cannot be used.

With respect to a site review, Mr. Johnston pointed out that section GC 3 Consultant, article 3.4 of the contract, requires the consultant to visit the site to determine if the work is proceeding in accordance with the contract documents. As well, under the Ontario Building Code, Mr. Garbutt had an obligation to review and ensure the construction was in accordance with the design that he intended. Additionally, as previously seen, there were provisions in subsection 9 of Division 7, Masonry of the contract specification for a mock-up or sample panel of the wall to be provided by the mason to ensure that the intended masonry was being installed with all appropriate anchorages, flashings, ties and generally in a manner as was intended by his drawings. That section is at Tab F of Exhibit #54, and although Mr. Johnston was originally of the view that such a mock-up while if it passes muster would remain as part of the actual building, section 9 of that specification required that the approved wall be retained until project completion and be removed when directed. Mr. Johnston, in the face of that specification, maintained that in practice the wall forms part of the building. In his view, a proper mock-up of wall would have avoided all of the deficiencies identified by him and would have clarified the various issues. He explained that the initiation for the mock-up wall must come from the contractor, and then the consultant is responsible for reviewing the mock-up wall.

Referring to the brickwork, Division 4 – Masonry at Tab F of Exhibit #56, makes reference to brickwork being based on an average unit cost of $300 per 1,000. It was Mr. Johnston's evidence that this was meant to indicate that the bricks were an allowance item, resulting in the situation that if brick acquisitioned was at a price less than $300 per 1,000 there would be a credit to the owners, and if in excess of $300 per 1,000 there would be an extra charge to the owner. Mr. Mueller pointed out that the actual contract documents do not list bricks as being among the allowable items. In any event, Mr. Johnston explained that assuming this to be an allowance item, the consultant should ensure that either the wall is designed to receive any kind of brick or the consultant should remain vigilant with respect to the type of brick that is actually being used. In this respect, Mr. Johnston noted that there is nothing in the record until September 21, 1993 going by way of a letter from Mr. Garbutt a warning to the plaintiffs that they may have bought the wrong kind of bricks.

Mr. Johnston then gave some significant evidence as to what Mr. Garbutt would have noted by way of deficiencies had he made regular site visits and review. With respect to the roof, Mr. Garbutt would have observed the following:

1) there was no slope;
2) the membrane flashings were not constructed as designed; the design requirement was for four plies up the parapet and two over and this was not done;
3) the first layer of insulation was adhered to the bitumen membrane;
4) the amount of ballast was incorrect to prevent floating;
5) the metal flashings or coping on the parapet was inadequate;
6) the height of the parapet was incorrect;
7) the parapet was not solid;
8) there was not the proper construction of drainage off the parapet;
9) the membrane should have been all the way across the parapet and down the front of the brick;
10) there was not sufficient attachment of the copings.

With respect to the wall, Mr. Garbutt during the course of site visits and review would have observed the following:

1) the 19 millimetre air space was not being maintained and where it exceeded that width, increased bonding and ties were not being implemented;
2) flashings were not being properly installed;

3) weep holes were not being properly installed;

4) alignment was not being maintained between the brick and the block and proper adjustable ties were not being used in that situation;

5) while there was good frequency of ties at the lower levels, there were no ties at all between the block wall and the brick wall from the roof beam up to the top of the parapet, a distance of some three feet. In this respect Mr. Johnston explained that the higher you get the more sensitive the wall is to ties;

6) there was no anchorage of the wall to the steel structure;

7) the control joints and expansion joints were inadequate;

8) no cold weather protection was being provided;

9) shelf plates were missing from two walls. (Mr. Johnston explained at this time that as between a consultant that is an architect and a consultant that is an engineer, in terms of site review there is a 90% overlap between their respective obligations.);

10) mortar fins existed between the two walls.

Mr. Mueller pointed out that although Mr. Garbutt is listed as a consultant in the contract documents, in his bill to the plaintiffs at Tab 43 of Exhibit #1, he referred to 25% of his total fee being allocated to "supervision". Mr. Johnston pointed out that there is a difference between a consultant and a supervisor. The accepted practice is that a consultant simply reviews the contractor's work, while a supervisor inspects and directs the work. Mr. Johnston made the point of explaining that in pointing out what Mr. Garbutt should have observed and considered on his site inspections, he was speaking from the point of view of Mr. Garbutt being a consultant rather than a supervisor.

Mr. Johnston expressed concern that in his letter of January 10, 1994 Mr. Garbutt denied that the masonry wall was designed as a cavity or veneered rain screen type. The presence of a 19 millimetre air space in Mr. Johnston's view inherently defines the exterior brick façade as a veneer application to the exterior side of a back-up concrete block inner wall. The Ontario Building Code then in force required that any wall with an air space under two inches be provided with weep holes immediately above the base flashing in veneered walls having bearing support.

The 1985 Ontario Building Code under which the building permit was issued stipulated in sub-section 5.4.5, Exterior Wall Cladding, that along with designing the wall "to shed water to prevent its entry into other components of the building assembly where there is a likelihood of some penetration" measures to return penetrating water to the exterior should be provided.

At Tab I of Exhibit #54 is Mr. Garbutt's letter dated January 10, 1994. At the bottom of the second page thereof under the heading Comments, he states the following: "Any remedial work suggested to essentially convert this wall into a veneer type (weep holes, vents, flashings, etc.) are ill advised because of the minimal width of the collar joint (approximately 12 millimetres) facilitating bridging and blockage by mortar droppings in an erratic unforeseeable pattern. The width of collar joint is well under that which is generally accepted for veneer construction (40 – 50 millimetres) and the 50 millimetre requirement for cavity walls." With respect to this comment, Mr. Johnston pointed out that Mr. Garbutt had specified a 19 millimetre air space, not a 12 millimetre air space, but agreed with Mr. Garbutt's comment to the extent that if the wall had been built with a 40 millimetre air space it would have been a lot easier to sustain.

Also in this letter Mr. Garbutt suggests that it is important to determine whether the collar joint has been filled in its entirety, or has been filled except for localized areas. Mr. Johnston pointed out with respect to this comment that the collar joint here was not filled nor according to the design was it intended to be filled. As well, had it been built as a solid wall which Mr. Garbutt says he intended, then no provision has been made in the drawings or specs for internal water protection, eg there was no parging called for nor installed on the inner surface of the block.

It was Mr. Johnston's evidence that had flashings and weepers been added to the wall as constructed, the wall would have provided a longer performance without water problems, but here the water is coming through and is rusting the ties which because of the bend in them could rust through. In his view, the problems with this wall cannot be rectified without installing flashings and weepers and without dealing with the block wall after the brick wall is removed.

It would be worthwhile to refer to specific paragraphs in Exhibit #54, and particularly the concluding paragraphs thereof, which say as follows.

> As to Garbutt's comments paragraph 4 in that last letter of January 10, 1994, his reference to a 12 millimetre (1/2 inch) air space is unfounded as his own specification calls for 19 millimetres (3/4 inch). Furthermore in the large majority of locations that we uncovered, the air space was in fact constructed closer to 40 millimetres which he admits by specific reference is the standard that he perhaps should have specified and attained if he were to now specify the veneer installation that he had originally intended. Ironically had the walls been fitted out with the specified flashing correctly, they were in fact constructed aside from tie back and structural reasons to perform at least moderately well and the excessive height structural deficiencies would unfortunately have remained undisclosed. Deterioration would manifest much later, we would predict. Above all, however, we would strongly disagree that the "collar joint" be fully grouted at this stage, a) without redressing the structural integrity and b) for the other reasons expressed above. This applies most specifically and most certainly with this brick.
>
> In addition it is our recommendation that any attempt to drain and vent this wall without reconstructing the back-up concrete wall and providing continuous means of redirecting incursive water back to the exterior would be unacceptable and would not solve the problems with the walls. If there were a breathing material that performs as Garbutt has indicated shedding water while having an ability to survive for many years and re-adhere the brick masonry, bridge and span the developed interstitial voids under the buffeting of severe wind loads, we would still suggest that there are serious shelf plate and lateral support structural deficiencies requiring rectification in the wall assembly throughout.

> We are concerned that there are material and long term risks attending the continued service of the walls without appropriate corrective measures. The timing of any failure and the inherent risk associated with that failure has not been nor may ever be quantifiable. While it seems that the porosity of the brick was always the Achilles heel in the assessment of wall detail and performance, and while we cannot endorse the manner in which the brick actually was installed, paradoxically it's very softness may have allowed not only absorption but also the essential re-drying from the exterior alone. The brick will ultimately suffer there's no doubt. We can only advise that means be developed to re-establish the structure in accordance with the Ontario Building Code and that action be taken to minimize the incursion of water in accordance with our suggestions.
>
> By re-stabilize, it is meant as per our several reports that
>
> - openings in the back-up wall are filled to provide an effective water shedding surface
> - flashings are provided at all shelf plates, through wall, and window/door openings
> - anchorage is provided to laterally support the inner four inch concrete block wall
> - an air barrier is attained or measures are taken to prevent moist interior air ex-filtration
> - non-corroding structurally effective masonry ties are uniformly installed per the OBC
> - structural measures are employed to comply with height limitations of veneer masonry.

Mr. Johnston then turned to Exhibit #55, Folio 2 entitled Building Envelope Review and Analysis, Roof and Parapet Design. Mr. Johnston explained that after his initial retainer to review and determine problems that might relate to the roof or the wall, he found that there were problems relating to both. He indicated, as has been indicated before, that it turned out that building permit drawings were not available from the City or any other source.

With respect to roof drawings prepared by Mr. Garbutt, he noted that there was a structural roof plan at Tab A of Exhibit #55, which did not indicate slopes or drains. As well there was a mechanical drawing of the roof for the placing of equipment which again did not show drains or slopes. Although he noted that two drains had in fact been added to the contract terms, there was no drawing to show these drains or no change order to indicate where they were located.

Having regard to Exhibit #56, Tab A and the drawings contained therein, it is clear that the first two drawings do reference an "irma system roof mate" by way of word reference. Also on the third of the small detailed drawings at Tab C, there are details consistent with an irma roof but nowhere in any of the drawings are any slopes or drains shown.

As pointed out by Mr. Johnston in Folio 2 page 2, the Canadian Roofing Contractors Association (CRCA) recommended in 1986 "positive drainage is a requirement for the protected membrane roofing". Mr. Johnston explained that positive drainage means an aggressive slope. In actual fact during the course of his inspection he saw negative slopes with standing water on the roof.

From all of his review, in terms of specifications and drawings, Mr. Johnston concluded that Mr. Garbutt did not intend that there be any slopes on the roof, notwithstanding the CRCA recommendation.

In his evidence Mr. Johnston pointed out that under the General Conditions, both the contractor and the consultant have an obligation to clarify any discrepancies as between the specifications and the drawings, but perhaps in clarification, the specifications govern. In this case, among the various records Mr. Johnston in his review could find no communications between the consultant and the contractor or change orders with respect to the wall and the roof.

Mr. Johnston explained that at this early stage, drawings and specifications were discovered to be in conflict and construction appeared to compromise or vary from the specifications, the drawings or both. There was no record available that demonstrated attention to inevitable conflicts either in the field, during construction or during scheduled progress meetings. The drawings and specifications were not revised to reflect the structural conditions. In the face of conflicts which needed clarification, the contractor should have spoken to the consultant. By way of example, Mr. Johnston pointed out that the specifications for the roof called for a slope while the drawings did not. The Contract at General Condition 3 provides that the consultant is the first arbiter in the face of conflicts.

Mr. Johnston additionally identified his major concerns in a report to the plaintiffs dated October 3, 1994 which is Tab F of Exhibit #55. His primary concerns following the bullets in Tab F, page 2 were as follows:

1) the specification called for built-up roofing membrane of felt and tar, not the more prevalent EPDM-type membrane;

9) no slopes or drains are noted on the steel drawing S-1 nor is thickening of concrete detailed to drain roof-accumulated water effectively. How is roof water evacuated efficiently consistent with irma design principles and the specified 2% slope;

10) despite the drawings calling for an irma roof application, the specification refers to a standard built-up exposed membrane roof with mopped-in insulation, not as is evidenced on site. Around the drains, expanded polystyrene insulation was evidenced, not the extruded styrene Roofmate as noted on the drawings.

At page 6 of Exhibit #55, Mr. Johnston discusses the roof disassembly previously described by Mr. Kendall. Essentially, the disassembly consisted of taking the top coping off the parapet for a distance and then stripping the roof layers to the membrane, pulling back the ballast then the filter sheet, and then the insulation.

As described by Mr. Johnston and the photographs, there was adhesion between the insulation and the hot bitumen. As far as Mr. Johnston is concerned it is incontestable that by the mid-

1980's the placing of the insulation on hot bitumen permitting it to adhere was not proper practice. It was accepted in the early '80's that this should not be done.

The photo at the bottom of page 7 shows the exposed parapet. Mr. Johnston noted that the bottom edge of the coping was approximately 8" above the membrane while Mr. Garbutt's detail required it to be approximately 16".

Underneath the coping, the plywood was flat rather than sloped to the roof as it should have been, and the plywood had evidence of dry rot and water stain. The pre-painted metal coping had the specified S-locks interlocking at the joined ends, but were not sealed as they should have been. The horizontal interlock with the interior parapet surface metal flashing was limited to a horizontal cleat type joint within barely 2" vertically of the gravel surface of the roof and it too was unsealed.

Mr. Johnston and Mr. Kendall opened a two foot strip of the roof from the parapet to the drain, a distance of approximately 16 feet in length. As previously described by Mr. Kendall, they then ran a water test. They determined there was no slope to the drain but rather a counter slope to the parapet. Water collected to a depth of 5/8" before it began to drain. As a result, in this area at least, the water never drains but develops mould and slime as depicted in the photo's at the bottom of page 8. This leaves the membrane exposed to water at all times and in those circumstances the bitumen, according to Mr. Johnston will decompose over time, losing its ability to respond to tensile forces. The bottom photo on the right hand side of page 8 shows the membrane on the top of the parapet extending no more than ½ way across whereas Mr. Garbutt called for the membrane to extend all the way across the parapet. As well, the photo shows that the wood blocking on the top of the parapet extends only to the brick wall whereas it should go to the outside edge of the brick wall. As well, the wood blocking was not attached to the brick as it should have been. Another deficiency was that there was a very small overlap of the exterior flashing drip edge over the brick masonry below. For Mr. Johnston, this unsealed edge became more disconcerting when no apparent measures could be identified that sealed off the open air space at the top of the masonry wall construction. Similarly, no sealing was applied to this leading edge of the exterior drip at the top of the brickwork, and this along with the loose layment of the wood blocking substrate prevented any effective top closure of the entire air space within the masonry envelope or wall construction below.

As Mr. Johnston testified, there was no conscious effort by either the masonry or roofing trades to seal off the parapet. Good roofing practice required closure of the parapet, as did the Ontario Building Code and indeed as did Mr. Garbutt in his drawings. It seems obvious that allowing convection currents to develop in the air space as a result of leaving the parapet open will result in driving rain into the parapet and wall.

As noted in his report, although there was no evidence of immediate impending failure in the roof section that he and Mr. Kendall exposed, Mr. Johnston was concerned that other areas of the roof might be closer to failure. Mr. Johnston was particularly concerned about the lack of slope which resulted in floating insulation and standing water. Mr. Johnston explained that the lack of slope and the flashing problem was related. Where the insulation abuts the side of the parapet, there is a freeze-thaw experience in the winter time which adversely affects the flashing at that point, and it is therefore most important to have a good transitional flashing at the juncture of the roof and the parapet.

Another water test was conducted by Mr. Kendall and Mr. Johnston at the northwest corner of the roof which is effectively the roof over the stairwell. The bricking of the stairwell column showed extreme efflorescence and staining. The roof in this area was flooded with a hose. The water got to the level of the gravel at which point it manifested itself on the exterior brick wall immediately below the soldier course. The water on the roof never did get to the top of the parapet. The fact that the water showed itself on the outside of the brick wall immediately below the soldier course, the bottom of which is at the level of the membrane, demonstrated that there was a roof failure with water flowing through the membrane at the bottom of the parapet.

As a result of these tests, Mr. Johnston recommended that a repair be instituted locally of part or all of the area over the northwest stair tower, which repair might, based on discovered extent of the failure, extend into the adjacent main building roof area. Because flooding in this area had resulted in substantial discolouration and disfiguration of the brick finish, Mr. Johnston suspected that the overall disfiguration of the northwest corner was the result of flooding in severe rain storms. He at that time did not believe that the same degree of roof membrane deterioration existed anywhere else as the damage to the brick appearance was at no place as severe as was noted locally at the stair tower. Mr. Johnston also recommended that at the same time repairs be effected to the roof hatch where the flashings appeared not to be in good condition. His recommendation for the roof repair was that it be done immediately and that at some point in time the entire roof be replaced, including the raising of the parapet to accommodate proper sloping in the roof. Also he recommended that the parapet construction be reassessed not only for an increased flood rim height but also for waterproof integrity of the top surface flashing. The coping had been installed with insufficient slope, insufficient exterior drip edge coverage on the brick, and insufficient waterproof membrane extension below the metal flashing typically around the building.

Mr. Johnston testified that it was obvious that any failure in the roof membrane in the area of the water test , if attributable to rupture or degradation of the drainage level, was complicated by construction of the wall and parapet where no obvious flashing or drainage provisions had been incorporated.

Based on the recommendations of Mr. Kendall and Mr. Johnston, the plaintiffs decided to proceed with the roof membrane repair at the northwest corner of the building, which repair was effected on December 5, 6 and 7 of 1994.

The photo on page 15 of the report shows ice on the top of the parapet coping, indicating that because of the lack of slope water is allowed to lie on the coping.

Mr. Johnston was asked to compare his findings determined during the repair of the northwest corner to his findings of the disassembly at the southeast corner of the roof. The comparisons were as follows:

1) in the northwest corner, the parapet membrane lay on top of the plywood, whereas in the southeast corner the membrane lay below the plywood;
2) in the northwest corner, the membrane went ¾ of the way across the top of the parapet while in the southeast corner it went ½ way across;
3) the vertical framing for the inside wall of the parapet was all wood in the northwest corner and concrete block in the southeast corner;
4) in the northwest corner there was no block underlying the plywood on the parapet, whereas in the southeast corner the layering downward was plywood, 2” wood, and concrete block.

Of some significance is the fact that in the northwest corner, only one ply of felt extended from the cant part of the way to the top of the parapet. The specifications on the other hand called for four plies at the cant with two plies extending up the side of the parapet wall. In our case, the four plies were turned on to the cant but quickly reduced to one ply which extended about ½ way up the side of the parapet, at which point an application of bitumen was serving as the flashing to the top of the parapet and over. It would appear from Tab D, Mr. Garbutt’s drawing of the parapet profile, that he is showing at least one or two felts with bitumen extending entirely up and over the parapet.

The photograph at page 16 of Exhibit #55, taken during the course of the roof repair in the northwest corner, clearly shows the wrinkling of the roof membrane. Compare this to the roof specifications at Tab E, section 6.2 which requires that the felts be laid without voids or wrinkles and firmly embedded in mopping of bitumen.

The photo’s taken of the juncture of the roof and parapet during the course of repairs, show very significant horizontal tears at the location of the cant. In one area where there is significant horizontal tear and delamination of the membrane flashing, a closer examination disclosed that the area behind the flashing was hollow, and that at one point the membrane flashing was spanning a distance of approximately 4”. This is faulty construction in that the membrane has nothing to adhere to. It was Mr. Johnston’s view that where the membrane had ruptured, felts were inadequately extended vertically at the parapet on to competent backing or substrate and were also laid up with an insufficient number of plies to properly extend from the horizontal base membrane, up the cant and the vertical parapet surface onto the horizontal finish of the parapet coping level.

For a clear demonstration of these tears, see the photographs on page 17.

During the course of this repair, Mr. Johnston determined as he expected that there was no slope at all to the underlying membrane and when the four felts were removed he observed no primer on the concrete floor, contrary to the requirement for such a primer under section 3.1.1 of Division 7, Tab E.

Referring again to the horizontal tears in the membrane as depicted in the photographs, Mr. Johnston pointed out that during the course of this work to repair the northwest corner, approximately 35 feet of parapet was exposed and the total tears discovered were approximately 24 feet. The tears in his view were being caused by the freeze-thaw effect at the parapet, the fact the membrane was spanning a hollow and the deterioration of the membrane once the water got in behind.

Mr. Johnston noted during the course of the repair and removing the insulation, that the joints as between the upper and lower layers of insulation were not staggered. It is bad practice not to stagger the joints. Staggered joints cause a sharing of the load which is a good thing.

Mr. Johnston noted that it was apparent that the roof of the stair tower had been done after the completion of the main roof, thus resulting in some 8 plies of felt building up at the connection. This caused a ridge along the length of the stair tower which effectively made a dam so that water could not drain from the roof above the stair tower. As a result, there was permanent standing water there.

The wood framing as installed in the parapet was neither adequately fastened nor sufficiently continuous to provide the solid backing necessary for any vertical roofing membrane flashing.

Mr. Johnston testified that while this roof repair was confined to one small portion of the roof it did in his view reinforce the concerns that were evidenced in his earlier investigation in October, 1994 namely:

1) there were no roof slopes, contrary to the recommendation of Dow Chemical and the recommendations of the Canadian Roof Contractors Association;
2) while two roof drains were included in the Appendix of Extras added to the contract in the negotiation stage, the fact that the drains were neither specified, dimensioned, detailed nor referenced on any of the plan documents led to inattention in how they were to function and to be served by the slopes that should have been identified on the drawings and achieved in the course of construction;
3) the elevations of the roof adjacent to critical parapet areas were neglected and failed to prevent back slope or ridging, resulting in the pooling of water at the bottom of the parapet, which standing water eventually caused failure of the membrane;
4) the actual location of the mechanical equipment on the roof was not shown on the drawings. The equipment sat on raised sleepers (long timber members) which in turn sat on the insulation. The equipment was located between the two drains and accordingly the timbers prevented flow of the water to the drains;
5) the wood framing used in the parapets did not even meet the intent of the limited details that formed part of the contract, and was used in lieu of the specified concrete block;
6) the parapet was deficient in both design and construction. The details referenced at Appendix 2D were insufficient in their provision of measurements, material explanation and construction direction, and failed to provide the necessary rigour to achieve under the contract a satisfactory parapet installation. It was Mr. Johnston's evidence that despite this deficiency of direction, even the poorest of tradesmen would recognize that the parapet constructed on this building nowhere nearly attained the height and protection that even the minimum detail provided would seem to indicate. No record of clarification was provided, no deficiency was reported in the documents, and no corrective measures

> were either documented or instituted. The flashing, that is the coping, failed to achieve adequate drip edge coverage, was not sealed in installation, and was susceptible to driven rain penetration. The primary membrane application reached only a part of the way across the top of the wall which had long-term implications for the integrity of the wood blocking, for prevention of incursive rain water gaining access to the upper sections of the exterior wall air space, and without an effective air barrier resulting in an inability to prevent any venturi effect with the envelope air space.

Mr. Johnston in his evidence then moved on to the general roof replacement which was effected in 1998. He explained that in his view the 1998 roof replacement simply gave the plaintiffs what they should have been given in 1986 and that it did not constitute an improvement. He conceded that what he thought to be a better membrane was used in 1998 as opposed to 1986 but it was one that became available after 1986 and was no more expensive than the other type of membrane.

It is to be noted that Mr. Johnston testified that the same pattern of systemic water penetration into the wall was observed following both the 1994 repair <u>and</u> the 1998 replacement which left him to conclude that a significant problem remained with the wall.

In installing the new roof, Mr. Johnston was able to give the roof 2% slopes by extending the height of the parapet and providing a drain in the northwest corner area.

The photo of the exposed parapet on page 22, taken during the course of the roof replacement, shows a 4” gap between the block wall and the brick wall. Mr. Johnston explained that while the increased distance between the two walls would require a stronger tie, there were in fact no ties in this area for a depth of three feet.

During the course of replacement, Mr. Johnston noted that approximately 75% of the lower insulation panels had adhered to the underlying bitumen, indicating a conscious intention during original construction, to adhere the insulation to the bitumen. As well, the photographs at page

23 of the roof with the ballast and insulation removed, show fissures in the membrane and a friable area which appears as a local decomposition of the surface membrane due to breakdown of the bitumen in the horizontal plane. Mr. Johnston noted that the deterioration did not appear to be going at that point in time beyond the top layers of felt, and constituted a slow decomposition of the felt material itself. Generally, the deterioration that Mr. Johnston observed over the entire roof at the time of the replacement was consistent with the deterioration that was seen during the repair of the northwest corner and supported his concerns that the roof could not live out its normal life.

Mr. Johnston noted that during the roof replacement, the original membrane was purposely left in position for the replacement and was reconditioned to become the air/vapour barrier of the new roofing application. The decomposition that was seen in the northwest corner failure was due to a thin membrane at the cant with poor adhesion, exposed to ice and water and his observations during the course of replacement in 1998 told Mr. Johnston that the same conditions threatened the entire roof.

Mr. Johnston's summary with respect to the roof is located at pages 25 and 26 of Exhibit #55. He notes in the preamble to his summary that as well as specific deficiencies identified during analysis and included in his several reports, more generally his conclusions can be summarized and attributed to all or combinations of inattention to design, general contract responsibility or individual trade practice. Had the various measures set out in nine bullets been properly provided, then the problems and deficiencies that he disclosed in his report and his evidence would not have occurred. It was his evidence that design, construction supervision and consultant's review were all inter-related and all contributed to the problems. He explained that sometimes the effect was cumulative. For example, the lack of specifications for the membrane could have been rectified if the contractor had consulted with the consultant. The intent of the roof specifications for example, was reasonably well expressed but was not detailed. In his view, it was not reasonably possible to isolate design or construction or consultant review, and to apportion the cause of the damage. In his view, this could not be done.

Mr. Johnston moved on to Exhibit #56, Folio 3 entitled ‘Building Envelope Review and Analysis, Wall Assembly and Design’.

Mr. Johnston noted that Mr. Garbutt’s drawings at Appendices A and B do provide an indication of over-all intent but do not show the inter-relationship of construction details.

Again, the wall drawings at Appendix C, while they show the intent are devoid of any real details. By reference to these drawings, and to his own sketch which he prepared in the courtroom and which was filed as Exhibit #57, Mr. Johnston explained the problem that was encountered when after starting the brick wall on the south side, the masons turned the corner and found that the brick course would not line up with the shelf plate on the beams on the east and west walls. He explained that one must drop the beams on the north and south facades in order to accommodate the 4” high shoes on the posts. The masons weren’t dealing with a 4” brick and as a result as they came around the corner, the plate on which the brick was to sit was in the middle of the next brick up. If one looks at Tab I, one can see an illustration of a north or south wall where the foot of the joist sits on the top of the flange and the floor then sits on top of the joist. It is obvious from Tab D where a drawing of the roof framing is included, that it was intended that there be shelf plates on all four sides. Then Mr. Johnston explained that the masons, starting the laying of brick on the south side, did calculations before starting to ensure that the course of bricks would properly end up on a shelf plate. Having done that, when they got around the corner, they found that the shelf plate on the east and west walls was effectively sitting in the middle of a course of bricks. Mr. Johnston explained that the problem that was encountered on site could have been resolved in a number of ways. He explained that the steel fabricator takes the building’s drawings and prepares his own shop drawings of the steel required, which shop drawings he sends to the contractor and the consultant for their approval. If the problem had been spotted, then it could have been resolved for example, by lowering the plate on the south side by 4” and then using a 16” beam on the south side and a 20” beam on the east side. In this way the plate levels on all four walls would match up. Another viable solution would be to use an angle rather than a plate, which would drop down from the bottom flange 4” and then jut out. Another solution would have been to construct the vertical plate in the I-beam and from that extend a horizontal plate at the proper height.

Filed as Exhibit #58 was another drawing illustrating the problem and possible solutions.

As we know from the evidence, the problem was not caught and once the steel arrived on site, the only possible solution consistent with the requirement for shelf plates would be to cut off the existing shelf plate, insert a vertical plate in the I-beam and then weld the horizontal plate to the vertical plate at the appropriate level. This in Mr. Johnston's opinion would have cost anything from $10,000 to $30,000 in 1986.

Referring to the wall plans at Tab C, it was Mr. Johnston's evidence that one would have expected the plate to be actually shown on the drawing rather than simply referred to. In fact, if one looks closely at Tab A, drawing 4, one can see the shelf plates drawn in at the locations where they actually ended up, that is at the bottom of the beam.

It is interesting to note that General Condition 34 (see Tab 2 of Exhibit #1) requires the contractor, prior to submission to the consultant, to review all shop drawings. He shall then submit the shop drawings to the consultant for the consultant's review, notifying the consultant in writing of any deviations in the shop drawings from the requirements of the contract documents. The consultant then reviews for conformity to the design concept.

Mr. Johnston explained that the '83/'84 code imposed an 11 metre height limit on walls without supports, and provided that if the height of the wall exceeded 11 metres, then there had to be a support every 12 feet. At a floor height of 3.2 metres, the brick wall height was well in excess of 11 metres. In any event, it appeared that the plans called for plates on both the south and east side and the north and west side as indicated at Tab A, Drawing 4.

It was the evidence of Mr. Johnston that it was never intended that the cavity between the brick wall and the block wall would be filled. That is evidence from the wall drawings at Tab C, particularly at the third drawing where the space between the two walls is white up to the level of the parapet and then appears to be filled in with horizontal lines, the conclusion being that the drawings where they intended to show filling did so. As well, at Tab F is the masonry specification forming part of the contract which under the heading 'Brick Work' directs the contractor to "maintain continuous 19 millimetre air space behind brick throughout the project". This is in conformity with the detailed drawings. There is nothing in the specifications or the drawings to indicate a solid wall or a filled collar joint. It was Mr. Johnston's evidence that the spandral beam interjected into the wall would prohibit this from being a solid wall under the code.

Mr. Johnston went on to explain that by very definition the 19 millimetre air space behind the face or façade brick units does not comprise a cavity wall as it is understood by the reference standards both in Mr. Garbutt's specification and relevant Canadian Standard Association documents. The air space simply does not attain the required minimum of 50 millimetres dimension recognized by the OBC, nor is there any insulation proposed in the cavity. The insulation was clearly designed and detailed in all references to be applied at the interior face of the interior concrete block. While a wall with a 19 millimetre air space cannot be a cavity wall, it must perform as one.

Mr. Johnston in his evidence identified critical elements that were neither identified nor annotated, either on the drawings or on the details, their presence having been dictated only by the written requirements of the specification. See page 7 of his report. These included the following:

1) no indication of two ounce copper fibreen flashing was shown at the slab level to prevent water incursion at the floor slab;
2) the shelf plate was neither drawn nor dimensioned, nor were there indications of what type and spacing of drainage provision was to be incorporated in the wall, nor how the means to achieve venting or drainage was to have been achieved, with or without continuous flashings;

3) the anchorage of the concrete block wall to the structure was not indicated on the drawings;

4) Article 2.2.6, Reinforcement, requires Number 9 imperial wire gauge truss type galvanized reinforcement for exterior walls, none of which is indicated on the drawings;

5) location and spacing of wall anchors were not detailed on the drawings in any manner;

6) no indication of flashing was shown at lintels, window heads or shelf plate intercessions.

Without indications as to where the flashing was to be installed, and how the shelf plate was to perform, it was impossible to establish how the masonry wall was to achieve its primary purpose, namely that of the 1983 OBC subsection 5.4.5, Exterior Wall Cladding, and article 5.4.5.1, ie, deterring water from entering into components of the building assembly and providing effective drainage to take any penetrating water back to the outside again.

In all of the circumstances, Mr. Johnston recommended to the plaintiffs further investigation in the form of the wall being at least partially disassembled at an appropriate location of apparent leakage to determine in actuality what measures had been taken.

The wall at the interior of the south elevation above the ceiling line of a fourth floor storage room was partially disassembled from the inside. The area was partially finished and was without a ceiling but was outfitted with drywall finishes that were reviewed and judged typical. Mr. Johnston explained that upon disassembly it could be seen that the polyethylene back of the drywall and in front of the insulation was loose and not sealed to the roof deck. He explained further that the Code did not require an air barrier at the time, nor did it require a vapour barrier as such, but there were some provisions on the Ontario Building Code for condensation control, and the National Code did in fact require an air barrier in 1985. On balance, good practice would require in 1985 an effort to control for exfiltration of air and less concern about vapour. The drywall if properly finished and painted could constitute a barrier to air and condensation, but it stopped short of the under part of the deck.

There was essentially no air barrier. Interior air was moving out of the system and outside air was coming in.

The test area above the window at the fourth floor was partially disassembled and photographed in stages for the record. The photographs at page 11 of Exhibit 56, show the polyethylene loosely fitted over the insulation but not sealed. It also shows that where the insulation was compressed or pushed away from contact with the polyethylene there resulted a soiled, dirty surface that Mr. Johnston presumed was indicative of air movement behind the polyethylene film where the insulation batt had been compressed enough to leave a free air channel between it and the film.

There was no concrete block in the 4” open space above the spandrel beam. The bottom photograph on page 11 shows staining on the insulation which indicates that dirty air is moving through the insulation leaving soil on the insulation surface. The photograph on page 12 of Exhibit 56 shows the mortar fins, a lack of filling of the head joints and white specks on the deck caused by condensation.

As well, at the opening, Mr. Johnston found fiberglass batts rather than rigid insulation called for by the drawings. The difficulty with fiberglass batts is that they are porous to air flow and water can get through.

At page 13 is a photograph which depicts the framing just above the window which shows a puddle of rust which is representative of an extensive collection of water on the horizontal member. The top photograph on page 14 shows the same opening and the same stain. This photograph also shows the lintel supporting the brick with the block lintel inside of that. Below the lintel is the stone surround. This photograph also shows the oxidation which Mr. Johnston observed all the way along. He could tell from the colour that this represented active oxidation. Mr. Johnston in his evidence expressed the opinion that the oxidation was caused by water incursion and collection as opposed to condensation. Water was therefore getting through the brick wythe, the air space and the block wythe, and then through the insulation. In Mr.

Johnston's view, the water was getting inside the concrete block and coming down between the block and the insulation. He looked for flashings and found that there were none. In his view, properly installed flashings would have prohibited the water from coming into the building.

Referencing the photograph of the exterior of the building at the bottom of page 14, Mr. Johnston explained that the area that was being investigated was the south façade above the window marked with an 'X' and immediately beneath the removed parapet finishes. With Mr. Kendall's assistance, Mr. Johnston projected a fine but moderate water spray on the face of the building with the hose being held approximately 8" from the façade. Mr. Johnston pointed out that the test was not intended to scientifically quantify any water incursion, but rather the intent was solely to establish if a light spray on the exterior façade could be seen to penetrate to the inside in even mild wind conditions which existed at the time.

Mr. Johnston observed that water did in fact penetrate the façade in a relatively short period of time, namely one minute and 15 seconds, and showed itself at the inside of the lintel angles above the stone surround as is indicated in the photograph at the bottom of page 15. Mr. Johnston observed that there was a clear flow of water which emanated as a rivulet from between the angles that formed the lintel framing at the inside and outside masonry at the head of the window opening. The water flowed across an active oxidation area to wet the stone header over the top of the window opening, before pooling on the drywall soffit at the head of the window below.

At about four minutes into the test, the first water could be seen entering through a mortar joint above the spandral beam as is depicted in the bottom photograph on page 16 of Exhibit 56. As the test proceeded, additional wetting continued to appear at the exposed interior face of the brick and several points of incursion on the brick were noted. Eventually, after about 12 minutes, the water drops could be seen subtended from the metal window head frame itself.

It was clear to Mr. Johnston that the location of the rivulets on the back of the brick followed a consistent pattern of discolouration, which indicated previous water penetration. See the bottom photograph on page 16.

As more flow and water penetration occurred, Mr. Johnston noted generally that the flow emanated from the top surface of the bricks and that there was no apparent crack or opening to the exterior in the mortar. In other words, as Mr. Johnston explained the water penetration was occurring at the uppermost plane of the brick surface where it met the mortar bed of the brick above. At no point during the testing did Mr. Johnston notice any flow from the often-incompletely mortar filled “header” or end joint of the bricks in the wall.

It was clear to Mr. Johnston that in a period of some four minutes the moving water had come down a distance of some 30 inches through the air space between the two wythes to the lintels and then through the lintels to the interior of the building. At 12 minutes into the test there was a collection of water on the soffit which began to drip down through the separation between the drywall and the window frame. Mr. Johnston testified that all this was consistent with Dr. Reed’s advice to him and what was being seen on the video.

Apart from the photo’s, Mr. Johnston saw water coming in at perhaps a dozen places on the lintel where there were corresponding rust marks as shown in the photograph at the bottom of page 15.

Referring to the photo at the bottom of page 16, Mr. Johnston explained that there were no visible cracks in the brick wall observed where there was either active water flow or old staining. He explained that the plane between the top of the brick and the mortar had become unbonded or partially debonded which permitted water to come through at that level, even in areas in which total bonding had not been lost. It was Mr. Johnston’s evidence that this is a natural phenomenon recognized in the industry where certain stresses can cause this debonding or partial debonding without there being any visible crack.

At this point, Mr. Johnston referred to his report dated May 16, 2002 to Mr. Mueller, which was marked as Exhibit #59. This report was a rebuttal of Mr. Quirouette's report dated May 10, 2002 wherein Mr. Quirouette critiqued the windows and window surrounds of the subject building. Mr. Quirouette's basic contention was that the leakage at the windows and particularly the top thereof was as a result of caulking break down. As well, apparently Mr. Quirouette attributed the problem at least partially to condensation.

On this latter point, it was Mr. Johnston's evidence that there could never be sufficient condensation in the wall or in or about the window to generate the kind of water flow that was observed. The spots on the underside of the roof deck previously referred to did show the existence of some condensation in the wall, but as Mr. Johnston noted it was globular and obviously had not flowed. Condensation occurs when warm moist air from the inside is introduced to the cold flutes. In Mr. Johnston's view, in this case condensation can be completely discounted as a source of running water in the wall.

With respect to Mr. Quirouette's theory relating to a deficiency in the caulking, by reference to Tab I, his drawing of the wall structure at a top window, it would be impossible for the observed water to be coming in between the lowest course of brick above the window and the steel lintel. To do so it would have to enter and then make its way <u>up</u> the back face of the lintel, and from there down through the separation between the window lintel and the back of the block lintel. Having regard to his spray test, Mr. Johnston pointed out that it would have been virtually impossible for water to accumulate on top of the window lintel in the times indicated to collect to the point of overflowing the top of the lintel plate. In so far as the water coming into the cavity between the bottom of the lintel plate and the top of the stone surround, having reference to the photograph at the bottom of page 15 where the rivulets of water are shown running down the back of the brick lintel, this was clearly not the case.

With respect to the window head surround, Mr. Johnston indicated that these are quite common in the Kingston area, dating back two or three hundred years when they were often used as lintels themselves and were sloped down because it was expected that water would get in. Here, although the stone header is flat, it would have presented no problem had the proper flashings and weepers been installed.

Again referring to Mr. Quirouette's critique, Mr. Johnston saw no design problem with the windows themselves nor was there anything of significance with respect to the manner in which the drywall was joining the window frame.

The window itself was an adequate, normal window though not the best available. Although there was only one seal between the window and the masonry and not two in higher priced windows, the one seal was quite adequate.

It was Mr. Johnston's evidence that the caulking in 1986 and 1992 was more than adequate in terms of maintenance and was in fact excessive. On his examination in 1994, he saw no sealant failures except a few very minor ones and he saw no water incursion as a result of sealant failure.

In Mr. Johnston's view, most water entered this building coming through the bed joints as seen by the testing in October of 1994.

In his report to the plaintiffs dated October 5, 1994, subsequent to his investigation, Mr. Johnston reported his findings among which were the following:

1) the cap joint to the underside of the spandral beam was mortar filled, not sealant or caulking filled as specified (a soft joint). In his evidence, Mr. Johnston explained that what he saw between the cap joint and the underside of the beam was a space at times filled with mortar and at other times not. In his view, it was certainly not a soft joint as required by section 2 of Division 4, Masonry, "to provide 6 millimetre continuous expansion joint caulked continuously between steel support angles and brick panels each floor." Although referring to support angles and brick panels, Mr. Johnston saw this specification as applying to the block wall, to the extent that the shelf plate or support angle lay along the bottom of the spandral beam.

Mr. Johnston in his report dated October 5, 1994 goes on to point out that he saw no evidence of lateral restraint or structural ties from the block wall to the spandral beam. In other words, there was no sign of any lateral support.

In explaining the advantages of a closed parapet rather than an open parapet as constructed, Mr. Johnston explained that by closing off the parapet, the venturi effect of air being sucked out of the wall and thus out of the building was reduced. This would limit exfiltration which is a good thing, in that it inhibits warm moist air being drawn into the wall cavity where it can cause rusting of the metal components. In our case, we have a closed air cavity which is open at the top which encourages exfiltration and condensation. There are no flashings or weep holes to either let the condensation out or let in air from the outside to help dry. As well, obviously if there is infiltration of water from the outside there is no way for it to get out in the absence of flashings and weepers.

Although the choice of brick has been criticized, Mr. Johnston explained that the impact of the porosity of the brick can be a positive one. With a lot of water coming in the brick can absorb some of it for later drying from the surface out.

Again, Mr. Johnston pointed out that in his investigation he determined that there were no ties between the block wall and the brick wall from the parapet down to the shelf plates, a distance of some four feet six inches.

Mr. Johnston testified that in his view, the provision for flashings and weepers in the wall would have solved the majority of the problems encountered. The problem with the wall as a air barrier and vapour barrier would still have been left unresolved. The problem with the ties would have been left unresolved. Mr. Johnston's observations in the fall of 1994 and early December, 1994 which we will hear about show that the ties did not do their job. The bond of the bed joints is gone. The ties serve two useful purposes. They transfer lateral pressure to the block wall, and simply put they hold up the brick wall. In this case, the brick wall has had to take lateral wind pressure on its own. As a result, it has lost its bonds at the bed joints.

Further investigation was conducted on December 7, 8 and 9, 1994 during roof repairs at the northwest corner of the building. Three or four bricks were removed from the façade at all sides of the building and at all floor levels. Locations were chosen that revealed the edge of the spandral beam at the perimeter of all floor levels, and it was determined that typically at the east and west façades, the shelf plate had been cut back flush to the edge of the lower beam flange. As well it was determined that there were no flashings or weeping to the exterior face of the brick. The void of the perimeter beams typically had been infilled with masonry flush to the edge of the interior masonry width. The head joint of the inner wythe where it met the underside of the perimeter beam was variously filled with mortar, bricks on edge, half blocks or indeed at some locations left completely unfilled. No measures were found or determined that provided any pressure equalization, weeping or flashing to the exterior of the face brick. It appeared that the specified air space was purposely constructed but with a range of width in the investigated locations from 19 millimetres (3/4") up to a full 40 millimetres (1 ½"), and later found to be up to 100 millimetres at west parapet level. It was clear that the back-up block wall abutted up to and was not engaged in the column indent. There were no tie ins or connections to the column discovered. This was on the north wall. (It appears therefore that the block wall is not supported at either the columns or the beams and there is a concern that it is not structurally sound.)

On the north and south walls the shelf plate was only stitch welded every eight inches or so to the bottom of the beam. As a result it did not act as a flashing and there was no flashing added as required by the specifications.

At one single location randomly selected on the west elevation, a larger irregular opening in the face brick approximately four feet square was removed to establish the nature of wall anchorage or ties from the brick façade to the back-up concrete block. At this particular location, just below the third floor between the windows, the air space was found to be 40 millimetres (1 ½") wide, and even at this width in many locations the accumulation of mortar squeezed out of the back of the bricks as they were laid up bridged the gap completely from brick to block. While the ties were placed appropriately, there were three different types. At the bottom of the right-hand photograph on page 20, Exhibit 56 can be seen the corrugated ties. They can also be seen in the photograph at the left. Because of the coursing, the tie had to be bent up from the block to the brick so that any water coming down the brick wall would go to the block. It is to be noted as well that according to Mr. Johnston's evidence, the bending of the tie results in a loss of strength and also breaks the galvanizing, making the tie prone to rusting.

In the photographs at the middle course and upper course of the blocks, one can see ladder type re-enforcers extending from the block wall which appear to be serving both as a re-enforcer of the block wall, and a tie between the block wall and the brick wall. It was Dr. Johnston's evidence that a truss type re-enforcer can serve a double purpose but not the ladder type re-enforcer. To make matters worse, the ladder re-enforcer at the top course was bent up from the block wall to meet the brick wall.

Mr. Johnston noted that a crimp tie bent three times as it was here was not appropriate above a 20 millimetre air space, and where the air space is more than 20 millimetres, one should go to a stronger tie.

There was also a small disassembly on the west elevation of the stair/lobby tower where in an opening large enough to reveal at least four ties, none were found. Opening further, an improperly installed adjustable tie was located. This was at the second floor level. A 32 millimetre (1 ¼") air space was measured behind the brick veneer.

In short, in three randomly selected locations, either no ties, the wrong ties or bent ties were found.

Investigation by Mr. Johnston established that there was air space between the block wall and the columns, and between the block wall and the beams, making it totally inappropriate as an air or vapour barrier wall.

The result of the removal of the shelf plates on the east and west walls was a 40-foot air column, interrupted only by the windows, something substantially different from the 12-foot air columns on the north and south side where the shelf plates were left in place. A 40-foot air column produced significant pressures in the wall. Firstly there is a chimney effect and secondly a venturi effect, so that the removal of the shelf plate has had not only a structural impact but also an air flow impact.

As well as removing the brick at sample exterior locations, Mr. Johnston had the area above the interior side of the window on the fourth floor opened further to permit a greater view of the interior of the brick façade than had previously been visible because of the concrete blocks that concealed his view in the investigation conducted two months previously. The opened area allowed him to view the brick wall immediately below the spandral beam of the south wall structure and exposed approximately 16” more of the wall immediately above the window opening.

On the inside surface of the brick wall shown in the photographs on page 21, Exhibit 56, Mr. Johnston observed in all some 22 water rivulet stains, where the water had flowed through the bed joint of the bricks. It should be noted that the photograph shows a relatively small portion of the inside of the brick wall. It was clear to Mr. Johnston that the staining was from bed joint penetration during rain storms. The weather was good at the time. It was clear that these were “old stains” providing what Mr. Johnston saw to be an indelible record of water incursion. The stains were typical of repeated occurrences.

As a matter of coincidence, on day 3 which would have been December 9, 1994, Mr. Johnston was in the cherry picker just completing the removal of the last brick on the south wall when a heavy sleet storm came up at about 3:30 p.m. He brought himself down and ran to the fourth floor and looked at the same area of the brick wall shown in the photographs on page 21, and noted that every one of the same bed joints were leaking.

If one looks closely at the brick and block lintels shown particularly in the top photograph on page 21, one can see rust on both.

Mr. Johnston noted that the width of the top edge of both the block lintel and the brick lintel was such that together, as they are in the configuration, they would contribute to an almost complete blockage and bridging on their own between the brick wall and the block wall, even without mortar droppings.

Among others, Mr. Johnston arrived at the following conclusions as a result of the wall investigation:

1) the width range of the air space from bottom to top was from 19 millimetres to 40 millimetres, and then even larger in the upper portion of the walls, at times expanding to some four inches;
2) while the OBC called for flashing and weepers wherever there was an interruption in the wall such as at windows, doors, shelf plates, there was no such flashing. The absence of flashings meant that the water coming in and making its way to the inside of the brick wall, in turn made its way to the block wall and through to the inner finishes, and leaked around the windows at the lintels. In short:
    - the water which was coming in was not being re-directed out
    - the water was coming through the bed joints
    - the brick itself was not a factor
    - the windows were not a factor
    - the windows did not leak; the walls did.
3) there were signs of much oxidation in the wall.

In Mr. Johnston's view, the bond between the top of the brick and the bottom of the mortar broke down over a period of time when sufficient lateral pressure had been applied to the brick wall. The ties were not sufficient to stop the bending and the bond simply let go. In short, the ties did not work and as a result the bond broke. In Mr. Johnston's view the ties were contrary to the specification, to the effect that Division 4, Masonry paragraph, Tab F, section 2.8 required the anchors and bonds to conform with the National Building Code and they did not.

Mr. Johnston noted that apart from that provision in the specifications there was no other reference to ties in any of the documents, specifications or drawings. There were no specific instructions given in the specifications and drawings, and in the circumstances the tie specifications should have been detailed.

Mr. Johnston noted that at no point in the areas that were investigated where the concrete block was removed at the fourth floor interior, was there any evidence of lateral support at the top of the concrete block panel where it connected to the beams. Nor as the block was being taken out did he find evidence of the specified reinforcement.

Mr. Johnston commented again on the fact that the ties that he observed had been bent at three places in order to connect between the block and the brick where the courses did not coincide. In his view, the lack of ties and the use of improper ties permitted cracking in the mortar joints.

In his summary at page 24 of Exhibit #56, Mr. Johnston notes that even after the roof replacement in 1998, leaks continued into the interior of the building at all elevations and floor levels as before. In his view, the absence of flashings in the original construction should have been detected by both the consultant and the mason, and he is of the view that in the original construction the mason Dick, unequivocally did not achieve a then-currently accepted standard of construction. The provision of flashing and venting was legislated and required performance. Mr. Johnston is also of the view that the mason was not well instructed and did not provide an acceptable mock-up.

Mr. Johnston is of the view that inter-related inattention between the consultant and the contractor resulted in the following deficiencies in documentation, execution, supervision, review and correction or seemingly a combination of:

1) failing to provide effective flashing within the wall at all shelf angles and horizontal window feature obstructions;

2) failing to provide means to shed incursive water back to the exterior once it penetrated the weather barrier;

3) failing to provide ventilation/drainage provisions despite a clear specification and OBC requirement to do so;

4) failing to specify required cross ties and anchorage provisions between the brick wall and the block wall according to the Code, by establishing clearly the type of construction and correspondingly acceptable ties; or if requiring compliance by specifying a "suitable" performance, reviewing the means of attaining that which was proposed and ensuring its execution;

5) failing to review and provide acceptable means of construction through the specified procedures of sampling and mock-up according to provisions of the CCDC 2 Contract;

6) failing to assess the implications of varying structural provisions for masonry support on different façades and failing to recognize structural implications of the OBC in corrective measures undertaken;

7) failing to provide field review detail interpretation and direction, including correction of errors or misinterpretations (in this respect see items 4-9, Summary of Roof Deficiencies, Folio 2, page 25), Exhibit 55.

Mr. Johnston was of the view that the following problems as yet uncorrected would not have occurred if the above-noted performance concerns had been corrected in design or construction:

a) inconsistent or inadequately specified masonry ties, inappropriately installed;

b) missing masonry ties in many randomly investigated locations, contrary to requirements of the OBC; lack of wall anchorage detail or instructions on how to achieve lateral wind resistance;

c) insufficient drainage and venting provisions, missing vents at specified centres;

d) missing all flashings, contrary to contract specifications and the OBC;

e) installing structurally unacceptable unsupported vertical heights of veneer masonry on the east and west elevations, contrary to OBC (the missing shelf plates);

f) inadequate parapet closures through wall flashing or solid construction achieving that closure;

g) deficient inner wall integrity and water shedding capability as an effective back-up;

h) not providing shelf plate or window flashings where plates or lintels provide closure of the air space and bridge to internal finishes; and

i) excessive mortar fins which although not specifically contrary to any governing legislation, in Mr. Johnston's opinion contribute to penetrating water transfer from the exterior wythe to internal finishes.

Once again, Mr. Johnston noted that without proper ties, let alone any ties, the brick wall relies on tensile strength or bond to counter the push and pull of the wind. In his view, the virtually unrestrained interstitial or capillary penetration of rainwater at the bed joint indicates clearly that the essential tensile bond is already compromised.

Mr. Johnston notes that while the upper spandrals are significantly exposed to larger wind forces than those at grade, paradoxically it is here that he found no tie backs to the back-up block wall. This is in violation of the National and Ontario Building Codes, and the loss of capacity in "shared load" with its intended inner wall partner, subjects the exterior brick wall to the continued strain of applied flexural loads in the face of the wind.

Mr. Johnston found four different types of ties in his examination. The first was a corrugated galvanized tie which was indeed intended to tie the brick façade to the back-up block wall. The other types of ties found were serving the dual purpose of a tie and a re-enforcer. One of these ladder ties was galvanized while the other was what is referred to as "bright finish" which finish is suitable only to protect against the effects of mortar, and not for protection against water. Yet another type of tie was found in the northwest corner.

As Mr. Johnston testified, he found corroded wall ties during the course of his investigation. For example the photo's at page 20 of Exhibit #56 depict the approximate four foot square area of the brick wall that was removed on the west elevation, just below the third floor windows. The ladder tie which was serving a dual purpose shown at the top of the opening, was as Mr. Johnston put it, “happily rusting away”.

In Exhibit #5, a small book of photographs, at the bottom left of page 4 thereof, is a photograph which depicts a corrugated tie between the block wall and the brick wall that is obviously rusting. This photograph was taken at a time when representatives of the defendant contractor were doing an inspection. Mr. Johnston was present and managed to retrieve the tie and an adjacent tie. Both ties were put into evidence as Exhibit #60.

It was Mr. Johnston’s evidence that the interpretation of structural performance and Code compliance on any wall of the subject building is that they are all compromised but the compromise is substantially more critical on the east and west walls. Not only are the masonry ties inconsistent, they are not fully corrosion resistant and they must perform across an air gap that exceeds its minimum specified dimension of 19 millimetres at only 25 – 30% of the wall’s ultimate height. At the top, this gap was measured at up to 100 millimetres and the thin narrow veneer of the brick wythe resting on the foundation is also more than 12% over the maximum height of 11 metres allowed by the Ontario Building Code. The Code requires a change in the material support and philosophy of structure for walls above this maximum height. This too has been neglected in design, execution and field review by those responsible in each discipline, and they are mutually responsible in Mr. Johnston’s opinion, as the conscious decision to remove a pre-planned structural shelf plate must have been considered by all parties involved in the construction.

Mr. Johnston went on to indicate as he did in his report that in his opinion there is a marked and material compromise in the essential elements of safety and performance of the composite masonry wall. He has a serious concern for the ability of the wall to perform its designed role safely and consistently. In his view, to rectify the situation, both structural and water shedding and venting problems must be resolved. He pointed out that once flashing and weepers were installed, the back-up block wall is the only protection for the interior of the building and that block wall is literally full of holes.

In his report, Exhibit #56, Mr. Johnston addresses several options that have been presented to resolve the wall problem:

1) siliconing the brick façade;
2) pressure grouting the air space;
3) exterior insulated finish system;
4) aluminium curtain wall;
5) remove veneer and replace brick façade.

For the reasons given at pages 28, 29 and 30 of his report, Mr. Johnston states that only the last-named proposal is an appropriate solution to correct all of the deficiencies and to retain at a reasonable cost the original design considerations for the building. The last proposal would involve the removal and discarding of the face brick, the repair of the back-up block wall, ties and structure, the introduction of venting and flashing, and the re-installation of new veneer.

The estimated cost of this solution is at page 31 of the report, and amounts to some $613,000, including an inflation value of some 5% per annum from 1994, for a total of 40% inflation on a base cost of $401,000. It should be noted that from the sub-total of $561,000 for 1994 cost and inflation, there has been deducted some $24,000 on account of parapet upgrades that have already been achieved in 1998.

Filed as Exhibit #61 was Mr. Johnston's report dated April 20, 2002 responding to the report prepared by architect Alexander Wilson for Mr. Garbutt, and dated April 8, 2002.

In his report, Mr. Wilson recommended stripping the brick veneer off at each floor level and at the windows, doors and parapets, to insert flashings and weep holes and to replace on the east and west walls the shelf angles which had been removed.

Mr. Johnston noted that Mr. Wilson agrees that there is a water incursion problem that must be remedied and that there is damage at the windows which must be repaired. We have already seen Dr. Reed's response to the Wilson report.

Mr. Johnston notes that Mr. Wilson says that there has been no cracking of the brick façade, whereas in fact there has been cracking. For example, on the north side at the fourth floor there is an open longitudinal crack of a length in excess of eight feet. As well, there is the extensive interstitial and unbonded interface cracks between brick and mortar caused by flexural deformation of the brick diaphragm and particularly on the east and west walls. Mr. Johnston notes that Mr. Wilson's inspection was cursory and did not involve viewing the exterior of the brick wall with a cherry picker.

In his report, Mr. Wilson notes that the basement and first floors are dry. Mr. Johnston explains that they are dry because:

a) Dr. Reed is collecting the water on the fourth floor; and

b) at the bottom of the building the wall provides a substantial rainwater sink or reservoir, and as a result the deterioration at the lower floors may appear to be less damaging. Regrettably this effect does not protect the lower floors entirely. There is evidence revealed by current investigative openings of the brick veneer that the 28 gauge wall ties are oxidizing completely through. This is occurring specifically at the first floor level in a location where the brick was recently removed by the defendant contractor's team. See the wall ties filed as an Exhibit.

Mr. Wilson in his report based on his observation of a "light drizzle" on the day of his inspection, estimates a collection of one-half to one litre of water at the window heads. We have Dr. Reed's estimate which is much more, but in any event, Mr. Johnston pointed out that rainwater coming through the brick veneer is not only running down to the window heads but indeed is running down the brick veneer piers between the windows all the way to the bottom of the wall. There is no flashing or weep holes at the foundation and as a result the bricks at the bottom of the wall are quite stained.

With respect to specific points made by Mr. Wilson in his report with respect to the roof, Mr. Johnston responded sequentially as follows. See his comments beginning at page 4 of Exhibit #61.

1) the whole building leaks, not only in very small quantities at the south facing windows on the three upper floors as suggested by Mr. Wilson;

2) the parapets were raised in 1998 to meet CRCA standards and to accommodate the proper introduction of 2% roof membrane slope to drain, amounting to a four and a half inch increase in roof height at the perimeter adjacent to the parapet;

3) the repair at the northwest corner in 1994 cured the problem of the efflorescence. The parapet was not raised at that time;

4) Mr. Wilson quotes from Folio 1 of Mr. Johnston, Exhibit 54 as indicating that the initial concept of the protected membrane roof was excellent but rather it was the detail and execution that was deficient. Mr. Johnston stands by his evidence that the concept of employing protected membrane roofing (pmr), the irma roof, was in Mr. Johnston's opinion excellent, particularly having regard to the fact that it was anticipated that two stories might be added to the building in the future. As Mr. Johnston notes in his report, "if installed correctly, the complete roof assembly could have been removed with ease.";

5) contrary to Mr. Wilson's statement, the design did incorporate a 2% slope to the roof drains consistent with the CRCA recommendations that were issued initially in 1978 and which should have been incorporated in the original construction. The roof assembly was wrongly specified as to roofing type but was certainly correctly specified as to slope, despite what was actually installed in the building;

6) Mr. Wilson in his report states, "the designer Donald Garbutt prepared roofing drawings and specifications which clearly showed design intent." Mr. Johnston is of the view that this is the most inaccurate statement that Mr. Wilson has offered. No architectural plan of the roof level was provided anywhere. The only roof drawings provided to Mr. Johnston from the record are:

    a) a structural diagram of roof framing which provides no information regarding slopes, drains, elevations, parapets or openings for roof equipment;

b) a mechanical drawing showing locations of HVAC units and some service lines but no roofing system detail definition or even curbs or support for the mechanical units. The drawing bears no resemblance to the ultimate and questionable configuration of that equipment at the roof, considering the final location of the roof drains that appear on no drawing whatsoever;

c) one of only three details of a large enough scale that should have provided direction on how parapets and roof to wall transitions were to be made. The detail is under-notated, unclear, shows no slope at the roof and is poorly thought out to the extent that because the parapet was so low no slopes to drain could reasonably be incorporated had they been contemplated at all. The lack of slopes clearly contravenes the insulation manufacturer's requirement and the Roofing Contractor's Association recommendations. In Mr. Johnston's opinion design intent is not provided when the only detail indicates only that there is a roofing membrane system that fails graphically or by notation to identify what the membrane consists of and how it is to be installed.

7) Mr. Wilson states, "design and construction of the main roof membrane was never a contributing factor in water infiltration." To this, Mr. Johnston responds that Mr. Kendall carefully inspected and found the parapet flashings deficient throughout the roof area and that they were indicative of flashings that were, in his opinion, inappropriate and deficient elements of practice that in other applications had led to failure. The re-roofing of the main roof was overdue considering the number of faults in detail, slope, standards compliance, specification, competent substrate and field quality that were determined on inspection. Deterioration of the built-up layers of felt and bitumen had already begun in 1994 adjacent to the more exposed areas at the parapet perimeter. As well, we know that the faulty roof hatch allowed for two significant floods and the back slope at the parapets contributed to the deterioration of the felts.

The following responses by Mr. Johnston deal with Mr. Wilson's comments with respect to the walls, and begin at page 7 of Exhibit 61.

8) Mr. Wilson says that the walls show minimal signs of deterioration. As pointed out by Mr. Johnston the bed joint bonds are already compromised as evidenced by water transition rivulets seen during his investigation, and the ties are either non-existent or have been in many cases compromised by rusting and material deterioration. The interior wythe of block is not capable of preventing water incursion or interior air exfiltration, or providing suitable lateral restraint in partnership with the brick.

9) Mr. Wilson says that he does not agree that the masonry design is deficient. Mr. Johnston notes that if, as Mr. Garbutt first maintained the wall was to be a solid masonry wall then the deficiency in design is in failing to ensure the structural steel is designed to the coursing of the brick in order to allow a continuous perimeter "solid masonry wall" at each floor, with appropriate flashings and soft joints to accommodate differential movements and deflections. Nowhere is this design clarified or is it sufficiently detailed to be perfectly obvious as to what specific masonry design was intended. No provisions were made to ensure lateral support nor to protect the interior finish of a solid masonry wall construction by parging, waterproof coating or other means. The specification does not support a solid masonry two wythe type construction and the details on the drawings are deficient. If Mr. Garbutt intended, as Mr. Johnston was advised 13 years after the fact, that he deferred to Mr. Roberts' air space design and a two wythe brick veneer type of wall with four inch concrete block back-up, then most importantly his design is deficient in that the same under-developed details failed to ensure that the back-up wall was competent enough to take the impact of the measures identified in the brick veneer specification (i.e., those measures that failed to be incorporated in the construction, such as weep holes, flashings, satisfactory anchorage, proper and consistent ties.) The specification is too vague and meager in detail and imprecise as to measures required to suit the design methodology.

10) Mr. Wilson states that deficiencies in the upper portion of the south facing wall were caused by workmanship factors and not by design factors or lack of proper supervision. Mr. Johnston responds that nowhere in the record are there construction progress review reports submitted by Mr. Garbutt. As a result Mr. Wilson cannot justify his position. From the record that we have, Mr. Garbutt missed almost half of construction progress

meetings and provided no progress documentation. Mr. Garbutt has alluded to being "there once or twice a week" but as Mr. Johnston understands it this was for the mechanical system installation and not for the purpose of inspecting in detail the wall, parapet or roof. Mr. Johnston points out that no record of the mock-up nor other on-site review of masonry construction has been provided by Mr. Garbutt.

From Mr. Johnston's investigation, the deficiencies are not solely in the south-facing wall, nor restricted to the upper floor. Rusting of masonry ties were found at Mr. Johnston's 1994 investigation at the second floor level and are currently found grievously under strength at the first floor level. Flashings are missing throughout the building top to bottom. Mr. Johnston testified that professionally Mr. Garbutt was obliged to review, report and attend for the total construction and not just the bottom half.

11) Mr. Wilson states that "the current 1997 Ontario Building Code allows controlled ingress of water…". In response, Mr. Johnston points out that the 1983 Ontario Building Code permitted no such latitudes. As well the 1983 Code provided that where there was likelihood of some penetration, drainage was to be provided to take water to the outside. Mr. Johnston notes that occurrences of rot are manifesting now at the fourth floor as a result of long-term water incursion and that serious deterioration of structural integrity and what Mr. Johnston refers to as blatant non-compliance with even the minimum Ontario Building Code structural competence requirements, are serious safety issues.

12) Mr. Wilson refers to a "modified design build construction contract". Mr. Johnston notes that this was not a design build construction contract wherein the contractor hires the consultant. Here, the consultant was contracted directly to the owner.

13) Mr. Wilson notes that "early problems with wick efflorescence and heightened water leakage were created by parapets which were too low and poorly executed counter flashings." Mr. Johnston agrees with this statement and notes that the parapet membrane breakdown concealed brick leakage or made up such a great proportion of the flow of water into the wall where there was a breakdown that the lesser but still present rain water incursion problems at all of the elevations noted was overshadowed. However, Mr. Johnston disagrees that the issue is solely workmanship and notes that proper detailing,

representative dimensioning, carefully reflected on drawings, finite material application instruction and clear "extense" and "joinery" would have clarified contract intent.

14) Mr. Wilson accuses Mr. Johnston of "trying to convince Drs. Lawless and Reed that their exterior brick wall was in imminent danger…". Mr. Johnston responds that he never used the term "imminent danger" and took care to identify "long term risks" that could be substantially reduced only with correction. He notes that the ties have already failed, are rusting prematurely and are deficient in their use and lack of consistent application. He notes as well that two of the four main walls of the building do not meet even the minimum Building Code Standards due to missing shelf angle plates.

15) Mr. Wilson in his report notes that "…on 30th of November, 1993 McAdam suggests that replacing the brick might not be such a bad idea after all…". Mr. Johnston in response notes that McAdam made this suggestion after he had continued his evaluation beyond the initial assessment phase that was represented by his October 11, 1993 comments.

16) Mr. Wilson notes that "simple improvements in the existing wall system" were never discussed nor seemingly "entertained". Mr. Johnston in response notes that others had suggested providing vent holes and/or vent holes and flashings as a means of correcting the wall system, and it was only after Mr. Johnston had done extensive destructive testing and veneer removal that he advised against only providing weep hole openings and flashings. He is of the firm opinion that the back up block wall is not competent to endure the exposure to induced air flows and water incursions that purposeful opening of the exterior brick veneer would bring.

17) Mr. Wilson in his report alludes to "the only water now entering the building was slowly dripping in at the window heads of the south windows on the top three floors". Mr. Johnston sees this as a gross generalization in the face of a single visit by Mr. Wilson in a light drizzle. For Mr. Wilson to suggest that rain is penetrating only through the brick above the windows and not through the brick on the balance of the wall is totally unfounded, unsupportable and contrary to the tests undertaken by Mr. Johnston. Once again Mr. Johnston points out that rainwater is coming through the bed joints of all the brick (not through mortar, nor through the brick itself). Some incursive water is diverted and collected by the window lintels and impervious framing, but 60% of the wall not

comprised of windows is absorbing the balance of rainwater penetration through brick elements not immediately above windows. Mr. Johnston sees this as a real concern.

18) Mr. Wilson in his report refers to “simple improvements to the existing wall system prepared by Tony Pascoal P.Eng”. Mr. Johnston in response notes that he has in fact very little problem with the explanation and rationale of what Mr. Pascoal originally suggested in his report of June 28, 1999, provided the following qualifications are acknowledged:

a) Pascoal recommends that flashings be introduced at all shelf angle locations, windows, doors and foundations. Mr. Johnston agrees that this was and still is a Code requirement but notes that installing these now without taking down the entire exterior brick veneer means sequentially removing part of the brick veneer to do the work and then re-installing or toothing the brick back into the wall. Johnston is of the view that re-inserting brick on such a large scale is not recommended by the Clay Brick Association of Canada specifications. It is simply not good practice as mortar bonds are impossible to achieve consistently. As well, because the structural lower steel spandral beam and back up interior wythe masonry is improperly prepared and not flashed at all nor in particular even masonry infilled at the 4” joist shoe opening, the entire spandral must be reconstructed to shed water that ties and mortar fins throughout the wall are conducting to that inner face. In any event, the Pascoal/Wilson recommendation would involve removing some 43% of the brick façade, which is time consuming, technically questionable and typically undesirable as it presumes an acceptable mortar match and overall blend in reconstruction. Even so, 57% of the still imperfect inner wythe would neither be accessible nor corrected. Johnston sees this solution as being impractical and hardly less expensive than full removal of the brick façade.

b) With respect to Pascoal’s recommendation that P.V.C. weep vents be installed at new flashings and below the shelf angles, Johnston points out that he would employ the P.V.C. vents only after the inner wythe was made into an effective air (not moisture) barrier, and after the surface was prepared to shed any incursive rain water onto flashings at each shelf plate.

c) Johnston agrees that new support angles on the east and west elevations, along with new soft joints horizontally at the shelf angle locations and new expansion joints need to be installed.

d) Mr. Pascoal and Mr. Roney recommend that new heli-fix stainless steel skewer type brick ties be installed where there is any potential concern of inadequate anchorage such as at the parapet locations. Mr. Johnston agrees that heli-fix anchors are excellent remedial type anchors and he supports their use in retro-fit or reclamation projects, but only if the changes that a rigid structural anchor such as this imparts to the wall and the building systems are verified by a structural engineer. He notes that even if partial masonry removal were a justifiable alternative, and he maintains that it is not, he presumes that normal adjustable rod anchors could be used where the brick is to be removed and replaced, and that heli-fix anchors would be used for the balance of the veneer masonry remaining in *situ*. This would amount to at least 57% of 2,400 minimum or in other words 1,370 anchors drilled and attached through the mortar joints. This would leave a patchy impression and noticeable pattern in the existing mortar joints, and together with the newly fashioned mortar joints would leave the building with an aesthetic problem.

Simply stated, Johnston can to some degree support the reclamation opinions that Mr. Pascoal endorses, provided the inner wythe is accessible to be reinstated to the competence required to sustain the outer veneer masonry when weep/vent holes are introduced carefully into the wall. In Johnston's view this is improbable and unless some other suggestion arises the brick exterior wall will have to be removed as he has consistently maintained.

19) Mr. Wilson quotes Mr. Pascoal as saying, "the walls themselves are not showing signs of distress such as visible bricks falling, brick cracking and mortar deterioration." Mr. Johnston in response notes that he has addressed the open porosity and capillary structure of the brick that precludes any spalling typical of denser and harder burned clay bricks.

Johnston notes however that there are more visible cracks that have been viewed during the current investigation. Particularly there is a large horizontal crack on the north elevation.

20) Mr. Wilson in his report notes that Mr. Pascoal acknowledges that the wall system at 275 Queen Street does not work and that concerns about the long-term durability of the wall system are legitimate. Mr. Johnston agrees.

21) The "simple improvements" suggested by Mr. Pascoal are as follows:

1) flashings to be introduced at all shelf angle locations, windows, doors and foundations;
2) P.V.C. weep vents at new flashings… below shelf angles;
3) new support angles along the east and west elevations;
4) new soft joints horizontally at each shelf angle location;
5) new expansion joints;
6) new heli-fix stainless steel skewer type brick ties to be installed;
7) detergent cleaning and soft bristle scrubbing of any efflorescence staining on the wall;

Mr. Wilson comments that all items except 6) and 7) were included in the original construction specifications or drawings. Mr. Johnston comments that he does not disagree that had items 1) through 5) been built into the wall, then leakage to the interior would in all likelihood have been less of a problem. However, Mr. Johnston points out that these items almost exclusively appeared in the specifications and <u>not</u> the drawings. That is,

a) no flashings were shown on any drawings save what appears to be a metal counter flash and coping at the parapet (but it is not named nor notated on the parapet detail);
b) no weep holes are noted anywhere on the drawings;
c) support angles appear to be shown at very small scale on the building sections without notation or definition, but are completely missing from the large scale details that could have defined them;
d) no soft joints are identified on any drawings;

e) no expansion joints are indicated on any drawing.

22) Mr. Wilson comments that "we entirely agree with Tony Pascoal's comments of simple improvements to the existing wall system." Mr. Johnston in response notes that without correction to the inner wall substrate Mr. Wilson's solutions alone would not remove the ultimate health and discomfort hazard, let alone prevent new problems associated with moist interior air ex-filtration and/or condensation within the wall that Mr. Johnston believes the Pascoal "simple improvements" would actually introduce. As he has taken pains to explain the inner wythe has not been prepared nor constructed to act as the primary air and weather seal of the building, and without correction the faults in its construction left unremediated, should vents be introduced into the exterior brick, can cause further problems within the exterior wall construction.

Filed as Exhibit 62 was Mr. Johnston's response to the investigation report of Mr. Quirouette dated April 23, 2002.

To begin with, Mr. Johnston noted that in preparing his report Mr. Quirouette did not interview any of the occupants of the subject building, nor did he inspect the inside of the building nor did he look at the roof.

Mr. Johnston notes that Mr. Quirouette in his report recommends certain ASTM testing for the windows but Mr. Johnston points out that the kind of test being recommended by Mr. Quirouette is put in place to determine if a window is leaking in its <u>assembly</u>. The ASTM testing is not meant for testing walls. In any event, Mr. Johnston points out that for his purposes there was no need to quantify the amount of water that was going into the building. As he pointed out, any amount of water leakage is bad.

Mr. Johnston in his evidence noted that Mr. Quirouette who is the defendant Emmons & Mitchell's witness said for the most part the problems encountered in the building are Mr. Garbutt's fault.

Mr. Quirouette in his report, article 4.4 notes that the wall is not a cavity wall and not a mass wall. Rather, Quirouette characterizes it as a masonry wall with more attributes of a mass wall than a cavity wall. Mr. Johnston points out that this is an incorrect assumption in that there is no documentation in the record that supports the mass wall or solid masonry with a mortar filled collar joint. Despite the specification to the contrary Mr. Quirouette still, according to Mr. Johnston confusingly considered that the designer intended an "exterior wall of the mass wall type".

With respect to the roof leaks commented upon by Mr. Quirouette, Mr. Johnston pointed out that the roof drains were not plugged and there was no drainage prevention. Mr. Johnston pointed out that the staining shown in photo's 9, 10, 11 and 12 of Exhibit 62 depicting the north side of the stair tower is indicative of failure of the roof membrane. He pointed out as well that the 1994 north west corner repair cured this problem.

Notwithstanding Mr. Quirouette's opinion as expressed in his report that the roof had "another 15 years of service with minimal maintenance, Mr. Johnston stood by his recommendation to the plaintiffs that the original roof assembly was unequivocally deficient and required replacement because of the following:

1) no slopes to drain, contrary to the specifications;
2) dangerous back slopes that failed to take into account the camber of the roof structure and allowed standing water within the roof assembly;
3) a health risk that standing water presented not only in the main roof disassembly strip but for the entire north west area that was totally undrained;
4) deficient installation practices demonstrated contrary to the insulation manufacturer and the CRCA (slopes, adhered insulation, edge flashings);
5) deficient parapet heights and flashings as identified by Mr. Kendall;
6) incomplete and half-executed membrane flashings under unfixed and unsealed parapet substrates, no leading drip edge seal, insufficient brick coverage;

7) parapets executed with flat surfaces, not drained to the roof collection area;

8) felt/bitumen membranes that require more slope, not less, to dry out and maintain their tensile strength properties;

9) complete membrane flashing failures at roof scuttle and north west stair/lobby tower areas;

10) totally inadequate detail and direction in contractual documentation;

11) clear and unclarified contradiction between drawings and specifications as to type of roof let alone standard of insulation;

12) unspecified drains as to type and drainage system design;

13) seriously deficient substrate execution and membrane flashing adhesion at diverse locations;

14) failure to provide let alone clarify the absence of CRCA standards reflected in drawings and details;

15) lack of independent roofing reports;

16) lack of base contract inspection reports by the consultant;

17) lack of primer application for membranes applied directly to concrete deck;

18) insufficient flood rim heights in general and most important at roof fitments, waterproofing cones and curbs;

19) a proven and documented failure of membrane flashings at locations remote from each other.

With respect to Mr. Quirouette's opinion that the single roof leak in the north west corner was not adequate justification to remove and replace the original roof system and that the new roof is "added value", Mr. Johnston makes the point that the new roof, contrary to Mr. Quirouette's assessment was simply an attainment of the originally specified roof that had not been provided by the general contractor.

Mr. Johnston notes that in his report Mr. Quirouette admits that in investigation procedures with Mr. Roney bed joints and head joints were found cracked and were visible at the top of the building when viewed from ground level. Mr. Johnston confirmed that the cracks are quite identifiable. He noted in particular that one crack on the north side above the upper windows has gone from 10 feet to a 20 foot open crack.

Mr. Quirouette in his article 5.10 notes that there are no expansion/control joints in the building and that that factor may have caused head and bed joints to crack. Mr. Johnston points out that the bed joint failures are universal and represent the major and continuing source of rainwater incursion. They were caused primarily by flexure in the brick veneer diaphragm unrestrained by appropriate metal ties properly and consistently installed; properly reinforced and well anchored back up structure (4” concrete block inner wythe); braced spandral beams designed to resist rotation at the bottom flange; measures taken at the east and west walls to allow for differential structure and thermally induced movements between an excessively high veneer brick wall sitting on the foundation and the three back up inner walls bearing separately on the steel structure at each floor and roof. These measures do not even touch on the subsequent aggravation ensuing from the lack of essential system elements like flashings and weep/vent holes.

Cross-examination by Mr. Garbutt

Mr. Johnston confirmed for Mr. Garbutt that with reference to page 20 of Exhibit #56, the extensive marking of the concrete block by mortar spreading from all the brick joints indicated that in at least 50% of the sampled area (approximately four square feet) mortar joints been squeezed full width or the aggregation of droppings had blocked the air space locally. Mr. Johnston explained that it was difficult to determine whether the mortar from the bricks actually adhered to the block.

Mr. Johnston confirmed that the bricks when exposed were dry, at least on the inside.

He confirmed as well that the masons from his observation of the construction video seemed experienced.

He confirmed that it was difficult to avoid bridging over a ¾" air space but it can be done.

Mr. Johnston confirmed that it appeared from the video that the masons were laying the brick overhand and that it is more difficult with this procedure to maintain the cavity. He agreed that this style of laying the brick is more indicative of a solid wall than a cavity wall but pointed out that when a solid wall is built the courses of both the veneer and the inner wall go up together. He confirmed as well that if one is constructing a solid collar joint, one would butter the inside face of the brick, unless the entire cavity was to be grouted afterwards.

Mr. Johnston did not note any major bulging or out of plumb of the brick wall.

From an architectural point of view, there are two types of wall; solid or cavity, with veneer being a type of cavity. On the other hand, structurally there are three types of wall; cavity, veneer and solid.

At Tab F of Exhibit #56 is Division 4 – Masonry. Mr. Johnston pointed out that conceivably there is a conflict between item 2.1.1 which directs that a continuous 19 millimetre air space be maintained behind the brick throughout, and section 4.1.4 which directs that the mason lay all masonry in full mortar beds and fill vertical joints solid with mortar. Arguably the vertical joints refer to the head joints rather than the collar joint. In any event, Mr. Johnston conceded that possibly in these circumstances the mason would look to the consultant for clarification.

Mr. Johnston conceded that there was no field evidence that the collar joint here was intended to be filled, but the air space did in fact go down to nil at one point on the south side in order to correct initial mis-alignment.

Exhibit #47 is the Masonry Code for the Canadian Standards Association (M78). At page 52, article 4.9.1.2(f) provides that bed joints shall be leveled back from the space to be grouted so as to avoid extrusion of the mortar into the grout space. Mr. Garbutt asked Mr. Johnston whether this would be equally applicable for a mortar filled solid wall. Mr. Johnston replied in the negative, saying that typically one would apply mortar to the collar joint as the wall went up.

At page 24 of M78, article 3.13.1 provides under the heading 'Drainage of Walls' that weep holes at least 10 millimetres in diameter shall be provided immediately above the base flashing in veneered walls having bearing support, and in cavity walls at horizontal spacing not exceeding 600 millimetres on centre. Mr. Johnston agreed that there was no mention of solid walls in these sections, but pointed out that solid walls were not meant for drainage.

Section 4.7.9, at page 46 of M78, provides that tooth joints shall not be used in sheer walls. Mr. Johnston conceded that a veneer wall does not qualify as a sheer wall, therefore using toothing as he observed is not contrary to the Code, but is still contrary to the Canadian Construction Association recommendations.

Having regard to section 6.1.1 of M78, at page 65, Mr. Johnston agreed that masonry veneer shall not be considered to be part of a wall when computing its strength or thickness, and from a structural point of view one only considers the thickness of the back-up wall which is supported floor to floor.

At page 13 of Exhibit #47, a panel wall is defined as a non-load bearing, exterior masonry wall having bearing support at each story. Mr. Johnston explained that it was difficult to say whether a particular panel wall includes the brick wall or is constituted only by the block wall. In his view, either or both could constitute a panel wall.

Mr. Garbutt then went to section 5.3.5.1, at page 62 of Exhibit #47, under the heading ‘Panel Walls’. Mr. Johnston expressed the view that our wall is a veneer wall and should not be considered as a panel wall.

For the purpose of his examination, Mr. Garbutt prepared what was entered as Exhibit #63, ‘Moments of Inertia for a Veneer Wall, a Cavity Wall and a Solid Wall’, which have relative values of 1, 2 and 10.5 respectively, with veneer being the weakest wall. In his calculations, Mr. Garbutt had the thickness of both the brick wall and the concrete block wall at four inches. Mr. Johnston pointed out that he has never used a four inch block for an inner wall, and is of the view that the minimum thickness of the block for the back-up wall should be six inches.

Mr. Johnston expressed the view that the minimum width of a filled collar joint to classify the wall as solid would be 10 to 14 millimetres. It could be smaller or greater than 19 millimetres, but not less than 10 to 14 millimetres. As a matter of interest, Mr. Johnston pointed out that the 19 millimetre specification comes from the thickness of the mason’s thumb which has to go between the brick wall and the block wall when the brick is being laid.

Mr. Johnston agreed with Mr. Garbutt that the bridging fins would provide support in compression but it seemed obvious to him that they were not intended to do so. He also conceded that the amount of bridging exceeded the spacing requirement for ties in the Code. Again, Mr. Johnston pointed out that if 50% of the mortar joints had bridged to the block wall, this could be satisfactory under the Code for compression but <u>not</u> for suction. He pointed out as well that this is not a faced wall under M78. A faced wall means a wall in which the masonry facing and backing are so bonded as to exert common action under load. That is not the case here.

Mr. Garbutt suggested to Mr. Johnston that in 16 years there has been no sign of buckling. Mr. Johnston disagreed. He pointed out there is open cracking at the corners and debonding of the brick and mortar as previously described.

Mr. Garbutt referred Mr. Johnston to Tab F of Folio 3, Exhibit 56, the Masonry specifications, section 2.2.1 which requires the mason to reinforce masonry walls with continuous masonry reinforcement in every second block course. Mr. Johnston conceded that from his limited review this appeared to have been done.

With respect to the windows, Mr. Johnston confirmed that he did not remove any window to establish installation details, nor did he review whether the installation was satisfactory to withstand lateral stress. From what he saw of the caulking, it was reasonably flexible and satisfactory. Mr. Johnston took the opportunity to reinforce his position that the leakage being experienced in this building is wall related and not window related.

Referring to Tab I, Folio 3, Mr. Johnston explained that making the joint between the block lintel and brick lintel watertight would solve nothing.

In response to a question from Mr. Garbutt as to why the bricks are not saturated, Mr. Johnston explained that they are drawing from the inside out. The particular brick used here is very forgiving and in any event the whole wall is acting like a water sink.

While conceding that flashing and weeping the lintels would cure some of the problems, Mr. Johnston pointed out that this is a patchwork remedy and in any event would involve taking out a great amount of brick, which would solve the water leakage at the window but not throughout the rest of the building. As well, with the weepers there will be an inflow of cold air to the block in the wintertime and it will blow right through the block wall in the spaces previously discussed.

Likewise, to Mr. Garbutt's suggestion to drill holes above the angles and inject foam, Mr. Johnston explained this would only address the window problems and not the rest of the building. Again, he pointed out that the whole wall leaks. Mr. Johnston pointed out that ties were rusting at the first floor level and those rusted ties were not over or under a window.

At Tab 1, Exhibit #1 is located Division 5 relating to structural steel. Section 1.3 requires the contractor to submit shop drawings which clearly indicate framing plans, bearing and anchorage details, etc., and section 1.10 requires the contractor to obtain written permission of the engineer prior to field cutting or altering of structural members.

At Tab D, Folio 3, the third drawing in is for roof framing, where a TJ (tie joist) is referred to. Mr. Johnston pointed out that a tie joist can be seen in the photograph at page 10, upper left hand corner in Folio 3. He explained that a tie joist is on the column line with both flanges attached, contributing to the stiffness of the frame.

At Tab 2, Exhibit #1 are the General Conditions (GC). GC 34 covers shop drawings; GC 34.3 requires the contractor to review all shop drawings prior to submission to the consultant; GC 34.5 provides that the consultant's review will be for conformity to the design, concept and for general arrangement only and such review shall not relieve the contractor of responsibility for errors or omissions in the shop drawings, or of responsibility for meeting all requirements of the contract documents unless a deviation on the shop drawings has been approved in writing by the consultant.

Division 7 of Tab 1, Exhibit #1 covers the roof. Section 2 provides for slope structures to drains at least 5m/100mm gradient. Mr. Johnston conceded that this may be an awkward way of specifying a 2% slope but it serves the purpose.

Mr. Johnston confirmed his earlier evidence that in 1994 he felt that the life expectancy of the roof was five years at best and indeed based on his investigation of the perimeter flashing he thought that the roof should be replaced imminently. The 1998 replacement was done in the absence of any indication one way of the other from the owners as to whether they would go ahead with adding more floors. The fact is however that the roof as presently constructed would be very receptive to additions. All that was intended to be done by the 1998 replacement was to give the owners the roof that they should have gotten in the first place.

Mr. Johnston confirmed that the Canadian Roofing Contractors Association issues recognized standards and it is expected that any member of the CRCA would abide by those standards.

Mr. Johnston conceded that the drawings with respect to the parapet sufficiently showed intent in terms of the parapet being solid and being of the appropriate height, but execution was not described in the drawings.

Mr. Johnston confirmed that the replacement of the roof was done to original specifications. The drains were not changed. He was not aware of a municipal by-law that required the installation of weir drains.

With respect to the flashings and weep holes, Mr. Johnston confirmed that the two go hand in hand and if there are no weep holes there are likely no flashings. Mr. Garbutt asked whether an architect who reviewed the contract documents and saw no weep holes would conclude that the wall was not constructed as a cavity or veneer wall. Mr. Johnston responded that the specifications call for flashings and weepers and therefore the architect would be looking for a veneer wall. An architect, looking at the wall and seeing no weepers, would advise the owner of his concern.

It was Mr. Johnston's evidence that in the proposed reconstruction he was not redesigning the wall system, he was simply designing the wall to make it work. The load will continue to be on the block wall only under Plan A, although there was an alternate Plan B which did go to a shared load. In Plan A of course, the back wall must be reinforced as previously discussed.

Mr. Johnston explained that in determining that this was a veneer wall, he did not start with that as a given fact and work backwards. Rather, as he put, he “followed the indicators to the route home.” The veneer wall was specified in the contract documents and he in fact found a veneer wall on site. He explained as well that if he had found in his investigation that it was the windows that were causing the trouble, it would be very easy to tell the owners that and suggest a manner in which the windows could be fixed.

In terms of reinforcing the back-up wall, Mr. Johnston has not yet determined the best manner of doing that. He agreed with Mr. Garbutt that reinforcement could consist of a sealed stud placed at the column, but that reinforcement does not solve the problems of the air holes in the back-up wall. Also, the placing of the stud presents a problem because it must be done from the inside of the building.

In terms of the possibility of grouting the air space, Mr. Johnston explained that this would not cure the bed joint bonding and there would still be water passing across the mortar fins. He does not recommend grouting. He explained that the whole wall is breaking. The block wall is probably absorbing half of the water that gets in through the brick veneer. The only reason the water is breaking through at the windows is because of the lintels.

Mr. Johnston confirmed once again that the concrete block did not enter into the column cavity. He did three openings and a parapet opening, and saw this to be the case and also the video clearly showed that the blocks did not go into the column, and that there was no anchorage.

Cross-examination by Mr. Nelson

Mr. Johnston first saw the building with Dr. Reed in August of 1994. He met with Dr. Reed for approximately two and a half hours and toured the building inside and out including the basement, the roof and each of the floors. He viewed the windows from the inside on the fourth floor. His best recollection is that during the meeting he looked at the three small drawings that have been put in evidence, and one or more of the other drawings.

At Exhibit #2, Tab 95 is Mr. Johnston's first report to Dr. Reed. He confirmed to Mr. Nelson that this report was based on his meeting with Dr. Reed and was prepared prior to obtaining a complete set of the materials. The report is dated August 29, 1994.

At Folio 1, Exhibit 54, Tab B is the extensive list of drawings and documents received by Mr. Johnston from Dr. Reed on September 23, 1994. Filed as Exhibit #64 was the covering letter from Dr. Reed. The construction video came some time later in November.

By way of his report dated August 29, 1994 Mr. Johnston provided Dr. Reed with his initial opinion based on what Dr. Reed remembered of the construction process and observations made at the time of the meeting.

Mr. Johnston confirmed for Mr. Nelson that at the time of his first observations it was apparent to him that it was an irma roof and that there were no flashings and weep holes visible in the brick veneer wall or set out in the drawings.

Mr. Johnston recalls seeing a roll of drawings. He recalls rolling the drawings out. There was not a lot of time available but he did see enough to get a general idea of the construction. He believes he probably would have seen a section drawing but cannot say that Drawing #4, Exhibit #11 was the section drawing that he saw.

At Tab 100, Exhibit #2 is Mr. Johnston's letter to Dr. Reed dated October 3, 1994. By this time Mr. Johnston had the drawings and specifications. Based on the drawings and the height of the building, Mr. Johnston's first supposition was that this was a veneer wall.

In this report, Mr. Johnston made the following points:

1) the specification called for a built-up roofing membrane of felt and tar, not the more prevalent rubber type membrane;

2) a revised parapet detail was added to Drawing 4, Sections, without scale or dimensions or details of incorporation into the contract;

3) the photographs indicate shelf plates at most perimeter beams as is called for on Drawing S1;

4) the presence of an air space (as is specified) and drainage/weep or pressure equalization holes (as is also specified) in masonry is very relevant to the nature and detail development of the brick and block exterior walls. The photographs seem to suggest an air space was maintained but Mr. Johnston points out that Dr. Reed had mentioned to him an intent to fill the joint between the brick and block with mortar. Mr. Johnston explained that this was information that Dr. Reed obtained from Mr. Garbutt. Mr. Johnston pointed out that the scale on Drawing #4, Exhibit #11 is too small to determine if a 19 millimetre opening is shown on it. Referring to the same drawing, and the revised parapet detail added to that drawing, Mr. Johnston confirmed that the parapet drawing does not show a cavity;

5) flashing of substantial quality (two ounce copper fibrene) is specified but none was evidenced in visual inspection or photographs. No flashing was detailed in any drawing. The nature and extent of shelf plate was not drawn nor flashing interface noted. Mr. Johnston confirmed in his evidence that there was no reference to weepers in the drawings either. He commented in passing that for a contractor to cut out a shelf plate without the engineer's instructions would be extraordinary;

6) extent of mortar was not detailed in any drawing, and despite the air space as specified, Mr. Garbutt's letter makes reference to "solid masonry" type construction of mortar filled collar joints;

7) the specification refers specifically to wall drainage (Division 4, article 8) and photographs provide no evidence of weeping or sealed cap joint to underside of steel support angles;

8) no slopes (nor drains) are noted on the steel drawings, nor is thickening of concrete detailed to drain roof accumulated water effectively. Mr. Johnston explained that two roof drains did eventually appear as an item on the pricing of the contract;

9) despite the drawings calling for an irma roofing application, the specification refers to a standard built-up exposed membrane roof with mopped in insulation, not as is evidenced on site.

Following this report it was decided to do an investigation of the building in October of 1994.

The testing was done on October 5, 1994 and at Tab 101 of Exhibit #2 is Mr. Johnston's report with respect to that testing.

In this report, Mr. Johnston begins by noting that during visual inspection of the interior of the wall, the majority of indications of water penetration was observed at the fourth floor windows at the head glazing stops and drywall soffits of window recesses. He notes that based on discussions with Dr. Reed, flooding patterns and damages at the interior appeared to originate at the exterior wall and migrate across interior floor finishes to the stair well. He notes as well that Dr. Reed told him that during severe storms of long duration, water penetration at lower floor windows occurred, but the majority of noticeable penetration at window heads occurred at the fourth floor and was known to accumulate in buckets.

In his introduction to the report, Mr. Johnston also says it is noteworthy to mention that an apparent blockage of the roof drains caused major flooding of the interior on one occasion, manifesting in major leakage through façade windows, stair lobby, window recesses and ultimately through the east stair well roof hatch where major flooding occurred. Mr. Johnston in his response to Mr. Nelson fairly well repeated the explanation that he gives in Exhibit 62 in response to Mr. Quirouette's reference to the plugging of roof drains. He explained that while drain blockage was initially considered, it was later confirmed not to be the problem.

At Tab 113 of Exhibit #2 is the summary of costs prepared by Dr. Reed. The total cost to repair the roof membrane in 1994 including the hatch was approximately $21,000, for which price the contractor also removed bricks from four or five locations on the wall.

Mr. Johnston confirmed for Mr. Nelson that the brick chose was known as 'John Price' and as far as he knew it was no longer available, notwithstanding Mr. Nelson's advice to him that the people doing the inspection for the defendants were able to replace the old brick with new. Mr. Johnston pointed out as a matter of interest that he had measured the new brick and it was smaller than the old. The brickyard in the Don Valley that produced this brick has not been in production for some 10 years now.

At Folio 3, Exhibit 56, page 14 is a photograph of a portion of the south wall, upper floors, where the spray test was carried out on October 5, 1994. Mr. Johnston confirmed that eight windows extend across the south façade in pairs. The distance from the roof to the top of the window is 48" and the windows themselves are five feet tall, so that it is nine or 10 feet from the top of the roof to the bottom of the windows.

On a drawing that will eventually be made an exhibit, Mr. Johnston showed the placement of the hose over the side of the roof, in line with the east edge of the second window in. He confirmed that he observed no cracks above the window but cracks further to the east corner.

He confirmed that a shelf plate was located about 20" above the top of the windows at the bottom of the spandral beam, and that there was a shelf plate about four feet below the bottom of the windows. By October 5, 1994 he had seen drawings showing the shelf plate above the windows and confirmed its existence on further testing on December 8, 1994.

Mr. Johnston confirmed that the stone surround head seemed to be acceptable and that he saw no cracked bricks. He saw no cracks in mortar joints in the test area. The caulking of the stone surround seemed acceptable.

The hose had been brought up through the back stairwell. Mr. Johnston conceded that he did not know the water pressure at 275 Queen Street, nor the water pressure in the one inch hose. He confirmed that the rate of flow was not calibrated in any way. For the test, he used a typical plastic nozzle from a garden hose.

The purpose of the spray test was to try to wet the surface of the wall above the second window in, in a moderate breeze to see if there was any incursion of water. The testing area was between the top of the second window in, and the roof. The hose hung down at a slight angle, and at a slight angle away from the brick.

It was Mr. Johnston's response to a question put to him by Mr. Nelson that the rate of wetting on the brick wall was substantially less than last Friday's rain storm.

Mr. Johnston was not aware of any standard that said that a water test should be done from the bottom up. He was referred to the 1105 Protocol by Mr. Nelson and quickly pointed out that that was a window test and not a wall test.

At Folio 3, Exhibit #56, Tab H is Mr. Johnston's report of the investigation and testing on October 5, 1994. At page 7 it is noted that the nozzle was aimed downwards and slightly toward the wall, for a slight angular incidence. This is in slight conflict with what he earlier told Mr. Nelson as to the angle of the hose, but it was Mr. Johnston's evidence here that he now seems to recall that it was the spray, not the hose, that was angled toward the wall. Nothing turns on this.

Mr. Johnston is aware that Mr. Quirouette has stated that the amount of water put on the wall was substantially more than would go on the wall during a rainstorm and Mr. Johnston strongly disagrees with that statement.

Mr. Nelson then turned to Tab I of Folio 3, Exhibit 56, Mr. Johnston's drawing of the cross-section of the wall at the fourth floor and roof that he prepared some time shortly after the water test on October 5, 1994.

The shelf plate that he confirmed in December of 1994 was above the window, attached to the bottom of the beam. It is shown as a dotted line on the bottom of the spandral beam.

On a copy of Tab I that went in as Exhibit #68, Mr. Johnston indicated where he first saw the water coming in during the testing, through the brick wall just above the top of the spandral beam. This spot matches the wet area at the mortar joint shown in the photograph at page 16 of Folio 3, where the water is seen coming down the inside of the brick wall. At this point, the nozzle was on a fine spray. The nozzle had been located approximately 10” from the wall and 10” above the mortar joint where the water was first seen to be penetrating.

Mr. Johnston is unable to say whether the water first viewed from the inside entering through the brick veneer at a point just above the top of the spandral beam eventually dripped down and somehow got passed the shelf plate attached to the bottom of the spandral beam. This he was simply unable to see in that he could not look from inside the building through the spandral beam. It is to be noted however that the shelf plate is spot welded to the bottom of the beam. That is, it is not a continuous weld. Mr. Johnston allowed for the possibility that perhaps water entered through the brick mortar joints at a point below the shelf plate.

At page 5 of Folio 3 it is noted that over the window opening would appear to be back to back steel angle lintels, supporting block on the inside and brick at the outside wythe and that the vertical legs were tight to each other and both aligned to fit into the air space visible at either side. The bearing elevation of each angle differed from interior to exterior wythes such that about one and a half inches of the brick lintel vertical leg was exposed below the underside of the block lintel horizontal leg. Mating of the back to back legs was sufficiently tight affording little space for business card stock to penetrate. Mr. Johnston in his evidence confirmed that all this was true. Mr. Johnston was asked whether it was possible for water to have penetrated the brick veneer by going through the top of the header and the bottom of the lintel, as drawn on Tab I. Mr. Johnston admitted to the possibility but he concluded from the rivulets that were coming down the inside of the brick wall that the water was coming through the brick.

Mr. Nelson presented to Mr. Johnson a note from Rigney Building Supplies indicating that the brick JP Niagara was available for delivery on four weeks' notice. Mr. Johnson maintained his position that he checked with the Toronto manufacturer in 1994 and the subject brick which is actually called Red Smooth JP 20 was not available.

At page 24 of Folio 3 Mr. Johnson states "The wall construction fails to keep precipitation out of the building interior and this result has not been caused by caulking failure." He stands by that evidence.

Mr. Johnson confirmed that as a result of the repair work in October of 1994, he sent to the plaintiffs a letter dated October 12, 1994 (Exhibit 2, Tab 103) enclosing his report of October 5, 1994 (Tab 101).

At page 9 of Tab 103 Mr. Johnson expressed to the owners concern for the integrity of the ties and that they will have to be revealed and reviewed to determine the essential competency of the wall. He goes on to say,

> If the masonry back up wythe and the brick lateral and supportive structure can be confirmed and endorsed by a professional engineer, then the operational and service competency can be addressed. On the other hand should it fail to pass muster we suggest the repair procedures may cost as much as pulling down the veneer and rebuilding.

Mr. Johnson explained that the engineer that he was referring to in that letter would be a structural engineer and he expected that ultimately a structural engineer would come on site and check the walls.

Mr. Johnson was aware that two structural engineers came on the site in 1999 on behalf of the defendant company. In fact Mr. Johnson monitored the test performed by the structural engineers Mr. Roney and Mr. Pascoal.

Mr. Johnson confirmed that he did not invite the defendants to review the roof replacement that was conducted in 1998.

In passing, Mr. Johnson noted that Mr. Roney's report of 1999 contained proposals similar to the recommendations made by Mr. Johnson.

Mr. Johnson, in his evidence, confirmed that in October 1994 he brought Mr. Kendall in to provide advice with respect to the roof. He readily conceded that he would defer to Mr. Kendall in roof matters. At Tab 102, Exhibit 2, is Mr. Kendall's report dated October 11, 1994 where Mr. Kendall identifies the roof as an inverted assembly employing two layers of three inch roof mate, built up felt water proofing membrane on concrete slab. The assembly also included a polyolifin filter blanket and approximately 2 ½ inches of crushed limestone ballast. Mr. Kendall notes that the specification was confusing in so much as it called for a conventional assembly as opposed to an irma style roof and he notes that this was clearly changed during construction. Mr. Johnson notes that while the specifications called for a conventional roof, the drawings in fact called for an irma roof.

At Tab 103, Exhibit 2 is Mr. Johnson's letter to the plaintiffs dated October 12, 1994 where he notes the following:

> The specification calls for an exposed membrane (standard BUR) roof while drawings indicate an irma (inverted roof membrane application as known in the U.S.A.) or protected membrane roof (PMR). No documentation resolving this conflict appears in construction progress records as provided.

Mr. Johnson explained that he and Mr. Kendall were of one mind with respect to the specifications calling for an exposed membrane and the drawings showing an irma roof.

In Exhibit 55, Folio 2 at page 4 thereof, Mr. Johnson again notes the discrepancy between the specifications and the drawings and notes as well that Article GC.1, sentence 1.6 of the General Conditions of the contract stipulate that specifications shall govern over drawings. Mr. Johnson noted that notwithstanding this conflict between the specifications and the drawings, the plaintiffs were not dissatisfied with the irma concept as dictated in the drawings.

There then ensued a discussion between Mr. Johnson and Mr. Nelson as to the meaning of the further provision of GC 1.6 which provides that, "Notwithstanding the foregoing, documents of later date shall always govern." Mr. Nelson tried to make the point that drawings of a later date would govern over earlier specifications according to this provision. However, as I read the provision it meant that later specifications would govern over earlier specifications and later drawings would govern over earlier drawings. Mr. Johnson conceded that a later drawing could override an earlier specification if properly processed.

Mr. Johnson confirmed for Mr. Nelson that while the drawings show no expansion joints, there is in fact one in the south wall. He pointed out as well that the specifications do call for expansion joints but do not specify where or how. Mr. Johnson explained that expansion joints are a matter of design.

Mr. Johnson confirmed that the drawings do not provide for flashing and weepers. He confirmed that the three small detailed drawings of June 27 do show a cavity between the brick wall and the block wall but that the section drawing #4 (Exhibit 11) dated July 30 does not show a cavity.

Mr. Johnson explained that the contract provides three methods of change. Firstly by site instruction. Secondly by notice of change and thirdly by a change order. The site instruction or notice of change can be issued by the consultant while the change order is used where there are cost implications and it must be signed by the owner. If one of those three things does not happen, then there is no agreement between the parties.

Exhibit 61 is Mr. Johnston's response to Mr. Wilson's report. At page 5 thereof Mr. Johnston notes that the only roof drawings provided to him from the record include one of only three details of a large enough scale that should have provided direction on how parapets and roof to wall transitions were to be made. Mr. Johnston notes that the detail is under-notated, unclear, shows no slopes at the roof and is poorly thought out, to the extent that because the parapet was so low, no slopes to drain could reasonably be incorporated had they been contemplated at all. Mr. Johnston also notes at page 5 that what is even more surprising is that Mr. Wilson, a practicing architect, could possibly confuse an irma roof with a standard exposed membrane roof that was specified by Mr. Garbutt. No clarification is provided and not one drawing shows drains, slopes, ballast nature or weight. Mr. Nelson then directed Mr. Johnston to Drawing No. 4 which is Exhibit 11, which is a blow-up of the parapet. Mr. Johnston confirmed that the blow-up is not very detailed and that the parapet is open to the extent that the wood is not attached to the wall and it's open to the air at the leading edge of the flashing.

Mr. Nelson referred Mr. Johnston to his Folio 1, Exhibit 54, and particularly page 5 thereof, where Mr. Johnston notes that "in our assessment solid masonry construction would have had serious problems as well, if it had been installed as Garbutt suggested in his letters was originally intended; for as we have identified in Folio 3, very little detail direction was provided in the wall sections and details of the contract drawings to ensure proper execution of the exterior brick wall by the contractor." Mr. Nelson suggested to Mr. Johnston that in that sentence Mr. Johnston was referring to Drawing No. 4 in Exhibit 11. Mr. Johnston corrected him and testified that he was referring to all drawings including the three small drawings. He reiterated that there is nothing later than Section 4 of Exhibit 11 (detail B) dated July 30, 1985 that impacts in any way on the wall detail shown. It has never changed.

Mr. Nelson suggested that the drawing detail was more consistent with a solid wall than a veneer. Mr. Johnston responded that it was impossible to tell because the scale was too small and that one must take direction from the specifications. The three drawings graphically depict a cavity but no drawings say that it should be 19 millimetres in width. It is the specs that say that. Mr. Johnston noted that Exhibit 9, the small drawing clearly shows an air space between the brick and the block which he circled in red. Mr. Johnston confirmed that the blow-up of the parapet does not show a cavity and confirmed as well that it shows no studs or dry wall either.

At page 6 of Folio 1, Exhibit 54, Mr. Johnston notes that "what was provided on the building was unmistakably a 'veneer wall' (see Folio 3), and notwithstanding the owners' particular choice of brick it was the responsibility of the designer to ensure that the design could function properly with that material." Mr. Johnston confirmed that was his evidence.

Mr. Nelson referred Mr. Johnston to Folio 3, Exhibit 56, page 7 thereof where he says the following, "In other words while it was clear that the details called for 100 millimetres (4") brick and a 90 millimetre (4") concrete block backed up by interior insulation and finish, there is very limited instruction or clarification on how to achieve the assembly effectively. The detailed drawings are very limited in explanation and direction to the extent that it becomes very difficult to establish how the exterior wall is expected to shed incursive precipitation." Mr. Johnston confirmed for Mr. Nelson that was his evidence.

Mr. Nelson referred Mr. Johnston to Folio 2, Exhibit 55, page 4 thereof where Mr. Johnston notes "in other words in all those drawings or the document record as we know it, roof slopes were not included nor intended, drains were not located nor defined anywhere and the provision for roof drains was covered in single note that neither defined where, what or how the drains were to be installed. It was left to the contractor to make allowances without professional direction." Again Mr. Johnston confirmed that that was his evidence.

Mr. Johnston confirmed that after the December, 1994 repairs to the roof his next involvement was with the 1998 replacement. He was asked to co-ordinate a tender, write contract documents and provide architectural consultant services. He explained that in replacing the irma roof in 1998 the original membrane was left and another membrane was placed on top. He confirmed that the original roof went on in the winter of 1986, that is January of 1986 and so was some 13 years old when it was replaced. Mr. Johnston confirmed that in the very beginning in discussions with his clients as to what impact the roof might have on the window leaks, they did think that it was a roof problem but further investigation led them to the conclusion that it was a wall problem. There was major leakage at all levels and all windows and it could not be just as a result of a few caulking problems.

Mr. Nelson pointed out that at page 27 of the examination for discovery of Dr. Lawless, it was suggested by counsel O'Donnell for the plaintiffs that the plaintiffs' position was that the roof should have lasted for 20 years so that to the extent that the plaintiffs got some years out of it, it has to be discounted because of that. Mr. Johnston conceded that he agreed with the concept of pro-rating but pointed out that while the plaintiffs got 13 years out of the roof it was not 13 years of perfect and effective use.

At Exhibit B (eventually made Exhibit # 115A) are the materials received from Amherst Roofing. At Tab 1 is a letter from Amherst dated January 17, 2000 addressed to Emmons & Mitchell. This is the letter in which Amherst in the person of Peter Hardy who had been hired by Mr. Johnston to install the new roof in 1998, wrote to Emmons & Mitchell advising that in doing so he found the existing membrane to be in good condition and the felt flashings also to be in good condition. Although disagreeing with Mr. Hardy's comments, Mr. Johnston was satisfied with the replacement work that he did in 1998 and commented that "he did an excellent job". Mr. Johnston added that he prepared the drawing for the 1998 replacement. That drawing is Exhibit # 69 and shows in great detail the slopes, elevations and parapet design. Mr. Johnston explained that the two existing drains were kept and as well a new drain was put in at the north west corner.

Referencing Exhibit # 69, Mr. Johnston explained that the existing parapet, that is the parapet existing prior to the 1998 replacement, is defined by a red marker. Mr. Johnston explained that the detail below the red line remained the same except for Helifix anchors that were put in just above the soldier course of bricks. Again referencing Exhibit #69 Mr. Johnston explained that he had originally drawn masonry anchors in above the red line and showed these on Exhibit #69 with short blue marker lines, but went on to explain that ultimately the helix ties went in beneath the red line at the location marked by the long blue line under the red line. He explained that the helix ties went through the mortar joints of the brick into the block wall. They were #9 stainless steel ties with the brand name Helifix, at two foot centers.

At Exhibit 115B, Tab 141, are the specifications for the roof replacement. Mr. Nelson pointed out to Mr. Johnston page 04200-3. Mr. Johnston conceded that in the parapet refinishing they did not use the ties as specified but testified that this was not a matter of concern as the ties had been properly shown in the detailed drawing, Exhibit #69.

At Tab 26, Exhibit 115A, is Mr. Johnston's site report dated December 4, 1998. Mr. Johnston notes in this report as follows, "All elevations were checked, masonry anchor points were complete and refill as directed in the soldier course joints at parapet level – the writer was advised anchor installation was completed December 1 and 2 but not reviewed by the writer; mortar fill points noted at proper centers for remedial Helifix type 9" st. steel anchors as were directed in lieu of that detailed." Mr. Johnston explained that he had had some discussion with the mason as to how the detail anchors would be installed and after further research he directed that Helifix anchors be used. These were verbal instructions and no written change order was prepared.

At Tab 19, Exhibit 115A, is a change notice relating to "upgraded waterproofing and insulation for exposed rooftop plenum enclosure per owners' request". Mr. Johnston explained that Caves Mechanical had suggested that the plenum be enclosed in an aluminium upgrade rather than black canvas. The cost of the upgrade was $4,531 and is reflected in the change order at Tab 15. The sum of $4,531 should be deducted from the claim for the installation of the new roof.

At Tab 139, Exhibit 115B, is the contract relating to the 1998 roof replacement. Mr. Johnston confirmed that this was a standard CCDC as revised in 1994. He explained as well that General Condition 1.1.9.4 which provides that "later dated documents shall govern over earlier documents of the same type", constitutes a change in the text. Our contract at General Condition 1.1.6 which provides certain directions in the event of conflicts finishes by saying, "Notwithstanding the foregoing, documents of later date shall always govern."

Mr. Nelson referred Mr. Johnston once again to Exhibit #61 which is Mr. Johnston's response to Mr. Wilson where at page 11 he said the following, "Rainwater is coming through the bed joints of all the brick (not through the mortar nor through the brick itself) as we point out. Some incursive water is diverted and collected by the window lintels and impervious framing to be sure, but 60% of the wall not comprised of windows is absorbing the rain water penetration through brick elements not immediately above windows. This is a very real concern." In response to Mr. Nelson, Mr. Johnston advised that he had made certain calculations at one point in time but did not know whether he still had them. He acknowledged that Mr. Mueller had indicated that there were no such calculations but explained that Mr. Mueller did not know of them at the time. Entered as Exhibit #71 were the calculations prepared by Mr. Johnston.

Mr. Nelson referred Mr. Johnston again to Exhibit #61, his response to Mr. Wilson, where at page 1, paragraph 3 Mr. Johnston says the following: "There is no doubt, despite qualifying, aesthetic or extolling virtue commentary from any of the professionals to date retained by the defendants (and in fact there is substantial agreement) that the building leaked shortly after it was completed and it continues to do so." Mr. Johnston confirmed that was his evidence. Mr. Johnston was referred to Exhibit #62, his response to Mr. Quirouette's report where at page 2 Mr. Johnston noted the following: "I personally have specified, supervised and witnessed window system failures in the field according to just such ASTM standard test procedures for the former Ontario Ministry of Housing and I am quite aware of the nature and scope of the flows encountered. In our opinion, a repeated, consistent and observable leakage at multiple windows during or after a rain storm was certainly sufficient to warrant further evaluation." Referring again to Exhibit #71, Mr. Johnston explained the calculations direct themselves to how much of the brick had to be removed if the Wilson recommendation was followed. Mr. Johnston calculated that 125.74 square metres would be removed out of a total of 292 square metres, leaving 167 square metres in place, with the result that 43% of the bricks would be removed from the south wall.

Re-examination by Mr. Mueller

Mr. Johnston explained that theoretically it would be possible for the shelf plates to act as flashing but it would take in his view an incredible amount of work. As well, they would have to be connected at the ends to increase stiffness. Mr. Johnston reiterated that Drawing No. 4 which was too small to find the air space and which had no dimension on it, was totally ineffective in architectural detail.

With respect to his spray test done with the hose, again Mr. Johnston explained that he was not trying to determine the actual amount of water coming in when conducting that test.

Plaintiff’s Reply Evidence from Dr. Reed

Approximately one week ago when there was a snow storm that dropped some six inches of snow, Dr. Reed accompanied by Mr. Mueller walked around the building and Dr. Reed went up onto the roof. From below Dr. Reed was able to see a two-foot length of snow that had accumulated on the overhang on the south side. Once he was up on the roof he could see that that two-foot length of snow accumulation was approximately in the middle of a total of a 20-foot accumulation. There was a similar 20-foot accumulation of snow on the north parapet. Both accumulations were on the inner width of the parapet and the outer width was clear. It was Dr. Reed’s evidence that at no time during the history of the building was there any complaint of water coming from melting snow on the parapet.

On the same occasion, one week ago, Dr. Reed made some observations with respect to snow on the heads of the stone window surrounds. Looking down from the roof, he observed snow on top of two fourth floor window surrounds at the west end of the building and similar accumulations on two windows on the north side. There was no other snow accumulation on the window surrounds.

With respect to Mr. Quirouette’s evidence wherein he commented that after very cold weather on warm spring days he would have expected condensation to develop with the warming of the cavity and thus creating moisture problems, it was Dr. Reed’s evidence that in his experience water never came into the building unless it was raining or unless very wet snow was falling. In his experience, no water ever came in during warm weather in the spring when snow was melting.

With respect to Mr. Pascoal’s evidence with respect to efflorescence at the ground level, possibly contributed to by de-icing salts, Dr. Reed pointed out that on the south side which is the main entrance there is approximately six feet between the sidewalk and the south wall with a significant height differential between the beginning of the brick courses and the level of the sidewalk. He did allow that there could be some splash of de-icing salt from the road that could hit the brick wall.

In discussing the transformer vault, reference was made to Exhibit 4, photographs 88 and 92, and Dr. Reed explained that the transformer vault is contained in that portion of the elevator shaft that was built after the main building was built. In other words, the transformer opening was part of the building that was put on after the main building. In addition, Exhibit 1, Tab 1-A indicates the possibility that that particular piece of construction may contain cavity insulation. Finally, he testified that the transformer area is probably the most protected area of the entire building.

Cross-examination by Mr. Nelson

As a matter of housekeeping, the entire “deficiency list” prepared by Dr. Reed, part of which has gone in as Tab 22 of Exhibit 1, was filed in its entirety as Exhibit 141. Dr. Reed confirmed that this is a list in his own handwriting dated September 3, 1986 of matters that were of concern.

Witness – Stephen Saxe

Examination-in-Chief

Mr. Saxe is a real estate appraiser. He is an accredited member of the Appraisal Institute of Canada, an associate member of the Institute of Municipal Assessors, an Ontario Land Economist and an associate broker with the Toronto Real Estate Board and the Ontario and Canadian Real Estate Associations. He has been working in the field since the mid-1960's and is well qualified to provide the court with his opinions relating to the value of the subject real estate.

Mr. Saxe's initial real estate evaluation report went in as Exhibit #72 and his letter of supplementary comments dated May 6, 2002 was filed as Exhibit #73.

Based on a combination of the direct comparative method and the income capitalization method, Mr. Saxe arrives at a fair market value of $720,000 after the remediation suggested by the plaintiffs' experts has been completed. Deducting from this figure the remediation cost of $613,000 leaves, in Mr. Saxe's opinion a fair market value of $107,000 as is.

At page 14 of Exhibit #72 is a list of the various sales looked at by Mr. Saxe, using the direct comparative method of evaluation. Using that method alone, he arrived at a market value of some $745,000.

At page 15 of Exhibit #72 is the income capitalization test calculations, which result in a capitalized value estimate of some $672,084 which is about $75,000 less than the evaluation using the comparative method of appraisal.

At page 16 of his report, Exhibit #72, Mr. Saxe addresses the impact of the building deficiencies on its fair market value. On his attendance, he observed water leakage around the windows on the fourth floor. At the time he could see both water coming in and signs of water having come in previously. He also saw second floor damage to the drywall on the upper areas of the wall and evidence of rusting through a hole in the wall. He observed that the leakage would not only deteriorate the materials but would raise a safety issue with respect to the possibility of the brick cladding falling away. He noted that in the market place if the building is known to be defective then there can be a problem with insurance and if he were to list the property for sale he would be under a duty to make full disclosure to any prospective buyers as to the problems that this building is facing.

Again at page 16, Mr. Saxe notes that the cost of remedial works estimated to be approximately $613,000 is considered to reflect the minimum discount from market value as estimated by the two methods that would be necessary to attract a purchaser to the subject building. He suggests that this ‘cost to cure’ estimate must be considered as a minimum due to the fact that most if not all prudent purchasers would be skeptical of cost information provided by the owner.

At Exhibit #73 is Mr. Saxe’s supplementary report wherein Mr. Saxe discusses the effect of the contradictory information coming from the various experts who have looked at this building and who have provided opinions as to what is required to bring it into proper condition. Mr. Saxe is of the view that one would expect a knowledgeable purchaser to review the various opinions in order to develop a conclusion as to the risks of premature failure of certain building elements, and on its direct (i.e. actual cash outlays for repairs) and indirect (i.e. interruption, suspension of rental income) financial performance over the projected investment horizon. Mr. Saxe notes that assuming the risk in such an investment could be accurately quantified and that a discount from what would otherwise be accepted as market value determined, the investor would still be faced with a <u>troubled</u> property stigma. By this he means the market’s concern that, a) repairs may not be properly completed and problems may occur, b) other structural problems not yet identified by any of the experts may exist, and c) the cost of remedial works in the event the purchaser undertakes this responsibility had been understated. In his view the only solution to the purchaser’s dilemma is to assume a worst-case scenario and that may render the building simply not saleable in its existing state.

Plaintiff's Read-Ins

Filed as Exhibit #3 was a list of questions and answers to be read from the examinations for discovery of Mr. Garbutt and the defendant company's representative, Mr. Mitchell.

Examination for Discovery of Mr. Garbutt

The examination for discovery of Mr. Garbutt disclosed the following. Mr. Garbutt is an engineer having received his engineering degree from Queen's in 1952 and his Master's in 1953. In the last 20 years he has worked as a private consultant on relatively small high-rise buildings.

On this particular project he retained Mr. Roberts as an architectural sub-consultant for the masonry and building envelope, as required by the then-existing Building Code.

Mr. Garbutt intended the wall to be a "solid masonry wall" but Mr. Roberts prepared a wall detail that showed a ¾" gap which Mr. Garbutt accepted when Mr. Roberts sent it to him.

At the time, Mr. Garbutt believed that there would not be a water penetration problem. The Building Code required that the exterior wall cladding shall be installed to shed water and as far as Mr. Garbutt was concerned the wall was designed to shed water.

While the brick chosen by the plaintiffs was a high porosity type brick, Mr. Garbutt admitted that he had never advised the plaintiffs that this might create a possible problem with water penetration.

Mr. Garbutt conceded that with the type of brick specified water getting through was supposed to have dripped down and out weepers.

Mr. Garbutt acknowledged that the brick ties should have been at an absolute minimum horizontal but preferably sloped down at an angle from the block to the brick.

While the brick and block was being installed, Mr. Garbutt attended at the site once or twice a week.

Mr. Garbutt agreed that the Code required one tie to be placed every four square feet and that the specifications called for brick ties.

Mr. Garbutt does not recall observing ties being placed at the 2nd floor level.

Mr. Garbutt acknowledged that there should not be water dripping in through and around the windows in the subject building.

Mr. Garbutt conceded that the mortar fins extending across the air space from the brick to the block was undesirable. As he put it, "If you call for an air space you don't want it mortared up." He explained that when the masons are doing the brick and block work, they should work so as to minimize the fins. The reason that a lack of fins does not have to be specified is that it is in the Building Code. The Code calls for back troweling.

Mr. Garbutt testified on discovery that to the extent that one can see at the various test openings excessive finning between the brick wythe and the block wythe, this was bad building practice on the part of the mason, Abe Dick Masonry and was contrary to the Code.

Mr. Garbutt agreed that his oral contract with the plaintiffs was to design the building and supervise the work, and that he would be paid a percentage of the construction cost.

Examination for Discovery of Murray Mitchell

Mr. Mitchell is the President of the defendant company, Emmons & Mitchell. The company has been in business since 1957 and specializes in constructing commercial/industrial buildings.

Mr. Mitchell acknowledged that the plaintiffs' building has had a water penetration problem.

As far as Mr. Mitchell knows, there has never been any discussion with the owners or Mr. Garbutt about flashings or weep holes.

Mr. Mitchell acknowledged that the first time his company knew there was a serious problem that could result in a law suit because of water penetration, was when they received the first lawyer's letter from MacPherson and Hogan in September of 1993.

Mr. Mitchell in his discovery agreed that there were no through wall flashings and no weep holes anywhere in this building.

Mr. Mitchell acknowledged that his site superintendent missed the fact that there were no wall flashings and weep holes because they were not shown on the drawings although they were called for in the specifications. At the same time Mr. Mitchell agreed that both the drawings and the specifications have to work together.

Mr. Mitchell agreed that there was nothing in the plans or the specifications to say that the roof was not to be sloped. As far as he is aware the roofing contractor did not ask his company or Mr. Garbutt about the slope or lack of slope, nor did Emmons & Mitchell ask Mr. Garbutt about the slope or lack of slope.

Mr. Mitchell agreed that there was nothing other than what is set out in the plans or specifications which would have told his company that there would be no weep holes. He agreed on discovery that there was nothing special about this building at all which would say to him "I don't even have to talk to the engineer, we are not going to put in weep holes." Mr. Mitchell agreed that there was nothing about the nature of the design of the building or the specifications and plans that would have indicated to him that there was no need to even talk to the engineer and that there simply would be no flashings.

Mr. Mitchell acknowledged that in September, 1986 and in 1992 his company recommended that the building be caulked to avoid a water problem, but acknowledged that his company did not visit the building before making these recommendations.

Mr. Mitchell acknowledged that in 1986 his company was paid $1,500 for the caulking and that in 1992 the plaintiffs made direct payment to the party who undertook the caulking.

Mr. Mitchell acknowledged that the drawings indicate there should be no roof slope, while the specifications mention a slope but in the context of a different roof system. He acknowledged that there was never a change in the roof design from the original contract documents, and the only addition to the contract was the addition of 3" in insulation.

According to his evidence on discovery, Mr. Mitchell understood that the roof was to have three drains in all, including one in the stairwell.

Mr. Mitchell conceded that the design of the roof from the plans and specifications was an inverted roof with the membrane on the bottom and the insulation on top being held down by stone ballast.

Again, Mr. Mitchell confirmed that it was Don Emmons of Emmons & Mitchell who suggested in 1986 and in 1992 that the building be caulked and that the recommendation was made on the assumption that caulking would have corrected the water penetration around the windows. After getting the plaintiffs' lawyer's letter in September 1993, the company concluded that caulking was not going to solve the problem.

Mr. Mitchell advised that his company first came to a conclusion as to what the basic problem was with the leaking when they started reading some of the experts' reports a few months before the discovery in July of 1999.

Mr. Mitchell acknowledged that the copper fibreen flashings called for in the specifications were not installed anywhere in the building. Mr. Mitchell conceded that if no flashings were being installed by the brick contractor, Abe Dick, then the question should have been put to the brick contractor at the time. Mr. Mitchell is unable to testify that either he or Abe Dick or anyone on behalf of Emmons & Mitchell spoke to the owners or Mr. Garbutt about flashings and were told to put none in. Mr. Mitchell conceded that the waterproof flashings would invariably have involved being placed above and around windows and doors.

Mr. Mitchell acknowledged that in the 23 years that he has been with the company, he has never built a building remotely like this particular one and put in no membrane flashings. He has never ever seen such a building properly built without waterproof membrane flashings of some kind.

Mr. Mitchell conceded that there should have been weep holes installed in 1985 and 1986.

Mr. Mitchell on discovery acknowledged that the angle irons were cut off the steel on the east and west sides and that the purpose of the angle irons was to support the brick. The examiner suggested to Mr. Mitchell that no-one from Emmons & Mitchell or Abe Dick had gone to the owners or to Mr. Garbutt to ask permission to remove the angle irons, to which Mr. Mitchell responded that there was nothing in writing. He was asked then whether it had been done orally and Mr. Mitchell responded, “I can’t confirm that.” Mr. Mitchell did concede that without the support angles on the east and west sides the building on those sides is 5 to 10 feet higher than the Building Code permitted without support.

Mr. Mitchell confirmed that at least in the parapet area there was a substantial lack of brick ties from what he was able to observe. He acknowledged that the parapet extends down from the top of the parapet to the top of the 4th floor windows. He acknowledged that the standard was one tie per 4 square feet, and believed that to be Mr. Garbutt’s specification. He acknowledged that that was standard by the Building Code.

Mr. Mitchell confirmed that Mr. Pascoal did not observe one brick tie in the parapet locations observed.

Finally, Mr. Mitchell acknowledged that in general construction practice the minimum standard to be expected by any contractor or sub-contractor, is conformance to the Ontario Building Code.

Witness – Donald Garbutt

Examination-in-Chief

Mr. Garbutt is a professional engineer and has been practising as such since 1953.

With respect to the subject project, Mr. Garbutt had been a patient of Dr. Lawless and was approached by the plaintiffs in his capacity as an engineer in 1984. At that time he was not aware that the plaintiffs had previously approached architect Roberts and in fact did not learn of that fact until the action was instituted.

Mr. Garbutt explained that after his retainer, the building was initially designed as a two-story building to accommodate the offices of each of the plaintiffs. Specifications were prepared for tender and the parties then discussed two different methods of proceeding. The first method available was one involving contract management where three individual bids on each part of the construction project would be obtained, employing a contract manager working on a fixed budget of $38-$40,000 per floor. The alternate method of proceeding was by way of lump sum tenders.

Initially, Mr. Garbutt felt that the project would proceed by way of contract management. He retained a contract manager who prepared estimates and at the same time Mr. Garbutt prepared detailed estimates of the work to be done for change purposes. At Tab 1A of Exhibit 1, at the back thereof, are both the contract manager's figures based on tenders and Mr. Garbutt's estimates which had been prepared on the basis that some time in the future two additional floors would be added to the original two floors of the building.

In due course, it was decided that the building would consist of four floors. It was Mr. Garbutt's evidence that the plaintiff owners then decided to call lump sum tenders and in due course received two lump sum tenders, the Emmons & Mitchell Construction Ltd. tender being the lowest. The contract was awarded to Emmons. Mr. Garbutt testified that as tendered it was a bare bones building and the tender came in under budget both with respect to Mr. Garbutt's previous estimates and the contract manager's estimates. Certain upgrades were incorporated and resulted in a final lump sum contract.

As a result of discussions with the chief building inspector for the City of Kingston, Mr. Garbutt decided that to be on the safe side with respect to then current regulations relating to the requirements of architects and engineers, he would retain at his own expense an architect before the building permit was issued. As a result he retained Mr. Roberts who produced certain drawings. The building permit was applied for and obtained.

Under the contract between the plaintiffs and Emmons, Mr. Garbutt was the consultant and thus responsible for general review of the project as it proceeded. Mr. Garbutt produced at Tab 1 of his Exhibit 74, a guideline for professional engineers providing general review of construction as required by the Ontario Building Code. The guideline sets out the professional responsibility and scope of work of the consultant charged with general review, and also sets out certain performance standards. It was Mr. Garbutt's evidence that when he undertook the project he was subject to the directions contained in Tab 1 of Exhibit 74 with respect to the scope of work and the performance standards. Mr. Garbutt explained that as well the contract itself at Tab 2 of Exhibit 1 set out certain responsibilities of the consultant by way of general review in various of the General Conditions, and particularly in General Conditions 3.4, 14.11, 25.1, 3.6, 3.9, 34.1, 34.3, 16.3 and 16.4

Certain of the obligations of the consultant are also set out in the specifications at Tab 1 of Exhibit 1 which specifications form part of the contract. In particular, Mr. Garbutt in his evidence referred to Division 1 Section 3.1. and Division 5.1.10.

With respect to these requirements and responsibilities as consultant charged with general review, it was Mr. Garbutt's evidence that he inspected the concrete and the structure as required by the municipality. He did not however check the contractor's layout work as he did not think that he as consultant was required to do so. Mr. Garbutt pointed out that James Sinclair was the general contractor supervisor, and Mr. Garbutt always felt that he should deal with the supervisor only in order to comply with General Condition 25.1 at Tab 2. He expressed in his evidence great confidence in the supervisor.

In his evidence, Mr. Garbutt pointed out that according to General Condition 3.6, the interpretation of the drawings and the specifications was the consultant's responsibility. As well, under G.C. 3.9 it was his obligation to review and take appropriate action upon the contractor's submittals, such as shop drawings, product data, etc. He noted that under the General Conditions, the Codes are primary and that the contractor was obligated to comply with all applicable codes. Perhaps of particular significance is Division 5.1.10 of the specifications, which obligates the contractor to obtain written permission of the engineer prior to "field cutting or altering of structural members". It was Mr. Garbutt's evidence that at no time was he approached by the contractor for permission to cut the support plates on the east and west walls, nor did he ever give such permission. In his evidence he indicated that this job could have been done in perhaps as little as a half hour and their removal would not be easily detectable.

Mr. Garbutt then in his evidence dealt with the masonry wall construction. It was his evidence that his design was for a solid masonry wall to provide wind bracing for the structural steel. The solid masonry wall was to consist of the two panel walls, block and brick, with in-filling in between. The drawings showed a four-inch block with no specific air space. To act as wind bracing, the four-inch block would not be sufficient. The specifications as written would have constituted a Code violation if the wall was constructed with a gap, and that is why the specification said solid wall.

As well, with respect to the masonry walls, Mr. Garbutt commented that Dr. Lawless's video taken at the start of the construction of the wall showed a collar wall with the collar in-filled and this was in accordance with Mr. Garbutt's intent.

Mr. Garbutt in his evidence relating to the roof testified as follows. When the specifications were drawn, they were drawn for a two-floor building. There was an intention to add two further floors at some point in the future, at which time the roof of the second floor would become the floor of the third floor. As a result, the roof of the fourth floor was eventually built flat with no slope. Mr. Garbutt felt the membrane as designed was adequate for a short period of time and thus was left as is. Mr. Garbutt had anticipated a life of some 15 years for the membrane. He commented that Mr. Johnston in his evidence felt in 1994 it had five years left which equated to a 13 year life span.

An irma style roof, which is the reverse of a standard built-up roof was used because it is more acceptable as a temporary roof.

The roof deficiencies eventually identified by both Mr. Kendall and Mr. Johnston were in fact identified during construction. See Tab 13 of Exhibit 1, referring to the break in the flashing on the corner of the roof, the hole in the flashing near the roof hatch, and the caulking on the flashing being poor. Mr. Garbutt testified that those three deficiencies were in fact supplied to Mr. Vinkel of the Emmons firm, which were in turn forwarded on to the roofer, Quintal.

With respect to the involvement of the architect, Mr. Roberts in the project, Mr. Garbutt pointed to Article 2.5.2.2(1) of the 1983 Ontario Building Code which provides that work done by an architect can only be reviewed by an architect. Mr. Garbutt never did receive any inspection certificates from Mr. Roberts and admits that he was probably remiss in not obtaining them. He is unaware whether or not Mr. Roberts filed any certificates with the municipality. As we know at this point in time, the building inspector is deceased and the City has lost all the documents relating to this project. Mr. Garbutt does not know whether Mr. Roberts did any inspections.

Again referring to the roof, Mr. Garbutt observed that the deficiencies subsequently noted by Messrs. Kendall and Johnston were in fact identified during the course of construction (see Tab 13 Exhibit 1) and should have been repaired by the roofer.

Again returning the walls, Mr. Garbutt noted that with respect to his Tab 16, Exhibit 74 the shelf plate does exist on the south and north walls at each of the floors. As we know, the shelf plates on the east and west walls were cut off.

Mr. Garbutt then went through various sections of Exhibit 47 which are the Canadian Standards relating to masonry design and construction for buildings. Section 2.1 defines a "panel wall" as meaning a non-load bearing exterior masonry wall having bearing support at each story. Section 5.3.5.1 provides that the thickness of a solid masonry panel wall shall be not less than 175 millimetres in actual thickness. Section 2.1 defines "solid masonry" as meaning masonry of solid or hollow units that does not have cavities between the wythes. Section 2.1 defines "cavity wall" as meaning a construction of masonry laid up with a cavity between the wythes which are tied together with metal ties or bonding units.

The point of Mr. Garbutt's reference to the various provisions of the Canadian Standards for masonry design was to illustrate that he intended a solid masonry wall and not a veneer wall, that is why he constructed the building to have a solid wall to be in conformance with the Code (the Canadian Standards). Again referring to the Standards, Mr. Garbutt pointed out section 5.4.6 which provides that where a cavity wall is non-load bearing, the total thickness of wythes and cavity shall be at least 230 millimetres. Section 5.4.3 provides that the width of the cavity in a cavity wall shall be not less than 50 millimetres and not more than 75 millimetres when tied with metal ties. Section 6.1.1 provides that masonry veneer shall not be considered to be part of a wall when computing its strength or thickness.

At Tab 9 of Exhibit 74, is an illustration of the dimensions of the brick and block used in the wall in terms of actual measurements. The brick is 100 millimetres thick, the block is 90 millimetres thick, and the space between the block and the brick as provided for in the specifications, Division 4 item 2.1.1 was 19 millimetres, for a total of 209 millimetres of thickness which, Mr. Garbutt says, meets the requirement for a solid wall but not for a veneer or cavity wall because the actual thickness is only 209 millimetres, and 230 millimetres is required for a cavity wall under the Code. Mr. Garbutt commented that these are non-load bearing walls but are used to resist wind.

Referring to Tab 10, Exhibit 74 Mr. Garbutt explained that the space between the block and the brick was set at 19 millimetres in order to accommodate the two vertical lintel angles, and thus to avoid cutting the masonry. Tab 10 of Exhibit 74 is Mr. Garbutt's drawing prepared in order to explain the 19 millimetre air space as specified. The joint thickness selected by the mason was 15 millimetres plus or minus three millimetres.

Mr. Garbutt referred to Mr. Johnston's evidence to the effect that 30 to 50 per cent of the mortar in any given horizontal brick joint actually bridged the space between the brick and the block, and he noted that if one was standing looking down between the brick course and the block course, the result of this bridging along the horizontal joints would appear as a solid joint in between the brick and the block. He did not however testify that he in fact at any given time looked down between the brick and the block and observed this situation.

At Exhibit 2 Tab 65, is a note which reads in part, "built as a solid masonry wall collar joints filled walls tied together with adjustable ties". Mr. Garbutt in his evidence was able to identify this as Mr. Emmons's handwriting and assuming that it is, Mr. Garbutt feels that this is confirmation that the wall was to be built as a solid wall in accordance with his instructions. In this connection, see Tab 64 which is a note from Mr. Garbutt to Mr. Emmons dated May 10, 1993 which reads in part, "See if the following can be determined: a) that the collar joint (between brick and block) is filled – there is no such thing as a one-half inch air space". Interestingly, Mr. Garbutt at the end of the letter notes that if the collar joint has not been filled and the wall subsequently not constructed as a cavity or veneer wall (flashing, weep holes, etc.) serious problems result.

Mr. Garbutt in his evidence commented that a solid wall without any cavity has no water to drain hence has no weep holes at the plate levels because there is no requirement for drainage.

Mr. Garbutt then referred to a number of reports received from the plaintiffs, from various experts during the period 1985 to 1996. These referrals were more by way of submissions than anything else. At Tab 19 of Exhibit 1, is the Mill Ross report dated November 14, 1986 which makes no comment with respect to the absence of weep holes or flashings, and Mr. Garbutt concludes that Mr. Mill at the time must in turn have concluded that this was a solid wall. MacAdam's report dated August 1, 1993 comments on the missing plates on the east and west walls and recommends that the plaintiffs contact their solicitors. MacAdam's report of September 25, 1993 recommends to the plaintiffs that they retain a roof consultant. InspecSol's report dated September 25, 1993 indicates that the roof is satisfactory, with some parapet problems and recommends a flooding test. MacAdam's report dated October 11, 1993 recommends the drilling of weep holes and comments that the costs of extensive demolition would be prohibitive. MacAdam's report dated November 30, 1993 comments that full correction would consist of removal and replacement of the brick. At that time, apparently Mr. Cromarty was retained who recommended pressure grouting. Mr. MacAdam in his report June 20, 1994 addresses Mr. Cromarty's recommendation.

At this point in time Mr. Johnston was on the scene and he performed destructive testing on the walls. Mr. Garbutt comments that Mr. Johnston found that wall ties existed but not in a degree required by a veneer wall in accordance with the Code. Mr. Garbutt notes that Mr. Johnston found the steel structure to be in good condition but with some pitting. The angle lintels over the windows were loose laid with a 'business card gap'. There was no mortar between the top brick and the underside of the bearing plate. There was no significant deterioration of the surface of the brick; no evidence of falling bricks, and the walls were plumb. The caulking around the stone surrounds was intact.

It should be noted that the above-noted comments were really made as submissions by Mr. Garbutt rather than evidence. He was simply commenting on the evidence that has already gone in, as he saw that evidence.

During the course of his evidence, Mr. Garbutt attempted to have admitted a letter from the Ontario Climate Centre addressed to him and dated May 29, 2002 relating to maximum wind speeds, and I ruled that absent the consent of counsel it was not admissible as such.

Discussing the performance of the wall as built, Mr. Garbutt referred to Tab 17 of Exhibit #74 which shows the rotation of the beams as referred to by Dr. Drysdale, and which apparently caused the horizontal cracking as observed by Dr. Reed in 1992. The cracking is shown in photographs at Tabs 25 and 26, Exhibit #74. Mr. Garbutt attempted to offer the opinion that this is where the water was getting in. several times during his evidence I cautioned Mr. Garbutt and he understood the caution that as a defendant in the action he was not entitled to offer an expert opinion as to matters that occurred after the completion of the work in connection with the causes of problems and the remedies available. I did instruct Mr. Garbutt that he was perfectly entitled to give me his evidence as an engineer during the course of construction as to why he took a particular view during construction and acted accordingly.

With respect to his counterclaim for approximately $6,500 remaining owing on his invoices, Mr. Garbutt testified that his agreement with the plaintiffs was verbal and it called for compensation to him for 5% of the actual construction cost of the building. He commented that additional costs were added to the contract after tender and these involved field review by him. He noted that Dr. Lawless testified that the plaintiffs did not supervise the work. Mr. Garbutt calculated that each $1,000 owing in 1986 is by reason of pre-judgment interest equal to $3,754 today, for a total of something in excess of $24,000 at the present time. He is also looking to add on to that number an inflation factor of some 1.544%.

With respect to the limitation defence, Mr. Garbutt's comments during the course of his evidence were primarily and essentially submissions on the issue. He noted that the one year limitation period under the *Engineers' Act* expired in the fall of 1987. He went through the chronology of the complaints made by the plaintiffs. He was first notified in writing by the plaintiffs' lawyers on September 16, 1993 at which time the lawyer complained about the design. There was then no action for a period of approximately 20 months. He was not being paid the balance of his fee. It was not possible for him to inspect and he was not asked to be involved in any of the destructive testing that was going on. The building inspector is now dead, as is Mr. Emmons. He had no dealings with either Mr. Emmons or the plaintiffs up until 1992. Presently the shop drawings and the municipality records are missing. He was never made aware of the re-caulking by Mr. Thompson in 1992, and did not know of the flashing re-caulking by Quintal in October of 1992. He was not invited to the destructive testing performed by Mr. Johnston nor to any of the roof investigations.

Cross-examination by Mr. Nelson

Mr. Garbutt is a professional engineer not an architect. The plaintiffs knew him to be an engineer. He did not tell the plaintiffs of his intention to retain Mr. Roberts who was both an engineer and an architect. In fact, Mr. Roberts had been trained as an engineer and was 'grandfathered' in as an architect during the transition period in the mid-1980's. Effectively there was no-one who had been trained as an architect as such, who was providing services to the plaintiffs.

Initially, it was planned to build a two-story building. Included among the drawings in Exhibit 8 which are the tender drawings, is an elevations drawing showing a two-story building. Mr. Garbutt testified that he in fact prepared a complete set of drawings for the two-story building but the only one left is the elevations drawing showing a two-story building, and he does not know where the others are. He does not know whether a set of tender drawings for the two-story building was delivered to the defendant Emmons.

From Exhibit 8, Mr. Nelson pulled the wall section drawing that had been sent to the defendant Emmons. Mr. Garbutt confirmed that he prepared this drawing and that he stamped it with his stamp. He conceded that the placing of his stamp on the drawing indicated his approval of the drawing. He accepted responsibility for the drawing although it was not in fact signed by him. The wall section drawing is dated March 1985. Mr. Garbutt confirmed that by that date the plaintiffs had gone to a four-story building and the wall section drawing is indeed for a four-story building. It was Mr. Garbutt's evidence that by that time as well the plaintiffs had told him that they might go to a six-story building and if so the roof of the four-story building would become the floor of the fifth floor. As a result, said Mr. Garbutt, he never intended that there be a roof slope, not with the two additional floors to be added.

Mr. Garbutt explained that the specifications had been prepared before Mr. Garbutt prepared the drawings for the four-floored building. They had been prepared by the construction manager. The specifications provided for a standard built-up roof. In fact the drawing shows an irma roof. Mr. Garbutt confirmed that every drawing in the tender documents describes an irma roof, which is the reverse of a standard built-up roof. With the irma roof the insulation lies on top of the membrane, and in his view it was a better roof to have where two more stories were to be added.

In any event, the specifications in existence as at March of 1985 called for a standard roof. The roof actually constructed was an irma roof and did not comply with the specifications.

Mr. Garbutt confirmed that his wall drawing does not show weep holes. He explained that he designed the wall as a solid wall and did not intend that there be weep holes, and thus none are shown on the drawings. On the other hand, the specifications call for weep holes in some locations. It is clear however that the tender drawings that went to the defendant Emmons did not call for weep holes.

As well, Mr. Garbutt confirmed that the drawings did not call for flashings, whereas the specifications did indeed call for flashings. Mr. Garbutt confirmed that in fact there are no flashings in the walls as built, which conforms with his drawings and is consistent with a solid wall.

Mr. Garbutt agreed with Mr. Nelson that the type of roof to be constructed is an issue of design, as are weep holes, flashings and control joints. Mr. Garbutt agreed that the type of wall is a design issue.

Again, referring to the tender documents, Exhibit 8 and particularly the window design, Mr. Garbutt agreed that the window design was prepared by him in terms of the size of the window and the fact of the surround. Mr. Garbutt explained that the contractor then provides the details of the window construction in shop drawings which we do not presently have.

Mr. Garbutt confirmed that the wall drawing shows a four-inch thickness of block and a four-inch thickness of brick, with no cavity in between.

Referring to the design of the parapet as shown on the drawings, Mr. Garbutt confirmed that this was his design. He confirmed that initially the irma roof design called for three inches of roof mate insulation which has troughs in the underside to allow for runoff. The thickness of roof mate was eventually increased to six inches and Mr. Garbutt conceded that when the increase was put in to effect, he did not add the additional three inches to the parapet drawing.

Referring to the structural drawings in Exhibit 8, Mr. Garbutt confirmed that they show shelf plates in the structural steel at each floor. Referring to his structural drawing, he pointed out that the shelf plates are called for at each floor on all four sides of the building, with the notation “8 inch by ¼ inch”. Mr. Garbutt explained that while he does not design how the beams will fit, he did plan that everything would in fact fit with the shelf plates on each wall.

Although Mr. Garbutt did not personally deliver a set of his tender documents to the defendant Emmons, he assumes that a set was in fact delivered to that defendant. He explained that while he was responsible for design, the plaintiffs decided to go for a lump sum tender, and they were to negotiate with the contractors for the lump sum.

With respect to supervision, Mr. Garbutt took the position that while he was in charge of field review, as previously testified, he did not take that to be responsibility for supervision of the work, although in his discovery on July 1999, at questions 220 and 221, he did use the word ‘supervision’ in describing his responsibilities. He explained here at trial that those answers were true in the context of his obligation for field review. He explained that ‘supervision’ to an engineer never means that he goes out to the job and actually supervises the work. He explained that the general contractor under the General Conditions has the absolute authority for supervision, and that Mr. Garbutt’s supervision was limited to field review.

Mr. Garbutt agreed that of the 26 site meetings, the plaintiffs appeared to have attended 20, and that he attended somewhat less than that number. He explained that he attended those site meetings where he had been advised in advance that there was an engineering problem involved.

Mr. Garbutt conceded that in the field he gave instructions to Mr. Emmons' supervisor, Mr. Sinclair and as previously stated had a great deal of confidence in Mr. Sinclair. Mr. Nelson asked Mr. Garbutt for his comment in the event that Mr. Sinclair testifies that he would never have cut the shelf plates without Mr. Garbutt's instructions. Mr. Garbutt responded that he would agree with that but that he does not mean that he instructed him to cut the shelf plates. Mr. Nelson suggested to Mr. Garbutt that he authorized the cutting of the shelf plates on the east and west walls, and Mr. Garbutt denied that to be the case. Mr. Garbutt could not recall when he became aware that the shelf plates had been cut on the east and west walls, but it was not during the course of construction. Mr. Garbutt commented that the contractor cut the shelf plates contrary to the drawings and contrary to the Certificate of Completion. He noted that there was no extra charge for the cutting and suggested that one should always "follow the money trail". He said whoever cut the shelf plates would not do it for nothing.

It was Mr. Garbutt's evidence that when the plaintiffs decided to put the project out to tender, they delivered the drawings and probably the specifications to the contractors, although in this particular instance Mr. Garbutt does not know whether when the drawings went to Mr. Emmons, the specifications went as well.

It was Mr. Garbutt's evidence that while he was aware of contradictions between the drawings and the specifications with respect to for example the roof, weep holes and flashings, he never went back to amend the specifications because the contractor Emmons was following the drawings and his instructions.

Referring to Exhibit 9, the three small drawings which are also contained at Tab D of Exhibit 55, Mr. Garbutt confirmed that these drawings were prepared by Mr. Roberts whom he had retained. They were prepared on Mr. Garbutt's instructions. Mr. Garbutt conceded to Mr. Nelson that the drawings show a gap between the concrete block and the brick. The gap is some ¾ of an inch wide which is the same width as the joint in the bricks, and Mr. Garbutt conceded that he approved the drawings.

Referring to the contract, Exhibit 1 Tab 2, fourth page in, Article A-1, Mr. Garbutt conceded that this article identifies him as the "consultant".

Mr. Garbutt conceded that General Condition 2.1 of the contract at the same Tab provides that during the progress of the Work the consultant will furnish to the contractors such additional instructions to supplement the contract documents as may be necessary for the performance of the Work. He conceded that General Condition 3.1 provides that the consultant will provide administration of the contract as described in the contract documents. He conceded that General Condition 3.2 provides that the consultant will be the owners' representative during construction. He conceded that General Condition 3.4 provides that the consultant will visit the site at intervals appropriate to the progress of construction, to familiarize himself with the progress and quality of the Work and to determine in general if the Work is proceeding in accordance with the contract documents. General Condition 3.4 goes on to provide that the consultant will not make exhaustive or continuous on-site inspections to check the quality or quantity of the Work. With respect to this General Condition, Mr. Garbutt commented that he did visit the site once or twice a week with respect to the structural steel and concrete work. Mr. Nelson pointed out that the Work referred to in the General Conditions is defined in those General Conditions as meaning the "total construction and related services required by the contract documents." In the face of this definition, Mr. Garbutt conceded that his obligation to visit the site was to review the total construction. He conceded as well that he had the power to reject work that was not in conformity with the contract. See General Condition 3.9 which states that the consultant will review and take appropriate action upon the contractor's submittals such as shop drawings, product data and samples in accordance with the requirements of the contract documents. See also General Condition 3.10 which provides that the consultant will prepare 'change orders' in accordance with the requirements of General Condition 11. General Condition 11 in turn provides that the owner, through the consultant, may make changes in the Work with the contract price and contract time being adjusted accordingly by written order and that no changes in the Work shall be proceeded with without a written order signed by the owner and no claim for a change in the contract price or change in the contract time shall be valid unless so ordered, and at the same time valued or agreed to be valued as provided in General Condition 12. Mr. Garbutt conceded those General Conditions and explained that that was the function of the meetings held on site.

With respect to 'change orders' Mr. Nelson referred Mr. Garbutt to Tab 48 of Exhibit 1 which is a handwritten document prepared by Dr. Reed which says in part, "Additional costs (plus or minus $200,000) over tendered cost are itemized by change orders. The majority of these were ordered and supervised by the owners, not by the engineer." Mr. Garbutt denied this to be the case. He noted that Dr. Lawless testified otherwise and said that there was no supervision by the owners. It was Mr. Garbutt's evidence that he supervised the change orders.

Mr. Garbutt conceded that he assumed the contract obligations as consultant, he approved progress claims, and he certified 100% completion of the project on July 31, 1986. Again referring to Tab 2 of Exhibit 1, General Condition 32.1 which provides that the owner and the consultant shall at all times have access to the Work, Mr. Garbutt conceded that he did in fact always have access to the Work.

With respect to the video taken by Dr. Lawless, it was Mr. Garbutt's evidence that he did not know that the video was being taken during the course of the Work and that Dr. Lawless never showed him the video during the course of construction. He did not learn of the video until this trial. At no time did the plaintiffs come to Mr. Garbutt with the video and point out something about which they were concerned. Mr. Garbutt does have a recollection that there was a conversation on site with one or both of the plaintiffs relating to the soldier course of brick and as a result of that conversation Mr. Garbutt gave orders to the contractor that the soldier course, which is the bottom course, be removed.

With respect to control joints, Mr. Garbutt conceded that this is a creature with a lot of different names, eg expansion joints. He explained that these joints permit two rigid surfaces to move independently. He conceded that the 'as built' drawings show one vertical control joint down the front of the south wall. Mr. Garbutt agreed that there are in fact no other control joints on the outside of the building. He conceded as well that the drawings do not provide for any expansion joints on the outside of the building. Mr. Garbutt conceded that control joints are an issue of design.

Mr. Nelson referred Mr. Garbutt to page 8 of his opening statement which has been filed, which notes in part that the specifications provided for soft-caulked exterior joints at the location of masonry bearing plates to allow for predicted movement at these joints, and that Emmons, more precisely its mason, sealed this joint with mortar rather than appropriate caulking, contrary to the specifications and good construction practice, and Mr. Garbutt did not catch the error. Mr. Garbutt considers this to be a construction deficiency. In his evidence here, Mr. Garbutt testified that he was not aware during construction that there had been discussions directly with the plaintiffs with respect to the control joint on the outside of the building. He testified that the joint that he is referring to at page 8 of his opening statement was the horizontal joint at each floor between the plate and the brick course. Mr. Garbutt explained that the drawings do not call for such a soft joint or control joint, but that the specifications do in fact call for such joints in the form of soft caulking. He referred to Exhibit 47, the Canadian Standard for Masonry Design, at page 15 thereof, section 3.2(e) which provides that drawings and specifications submitted with the application to build a building incorporating masonry shall indicate the details and location of control joints. Mr. Garbutt confirmed that the specifications at Tab 1 of Exhibit 1 were prepared for a two-story building. He was referred to Division 4, Masonry, section 2.2.2, which dictates, "provide control joints as dictated by structural design." He confirmed that his drawings did not call for any such control joints. He was referred to the same Division 4, section 2.1.1 which dictates, "provide 6 millimetre continuous expansion joint caulked continuously between steel support angles and the brick panels each floor." Mr. Garbutt confirmed that these were the control joints referred to in his opening statement and confirmed as well that the drawings did not show any such control joints. He reiterated that these particular joints were mortared rather than caulked and that he simply missed it on his inspection.

Mr. Garbutt confirmed as true his evidence on discovery where he testified that he was on the site once or twice a week while the block and brick were going up, but testified here at trial that that did not mean that he was present on the site every day.

Mr. Garbutt in his evidence here confirmed that he saw the ties being installed between the brick and the block courses. He confirmed that he never complained to the contractor with respect to the ties.

He confirmed that this project was proceeded with during the winter of 1985-1986. He may have had a conversation with the plaintiffs as to the advantages or disadvantages of winter construction. He was generally not fussy about winter construction and testified that the plaintiffs wanted to continue through winter.

At the completion of the Work, Mr. Garbutt prepared a list of deficiencies. He conceded that the fee of 5% of construction cost was a little on the low side but that was the deal and he stayed with the job until the end. He referred to Tab 12, Exhibit 1 as a deficiency list that he prepared, dated June 5, 1986. The building was totally completed by July of 1986 and the deficiencies which included some six masonry deficiencies were corrected by the end of July, 1986. It should be noted that one of the deficiencies that Mr. Garbutt noted under Masonry as of June 5, 1986 was "in general, poorly done."

At Tab 21 of Exhibit 1 is a document dated November 26, 1986, headed "Items to be completed". Mr. Garbutt did not know who prepared this document, and does not recall seeing it at the time.

At Tab 22 is a document relating to the first floor, which notes in part, "brick unsatisfactory". Mr. Garbutt testified that this document was not prepared by him.

It was Mr. Garbutt's evidence that his work on the project came to completion at the end of July, 1986 when he was last involved with the project. He was not involved again until September of 1993 when he received the plaintiff's lawyer's letter.

Referring back again to the ties that he had observed being placed while he was on site, he explained that there were two types being installed; firstly, corrugated ties were being installed on all four sides of the building where the wall was "solid"; while at the main entrance where there were cavity walls, cavity ties were being installed.

Prior to continuing with Mr. Mueller's cross-examination, there was put into evidence as Exhibit 76 the original of Tab 65, Exhibit 2 which allegedly is a note written by Mr. Emmons which is difficult to read.

Cross-examination by Mr. Mueller

Mr. Garbutt confirmed that his original design concept for wall was for solid masonry with no air space.

Mr. Garbutt agreed that what was constructed was a hybrid wall which had more aspect of a solid wall than a veneer wall. He explained that the wall has a honeycombed interior but we are not presently aware of the extent of the honeycomb. It is, as he described it, a wall with many cavities and fins, but one is unable to say what the proportion of fins to air space is.

Mr. Garbutt confirmed that he is familiar with the Codes and has been since 1957, but he has not sat on any of the committees that write the Codes, as Dr. Drysdale has done.

In terms of his qualifications, Mr. Garbutt explained that he has a general engineering licence which authorizes him to work in several fields. He does not have a special structural specialty, as does Mr. Rooney. In any event, Mr. Garbutt agreed that the plaintiffs had a right to expect from him a high level of quality, both with respect to his services as they related to structural engineering and otherwise. He explained that in the mid-1980's he was doing $25 - $50 million worth of construction value per annum, in both design and field review. At the present time, his work is primarily confined to consulting, as his eyes do not permit him to do drawings.

Mr. Mueller then turned to sections 34 and 35 of the *Professional Engineers Act*, and the Regulations under the *Professional Engineers Act*, section 88.1 which is contained at Tab 10 of Exhibit 1.

Section 34 provides that it is a condition of every Certificate of Authorization that the holder of the Certificate shall not offer or provide to the public services of professional engineering unless the holder is insured in respect of professional liability in accordance with the Regulations. That section and section 35 went in as Exhibit 77. It was in force in 1984. The Regulations at Tab 10 of Exhibit 1, section 88 thereof, provide that the minimum requirements for the holder of the Certificate of Authorization with respect to professional liability insurance are a policy limit for each single occurrence of not less than $250,000 and $500,000 per year in the aggregate. It appears that that Regulation came into force in January of 1985. Although the drawings with respect to this project were made in 1984, the actual construction occurred in 1985-1986, and Mr. Garbutt conceded that during the course of the construction of this project, notwithstanding the provisions of the *Professional Engineers Act* and the Regulations, he had no insurance. He explained that he was somehow under the impression that his Professional Engineers Association had told him that despite the general requirement, he was in a position to take out a single project policy and only needed to do so if his client specifically asked him to do so. Mr. Garbutt understood the nature of claims made to policy, and while this particular claim was made in 1995 at that point in time he had a company insurance policy but no personal coverage. The bottom line is that he had no insurance in 1985-1986 and first made that fact known to the plaintiffs' solicitors in 1996-1997.

Exhibit 56 is architect Paul Johnston's Folio 3 relating to the wall assembly. Mr. Garbutt confirmed that his drawings are contained at Tabs A and B. Not all of his drawings are contained at these Tabs. He confirmed that Mr. Roberts' drawings (Exhibit 9) are contained at Tab C.

Referring to the drawings, Mr. Garbutt first told Mr. Mueller that when he first received Mr. Roberts' drawings, he, Mr. Garbutt appreciated that the solid walls had changed to cavitied walls on those drawings. His words to Mr. Mueller were, "Yes, I understood that to be the case." He confirmed for Mr. Mueller his evidence at examination for discovery, July 9, 1999 questions 34 - 37 that while he, Mr. Garbutt intended it to be a solid masonry wall, the walls shown in Mr. Roberts' drawings were different in that there was a ¾ inch gap between the block wall and the brick wall. Question 37, "When he sent that to you, you accepted that?" Answer, "That is correct." Here at trial, while Mr. Garbutt acknowledged that Mr. Roberts had put into the drawing a ¾ inch gap between the block wall and the brick wall, Mr. Garbutt testified that a ¾ inch gap was standard in a solid wall. He explained here that the gap was needed to accommodate the vertical arms of the plates and it was intended that the ¾ inch gap be filled in entirely.

Mr. Garbutt was referred again to his examination for discovery, Questions 102 – 106, where he agreed that the brick ties should have been minimally horizontal but preferably down on an angle from the block to the brick. At discovery, he explained that the ties between the brick and the block wall should be preferably down and the purpose of that was so that the water could not meander from the brick tie to the block.

Mr. Garbutt conceded that the specifications called for a 19 millimetre air space between the block wall and the brick wall. He explained that 19 millimetres equals ¾ of an inch, which would be <u>twice</u> the thickness of the mortar joint of 3/8 of an inch. Mr. Garbutt conceded that if the specifications called for an air space, one would not want mortar in the air space. He conceded that avoidance of mortar in an airspace can be achieved by the careful mason in various ways, which will avoid mortar bridges. Mr. Garbutt confirmed that at examination he had testified that it was bad building practice by Emmons' masonry contractor, A. Dick, to permit the mortar bridges that it did, although here at trial he toned it down somewhat by saying the practice was “not the best”.

Mr. Garbutt conceded that the object during the course of construction is to read the plans with the specifications. He conceded that what is called for in one is called for by all. See General Condition 1.2 which states that the contract documents are complementary and what is required by any one shall be as binding as if required by all. A particular requirement need not be shown in both the plans and the specifications, but if it is shown in both it is even clearer. He conceded that Mr. Roberts’ plans showed an air space between the block and the brick, and that the specifications called for a 19 millimetre air space, or ¾ of an inch, to be maintained. He pointed out however that the air space in Mr. Roberts’ drawings is about the same width as the width of the joint between the bricks.

Mr. Garbutt conceded that the specifications at Tab 1 of Exhibit 1, Division 4, Masonry, article 2.1.1 required that a continuous 19 millimetre air space be maintained behind the brick throughout the project. He acknowledged to Mr. Mueller that the Roberts’ drawing showing an air space and the specification requiring a 19 millimetre air space behind the brick, might lead, in the absence of clarification, the mason to think that he, Mr. Garbutt, wanted the space between the brick and the block maintained, unless he, the mason received instructions to the contrary. In this respect Mr. Garbutt could not recall if he gave any such instructions. He said however that an intelligent mason would normally come to him for explanation in those circumstances.

Mr. Garbutt then discussed with Mr. Mueller the terms ‘inspection’, ‘supervision’, ‘review’ as between which Mr. Garbutt has differentiated during the course of his evidence. Mr. Garbutt explained that his was an oral contract with the plaintiffs. He conceded that his invoices to the plaintiffs refer to “supervision” and acknowledged Dr. Reed’s evidence that that is what the plaintiffs expected from him. He explained that the expression “supervision” was changed in the 1983 Code to the word “review”, in order to avoid misunderstanding. To him, the two different words were like the sun and the moon. He was unable to agree with Mr. Mueller that to the lay man “supervision” would be more onerous than “review”.

Mr. Mueller was referred to Tab 43, Exhibit 1, his invoice of January 3, 1986 to the plaintiffs, which refers in part to “supervision, 25% of total, 67% complete, $6,388.50”. The other part of his invoice refers to “design, 75% of total fee, 100% complete, $28,605.” Mr. Garbutt confirmed that although he refers to “supervision” in his invoice, he now says that what he was doing was “review” as the consultant. He acknowledged that during the course of his examination for discovery, at Question 220, the exchange as follows occurred:

> Q. And that the contract was, albeit oral, that you would design the building and supervise the work to whatever extent one might think?
> A. Yes, but not…
> Q. I am not getting into the extent to which you would supervise, but that was the deal?
> A. That was the deal.

Mr. Garbutt here at trial testified and I believe appropriately so that he had not during the course of that exchange at examination for discovery been allowed to address the extent of the “supervision” that was required of him under the contract.

Mr. Garbutt conceded that even after the Code changed the requirement from “supervision” to “review” he continued to use the word “supervision” and probably used that word when he was telling the plaintiffs what he would do under the terms of his contract with them.

Turning then to the documents, at Tab 2 of Exhibit 1, Article A-2 of the Contract relating to contract documents identifies the first contract documents as “specifications and details, dated June 27, 1985 as prepared by N.D. Garbutt, P.Eng. and signed by Emmons & Mitchell Construction Ltd.” Mr. Garbutt agreed that the specifications referred to in Article A-2 of the Contract are the specifications located at Tab 1 and signed by Emmons & Mitchell and dated June 27, 1985. Mr. Garbutt conceded that those specifications came from his office.

With respect to the reference to “details dated June 27, 1985” referred to in Article A-2, Mr. Garbutt conceded that the details being referred to are Mr. Roberts’ three small drawings

(Exhibit 9) located at Tab C of Exhibit 56, which bear the same date, June 27, 1985 and the signature of Emmons & Mitchell Construction Ltd.

With respect to the specifications, while conceding that they came from his office, Mr. Garbutt explained that they were prepared by a Mr. Jewel who was his specification writer at the time. Mr. Jewel had considerable experience in contract management and they were prepared as part of the contract management approach which was first being considered. In any event, Mr. Garbutt testified that he felt at that time that the specifications were consistent with his drawings. He conceded as well that the contractor needed both specifications and drawings in order to construct the building.

Mr. Garbutt confirmed that General Condition 1.6 (Tab 2, Exhibit 1) provides that in the event of conflicts between contract documents, specifications shall govern over drawings. He conceded as well that General Condition 25.5 provides that the contractor shall review the contract documents and shall promptly report to the consultant any error, inconsistency or omission he may discover.

With respect to General Condition 25.5, Mr. Garbutt conceded that his drawing which showed an irma style roof was patently inconsistent with the specifications that called for a standard built-up roof. Notwithstanding this inconsistency, Mr. Garbutt could not recall if the contractor ever discussed it with him.

Mr. Garbutt explained that the General Conditions have been in existence since the early 1970’s and are modified periodically. They are in a sense the ‘bible’ of the construction industry, and are prepared by the Canada Contract Documents Committee.

With respect to the specifications, Mr. Garbutt advised that they were available to him in the field office upon his attendances at the site. He would refer to them in order to perform his job obligations, and in addition to the specifications he relied on “fixed notions” which he conceded can at times be very dangerous. While carrying out his responsibilities, Mr. Garbutt also had available to him the General Conditions.

Mr. Mueller asked Mr. Garbutt why, having accepted Mr. Roberts' drawings with the air space between the block wall and the brick wall (Exhibit 9) did he not show flashings and weepers in his drawings. He explained that in a two-story building the absence of weepers and flashings was "okay", but once they had gone to a four-story building the best way to go was a solid wall which did not require flashings or weepers. Mr. Garbutt conceded that he was aware that his specifications called for weepers and flashings, and explained further that the two go together. When one is called for, the other is automatically called for as well. In other words, weepers and flashings must go together.

Referring to Exhibit 47, the Canadian National Masonry Standard, Mr. Mueller pointed Mr. Garbutt to Article 3.16.1, which requires flashing to be installed in masonry and masonry veneer walls in various locations, including beneath the weep holes, and directed Mr. Garbutt to Mr. Drysdale's evidence that this section applied to both solid and veneer walls. Mr. Garbutt appeared to agree with this proposition. Mr. Mueller pointed Mr. Garbutt to the fact that it is his evidence that what he intended was a solid wall and at the same time his specifications called for flashings and indeed the Canadian Standards call for flashings. Again, he appeared to agree.

Mr. Mueller pointed Mr. Garbutt to section 3.13.1 under the heading, 'Drainage Walls', which provides that weep holes at least 10 millimetres in diameter shall be provided immediately above the base flashing in veneered walls having bearing support, and in cavity walls at horizontal spacing. Mr. Garbutt responded that he provided for flashings, thinking it was going to be a solid wall.

Mr. Mueller put to him that he actually had in mind a panel wall, and he agreed, saying that panel walls had been utilized under the Code since the 1960's. He then testified that in his mind, rightly or wrongly, a solid wall did not require flashings and weepers, notwithstanding that flashings and weepers were called for in his specifications. It was his evidence that he had authority to change the specifications. He conceded that there was no written change order to the specifications. He conceded that it would be good practice to have such a change order in writing, and that if there was not a change order there could be disputes. He then testified that he verbally authorized the contractor Emmons not to use flashings. He admitted that there was no writing in this respect, but that writing was required by General Condition 2.2, which provides that additional instructions must be by way of drawings or in writing. Mr. Garbutt then conceded that he was not sure if he gave any instructions at all to the contractor to eliminate the flashings. He can't remember.

Mr. Garbutt was asked whether he recalled any discussions with Emmons with respect to the wall being solid or having a space between the block and the brick. He responded initially that he instructed Emmons that the wall should be solid, and that discussion would have been had with Mr. Sinclair. He then backtracked to some extent, saying he believed he did have such a conversation but cannot specifically remember such a conversation. He said, "It had to have been done." He does not recall the name of the masonry supervisor. He confirmed in his evidence that at one point he went to Emmons and asked him to find out how the wall had been built. He eventually received the note from Emmons previously referred to, indicating that the wall was solid. Mr. Mueller put it to him that we now know that the collar joint had not been filled and Mr. Garbutt agreed with that proposition. As well, he agreed with the proposition that the statement that he received from Emmons to the effect that the collar joint was filled and that adjustable ties had been used was inaccurate.

Mr. Garbutt knows at this point that the collar joint was not filled and there were no adjustable ties between the bricks and the blocks.

Mr. Garbutt was referred to Exhibit 34, Dr. Drysdale's comments with respect to the wall. At bullet three, Dr. Drysdale comments that the thickness of 210 millimetres is satisfactory for the top three stories but must be 290 millimetres for the bottom story because the building height exceeds 11 metres. Mr. Garbutt appreciated that Dr. Drysdale relied on section 5.3.3.2 of the National Masonry Design Standard (Exhibit 47) in arriving at this conclusion, but disagreed with the conclusion. Although the height of the wall does exceed 11 metres by some 1.8 metres, Mr. Garbutt was of the view that section 5.3.3.2 of the Standard did not apply because the wall was supported at every floor.

Mr. Garbutt was referred again to Exhibit 34 wherein Dr. Drysdale expresses the opinion in bullet five that the walls are part of the load-bearing system, and load-bearing walls are not panel walls. Mr. Garbutt persisted and testified that the wall was not holding up anything such as floors; they carried only wind loads. He conceded that he had a difference of opinion with Dr. Drysdale which was fundamental.

Eventually, Mr. Garbutt clearly conceded that what he wanted was a solid wall, and that he did not get one. It was his evidence that neither did he get a veneer wall. What he got was a bastard wall.

Mr. Garbutt was referred to the specifications, Division 4 and particularly section 2.2.3 which requires all block walls to be bonded at intersections. It was his evidence that this meant that they were to be bonded at the corners where one wall meets another.

He was referred to section 2.2.5 which requires anchors to be hot-dip galvanized to suit the conditions. It was his evidence that this simply directed that if anchors were going to be used, they had to be hot-dipped. It did not mean that the block wall had to be anchored to the vertical columns.

At Tab 106 of Exhibit 2, Mr. Johnston's report, at page 4, it is noted that the backup four inch masonry wall butted up to but did not engage the vertical columns, and that no tie-ins or connections to the columns were discovered. Mr. Garbutt agreed with Mr. Johnston's observation and said that there was no need for any kind of a tie-in insofar as the block wall was tied to the brick wall which passed in front of the vertical columns, and thus the entire wall was satisfactory for wind forces.

Mr. Johnston makes the same comment in Exhibit 56 at page 23 of his report, and Mr. Garbutt in his evidence gave the same response.

With respect to parging of the inner block surface, where the wall is intended to be a solid wall, Mr. Garbutt explained that such parging was required in the early 1960's when insulation was being placed directly against the block wall, but here it was not necessary to parge.

Mr. Garbutt was referred to Tab 106 of Exhibit #2, page 4, again Mr. Johnston's report, wherein Mr. Johnston observes that the infill masonry at the interior block wall to the underside of the spandrel plates were completed with brick, cut block or block pieces as suited the dimension, and generally poorly mortared. Mr. Garbutt responded that his intention would have been for an expandable filler between the top of the block wall and the bottom of the spandrel plate, in order to stop air flow. He conceded that without such filler and with weep holes, air could come into the building and might affect the heating.

Section 9 of Division 4 of the specifications (masonry) provides for a job mock-up where a sample brick wall is constructed of mortar and brick. The units are to show colour, range, texture, bond, jointing for approval. If the sample wall is not approved it is to be removed and reconstructed until approval is received. The approved wall should be retained until project completion. Mr. Garbutt confirmed that this mock-up is to be constructed by the contractor and in the normal course, he, the consultant would be able to look at the mock-up to determine whether the air gap is correct and the ties are properly in place, etc. Mr. Garbutt conceded that no mock-up as such was constructed by the contractor, but that the video of Dr. Lawless did show one of the initial pieces of wall construction. However, the job minutes do not refer to any mock-up and Mr. Garbutt explained that in practice the so-called mock-up is built as part of the building. He conceded that they did not go by the specification on this particular occasion, nor is he aware of any agreement with the contractor whereby the initial wall construction would be treated as the mock-up. However, he did treat it as a mock-up. He explained that while he did not stand there and watch them build the particular portion of the wall that is shown on the video, he was eventually present and told the contractor to take it down because the soldier course was out of plumb. It was his evidence that he observed at that time the distance between the block wall and the brick wall to be something less than 19 millimetres, and he observed the gap to be fully filled with mortar. He believes Dr. Lawless's video shows it filled in, but there is some dispute over this.

Tab 49 of Exhibit 1 is a site meeting minute dated November 6, 1985 which refers to a section of the brick wall that had been laid 'yesterday', being rejected, and noting that the masonry contractor, Abe Dick will rectify. Mr. Garbutt conceded that this was an accurate minute but that it was not necessarily the same piece of wall that we have been discussing.

Mr. Garbutt conceded having regard to Tab 29 of Exhibit 1, the Emmons invoice dated November 29, 1985, that they were well into the brick work by that time and that by Tab 31 dated January 29, 1986, again the Emmons invoice, the masonry work was 90% completed. Still, Mr. Garbutt was unable to say that the section of brick wall being rejected in Tab 49 was the section he considered to be a mock-up. He conceded that he did not abide by the General Condition that the piece torn down be reconstructed until approved.

Again with respect to the weepers and flashings which Mr. Garbutt believes he instructed the contractors to remove from the walls, he conceded that had he known the wall was not a solid wall he would not have authorized the removal.

Mr. Mueller reviewed with Mr. Garbutt correspondence between Emmons and Dr. Reed, dated September 11, 1986, copied to Mr. Garbutt (Tab 17), Exhibit #1 and Emmons' correspondence with Abe Dick Masonry dated September 15, 1986 (Tab 19), and Emmons' letter to the plaintiffs dated November 20, 1986 (Tab 20), all of which refer to deficiencies in the masonry. Mr. Garbutt acknowledged that he knew at the time that the walls were not attractive from the plaintiffs' point of view and was aware that a credit was being given from the masonry contract to the plaintiffs.

Referring to his list of deficiencies dated June 5, 1986 (Tab 12), which notes that the masonry in general was poorly done, Mr. Garbutt acknowledged that he knew there were general problems. He conceded to Mr. Mueller that knowing that one of the sub-trades was not getting the work done properly, he would think it advisable to advise the contractor to be a little more vigilant. He agreed to some extent with Mr. Mueller's proposition that a prudent consultant should be more vigilant with someone who has done the work poorly.

Mr. Mueller then reviewed with Mr. Garbutt the duties of the consultant under the General Conditions of the Contract, having extracted from him an acknowledgement that the consultant's duties under his personal contract with his clients could conceivably be higher than the duties under the General Conditions.

In any event, referring to the General Conditions in Tab 2 of Exhibit 1, Mr. Mueller pointed out the following General Conditions which Mr. Garbutt conceded fairly stated his obligations under the Conditions.

- General Condition 3.4 – the consultant will visit the site at intervals appropriate to the progress of construction to familiarize himself with the progress and quality of the work and to determine in general if the work is proceeding in accordance with the contract documents; however the consultant will not make exhaustive or continuous on site inspections to check the quality or quantity of the work. In this respect Mr. Garbutt agreed that he had to be on the site a couple of times a week to "handle all of the disciplines including architectural and structural". He noted that the wall was in effect a hybrid of both architectural and structural having regard to the wind force requirements, and that there was what he referred to as a 'patent conflict' between him and Mr. Roberts, the architect with respect to the wall. At Tab 11 is a sample General Review Commitment Certificate which lists both the engineer's obligations and the architect's obligations under the *Professional Engineer's Act* and the *Architect's Act* with respect to a project. Under the engineer's obligations it is noted that he shall make periodic visits to the site to determine on a rational sampling basis whether the work is in general conformity with the plans and specifications for the building, and shall record deficiencies found during site visits and provide the client, the contractor and the owner with written reports of the deficiencies and the actions that must be taken to rectify the deficiencies. Mr. Garbutt agreed that this was similar to the obligations under the General Conditions with the added component of 'rational sampling.'

There then ensued a discussion with respect to architect Roberts. Mr. Garbutt admitted that he hired him and paid him, and took responsibility for the work that he was performing as a sub-consultant. He conceded that he was probably remiss in not obtaining Mr. Roberts' Certificates, and in fact does not know if Mr. Roberts inspected or not. Mr. Mueller asked Mr. Garbutt whether with respect to the air gap, the number and quality of ties, the flashings and the weepers, would he expect Mr. Roberts to have looked into these matters as opposed to himself. Mr. Garbutt responded that normally he would have covered these items on his own. He commented that he did not trust others whom he had delegated, however he agreed with Mr. Mueller that he thought the primary responsibility for these matters and for inspections under the Code lay with Mr. Roberts. He conceded that while he was entitled to rely on Mr. Roberts to inspect the wall, he did not in fact rely on him entirely. He did expect Mr. Roberts to attend once or twice a week for the purpose of checking the gap and the ties etc.

Mr. Garbutt conceded that he never talked to Mr. Roberts about the quality of the work being performed but only progress payments. He does intend to pursue him in an action for negligence in the event this particular law suit goes bad for Mr. Garbutt and has put Mr. Roberts on notice of that fact.

Mr. Garbutt acknowledged that while it was necessary for him and Mr. Roberts to work together as a team, they never did sit down and talk about the wall. Mr. Garbutt trusted him to do his job (this is somewhat contrary to his previous testimony).

Referring to the shelf angles, which had been cut off on the east and west walls, Mr. Garbutt explained that it would have been easy not to have seen this as it could have been done in approximately half an hour. He explained that the steel was in place before the brick work started and it may have been that the shelf angles at each floor were cut off as the masonry reached each floor. He conceded that if all of the shelf angles were taken off at the same time at all floors, before the wall was started, then that could have been observed. He conceded Dr. Lawless had testified that the cut off plates had been left on the ground (see Exhibit 14), some of which had been picked up by Dr. Lawless, but his evidence was that he, Mr. Garbutt, had not seen them. Mr. Garbutt did testify that during the course of construction it was evident to him that no weepers were being provided in the walls.

At Tabs 46 and 47 of Exhibit 1 are some rather disagreeable letters between Mr. Roberts and Mr. Garbutt. In his letter dated February 25, 1987 Mr. Roberts is looking for payment of his invoice and is complaining about the fact that Mr. Garbutt had sent <u>him</u> an invoice. In his letter to Mr. Roberts dated February 27, 1987 in reply, Mr. Garbutt says the following,

> Thank you for your letter of February 25, 1987 in which you admit that you performed no work beyond the original base contract. Your current invoice is based on the final contract amount. The difference between final and base contract amounts at 1% representing essentially the amount of your current invoice, for which you performed no services. You were not requested to do services beyond the base contract stage due to time delays on your part at the base stage.

Mr. Garbutt explained in his evidence that the base contract stage referred to in his letter meant the work that was required to be done for the building permit. In the face of that, Mr. Mueller asked him whether in effect Mr. Roberts had been cut off from the project after the base contract stage, and Mr. Garbutt denied that to be the fact. He then conceded that from the point of view of the General Conditions, Mr. Roberts was not required to inspect but he did have an obligation under the Building Code which was similar in scope

Carrying on his discussion with Mr. Garbutt with respect to his obligations as consultant under the General Conditions, the following General Conditions were pointed out after Mr. Mueller pointed to Definitions 13 and 14 under the General Conditions, which refer to Certificates of Substantial Performance and Total Performance to be issued by the consultant.

- General Condition 14.3 provides that the consultant will after the receipt of an application from the contractor for a Certificate of Substantial Performance of the work, make an inspection and assessment and notify the contractor of his approval or the reasons for his disapproval of the application. When the consultant finds that substantial performance of the work has been reached, he will issue a Certificate noting the date of substantial performance.

- General Condition 14.4 provides that immediately following the issuance of the Certificate of Substantial Performance of the Work the consultant will issue a Certificate for Payment of Hold Back Monies.

- General Condition 14.7 provides that the consultant will after the receipt of an application from the contractor for payment upon Total Performance of the Work, make an inspection and assessment of the work to verify the validity of the application, and will thereafter notify the contractor of his approval or the reasons for his disapproval of the application. When the consultant finds that Total Performance of the Work has been reached, he will issue a Certificate of Total Performance of the Work and certify for payment the remaining monies due to the contractor under the contract, less the hold back monies, which are required to be retained. In this regard, Mr. Garbutt conceded that in practice no Certificate of Total Performance is ever issued, and conceded in this case, referring to Tabs 26 through 37, only ordinary monthly Certificates were issued by Mr. Garbutt and that he was in fact never asked for a Certificate of Final Completion nor did he ever issue one.

- Mr. Mueller pointed Mr. Garbutt to General Condition 3.8 which provides that the consultant will have authority to reject work which in his opinion does not conform to the requirements of the contract documents, and Mr. Garbutt agreed with that provision.

- He agreed as well with General Condition 33.1 which provides that defective work rejected by the consultant as failing to conform to the contract documents must be removed promptly and replaced. Mr. Garbutt agreed that the procedure was to inspect, reject and replace.

- Mr. Garbutt acknowledged as well General Condition 14.10, that no payment made by the owner under the contract shall constitute an acceptance of work which are not in accordance with the requirements of the contract documents, and noted that this meant that even if the consultant makes a mistake and misses defective work and it gets paid for, the contractor still has to make it right.

Mr. Mueller then discussed with Mr. Garbutt the parapet, which is shown in the third of the three small Roberts' drawings at Tab D of Exhibit 55 (Exhibit 9). Mr. Garbutt agreed that he was not present when the parapet was constructed but that he inspected it afterwards. He conceded that at the time of his inspection he would not be able to see what was underneath the flashing.

Mr. Mueller discussed with Mr. Garbutt what he saw to be some seven problems relating to the parapet.

1) Mr. Garbutt agreed with the following: the third Roberts' drawing shows a width of 200 millimetres (eight inches) across the top of the parapet from which measurement it could be determined that relatively it was intended that the parapet rise up an equal distance above the gravel, that is 200 millimetres or eight inches. Mr. Roberts' drawing was done at a time when it was intended to place three inches of insulation on the roof as opposed to the six inches that was eventually placed. Mr. Garbutt does not recall any communications with anybody with respect to reducing the height of the parapet by adding insulation. It was however not his intention to decrease the height of the parapet by the addition of the insulation. In any event, as it turned out while the drawing shows a parapet height of some eight inches, it was actually built to a height of two inches above the gravel. He conceded that this would have been visible to the consultant, but he did not convey this fact to the contractor.
2) At Exhibit 55, Tab H is Mr. Kendall's report containing photographs five and six, showing the top of the parapet. Picture five shows a space between the horizontal board and the soldier course of bricks that was never intended, having regard to the third drawing at Tab D, which shows no such space. Mr. Garbutt agreed that no rational

tradesman would want to leave that kind of an opening and conceded that this deficiency could not have been observed after the parapet was completed. He conceded that any moisture getting in through such an opening on the north and south walls would fall down to the floor plate, but could go straight to the bottom of the building on the east and west walls where there were no floor plates.

3) Photo five shows the membrane stopping some one third to one half way across the top of the parapet, which Mr. Garbutt conceded was poor workmanship.

4) Photo six shows that the metal turndown extends only approximately ½ inch below the top of the brick, as opposed to a good two inches recommended by Kendall.

5) Division 7 of the specifications for the roof, section 3.6.1 provided for the carrying of four-ply felt and asphalt membrane flashings up the surface of the wall to a point not less than 200 millimetres (eight inches) above the surface of the roof. Mr. Garbutt admitted that this was as required by the specifications and would have been better than what was actually constructed.

6) Mr. Kendall testified that the coping was flat rather than sloped toward the roof, for the purpose of draining water, and Mr. Garbutt agreed with that.

7) Mr. Garbutt acknowledged that Mr. Kendall had said that the horizontal and vertical coping should have been locked, and Mr. Garbutt agreed with that proposition.

Finally, Mr. Garbutt agreed that the construction of the parapet was a departure from the specifications and plans, and that it represented simply bad practice in general.

There then ensued between Mr. Mueller and Mr. Garbutt a discussion with respect to the roof. At Tab B of Exhibit 55, are found two Canadian Roofing Contractors' Association specifications, 1) from 1978 until the 1986 revision, and 2) the 1986 revision. Mr. Garbutt acknowledged that the 1978 specifications "strongly recommended minimum slope for proper drainage is 1:50" and that the 1986 revision in August provided "positive drainage is a requirement for the protected membrane roofing." Mr. Garbutt conceded that the irma roof is a protected membrane roof and that he was aware of the Canadian Roofing Contractors' Association specifications during the course of this work. He testified as well that he was aware that the insulation was not to be laid into the hot bitumen, so that in total according to his evidence he was aware that for irma roofs there should be positive drainage and also the insulation should not be allowed to adhere to the underlying bitumen.

At Tab C of Exhibit 55, is a specification with respect to organic felt membrane which was the type of felt used on the project, which calls for a minimum slope of 1:50, which is equal to a 2% slope, and Mr. Garbutt agreed with this proposition.

Mr. Mueller then turned to Tab K of Exhibit 55, which is the promotional and specification material from Dow for the roofmate insulation to be used with a protective membrane roof. Mr. Garbutt confirmed that it was this insulation that he specified. Mr. Mueller pointed to page three of Tab K, under the heading “Design Considerations”, which provides that membrane flashings should be extended well above the expected high water level, and that roof drains should be at the low points in the roof. Mr. Garbutt said he knew all of this at the time. Turning to page four of Tab K, Mr. Mueller pointed to the fact that Dow recommended that if a bituminous membrane is used (as was the case here) the final flood coat must be allowed to cool before applying the insulation loose, and that it is the roofer’s responsibility to ensure that bonding of the insulation to the bitumen does not occur. Mr. Garbutt acknowledged that he knew this at the time of construction, and that he would have expected a roofer to have known all of this, particularly one that was a member of the Roofers’ Association. Mr. Garbutt acknowledged that the specifications for the roof at Division 7 never changed throughout the project.

Then some difference of opinion arose among counsel as to whether or not Tab K, the Dow material, was effective at the point in time when this roof was being constructed in 1985-1986. I have resolved that dispute. Mr. Kendall testified for the plaintiffs. He is a roofing expert. My notes of his evidence (page 3 of the typewritten summary, page 7 and page 26 of the summary which was evidence under cross-examination) clearly show that on Mr. Kendall’s evidence the specification was in force in the early 1980’s, well before the construction of this roof.

Again, Mr. Garbutt conceded that during the course of the project he knew of the CRCA recommendation for a 2% slope in an irma style roof and knew that adherence of the insulation to the bitumen was advised against by Dow.

Mr. Garbutt acknowledged that in his specifications, Division 7 – Thermal and Moisture Protection, he adopted the standards of the CRCA Roofing Specifications. Section 2 of Division 7 under the heading, 'Roof Drainage' says to provide all roofs with roof drains and for inverted roofs, "slope structures to drains at least 5m/100mm gradient". That is an obvious mistake and would result in an almost vertical slope. It likely means 5 millimetres, that is 5mm/100mm gradient for a 2% slope. Mr. Garbutt testified that when the contractor read that particular specification, the contractor should have sought his guidance. He has no recollection of the contractor doing so. He says had the contractor inquired, he would have told the contractor to make the roof flat because it was temporary and waiting for two additional floors. In any event, Mr. Garbutt conceded that this particular specification was never withdrawn.

With respect to section 3, for the standard built-up roof, Mr. Garbutt conceded that apart from the reversing of the positions of the membrane and the vapour barrier, the section is for the most part equally applicable to irma and built-up roofs.

Mr. Garbutt acknowledged Mr. Kendall's and Mr. Johnston's evidence that in order to provide the required roof slope, the slope could have been put into the roofing materials as opposed to the slab. He acknowledged that could have been done in 1985-1986 but it would have been expensive. Mr. Garbutt acknowledged that he anticipated there might be two additional floors added to the building some time in the future, and he treated this as a design requirement and was thus concerned that if the roof slab was sloped and eventually became the fifth floor, then the fifth floor would be sloped. He acknowledged that if such was the case, what could in the future become the fifth floor could have then been flattened out by a pouring of further cement, but that in such a case the underlying steel structure would have to be strengthened.

Mr. Garbutt acknowledged that he had no discussions with the plaintiffs as to when they might want to add the fifth and sixth floors; they simply indicated to him that they would like the option.

Mr. Garbutt did acknowledge that Mr. Kendall in his evidence expressed the belief that no one would have been comfortable leaving organic felts in place without the appropriate slopes for more than six months because of anticipated water damage to the organic felts.

Mr. Garbutt explained that he did not inspect the roof work while it was in process. As well he acknowledged Kendall's and Johnston's evidence that the purpose of avoiding adherence of the insulation to the membrane is that in the event of flooding, the insulation rises, pulls away from the membrane and in the process damages it. Mr. Garbutt agreed that this could happen over time, but there would be no such problem if it was only a temporary installation. He agreed that long-term, one did not want insulation imbedded in the bitumen. He said it was a judgment call on his part.

Mr. Garbutt acknowledged Mr. Kendall's evidence that a good irma roof with slope could last 30 years. He acknowledged that in this case a new roof was installed after 13 years but he would not acknowledge that it had been established that the membrane at the end of 13 years was totally defunct, as was the judgment of Mr. Kendall.

Again, Mr. Garbutt acknowledged that he never inspected the roof work while the work was actually being done, and the first time that he saw it was on the video. He expressed the opinion that it probably took no more than two or three days to complete the roof work. He does not recall requesting the contractor to notify him when the roof work commenced. He conceded that the video shows substantial adherence of the insulation to the bitumen and conceded that from the standpoint of attempting to avoid that adherence, it was a bad job.

At Tab 13, Exhibit 1 is a letter from Emmons to Quintal, the roofer, dated September 4, 1986 setting out the following deficiencies:

1) break in flashing on corner near rust check;

2) hole in flashing near roof hatch;

3) caulking on flashing is poor.

Mr. Garbutt explained that this letter went to Quintal as a result of Finkel and he doing an inspection of the roof and noting these deficiencies. He conceded that these problem areas in September of 1986 were generally the same problem areas observed by Kendall in subsequent years. It was his evidence that at that time, September 4, 1986, the deficiencies noted were in fact repaired. With respect to the deficiencies noted in September, 1986 Mr. Garbutt has no recollection of whether he brought them to the attention of the plaintiffs. They may have been mentioned in site meeting minutes but he cannot recall. Having regard to Mr. Kendall's findings, he is of the view that the deficiencies noted in the September 4, 1986 letter were repaired and other problems developed in the same areas.

There then ensued another discussion between Mr. Mueller and Mr. Garbutt with respect to the shelf plates.

Again, Mr. Garbutt conceded that the requirement for the shelf plates at each floor were shown on his drawings and in his specifications. Indeed, it is clear that initially the contractor did install the steel with shelf plates at each floor on all four sides of the building. He conceded as well that they were a Code requirement for a panel wall, and indeed that the masonry design requirements of the National Standard of Canada required the plates for all types of walls.

It was Mr. Garbutt's evidence that he had no knowledge whatsoever that the plates had been removed on the east and west walls of the building, and would never have permitted their removal. He went so far as to agree that he could have lost his "ticket" if he had permitted the removal of the plates without notifying the appropriate building authority.

Mr. Garbutt agreed that architect Paul Johnston, in his report Exhibit 56, relating to the building envelope gave a fair description and explanation at page four as to what happened with respect to the removal of the plates. The explanation is this. On the north and south sides of the building, the perimeter beams at every floor would need to be depressed to accommodate the ends of the intermediate steel floor joists that ran from the north side to the south. These particular joists have an end reinforcement or "shoe" to handle the forces at their anchorage points. On this particular building, they are four inches high. In other words the steel beams on the north and south sides of the building would need to be four inches lower than the elevation of the steel deck and concrete floor slab of the typical floor structure. There is no detail that shows that any of the steel beams were lowered to accommodate the shoe below the deck. As a result, as the courses went up the last brick course underneath the plate on the north and side sides of the building fit properly under the plate but did not fit under the plate on the east and west sides of the building, but rather bumped up against the plate on the east and west sides of the building. As a result, by that point in time, the plates had to be cut off on the east and west sides of the building.

Mr. Garbutt agreed with Mr. Johnston's evidence that the problem could have been properly remedied either in the structural steel shop or in the field. Mr. Garbutt acknowledged that Division 5 of the specifications, section 1.3 required the contractor to submit shop drawings which should have dealt with the problem. As well, section 1.10 provides that the contractor shall obtain written permission of the engineer prior to field cutting or altering of structural members. As well, section 7 of Division 5 – Metals, provides that before beginning fabrication, the contractor shall field check all dimensions and shall report any major variance from the plans to the contractor's consultant. Again, General Condition 11 relating to changes in the work provides that any changes in the work have to be by written order signed by the owner or his consultant.

Mr. Garbutt repeated that at no time did he ever tell Dr. Lawless that the building was okay without the plates, because he was not aware that the plates had been cut off. He theorizes that someone had to pay for the removal of the plates, and yet there was no change in the contract price.

Mr. Mueller and Mr. Garbutt then had further discussions about the air gap between the brick and the block walls.

Mr. Garbutt acknowledged once again that Division 4, section 2.1.1 relating to the brick work required that a continuous 19 millimetre air space be maintained behind the brick throughout the project. He conceded once again that he cannot recall ever looking at the gap between the brick and the block walls, except with respect to what he treated as the original mock-up, and what he saw in the video. He acknowledged that Mr. Johnston's measurements of the gap between the brick and the block went from ¾ of an inch at the bottom to 1 ½ inches at the top. He acknowledged that according to Mr. Roney, the gap exceeded one inch even at the second floor.

A further discussion was had with respect to the ties.

Division 4, section 2.1.1 required that "units shall be tied every other course as per drawing". Mr. Garbutt explained that the ties were to be two courses apart in the brick courses, vertically, and as far as horizontal distance between the ties, there should be a tie every four square feet between the brick and the block. He acknowledged as well that section 2.8 of Division 4 required that anchor and bond masonry conformed with the National Building Code. In passing, he confirmed that Division 4, section 6 required that the wall flashing be copper Fibreen.

Again relating to the ties, Mr. Mueller and Mr. Garbutt then reviewed Exhibit 53 which is a collection of Canadian Building Digests issued by the National Research Council, and Technical Notes issued by the Brick Institute of America. Mr. Garbutt acknowledged that these papers were a tool for the construction industry.

Technical Note 21 revised, issued by the Brick Institute of America at page four thereof under the heading 'Bonding' suggests that there should be at least one metal tie for each 4 ½ square feet of wall area, that the maximum vertical distance between ties should not exceed 24 inches and the maximum horizontal distance should not exceed 36 inches. Mr. Garbutt acknowledged this as a good general indication of the industry practice, and noted that it was not as stringent as his requirement of one tie every 4 square feet.

The last Technical Note, 44B, again issued by the Brick Institute of America at page two, Figure 1 shows a corrugated tie which is the equivalent of our strip tie, Exhibit 38. Mr. Garbutt noted that it is relatively flexible as compared to our Exhibit 36 which is quite stiff end on end (compression), and suitable for a “wind tie”.

At page two of this Technical Note, it is noted that corrugated ties are typically used in low rise residential veneer over wood frame construction, and are not recommended for construction incorporating brick veneer over steel studs, masonry backed cavity walls, multi-wythe walls, or grouted masonry walls. Mr. Garbutt conceded that what the Technical Note was saying was that if one were to take the mortar bridges out of our wall, then corrugated ties should not have been used.

At page three of the Technical Note is shown a typical ladder type tie. Mr. Garbutt concedes that subsequent investigation found the ladder tie in the upper reaches of our wall, but that generally what was found in our wall were corrugated ties.

At page four of this Technical Note, Figures 5 and 6 show examples of adjustable ties. Mr. Garbutt explained that adjustable ties were meant to accommodate the varying levels of joints as between the brick courses and the block courses. If the joints are out of kilter with each other, they can effectively be gotten back into kilter by using adjustable ties.

Relating to the finding in our building, Mr. Garbutt acknowledged that bent ties were found, that is ties were bent between the brick and the block to accommodate the fact that the joints were out of kilter. He conceded that if the ties got to a point where they were acting against wind forces, then the fact that they were bent would be a negative factor, but again it was his evidence that he felt that a solid wall was being built. He testified that if in fact he had been on the site continuously and had viewed day by day the gaps in the wall, he would not have accepted it.

Mr. Garbutt acknowledged the previous evidence of the plaintiffs' experts that there were no ties in the top 30 to 40 inches of the wall, and that the location of ties was skimpy elsewhere.

Mr. Garbutt was asked whether he agreed that whatever inspection practice was required of the consultant, the consultant during the construction project would be looking for ties in the wall. Mr. Garbutt responded that if what was being built was a cavity wall, that would be true but it was not true in the case where a solid wall was being constructed. Mr. Garbutt was then confronted with his examination for discovery testimony where the following exchange took place:

Q. *You and I understand one per four square feet?*
A. *Four square feet, yes.*
Q. *On average. Did you happen to notice whether or not that standard had been adhered to by Abe Dick Masonry?*
A. *They appeared to be adhering to it, yes.*
Q. Did you actually look for brick ties?
A. Of course you look for brick ties.

Now, this evidence is completely contradictory to the evidence that Mr. Garbutt gave on this narrow point at trial. It has to reflect on his credibility to some extent, particularly to the extent that he admitted here at trial that those answers at the time were true.

The discussion then turned to control joints and expansion joints.

Section 2.1 of Division 4 provides for six millimetres of continuous expansion joint, caulked continuously between the steel support angles and the brick panels each floor. Mr. Garbutt acknowledged that this was called for by the specifications, that the contractor failed to comply with the specifications and that he simply missed their non-compliance. He says he looked at the wall but it did not register. When pushed, he testified he saw they were not there but did not realize the significance of it.

Section 2.2 of Division 4 requires the contractor to provide control joints as dictated by structural design. Mr. Garbutt conceded that there were no control joints shown on the drawings. He explained that they were in the specifications and he did not anticipate any problems. He does not recall whether he had any discussion with the contractor with respect to vertical control joints. He acknowledges Mr. Roney's evidence that he recommends control joints at the corners.

A discussion then ensued with respect to weepers and flashings. As previously noted, at Division 4 section 6 of the specifications the wall flashing was to consist of copper Fibreen. Having regard to Exhibit 47, the Masonry Standards and particularly sections 3.13.1 and 3.16.1, Mr. Garbutt agreed that flashing is required in masonry and masonry veneer walls at various locations; that is, whether solid or cavity. Mr. Mueller reminded Mr. Garbutt of his previous evidence that automatically where there is a flashing there must be a weeper. Mr. Garbutt agreed with the caveat that there must be a weeper if there is something to drain. He then testified that he probably gave verbal authorization to the contractor to leave out the Fibreen flashings. He simply does not remember.

With respect to the mortar fins extending from the brick wall to the block wall, Mr. Mueller referred Mr. Garbutt to Exhibit 53, Technical Note 21C, which provides that it is vital that the cavity be kept clean of mortar droppings because if mortar forms into the cavity it may form bridges for moisture passage, or it may fall to the flashing, blocking the weep holes. Mr. Garbutt agreed with that statement, provided it was referring to a cavity wall. He, on the other hand, was building a solid wall and in his view the contractor mis-constructed his solid wall. At page two of Technical Note 21C, recommendations are made as to how to avoid mortar fins between the two wythes. Again Mr. Garbutt agreed with the recommendations provided one was building a cavity wall.

Mr. Mueller and Mr. Garbutt then had a discussion with respect to Mr. Garbutt's theory as illustrated by his drawing at Tab 12 of his Exhibit Book 74, where he suggested that looking down on the cavity with the mortar bridges extending between the brick and the block, one might assume it was solid infill. Mr. Mueller suggested that if one simply looked at the last course of brick and block laid, one would immediately recognize that the bridges or fins are not solid along the entire length of the course. Remember Mr. Johnston's evidence that the fins bridged approximately 30 to 50% of the wall. Mr. Garbutt would not agree but Mr. Mueller makes his point.

Mr. Mueller and Mr. Garbutt then discussed the fact that in Mr. Garbutt's design there was no anchor of the block wall panel to the vertical steel columns or the horizontal steel beam at the top. Mr. Garbutt agreed that that was the design and the manner in which it was built because he believed he was building a solid wall, and that the bricks went past the upright columns. He conceded that when retrofitting one must follow the present Code rather than the Code under which the original building was constructed. He appreciated that both Dr. Drysdale and Mr. Johnston take the position that if the brick wall is taken down then the block wall has got to be brought up to Code, and will not be allowed to be free standing.

Referring to Exhibit 29, Dr. Drysdale's report, Mr. Garbutt was aware of the fact that Dr. Drysdale identified as a construction fault a lack of positive connection of masonry to the steel frame, and at page five of the report said that the top two feet of the brick must be attached where there is no block back-up wall. Mr. Garbutt was aware as well that at the top of page six of his report of May 6, 2002, Exhibit 29, Dr. Drysdale commented that there is no provision for attaching the masonry to the steel frame, and that relying on friction is a very doubtful practice as axial load will be highly variable, including the potential for no axial load. In addition, wall anchorage to satisfy earthquake loading conditions must resist very high loads. While Mr. Garbutt did not agree with Dr. Drysdale's comments, he did realize that Dr. Drysdale was talking about the fact that the block wall was free standing.

At page one of the report, Dr. Drysdale comments that the stability of the building against lateral loading depends on the infill concrete block masonry being built tight to adjacent columns and to the underside of the building edge beams. Mr. Garbutt acknowledged that that was what Dr. Drysdale was saying, but did not adopt his opinions.

Mr. Garbutt was also aware that Mr. Roney's engineering report prepared for the defendant Emmons dated April 24, 2002 which has not yet gone into evidence, notes that at a test opening, small areas of the interface between the top of the concrete block wall and the underside of the perimeter steel beams were viewed. Roney notes that the construction of this interface was found to consist of a mortar joint in one case approximately two inches thick between the top of the masonry wall and the underside of the steel beam, and that no other anchors or mechanical fastenings between the two elements was noted. Mr. Garbutt was also mindful of the evidence of Mr. Johnston who had said that that particular space was sometimes empty, sometimes mortared, and sometimes filled in with brick.

As well, Mr. Garbutt was aware that Mr. Roney in his report at page 15, commented that the governing standard CAN 3-S304 requires that masonry walls be anchored to their lateral supports, using metal anchors or "other approved construction systems". Mr. Garbutt asked Mr. Mueller to bear in mind that both Dr. Drysdale and Mr. Roney were talking about veneer walls and not a solid wall as he had intended. He agreed with Mr. Mueller that his attitude is, if Roney and Drysdale are right and a retrofit is required, then don't blame him.

Finally, Mr. Garbutt acknowledged that Mr. Roney in his report at the same page 15 comments that it appears that the intent of the detail shown in the contract documents was to use the bond achieved by the mortar joint between the wall and the steel to provide the necessary lateral support to the wall, and that this method is somewhat unconventional, although an analysis of the theoretical structural bond formed between the mortar and the steel indicates that such a bond would provide adequate lateral resistance to the lateral wind loads specified in the OBC.

Entered as Exhibit 78 was Mr. Mueller's sketch of the block wall referred to during the course of his cross-examination of Mr. Garbutt.

Discussing then as-built drawings, Mr. Garbutt agreed that the specifications, Tab 1 of Exhibit 1, Division 1, section 11 did call for as-built drawings. Mr. Garbutt agreed that as-built drawings are something different from shop drawings. The as-built drawings are meant to indicate all of the changes that have been incorporated during the course of construction, and to reflect the building as actually built. He explained that the only one who can prepare as-built drawings is the contractor, and such as-built drawings were never turned over to either him or to the plaintiffs. He was asked by Mr. Mueller why he did not enforce the specifications and require delivery of the as-built drawings. Mr. Garbutt responded that he does not know if he asked the contractor or not. He did agree that the consultant is required to get the contractor to do what he is supposed to do.

Again, discussing the wall, Mr. Garbutt testified that he next looked at the walls subsequent to the completion of the project, in the fall of 1993 when he received the letter from the plaintiffs' lawyers. Looking at the wall at that time he saw no weepers and did not anticipate seeing any because he had intended to construct a solid wall. He was reminded of his previous testimony that he treated the shelf plates as flashings, and agreed that if the water had gone in through the bricks as described by Mr. Johnston, then the water would go to the shelf plates, but again if it was a solid wall, it could not do so.

Referring to Division 1 of the specifications, section 12 which requires the contractor to submit to the owner two operating manuals containing *inter alia* recommended maintenance procedures, Mr. Garbutt acknowledged that he is now aware that what the plaintiffs were given at the time were four to five pages of miscellaneous papers. He is aware now that that is what was presented to them as an operating manual, but he was not aware at the time of what they received. Having regard to his original recommendation to them with respect to the application of a sealer to the outside of the bricks to relieve the problems of leakage that they were experiencing, he acknowledged that there was no such direction in the so-called operating manual, but commented that there need not necessarily be.

At Tab 71 is a letter from Mr. Garbutt dated September 21, 1993 to the plaintiffs' then counsel. In his letter, Mr. Garbutt comments that the brick selected by the plaintiffs was probably the most porous one available, and as a result must be accompanied by a comprehensive regular compensating maintenance program to prevent penetration of moisture. He agreed that this was an oblique reference to the sealer. He agreed with Mr. Mueller that he had never told the plaintiffs prior to this letter of September 21, 1993 that what they had chosen was the most porous brick available. He agreed with Mr. Mueller that the problem with the wall cannot be solved now with a sealer because we know now that we are not dealing with a solid wall.

At Exhibit 2, Tab 64 is a letter from Mr. Garbutt to Mr. Emmons dated May 10, 1993. It begins by referring to "our telephone conversation of May 7, 1993 re masonry work". Mr. Garbutt confirmed that he indeed had that telephone conversation with Mr. Emmons. In Item 2 of the letter, Mr. Garbutt says, "See if the following can be determined: a) that the collar joint (between brick and block) is filled – there is no such thing as a ½ inch air space." Mr. Mueller suggested to Mr. Garbutt that Mr. Emmons in the telephone conversation must have told him that the wall had been built with an air space of ½ inch. Mr. Garbutt had no such recollection and did not know why in his letter to Mr. Emmons, he mentioned that there was "no such thing as a ½ inch air space". It was his evidence that at the time he knew that a solid wall had been built (at least in his own mind). Mr. Mueller asked him why, if he knew that, he was asking Mr. Emmons to see if the collar joint was in fact filled. Mr. Garbutt explained that he "didn't want to be shot in the back", he wanted to know what Mr. Emmons would say. Mr. Mueller suggested to him that at the time he had no idea as to how the wall had been built, and Mr. Garbutt denied that to be the fact.

In his letter, Mr. Garbutt asks Mr. Emmons as well to see if it can be determined what type of masonry ties had been used. Mr. Mueller suggested to him that he was asking this because he did not know. Again Mr. Garbutt denied that to be the case. He knew they were corrugated ties because there were bundles of them on the floor during the course of the project.

In the closing paragraph of his letter, Mr. Garbutt tells Mr. Emmons the following, "If the collar joint has not been filled and the wall subsequently not constructed as a cavity or veneer wall (flashing, weep holes etc.) serious problems result." Mr. Garbutt confirmed that that was his belief at the time. Mr. Mueller asked Mr. Garbutt why he would make such a comment if he knew at the time that the cavity was filled. Mr. Garbutt responded, "I told them to fill." Mr.

Mueller correctly pointed out to Mr. Garbutt that that was different from knowing that it was filled, and Mr. Garbutt agreed. He then explained that at the time he knew it was filled "in my own mind."

There was filed as Exhibit 76 three sheets of paper; they contain better copies of Tabs 64, 65 and 66 in Exhibit 2. Referring to both Tab 65 and Exhibit 76, it appears that at some point in time and likely in response to Mr. Garbutt's letter dated May 10, 1993 Mr. Emmons wrote the following note:

> Primeau – J.P. Price – Antique
> Built as a solid masonry wall, collar joints filled, walls tied together with adjustable ties.

It should be noted that the other thing that Mr. Garbutt asked Mr. Emmons to determine was the type of brick selected by the plaintiffs. Clearly, the note identifies Primeau who was the masonry contractor and J.P. Price Antique which was the type of brick selected by the plaintiffs. The note then goes on to answer the questions posed by Mr. Garbutt, alleging that the wall was built as a solid masonry wall with the collar joints filled and that the walls were tied together with adjustable ties.

It was Mr. Garbutt's evidence that he did not receive this note as such but it is quite possible that Mr. Emmons after Mr. Garbutt's letter to him on May 10, 1993 telephoned him back and gave him these answers. Mr. Garbutt agreed that if he had received those answers at that time he would have been comforted.

At Tab 66 and as part of Exhibit 76, is a fax to Mr. Emmons from Mr. Garbutt dated May 13, 1993 which says the following:

> wall construction
> 4" brick 4" concrete block} solid wall no air space
> indicated

With respect to this note to Mr. Emmons, Mr. Garbutt explained that he went to the plan and told Mr. Emmons what the plan provided. [If Mr. Garbutt did indeed intend that there be a solid wall constructed, he certainly did not do a very good job ensuring that a solid wall was in fact constructed.]

At Tab 74 is engineer MacAdam's letter to the plaintiffs dated October 11, 1993, expressing the opinion that water is entering the building through the masonry by penetrating the exterior wythe of face brick through cracks and through the mortar joints, and that due to the absence of wall flashing, vents or weep holes, the water then penetrates the inner masonry wythe into the interior of the building. At Tab 76 is a letter from the plaintiffs' lawyers to Mr. Garbutt dated December 23, 1993 enclosing Mr. MacAdam's reports, included the aforesaid report. At Tab 77 is Mr. Garbutt's response to the plaintiffs' solicitor dated January 10, 1994, telling him in part that MacAdam had it all wrong in that a cavity wall had not been constructed, and that the wall had been designed and built as a solid masonry wall.

In the same letter, Mr. Garbutt suggests to the plaintiffs' lawyer that:

> should collar joint filling prove to be required, the procedure consists of anchoring the brick to the back-up and pressure grouting the joint. Once the integrity of the collar joint has been ensured, the wall will conform to the requirements of the Codes in effect at the time of construction." Mr. Garbutt acknowledged here that he knew at the time the danger of pressure grouting, the danger was that the wall would blow out and that is why he added the proviso that the brick wall would have to be anchored to the back-up block wall to prevent such a blow out.

In his letter of January 10, 1994 Mr. Garbutt also commented at the bottom of page two that

> any remedial work suggested to essentially convert this wall into a veneer type (weep holes, vents, flashings etc.) are ill-advised because of the minimal width of the collar joint (approximately 12 millimetres) facilitating bridging and blockages by mortar droppings in an erratic unprocedable pattern.

Mr. Mueller pointed out that Mr. Garbutt was saying that it would be ill-advised to convert to a veneer wall but that that was exactly what the remaining defendants are saying is the solution here. Mr. Garbutt acknowledged that that was what the defendants' experts were suggesting but testified that he still thought they would be ill-advised to take that route.

At page two of Mr. Garbutt's letter, paragraph three, Mr. Garbutt advises plaintiffs' counsel as follows:

> Until factual data is obtained from field investigation, all reports from this point are speculative and non-contributive. I would recommend that your clients contact Mr. Rob Cole of Inspec-sol to perform the field investigation work.

Mr. Garbutt agreed with Mr. Mueller's suggestion that what he was saying at that time was that until there was field investigation, they would not know what the problems were. He agreed as well that it would have been irresponsible for the plaintiffs to sue without further investigation at that time.

Again referring to page two paragraph three of Mr. Garbutt's letter of January 10, 1994 to plaintiffs' counsel, Mr. Mueller pointed out that the letter reads in part

> It is important to determine whether the collar joint has been filled in its entirety, or, has been filled except for localized areas (and, if so, to ascertain the extent and location of these areas).

Mr. Mueller inquired of Mr. Garbutt why, if he had from Mr. Emmons the information received in May of 1993, he was suggesting in January of 1994 that it was important to determine whether the collar joint had been filled in, in its entirety. Mr. Garbutt responded that in his letter of January 10, 1994 he was simply suggesting that perhaps a few areas had not been filled in.

In his evidence, Mr. Garbutt agreed that it was prudent for the plaintiffs to write to him and inquire as to how the building had been constructed. He agreed that they might have more reason to rely on him than Mr. MacAdam and wondered at the time why they had never contacted him in seven years.

At Tab 80 is a letter from architect Cromarty, addressed to Emmons, confirming that there will be a meeting with Emmons, Vinkel, Garbutt, Roberts and representatives of the steel company and the masonry company, on April 20, 1994. The letter is dated March 31, 1994. The meeting was to be held without prejudice and was intended to assist the parties in order to better understand how the building was actually built, and what and why any changes were made to the contract documents. Mr. Garbutt confirmed that he never went to the meeting but rather sent Mr. Roberts as his delegate. At Tab 82 is a letter from Mr. Cromarty bearing the same date to GeoTech Contracting Ltd., observing that

> it was apparently intended that this building was to be constructed with a solid masonry wall, but it appears was actually built as a bastard type cavity wall as the collar joint was not filled and the wall was built without weep holes, flashings etc.

It was Mr. Garbutt's information that he did not receive that particular allegation at that time nor did he receive any report back from Mr. Roberts following the meeting of April 20, 1994.

At Tab 84 is a fax from Mr. Garbutt to Cromarty dated April 19, 1994, acknowledging receipt of a copy of the letter dated March 31, 1994 to Emmons & Mitchell, and advising Cromarty that he was already on written record regarding the matter and saw no point in attending the meeting.

At Tab 88 is a letter from Cromarty to the plaintiffs dated May 17, 1994 which comments in part,

> Although the specifications call for cavity type wall, or flashing over windows etc., the structural consultant claims that the building was designed for an exterior wall of solid masonry. The drawings tend to indicate solid wall construction without cavity flashing, cavity weepers or flashing at lintels.

Mr. Garbutt confirmed that he had previously said that in his letters.

At Tab 94 is a letter from the plaintiffs' solicitors dated August 19, 1994 inviting Mr. Garbutt to a meeting in that the plaintiffs

> feel that they have sufficient information regarding the cause or causes of the water problems at their building at 275 Queen Street, the alternatives for remedial work and the cost estimates for such work and they would like to proceed with remedial work as soon as possible.

Also invited to the meeting was Mr. Emmons. At Tab 96 is Mr. Garbutt's response dated September 19, 1994, acknowledging the letter of August 19, 1994 and requesting a number of items, including the purpose of the meeting and the proposed agenda, and the plaintiffs' expert reports.

Turning once again to Exhibit 53, a collection of building digests, and referring to Canadian Building Digest No. 6, and particularly page two thereof, Mr. Garbutt confirmed with Mr. Mueller, the Digest comment that in most instances of rain penetration of brick walls, leakage takes place between the brick and the mortar; only under unusual circumstances does rain pass through the brick or through the mortar. As well Mr. Garbutt agreed with the Digest comment that unbonded areas between brick and mortar, which usually are the cause of leakage in brick walls, may result from faulty construction techniques in which insufficient mortar is used to form the joint or from an unsuitable combination of brick and mortar. In the latter case, a tight bond between the brick and mortar is not obtained when they are brought together in the construction of the brickwork. Mr. Garbutt agreed as well to the following comment to the effect that failure in the design of a building to provide for accommodation of differential movements between its parts may lead to cracking of the masonry. Rain may then enter where the masonry is opened up as a result of the stresses placed on it. Mr. Garbutt agreed as well with the following comment in the Digest to the effect that differential movements between frame and masonry are a possible source of cracks in the masonry. The flashing in the wall at the spandral beams for example has sometimes provided a cleavage plane for the relief of stresses in the masonry, resulting in cracking of the wall.

Mr. Garbutt did not agree with the Digest's comment that most of the rain which penetrates unit masonry passes through unbonded areas between unit and mortar, which usually can't be seen and where the wall appears perfectly sound. Mr. Garbutt's position is that the water comes through cracks that are noticeable, and in that respect he agreed with Mr. Roney's opinion. He acknowledged that Dr. Drysdale opined that the water was coming through "invisible cracks".

Mr. Mueller referred Mr. Garbutt to Exhibit 56, Mr. Johnston's report relating to the wall, and particularly page 16 thereof relating to the water spray test where Mr. Johnston observed water flow emanating from the top surface of the bricks where there was no apparent crack or opening to the exterior, and the water penetration was occurring at the uppermost plane of the brick surface where it met the mortar bed of the brick above. Mr. Johnston notes that during the limited time of the test he did not notice any flow from the often incompletely mortar filled header or end joint of the bricks in the wall. Mr. Garbutt acknowledged that that was what Mr. Johnston was saying.

Mr. Mueller referred Mr. Garbutt to pages 21 and 22 of Mr. Johnston's report, Exhibit 56, where Mr. Johnston notes that the area above the interior side of the window on the fourth floor had been opened further to permit a greater view of the interior of the brick façade than had been previously visible in that the concrete block had concealed their view in the investigation conducted two months previously. This opening allowed them to view the brick immediately below the spandral beam of the south wall structure and exposed approximately 16 inches more of the wall immediately above the window opening. Mr. Johnston noted that the interior surface of the brick displayed obvious water flow markings where rain water had penetrated, and that these stains were evident as permanent indications of the flow, indicating as well that most rivulets started at or had as their source the upper plane of the brick. Mr. Garbutt conceded that, referring to his Tab 25 of his Exhibit 74, the same area being viewed by Johnston is shown in his photograph of the southeast corner at the top right hand corner. As well, Mr. Garbutt conceded that the only person who has testified about incursion of water in an actual rainstorm is Mr. Johnston, who while conducting an inspection encountered a sudden rainstorm and going to the fourth floor, he observed leaking from the bed joints.

Mr. Garbutt confirmed once again that during the course of construction he looked at the joints at the shelf plates on the north and south sides of the building, and simply missed the fact that they were mortared rather than caulked as required. He explained that the plate is located in the joint and the caulking is placed between the bottom of the top brick and the top of the bottom brick with the plate in between the two courses. He explained further that usually a space is left as the courses go up and that particular joint is caulked afterwards. He says he probably viewed this from a scaffold. He looked and he missed it. He was asked by Mr. Mueller why he did not check the same joints on the east and west sides where the plates had been cut off and his response, which I really did not quite understand, was that he had not done it on the north and south side so he had no reason to do it on the east and west sides. In passing, he noted that the plate is located five courses above the top of the window.

Mr. Garbutt confirmed that there was no contact between himself and the plaintiffs between 1986 and 1993. He confirmed as well that he has now learned that there were one or two heavy incursions of water at the roof hatch in 1992 and that up until that point in time, from 1986 to 1992 there had been some small amount of leakage in the building on a relatively constant basis. He confirmed that when he found out about the fairly continuous leakage into the building in 1992 it was his recommendation that the brick be sealed and he confirmed that that is what he would have suggested as a remedial step if he had been asked for his opinion at any time between 1986 and 1993.

With respect to the 'as-built' drawings, had they been produced to him he would have looked at them and having looked at them, if he had seen that the shelf plate had been cut off the east and west walls, then "all hell would have broken loose". He confirmed that in the normal course, he would receive the 'as-built' drawings and communicate with the owners, the plaintiffs, but he does not believe that he pressed the contractor for 'as-built' drawings, nor did he discuss the absence of the drawings with the plaintiffs.

With respect to the block wall which as we now know was designed to be and is 'free-standing', Mr. Mueller suggested that while Mr. Garbutt's theory that the bricks passing in front of the vertical column would prevent any movement of the block wall behind it in connection with wind forces applied directly to the brick wall, the same could not be said for the suction forces created on the opposite wall by the same wind forces. Mr. Garbutt conceded that wind blowing on the west wall could create a suction force on the east wall, but he disagreed that clips would be needed between the block wall and the structural frame to protect against that wind suction. It was his view that the magnitude of such wind loadings is such that there is probably no issue. He commented that there is obviously <u>not</u> an issue because there has been no cracking. He did however concede that the block wall would be more resistant to head on wind than to the suction forces.

With respect to the stone surrounds, Mr. Garbutt recalled the allegation during the course of the trial that someone had warned against the risk of water entering the building at the location of the stone surrounds. Mr. Garbutt testified here that he had no recollection of anyone coming to him with such a warning. He was aware that the stone surrounds were caulked in 1986 and the caulking renewed in 1992, and that at all material times it would have been in very good condition. His evidence was that it was not tenable that water got into the building through the surrounds, and in that respect he agrees with Mr. Johnston's theory of entry. He does not agree with Mr. Quirouette's theory of entry and does agree with Mr. Johnston.

With respect to the cost of repair, Mr. Garbutt agreed generally with the proposition that usually it is much more expensive to repair a deficiency <u>after</u> rather than <u>during</u> construction, and that in any event the cost itself of remedial work in and by itself increases with time.

Mr. Mueller reviewed with Mr. Garbutt the fact that General Condition 25.1 provides that the contractor shall have complete control of the work and shall effectively direct and supervise the work, so as to ensure conformance with the contract documents, and General Condition 26.1 provides that the contractor shall employ a competent supervisor and necessary assistants, who shall be in attendance at the place of the work while work is being performed. Mr. Garbutt during this conversation confirmed that it is not unusual for owners to attend site meetings, particularly in the private sector, that is where private money is being spent on the construction, although attendance of the owners is not mandated.

Mr. Mueller then reviewed with Mr. Garbutt a few comments made by Mr. Garbutt in the notes that he had prepared for the purpose of giving his evidence, and which he was allowed to refer to during the course of his evidence. A review of the notes disclosed the following.

Before construction started, the plaintiffs advised Mr. Garbutt that they did not want their building to be covered with efflorescence, as was the case in the construction of the Whig-

Standard building, which was a veneer building. That instruction from the plaintiffs was tied in with Mr. Garbutt's intention to build a solid wall, rather than a veneer.

Mr. Garbutt referred in his notes to an article which said in effect that solid masonry walls have no vented draining space but are filled with mortar, and their resistance to rain penetration is not as good as in rain-screen construction, and as a result vertical and horizontal joints are required to accommodate movement in a solid wall construction. Mr. Garbutt testified here that he agreed with the passage in 2002, but in 1986 it was believed in the industry that solid wall construction had good resistance.

In his notes, Mr. Garbutt also had reference to a comment that the selection of remedial approaches must be directed to the true cause of the problem, so as to ensure that there is no misdiagnosis. In his evidence here, he agreed with that proposition and agreed that it was inappropriate to remediate without a full understanding of the problem after close investigation, involving all concerned.

In his notes, Mr. Garbutt commented that in his view there was no effort made here to keep the collar clear, and in his evidence here he confirmed that as a correct statement, and said in his evidence that it was not the mason's apparent objective to keep the collar clear.

In his notes, Mr. Garbutt comments that Dr. Drysdale said that the beams would rotate under loading after the masonry was completed, and that this causes the mortar joint to open under stress, and that these cracks are shown in his photographs in his Exhibit Book, Exhibit 74 at the bottom of the spandrel locations. In his evidence, Mr. Garbutt confirmed his thoughts on this point and confirmed that he viewed the joint under the plate as being the primary entry point for rainwater. In this respect, Mr. Mueller reminded Mr. Garbutt of Mr. Johnston's evidence and Mr. Garbutt responded that he had no dispute with anything that Mr. Johnston had to say about the problem, although when reminded by Mr. Mueller that Mr. Johnston saw water coming in through the brick at five courses below the plate, Mr. Garbutt maintained that he believed that only the bond of the joint below the plate was affected by the rotation. He explained that as the bricks were put in underneath the plate they had to be put in horizontally rather than vertically, and as a result very little if any bond was achieved.

Mr. Garbutt agreed that any water coming in on the east and west walls cannot be related to the plates because there are none.

Finally, Mr. Garbutt explained that he was not disputing the fact that water was coming in through courses below the plate.

Mr. Garbutt's Reply Evidence

At Tab 19, Exhibit 74, Mr. Garbutt's Exhibit Book, is a diagram drawn by Mr. Garbutt which he testified illustrates what happens where the collar joint between the brick course and the block course is clear. The diagram shows water coming in to the collar joint underneath the plate and on top of the brick and then running down to the next plate, collecting there, saturating the brick and causing brick deterioration by freeze/thaw. Mr. Garbutt pointed out that at least in his view his photos at Tabs 25 and 26 illustrate the effect.

Mr. Garbutt agrees with Mr. Johnston's statement that the porosity of the brick itself has been a benefit in that because of the matrix effect of the mortar bridges, the water does not get all the way down to the next plate but rather the pressure inside the wall pushes the water out through the porous bricks. It was Mr. Garbutt's evidence that in this respect it would be counter-productive to put weepers in, thus letting the air pressure out through the weepers.

Mr. Garbutt pointed out that although he agreed with Mr. Johnston with respect to Mr. Johnston's determination as to what is wrong with the building, he disagreed with Mr. Johnston's opinion with respect to remedial work. In particular, he disagrees with Mr. Johnston's proposal to install flashings and weep holes. It was Mr. Garbutt's position that the system has worked for 17 years, so why change it. He suggested that it would be better to inject foam into the cavity above the window for a depth of five courses, from underneath the plate down to the window lintel, and then seal and re-caulk between the stone surround and the lintel, all as shown in Tab 20 of Exhibit 74, which is a drawing prepared by Mr. Garbutt to illustrate the suggested remedy.

It was his further evidence that the alternative to the above-noted procedure would be to remove the five courses of bricks over the top of each window, install new flashing to the stone surround, re-install the bricks with two weepers over each window. It is to be noted that both of these alternative remedial efforts would be confined to the top of each of the windows.

Tab 22 of Exhibit 74 illustrates that the latter remedy of removing and re-installing the brick would involve only 3% of the entire brick wall.

Mr. Garbutt disagrees with the proposal to flash the entire plates and to install weepers, again on the basis that this would reduce the inside pressure in the matrix, which would defeat the system as it presently works, that is expelling moisture through the bricks by reason of interior pressure.

Mr. Garbutt wanted to make it clear in his evidence that he was not disputing the experts' theories as such, and was not disputing the suggested remedies generally, he was simply saying that in this particular building there is a system that has been working for 17 years with pressure from the inside, so why change that system.

With respect to the costing of the various remedies, Mr. Garbutt notes that Mr. Roney costed on the basis of M-78, the old Code, while Dr. Drysdale costed on the basis of the current Code. In Mr. Garbutt's view, both calculations must be done in order to determine the factor of betterment to the building. According to his evidence, both calculations should be done but both should be done on the same basis. He agrees that the only way to do the calculations is to assume a cavity wall, because one cannot evaluate what he calls a 'matrix' which as he explained it is the honeycomb effect resulting from the mortar bridging.

At Tab 2 of Exhibit 74, Mr. Garbutt has included section 4.1.1.4(1) of the 1983 Code that was in effect at the time. That section, under the heading Structural Loads and Procedures, provides that buildings and their structural members shall be designed by one of the following methods

a) standard design procedures and practices provided by this Part; or

b) one of the following three bases of design

   i) analysis based on generally established theory;

   ii) evaluation of a given full-scale structure or prototype by a loading test; and

   iii) studies of model analogues.

Mr. Garbutt suggested that both Mr. Roney and Dr. Drysdale were relying on section 4.1.1.4(1)(a), whereas he was suggesting that they might well employ (1)(b)(ii) and evaluate the existing wall by some appropriate load testing. He makes this concession while acknowledging that Mr. Johnston wanting to replace the entire façade has some logic to it, to the extent that we do not know what is behind the bricks.

Referring to Mr. Mueller's cross-examination with respect to the roof and the slopes etc., Mr. Garbutt pointed out that initially the building was designed for two floors, and then was extended to four floors with the anticipation that two more floors would be added some time in the future. He explained that having regard to the anticipated addition of two floors, there were a number of items involved and to be considered including structure, mechanical and electrical. He explained that the Codes on average change every seven years, for example in the past the 1983 Code was revised in1990 and the 1990 Code was revised in 1997. Having regard to these revisions, he explained that generally then the mechanical and electrical installations installed in one year would not meet Code requirements in 14 years from installation. He reasoned then that allowing for a 15-year life for the roof was not unreasonable, and that to have gone to a permanent roof initially would have been almost self-defeating.

Witness – John Wilson

Examination-in-Chief

Mr. Garbutt called as his architectural witness John Wilson, who has been a practising architect since 1980. His c.v. was marked as Exhibit 79. He has significant experience in renovation work. While he has worked on many buildings that had envelope problems, he has not run into a situation where the building wall had been designed as a solid wall and turned out to be a cavity or veneer wall, although he suspects that occurs quite often. He has never dealt with a situation where the floor ledge plates have been cut off as in our case, nor has he dealt with a situation where there have been insufficient ties or inappropriate type of ties installed. He candidly admitted that if he had been faced with these kinds of problems, he likely would have turned them over to a structural engineer.

I found Mr. Wilson to be qualified in the field of architecture and qualified to provide me with opinion evidence on matters architectural.

Filed as Exhibit 80 was Mr. Wilson's report dated April 8, 2002. The report is titled, 'Assessment of Issues Relating to Liability'.

Mr. Wilson prepared his report with the following understanding of the factual contentions in the case. Mr. Garbutt was retained by the plaintiffs to design a four-story building which was to carry two additional floors imminently after completion of construction. In September 1986 there were already complaints of water leakage above the windows and the doorways. The windows were re-caulked but this did not do any good. In 1994 Mr. MacAdam and then Mr. Johnston, the architect got involved for the plaintiffs. Mr. Johnston prepared a series of studies relating to both the roof and the walls and got other specialists involved. Certain recommendations were made in 1994. It appeared that the construction of the parapets was very poor. The roof membrane itself was in good condition but was in bad condition where it went over the parapet. In Mr. Wilson's view Mr. Johnston rightly saw the prime reason for water getting into and down between the brick and block. In Mr. Wilson's view, Mr. Johnston removed bricks for destructive testing in "strange areas."

On March 20, 2002 Mr. Wilson visited the subject building with his partner Ken Dantzer to review the condition of the building envelope and "to gain a first-hand understanding of the amount of water penetrating into the interior." On the day in question, exterior temperature was approximately five degrees Centigrade, with a light drizzle of rain. He and his partner walked around the building and his partner took photographs. He had read Mr. Johnston's reports and unexpectedly Mr. Wilson saw an attractive building, mostly without efflorescence; it appeared fresh and attractive with no sign of spalling or physical cracks in the brick.

He then viewed the inside of the building with the plaintiff's wife, Mrs. Reed.

The basement appeared to be completely dry.

The tenant of the first floor expressed no problems since 1989.

Mr. Wilson observed leakage in the second floor premises. He observed water damage at the windows, particularly on the south side, and damage to the drywall at the edges of the windows, again primarily on the south side. There was no staining of the walls in general or the carpet. The water damage was confined to the windows.

On the third floor, Mr. Wilson observed water damage at the windows and the drywall abutting the windows. There was no damage to the ceiling tiles that he could observe.

He observed the most water damage on the fourth floor, particularly on the south side but also on the north side. The water damage appeared to be relegated to the window locations. There was no sign of roof leakage. On the fourth floor he observed a slow drip of water at the heads of the windows, which was being controlled by an eavestroughing arrangement. He observed constant dripping and this in a situation where there was, as he described it, a light drizzle of rain.

He concluded, in his own words, “no doubt about it, the building leaks.”

He inspected the roof. He was aware that the parapet had been raised and the flashings corrected. He was aware that a new roof had been installed in 1998. It was his information that the existing membrane had been used and the irma roof retrofitted with sloped insulation, covered with membrane bitumen and then with ballast. With respect to the roof, Mr. Wilson was of the opinion that the original roof could simply have been patched and properly flashed into the new parapets at a modest cost.

With respect to the roof, Mr. Wilson concluded that there had been a real problem from 1985 through to 1994 with water from the roof entering into the cavity through the flashings, running down the cavity and coming into the building at the lintels above the windows. The leaks were not through the roof itself. In Mr. Wilson’s view, when the parapets were raised and the flashings fixed, the majority of the problems were solved.

With respect to the wall, Mr. Wilson does not agree with Mr. Johnston that the masonry design is deficient. Mr. Wilson is of the opinion that Mr. Garbutt prepared original drawings and specifications, which met the requirements of the National Standard Council of Canada, Masonry Design and Construction of Buildings, and thus the masonry requirements of the National Building Code. In Mr. Wilson’s view, problems in the masonry wall are only apparent at the heads of the south-facing windows on the second, third and fourth floors, and the leaking problems there are probably due to sub-standard workmanship on the part of the masons who failed to continue with the workmanship standards originally inspected and approved for the lower floors by Mr. Garbutt.

With respect to the roof, Mr. Wilson does not agree with Mr. Johnston that the roof had originally been poorly designed and specified. In Mr. Wilson’s view Mr. Garbutt prepared roofing drawings and specifications which clearly showed design intent as is the norm in a modified design/build construction contract. In Mr. Wilson’s view, the early problems with brick efflorescence and heightened water leakage were created by parapets which were too low, and poorly executed counter flashings. This is a matter of sub-standard workmanship and not design. In Mr. Wilson's view, the design and construction of the main roof membrane was never a contributing factor in water infiltration.

Mr. Wilson noted that with respect to the wall, Mr. MacAdam in his letter to the plaintiffs dated October 11, 1993 indicated that while a full, proper and permanent solution would involve extensive demolition and reconstruction, it would be "prohibitively expensive, and we would not suggest it seriously." Mr. Wilson notes that in his letter to the plaintiffs of November 30, 1993 Mr. MacAdam re-thinks the replacing of the brick, and says that it might not be such a bad idea after all. Mr. Wilson notes that the system that Mr. Johnston suggested in his final report was to remove the face brick, rectify the back-up structure and re-install new brick at the exterior face of the building. Mr. Wilson wonders why, after correcting the problems with the roof parapets and flashings, Mr. Johnston did not even suggest the installation of proper counter-flashing at the offending window heads, except as part of a wholesale program of complete brick removal and replacement.

Mr. Wilson had reviewed Mr. Pascoal's letter dated June 28, 1999, prepared for the defendant Emmons, and Mr. Wilson found "its historical perspective and constructive suggestions very refreshing." Mr. Wilson testified that in his report, Mr. Pascoal noted that in the 1980's, thin wall masonry was in a transitional phase from solid to veneer walls, and that standards became local.

Based on the descriptions in Mr. Johnston's reports and the destructive testing that has been done, Mr. Wilson feels that the subject wall is not a mass wall with the brick actually bonded to the block, and at the same time it is not a classic rain-screen wall with two inches of clean air space maintained between the bricks and the blocks. In Mr. Wilson's view, the wall as-built is a hybrid between the two. He sees a brick rain-screen with airspace ranging from 19 millimetres (3/4 inch) to 2 ½ inches. He sees metal ties between the brick and the block, which sometimes have become solid by reason of mortar fins. It was Mr. Wilson's view that looking at the drawings and the photographs, he thinks that ours is a very good wall, which meets Code in that it exhibits principles of design to keep water out and maintain pressure equalization. The brick is in good condition and there is good bonding between the brick and the block by reason of mortar fins.

In his report, Exhibit #80 and in his evidence, Mr. Wilson agrees with Mr. Pascoal that the wall system does not work perfectly and that concerns about the long-term durability of the wall system are legitimate. Mr. Wilson notes however that instead of recommending a "hideously expensive and unnecessary brick replacement program", Mr. Pascoal suggests retrofitting the wall system to meet current standards and proposes remedial work which in Mr. Wilson's opinion is "the most cost-effective, elegant and least likely to cause disruption to the day-to-day functioning of the building." Those recommendations of Mr. Pascoal with which Mr. Wilson entirely agrees are as follows:

1) waterproof membrane flashings to be installed at all shelf-angle locations, at all window and door locations and foundation perimeters;
2) PVC grade weep vents be introduced over new flashings, plus at all locations just below shelf-angles at the top of each floor to increase air circulation in the cavity;
3) new support angles at acceptable intervals be installed along the east and west elevations at each missing location;
4) new soft joints be saw cut and caulked horizontally at shelf-angle locations;
5) new expansion joints be saw cut and caulked vertically at required strategic but aesthetically acceptable locations;
6) new helifix stainless steel skewer type brick ties be installed where any potential concern of inadequate anchorage, such as at parapet locations and where remedial work requires saw cutting and brick replacement; and
7) detergent cleaning and soft bristle scrubbing of any efflorescence staining on the wall.

Mr. Wilson notes that all items except 6 and 7 were included in the original construction specifications or drawings, and Mr. Wilson suspects that if these features had been built into the original work as specified, water leakage problems would be non-existent.

Cross-examination by Mr. Nelson

Mr. Wilson knew Mr. Garbutt to be an engineer with specialties in structural, mechanical and electrical work. As an architect, Mr. Wilson would retain an engineer to do the structural, mechanical and electrical work involved with a project, although Mr. Wilson would have general knowledge of those systems. What the architect brings to the process is the co-ordination of the various design elements. The architect determines the client's needs, ensures that zoning by-laws are complied with, ensures that the design of the building fits the site, and that the building as created meets the needs of the clients. The architect is in charge of co-ordinating the input of the various specialists, and brings his skills to the intersection of the various disciplines.

Mr. Wilson agreed with Mr. Nelson that if a client decides not to engage an architect, he may be missing a certain element that goes into construction, and explained that some types of buildings require both the engineer and the architect, for example buildings which will house large numbers of people such as schools and restaurants, and where safety factors are of greater importance.

Mr. Wilson confirmed that once an architect is engaged he has a duty to the client as does the engineer, but at the same time both have a duty to the public. The duty to the public involves safety, health and welfare. With respect to the subject building, based upon his view of both the outside and the inside, Mr. Wilson was of the view that there was nothing about its condition that affected the safety, health and welfare of the public. He confirmed that if the architect or the engineer comes across a situation which concerns the health, welfare or safety of the public, he is under an obligation to bring that concern to the attention of the client and if the client (owner) decides to take no action with respect to that threat to public safety, welfare or health then the architect or the engineer would be under an obligation to go to the public authority, in this case the City of Kingston. In this respect, Mr. Nelson went through with Mr. Wilson a list of approximately a dozen engineers and architects who have been involved in this litigation and in the construction and remedial work, and Mr. Wilson confirmed that all would have a duty to go to the municipal authority in the event that any of them felt there was a threat to the public that the owner was not prepared to deal with. As far as he is aware, none of the dozen engineers and architects have reported this matter to the City of Kingston.

At Exhibit 2, Tab 111 is Mr. Johnston's report to the plaintiffs dated July 16, 1998 which notes in part that the

> work regime is related strictly to roof and parapet only with wall correction restricted at the suggestion of your solicitor to the parapet interface required only for proper roofing execution. No correction otherwise of the wall is contemplated nor included in the scope of the work.

Mr. Wilson agreed with Mr. Nelson that if the engineer or the architect had a concern with public safety he could not delegate to the lawyer to determine whether or not proper remedial work would be proceeded with.

In Mr. Wilson's opinion, having regard to a visual examination only which saw no spalling, cracking, binding of doors etc., the building appeared to be safe and secure.

Again, Mr. Wilson confirmed that during his visual inspection of the building he saw no evidence of any water leakage in the basement.

He was aware that the shelf plates had been cut off on the east and west walls.

He was aware that there is a gap between the brick and the block.

He agreed with Mr. Nelson that if water was getting into the east and west wall cavities, there were no shelf plates to interrupt the water flow inside that cavity.

Mr. Wilson confirmed that during his inspection of the subject building, he did not see any deterioration caused by water on the first floor.

With respect to the second floor, Mr. Wilson in his report notes that he saw some slight condensation damages to the gypsum wall board at the edges of the south facing windows, and that the third window from Barrie Street on the south side had a slow leak at the upper corner. He saw no damage by staining on the carpet or on the laid in ceiling tiles.

In his *viva voce* evidence, Mr. Wilson explained condensation, which occurs where the gypsum meets the aluminium frame and at first testified that he saw no other incursion of water other than from the condensation. He then corrected himself, I believe at my prompting, and added that he did see a slow leak in the third window from Barrie Street at its upper corner.

With respect to the third floor, Mr. Wilson noted water infiltration and condensation on the south facing windows, which caused the delamination of the edge of the wallpaper from the gypsum wallboard adjacent to the windows. In his report, he noted that the east and west facing windows showed no signs of leaking. In his report, Mr. Wilson makes no mention of the north wall.

With respect to the fourth floor, Mr. Wilson observed slow dripping water infiltration at the heads of the south facing windows which was being collected into an eaves trough set up which in turn discharged the water through surgical tubing into canisters on the window sills. In his report, Mr. Wilson noted that the north and east windows did not appear to leak. In his report he makes no mention of the west windows at all. In his *viva voce* evidence he conceded that the windows on the south side leaked dramatically compared to the rest, and he estimated that the total volume of water collected on a rainy day in the circumstances that he noted might be in the order of .5 to 1.0 litres. Mr. Wilson did notice two stained ceiling tiles in the public corridor and three stained ceiling tiles in Dr. Reed’s private office that appeared to be old and dry, suggesting according to his report that they might have been caused by condensation drips from mechanical piping above the ceiling. As well, he noticed a small amount of unpainted gypsum wall board patching in the main corridor outside Dr. Reed’s office.

Mr. Wilson was on site for approximately an hour and he acknowledges that Dr. Reed took a video later that night showing much more water infiltrating the building than Mr. Wilson saw on the occasion of his visit.

In his report at page 5, and confirmed during the course of his *viva voce* evidence, Mr. Wilson saw no evidence of "wetted walls and severe staining and discolouration of the exterior brick surfaces and finishes, nor of deterioration and damage to interior finishes or severe efflorescence or large volumes of rain water penetration." These were claims that were being made by the plaintiffs. I took him up on his statement in his report that there was no deterioration or damage to interior finishes, and he confirmed that there was indeed damage that he noted to the wallpaper and wall board at the windows. He confirmed as well that he saw some slight efflorescence on a planter near the main entrance.

Entered as Exhibit 82 was Mr. Wilson's drawing of a window.

In his report, Mr. Wilson notes that "the single biggest improvement to the building envelope since Paul Johnston's initial report was that the roof parapets were heightened. We are of the opinion that the original roof could simply have been patched and properly flashed into the new parapets at a modest cost." In this respect, Mr. Wilson acknowledged that as far as he could tell, the construction of the parapet was defective and that the counter flashings (felts) at the parapet were cracked. He was of the view that after those repairs the majority of the water problems in the wall were solved, and he went so far as to say that with this work 90% of the water problems were solved.

At page 6 of his report, Mr. Wilson suggests that what he sees to be a minimal leaking problem is probably due to sub-standard workmanship on the part of the masons who, for whatever reasons, failed to continue with the workmanship standards originally inspected and approved for the lower floors by Mr. Garbutt. He expresses the opinion that the deficiencies in the upper portion of the south facing wall were caused by workmanship factors and not by design factors or lack of proper supervision. He confirmed to Mr. Nelson that his information was that Mr.

Garbutt inspected the lower floors. He does not know where he got that information but confirmed his opinion that the problems in the south wall were caused by poor workmanship.

Mr. Wilson knows engineer Pascoal with whom he has worked on two projects, and at page 9 of his report endorses Mr. Pascoal's recommendations in terms of remedial work.

Mr. Wilson confirmed Mr. Pascoal's view that during the 1980's masonry walls were in a state of transition between the traditional solid wall where the wythes were bonded together to rain-screen or veneer or cavity walls. In answer to questions from me, Mr. Wilson confirmed that the transition as such was a matter of choosing one or the other or a hybrid thereof. He confirmed for me that through the 1980's, indeed as is the situation at the present time, a wall was either a solid wall, a veneer wall, or a hybrid of the two.

Mr. Wilson confirmed that the contract drawings show no flashings or weepers. He disagreed with Mr. Nelson's suggestion the support angles were shown on a very small scale on the building sections. It was his evidence that the support angles were there clearly on the drawings and that indeed the steel was delivered with support angles. He pointed to the structural drawings prepared by Mr. Garbutt, Exhibit 9, showing the shelf angles and I had Mr. Nelson circle the significant part of the drawing with a highlighter.

Mr. Wilson confirmed that no soft joints were identified on the original drawings, but were identified in the specifications.

He confirmed as well that no expansion joints were shown on any of the drawings but pointed out that there was effective reference to expansion joints by reference to appropriate standards in the specifications.

Cross-examination by Mr. Mueller

Mr. Wilson confirmed that he had read Mr. Johnston's rebuttal of his report which is Exhibit 61. As well, he has seen Dr. Reed's video with respect to the infiltration of water on the evening of Mr. Wilson's visit, after he had gone. He is not aware of Dr. Reed's *viva voce* evidence with respect to the number of things, which Dr. Reed says Mr. Wilson did not see on the occasion of his visit.

Mr. Wilson agreed that he is not a roofing specialist, like Mr. Kendall. He has never run into a situation where a wall has been designed as a solid wall and turned out to be a veneer. He has never seen a situation where the veneer wall has been defective in terms of spacing of ties and type of ties used. He has never experienced a situation where wall plates have been cut off.

He confirmed that generally he is of the opinion that the problems are attributed to the contractor rather than Mr. Garbutt as designer.

Mr. Wilson in his report lists some 23 documents that he has reviewed for the purpose of preparing this report. He agreed with Mr. Mueller that he has not seen before now Tab 64, Exhibit 2 which is Mr. Garbutt's memo to Mr. Emmons asking him to determine whether or not the collar joint had been filled. He confirmed that he has not seen before now Tab 71, Exhibit 2, Mr. Garbutt's letter of September 21, 1993 to the plaintiffs' lawyer, suggesting that the remedy would be to seal the brick wall. Mr. Wilson explained that he has seen at some point in time Tab 75 which is Mr. MacAdam's letter to the plaintiffs dated November 30, 1993, but confirmed that it is not on his list, nor is Tab 77, Mr. Garbutt's letter to the plaintiffs' solicitor dated January 10, 1994 explaining that Mr. MacAdam is in fundamental error in that the masonry wall was designed as a solid wall and not cavity. Mr. Wilson acknowledged that he has seen this particular letter since completing his report and indeed has had ongoing consultation with Mr. Garbutt after the preparation of his report.

Finally, Mr. Wilson confirmed that he has not seen Tab 78, Mr. MacAdam's critique of Mr. Garbutt's letter dated January 10, 1994.

Mr. Wilson confirmed that his view of the inside of the walls has been confined to photographs and whatever he could see by a walk around at ground level, while the test cavities were open. At page 3 of his report, Exhibit 80, he notes that from his observations during the site visit on March 20, 2002, from the exterior the brick work appears to be in good structural condition with no signs of cracking or spalling. He acknowledged that he is now aware of a crack some 20 feet long on the north side of the building, and thinks that if he had been at a higher level during his walk around he would have seen it. He is now aware as well of other cracks in the building as shown in Mr. Garbutt's document book, Exhibit 74, in photographs at Tabs 25 and 26.

There then ensued a discussion between Mr. Mueller and Mr. Wilson which clarified for Mr. Wilson what Mr. Johnston had been saying in his reports, and particularly with respect to Mr. Johnston's observations of moisture coming through "invisible" cracks and leaving archaeological evidence of such. As it turns out, Mr. Wilson assumed that Mr. Johnston was talking about invisible cracks in the brick, which he could not understand. It was his evidence here when Mr. Mueller explained that Mr. Johnston was talking about invisible cracks in the bed joints between the bricks, that that made perfect sense to him and that this kind of "invisible" crack in the bed joints has been known for a long time in the literature.

Mr. Wilson conceded that he agreed with Messrs. Johnston, Garbutt and Drysdale that the water entered the building through the cavity as opposed to the windows themselves, and conceded that Mr. Johnston rightly identified the source of the water incursion.

At page 7 of his report, Exhibit 80, Mr. Wilson asked the question,

> Why was it – that after correcting the problems with the roof parapets and flashings and realizing the only water now entering the building was slowly dripping in at the window heads of the south windows on the top three floors – that Mr. Johnston did not even suggest the installation of proper counter flashings at the offending window heads except as part of a wholesale program of complete brick removal and replacement.

In this respect, Mr. Mueller asked Mr. Wilson whether he agreed that the water getting in to the cavity was coming in through the entire length of the building and not only at the windows. Mr.

Wilson responded that the whole principle of a rain screen wall was to allow water in and then drain down the cavity and to drain out in a controlled way. He agreed to the extent that the window is there with a lintel above, the lintel will catch the flow. Mr. Wilson at this point noted that there is no way of knowing the extent to which the water is going over the mortar fins to the block wall. He noted that there is an enormous vacuum with respect to what is going on inside the building wall, other than at the windows.

Mr. Wilson acknowledged that he was aware of corrosion to existing ties (Exhibit 60). He noted that they were ties that had been exposed to water coming in through the parapets for some 10 years from 1985 to 1995. With respect to the bending of the ties, he agreed that the bending would create a significant compromise with respect to the ties being able to withstand a lateral force (wind).

Mr. Wilson seemed to agree with Mr. Mueller's proposition to say that water was entering at the windows does not address what is happening in the wall between the windows. Mr. Wilson did note that the best work to date was the correction of the parapets and the flashings at the parapets, and he conceded that even after this work water was still getting in.

With respect to the second floor, Mr. Wilson conceded that the video showed much more damage than he saw on his walk around.

With respect to the temperature on the day in question, Mr. Wilson stood by his approximation of 5 degrees Centigrade, although he conceded that he had no thermometer with him at the time. He was not aware that there was evidence from some source that indicated the temperature to be zero degrees Centigrade as taken from the car thermometer.

Mr. Wilson was advised by Mr. Mueller that it was the evidence of Dr. Reed that the north wall leaks from time to time and Mr. Wilson was not in a position to dispute that.

With respect to the fourth floor, Mr. Wilson agreed that the video showed much more delamination of the wallboard surrounding the windows than he observed on his site visit, and also is aware now of the evidence that some of the windowsills had to be replaced.

At page 5 of his report, Exhibit 80, Mr. Wilson at the bottom thereof notes that the new roof did not incorporate a roofing substrate slope to the roof drains, a feature which Mr. Johnston had earlier insisted was necessary to ensure the longevity of the roof membrane. In his evidence, Mr. Wilson conceded that he now knows to be an inaccurate statement.

With respect to the wetting of the exterior walls, Mr. Wilson conceded that he has not had an opportunity to observe following a rainstorm the extent to which the subject walls glisten in comparison to adjoining buildings.

Mr. Wilson confirmed his understanding that the major source of water flow into the parapet was at the base flashing level and that the repair of the parapet fixed that particular problem. He was aware as well that when the main roof was removed, it was noted that there was widespread adhesion of the insulation to the bitumen.

Mr. Wilson confirmed to Mr. Mueller that it was his understanding that the roof had drained well from the outset. He then conceded that he was not aware that there was evidence that there was a back slope directing the water to the membrane at the bottom of the parapet. He conceded that if such was the case, this would be bad building practice in that it would have the effect of draining the roof into the parapet and thus into the cavity.

Mr. Mueller asked Mr. Wilson whether he was aware that Messrs. Johnston and Kendall testified in effect that the adhesion of the insulation to the bitumen had produced ponding that deteriorated the membrane and that they were of the view that if there was not a full job, there might well be a repetition of the type of flooding that had occurred at the northwest corner and the roof hatch. Mr. Wilson said he was not aware of that evidence but found it to be an interesting theory.

In his report at page 7 Mr. Wilson refers to the parapets and counter flashings being properly reconfigured between 1995 and 1998. He conceded that he was not aware that the only work that was done in 1995 was with respect to the northwest corner and the hatch. He conceded that he did not realize that the major work to the parapets and the roof had not been done until 1998. He agreed with Mr. Mueller that he misunderstood what the situation was.

In his report at page 6 Mr. Wilson states the opinion that the minimal leaking problem is probably due to substandard workmanship on the part of the masons who failed to continue with the workmanship standards originally inspected and approved for the lower floors by Donald Garbutt. He is of the opinion that the deficiencies in the upper portion of the south facing wall were caused by workmanship factors and not by design factors or lack of proper supervision. In this respect, he explained to Mr. Mueller that he was referring to bad counter flashing, leaking parapet, shoddy work, inadequate ties, etc. He referred to the fact that he also knows at the present time that the cavity opening was like a funnel, narrow at the bottom and wide at the top. To use his words, “I don’t know what is going where.” He noted as well, if one looks at the brick wall itself, it appears to be plumb.

Mr. Wilson was aware of the specification, which calls for a mock-up wall. He explained that the mock-up wall may be free standing but it is usually built as a corner of the real wall and the consultant and the contractor agree then that the standards expressed in the corner of the real wall are the standards that will be followed. In this respect he agreed that communications between the consultant and the contractor are very important in the construction of the building. He agreed that in the perfect world, the consultant if he is not satisfied with the mock-up and requires that it be torn down, should look again once it is rebuilt.

At the bottom of page 6 and top of page 7 of his report, Mr. Wilson notes that Mr. Garbutt prepared roofing drawings and specifications, which clearly show design intent, “as is the norm in a modified design/build construction contract”. In this respect he testified that he is aware that this was a stipulated price contract, and that by a design/build construction contract he meant that many changes would have been made, for example a change from a built-up roof to an irma type roof. He acknowledged that the drawings and the specifications were inconsistent with respect to the roof, and now recognized that the specifications bear a later date than the drawings. Mr. Mueller put it to Mr. Wilson that the specifications, Division 7, Thermal and Moisture Protection refer to C.R.C.A. Roofing Specifications, and the sloping of the roof to the drains. Mr. Wilson responded that the manufacturer of the insulation on the irma roof provided that there did not in fact have to be slopes. Mr. Mueller then put Exhibit 55, Tab K to Mr. Wilson, which is the Dow literature, which says that with the use of their insulation with an irma style roof, there must be a slope. Mr. Wilson then conceded that given the specification in Division 7 incorporating CRCA specifications and expressly requiring a slope, he would expect that there would be a slope. As he put it, there would be a reasonable expectation that there would be a slope in the roof to the drains. He went on to concede that absent any discussions between Mr. Garbutt and the contractor, he would have expected a slope to the drains of 2%.

In the second paragraph of his report at page 7, Mr. Wilson alleges that Mr. Johnston and other consultants spent a great deal of time trying to convince the plaintiffs that their exterior brick wall was in imminent danger of having something bad happen to it. In support of that allegation, Mr. Wilson refers to a portion of Mr. Johnston's report which notes that he is concerned that there are material and long term risks attending the continued service of the walls without appropriate corrective actions, and that the timing of any failure and the inherent risk associated with that failure has not been nor may ever be quantifiable. In Mr. Wilson's view, these were very strong words that were raising a red flag and setting off alarm bells.

With respect to wind gusts, Mr. Wilson agreed that a one-in-100 year wind factor is very unpredictable, but in his view the building would still be standing after a one-in-100 year wind gust.

Mr. Wilson, if he was turning this matter over to a structural engineer for consideration, he would want to tell the structural engineer about the corroded ties, the infrequency of the ties, the fact that they were bent, the fact that they were strap ties and not adjustable ties, the fact that the shelf angles had been cut off and the fact that immediately below the shelf angles, there was mortar rather than caulking.

Interestingly, at this point Mr. Wilson testified that he was aware that the block wall was mechanically attached in some way to the upright columns and that at the top while there was no mechanical attachment to the beam, there was a mortared joint between the block and the undersigned of the beam. When he was advised by Mr. Mueller that there was no mechanical attachment of the block wall to the upright columns and that the space between the top of the block wall and the bottom of the beam varied from loose bricks, mortar to nothing at all, he confirmed that he would in turning this matter over to a structural engineer, make those facts known to the structural engineer. He added as well that if he was designing the block wall he would not design it that way.

Mr. Wilson acknowledged that he was aware that Dr. Drysdale and Mr. Roney have each done an analysis of the capacity of the wall to withstand wind forces. He took this opportunity to comment with respect to Dr. Drysdale that there can be a difference between academic qualifications and practical qualifications.

Interestingly, Mr. Wilson disclosed that he was not aware to this point in time that Mr. Garbutt had always intended a solid wall, and while he and Mr. Mueller discussed the frequency of site visits to fulfil the consultant's obligations, he conceded that if Mr. Garbutt always intended a solid wall, Mr. Garbutt would have found out sometime during the construction that the wall was not going up as a solid wall.

With respect to Mr. Wilson's comment at page 6 of his report that section 5.6.1.1(2) of the current 1997 Ontario Building Code allows controlled ingress of water as long as it meets certain stipulations, he agrees with Mr. Johnston's comment at page 9 of his responding report, Exhibit 61, that the 1983 Code provided that "exterior wall cladding shall be so installed that it sheds water to prevent entry into other components of the building assembly. Where there is likelihood of some penetration, drainage shall be provided to take water to the outside."

Mr. Wilson acknowledged that to do many of the remedial works recommended by Tony Pascoal, bricks would have to be removed from the wall in selected locations. In terms of cosmetic problems, Mr. Wilson commented that he had seen many photographs where bricks had been taken out and replaced, and one could not see where that had been done. It was his view that the bricks could be matched in colour. As well the mortar can be matched. In any event, he agrees with Mr. Pascoal's seven points as set out at page 9 of his report, Exhibit 80.

With respect to the one-in-one hundred year wind gusts, I believe that Mr. Wilson conceded that if that was the only concern, then the engineer or architect would be less likely to go to the building inspector. Mr. Wilson commented however that even if that was the only concern, he would certainly warn the clients and ask their advice as to how they wanted to proceed. He is aware that only Mr. Roney and Dr. Drysdale have done official reports on wind loads, and to this extent he conceded that they are the only two in a position to go to the municipality, perhaps other than Mr. Johnston who took instructions from Dr. Drysdale.

Re-examination by Mr. Garbutt

With respect to his comment to Mr. Mueller relating to practical versus academic qualifications, Mr. Wilson explained that in practice the engineer may be more intuitive in terms of the Code whereas the academic may be more "pure".

With respect to the width of the air gap varying from bottom to top, Mr. Wilson offered one explanation that at any given floor the block wall may be moved inward on the slab. He is unable to definitely say that the wall is out of plumb.

<u>Defence of Emmons & Mitchell</u>

At the opening of his case, Mr. Nelson sought to enter into evidence records of the Bank of Nova Scotia relating to the mortgage on 275 Queen Street. The admission of the records was objected to by Mr. Mueller. I decided in these circumstances to hold a *voir dire* to determine the admissibility of the records pursuant to section 35 of the *Ontario Evidence Act* or at common law. Mr. Mueller initially objected on the basis that no appropriate notice had been given under section 35 of the *Act*, that the documents were not proven to be business documents prepared and maintained in the ordinary course of business and that they were in any event privileged.

On the *voir dire* I heard from two employees of the Bank of Nova Scotia, Eric Gault and Donald Cadwell. It appeared that the mortgage was initially taken out with Victoria & Grey which eventually merged with National Trust which in turn was eventually acquired by the Bank of Nova Scotia in June of 1997. As it turned out, Mr. Cadwell was a common factor throughout the piece; he started with Victoria & Grey in May of 1979 and continued on with National Trust when the merger occurred and then continued with the Bank of Nova Scotia when the Bank acquired National Trust.

In all of the circumstances, the Bank records were marked as Exhibit 83 and ruled admissible as business records pursuant to section 35 of the *Ontario Evidence Act*. Subject to any issue that might arise with respect to a specific document, Mr. Mueller conceded that generally the documents were relevant and he in the circumstances waived his client's claim for privilege.

Witness – Eric Gault

Examination-in-Chief

At page 221 of the Bank documents is a statement of income and expenses for the subject building for 1996, showing gross income of $166,000 with expenses of $13,000 for a net income of $152,000. It was Mr. Gault's belief that this would have been prepared by one of the doctors. This document refers to a maintenance expense of some 2%.

The mortgage was renewed in 1996 for two years. At page 215 of the Bank documents is a letter from Dr. Reed to Mr. Cadwell dated June 24, 1996 requesting a renewal term of one year at a fixed interest rate. In connection with the application for renewal, at page 216 is an application prepared by Mr. Cadwell which notes that the owners are both dentists and have a good outside net worth as shown by their attached financial information. Page 217 is a further portion of the application and notes that the security carries itself quite well and that outside support of the doctors has not been required to that point. It states further that the doctors' outside positions would enable them to step in if any cash flow problems were to develop. The application prepared by Mr. Cadwell notes that the net income for Dr. Reed in 1995 was $224,000 and for Dr. Lawless was $514,000.

It should be noted that I made it clear to Mr. Nelson that other than with respect to financial statements actually signed and delivered by the plaintiffs to the Bank, I was prepared to receive other documents in Exhibit 83 for the purpose only of establishing the basis upon which the Bank proceeded with its lending, and not for the truth of anything contained in the documents.

Beginning at page 190 of the Bank documents is Dr. Reed's financial statements for the year ended March 31, 1995, showing net income for 1995 of $224,000, out of a gross of $592,000. The statement shows for 1994 net income of $290,000 out of a gross of $553,000. The statement also shows that no rent was paid by Dr. Reed in 1994 although $45,000 in rent was paid in 1995. The witness confirmed that the information at page 217 of the documents appears to come from

Dr. Reed's financial statements. He confirmed that the records do not appear to contain a similar financial statement for Dr. Lawless.

At page 164 of the Bank documents, is a credit application for renewal of the mortgage from 1998 for a period of three years. The application is prepared by Mr. Cadwell. At page 165 Mr. Cadwell notes that the building is in good condition and that while the security was valued at $1,250,000 during the June 10, 1996 Renewal Credit, the market for office space buildings has deteriorated over the past two years or so. It is noted that appraiser McGugan was retained to prepare an appraisal. Based on his appraisal the value of the security has been adjusted to $665,000. Page 156 is part of the appraisal, as is page 153. In this respect I reminded Mr. Nelson again that I was not prepared to accept this Bank record or the appraisal report as evidence of the value of the building at the time. It was hearsay and if the fair market value of the building in 1998 is of any significance, then first hand evidence would have to be led in that respect.

At page 109 is an inspection report prepared by Mr. Cadwell in May of 1999, noting that the building is in good condition.

At page 104 is a letter from Dr. Reed to Mr. Cadwell in connection with the mortgage review and advising Mr. Cadwell that his personal net worth and personal income tax returns have not substantially changed.

At page 93 is a letter from Mr. Gault to the plaintiffs dated April 18, 2001 advising that the mortgage matures June 1, 2001 and requiring certain financial information in order to permit the Bank to re-evaluate the property. It was Mr. Gault's evidence that eventually a response came November 20, 2001 and it is located at page 83 of the Bank documents.

At pages 84 and 85 is the Summary of Personal Finances (SPF) signed by Dr. Reed and dated November 20, 2001. Dr. Reed in signing the document certifies that the information contained therein is accurate. In the statement Dr. Reed shows a net worth of $381,000 and the properties owned by him and Dr. Lawless valued in the amount of $1,450,000 with mortgages of $657,000. The statement shows employment income including his spouse's to be $250,000 with total annual expenses of $66,300. At the bottom of the statement Dr. Reed indicates that he has not gone bankrupt, that he or his spouse are not involved in any law suits, that they have no judgments filed against them, and that they have not guaranteed any other loan obligations.

In the same connection is Dr. Lawless's SPF at pages 72, 73 and 74 of the records. His statement is dated November 12, 2001. In his statement of real estate, Dr. Lawless lists real estate interests of some $1,612,000 with mortgage amounts of $404,000 for a net real estate value of $1.2 million. As part of the statement, Dr. Lawless gave his income tax return for the year 2000, which shows net income of $276,000. Again, at the bottom of page 72 of his SPF, Dr. Lawless denies bankruptcy, involvement in any law suits, having judgments filed against him or guaranteeing other loans.

At page 65 is the Commercial Credit Application wherein Mr. Gault proposed a renewal of the doctors' mortgage, which he forwarded on to Ottawa Branch by letter at page 63. In that letter, he notes that he has received information revealing a total income for Dr. Lawless of $381,000 and $251,000 for Dr. Reed, and noting that they can more than cover the shortfall on the monthly payments as well as keeping taxes current.

As a result, the mortgage was renewed for one year to June of 2002.

At page 41 is the letter from Mr. Gault to Dr. Reed advising that the mortgage will be maturing June 1, 2002.

At page 54 is Dr. Lawless's SPF dated May 7, 2002 wherein he shows total real estate of $1,110,000 against mortgages of $490,000. It should be noted that with respect to 275 Queen Street, he estimates his interest in that building to be $400,000. He shows a net income of $300,000 against expenses of $130,000 and he indicates at the bottom of the page that he is not involved in any law suits.

At page 42 is Dr. Reed's letter to Mr. Gault dated May 2, 2002 enclosing his SPF which starts at page 43. Dr. Reed in his statement shows assets of $575,000, liabilities of $198,000 for a net worth of $377,000. He lists total real estate of $1,475,000, but in doing so appears to include both his interest and Dr. Lawless's interest in the partnership properties. Against the $1,475,000 he notes mortgages of $618,000. He lists income of $250,000 and notes that he is not involved in any lawsuits.

At page 35 is Mr. Gault's inspection report dated May 14, 2002. He notes that from the outside the building appears to be in good shape. Filed as Exhibit 84 was a photograph taken by him that date.

At page 32 is a report prepared by Mr. Gault with respect to the renewal. He notes that the building is maintained in good condition. He notes as well that during the June, 1996 renewal the security was valued at $1,250,000 for the two buildings, and was reduced to $830,000 in the June, 1998 renewal, and at this point in time the appraisal is down to $826,000 for both buildings. In this report, Mr. Gault notes the doctors' net worth and incomes.

At page 28 is a memo from Mr. Gault to Head Office with respect to the anticipated renewal. It is dated June 10, 2002. It notes that Drs. Reed and Lawless have made a proposal to pay down the mortgage by $250,000 to $270,000 from personal resources, in return for which the Bank would release its charge over the garage portion of the property and take a reduced mortgage for a five-year term. Mr. Gault explained that Dr. Lawless had made this request and Mr. Gault was eventually able to obtain approval for the renewal on this basis (see page 15) which saw the mortgage reduced from $605,000 to $371,000 upon a pay down of $234,000. This of course resulted in a reduced monthly payment.

Cross-examination by Mr. Mueller

Under cross-examination Mr. Gault acknowledged that at no time did he ever ask either of the plaintiffs anything about the physical condition of the building or whether there might be something wrong with it. Throughout he found the plaintiffs to be totally honest.

With respect to the question in the Summary of Personal Finances to the effect, “Are you or your spouse involved in any law suits or claims”, Mr. Gault explained that the object of that question is to determine whether or not the borrowers have any actions against them.

Mr. Mueller then went through an exercise with Mr. Gault where the two of them compared the information that was being provided in the Summaries of Personal Finances, particularly those delivered in the spring of 2002, with the evidence that was given by the plaintiffs during the course of this trial. To facilitate this exercise, Mr. Mueller produced transcripts of the evidence of Dr. Reed taken May 15, 2002 and the evidence of Dr. Lawless taken May 17, 2002. Without going into significant detail with respect to each of the comparisons, it is fair to say that for the most part the evidence given by the doctors with respect to their financial resources during the course of their evidence here at trial was consistent with the information that they were giving to the Bank in the spring of 2002. It was Mr. Gault’s evidence that he found no substantial differences between the doctors’ statements of personal finances as submitted to the Bank and their testimony during the course of this trial.

This witness confirmed that the doctors have not asked for a $600,000 to $700,000 mortgage to assist in remedying the building, and he confirmed that if they made that request it would be a matter of pretty serious concern for the Bank.

Re-examination by Mr. Nelson

Mr. Nelson pointed out to Mr. Gault that at page 65 of Dr. Reed’s transcript from trial, taken May 15, 2002, the following exchange took place:

> Q. Okay, and this mortgage, did it come up for renewal, how did it work, what were the… this mortgage, was it a five-year mortgage?

A. It's a five-year term, it has been a five-year term, more recently I think it was a three-year term.

Mr. Gault confirmed that as of May 15, 2002 it was still a one-year term and he was unable to say with any certainty when the five-year term was first raised by the plaintiffs. He does know that he sent his letter to Toronto on June 10, 2002 confirming that the plaintiffs were looking for a five-year term. As well he was able to confirm that in 1998 the mortgage was renewed for a period of three years, and then in 2001 for a one-year term.

Mr. Nelson pointed out to Mr. Gault as well that on page 66 of the transcript, the following exchange took place:

Q. Okay, and are there rights to make capital payments when the term comes up?
A. That's correct.
Q. See that's what I thought. When you testified you said to the court that the mortgage had been $1,100,000. I think you've got it down to about $600,000 now.
A. I think it will be $618,000 very shortly.
Q. Okay, because you are about to make another capital payment?
A. We have to the best of my knowledge never made a capital payment.

Mr. Gault, with this having been put to him, replied simply that the plaintiffs had not in fact made any capital payment as of May 15, 2002.

Mr. Nelson pointed out to Mr. Gault the Summary of Personal Finances of Dr. Reed at page 43 of the Bank documents, where he indicates that he has no personal loans, while in his transcript at page 24, the following exchange takes place:

Q. Do you still have practice loans?
A. I do.

Mr. Gault explained that very often banks distinguish between personal loans and loans that are attributable to the profession, or if you will the business of the profession. In this regard he did not consider Dr. Reed's evidence necessarily inconsistent with the Summary of Personal Finances.

I invited each counsel to indicate to me by way of submissions what they thought the significance of the evidence of the Bank manager was.

Mr. Nelson was of the view that it went to the issue of resources and the ability of the doctors to fund repairs. As well he submitted that the evidence went to the credibility of the doctors in that they both certified that they were not involved in a law suit in their respective Summary of Personal Finances, and that Dr. Reed testified during the first portion of the trial that they had never made a capital payment on the mortgage. He did along with Dr. Lawless a few days following the adjournment of the trial make such a capital payment.

Mr. Mueller submitted that the ability to finance or inability to finance was only one factor for the plaintiffs to consider in determining whether to go ahead and perform the repairs to the wall. It was Mr. Mueller's submission that the cross-examination of the Bank manager showed that what the plaintiffs testified to at trial was entirely consistent with the Bank records.

Witness – Stephen Murphy

Examination-in-Chief

Mr. Murphy is the supervisor of the building section for the City of Kingston. He is involved in both building inspections and enforcement of property standards, and has held that position for some five years. Before that he was a building inspector for the Township of Kingston.

He confirmed for the court that he was under subpoena by Mr. Nelson which subpoena required him to bring any documents relating to this project such as drawings, orders, complaints and inspection reports. In response to the subpoena he had conducted a search of the City archives and the current files, and he advised that the result of the search revealed no complaint made by Dr. Drysdale, Mr. Johnston or Drs. Reed or Lawless with respect to the subject property.

Reference was made to Exhibit 1, Tab 7 which is the building permit dated August 28, 1985. It was issued by Jack Wright who is no longer employed by the City. His whereabouts are presently unknown by this witness.

Cross-examination by Mr. Mueller

The witness confirmed that he has never been notified of any situation unless the situation is unsafe or constitutes an immediate danger. He agreed generally with Mr. Mueller's suggestion that he would not expect to receive from interested parties notice of repairs that were being considered over a lengthy period of time.

In response to questions that I put to the witness, he advised that he suspects that there are documents relating to this project somewhere, but they have yet to be found. Apparently, the search requested by Mr. Nelson continues and there may yet be results from whoever is conducting the search. The witness undertook to advise counsel in the event that any documents were uncovered.

Witness – Chris Roney

Mr. Roney is 40 years of age. He has been a licensed engineer since 1993. His expertise is in the structural field. He has been involved in a number of masonry investigations. Because of his relative youth, he has not designed any building to the 85/86 Code. He has not been personally involved in a situation where the wall was designed to be solid but was built as a cavity. He believes that he has been involved in perhaps a couple of situations where remedial ties had to be implemented.

I found Mr. Roney to be qualified as an engineer and more specifically a structural engineer and qualified to provide me with his opinions in that respect.

Examination-in-Chief

Mr. Roney confirmed that the obligation of the engineer if he encounters an unsafe condition which threatens the safety of the public is to report the matter to the client for proper action, and if no proper action is taken by the client then to report to his own profession and to the municipality. Although there have been some seven engineers involved with this project, he is not aware of any such reports being made. In his view, the building is safe.

The following reports of Mr. Roney were entered as exhibits: Exhibit 86, report dated July 2, 1999; Exhibit 87, report dated April 24, 2002; Exhibit 88, report dated May 14, 2002; Exhibit 89, report dated October 21, 2002.

Mr. Roney first inspected the property in 1999 and the results of that inspection are incorporated in Exhibit 87, his report of April 24, 2002. As I understood his evidence, what he has done is to incorporate Exhibit 86, his report of July 2, 1999 into Exhibit 87, his report of April 24, 2002. He explained that during his 1999 inspection, he looked at two interior test openings and then in 2002 he looked at four exterior test openings and twelve small bore holes where a boroscope was used.

The purpose of Exhibit 87 (April 24, 2002 report) was to assess the condition of the exterior wall and compare it with the original design and specifications to determine whether or not it was built in accordance with the design and specifications, and further to assess the structural integrity of the wall.

In April of 2002 the only reports that Mr. Roney had available to him were those of architect Johnston. Mr. Roney saw the scope of his report to be to examine the exterior and the interior of the building for the purpose of addressing the structural problems as raised by Mr. Johnston, and determining the validity of those alleged problems. In this regard Mr. Roney did not prepare any structural analysis of the superstructure as had Dr. Drysdale.

For the purpose of writing the report, Mr. Roney reviewed all the available drawings and specifications in attempting to interpret and determine the intention of the designs and specifications. He then conducted a field review and looked into certain wall openings to determine how the wall was actually built as compared to the drawings and specifications.

The Code in effect in 1985 was the 1984 OBC which in turn references M-78, the National Standard of Canada for Masonry Design (Exhibit 47).

Put into evidence as Exhibit 90 were a set of drawings that Mr. Roney had obtained from Emmons & Mitchell. There are seven in all, and include the following: Site Plan No. 1 – Garbutt; Elevations No. 2B – Garbutt & Roberts; Plans No. 3 – Garbutt; Typical Floor 3A – Garbutt & Roberts; Sections No. 4 – Garbutt; Sections No. 5 – Garbutt; Structural S1.

I was advised that these drawings are found in Exhibit 10 and two of them are contained in Exhibit 11.

In addition, Mr. Roney had the three small drawings that are contained in one of the Exhibit books, Exhibit #56, Tab C, and are in as an Exhibit unto themselves. They are the drawings of Mr. Roberts showing details and are attached as Appendix A to Mr. Roney’s report, Exhibit 87.

Mr. Roney went on to explain, referencing the Sections No. 4 drawing (which is also at Tab A of Exhibit 56), that it shows the north and south walls connected by the joists and the webbing which forms the underside of the floor slab. Mr. Roney noted that on this drawing the brick and block and stud are shown only by lines, so one cannot see any details.

With respect to Sections No. 5 (also seen at Tab A of Exhibit 56), this in effect shows the building cut down the long length and shows the east and the west walls as opposed to drawing No. 4 which is a cut through the short side showing the north and south walls. Mr. Roney noted again that details are wanting in this drawing. He noted however that the details are shown in the three small drawings in Appendix A to his report (the Roberts drawings), although these three small detail drawings are not referenced at all in the large drawings.

The first small drawing in Appendix A shows the perimeter beam with no plate drawn to the underside of the beam. Mr. Roney took this to be a typical floor detail at the east and west walls.

Drawing number two in Appendix A would be the first floor detail at level two as shown on drawing Section No. 5.

Drawing number three would be the same east and west walls detailed at roof level.

In his report, Mr. Roney notes that the three detailed sketches illustrate a wall construction consistent with that specified on drawings four and five. The wall construction is noted to consist of four-inch block, four-inch brick and an interior insulated steel stud wall. The details show an unidentified gap between the wythes of brick and concrete block. It is not clear whether this gap, which was drawn at about the same thickness as a mortar joint, was intended to be filled solid with mortar to form a collar joint, or left open. Mr. Roney notes that in his opinion the detail could have been interpreted either way. However, in his *viva voce* evidence, Mr. Roney in referring to small drawing number two notes that the cross hatching which goes through the horizontal brick mortar joints indicating a solid joint, does not go through the space between the brick and the block.

As Mr. Roney notes in his report, Division 4 section 2.1.1 of the specifications calls for a continuous 19 millimetre void space to be maintained behind the brick throughout the project. Though no scale is specified for the detail sketches, the joint illustrated between the brick and the block is about the same width as the mortar joint between bricks which is only approximately 10 millimetres, so that the joint as between the brick and block as illustrated is clearly not as wide as 19 millimetres. It was Mr. Roney's view that given the lack of any notation, hatching or identification of the gap and since the specifications called for a void space behind the brick, it would be reasonable for a contractor to interpret that the intent of the contract documents was that the two wythes of masonry are to be constructed with a void space between them.

Referring to drawings four and five, the north and south walls and the east and west walls respectively, Mr. Roney noted that the joists in this building extend from the north to the south walls. There are no joists extending from east to west, and as a result the beam is flush with the floor. With respect to the north and south walls, there are four inch shoes on the joists that stand on the beams, and therefore the beams are four inches lower on the north and south walls to accommodate the joists. However, Mr. Roney notes that no detail drawing was provided to illustrate the condition at the north and south exterior walls.

With respect to the masonry wall ties, Mr. Roney noted that there are three possible references to same in the specifications, Division 4, Exhibit 4, Tab 1:

a) section 2.1.1 provides under the heading 'Brickwork' that units shall be tied every other course as per drawings. Mr. Roney notes that there is nothing in the drawings relating to ties.

b) section 2.2.8 provides "anchor and bond masonry to conform with National Building Code"; and

c) section 2.2.6 provides under the heading 'Reinforcement', "Duro-O-Wall Truss manufactured by Durowall" or "block truss manufactured by Block Lock Limited, Wire 9

> Imperial Gauge deformed bright finish for interior wall location, galvanized for exterior wall location". Mr. Roney explained that possibly this refers to a horizontal ladder type tie to be used to tie the brick to the block. Filed as Exhibit 91 was an excerpt from a magazine relating to this type of tie. He did however stipulate that it was speculation only as to whether section 2.2.6 applied, but these ladder type ties are sometimes used to tie the brick to the block.

Mr. Roney pointed out in his report that M-78 (Exhibit 47) contains prescriptive requirements for type, size and spacing of masonry ties, and that ties are a very common element in masonry walls. He expresses the view that if the ties are simply required to secure the brick to a back-up wall where "the brick wall is simply along for the ride" then light strap ties are suitable. If, however, the ties are required to bond the wythes such that they exert common action under load, then M-78 prescribes a heavier tie system. In any event, if the designer required that the wythes be structurally bonded, then he would be required to explicitly specify the type, size and spacing of the ties. Without such information, opines Mr. Roney, it would be expected that the contractor would employ strap ties or light gauge joint reinforcement ties, since these ties were the most common type in use at the time of construction. Mr. Roney noted that the drawings show nothing with respect to ties, and that the specifications make only general reference to ties. It was his opinion that in the circumstances the contractor would assume that he could use strap ties. I asked Mr. Roney the question, why would the contractor assume, why would he not ask, and Mr. Roney said he had no answer to that question.

Referring to section 4.3 of his report, Movement Joints, Mr. Roney noted that section 3.1.9 of M-78 requires that control joints shall be provided in masonry when necessary to relieve excessive temperature and shrinkage stress, and section 3.2(e) provides that drawings and specifications to build a building incorporating masonry shall indicate the details and location of control joints. In this respect, Mr. Roney pointed out that section 2.2.2 of the specifications provides "provide control joints as dictated by structural design." Mr. Roney noted that in the face of this, the drawings show no control joints and none were provided other than a vertical expansion joint on the south side of the building.

At the same time, section 2.1.1 of the specifications provides "provide six millimetre continuous expansion joint caulked continuously between steel support angles and brick panels each floor." Mr. Roney noted that there was nothing in the drawings relating to expansion joints, but noted that specifications sometimes call for things that are not really required and sometimes, depending on the circumstances, contractors will go by the drawings notwithstanding the specifications. Mr. Roney agreed that the specification is clear but on the other hand he would have expected to see it on the drawings as well. [I think Mr. Roney is stretching on this point.]

We know of course that there is no expansion joint between the steel support angles and the brick panels at each floor, and that what is there is a solid mortar joint.

With respect to the anchorage of the block wall, Mr. Roney noted that drawings four and five show that that wall sits directly on the concrete floor slab, and extends to the underside of the steel beam. He notes as well that the small drawings show that. He notes that neither the specifications nor the drawings specifically address the anchorage of the top of the block wall to the underside of the beam. He looks to the small drawings, particularly drawing number two, which shows the hatching extending beyond the top of the block to the underside of the steel beam, and suggests that that drawing appears to require a mortar joint. He concludes that the intention was that there by a mortar joint at the top of the block extending to the underside of the beam. As we have seen, such was not consistently the case. Mr. Roney notes that section 3.6.1 of M-78 provides that masonry walls shall be anchored to their lateral supports by either metal anchors or "other approved construction systems", and it was his evidence that the mortar joint could constitute an "other approved construction system".

With respect to the shelf angles, at the time Mr. Roney prepared his report he assumed that the typical floor to floor height was three metres. It is actually 3.2 metres. In any event, while there are no shelf plates on the east and west walls, he is of the view that section 6.2.3 of M-78,Exhibit #47 requires only a shelf plate at the fourth floor of this building. Section 6.2.3 provides that "unit masonry veneer more than 11 metres above the top of the foundation wall shall bear on masonry, concrete or other non-combustible bearing supports spaced not more than 3.6 metres vertically." In other words, it is his opinion that only the height above 11 metres need be supported independently of the foundation. The total height here is about 13 metres, so that in his opinion only two metres need by supported.

With respect to the plates, Mr. Roney goes on to note that detail drawing number one does reference a plate with the notation Pl. 8 x ¼ inch. As well, on drawing S-1 there is a notation for all perimeter beams to have the plates. However he notes that the plate itself is not illustrated as such in the detailed drawings. He notes that the sections on drawing four appear to show plates on the underside of the perimeter beams extending towards the exterior face of the building, but notes that these plates are shown supporting the exterior brick wythe at each floor level, at the north and south walls of the building, and no such plates are shown in drawing number five which illustrates the condition at the east and west exterior walls. As such, he notes there is an apparent contradiction between drawing S-1 which calls for plates at all of the perimeter beams and detail drawings which specify a plate but show none, and the cross section of drawing five which neither shows nor specifies any plate.

That then brought Mr. Roney to an analysis of his field review which took place in both 1999 and 2002. He was not present when the interior test openings were made in 1999 but was present through part of the making of the exterior test openings in 2002. With respect to the exterior test openings in 2002, he noted that the bricks had to be sawed out and that at least two ties had been cut while the openings were being made. In this respect, Mr. Roney looked at the two ties which comprise Exhibit 60. With respect to the first tie that had broken into two, he noted abrasions on the tie and speculated that it had been cut. As well, he saw corrosion on the end of the tie which he assumed was in the block wall. With respect to the second tie he was unable to say which end went into the brick as opposed to the block wall. He did see abrasions on both ends and surface corrosion on one end.

In his evidence, Mr. Roney covered the two interior test openings that he observed in 1999. Opening number one was on the south side, second floor where the drywall, insulation, plastic and eventually the blocks were taken out of an area approximately a metre long and half a metre high from the window lintels to the underside of the beam. The second interior test hole was on the east wall, second floor, again over a window.

In test hole number one, Mr. Roney observed a mortar joint between the top block and the underside of the beam. As well, he measured a void of some ¾ inches between the brick and the block wythes. He noticed in this void numerous mortar fins but no accumulation of mortar at the lintels. He noted that the mortar joint between the top of the brick and the underside of the plate was solid. He noted minor corrosion on the top of the window lintels. There were no flashings, no weep holes and no ties observed.

In test opening number two on the east side of the building, the steel beam was tight to the underside of the floor. The steel plates had been cut off the beam. There was a mortar joint between the top block and the underside of the steel beam. Again there was a void between the brick and block wythes, except for mortar fins. The width of the void ranged from 7/8 inch to 15/16 inch, or 22 to 24 millimetres. Again, no flashings, ties, soft joints or weep holes were found at this test opening.

Mr. Roney then reviewed his findings with respect to the test openings in 2002. These were openings on the exterior of the walls as opposed to the inside of the walls as in 1999. There were some four exterior openings made. The locations were selected by Mr. Roney and the openings were actually created by workers including one from the defendant Emmons & Mitchell and perhaps one from the masonry contractor.

The first exterior opening was at the southwest corner of the building, at the first level above the street. The opening extended around the corner so that the interior of both the south wall and the west wall could be observed.

Test opening number two was near the roof of the building on the north wall, adjacent the most easterly window.

The third test opening was at the level of the fourth floor, at the northeast corner. Both the interior of the north wall and the east wall could be observed.

Finally, the fourth test hole was at the parapet level on the east wall.

Also used by Mr. Roney during the course of his investigation was a boroscope which could be inserted through a hole drilled in the wall in order to observe, albeit in a limited fashion, the interior of the wall, that is between the brick wall and the block wall, and a wall scanner which is essentially a metal detector used to pick up the location of ties.

With regard to test opening number one, Mr. Roney observed that the beam on the west wall was up tight to the floor. The beam on the south wall was four inches lower to accommodate the four-inch joist shoe. The void in the west wall between the brick and the block was 7/8 inch -– 22 millimetres to 1 5/16 inch – 33 millimetres. The void on the south wall was ¾ inch – 19 millimetres below the second floor, and 3/8 inch to1/2 inch – 10 to 13 millimetres above the second floor. It could be observed on the west face of the building that the shelf plate had been cut off. If it had not been cut off it would not have lined up with the brick coursing. On the south side of the building the shelf plate did line up with the brick coursing.

Mr. Roney noted on the south wall a rowlock course of bricks extending inwards and resting upon the concrete floor slab. At the west wall, the rowlock course had been cut off flush with the inside face of the bricks and did not extend into the floor slab. A rowlock is a brick laid on its edge so that its end is visible. A row of 9 gauge joint reinforcement was encountered at the test opening on the west wall, approximately 13 inches below the finished floor. As well, a strap tie was encountered approximately 34 inches below the finished floor, and 21 inches from the corner of the wall. Additional strap ties were located using a wall scanner. They were spaced vertically between 16 and 22 inches, and horizontally from 11 to 19 inches on centre. Accumulations of mortar were found on top of the observed ties. At the south wall a strap tie was encountered in the test opening, and additional ties were located with the scanner. Mr.

Roney noted that the ties were bent down from the brick to the block to accommodate the coursing. In the south wall, Mr. Roney observed one strap tie in the opening, and additional ties were detected using the wall scanner.

At test hole number two on the north wall near the roof, Mr. Roney observed the shelf plate at the bottom of the beam extending into the brick wythe. He observed a mortar joint between the top of the block and the underside of the beam, but noted that on occasion brick units had been used at the top between the block and the underside of the beam. The void between the brick wall and the block wall was measured at 1 3/8 inches or 35 millimetres. Mr. Roney observed over the window lintel a strap tie embedded in the brick but not in the block. It was in good shape but effectively was doing nothing. He noticed as well helifix stainless steel ties from the brick down into the wood blocking above the steel beam. He was told that these were remedial ties inserted by Mr. Johnston. Mr. Roney observed no other ties in the vicinity of the test opening, and the nearest ties other than above-stated were located by the wall scanner some 53 inches below the beam. Mr. Roney observed as well a horizontal crack where the plate was embedded in the mortar joint between the brick courses.

In test opening number three at the northeast corner, fourth floor, Mr. Roney was able to observe on the east wall that the shelf plate had been cut flush with the beam flange. The area between the top and bottom flanges of the beam was found to have been infilled with a combination of concrete block and brick. The void between the brick and the block was 7/8 inches – 22 millimetres to 1 ½ inches – 29 millimetres. On the north wall Mr. Roney observed joint reinforcement some 12 inches below the fourth floor level in the opening itself. With the scanner he was able to locate more horizontal joint reinforcement 16 inches above the opening and 16 inches and 32 inches below the opening. These joint reinforcements are of the ladder type and lie along the length of the brick course. On the east wall 9 gauge horizontal joint reinforcement was encountered in the brick at the mid-height of the perimeter steel beam and approximately 16 ¾ inches below this a strap tie was encountered with others detected by the scanner. At this opening, Mr. Roney also observed a vertical mortar joint provided between the north end of the concrete wall and the upright steel column.

At test opening number four, at the parapet level on the east wall, Mr. Roney observed brick and concrete block infill in the flange of the beam. The void between the brick wall and the block wall was 1 3/8 inches – 35 millimetres.

Entered as Exhibit 92 were all of Mr. Roney's drawings relating to his observations of the test holes.

Mr. Roney then went on to discuss the conclusions that he had reached based on his review of the drawings and the specifications and his on-site observations.

He concluded that the drawings and specifications suggest that the designer wanted an unfilled void between the brick and block walls and that this void space was provided by the contractor, albeit with frequent mortar bridges and mortar on the ties. The wall as built is in general conformity with the drawings and specifications but the width of the void varied. Mortar fins were undesirable but not of any structural significance.

He concluded that the exterior face of the walls is in good condition. Inside, the vertical joints between the blocks are not as good as they might be but the horizontal joints are satisfactory.

With respect to the ties, he observed two types; the strap ties and the ladder type horizontal ties. It is his opinion that both would satisfy the standards at the time where "the brick is just along for the ride" and not sharing a load with the block wall. He notes that no special ties were required in the drawings or the specifications. He opines that the type of ties being used would have been visible as the walls were laid up in the course of construction. If the designer wanted to create a wall that saw a sharing of loads, that is both the bricks and the blocks bearing the loads, then in his view he would have specified special ties. He noted again that many of the strap ties that he observed were bent to accommodate for coursing as between the bricks and the blocks. He noted that the standards in 85-86 were silent with respect to the issue of bending but later standards do not permit the bending of strap ties in that it reduces strength on compression.

It was his opinion that it was relatively common practice in 85-86 to bend the ties to accommodate the courses. In view of the fact that he was not licensed to practice as an engineer until 1993, he was asked how he might know the practice in 85-86 and he commented that he does a lot of work on buildings that were constructed prior to his license. In any event, in his conclusions he finds that in some areas uncovered during his test openings, he did not find ties at the required spacing. Assuming the conditions at the test openings are representative of the conditions elsewhere in the wall, he has concluded that additional remedial tying of the wythes is necessary at certain localized areas. In conclusion, on the issue of ties, he is of the view that the type of ties implemented are in accordance with the designs and specifications, and he is of the view that if the designer did not want those types of ties used, then he should have so indicated.

With respect to the movement joints, Mr. Roney explained that he considers a control joint to be a joint which provides for the relief of the stresses of expansion or contraction between two bodies. The control joint is usually a space that is caulked. He sees an expansion joint as specifically allowing the building frame to flex without imposing a load on the brick. Again, the expansion joint is usually a caulked space so that they are essentially the same thing but they address different phenomena. As far as the brick wall was concerned, he observed some cracking on the exterior, most notably at the top in the brick joints themselves. All the horizontal cracking in test hole number two on the north wall near the roof coincided with the shelf plate location, but he considered the cracking relatively minor for a building without control joints of this age. He sees the responsibility for control joints to be that of the designer. He notes again that the specifications call for a continuous expansion joint between the underside of the shelf plate and the top of the brick panels at each floor, but says that the introduction of such joints is not required given the design and performance of the building. It is his opinion that such joints are typically provided to allow for differential movements between the masonry and the building frame, but such a concern primarily relates to concrete frame buildings, and steel frame buildings such as ours are dimensionally stable and anticipated stresses developed by normal volumetric changes in the brick and frame would be small.

With respect to the heading ‘Bearing Support and Anchorage for Exterior Masonry Walls’ (section 6.4 of his report dated April 24, 2002, Exhibit 87), Mr. Roney is of the view as previously expressed that the drawings show a mortar joint between the top of the block wall and the undersurface of the steel beam. He concludes that this is what the designer intended, to achieve lateral support. It was his evidence that he carried out an analysis that resulted in his concluding that there was sufficient potential for bond between the top row of blocks and the underside of the steel beam to provide for adequate lateral support. He explained that “bond” is a chemical adherence. His analysis indicates that the block wall is not over-stressed by the axial loading. In conclusion he is of the opinion that the block wall was built in accordance with the drawings, that is a mortar joint at the top between the top block and the underside of the steel beam, and that there is sufficient lateral support at the top of the block walls.

With respect to the brick wall, Mr. Roney concedes that it was a major decision to cut the shelf plates on the east and west walls, a decision that was not to be made lightly. [One wonders why he is of this view when he feels that only the top two metres of the wall as constructed need be supported by a shelf plate.] He then went on to explain that the entire east and west walls are free to expand whereas the north and south walls are confined by the plates. He sees the problem then to be at the corners where differential movements without control joints may cause cracking, which indeed he observed. He did however carry out an analysis to determine that the full height of the east and west walls as built is structurally acceptable, provided that strategically placed vertical control joints are provided at the corners of the building.

In section 6.5 of his report, April 24, 2002 (Exhibit 87) Mr. Roney addresses the structural analysis of the exterior masonry walls that he has carried out in order to determine whether the walls as designed and constructed can safely resist the vertical and lateral wind loads specified in the Ontario Building Code in effect at the time of construction of the building.

In his report of April 24, 2002 (Exhibit 87) he states, “Our analysis has determined that the walls as constructed can safely resist the out of plane lateral and anticipated vertical loads specified in the OBC with the exception of the portion of the east and west walls above the level four floor.

The concrete block wythe at this latter area is overstressed when subjected to a combination of lateral wind and vertical gravity loads by approximately 39%." Mr. Roney explained that the analysis which brought that result was based on an assumption that the between floor height was three metres. It was eventually brought to his attention by Mr. Nelson on May 2nd, 2002 that the as-built height between floors was 3.2 metres. See Exhibit 93 which is the letter from Mr. Emmons to Dr. Brian Reed dated August 14, 1985 relating to the change in height.

Exhibit 88 is Mr. Roney's report dated May 14, 2002 setting out his various calculations contained as Appendix B. These are the calculations that are referred to in his report of April 24, 2002, Exhibit 87, but which are not actually attached to or contained in that report. In Exhibit 88, in his covering letter, Mr. Roney suggests that his calculations at Appendix B in his report of May 14, 2002 be inserted as Appendix B to his April 24, 2002 report.

As well, in his covering letter, Exhibit 88, he asks the reader to note that his initial calculations used in the preparation of his report were carried out using the current version of the Ontario Building Code (1997) and CSA Standard 1994. He notes that the use of the modern Code and Standards was dictated by the time constraints imposed by the court since the analysis could be carried out more efficiently using the modern analysis tools available to him today which are tailored to the present Code. He notes that following the submission of his April 24, 2002 report, he had the opportunity to recalibrate his analysis of the walls using the Code in effect at the time that the building was constructed, that is OBC 1984 and CSA Standard M-78.

Mr. Roney notes as well in Exhibit 88 that his initial calculations were based on a three metre floor to floor height for the building, and having regard to Mr. Nelson's direction of May 2nd, 2002 and Mr. Emmons's letter to Dr. Reed dated August 14, 1985, he has adjusted his calculations to reflect the 3.2 metre height.

In Exhibit 88, Mr. Roney notes that his updated calculations have determined that the level of theoretical overstress previously reported with respect to the upper portion of the wall is less than the level given in his report of April 24, 2002. He sees the overstress now to be 6% versus 39%.

He notes that the revision to the theoretical overstress is largely due to the very different design philosophy used in the earlier Standard. He notes however that the increased wall height has pushed the brick wall beyond the unbraced height limitations of the Code. He sees this as easily and economically addressed through the use of the same type of ties recommended in his report of April 24, 2002. Therefore he says the only change to the recommendations contained in his April 24, 2002 report is the addition of a row of remedial wall ties near the bottom flange of the perimeter beams and the east and west walls at <u>each floor level</u>. He says the approximate number of additional ties not formerly specified is 110 and he estimates the additional cost would be in the order of $1,100.

Mr. Roney in his evidence further explained that in his original calculations, although he used the provisions of the 1985-86 Code (that would be OBC 1984) and CSA Standard M-78, the software package that he used to assist him in making his calculations had incorporated the 1997 Ontario Building Code and the 1994 CSA Standard. In order to revise his calculations, he had to alter the floor to floor height from three metres to 3.2 metres and he had to develop in-house software to assist him in his calculations rather than use the commercial software known as Masonry LSD 95. He explained that before going into engineering he was a computer programmer. He provided no details of the software program that he used in his fresh calculations. In any event, his bottom line conclusion was that the floor to floor height of 3.2 metres as opposed to three metres pushed the height of the wall beyond the height limitations in the 1984 Code and so additional ties were required in the form of an additional row of remedial ties at each floor level near the bottom flange of the beams.

Mr. Roney then proceeded to review his calculations in Exhibit 88. He started out by again pointing out that the calculations were carried out in accordance with the 1984 edition of the OBC in effect at the time of the construction. The 1984 OBC of course specifies that buildings made of plain or reinforced masonry shall conform to CSA Standard M-78 and Mr. Roney confirms that that standard has been used in his analysis.

Mr. Roney notes that the scope of his analysis was limited to the structural issues raised by Mr. Johnston, and deals primarily with the structural integrity of the exterior masonry walls when subjected to out of plane forces due to wind and earthquake. He explained that 'out of plane forces' means in and out forces.

He explained that while wind and out of plane seismic forces were calculated by him using the procedures contained in the OBC and the Supplement to the National Building Code of Canada, his analysis determined that wind forces governed the analysis of the wall panels because those forces were greater than seismic forces. At sheet number one of his calculations he calculates that design pressure with respect to wind is .70 kPa and that the design pressure with respect to seismic forces is .082 kPa for the brick wythe and .054 kPa for the block wythe.

At sheet number 2, Exhibit 88 is a list of the material properties of the elements that made up the wall, that is the concrete block and the brick.

At sheet number 3 is an analysis of the distribution of lateral loads between the wythes. Mr. Roney's calculations indicated to him that of a total lateral wind load of .70 kPa the brick takes .39 kPa and .31 kPa is transferred to the block wall. It was Mr. Roney's view that this positive load transfer from the brick wall to the block wall takes place as a result of the mortar fins and the strap ties that are covered with mortar.

Sheets four to eight relate to the north and south walls, and involve two cases. Case number one relates to the top of the wall where the wind forces may be a little higher, and the axial load less. Case number two relates to the bottom of the walls where the wind force is a little less and the axial forces are higher. The axial force of course is the downward force. The two cases are set out on page number four with respect to the block wythe and the brick wythe.

Page five relates to the analysis of the concrete block wythe and the top floor and page six relates to an analysis of the concrete block wythe at the first floor.

Page seven relates to an analysis of the brick wythe at the top floor and page eight relates to an analysis of the brick wythe at the first floor.

It was Mr. Roney's evidence that in all cases with one small exception the stresses calculated on the block wall were within allowable Code stresses. The one exception is noted on page six where at the bottom floor Mr. Roney calculated a one-third percentage overstress on the block wall. The actual stress calculated was 2.43 MPa and the allowable stress was 2.40 MPa. Mr. Roney considered this a negligible overstress.

The same process was repeated with respect to the east and west walls, and the analyses with respect to these walls are set out at pages nine, 10, 11, 12, 13 and 14. With respect to the east and west walls, Mr. Roney introduced a third case between the third and fourth floors.

The analysis with respect to the east and west walls indicated to Mr. Roney that in all cases other than one, the stresses on the wall were within those allowed by the Code. The one exception with respect to both of the walls was at the top of the wall, mid-point between the fourth floor and the roof, where there was 6% overstress on the block wall.

At sheet 15, Mr. Roney addresses the lateral support at the block walls. This is a calculation of the bond between the top block and the underside of the steel beam by way of mortar joint. He explained that the worst case reaction at the top of the block wall was R=0.51 KN/M. He calculated the ultimate bond strength between the block and the beam at 1.43 MPa's. He applied a factor of safety of 5 which effectively reduced the allowable bond stress to .29 MPa (1.43/5). It should be noted that the 1.43 MPa comes from publications relating to bonds. Dividing the allowable bond stress of .29 MPa over the area of contact resulted in an allowable bond strength of 7.54 KN/M which was significantly greater than the worst case reaction. His conclusion was then that the mortar joint at the top of the wall panels provided adequate lateral support. [That is assuming that there was a uniform mortar joint throughout the block wall. I have heard evidence to the contrary.]

It was Mr. Roney's view that the overstress in the block wall at the top of the building could be addressed by means that would get the brick and the block wythes to work together as a bonded wall.

As is noted in Exhibit 88, the increased wall height (from three to 3.2 metres) has pushed the brick wall beyond the unbraced height limitations of the Code. Mr. Roney expressed the view that this is easily and economically addressed through the use of the same type of ties recommended in his report, Exhibit 87, and he states that the only change to the recommendations contained in his report, Exhibit 87, is for the addition of a row of remedial wall ties near the bottom flange of the perimeter beams at the east and west walls at each floor level, the approximate number of additional ties not formerly specified is 110 and he estimates that the additional cost would be in the order of $1,100.

At item 3.0 of his calculations, he again notes the theoretical overstress of approximately 6% in the concrete block wythe at the upper floor of the east and west walls, and suggests that sheer studs between the brick and the block wythes would permit appropriate composite action to remedy the situation.

Returning to Exhibit 87, his report of April 24, 2002, Mr. Roney covered in his evidence his recommendations for remedial work.

Section 7.1 of his report relates to the remedial masonry wall ties:

a) he recommends that at the north and south walls 6 millimetre helifix ties be installed at a rate of one per four square feet of wall surface area over the entire wall, from an elevation approximately four feet above the level four floor up to the top of the perimeter steel beam (at the roof). This is in relation to test opening number two where he detected only minimal tying of the wythes. Four hundred ties would be required for this exercise.

b) at the east and west walls he recommends the same ties be installed at a rate of one per square foot of wall surface area over the entire surface area from a point 16 inches above the level four floor, up to the top of the perimeter steel roof beam (the roof). He notes that the purpose of these ties is to structurally bond the brick and block walls together so that they exert common action under load. He notes that these ties are required to correct the design deficiency discussed in section 6.5 of his report, and not the responsibility of the contractor, and he approximates 600 ties will be required.

c) he recommends a horizontal row of ties approximately two inches below each floor level and at the roof level, spaced at 24 inches on centre. The purpose is to provide a reliable point of lateral support to the brick wythe at these locations. Again, he notes that these ties are required to address a potential design deficiency, and are not the responsibility of the contractor. Approximately 300 ties are required. It should be noted that initially Mr. Roney indicated to me that these ties were for the east and west walls only, but on reconsideration he testified that they would go into all four walls.

With respect to these remedial ties, Mr. Roney explained that they are installed from outside the building. A hole is drilled in the brick wall. The ties are self-tapping into the block wall and adhered to the outer brick wall by epoxy which up close is visible.

Section 7.2 of his report, Exhibit 87, relates to the remedial work involving control joints. In this respect, Mr. Roney notes that the shelf plates have been cut off the east and west wall beams so that they can expand and contract differently from the north and south walls where the plates are in place. As a result cracking can and has occurred at the corners. His recommendation is to put in one control joint at each corner, always on the north or south wall because the plate ends before the corner. He recommends that after the control joint is installed, then additional ties should be put on each side of the joint to secure the brick every 24 inches on both sides of the joint.

In conclusion, Mr. Roney opined that his analysis has determined that the walls as constructed can safely resist the out-of-plane lateral and anticipated vertical loads specified in the OBC with the exception of the portion of the east and west walls above the level four floor which he sees to be overstressed now by 6% as opposed to the 39% referred to in Exhibit 87.

The costs of the remedial work as suggested by Mr. Roney will be addressed by Mr. Pascoal.

Mr. Roney then moved on to his report, Exhibit 89 which is a rebuttal of Dr. Drysdale's calculations in Exhibit 31, and a rebuttal of Dr. Drysdale's comments related to Mr. Roney's May 14, 2002 calculations appended to Mr. Roney's April 24, 2002 report (Exhibits 88 and 87).

Referring firstly to Dr. Drysdale's Appendix G, Exhibit 31, Mr. Roney noted that it consists of a lateral load analysis of a one-floor high section of wall and is not an analysis of the as-built or as-designed walls. The analysis is of a wall consisting of a wythe of block and a wythe of brick tied together using remedial ties only. In Mr. Roney's view, the analysis does not attempt to consider the effect of the ties which were provided in the original construction. In Mr. Roney's view this omission, likely done for simplicity of analysis and due to the effect that Dr. Drysdale did not carry out any personal observations of the existing ties or wall construction, results in an overly conservative analysis. In Mr. Roney's view, the calculations of Dr. Drysdale do not support the claim that he made in his May 6 report that the walls do not meet the requirements of the Codes and Standards in effect at the time of construction. As well, Mr. Roney notes that all of Dr. Drysdale's calculations were based on CSA 94 rather than CSA 78 which was in place at the time of construction. Mr. Roney points out that his calculations were meant to determine whether the wall as built in 1985-86 met the standards in place at that time. Mr. Roney goes on to criticize that the Drysdale analysis is performed strictly using lateral wind loads and that Dr. Drysdale has chosen to omit the effect of axial compression on the walls, that is downward forces on the wall, which has the tendency of increasing the ability to withstand lateral forces.

Mr. Roney notes that the analysis presented by Dr. Drysdale consistently suggests that the maximum bending moments occur at the floor elevations and approximately mid-way between the floors. Mr. Roney says the predicted tensile stresses at these locations are significant and if correct would result in cracks at these locations. However his observations of the actual buildings did not reveal the presence of the predicted cracks. He notes that Dr. Drysdale has stated that the cracks would be invisible, but Mr. Roney says that given the stress levels predicted by the Drysdale analysis, it is inconceivable that after 16 years of service the cracks would still be invisible. It should be noted that this is contrary to the plaintiffs' evidence and indeed to the evidence of architect John Wilson called by Mr. Garbutt.

Mr. Roney then proceeded to review in some detail Dr. Drysdale's comments relating to Mr. Roney's calculations of May 14, 2002 (Exhibit 88), which comments are contained in Exhibit 32. There is no good purpose to be served in going through each of Dr. Drysdale's comments (there are 14 in all) and Mr. Roney's response to each of these 14 comments as contained in Exhibit 89. The comments of Dr. Drysdale (some 14 in all) are there to be read in Exhibit 32 and Mr. Roney's responses with respect to each of those comments are there to be read in Exhibit 89. A consistent theme that runs through each of Mr. Roney's responses is that Dr. Drysdale uses CSA Standard 1995 adopted by OBC 1997 in making his determinations, whereas Mr. Roney says that CSA Standard 78 as reference in the 1983 OBC should be used for the purpose of analysis, bearing in mind that the purpose of Mr. Roney's analysis was to assist the court in determining whether or not the as-built construction of the building in question met the standards of the day – not whether it meets today's standards. Notwithstanding this consistent theme, Mr. Roney did concede that even the limited remedial work suggested by him in the form of ties and control joints constitutes an up-grade and as a result the present Code and present CSA standards apply to the up-grade. It should be noted that with respect to Dr. Drysdale's item 14 where he says that the masonry is so rigid that tying the brick to the bottom flange of the beam and to the top will produce cracking simply because of movement of the beam, Mr. Roney responds that examination of the condition of the masonry at the test openings revealed no such cracking after 16 years of service, except for a horizontal crack found at the north wall coincident with the plate on the bottom flange of the perimeter steel beam at roof level. He notes that it is not clear whether this observed crack was related to lateral movement of the beam. He notes further that if such lateral movement was the cause, he would have expected to encounter similar cracking elsewhere. This response of course completely discounts the "invisible" cracks referred to by Dr. Drysdale.

In his closing remarks in response to Dr. Drysdale's critique of his calculations in Exhibit 32, Mr. Roney in Exhibit 89 notes the following,

> No model is a perfect predictor of the actual mechanisms and stresses present in a real building. All models must be carefully developed to account for the actual conditions present and the results must be verifiable in the actual structure. If the analytical results are not consistent with the observed performance of the building, then the model cannot be considered accurate. This concept is perhaps best stated in the following excerpt from the National Research Council of Canada's Institute for Research in Construction: 'The Code recognizes that the real world is the best test laboratory and that current scientific knowledge cannot necessarily explain in detail the extent to which various mechanisms might be at work in any particular situation. Consequently many Code provisions describe constructions that have demonstrated effective performance over time whether or not we understand exactly why. As we learn more about building performance, the requirements of the Code are updated to reflect that understanding. Great care must be taken however to ensure that scientific evidence is verifiable. Performance as exhibited by a computer model may not in of itself be adequate to support a requirement or the elimination of one.'

That said, Mr. Roney notes that if Dr. Drysdale's analysis was correctly predicting the stresses within the walls of this building there would be significant cracking and stress clearly visible at the face of the walls. Horizontal cracks would be evident, particularly at mid-height between flights, most notably between the window openings. Mr. Roney finds no such evidence of such cracks. Again, he completely discounts the evidence with respect to invisible cracks from Dr. Drysdale, and the first-hand evidence of water entering the cavity where there is no visible crack, that is water entering at the bottom of the mortar joint.

Cross-examination by Mr. Mueller

To begin with, Mr. Roney acknowledged that Part 5 of his report dated July 2, 1999 (Exhibit 86) which deals with the field review that he conducted in 1999 with respect to two interior test openings, has been transferred as a piece into Exhibit 87, his report dated April 24, 2002.

In Exhibit 86 under the heading 'Conclusions', Mr. Roney had written that having inspected the wall through two inside holes, he found that the as-built construction revealed at those test openings was in general conformity with the construction drawings and specifications with the exceptions that

a) weep holes were not provided;

b) no expansion (soft) joint was provided between the top of the brick panels at each floor and the underside of the steel plates; and

c) the steel plate at the east and west walls had been cut off and does not therefore provide bearing support for the brick.

Mr. Roney acknowledged that today, being more familiar with the specifications than he was in 1999, he would add to the above-noted list, flashings. He confirmed that these were his conclusions when he was corresponding with Emmons & Mitchell directly on July 2, 1999. Mr. Mueller put it to Mr. Roney that having reached those conclusions, he then in his report to the defendants' lawyers dated April 24, 2002 (Exhibit 87) concluded that the contractor had not failed in any respect. Mr. Roney did not agree with that suggestion.

Mr. Mueller referred Mr. Roney to his July 2, 1999 report once again where he comments that Mr. Johnston in his report states that the use of the galvanized corrugated metal strap ties is not appropriate. Mr. Roney agrees with Mr. Johnston in that the current (and in fact the previous) editions of the OBC and Masonry Standards do not permit this type of tie in a building of this type. He notes however that at the time during which the building was designed and constructed the OBC of the day specified that buildings and their structural members made of masonry must conform with M-78 and that article 6.2.6 of that standard explicitly permits the use of galvanized strap anchors found by Mr. Johnston for the tying of masonry veneer such as that used in this building. [We must bear in mind in this regard Mr. Garbutt's evidence that he designed this wall as a solid wall and not as a veneer wall.] Mr. Roney conceded that if the intent was to have the brick wythe and block wythe combined to resist forces together then the 85-86 Code required stronger ties than were used.

In his report of July 2, 1999 Mr. Roney notes that the responsibility for the omission of flashings is not entirely clear and suggests that it is certainly an item which should have been clearly detailed on the drawings but this was not done.

In the closing paragraph of his report of July 2, 1999 Mr. Roney identifies the lack of weep holes (and thus the lack of flashings) and the absence of the expansion joints between the top of the brick panels and the underside of the steel plates as being deficiencies on the part of the contractor but in his view not uncommon in construction of this type. He notes that the deficiencies can be easily rectified. He notes as well that his firm has encountered past situations in other buildings where the bearing support for the brick veneer was not provided at the intervals required by the Building Code and his firm has devised methods of correcting such a condition in an economical fashion. Mr. Mueller asked Mr. Roney whether in writing those words he was thinking of replacing the support plates that had been cut off the beams. He first responded negatively and then said, possibly yes. [It is interesting to note that Mr. Roney in his first letter to Emmons & Mitchell did not express the view that the support plates were not required at every floor in any event, according to the Code.]

Finally, with respect to Exhibit 86 Mr. Roney confirmed for Mr. Mueller that there had been no drafts prepared of that report.

With respect to Exhibit 87, his report dated April 24, 2002 to Mr. Nelson, Mr. Roney acknowledged that there had been two drafts prepared by him prior to the final report, both of which had been sent to Mr. Nelson. He denied that there had been any letters between Mr. Nelson and himself relating to the drafts. At this point in the cross-examination Mr. Nelson objected on the basis of solicitor-client privilege. Mr. Mueller took the position that he had some authority which would permit him to get into this area on cross-examination and I invited him to provide me with those authorities. He thus stood down this portion of his intended cross-examination.

Mr. Mueller noted that in his examination-in-chief Mr. Roney had not been asked anything with respect to the General Conditions attached to the Contract. Mr. Roney agreed that he had some familiarity with the standard General Conditions but that contract administration was not one of his office functions.

Mr. Roney agreed that in the world of construction there is sometimes misunderstanding of intent as between the contractor and the designer. He agreed that there may be at times ambiguity as between the specifications and the drawings that needs sorting out as between the contractor and the designer. He was aware that the general conditions provide mechanisms whereby the contractor can request instructions and the designer can give instructions, eg, see GC 2.1 which provides that during the progress of the work the consultant will furnish to the contractor such additional instructions to supplement the contract documents as may be necessary for the performance of the work. Mr. Roney conceded that in his own practice there are instances when contractors come to him and ask for clarification of the contract documents, and Mr. Roney provides instructions for the purpose of resolving the questions put by the contractor.

Mr. Roney was familiar with GC 25.5 which provides that the contractor shall review the contract documents and promptly report to the consultant any error, inconsistency or omission that he may discover, and indeed Mr. Roney has experienced that request.

Mr. Roney agreed that inconsistencies between the drawings and the specifications can occur so as to require clarification. He was aware of GC 1.2 which provides that the contract documents are complementary and what is required by any one shall be as binding as if required by all. He understands that to have always been the rule, as is his understanding with respect to G.C. 1.6(c) which provides that in the event of conflicts between contract documents, the specifications shall govern over drawings.

It was Mr. Roney's evidence that he is familiar with General Conditions 11 and 12 which provide that the owner through the consultant may make changes in the work by written order and that no changes in the work shall be proceeded with without a written order signed by the owner. Mr. Roney knew this to be the normal practice.

In the face of those General Conditions, Mr. Mueller reviewed with Mr. Roney portions of his report dated April 24, 2002, Exhibit 87:

a) at the bottom of page 3 Mr. Roney notes that the three detail sketches show an identified gap between the wythes of brick and concrete block and that "it is not clear whether this gap which was drawn at about the same thickness as a mortar joint was intended to be filled solid with mortar to form a collar joint or left open. In our opinion, the detail could have been interpreted either way." In this respect Mr. Roney agreed that the detail was ambiguous and not entirely clear one way or the other, although in his report he went on to rely on the absence of hatching and noted that it would be reasonable for a contractor to interpret the intent of the contract documents to the effect that the two wythes of masonry were to be constructed with a void space between them. Mr. Roney did go so far as to admit that absent the specifications that called for a 19 millimetre air space, the situation perhaps required clarification by the contractor.

b) At the top of page 4 of his report, Mr. Roney notes that the specifications call for a 19 millimetre void space between the brick and the block, and that the joint illustrated on the small detail sketches is clearly not as wide as 19 millimetres. As such he says, there appears to be a slight conflict between the drawings and the specifications. Still in the face of this, Mr. Roney was not prepared to concede that the reasonable thing for the contractor to do in the circumstances is to meet with the consultant to clarify.

c) Again at page 4 of his report, Mr. Roney notes that the beam designation and the elevation of the steel deck in relation to the beam is consistent with the conditions at the east and west exterior walls, as illustrated in section E on drawing five, but no details were provided to illustrate the condition at the north and south exterior walls, the construction of which would differ due to the lower elevation of the perimeter steel beams to account for the joist shoes which are four inches deep. In this respect, Mr. Roney did concede that a consultation was required as between the contractor and the consultant.

d) As well, at page four of his report, Mr. Roney notes that the specifications require that "units shall be tied every other course as noted on the drawings" but that the drawings contain no notes regarding ties. He concedes that in this situation the contractor should have required clarification.

e) In his report, Mr. Roney notes that the specification reference to ties at every other course presumably is referring to the concrete block course although it is under the heading 'Brick work', because ties at every other course of the brick work would result in an excessive number of ties. Again, there should have been clarification by communication between the contractor and the consultant.

f) In his report, Mr. Roney notes that if the ties are simply required to secure the brick to a back-up wall as is the most common case, then light strap ties are suitable. If however the ties are required to bond the wythes such that they exert common action under load, the CSA Standard prescribes a heavier tie system. Mr. Roney seemed to agree that the contractor would not know what kind of tie to use without being told by the consultant. The contractor does know that he can go to his supplier and buy all sorts of ties, and Mr. Roney agreed that if he did not know which type to buy he would go to the consultant.

g) Under the heading 'Movement Joints' in his report Mr. Roney notes that the specifications call for "provide control joints as dictated by structural design". However the drawings do not indicate any control joints nor were any provided except for one on the south face of the building. Mr. Roney agreed that the contractor would expect them to be shown on the drawings. In my view the obvious conclusion is that if they are not on the drawings the contractor should have asked.

h) In his report, Mr. Roney notes that Division 4, section 2.1.1 of the specifications calls for a 6 millimetre continuous expansion joint, caulked continuously between the steel support angles and the brick panels at each floor, and that no such soft joint is shown on the drawings or detail sketches. I believe Mr. Roney conceded that this is an ambiguity as between the specifications and the drawings, and one that should have been checked out by the contractor, although he did not say so in so many words.

In his report, Mr. Roney notes that in practice contractors tend to follow and give precedence to the drawings, since the specifications frequently reference types of construction and/or materials which may not apply. He notes that if movement joins were to be a requirement of the construction contract, they should have been explicitly detailed in the drawings. Mr. Roney acknowledged that this is contrary to the General Conditions.

i) In his report, Mr. Roney notes that the three detail sketches illustrate what appears to be a joint between the block and the underside of the steel beam, and suggests that the intent was simply to provide a mortar joint between the top of the block and the underside of the beams. In this respect Mr. Roney conceded that neither the specifications nor the big drawings indicate a mortar joint between the block and the underside of the beam, and that he is relying on the small drawings to arrive at that conclusion. In fact, only one of the three small drawings has the cross hatching extending into the space between the top block and the underside of the beam. Mr. Roney conceded that it is not the clearest illustration and conceded that there was a need for a meeting between the consultant and the contractor on this item.

j) In his report, Mr. Roney notes that section 3.6.1 of M-78 requires that masonry walls be anchored to their lateral supports, where these supports are other than masonry, by metal anchors or “other approved construction systems”. It was Mr. Roney’s evidence that the mortar joint between the top of the block and the underside of the steel beam could be considered an “other approved construction system”.

It would be worthwhile setting out section 3.6.1 of the National Standard of Canada, M-78. It is as follows:

<u>Anchorage</u>

3.6.1 Masonry walls and partitions shall be anchored to their lateral supports where these supports are other than masonry by:

a) metal anchors spaced vertically at not more than 4 times the normal thickness of the wall or partition where lateral supports are vertical, or not more than 2 millimetres o.c. (on centre) horizontally where the lateral supports are horizontal; or

b) other approved construction systems.

In due course it was agreed by all parties that the reference to 2 millimetres in 3.6.1(a) should read 2 metres.

With respect to section 3.6.1, Mr. Roney agreed that there are two types of lateral movement possible, that is in and out, or side to side, and that the section would seem to indicate that the designer has a choice between anchoring the masonry wall to the vertical columns on either side or to the horizontal, that is to the floor slab and to the underside of the beam.

Mr. Roney did agree without reservation with the proposition that there must be some anchor of the block wall to the frame. It was suggested to him that if the conclusion is that the mortar joint cannot serve to stabilize the block wall against lateral forces, then the outside wall will have to be removed to repair the inner. Mr. Roney did not agree and testified that it could be done from inside or from outside by locally removing a relatively few number of bricks.

Mr. Roney pointed out that on the east and west walls, which are the short walls, there is but one column of block that extends from the south to the north wall. On the north and south walls however, there are four columns of block, one between every second window.

With respect to the mortar joints between the top of the block wall and the underside of the beams, Mr. Roney conceded that he has seen these mortar joints at seven points only, and the extent of the mortar joints that he has so observed is approximately 10 feet or 1% of the total of 1,000 feet of block to beam underside throughout the building. In this respect Mr. Roney was not familiar with Mr. Johnston's observations that sometimes there was a mortar joint, sometimes the joint was void, sometimes there were brick and block pieces loosely inserted into the void, and sometimes those pieces were mortared in. Mr. Roney noted that he had been given a limited number of test openings to work with. He is for the purpose of his analysis assuming that the openings that he saw are representative. Mr. Mueller suggested that the alternative would be to not take chances and take down the whole outer brick wall so that one could determine whether the block wall is properly anchored to the frame. Mr. Roney's response was that there are further investigations that could be done before such drastic steps are taken. For example, one can look at the exterior of the building; it has been there for 16 years under wind forces and there appears to be no signs of distress with respect to the anchorage. Secondly, one must ask how critical the wall is that we have. His calculations show that the bond is well in excess of the maximum loads.

With respect to the shelf plates, Mr. Roney conceded that there is no doubt from the plans that it was the intent of the designer to have the shelf plates on each and every floor. In this respect, Mr. Roney maintained his position that technically section 6.2.3 of M-78 says that you need put a plate in only after you have reached 11 metres of height. Mr. Roney agreed with Mr. Mueller that in giving that evidence he used the expressions "technically" and "strictly reading". He agreed with Mr. Mueller that what he is doing is inserting the words at the beginning of the section, "The portion of", so that the section in effect reads, "The portion of the unit masonry veneer more than 11 metres above the top of the foundation wall shall bear on masonry, concrete or other non-combustible bearing supports spaced not more than 3.6 metres vertically." [I don't think I agree with his interpretation of this section. If that is what the section meant, then once you got to 11 metres of height, if the first 11 metres was okay without any additional support, then the second 11 metres should be equally okay without anything other than the one shelf plate at the 11 metre height. In other words, if you allow Mr. Roney his interpretation, you would have a shelf plate at every 11 metres of height. What is good for the first 11 metres is good for the second 11 metres, and the third 11 metres etc.] In any event Mr. Roney pointed out that section 6.2.3 applies to a veneer wall and if it is not a veneer wall that we are talking about then the section does not apply. Mr. Roney did agree that more shelf plates are better in that they cannot be harmful and they may be of some benefit.

At page six, middle paragraph of his report of April 24, 2002, Mr. Roney points out another contradiction. Though a plate is specified in the notes on the detail drawings and on drawing S-1 for all perimeter beams at each floor above level one and at the roof, no such plate is <u>illustrated</u> in the detail drawings. The sections on drawing four appear to show plates on the underside of the perimeter beams extending towards the exterior face of the building, and plates are shown supporting the exterior brick wythe at each floor level at the north and south walls of the building. Mr. Roney in his report notes however that no such plates are shown on drawing number five which illustrates the condition at the east and west exterior walls. As such, he notes there is an apparent contradiction between drawing S-1 which calls for plates at all of the perimeter beams, the detail drawings which specify a plate but show none, and the cross section of drawing five which neither shows nor specifies any plate. [This obviously seems another situation where the contractor should have conferred with the consultant.] Mr. Roney confirmed again that the plates were cut off the beams on the east and west walls because the courses of brick would not mesh properly with the plates. Mr. Roney agreed that the steel fabrication shop drawings should reflect the contract documents, and the fabricator should have been aware that the plates would come in at different levels. Mr. Roney confirmed that it is the obligation of the contractor to do field measurements. It was suggested to Mr. Roney that typically if the contractor raised it the consultant might throw the problem back at the contractor and say, ‘you fix it.’ Mr. Roney considered this to be a matter of speculation.

Mr. Mueller discussed with Mr. Roney at some length the strap ties, Exhibit 60, which exhibit some damage by rust. Mr. Roney agreed that the oxidation which is what rust is, has the effect of reducing the thickness of the product. He noted that the volume of rust is much greater than the metal itself in the sense that one millimetre of metal can produce 16 millimetres of oxidant. Mr. Mueller invited Mr. Roney to test with his fingers the two ties and invited him to find at the spot where they had obviously rusted the material was thinner than where it had not rusted. Mr. Roney was unable to confirm Mr. Mueller’s tactile abilities.

Again with respect to the shelf plate being cut, Mr. Roney confirmed that from his observation the shelf plate had been cut off each beam on the east and west walls by a continuous torch cut along the entire length of the beam adjacent the edge of the flange. Indeed, the portion of the plate that had been intermittently welded to the underside of the beam remained in place after the cutting.

An interesting discussion ensued between Mr. Mueller and Mr. Roney with respect to the ties that Mr. Roney actually observed by eye in his test holes, and those that he observed through his borescope in the area surrounding his test holes. In this respect, Mr. Roney explained that with the borescope he can see 360 degrees within an eight inch radius of the borescope head. In all he bored some 14 holes in which to insert his borescope. Bear in mind that this is with respect to the test openings made in 2002. No straps were observed by Mr. Roney in his 1999 investigation of the test openings made from the inside.

In any event, it was the sum and substance of Mr. Roney's evidence that with respect to the four test holes in 2002 in all he visually observed two ties in test opening number one, and observed through his borescope five or six more. In test hole number two he observed one strap tie. In test hole number three he observed one strap tie, and in test hole number four he saw none. In test holes two, three and four he was unable to observe any ties with his borescope.

With respect to the ties that he saw with his borescope, he viewed them from the bottom for the most part and only one did he view from the side. He viewed none of them from the top. Notwithstanding this he is prepared to say that there was significant mortar on top of each of the strap ties.

Mr. Roney was able to confirm that page four of Exhibit 5 as a sheet of photographs of his test hole number one where he observed a strap tie and the photograph shows no mortar on top of the strap tie. Mr. Roney speculated that the mortar must have been knocked off in the opening procedure, although he did not actually see that happen. [I think Mr. Roney is stretching a point when he says that there was mortar on top of all of the strap ties. This of course is a very important feature of his calculations because the mortar turns the tie into a useful means of transferring the forces from the brick to the block, whereas on their own they are of no great consequence.]

Mr. Mueller backtracked for a moment to Exhibit 87, Mr. Roney's report dated April 24, 2002, page six thereof, whereof Mr. Roney notes that section 3.6.1 of M-78 requires that masonry walls be anchored to their lateral supports, where the supports are other than masonry, by metal anchors or "<u>other approved construction systems</u>". Mr. Roney explained that the approval must come from the design engineer in accordance with appropriate engineering principles and Code provisions.

With respect to page 10 of his report, relating to the ties found and the fact that they were found with an accumulation of mortar on top, Mr. Roney confirmed that the ties that he found appeared to be bent down from the brick to the block. He conceded that in terms of water transmission to the block wall this was less preferable than if the ties had been bent from the brick up to the block wall.

With respect to the detection device that he used to determine the presence of ties, Mr. Roney confirmed that unless he could see such ties with the borescope, their detection by the scanner did not disclose whether they were bent or straight, or whether indeed they were actually connected between the two wythes.

At page 11 of his report, referring to test opening number two made in the north wall at the approximate elevation of the perimeter roof beam, Mr. Roney noted that he observed a small horizontal crack in the mortar joint where the steel plate at the underside of the beam was imbedded in the wall. Mr. Mueller asked Mr. Roney whether this would be consistent with Dr. Drysdale's comments in Exhibit 32 where he spoke of relatively small deflections of the beam, causing cracks. Mr. Roney responded that he did not believe that the deflection of the beam caused this particular crack, because he did not see similar cracks elsewhere. He conceded however that what he saw by invasive investigation was approximately 1% of the perimeter.

At the top of page 13 of his report, Exhibit 87, Mr. Roney again notes that he observed frequent mortar droppings within the void space, primarily at the tie locations. He agreed that these mortar droppings could convey water. He conceded that in his report he was not addressing water problems, either as they may be created by the mortar covered ties, or the mortar fins themselves.

Again at page 13 of his report, Mr. Roney makes the comment that the increased width of the void space, which varied from the 19 millimetres specified, was not structurally significant, assuming the intent was to provide a veneered wall; that is assuming it was not intended that the outer wall convey force to the inner wall. Mr. Mueller suggested to Mr. Roney that if Mr. Garbutt had intended a solid wall, then this would constitute a major discrepancy. Mr. Roney did not understand the question. Mr. Mueller asked Mr. Roney whether he knew that Mr. Garbutt's evidence was that he intended a solid wall, and indeed thought that he was getting a solid wall. Mr. Roney conceded that he had heard some passing reference to that effect. Mr. Mueller suggested that in any event Mr. Garbutt's stated intent is different from what he, Mr. Roney, saw. Mr. Roney agreed that it is different from what anyone reading the drawings would have interpreted. Mr. Roney agreed that if one accepted Mr. Garbutt's evidence then obviously what was constructed was not what was intended.

Again at page 13 of his report, Mr. Roney notes that if the ties were required to structurally bond the wythes then it would have been the designer's responsibility to explicitly specify the tie requirements. He notes that no special tie requirements were indicated in the contract documents. Furthermore, he says, the ties employed by the contractor were clearly visible during the course of construction and the designer of the wall would have had ample opportunity to inspect its construction and tying. Based on these facts, it is evident to Mr. Roney that the designer was satisfied at the time of construction with the type of tie system being used. Mr. Roney agreed with the alternative suggested to him by Mr. Mueller, and that is, assuming Mr. Garbutt intended a solid wall then he was extremely lax in his inspections, and with proper inspections he would normally have noticed what was going on. Mr. Garbutt either approved of what he saw, or his level of inspection was not sufficient. If his inspection was not sufficient to pick up the fact that what was being constructed was not what he intended, then this would have constituted a marked departure from what would have reasonably been expected of Mr. Garbutt in terms of his inspections.

Again at page 13 of his report, Mr. Roney notes that he observed many of the corrugated strap ties were bent as they crossed the space between the wythes. This was done to compensate for the differences in the coursing between the brick and the concrete block wythes. While M-78 did not prohibit or limit this practice in any way, later editions of the Standard have since recognized that bending of the ties tends to reduce their compressive strength. Mr. Roney in this regard noted that helifix ties have much better compressive strength than strap ties, even if the helifix tie is bent.

Mr. Roney went on to explain that compressive strength involves being “pushed against”; tensile strength involves on the other hand “being pulled apart”. He explained that when wind is pushing against one side of the building (which involves compression strength), then the other side of the building may be subjected to suction which brings into play the tensile strength of the tie. He explained further that the Helifix tie is better in tension simply because there is more steel in it.

Mr. Roney conceded once again that apart from the few buildings he has looked at constructed in 85-86 he has no practical knowledge of the situation at that time. He was referred to the last Technical Note 44-B in Exhibit 53. At the bottom of page two he confirmed that he observed ladder ties in the construction of our wall. He noted the comment at the top of page three of 44-B that “adjustable tie systems were initially developed to accommodate the use of face brick that did not course out vertically with interior masonry wythes” and confirmed that figure six on page four illustrates adjustable ties meant to accommodate variations in coursing. He confirmed as well that in Table 3 of the Bulletin, it is clear that corrugated strap ties are not recommended for brick veneer walls.

With respect to the remedial work presently being recommended by Mr. Roney, Mr. Roney conceded that his analysis assumes that there is already in place within the wall a tie system that is capable of transferring forces from the brick wythe to the block wythe. He concedes that if the ties are not in fact there, then his calculations are not correct.

With respect to the wind force being used in his calculations and indeed in Dr. Drysdale’s calculations, Mr. Roney explained that these values represent one in 10 year winds, that is in any given year there is a one in 10 chance that that particular wind force will be exceeded. On the other hand, as he explained, a one in 10 wind may not be experienced in a span of 100 years. With respect to such wind forces, Mr. Roney conceded that he is relying in part on an existing tie system that includes bent ties which were not designed to be bent. As he confirmed earlier the helifix ties which he proposes to use in his remedial work are stronger both in compression and tension than the strap ties. While Mr. Roney conceded that he was relying on the compatibility of two different tie systems, he pointed out that there was more to it than that, in that the mortar accumulation on the strap ties caused them to act as very good transfers of load. At the same time, he concedes that the stiffer helifix ties attract the loading forces of the winds more than the strap ties, and take more of the force than the strap ties. He conceded the general principle that wind will tend to be impacted on stiffer elements.

It is Mr. Roney's opinion that he can depend on the fact that lateral loads are being shared between brick and block at the present time because there are no observable signs of stress. It is his opinion, contrary to Dr. Drysdale, that the present tie system would be compatible with the remedial tie system, notwithstanding that Dr. Drysdale says that the deformity of the strap ties and the lack of compatibility with the helifix ties negates Mr. Roney's remedial plans.

At page 14 of his report, Mr. Roney notes that assuming the conditions at the test openings are representative of the conditions elsewhere in the wall, he has concluded that additional remedial tying of the wythes is necessary at certain localized areas. In this respect Mr. Roney again conceded that he is assuming that what he is seeing in his test openings is a representative sampling.

In his report, he goes on to say that remedial ties shall be provided at those areas identified by the test openings as lacking in adequate ties and that typically this shall apply above the fourth floor level on the east and west walls and above the mid-height point of the fourth floor windows. He concedes that this is based on a sampling, and he concedes that there is a risk that the sample is not consistent throughout.

Again at page 14 of his report, Mr. Roney notes that a visual review of the exterior brick work reveals the wall has exhibited only relatively minor cracking in the mortar joints between the bricks. He concedes that this is a significant factor in his analysis, and he concludes that there is no indication of unusual stress in the wall. When it was pointed out to Mr. Roney that the 20-foot crack on the north wall near the shelf angle at the top of the wall was initially eight feet and is now 20 feet, Mr. Roney stated simply that he had not observed the crack change from eight feet to 20 feet, although he accepted Mr. Mueller's assurance that the evidence in the Johnston report was that it propagated from eight feet to 20 feet by the time of trial. Mr. Roney conceded that cracks propagate because of stress.

At Exhibit 74, Mr. Garbutt's Exhibit book, Tab 25, there is noted a crack on the southeast corner of the building which has been caulked closed. It is at the bottom of the spandral beam where the plate is. Mr. Roney was aware of that crack.

At Tab 26 of the same Exhibit is a crack at the northeast corner, again at plate level, and Mr. Roney confirmed that this was the crack that he was talking about.

Mr. Roney was not aware of Mr. Johnston's evidence that in December of 1994 during the rain storm he observed from an inside opening water flowing out of a series of bed joints in the brick wall.

Mr. Roney was referred to Mr. Johnston's report, Exhibit 56 at pages 21 and 22, accompanied by photographs where Mr. Johnston notes that the opened area allowed him to view the brick immediately below the spandral beam of the south wall structure, and exposed approximately 16" more of the wall immediately above the window opening. Mr. Johnston in his report notes that the interior surface of the brick displayed obvious water flow markings where rain water had penetrated. Mr. Johnston noted that these stains were evident as permanent indications of the general extent of the incursive flow, indicating as well that most rivulets started at or had as their source the upper plane of the brick. Having pointed out this portion of Mr. Johnston's report, Mr. Mueller then pointed out to Mr. Roney that it was Mr. Johnston's evidence that later in

December of 1994 during a rain storm he actually saw water coming in through the bed joints as he had inferred from the archeaological evidence. Mr. Mueller put it to Mr. Roney that if you accept the testimony of Mr. Johnston, it is clear that there was a debonding of the brick to allow the water to pass through. In response, Mr. Roney suggests that there were many other mechanisms for the entry of water. He noted that there was no way of knowing whether this was a series of single separate drips or whether there was a collection of water on the horizontal plane that formed a drip. He felt that one would need a significant crack for a fast flow of water between the joints, unless the water accumulated over a broad horizontal area and then formed one rivulet. He felt a heavy flow would be more indicative of a visible crack.

Mr. Mueller pointed Mr. Roney to the first Building Digest in Exhibit 53, which at page 6-4 says, "[M]ost of the rain which penetrates a unit masonry passes through unbonded areas between unit and mortar. These usually cannot be seen and the wall appears perfectly sound." Mr. Roney was unable to comment on the extract from the Digest.

With respect to his comments relating to movement joints, in Exhibit 87 at page 14, Mr. Roney explained that when he said that he was reluctant to introduce movement joints into the building which was in equilibrium, he meant that the brick was under compression and was protecting against tensile forces.

In discussing the mortar joint between the top of the block wall and the underside of the spandral beam, Mr. Roney confirmed that he did not use a microscope to examine that joint. He was referred to Mr. Johnston's report, Exhibit 56, page 19 where Mr. Johnston in his investigation notes that the "head joint of the inner wythe where it met the underside of the perimeter beam was variously filled with mortar, bricks on edge, half blocks or indeed at some locations left completely unfilled." In this respect, Mr. Roney did concede that he saw some bricks between the top of the block wall and the underside of the beam in hole number two, but in such cases the mortar joint was between the brick and the beam.

Mr. Mueller pointed out to Mr. Roney Mr. Johnston's report at Exhibit 2, Tab 106, page four, where Mr. Johnston said as follows, "[S]ignificantly, no flashings were observed at any of the spandral plate locations on any façade, and infill masonry at the interior wythe to the underside of the spandral plates were completed in brick, cut block or block pieces as suited the dimension, and generally poorly mortared in (some locations with no mortar at the top surface)." Mr. Roney responded that in the two inside openings in 1999 and the four outside openings in 2002 he found nothing similar to what Mr. Johnston was describing and found to the opposite that there was a good mortar joint between the top of the block wall and the underside of the beam. Again, Mr. Roney conceded that he has seen but 1% of the mortar joints between the top of the blocks and the underside of the beam.

At page 15 of his report, again discussing the anchorage of the concrete block wythe, Mr. Roney states the following.

> It appears that the intent of the detail shown in the Contract Documents was to use the bond achieved by the mortar joint between the wall and the steel to provide the necessary lateral support to the wall. This method is somewhat unconventional, though an analysis of the theoretical structural bond formed between the mortar and the steel indicates that such a bond would provide adequate lateral resistance to the lateral wind loads specified in the OBC.

There ensued a long discussion between Mr. Mueller and Mr. Roney. They discussed the fact that the steel goes up first, the floors are poured and the beams therefore deflect before the walls go up. The blocks are fixed to the floor slab with a mortar joint. Mr. Roney conceded that there would be plastic shrinkage of the mortar joint between the top of the block and the underside of the beam, but maintained that this shrinkage did not cause the mortar to pull away. In any event the shrinkage is negligible and the effect on the bond would be negligible. It was Mr. Roney's view that steel and mortar are very compatible in terms of expansion and contraction, and gave as an example steel reinforced concrete that is used worldwide in construction. Mr. Roney throughout the conversation maintained that it was a reliable bond. He reminded Mr. Mueller that in doing his calculations he assumed that the bond existed between only 50% of the top of the block wall and the beam. In other words, he took only 50% of the available area of contact, notwithstanding that it appeared to him in his observations that the contact was full between the block and the beam. Mr. Roney did agree that it can be difficult for the mason to fill the entire width of the top of the block with mortar.

At page 15 of his report, Mr. Roney noted that the subject mortar joint at one point was approximately two inches thick and that no other anchors or mechanical fastenings between the two elements were noted. Mr. Roney agreed that the wider the gap between the block wall and the beam, the more shrinkage occurs.

Further at page 15 of his report, Mr. Roney noted that “[T]ypically, lateral support is provided to non-load bearing masonry walls such as this, using short lengths of clip angles welded to the steel beams and positioned so as to periodically restrain each side of the wall against lateral pressures and suction."” Mr. Roney explained that by this he meant that typically there would be a clip at the top of the block wall on both sides of the wall anchoring it to the steel beam.

Again at page 15 of his report, Mr. Roney notes, as we have seen before, that “[I]t appears that the intent of the detail shown in the contract documents was to use the bond achieved by the mortar joint between the wall and the steel to provide the necessary lateral support to the wall.” It came as some surprise to Mr. Roney when Mr. Mueller advised him that Mr. Garbutt had testified that that was not his intent at all. Mr. Roney conceded that if it was not Mr. Garbutt’s design intent to have a mortar joint between the top of the wall and the underside of the beam and it was not his intent to have the wall built with a gap between the two wythes, then obviously the building was not built as intended. Mr. Mueller pointed out, to be fair to Mr. Garbutt, that he intended a solid wall and in terms of lateral support intended that the brick would be facing across the vertical columns. Mr. Roney understood that, but questioned how that might work on the east and west wall where there were only two columns, one at each corner, and in his view the span was much too large. As well, with Mr. Garbutt’s theory of the bricks pressing against the vertical columns, there would be a problem on the opposite wall with suction.

Mr. Roney explained once again that his analysis which found that the concrete block wythe of the east and west walls above the level four floor is overstressed by approximately 39% when subjected to a combination of lateral wind and vertical gravity loads, was the result of an analysis using the National Standard of Canada for Masonry Design for 1994. He explained as well that in determining the loadings, he used the 1983 Code which incorporated National Standard of Canada for 1978 (M-78). That is, in determining the forces that the building could be expected to be called upon to resist, he used M-78. In determining the capacity of the wall to resist, he used M-94.

There followed a discussion between Mr. Mueller and Mr. Roney of Exhibit 88 which is Mr. Roney’s report to Mr. Nelson dated May 14, 2002. Mr. Roney in his report notes that he has assembled his calculations to be inserted as Appendix B to the April 24, 2002 report, Exhibit 87. In Exhibit 88, Mr. Roney notes,

> Please note that our initial calculations used in the preparation of our report were carried out using the current version of the Ontario Building Code (1997) and the CSA Standard S.304.1-94 that it references. The use of the modern Code and Standards was dictated by the time constraints imposed by the court, since the analysis could be carried out more efficiently using the modern analysis tools available to us today which are tailored to the present Code. Following the submission of our April 24 report, we had the opportunity to recalibrate our analysis of the walls using the Code in effect at the time that the building was constructed (OBC 1984 and CSA Standard S.304 M-78).

Again Mr. Roney explained that for the purposes of Exhibit 87 he used the 1984 Code to determine the loading and M-94 to determine the capacity to resist. Then for the purpose of his Exhibit 88 calculations, he went back to M-78 to determine the capacity to resist. As he notes in Exhibit 88, his initial calculations were based on a three metre floor to floor height for the building, and on May 2, 2002 he received from Mr. Nelson a copy of a letter addressed to Dr. Reed from Emmons & Mitchell dated August 14, 1985 which indicated that the floor to floor height was 3.2 metres.

Mr. Roney explained that with the change in height, floor to floor, and the use of M-78 to determine the capacity to resist the wind forces, the overstress of 39% in the block wythes in the east and west walls is reduced to 6%, as reflected in Exhibit 88. Mr. Roney conceded that as it turned out, the calculations which Mr. Mueller had sought with respect to the original report,

Exhibit 87, and the 39% overstress, were not in fact received but rather what were received were the new calculations, Exhibit 88 which showed only a 6% overstress in the east and west walls.

Mr. Roney confirmed for Mr. Mueller that his decision to recalculate for a 3.2 metre height and for M-78 provisions, was his own decision and he was not prompted to do so by Mr. Nelson or by any other person. Mr. Roney then tendered as Exhibit 95 all of his calculations, including the old calculations which produced the 39% overstress and that was marked as Exhibit 95.

At page 16 of Exhibit 87, when addressing the cutting of the plates from the beams on the east and west walls, Mr. Roney noted that it was inconceivable that the decision to remove a load bearing element such as this would be made by the contractor without consulting the designer. Mr. Roney conceded that it would be equally inconceivable that the engineer would direct that the plates be cut off when he has filed with the municipality his stamped drawings showing the existence of the plates. To this Mr. Roney added a caveat that it would be inconceivable unless the engineer felt after further analysis that the plates were not necessary. Mr. Roney confirmed with Mr. Mueller that if one accepts that Mr. Garbutt did not intend that there be a gap between the brick wythe and the block wythe, then he did not meet reasonable standards of site review. Mr. Roney conceded that in the event Mr. Garbutt, upon further reflection, had decided that the plates were not required on the east and west walls, and in the event he gave instructions in that respect to the contractor, then this would constitute a material change and new drawings would have to be filed with the municipality showing that the plates had been removed. It should be noted that Mr. Mitchell, when examined for discovery in July of 1999, testified as follows, at page 50 of the transcript, Question 331:

> Q. And they were removed and nobody, be it Emmons & Mitchell or Abe Dick, went to the owners or the engineers to ask for permission to remove that?
>
> A. It's not in writing, no.

Question 332:

> Q. Was it done orally?
>
> A. I can't confirm that.

Question 333:

> Q. If you do later take the position, I want to know exactly what was said, in other words, the consequences of it. You know, you go to the doctor and he says, I'm going to give you this pill, there's side effects maybe, so I mean, the

consequences mainly that there be no support now from the angle iron as I call it on the east and west side, and indeed this is now going to be what, five to 10 feet higher than the Building Code permitted.

A. Correct.

Q. Without support. Do you understand?

A. Correct, yes.

It should be noted that at that point in the discovery, then counsel for Emmons & Mitchell undertook that if they got any information in that respect that it would be reported to then counsel for the plaintiff.

At page 17 of Exhibit 87, Mr. Roney notes that out of plane lateral wind loads were distributed to the wythes in proportion to their relative stiffness, and vertical loads consisted of self weight of the wall plus live loads transferred to the block wythe through the mortared joint between the top of the wall panels and the beams. Mr. Mueller suggested that the 39% overstress as originally calculated meant that the wind loading compared to the wall's ability to resist had been exceeded by some 39%. Mr. Roney agreed. He did not agree however that this necessarily meant there was a risk of collapse, nor that it represented significant structural distress.

With respect to this particular building, Mr. Roney does not feel that there is sufficient risk involving the building at the present time to report it to the municipality. He does however feel that remedial work should be done now. He confirmed for Mr. Mueller that if a one in 10 wind can cause cracking, that would not be for Mr. Roney an immediate concern that would cause him to go to the authorities. Mr. Roney confirmed that if Dr. Drysdale perceives the risk as being greater than Mr. Roney does, and the building being much more under-designed than Mr. Roney thinks, but that a one in 10 wind will not bring it down, then Mr. Roney is of the view that in those circumstances one advises the client but does not go to the municipality.

Mr. Mueller reviewed once again with Mr. Roney his findings with respect to the ties. Mr. Roney agreed that in some places the ties were there with appropriate frequency; in other places they were infrequent and still other places there were none. Mr. Roney agreed that based on the one area where appropriate ties were placed, this showed that the contractor knew how to do it when he wanted to do it. He agreed as well that the variability with respect to the tie situation may indicate a lack of supervision by the contractor and/or the mason and/or supervision by the consultant. He agreed that to the extent that the court might conclude that the situation with respect to the ties was variable, that is indicative of a lack of supervision on the part of the contractor and mason and a lack of proper review on the part of the consultant.

Again with respect to the ties, Mr. Roney confirmed that the strap ties which were apparently used would be acceptable if the brick wall was only "along for the ride" but that in the event the brick wythe and the block wythe were intended to act together to resist forces, then the ladder tie would be appropriate and the strap tie would not be appropriate.

At page 18 of Exhibit 87, when discussing the remedial masonry wall ties, Mr. Roney says, "We suspect that the lack of ties is a localized condition, since the brick has performed relatively well given its age." In this respect, Mr. Roney agreed that the use of the word 'suspect' means that there is a lack of knowledge, one way or the other.

Further on page 18, Mr. Roney states,

> Furthermore, we recommend that a horizontal row of ties be provided approximately two inches below each floor and at the roof level, spaced at 24 inches on centre. The purpose of these ties is to provide a reliable point of lateral support to the brick wythe at these locations, since our analysis requires such support at these points. These ties are required to address a potential design deficiency, and are not therefore the responsibility of the contractor.

Mr. Roney conceded that if the shelf plates had been located at the east and west walls, then he would not have found the overstress that he reports.

In all, Mr. Roney's remedial work would require 1,600 helifix ties. They are put in place by drilling a hole through the mortar between the bricks (almost the entire width of the mortar) and then drilling a pilot hole in the block wall. The helifix tie is self-tapping. It is then inserted through the hole, into the void and into the pilot hole in the block wall, and screwed in. The tie in the brick wall is epoxied and then a mortar patch put on the outside. It was Mr. Roney's evidence that if proper care was taken to match the mortar, the patching would not be very noticeable.

Referring to Exhibit C, Dr. Drysdale's response to Mr. Roney's report dated October 21, 2002 (Exhibit 89) which in turn is a response to Dr. Drysdale's comments in Exhibit 32 and his calculations filed as Exhibit 31; in Exhibit C Dr. Drysdale referring to the greater wall heights notes that greater wall heights produce more bending moment due to wind pressure, resulting in larger tensile stresses (cracking stresses) and that the increase is proportional to the square of the ratio of the 2.6 metre versus 2.4 metre wall height, which works out to be 1.17, that is 17% more bending with the corresponding 17% increase in tensile (cracking) stress. Mr. Roney agrees with this comment.

In Exhibit 88, section 2.0 under the heading 'Discussion', Mr. Roney notes that his field observations were key to confirming the actual conditions at the wall, thus permitting the reliable use of his analytical model. In this respect, Mr. Roney agreed with Mr. Mueller that his observations included the following:

a) few visible cracks on the wall;

b) a mortar joint between the top of the block wall and the underside of the beam;

c) the existence of some ties.

Mr. Mueller and Mr. Roney then discussed Exhibit 89, which is Mr. Roney's report dated October 21, 2002, responding to Dr. Drysdale's calculations, Appendix G, Exhibit 31 and Dr. Drysdale's comments (Exhibit 32) relating to the May 14, 2002 calculations of Mr. Roney, Exhibit 88. Mr. Roney in his *viva voce* evidence confirmed the comment in his report, Exhibit 89, criticizing Dr. Drysdale's calculations because he, Dr. Drysdale, did not attempt to consider the effect of the ties that were provided in the original construction. Mr. Mueller pointed out to Mr. Roney that in Exhibit C, page one, item a), Dr. Drysdale points out that no analysis was carried out on the as-built wall because the ties used in the as-built construction are completely inadequate. He points out that the corrugated metal strap ties were never intended for cavities over 25 millimetres, and in addition regardless of cavity width, when bent to fit into the mortar joints they have negligible capacity. Mr. Roney countered that if the tie system as it presently exists was completely inadequate, then one would see significant cracking.

Mr. Mueller suggested to Mr. Roney that it came down to this; Mr. Roney believed that the ties in the wall could be used as part of a remedial system to withstand wind loading, and Dr. Drysdale does not agree. Mr. Roney responded that if all we had in the wall were the ties, he would agree that they are too flexible to share the load, but he is of the view that what we have in the wall is a stiff mechanism that can resist wind forces, and he concludes this on the basis of the fact that there are no cracks observed in the building. I understood from Mr. Roney's evidence in this regard that he was referring to the fact that there are mortar bridges between the block wall and the brick wall and as well the strap ties are covered with mortar which stiffens them up.

Mr. Mueller pointed out to Mr. Roney, Dr. Drysdale's comments in Bullets 9 and 10 of Exhibit 32. There Dr. Drysdale points out that Mr. Roney's calculations that the wind pressure is distributed between the brick and the block wythes of masonry according to calculated EI values representing section stiffness is only valid if,

a) neither wythe cracks (cracking of the brick was reported);
b) the wythes have identical support conditions and are of the same height (neither of these conditions is met);
c) ties connecting the wythes are sufficiently stiff to force the wythes to deflect equally (very flexible ties were used);
d) additional bending due to eccentric axial loads is not applied to one or more wythes (eccentric axial load is applied in indeterminate and inconsistent ways to both wythes);
e) additional stresses are not introduced by restraining deformations due to moisture and temperature changes (expansion or contraction), and in plane forces caused by the infill resisting lateral loads (all of these stresses do exist).

Dr. Drysdale in Bullet 10 goes on to note that the analysis presented by Mr. Roney has all of the above implicit assumptions, and that violation of any one of these renders the analysis incorrect. Dr. Drysdale is of the view that all of the assumptions are violated, and that the analysis cannot be used as the basis for remedial measures. Alternatively, says Dr. Drysdale, if it is assumed that the brick is not cracked (contrary to reports) over the story height, and that the mortar bond is not weak (contrary to reports), the analysis in Appendix E from the Drysdale report results in tensile stress of .62 MPa compared the Code value of .275 MPa.

Mr. Mueller also pointed out to Mr. Roney Dr. Drysdale's comments in Exhibit C, page one, to the effect that even if it had been found that calculated tensile stresses were not high enough to cause concern, remedial ties would still be required over the entire building. However, Dr. Drysdale's calculations show that introduction of remedial ties is not sufficient and other strengthening or building measures are required because stresses are greater than the tensile capacity of the masonry. Therefore, according to Dr. Drysdale, rebuilding of the concrete block wall using a larger size block or other strengthening measures is required to ensure that sufficient resistance to wind pressure is provided. To all of this, Mr. Roney replied that he acknowledged that that was what Dr. Drysdale was saying, but obviously he did not agree, and he did not agree in particular that the block wall had to be rebuilt.

Mr. Mueller pointed out to Mr. Roney that again at page one of Exhibit C Dr. Drysdale commented that the brick masonry cladding and the concrete block masonry infill must be positively attached to the structural frame. Dr. Drysdale notes that reliance on friction was not accepted in the design Code as sufficient, either at the time of the original design, or now. Retrofit to properly attach the masonry to the frame so that it cannot be pushed into the building or pulled off the building is necessary. Also says Dr. Drysdale, proper attachment to the steel frame is needed for the infill masonry to help stiffen and strengthen the frame to resist lateral load. Positive mechanical attachments such as strap anchors or rods extending from the columns into the mortar bed joints and clip angles extending down from the undersides of the beams on both sides of the block walls are commonly used. Mr. Roney conceded that if it could be said that the mortar joints between the block and the beam cannot be relied upon, then something has to be done to anchor the block wall to the frame.

Referring to his report, Exhibit 89, Mr. Roney explained that his purpose was to determine whether or not the building was properly constructed at the time and could resist the stipulated loads at the time. He has concluded generally that the building cannot resist the loads and needs ties and control joints. He agrees that when remedial work is decided upon, then the remedial work must be performed in accordance with the current Code.

In this respect, Dr. Drysdale in Exhibit C, paragraph b) says the following:

> The 1978 edition of the Masonry Code applies to checking the original design and the 1994 edition applies to any retrofit. Mr. Roney acknowledges that some retrofit related to supplying additional ties is necessary. In addition, because he had himself opened at the walls, looked at photographs and should have evaluation the deformational incompatibilities between the existing ties and their inadequate strength, he should have known that more extensive retrofit was required. However, as I pointed out in Exhibit 32 (second last bullet) the problem with the design exists regardless of which edition (ie 1978 or 1994) of the Masonry Code is used.

In Exhibit 32 in his second last bullet, Dr. Drysdale notes that with respect to Mr. Roney's decision to judge capacity based on M-78, there is not much difference with regard to flexural cracking as the allowable tensile stresses of .19 MPa for brick masonry and .16 MPa for block masonry compare to .18 MPa and .11 MPa respectively from M-94, but corrected for ultimate limit state factors. Also points out Dr. Drysdale in the second last bullet (number 13 as marked), for load bearing wythes the minimum tie requirement was 5 millimetres diameter or equivalent (section 3.7.3.2) which is not satisfied by either the existing joint reinforcing, the existing corrugated strip ties, or the proposed helifix ties. In this respect Mr. Roney conceded that in terms of capacity there is not much difference with regard to flexural cracking as between M-78 and M-94.

In his report of October 21, 2002, Exhibit 89, Mr. Roney notes that the material properties that Dr. Drysdale used in his analysis are based on the 1994 edition of the CSA Standard. Mr. Roney notes that the values chosen are basically correct though by Dr. Drysdale's own admission they are conservative. Mr. Roney goes on to say that they are however not the appropriate values to use since they are from the current edition of the Standard, not the Standard in effect at the time of construction. Mr. Mueller pointed out to Mr. Roney Dr. Drysdale's response to this in Exhibit C, at page two, paragraph c) which states that the conservativeness referred to by Mr. Roney relates to compressive strength which in turn relates to stiffness of the brick. Using higher values for compressive strength will give higher stiffness values for the brick veneer which will result in it taking a larger share of the bending due to wind pressure. This would cause higher tensile stresses in the brick veneer and indicate a greater susceptibility to cracking. Mr. Roney conceded that Dr. Drysdale was correct in this analysis.

In Exhibit 89, pages one and two, Mr. Roney criticizes Dr. Drysdale for using strictly lateral wind loads in his analysis. Mr. Roney states that Dr. Drysdale has chosen to omit the effect of axial compression on the wall because it is extremely difficult to quantify the amount of axial load transferred to the masonry. Mr. Roney is of the opinion that to omit the effect of axial load is extremely conservative. In his report, Exhibit C, now marked as Exhibit #132, Dr. Drysdale responds that not only are axial (vertical) loads difficult to predict but it is certain that except for self weight, no axial load exists in some cases. For example, no applied vertical load exists on the concrete masonry wall under the steel beam near the columns. At this location the steel beam does not deflect and cannot close the gap between the bottom of the beam and the top of the block wall caused by shrinkage of the concrete masonry. Mr. Roney conceded that Dr. Drysdale was disagreeing with him. Mr. Roney maintained that Dr. Drysdale is wrong in suggesting that there is no load at the end of the wall. The block wall distributes the load from the centre to the sides of the block wall.

Mr. Mueller brought to Mr. Roney's attention the last portion of paragraph d) of Exhibit C where Dr. Drysdale notes Mr. Johnston's observations with respect to the variability of the gap between the top of the concrete block wall and the underside of the beam, all of which Dr. Drysdale says would tend to apply the compression at a large eccentricity along the face of the wall, causing large bending stresses and tension. Mr. Roney responded that if that was the case, then the eccentricities would show themselves as cracks on the outside of the wall.

At page three, items 1 and 2, Dr. Drysdale criticizes Mr. Roney's calculations and says that his method of distribution of the load between the brick and the block wythes requires the walls to be forced to deflect equally, and he notes that Mr. Roney should have seen the existing ties, bent or not, do not satisfy the condition of very stiff ties which are needed to satisfy the condition that the walls deflect equally. Also, Dr. Drysdale notes that Mr. Roney failed to take into consideration the natural rotation of the beam caused by floor loading, which results in the beam bearing along one face of the wall thus causing an eccentricity of force. Mr. Roney did not agree that the beams rotated.

At page two of Exhibit 89, Mr. Roney notes that given the stress levels predicted by the Drysdale analysis, it is inconceivable that after 16 years of service, the cracks would still be invisible.

Mr. Roney agreed that if in fact there was a Code wind, one in 10, in the last 16 years during the life of the building, it is fair to say that one would expect cracking in the brick wall, if Dr. Drysdale is correct.

It was Mr. Roney's evidence and he maintained that given the stress levels predicted by Dr. Drysdale's analysis, it is inconceivable that after 16 years in service the cracks would still be invisible. At page three of Exhibit C, Dr. Drysdale noted that masonry is very stiff and very brittle in tension. Because of this, with a 2.6 metre vertical span from the floor to the underside of the steel beams, a one to two millimetre deflection is large, and that naturally the size of cracks associated with such a small deflection are microscopic (that is, cannot be seen with the naked eye). Therefore, Dr. Drysdale says, cracking can and does exist without being visible, and at bullet 14 of Exhibit 32, Dr. Drysdale notes again that masonry is so rigid that tying the brick to the bottom flange of a beam and to the top will produce cracking simply because of movement of the beam. Mr. Roney did not agree.

In his report dated November 18, 2002, Exhibit C, Dr. Drysdale notes that the cracks associated with such a small deflection are microscopic. Mr. Roney confirmed that he did not use a microscope in his examinations.

At the top of page 5 of Exhibit C, Dr. Drysdale notes the following:

> Mr. Roney takes comfort in the fact that the building has survived for 16 years. That is like taking comfort from pulling the trigger on a 6 shooter 3 times without firing as evidence that the next 3 times will also be safe. Neither Mr. Roney, the owners, nor anyone else has the right to accept a significantly increased probability of failure where lives may be endangered. Mr. Roney is on very dangerous ground when he uses the obvious current survival as proof of adequate capacity.

In the face of that comment, Mr. Mueller asked Mr. Roney whether he agreed that if in fact Dr. Drysdale calculations with respect to wind loadings are reasonable, then it would be dangerous to rely on 16 years (survival) as being definitive as to what the building may show in the face of a Code wind. To this question, Mr. Roney replied that he believed that after 16 years, there would be sufficient obvious damage to the building to show overstress, although he agreed that a Code wind was a factor in his opinion.

Mr. Mueller asked Mr. Roney whether if one assumes that Dr. Drysdale is correct with respect to his calculation of wind forces, and is correct that there has not been a one in 10 wind during the last 16 years, does Mr. Roney agree that it is dangerous to rely on the existing state of the building because it has not yet been exposed to a one in 10 wind. Mr. Roney did not agree. Given his knowledge of gusts over the last 16 years, there have been significant wind forces applied to the wall which fell short of a one in 10 wind but in his view even the lesser winds would cause signs of overstress. He agreed with Mr. Mueller that a one in 10 wind is not a gust, but rather an average over a one hour period of time.

At page four, Exhibit 89 is Item No. 3, Mr. Roney's comments with respect to Dr. Drysdale's critique of Mr. Roney's calculations in Exhibit 32. In Exhibit 32, Dr. Drysdale pointed out that for wind loading, the pressure coefficients and gust factors used have been misinterpreted. He notes that for internal pressures the use of Cpi = 0.3 as used by Mr. Roney would require a category one building which in turn requires uniformly distributed openings accumulating to less than .1% of the total surface area. Dr. Drysdale points out that the building does not have uniformly distributed openings, and that the percent of area is roughly 9%. In Dr. Drysdale's view, category two from the 1995 National Building Code is the appropriate category, and in that case Cpi = .7, assuming that the glass is adequately designed for wind. In this respect Mr. Roney confirmed that in his response, Item No. 3, Exhibit 89, he relied on the National Building Code of Canada 1980, which specifies that for buildings in which the openings are uniformly distributed in all four walls, the internal pressure coefficient Cpi is .3. Dr. Drysdale in his Exhibit C, page five Item J notes that Mr. Roney omitted to mention that the coefficient of .3 only applied to buildings with uniformly distributed openings which is not representative of the subject building where the windows are more concentrated along two faces. It was Dr. Drysdale's opinion that the correct application of the 1980 NBC gives the same coefficient of .7 used in Dr. Drysdale's analysis. Mr. Roney agreed that in his analysis he did not consider the windows to be openings within the meaning of the 1980 Code.

Dr. Drysdale goes on in Exhibit C, Item J to say, "nonetheless even if the windows were not thought of as openings and the 0.3 coefficient was incorrectly used, tensile stresses above those allowed in either 1978 or 1994 would still exist." He notes that the .4 difference in coefficients would amount to a 16.3% decrease in tensile stress which is more than cancelled by the 17% increase due to using the altered wall heights.

In this respect, Mr. Roney maintained his position that the windows are sealed and are designed for the same wind pressure as the cladding, and therefore should not be considered as openings. Mr. Mueller put it to Mr. Roney that notwithstanding in his response in Exhibit 89, Item 3, Mr. Roney confined his critique of Dr. Drysdale to Dr. Drysdale using the wrong Code. Mr. Roney is going even further now and saying that even if the 1995 Code is used, .3 is the correct factor. Mr. Roney agreed with that suggestion, again maintaining his position that the windows are not openings either the 1980 NBC or the 1995 NBC.

In his report, Exhibit 89, in Item 7, Mr. Roney responds to Dr. Drysdale's bullet 7 in Exhibit 32, where Dr. Drysdale notes that Mr. Roney's calculations do not make any adjustments for the effect of windows which reduce the section resisting wind pressure. Dr. Drysdale notes that for the 2.5 metre window spacing along the south wall, stresses due to bending caused by wind pressure need be increased by the ratio 1.56, and that the masonry between the windows on the east wall is even more critical, whereas on the north and west it is less critical. In his response, Mr. Roney says that Dr. Drysdale is correct in that the presence of window openings reduces the effective cross-sectional area of wall, resulting in increased bending stresses between the windows. However, Mr. Roney notes that in this analysis Dr. Drysdale did not take into consideration the axial forces that are at play and which have already been discussed.

In Item 8 of Exhibit 89, Mr. Roney notes that Dr. Drysdale in Exhibit C, Item O has noted that because of the frogged bed plane (indentation) the full brick cross section does not resist bending, and Dr. Drysdale notes that accounting for this, thus increases tensile stress in the brick by about 10%. In his response, Mr. Roney notes that Dr. Drysdale has neglected the fact that this same indent has a dramatic favourable impact (33%) on the axial compressive stresses, which has the effect of lowering the magnitude of the tensile stresses, and that the virtual absence of horizontal cracking of the structure indicates the presence of this overall beneficial effect. Dr. Drysdale in Item O of Exhibit C notes that even using Mr. Roney's numbers, 10% increase on his .18 MPa tension for bending, is .018 MPa which exceeds the counteracting effect of 33% on his .03 MPa compression due to vertical load. Dr. Drysdale notes that the relative effect of increased tensile stress is more pronounced using Dr. Drysdale's calculations, and that there is indeed a net increase in tensile stress as a result of accounting for the indent in the bottom of the brick. Mr. Roney's only comment was that he has not checked Dr. Drysdale's calculations.

In Exhibit 89, Item 9, Mr. Roney notes that Dr. Drysdale in Exhibit 32, bullet 9 states that the wind pressure is distributed between the brick and the block wythes of masonry according to calculated EI values representing section stiffness, and that this is only valid if a number of criteria are met. We have already reviewed those criteria. Mr. Roney conceded that he does not dispute that these criteria have to be satisfied before an analysis of distribution of wind pressure as between the brick and the block can be relied upon. In this respect, Mr. Mueller pointed out to Mr. Roney Dr. Drysdale's comment in Exhibit C at Item P that the ties may be either in tension or compression, so that any mortar droppings are unlikely to be able to benefit both conditions.

Dr. Drysdale expresses the opinion that reliance on random mortar droppings to provide sufficient strength and stiffness of ties is wildly optimistic. Notwithstanding Dr. Drysdale's comments, Mr. Roney stood by his opinion that he observed mortar on the ties which thus provides stiffness to the ties and thus transmits the forces between the brick wythe and the block wythe. As well, he relies on the fact that he observes no damage to the building.

With respect to Dr. Drysdale's second bullet in Bullet 9, Exhibit 32 which requires that in order for the analysis of distribution to be correct, the wythes must have identical support conditions, and be of the same height, in this respect, Mr. Roney did not agree that there was any difference in the support conditions as between the brick wythe and the concrete wythe. He did agree that there was a difference in height, the block panel being 2.6 metres in height and the brick 3.2 metres, representing approximately a 25% difference in height. To counter this, he asked that it be remembered that there was infill in the flange with a ladder reinforcement between the brick and the infill. He observed this at test opening number three at the fourth floor, and at the southwest corner, second floor in test opening number one. These observations are confirmed in his report, Exhibit 87 at pages 10 and 12.

With respect to his suggestion for remedial ties, Mr. Roney agreed that the preponderance of his suggested remedial ties would be located at the top of the building, as opposed to the lower floors.

With respect to Dr. Drysdale's Bullet 10 in Exhibit 32 wherein he says all of the assumptions noted in Bullet 9 are violated, Mr. Roney in Exhibit 89 responds that Dr. Drysdale's results show that the percentage of lateral stress borne by the brick ranges from 52 to 61% as compared to Mr. Roney's calculation of 56%. Dr. Drysdale in Exhibit C, Item Q notes that for walls that only span between the floor and the bottom of the steel beam, the presence of rigid ties (from retrofit), equal wall heights and nearly similar support conditions, does produce results that are in the same range, but notes that for the case with the veneer extending .6 of a metre beyond the height of the block wall, this is not the case. In addition Dr. Drysdale in Item Q Exhibit C notes that it is important to remember that both he and Mr. Roney assumed that the top of the block wall is free to rotate. Dr. Drysdale notes that if Mr. Roney wanted to include axial load on the block wall from the steel beam, the end condition would have to be altered which will have a very significant impact on the calculated distribution of load. Dr. Drysdale notes that to give some idea of the effect, a wall with ends restrained against rotation (by being clamped between the floor and the underside of the steel beam) will have only 20% of the deflection for a similar wall under the same loading but with the ends not restrained against rotation. Mr. Roney agrees with that calculation, but notes that if one assumes that there is some ability on the part of the brick itself to resist forces, then there is less force on the block wall.

Cross-examination by Mr. Garbutt

During his cross-examination of Mr. Roney, Mr. Garbutt tendered five pages of illustrations and calculations which were marked Exhibits 98 through to 102. Although these sheets had not been produced before, I permitted them to be filed as exhibits in the face of Mr. Garbutt's submission that they would facilitate his cross-examination.

After a discussion of the provisions of M-78, Mr. Roney agreed with Mr. Garbutt that for the purpose of his calculations, he assumed that the wall as constructed was either a veneer wall or a cavity wall within the meaning of M-78, Exhibit 47, even though the wall thicknesses did not meet the criteria of M-78 for a cavity or a veneer wall. In this respect Mr. Roney advised Mr. Garbutt that at the time of his investigation he had no information from Emmons & Mitchell that the wall had been designed and constructed as a solid wall.

Exhibit 99 was a drawing similar to the second of the small drawings. Mr. Garbutt reminded Mr. Roney that Mr. Roney had testified by reference to the second of the small drawings that insofar as there was no hatching through the void between the brick wythe and block wythe, he concluded that it was intended to be an open space, and that although there was no cross hatching between the flanges of the beam, it was in fact filled with masonry. Mr. Roney confirmed this to be the fact and confirmed that he observed concrete block in all of the flanges that he observed, except in test hole number three on the east face where he observed both concrete and block.

Mr. Roney admitted to the possibility that the mason had sought and received instructions to infill the space between the flanges, notwithstanding that on the small drawing it appeared to be a void.

Mr. Garbutt referred Mr. Roney to the specifications, Division 4 where at section 2.2.6 the following specification is made: “Reinforcement: Dur-o-wal truss manufactured by DuroWall or Blok truss manufactured by Block Lock Limited.” Mr. Garbutt also brought to Mr. Roney’s attention section 2.2.1 which provides as follows: “Reinforced masonry walls indicated with continuous masonry reinforcement in every second block course.” Mr. Roney appeared to agree that these provisions specified the type of reinforcement and the location of the reinforcement. He agreed with Mr. Garbutt that reinforcement is sometimes used to bond the brick onto the block, for example in a veneer wall. Mr. Roney agreed that ladder joint reinforcement can also be used to resist expansion and contraction, and to provide sheer resistance.

Referring to Exhibit 100, Mr. Roney agreed that Mr. Garbutt’s illustrations correctly identified the location of the ladder joint reinforcements in test holes one and three, and agreed that those locations which are unlabeled as to test holes were as per the requirements of the specifications.

Mr. Garbutt then referred Mr. Roney to section 3.7.3.3 of M-78 which provides that where the space between metal tied wythes is filled with mortar or grout, the allowable stresses and other provisions for masonry bonded walls shall apply, and where the space is not filled with mortar or grout, such walls shall conform to the requirements for cavity walls. Mr. Garbutt suggested to Mr. Roney that to meet that requirement, strap ties are acceptable if the cavity is to be filled. Mr. Roney disagreed. It was his view that for bonding purposes heavier tie requirements had to be met, similar to cavity walls, whether the wall was filled or not.

Mr. Garbutt referred Mr. Roney to the masonry specifications, section 2.1 which requires that there be provided a 6 millimetre continuous expansion joint caulked continuously between steel support angles and brick panels each floor. Mr. Roney agreed that this referred to a soft joint, and the evidence is clear that it was not provided.

Mr. Roney confirmed for Mr. Garbutt that in his view there was a transfer of forces between the wythes having regard to the condition of the wall. He was aware of Mr. Johnston's evidence that 30% to 50% of the length of the brick joints bridged the collar joint. He agreed that it was important to know what was in the walls so as not to destroy what was there by way of remedial work.

There then ensued between Mr. Garbutt and Mr. Roney a discussion with respect to calculations made by Mr. Garbutt with respect to mortar fin stresses and with respect to certain portions of Mr. Garbutt's exhibit book, Exhibit 74. In Exhibit 101, Mr. Garbutt purported to have worked out the mortar fin stresses for a typical panel on the north and south wall. As well, Mr. Garbutt referred Mr. Roney to his Exhibit 102 which purports to be a calculation of fin sheer connection. Mr. Roney appeared to have an understanding of what Mr. Garbutt was attempting to calculate in terms of mortar fin stresses and the fin sheer connection between the two wythes, but noted that he was not certain that such a simplified approach was sufficient. He had not seen these calculations before and this type of calculation was not within the work that he had performed. He stated therefore that he would not want any of his answers to mean that he accepted Mr. Garbutt's calculations. Mr. Roney was prepared to concede that assuming the fins as noted by Mr. Johnston extended completely across the void and connected the brick wall to the block wall, then there would be some sheer transfer potential across the wythes.

Mr. Garbutt then put the question to Mr. Roney, why would not the wall as it sits now be adequate without any remedial work. Mr. Roney responded that in his analysis he did not rely on any sheer transfer between the wythes, he simply was not comfortable in attempting to calculate the sheer capacity of the mortar bridges between the wythes.

With respect to field review, Mr. Roney agreed with Mr. Garbutt that when field review is done at a particular point in time and the construction appears satisfactory, there is no guarantee that when the consultant is not there the construction will continue in the same fashion. Mr. Roney agreed as well that the consultant is entitled to rely on what he sees at any given time, in terms of the work being carried on in accordance with the contract documents.

Finally, Mr. Roney agreed with Mr. Garbutt's suggestion that in the event remedial work involved removing bricks over the windows to the height of the block wall behind, then anchors could be installed through those openings to anchor the block wall to the beam.

Mr. Roney agreed that anchors could be installed between the block and the beam through the opening provided by the removal of the bricks above the windows.

Further cross-examination by Mr. Mueller

In Mr. Roney's view, the wall is indeed load bearing and therefore it is not a panel wall when taking an axial bearing as it is now.

Re-examination by Mr. Nelson

The cardboard box that Mr. Roney has been using as a model was numbered Exhibit 103.

Referring to Exhibit 87, Mr. Roney confirmed that the test holes referred to therein were selected by him and that it was not unusual to analyze a building by opening test holes. In his view, it was appropriate to use the number of test openings that he used, notwithstanding that they exposed only 1% of the lineal perimeter. In Mr. Roney's view, such test holes are a common diagnostic tool.

In re-examination, Mr. Roney confirmed his findings at the test holes, all as set out in Exhibit 87.

Witness – Robert Vinkle

Examination-in-Chief

As was eventually disclosed on cross-examination, Mr. Vinkle is presently employed by Emmons & Mitchell Construction (2000) Limited and indeed he is one of three principals of that new company, the other two principals being the sons of Mr. Emmons and Mr. Mitchell. Mr. Vinkle explained that by the year 2000, more specifically June of 2000, neither Mr. Emmons nor Mr. Mitchell had any further shares in the old company. Their shares were being redeemed annually for a significant period of time. As a result, as of June 2000, the new company was formed, Emmons & Mitchell Construction (2000) Limited. It was Mr. Vinkle's evidence that the original company is still active, although all contract work is now done by E&M (2000).

Mr. Vinkle started with the original company in 1973 as an apprentice, and worked as a carpenter from 1973 to 1977. In 1977 he moved into the company office as a junior estimator. He has been since the early 1990's the chief estimator. He explained that as the estimator the drawings come to him. He reviews them; he sources out the best numbers for the tenders, both with respect to materials and sub-trades; he puts the tender together and presents it by closing date. He explained that his work as chief estimator is different from that of junior estimator, to the extent that when he was a junior estimator the senior partners Emmons & Mitchell were watching his work closely.

Mr. Emmons died in October of 2000 and Mr. Mitchell who is now 80 retired in 1985. Mr. Mitchell was not involved with this project. He is presently not enjoying good health.

Mr. Vinkle became involved with this project in April of 1985 when Mr. Garbutt brought drawings to him and asked him if his company would be interested in bidding. Initially Mr. Garbutt presented only the drawings and not the specifications. Mr. Vinkle then proceeded to obtain quotes from the trades. Mr. Emmons oversaw the project; in due course he was the go-between between the plaintiffs and E&M.

The project supervisor was Jim Sinclair who acted as field supervisor on this project. He is now retired. Mr. Sinclair was the full-time supervisor on this project and had no other projects on the go. He was on the site from August 12, 1985 to July of 1986. He was very good at keeping notes and kept a diary during the course of this project. Mr. Vinkle does not know what happened to the diary.

The project was turned over to the plaintiffs in June or July of 1986, and the final billing was in September of 1986.

At Tab 2 of Exhibit 1, is the Contract dated August 12, 1985. Attached to it is Mr. Vinkle's letter to Dr. Lawless dated May 8, 1985 which notes in its opening paragraph, "We have compiled a price to complete all work as outlined on drawings prepared by N.D. Garbutt, Professional Engineer numbered 1, 2A, 2B, 3, 4, 5, S-1, ME-1, ME-2, ME-3, ME-4, ME-5 and ME-6." Mr. Vinkle confirmed that these were the drawings that he received from Mr. Garbutt. The letter is sent to Dr. Lawless because Mr. Garbutt suggested that that be the case. As a result Mr. Vinkle at that point was negotiating directly with the plaintiffs.

As of May 8, 1985 Mr. Vinkle did not yet have the specifications. The third page of the letter dated May 8, 1985 quotes a price of $621,749, including the roof. Mr. Vinkle confirmed that at that point it was an irma roof that had been priced; an irma roof was shown on the drawings.

At Tab 1A is a letter from Mr. Vinkle to the plaintiffs dated May 22, 1985 relating to various allowances. Mr. Vinkle's letter refers to a letter from the plaintiffs which he does not have. He suspects that the plaintiffs had reviewed the tender and were negotiating with respect to certain items.

Also at Tab 1A of Exhibit 1 is a minute of an office meeting with the plaintiffs and the electrical contractor, K.R. McGowan Electric. The purpose of that meeting was to discuss electrical issues. The plaintiffs were there together with McGowan and Vinkle but Mr. Garbutt was not present.

Tab 1A is a letter from Mr. Vinkle to the plaintiffs setting out confirmation of certain approved changes. Mr. Vinkle's letter refers to a plaintiffs' letter which he does not have.

At Tab 1A is a draft contract dated June 27, 1985. The contract was not in fact signed at that time. The draft contract shows Emmons & Mitchell as being the consultant. Beside the word consultant is a notation "our consultant" which according to Mr. Vinkle was written in by one of the plaintiffs.

Following the draft contract at Tab 1A is a letter dated June 27, 1985 which refers *inter alia* to a change in the roof insulation to six inches at an added cost of $5,400. Mr. Vinkle explained that this change was at the request of the owners and to this point in time he was still dealing directly with the owners.

Page six of this "letter" which is titled Appendix A, Contract Price Summary notes that an additional item is an elevator capable of going two additional floors with a larger pump and additional material, for an additional cost of $13,500. Mr. Vinkle explained that the original elevator was only good for four floors, and the owners had anticipated adding two more floors and therefore paid an additional $13,500 for the ability to extend the elevator without major work.

Article A2 of the draft contract dated June 27, 1985 refers to specifications and details dated June 27, 1985 as prepared by Mr. Garbutt and signed by Emmons & Mitchell. Mr. Vinkle explained that by this date, June 27, 1985, Mr. Garbutt had brought in the specifications and had brought in the three small drawings which he assumed were prepared by Mr. Garbutt's office. Article A2 also refers to drawings prepared by Roberts Engineering. Mr. Vinkle was familiar with Mr. Roberts and indeed there were drawings with Mr. Roberts's name on the title block.

Mr. Vinkle identified the specifications at Tab 1 of Exhibit 1 as the specifications which he and Mr. Emmons signed.

Eventually, the contract was signed on August 12, 1985. The contract provided *inter alia* that Mr. Garbutt would be the consultant and indeed he was throughout.

Between the end of June and August 12, 1985 the plaintiffs were in the process of getting City approvals and were also negotiating with the company with respect to small details. It was Mr. Vinkle's evidence that the plaintiffs looked after site plan approval. The work started August 12, 1985 without a building permit, but the company was allowed by the City to dig the hole.

On site, Jim Sinclair was in charge of the construction. He in turn reported to Mr. Emmons. During the course of the project Mr. Vinkle's involvement was with respect to the site meetings that were held every two weeks. He was responsible for keeping the minutes. It was his evidence that there were 25 meetings in all. The plaintiffs attended 23 of the meetings. Mr. Vinkle attended 20. Mr. Emmons attended five and Mr. Garbutt attended 12. The purpose of the site meetings was to report the progress of the project and any problems encountered. The plaintiffs would add their comments if they were not satisfied with any particular aspect.

With respect to change orders, Mr. Vinkle testified that they were prepared by him and by Mr. Emmons. At Tab 48 is a document that was prepared by Dr. Reed which states in part, "additional costs over tendered costs are itemized by change orders. The majority of these were ordered and supervised by the owners, not by the engineer." Mr. Vinkle was able to confirm that changes were ordered by the owners. He was unable to confirm the extent to which they may have supervised the changes.

The project was essentially completed by the summer of 1986 and Mr. Vinkle's role at that point in time would be getting letters of deficiencies out to the various trades.

Mr. Vinkle identified at Tab 12 Mr. Garbutt's preliminary list of deficiencies dated June 5, 1986 and at Tab 22 the list of deficiencies received from the plaintiffs with respect to the first floor, the basement and the exterior.

At Tab 13 is a letter from Mr. Vinkle to the roofing contractor dated September 4, 1986 setting out deficiencies including a break in flashing on the corner near Rust Check, a hole in the flashing near the roof hatch and caulking on the flashing generally being poor. It was Mr. Vinkle's evidence that these deficiencies had been pointed out by Messrs. Reed and Lawless. At Tab 14 is Mr. Vinkle's letter to the glass contractor dated September 4, 1986 setting out certain concerns that had been brought to Mr. Vinkle's attention by the plaintiffs, including window leaks and the third and fourth floor.

At Tab 15 is a letter from Mr. Vinkle addressed to Emmons & Mitchell which was simply meant to record the work required to be done by Emmons & Mitchell. At Tab 16 is Mr. Vinkle's letter to the masonry contractor dated September 4, 1986. Again Mr. Vinkle testified that this letter was as a result of the plaintiffs' deficiency list which he received on September 3, 1986. He noted that the deficiency list of Mr. Garbutt at Tab 12 is dated June 5, 1986. That deficiency list which addresses both masonry and windows had been sent to the appropriate trades at that time.

At Tab 17 is a letter from Mr. Vinkle to Dr. Reed dated September 11, 1986 to let Dr. Reed know that he was dealing with the deficiencies, and to request their final draw of $10,475.

At Tab 18 is a letter written by Mr. Emmons dated September 15, 1986 to the masonry contractor, attempting to deal with the owners' dissatisfaction with the masonry work.

At Tab 19 is a letter from Mill & Ross Architects to Dr. Lawless. Mr. Vinkle was not aware at the time that Mill & Ross had been retained by the plaintiffs.

At Tab 20 is a letter written by Mr. Emmons to the plaintiffs dated November 20, 1986 wherein he is attempting to negotiate a credit with respect to the poor masonry work.

Entered as Exhibit 104 is a letter from Mr. Emmons to the plaintiffs dated October 2, 1986 which notes in its opening remarks that the company has taken the original deficiency list and broken it down so that the various parts applied to the applicable trades involved. Mr. Vinkle confirmed that this was the process that was being followed. The letter also notes that it is enclosing a number of items "in binder form for your records." Mr. Vinkle confirmed that this was standard procedure on the part of the company. The letter notes that with respect to roofing and flashings, the items have been completed and the roof guarantee is attached. At Tab 23 is the roof guarantee that was attached. It is a two-year guarantee with respect to leaks and a one-year guarantee on the metal flashings, as required by the specifications.

At the last page of Exhibit 104 is a reference to one set of architectural drawings marked as to underground service locations being enclosed. Mr. Vinkle explained that these were as-built drawings for underground services and he was unable to assist as to whether there were any as-built drawings for the remainder of the project.

At Tab 20 is Mr. Emmons's letter to the plaintiffs dated November 20, 1986 confirming the credit with respect to the masonry work. Mr. Vinkle was not involved in negotiating this amount.

Filed as Exhibit 105 is a letter from Mr. Emmons to the masonry contractor dated January 15, 1987, which again refers to the credit back to the owners for the poor masonry workmanship.

Mr. Vinkle in his evidence confirmed that he was not involved after the project completion with respect to any issues relating to water problems.

There then followed a discussion between Mr. Nelson and Mr. Vinkle with respect to the caulking of the windows. Division 7 of the specifications under the heading 'Caulking' provides as follows: "Caulk all exterior joints between different materials and around all openings, doors, windows and fittings as required to provide a weather tight seal using one or a combination of the following sealants…" It was Mr. Vinkle's view that this specification required as part of the contract only the caulking between the metal frames of the windows to the stone surrounds, and did not include caulking as between the stone surrounds and the adjacent brick wall. He believes that in 1986 caulking was applied between the metal frame of the windows and the stone surrounds as part of the contract, and as well he believes that the company's caulking contractor caulked between the stone surround and the masonry and charged extra. He believes further that in 1992 the joint between the stone surround and the masonry was again caulked and it was charged for.

Mr. Vinkle had no discussion with the owners between 1986 and 1992. He was aware of the lawyer's letter that was sent to the company September 16, 1993.

It was Mr. Vinkle's evidence that as far as he knows, the company was not advised in 1989 that renovations were done to the building to accommodate the flower shop on the first floor. The company was not advised of the roof repair in 1994. The company was not advised with respect to the installation of a new roof in 1998, and the company was not aware of any investigation done with respect to the building prior to 1998 by the owners' agents.

At Tab 39 is a statement of account to the plaintiffs from E&M dated October 7, 1986. One of the items is "charge for caulking of windows throughout, $1,500." Mr. Vinkle believed this to be the caulking between the stone surround and the masonry.

Cross-examination by Mr. Garbutt

Referring to Tab 36, which is a statement of account from the defendant company to the plaintiffs and refers to change order number one relating to the brick, Mr. Vinkle explained that when such a change order came in it would be reviewed by Mr. Emmons and himself. With respect to this particular change order Mr. Vinkle explained that initially the contract had provided for a brick allowance of $300 per 1,000; as it turned out the type of brick that the plaintiffs wanted was more expensive and thus called for $10,884.50 extra. As far as Mr. Vinkle knew, the change order did not in any way relate to the increased height of the building.

Mr. Vinkle confirmed that where the sub-trades normally supplied shop drawings, then they would come into the company office at which time Mr. Vinkle and Mr. Emmons would process the shop drawings. Mr. Vinkle recalled no change order from the steel fabricator, Christmas Steel relating to the cutting of the plates, or from any other person for that purpose.

Mr. Vinkle testified that Mr. Sinclair did not consider that he had any leeway and that in practice he would not authorize any kind of a change without contacting the office. Mr. Vinkle is not aware of any charge levied by Christmas Steel relating to the cutting of the plates.

With respect to Mr. Sinclair's diary, it was Mr. Vinkle's evidence that it was not the company policy to retain the supervisor's diary.

Cross-examination by Mr. Mueller

I have already set out the history of the company and its transition to Emmons & Mitchell (2000) Limited at the beginning of Mr. Vinkle's examination-in-chief. This information was in fact extracted by Mr. Mueller on cross-examination.

Mr. Vinkle confirmed for Mr. Mueller that the tradition of the defendant company has been to produce first class work and that was always a corporate objective.

Although Mr. Mitchell retired in 1985, Mr. Emmons was around until he died in the year 2000. Until his death, he was coming in to the office on a partial basis.

Mr. Vinkle explained that by the time he prepared the draft contract at Tab 1A dated June 27, 1985 he did in fact have the specifications and the three detail drawings.

Filed as Exhibit 107 was the Contract between Emmons & Mitchell and Abe Dick Masonry Limited dated August 22, 1985, together with two letters, one from Abe Dick and the other from E&M relating to the Contract. Mr. Vinkle has signed the Contract as Secretary-Treasurer of

E&M, and indeed he prepared the Contract. The Contract itself refers to the Prime Contract dated August 12, 1985, and notes that the Prime Contract includes the work to be performed under this particular Contract between E&M and Abe Dick, "hereinafter referred to as the sub-contract work in accordance with plans and specifications and addenda prepared by Mr. Garbutt, hereinafter called the consultant."

Under the heading 'Work to be Performed', the subcontractor is to supply all material, labour, tools and equipment necessary for the proper performance of the subcontract work pertaining to the following portions of the Prime Contract, namely "all labour and materials to complete all masonry including cavity wall insulation at elevator shaft, stone surrounds at windows and planters, all as per plans, specifications and to owner/engineer approval."

Mr. Vinkle was able to advise that E&M had worked quite often with Abe Dick prior to this particular project and continued working with Abe Dick afterwards and to the present time.

Mr. Vinkle confirmed that in referring to the plans and specifications in the subcontract, the reference was to the head contract documents.

The subcontract provides that the subcontract work shall be performed in accordance with the plans; specifications and other documents described in Appendix A attached. Although Appendix A is attached to the subcontract, the page is blank. Mr. Vinkle explained that it was not the practice of E&M to fill out this Appendix.

Mr. Vinkle confirmed that the plans and specifications were sent out to Abe Dick by his office staff on his instructions.

Mr. Vinkle confirmed that he was never advised by Abe Dick or by the roofing contractor that they lacked any contract documents, nor did either the mason Abe Dick or the roofer ever enquire of Mr. Vinkle with respect to the interpretation of the contract documents. Nor did either Mr. Sinclair or Mr. Emmons ever advise Mr. Vinkle that there had been such enquiries.

Article 9 of the subcontract provides that the contractor may make changes by altering the subcontract work but no changes shall be made without a written order from the contractor. Mr. Vinkle acknowledged that if there were to be any changes they had to be "papered" and that as a result if there was any change in the head contract that change would be replicated in the subcontract.

Article 15 of the subcontract relates to waiver as of the date of total performance of the Prime Contract, and as set out in the Certificate of Total Performance of the Prime Contract. Over and above this contractual waiver, Mr. Vinkle confirmed that he did not personally provide any waiver to the mason, nor is he aware of any such waiver or release having been provided to the mason by any other person in E&M. He has not seen any such waiver or release on any piece of paper in the files.

It was Mr. Vinkle's evidence that he never conveyed to the masonry contractor, nor did anyone else to his knowledge convey to the masonry contractor, that the masonry contractor would be involved in litigation. Nor was Mr. Vinkle aware that the plaintiffs' then lawyer had written to E&M's lawyer advising that in due course the masonry contractor may be involved in litigation.

At Tab 49 is the Minute dated November 6, 1985; section 10.1 notes that a section of the brick wall that had been laid yesterday was rejected and that Abe Dick would rectify. Mr. Vinkle confirmed that was his Minute and also confirmed that he did not actually see the wall nor look into it at all.

At Tab 51 is the Minute dated January 17, 1986 where in Item 2 it is noted that the major portion of the masonry had been completed on Thursday, Friday and Saturday. Mr. Vinkle commented that he was probably told that by Mr. Sinclair.

At Tab 54 is the Minute dated February 12, 1986, where Item B notes that the masonry walls will be cleaned and pointed as necessary as soon as weather permits. Mr. Vinkle noted in his evidence that this was efflorescence that he actually observed himself. At Tab 58 is the Minute dated April 9, 1986. Item 20.3 (iii) notes "have Primo complete the brick cleaning and wash down so we can have a final inspection and acceptance by the owner" (Primo was the field supervisor for Abe Dick Masonry Contractor). Mr. Vinkle explained that this again referred to efflorescence. Item 20.3 (ii) notes that "Jim Sinclair will check with Primo on any windows missing the window water drip and this will be corrected." Mr. Vinkle explained that this item related to the window stone surrounds and that the required cut in the bottom of the heads and sills were missing in some of the units. This was a cut that was made in order to collect water that collects as drips at the edge of the stone surround.

At Tab 59 is the Minute for April 25, 1986, which notes that Mr. Emmons is to call Abe Dick Masonry and set up a meeting once all the work has been completed. In this connection Mr. Vinkle was aware of the dissatisfaction of the owners re the masonry work. Mr. Vinkle noted that the brick was not square and in his view was hard to lay.

Mr. Vinkle acknowledged that eventually there was a credit to the owners with respect to the masonry work and that something was subtracted from the masonry contract on account of bad appearance.

At Tab 61 is the Minute for the site meeting number 25 on July 3, 1986 at which the plaintiffs, Abe Dick and Don Emmons were present. The Minute notes that a site inspection was carried out by all present to determine the method and the procedure to follow to remove mortar, stains and discolouration in order to obtain a more uniform appearance to the overall job. The Minute notes as well that Mr. Dick agrees to an overall clean down of the entire building using a solution of his choice. It was noted that exterior light fixtures would be covered and protected and if he felt necessary the window frames would be covered. Mr. Vinkle agreed with Mr. Mueller that it was clear to him that the complaints of the plaintiffs with respect to the walls were being made on an aesthetic basis as opposed to a functional basis.

At Tab 12 is Mr. Garbutt's deficiency list dated June 5, 1986, wherein he notes that the masonry, in general, is poorly done. Mr. Vinkle conceded that he did not dispute that characterization.

At Tab 16 is a letter from E&M to Abe Dick dated September 4, 1986, asking Abe Dick to review the deficiencies. Mr. Mueller asked Mr. Vinkle why E&M was still asking the mason to fix deficiencies when Mr. Garbutt had put his deficiency list in, in June. Mr. Vinkle explained that these particular deficiencies came from the plaintiffs' reports the day before, but he did agree that the nature of the plaintiffs' deficiencies were similar to Mr. Garbutt's.

At Tab 17 is a letter from Mr. Vinkle to Dr. Reed dated September 11, 1986 indicating that Mr. Emmons is reviewing the masonry problems and will contact Dr. Reed to set up a meeting. Mr. Vinkle agreed that the problems referred to were aesthetic.

At Tab 18 is a letter from E&M to Abe Dick dated September 15, 1986 noting that since their last meeting with the plaintiffs it was agreed to try a complete cleaning of the brick work to try to remove mortar splatters and improve the overall appearance of the brick. Mr. Emmons notes that this has now been completed and that at a subsequent meeting with the plaintiffs, they still feel the overall appearance to be unsatisfactory. Again, Mr. Vinkle confirmed that the dissatisfaction was with respect to the appearance of the masonry.

Mr. Vinkle was able to confirm for Mr. Mueller that he had no discussion whatsoever with Mr. Sinclair or Mr. Garbutt, or with the masonry contractor or the roofing contractor, with respect to:

a) the cutting of the support plates on the east and west walls;
b) the window leaks;
c) whether or not the wall was to be solid or cavity;
d) the roof slope;
e) the type and placement of ties;
f) the 6 millimetre soft joint called for in the specifications;
g) weepers and flashings.

Referring once again to Tab 12, Mr. Garbutt's list of deficiencies, June 5, 1986, Mr. Garbutt lists "caulking between frames and stone surrounds." At Tab 14 is a letter from Mr. Vinkle to Fort Glass Incorporated, the glass subcontractor, setting out certain deficiencies as follows:

a) all caulking should be checked as general poor workmanship;

b) fourth floor window leaks onto sills;

c) third floor window leaks.

Mr. Vinkle confirmed that he received this list of deficiencies from the plaintiffs and indeed viewed them as deficiencies. He has no idea whether the leaks at the fourth and third floors were fixed. At Tab 39 is the statement of account from E&M to the plaintiffs dated October 7, 1986. It includes a "charge for caulking of windows throughout, $1,500". Mr. Vinkle confirmed that this was prepared under his direction. It was his position that this charge referred to caulking from the surround to the bricks, notwithstanding that it refers to "windows". At Tab 40 is an account to the plaintiffs from E&M dated November 20, 1986 which again refers to a charge for caulking of windows throughout of $1,500. With respect to this item, Mr. Vinkle conceded that he never did go onto the site to determine what was being caulked and was not being caulked. Mr. Vinkle was reminded of Dr. Reed's evidence that this caulking referred to caulking from the window frame to the surround. Mr. Vinkle maintained his position that it referred to caulking from the surround to the brick, notwithstanding that he did not see it done. It was his understanding that this work was done by E&M's caulker.

At Tab 21 is a list of items to be completed, dated November 26, 1986. It speaks of ensuring that water problems in the "brick around middle window on fourth floor north side and brick around rear exit are resolved completely to owners' satisfaction and to correct water leaks around all exterior doorways." Mr. Vinkle is not aware who created this particular document. He has never seen it before and he does not know what was done to address the concerns listed.

With respect to Dr. Reed's evidence that after the caulking was done in 1986, the problems seemed to be corrected but then the problems came back and there was a further meeting on site, Mr. Vinkle knew nothing of that meeting.

Referring to the specifications, section 7.6.1 which calls for caulking of all exterior joints between different materials, Mr. Vinkle again expressed the view that this specification refers to caulking between the window frame and the surround, and that the stone surrounds in his view are part of the masonry.

At Tab 13 is a letter from Emmons & Mitchell to the roofing contractor dated September 4, 1986 listing deficiencies including a break in the flashing on the corner near the Rust Check building; a hole in the flashing near the roof hatch; and the fact that the caulking on the flashing is poor. Mr. Vinkle had no knowledge as to whether or not these deficiencies were rectified. He assumed that they were.

With respect to Exhibit 104, Emmons & Mitchell's letter to the plaintiffs dated October 2, 1986, Mr. Vinkle agreed that Mr. Emmons in this letter was trying to convey to the plaintiffs that the deficiencies had been rectified. Interestingly, that letter lists just about all of the trades but does not list the masonry work as being completed. In that respect, we have Tab 18 which is Mr. Emmons's letter to Mr. Dick saying that the cleaning has not satisfied the plaintiffs. We have as well Tab 20, which is a letter from Mr. Emmons to the plaintiffs confirming that they are looking for a $12,000 credit on the masonry deficiencies, and we have Tab 45, which is Mr. Emmons's letter to the plaintiffs dated January 15, 1987 confirming that they have agreed on a $9,000 credit for the masonry. Again, Mr. Vinkle was able to confirm that this settlement was with respect to the aesthetic problems, not any functional problems. Exhibit 105 which is Mr. Emmons's letter to Abe Dick dated January 15, 1987 confirms that of the $9,000 credit, Emmons & Mitchell will pick up $1,000 and Abe Dick will pick up $8,000. Three letters then went into evidence relating to the credit, as Exhibit 108. There is a change order dated December 31, 1986 deducting $8,000 from Abe Dick's contract, a letter dated January 28, 1987 from Abe Dick to E&M enclosing a credit note in the amount of $8,000, and there is the credit note itself. Mr. Vinkle confirmed that this particular change was all properly formalized [it is interesting that an $8,000 credit constituting a change to the masonry subcontract and as a result a change to the main contract, was extremely well papered while something as critical as the cutting off of the plates on the east and west sides is not papered at all].

At Tab 23 of Exhibit 1 is the roofer's guarantee, which certifies that the roofer has applied the roofing and flashing membrane in accordance with the specification requirements.

At Exhibit 2, Tab 64 is Mr. Garbutt's letter to Mr. Emmons dated May 10, 1993 requesting Mr. Emmons to determine,

a) that the collar joint is filled;

…

b) to determine the type of masonry ties used.

Mr. Vinkle testified that there was no discussion between him and Mr. Emmons with respect to this enquiry, and he recalls no discussion whatsoever up until 1993 about there being a void between the brick and block wythes. He was able to identify Tab 65 as being in Mr. Emmons's handwriting, wherein he identifies Primo, the masonry field supervisor, the type of brick and the fact that it was built as a solid masonry wall with filled collar joints, and that the walls were tied together with adjustable ties. Mr. Vinkle agreed that in the face of the enquiry it would be logical for Mr. Emmons to go to either Primo Santin, the masonry field supervisor, or Mr. Sinclair, E&M's site supervisor.

With respect to Mr. Harding of Amherst Roofing who did the remedial work on the roof for the plaintiffs, and who apparently will give evidence for the defendants, contrary to the interest of the plaintiffs, Mr. Vinkle confirmed that Amherst Roofing does work for E&M, is still bidding on E&M's projects and is still getting work from the company.

Re-examination by Mr. Nelson

With respect to Exhibit 107, the masonry subcontract, article 1, 'Work to be Performed', closes with the expression "and to owner/engineer approval". Mr. Vinkle confirmed that that provision is in all of their subcontracts.

Witness – Primo Santin

Examination-in-Chief

Mr. Santin is 67 years of age. He was born in Italy and was taught the masonry trade by his father. He came to Canada in 1953 and began working with Abe Dick in 1956. He has been with that company ever since. The company has its head office in St. Catharine and a field office in Kingston. Mr. Santin started working in Kingston in 1960 and moved to Kingston permanently in 1965. On his first job in Kingston he was a field foreman and by 1985 with some 25 years experience in Kingston, he was a field supervisor. The job of field supervisor, he explained, is to make sure that the job is correct and done right. At any given time the field supervisor may have eight or nine jobs on the go. His responsibility is to check the particular job every day. He goes to the job site and checks with his foreman and asks if there are any problems. He then goes to the general contractor's site supervisor and asks if there are any problems.

Prior to 1985 Mr. Santin had worked on other E&M projects, and considered E&M one of the best companies in the business.

With respect to this particular job, Mr. Santin recalls getting a drawing which he sent on to head office to price out, and eventually Abe Dick submitted the successful bid. In the bid, Abe Dick included so much per thousand bricks as an allowance. The way it worked was that if the bricks cost more than the price per thousand, then E&M and thus the owners would pay the additional cost. On the other hand if it turned out that the cost of the brick was less than the allowance then E&M and thus the owners would get a credit.

Entered into evidence as Exhibit 109 was Mr. Santin's time book from this particular project.

Mr. Santin explained that with respect to his role on this particular job, he would check at the site every day to see if there were any problems. He would check with both his foreman Ronnie

McCoy and with Mr. Sinclair, Emmons & Mitchell's site supervisor. He would ask both of them if there were any problems and if there were, they would be corrected.

The masonry work started approximately November 1, 1985 and continued on to the end of April, 1986. During this space of time, Mr. Santin had approximately six other projects on the go.

With respect to the brick, Mr. Santin was aware that "John Price Ontario size" was used. He explained that Ontario size is a bigger brick than normal, resulting in a height of 8.5" for every three courses, including three mortar joints, as opposed to approximately eight inches in height for every three courses of ordinary brick. He noted that John Price is used normally for interior work and is not as good for exterior work. He recalls telling Dr. Reed and Mr. Sinclair that the bricks were difficult to achieve a perfect job with.

Mr. Santin confirmed that the concrete block is eight inches high with one mortar joint.

When he discussed the brick with Messrs. Reed and Sinclair he got no instructions other than this was the brick that they wanted laid.

It was Mr. Santin's evidence that the only drawing he ever received and saw was Drawing Number Four from Exhibit 9, which is a section drawing. Mr. Santin was asked what type of wall he was constructing on this project. He responded that it was not a cavity wall because there was no insulation between the brick and the block wythes. It was what he referred to as a "shell wall" or a "cool wall".

Mr. Santin was asked whether there was a gap between the wythes and he responded that all the time they were coming up with 3/8 inch to ½ inch between the brick and block wythes. He explained that in a cavity wall with insulation there is a lot more space between the wythes. He explained that in this particular instance, the mason who is laying the brick has to grab it at the top and sides, and therefore he needs room for his fingers, thus is a void created between the brick wythe and the block wythe. It was his evidence that he observed the space between the two wythes. He asked Mr. Sinclair if there was any complaint and Mr. Sinclair responded in the negative.

Mr. Santin explained that the masons proceeded by laying two courses of block from the inside of the building, and then from the outside of the building, on scaffolding, laid courses of brick to the top of the laid block.

With respect to connections between the brick and block wythes, he explained that where the brick and block courses came up even, then a Dur-o-wal or ladder type tie was laid between the two wythes along the length of the wall. Some three or four courses of block were sometimes required before the joint was even with a brick course joint. Mr. Santin explained that if the block and the brick are standard sizes, then the courses match and in those circumstances a tie is placed every two blocks. He explained that with the Ontario size brick the courses did not match all the time and if the courses did not match then his workers placed a strap tie across the block and the brick wythes. In that respect, sometimes they had to bend the strap ties in order to connect the two wythes. It was Mr. Santin's evidence that he actually saw Dur-o-wal ties and strap ties being installed in the walls. Again he explained that every four courses of block came even with a course of brick and a Dur-o-wal ladder style tie was placed in that event. Otherwise, strap ties were installed every 16" of height. He explained again that it was necessary to bend the strap ties in that the brick course was higher than the block course.

During the course of the masonry work, a period of some six months, no one every approached Mr. Santin and either complained about the work or said it was okay. It was Mr. Santin's evidence that Mr. Sinclair approved the placing of strap ties where the courses did not meet, and advised Mr. Santin that if there was any problem in this respect he would get back to him. He never did get back to him. At this point, Mr. Santin gratuitously commented that the brick has been there for 16 years and that if it was not properly tied then it would have fallen away.

Mr. Santin testified that he actually saw mortar droppings falling into the cavity. He explained that it is a natural thing to occur when one puts the brick on top of the mortar as the courses are laid. It was his evidence that he actually saw mortar falling onto the top of strap ties and that his masons would not clean the mortar off the strap ties. [I observed Mr. Santin closely as he gave this evidence. It did not have a ring of truth to it and he appeared for a moment rather nervous in giving the evidence as opposed to some of his other evidence which he gave with great assurance.]

With respect to mortar joints, Mr. Nelson asked Mr. Santin whether the block came in contact with the steel <u>beam</u>. Mr. Santin replied in the affirmative and then drew a sketch, Exhibit 110, which brought home to me that he did not really understand the question. What he drew was a top view of the steel column, and testified that he had been instructed to make the block wall tight inside the flanges, and that this instruction had come from Mr. Sinclair. The flanges for the beam of course are in the horizontal plane and not in the vertical plane, and with respect to the top of the block wall, there is no way to get it "inside" the flanges. As well, it should be noted that other evidence has indicated that the ends of the block walls were <u>not</u> placed inside the flanges of the vertical columns.

With respect to control joints, Mr. Santin explained that there was only one installed and that was on the Queen Street side (the south wall). Mr. Sinclair had given him instructions to put it in the middle of the building, and told him that the plaintiffs did not want any control joints. Also Mr. Sinclair told him after the control joint was installed that the plaintiffs were not happy with it.

With respect to windows, they were installed after the masonry work was completed. Abe Dick did not install the windows. Abe Dick did install the stone surrounds and in that respect Mr. Santin told Mr. Sinclair that the top of the surround would constitute a water trap. Mr. Santin testified that he had never seen this before. There was a space between the top of the stone surround and the underside of the steel angle, which allowed water to enter the wall. It was Mr. Santin's evidence that he spoke to Mr. Sinclair and Mr. Sinclair in due course told him that he had talked to both the engineer and the plaintiffs and that they wanted the stone surround. Mr.

Santin attempted to draw a sketch to illustrate his concern. It went in as Exhibit 111 and I did not find it helpful.

Cross-examination by Mr. Garbutt

Mr. Garbutt discussed with Mr. Santin the dimensions of the “Ontario brick”. Mr. Garbutt referred to his Tab 9 in his exhibit book, Exhibit 74, which illustrates the brick used as compared to the normal brick. Unfortunately Mr. Garbutt’s measurements were in millimetres. He and Mr. Santin attempted to convert from millimetres to inches which Mr. Santin preferred. I think they agreed that the brick used was 8 ¼ to 8 ½ inches long, four inches wide and 2 ¼ to 2 3/8 of an inch high.

Mr. Santin agreed that the standard mortar joint is approximately 3/8 of an inch. Mr. Santin was asked whether he ever gave instructions to vary the thickness of the mortar joint, and he explained that sometimes when the workers hit the steel angle they would have to vary the thickness of the mortar joint. As well, sometimes the brick was curved so that the thickness of the mortar joint differed. In his view, the thickness of the mortar joint varied because of the unevenness of the brick. In this respect, Mr. Santin explained that the practice is to have a “brick stick” which is a length of wood on which the coursing is first figured out before the actual laying of the bricks commences. The purpose of figuring out the coursing is to ensure that the masons get to where they want to get as they lay up the courses. The stick is marked with the proper coursing and then held against the courses as they go up. Mr. Santin maintained of course that if the brick had not been of a different size, then the masons would have been okay.

Mr. Santin then disclosed that he went back to the building last week to measure the brick coursing and found that the height of three courses including three bricks and three mortar joints was 8 ½ inches.

Mr. Santin explained that it would have been difficult to vary the joints in the block wall in order to accommodate the ladder tie, and to attempt to reduce the joint between the bricks might result in there being no mortar left between the bricks.

Mr. Santin's discussion with Mr. Sinclair about the top of the window surround occurred when they hit the first window level. At that point he told Mr. Sinclair that the header would be a water trap. Mr. Sinclair eventually came back and said, that's what they wanted. Oddly Mr. Santin could not recall that the stone surrounds were part of his contract until he was referred to Exhibit 107 in that respect.

Mr. Santin was able to confirm that at the time he had his discussion with Mr. Sinclair about the window surrounds, all of the surrounds were on site. He does not remember who ordered the surrounds. Then he recalled that Emmons & Mitchell told Abe Dick to order the window surrounds. Mr. Garbutt suggested to Mr. Santin that it would appear that he saw no problem with the surrounds when they were being ordered. Mr. Santin responded that he looked at the surrounds at the time but then started thinking about them. It was Mr. Santin's view that the problem as he perceived it could not have been remedied by flashing because the flashing would still be sitting on top of the stone surround and water could get in underneath.

In the final analysis, Mr. Santin testified that he did not know who ordered the stone surrounds, he simply went to the site one day and saw that the stone surrounds had arrived and that the top surround was flat rather than equipped with a ledge that he felt would have stopped water penetration.

Mr. Santin was not aware whether or not a shop drawing had ever been prepared with respect to the stone surrounds.

It was his evidence that the width of the space between the top of the stone and the underside of the window steel lintel was approximately 1/8 inch. He did not take the opportunity on his visit to the site last week to have a look at that space.

With respect to his discussion with Mr. Sinclair about the problem that he saw with the top of the stone surround, it was Mr. Santin's evidence that he told Mr. Sinclair to get a different stone surround. He then testified that one did not need any shop drawing for the stone surround because it was a standard item, and it was bought in accordance with the drawing. He does however recall some dispute between Emmons & Mitchell and Abe Dick with respect to the stone surround.

Mr. Garbutt produced to Mr. Santin Drawing Number Four, Exhibit 11 which shows the window surround and which is completely devoid of any detail or size relating to the window surround.

Cross-examination by Mr. Mueller

Mr. Mueller brought to Mr. Santin's attention portions of Mr. Johnston's report, Exhibit 56. Mr. Santin was not aware of Mr. Johnston's comment at page 14 that:

> The rust marks and pattern were high enough on the angle leg to infer a flow from above and not simply through the horizontal joint between the stone header and steel; equally there were no stains from the horizontal window to stone joint that was recaulked twice.

Nor was Mr. Santin aware of Mr. Johnston's comment at page 15 of his report as follows:

> There was a clear flow of water which emanated as a rivulet from between the angles that formed the lintel framing at the inside and outside masonry at the head of the window opening. The water was clearly visible on the back of the exterior steel angle and flowed across an active oxidation area to wet the stone header at the top of the window opening before pooling on the drywall soffit at the head of the window below.

Nor was Mr. Santin aware of Mr. Johnston's comment at page 16 as follows:

> As more flow and water penetration occurred, it was noted generally that the flow emanated from the top surface of the bricks and there was no apparent crack or opening to the exterior in the mortar at this location. In other words, the water penetration was occurring at the uppermost plane of the brick surface where it met the mortar bed of the brick above. At no point during the limited time of our test did we notice any flow from the often incompletely mortar-filled header or end joint of the bricks in the wall.

Mr. Santin confirmed for Mr. Mueller that the Abe Dick company is still active.

Mr. Santin confirmed for Mr. Mueller that notwithstanding what Mr. Vinkle said about sending the drawings and the specifications to the various subs, all that Mr. Santin received was one drawing as previously identified. There were no specifications given to him. Mr. Santin was referred to the specifications at Tab 1, Exhibit 1 and it was his evidence that when Abe Dick tendered, it did not have the specifications. Its bid was based solely on the drawing. Mr. Santin did not think that Abe Dick ever saw the specifications. Mr. Santin recalls questioning Mr. Sinclair as to where the specifications were and Mr. Sinclair told him that they were supposed to come.

Then Mr. Santin surprisingly testified that once his men were on the job and working at the masonry he still did not have the specifications. He was asked how the work could be proceeded with without specifications and he explained that they went by the drawings. He remembers questioning Mr. Sinclair often to the effect that "what do I do here and here, I have no specs." He never went to the site office to look for or to look at the specifications. When he went to the site he spoke to his foreman and to Mr. Sinclair and they indicated to him that there were no problems.

Mr. Santin conceded that on most jobs he is given both the drawings and the specifications but here one drawing was handed to him and he sent it on to head office at St. Catharines. No specifications were given to him. After the job was priced, Mr. Santin asked Mr. Sinclair where were the specifications and Mr. Sinclair told him they were coming.

The masonry subcontract was entered into on August 22, 1985 and work did not begin until the first of November. Mr. Mueller asked Mr. Santin what he did in that two-month period in an attempt to get the specifications. Mr. Santin advised that he was busy with other jobs at that time. In the week before the masonry work started, he did not ask for the specifications. His discussion with Mr. Sinclair about the specifications took place at the time of tendering.

Mr. Santin confirmed that while the masonry work was going on he never had the specifications and he does not recall asking for them during the work. He never asked his head office whether they had the specifications. He explained that in 99% of the cases, if head office has the specifications they send them to him. In this case they did send back to him the one drawing that he had received and sent on to head office.

It was Mr. Santin's evidence that he was satisfied to complete the masonry work on the basis of the drawing alone, and if there had been any problems Mr. Sinclair would have advised him or would have advised his foreman. In any event, Mr. Santin was quite certain that he never had the specifications in his hands, he never saw them during the course of the work and he does not know whether or not his foreman McCoy had them or saw them. He did not have any discussions with Mr. McCoy as to how to go about the work. Mr. McCoy would have gone by what he had and he does not know whether Mr. McCoy had the specifications.

Mr. Santin confirmed that he had one drawing only, that at no time did he have or did he see the three small drawings showing the cavity. He saw those drawings yesterday for the first time. He confirmed that the wall was built without specifications and without any drawings other than the one big drawing, Drawing Number Four of Exhibit 9.

Mr. Santin admitted that he was running a lot of jobs at the time and had a lot of responsibility. He would come to the site daily, he would talk to his foreman and ask if there were any complaints, and he was being told there were none. He would then speak to Mr. Sinclair and ask if there were any problems and Mr. Sinclair did not bring any problems to his attention. At no time did he receive any complaints from the engineer. Mr. Santin did indicate that he was more concerned about this job than all of the others, because the masonry work was being performed in the winter time which was not the case with the other jobs with which he was involved.

Mr. Santin was never invited to the site meetings.

Mr. Santin confirmed that he saw his men installing ladder ties and strap ties.

Mr. Santin confirmed for Mr. Mueller that there would be an approximate ½ inch space between the brick and the block wythes in order to accommodate the mason having to hold the brick as he put it into place. Mr. Mueller brought to Mr. Santin's attention that there was evidence that there were spaces between the two wythes of up to 1 ½ inches, and that in any event the spacing varied throughout the wall. Mr. Mueller asked Mr. Santin if he had ever seen spacing more than ½ inch. After some discussion about the block possibly being laid as much as an inch past the outer edge of the flange of the beam, and Mr. Santin conceding that he sometimes saw spacing more than ½ inch or more than the ¾ inch specified by the specifications, Mr. Santin finally disclosed that his foreman had told him during the course of construction that the structural steel was out of plumb. Mr. Santin went to Mr. Sinclair and pointed this out to him, and pointed out the wider spacing between the wythes, and Mr. Sinclair said that it was too late to do anything. There was no discussion between Mr. Santin and Mr. Sinclair about going to Mr. Garbutt to discuss the size of the gap between the wythes. Effectively what Mr. Sinclair said was, "the steel is up, the floors are poured, what can he do, tell us to tear it down." It was Mr. Santin's position here at trial that before his men start the work the steel work is supposed to be plumb. He then testified that he talked to Mr. Sinclair before the construction started and told him to make sure that the steel work was plumb. In the final analysis, Mr. Santin did not speak to Mr. Garbutt at any time about this problem. He does not know whether Mr. Sinclair ever spoke to Mr. Garbutt.

Mr. Mueller put to Mr. Santin that he knew that Mr. Garbutt had designed a solid wall and that he believed a solid wall was being constructed. Mr. Santin disagreed and said that the drawing that he had did not call for a solid cavity, and that no one ever brought to his attention the fact that anyone wanted it to be solid. It was his evidence that he came to the site every day and sometimes spent an hour or two hours looking at the work as it was performed.

Drawing Number Four, Exhibit 9 was presented to Mr. Santin, and it was suggested to him that the drawing does not show a cavity between the brick wythe and the block wythe (that is certainly the fact of the matter). Mr. Santin responded that if the designer wanted a solid wall with no cavity then he should have said so on the drawing. He explained that in any building such as this, the cavity is never solid. If it is solid, the moisture never gets out. In this respect Mr. Santin agreed with Mr. Mueller that the void in a cavity wall is usually two to three inches and it accommodates insulation being placed between the two wythes. Mr. Santin seemed to agree that a 19 millimetre or ¾ inch space between the wythes was too narrow for a cavity wall but seemed not to agree that 19 millimetres was too narrow for a veneer wall.

Mr. Santin confirmed that while Drawing Number Four, Exhibit 9 shows no gap but does not require in writing a filled joint between the wythes, he did not seek any instructions from Mr. Garbutt.

With respect to Ron McCoy, the Abe Dick foreman, Mr. Santin disclosed that he is retired now but living near Kingston. Mr. Santin phoned him at one point to tell him that he would like him to come and answer some questions. First Mr. McCoy said he would do so and a meeting was scheduled, however the day before the meeting Mr. McCoy called and said he did not want to have anything to do with anybody.

Mr. Santin confirmed that most of the Abe Dick men working on the project were consistently there throughout the project and that at no time did any of the Abe Dick workers, including the foreman, come to him and say that they were uncertain as to what to do.

With respect to the shelf plates, Mr. Santin confirmed that he saw them on all four sides of the building. He was never aware that they disappeared from the east and west sides of the building. He confirmed that Abe Dick is not in the business of cutting shelf plates and they had no cutting equipment with them. Mr. Santin was never made aware of the fact that they were cut off.

With respect to the ties, Mr. Santin recalls his discussions with Mr. Sinclair and he has already given his evidence in that respect and he said that he had nothing to add.

With respect to the joint between the brick and the underside of the plates, it was Mr. Santin's evidence that Mr. Sinclair told him to put a mortar joint there. It was his evidence that it would have been easy to put a caulked joint, but he asked Mr. Sinclair what he wanted between the brick and the underside of the shelf plate and Mr. Sinclair told him he wanted a mortar joint.

With respect to the area between the top of the block and the underside of the beam, Mr. Santin testified that they filled this solid with mortar. Indeed, Mr. Santin had asked Mr. Sinclair whether this space was going to be left empty, and Mr. Sinclair told him that he wanted a solid mortar joint there. Mr. Santin is not aware that some witnesses found the joint in this location to be variable.

Mr. Mueller pointed out to Mr. Santin that the specifications called for weepers and flashings, and none were installed, and suggested to Mr. Santin that he knew the building had a cavity, to which Mr. Santin agreed. Mr. Mueller suggested that it was basic construction that if there is a cavity it must be drained by weepers and flashings. Mr. Santin responded that if it was a clear cavity wall, then Mr. Mueller was right, but this was not a cavity wall, it was a “standard” wall. Mr. Mueller suggested that the wall had a gap, to which Mr. Santin responded that every wall had a gap. Mr. Santin agreed with Mr. Mueller that in windscreen walls, weepers and flashings are installed. Again Mr. Mueller asked Mr. Santin why weepers and flashings were not installed here, and Mr. Santin’s simple answer was, “I don’t know.” He then said that they are not put in, in this type of a construction. Mr. Mueller asked Mr. Santin whether he discussed the question of weepers and flashings with Mr. Sinclair. Mr. Santin said he recalled that there was some discussion but he could not remember the detail. He explained that it was hard for him to go back and remember such discussions.

Mr. Santin does not remember whether Mr. Sinclair knew that he did not have the specifications.

Mr. Santin was familiar with the concept of a mock-up and said that in this case they did build a mock-up; they built a corner and then called Dr. Reed and “they approved everything”. Mr. Mueller suggested to Mr. Santin that the area that he was describing was the area with the soldier courses which were rejected, and Mr. Santin agreed. He then conceded that the soldier course had been complained about and he had told whoever was there that if they would approve the joints then the soldier course would be fixed. As far as Mr. Santin knows, the mock-up was taken down, it was rebuilt and his foreman told him that everything was approved. Mr. Santin did offer that he did not care for the coursing joint that the plaintiffs had picked.

Mr. Santin explained to Mr. Mueller that as far as he was concerned, a ‘cool’ or ‘shell’ wall was the same as a veneer wall.

In discussing the ties with Mr. Mueller, Mr. Santin commented that at the top of the building they had a problem tying the brick in. He asked Mr. Sinclair what to do and Mr. Sinclair, after checking the drawing, said a weld or anchor was not called for. As I understood the evidence, at the top I-beam and above there was nothing to anchor the brick to the steel. It is usually done with clips that extend out from the flange of the I-beam into the brick coursing. Mr. Sinclair checked the drawing and said none were called for. In his view, Mr. Santin thought this was improper construction. He was not aware that the parapet has been completely rebuilt.

Mr. Santin admitted that he had been shown some of the specifications, perhaps one page thereof yesterday, and had also been shown the small drawing which said that the ties should be installed every two courses. He confirmed that he saw the reference to Dur-o-wall ties in the specifications that he saw yesterday. He is sure that his men used Dur-o-wall ties; he is sure that they did not use adjustable ties. He had no discussions with Mr. Sinclair about using adjustable ties. He had been advised by Mr. Sinclair to use Dur-o-wall ties and straps. He knew that adjustable ties are used when the coursing is not on. He does not believe that there were any adjustable ties on site. He is aware that they are sometimes used on cavity walls and on veneer walls when the wythes have to be tied together. His foreman McCoy told him that Sinclair had told McCoy to use strap ties when the Dur-o-wall ties could not be used.

The strap ties that he saw installed were bent both ways, both from the brick down to the block and from the block down to the brick.

Mr. Santin was able to confirm for Mr. Mueller the fact that there are techniques available to avoid mortar droppings such as beveling and using a piece of wood inserted between the wythes that can be pulled out. As well the excess mortar can be smeared against the back of the brick wall. Mr. Santin explained that this is done in cavity walls where insulation is to be placed between the two wythes. Mr. Mueller asked what there was about a veneer wall that was less important in terms of allowing mortar to drop. Mr. Santin responded that it was “just a brick veneer to block” and that a cavity wall was not called for here, whereas a cavity wall was called for at the stairway and it was different as night and day from the overall wall.

Mr. Santin does recall that there was some kind of an argument relating to the stone surround but he cannot remember the details thereof.

Mr. Santin confirmed for Mr. Mueller that any correspondence between Abe Dick and Emmons & Mitchell would go to or from Abe Dick’s St. Catharine’s address. Mr. Santin agreed that if in fact Mr. Vinkle sent out the plans and specifications to Abe Dick, then they would have gone to the head office in St. Catharines, but Mr. Santin maintained that all that he ever had was one drawing that had been given to him.

At Tab 64 of Exhibit 2 is Mr. Garbutt’s note to Mr. Emmons dated May 10, 1993 looking for confirmation that the collar joint was filled, the type of brick used, and the type of masonry ties used. Tab 65 is Mr. Emmons’s note with respect to these enquiries. Mr. Santin has no recollection of being asked by Mr. Emmons or by anyone else with respect to these questions. He never saw Tab 64 before and no one has ever asked him for this information. No one questioned him with respect to the job after it was finished. Indeed, he never heard about this litigation until approximately a year ago.

Mr. Santin confirmed that he would not have answered those questions in the manner in which Mr. Emmons noted because he was aware that there was a space between the wythes and that there were no adjustable ties used. He does not know who would have given these answers to

Mr. Emmons other than Mr. Sinclair or Mr. McCoy. It is not possible that his head office asked him and that he responded to head office who in turn passed it on to Mr. Emmons.

Mr. Santin acknowledged that Mr. Abe Dick knows that he is here. Mr. Santin mentioned it to him last week. He does not know whether Mr. Dick is aware of the law suit.

Mr. Mueller asked Mr. Santin whether he understood that if the court finds the masonry was not properly done, then he might be criticized. Mr. Santin noted in response that Abe Dick was working for the general contractor and they never complained. Mr. Santin is not aware that Mr. Mueller sent a letter to Emmons & Mitchell saying that if the plaintiffs obtained judgment and could not collect from Emmons & Mitchell, then they would sue Abe Dick. Mr. Nelson has never advised him that Abe Dick is still at risk with respect to this project.

There then took place an exchange between Mr. Mueller and Mr. Santin with respect to a telephone call that Mr. Mueller placed to Mr. Santin in May of this year. Mr. Santin professed not to recall the contents of the discussion.

With respect to the credit note of $8,000 between Abe Dick and Emmons & Mitchell, Mr. Santin did not hear of this until some six months after the work was done. He was not involved in negotiating the credit note. Mr. Santin did agree that any dispute in 1986 relating to the masonry was cosmetic in nature.

At Tab 49 of Exhibit 1 is the Minute dated November 6, 1985; section 10.1 references a section of the brick wall that had been laid the day before and had been rejected, noting that Abe Dick will rectify. Mr. Santin confirmed that that was the section involving the faulty soldier course. He does not recall whether or not this Minute would have been sent to him at the time. Sometimes such documents were sent to him, sometimes they were not.

It was Mr. Santin's evidence that there was no arrangement between himself or the Abe Dick company and E&M with respect to his coming forward to give evidence, and he was not being paid any money beyond ordinary witness expenses to appear.

At Tab 58 of Exhibit 1 is the Minute dated April 9, 1986, which notes that "Jim Sinclair will check with Mr. Santin on any windows missing the window water drip." Mr. Santin remembered this particular item and this item is different from the item that concerned him about the leaking at the top of the window surround.

Re-examination by Mr. Nelson

With respect to the steel work being out of plumb, Mr. Nelson asked Mr. Santin if he had ever measured the steel. Mr. Santin explained that he checked from floor to floor. He explained that his men were working on the floor below in terms of constructing the wall while the steel and the slab floors were going up above them. With respect to the out of plumb steel, Mr. Santin in response to Mr. Nelson's question, testified that he knew that the steel was out of plumb. He never expected that it would be so out of plumb at the top of the building. He never expected it to be as far out of plumb as it was at the top.

Witness – Peter Harding

Examination-in-Chief

Mr. Peter Harding is the owner of Amherst Roofing and did the new roof for the plaintiffs in 1998. Mr. Mueller was complaining about the fact that subsequently he gave a report to Emmons & Mitchell and handed over the contract documents. In any event, I eventually ruled that Mr. Harding could be called as a fact witness and within reasonable limits could be called upon during his evidence to express opinions as a roofer, not as an expert witness.

The two volumes of materials received from Amherst Roofing, previously marked as Exhibit B, went in as Exhibits 115A and 115B.

As previously noted, Mr. Harding is the owner of Amherst Roofing. It was a company started by his father and Mr. Harding has been personally involved in the company since 1981.

At Exhibit 115, Tab 103 is the letter from Mr. Johnston to Amherst Roofing requesting that the company bid on the installation of a new roof on the subject premises. The letter sets out the work to be done.

At Tab 79 is the letter from Mr. Johnston to Amherst indicating that their bid was provisionally acceptable and inviting Amherst to enter into discussions toward a contract. At Tab 80 is the agenda for the meeting which was scheduled for September 23, 1998. Present at the meeting were Mr. Johnston, Mr. Harding, Messrs. Reed and Lawless, and a Mr. Emmons of Cave's Mechanical (this was Harding's mechanical sub and this Emmons is related to the Mr. Emmons in Emmons & Mitchell).

Mr. Harding noted that at this meeting Dr. Reed expressed concerns with respect to the work being done over his offices and wanted Mr. Harding to understand that his office facility beneath the roof was not unlike a hospital where operations were being carried out. As well, during the tender process Mr. Harding asked Dr. Reed if there were any leaks in the roof. He confirmed that there were and showed him leaks under the vent stack.

At Tab 76 is a letter dated September 24, 1998 from Mr. Johnston to Mr. Harding advising that the plaintiffs are prepared to enter into a contract with Amherst Roofing.

The roofing work started on November 2, 1998 and stretched on longer than had been anticipated. The roof itself was completed in 1998 and the job shut down for a time. The flashing was then done in 1999. During the course of the job, Amherst took instructions from Mr. Johnston and Mr. Kendall and sometimes had communications with Dr. Reed.

From time to time, Mr. Harding would receive inspection reports from Mr. Johnston during the course of the work. At Tab 33 is a site report dated November 2, 1998 from Mr. Johnston wherein Mr. Johnston notes that he had reviewed the exposed built-up membrane and noted a number of observations that he made with respect to same as follows:

a) in the large roof area, the asphalt quantity and coverage at the membrane surface appeared consistent; in some undisturbed areas soiling had occurred due to the essentially flat substrate;

b) in one location the top asphalt coat which had not adhered to the lower insulation panel exhibited three fish mouth fissures of three to four inches in length adjacent to a two-square foot irregular shaped area which was crazed and randomly cracked with intersecting ragged fissures, 3/8 to ½ inch long over the whole area;

c) some smooth edge craters, 3/16 of an inch deep and six to 10 inch in irregular diameter were noted in the exposed membrane asphalt finish, indicative of hollows captured or impressed in the flood coat in the laying of the original lower insulation panels;

d) other local depressions or craters were noted which exhibited crystalline shiny sharp speared edges indicative of part or chunks of the asphalt finish breaking free when the insulation panel is pried up;

e) in a few locations, removal of the lower insulation panel cleanly left slight asphalt ridges of material along perimeter edges, compressed when panels were mated together.

It was Mr. Harding's evidence that he saw basically the same thing that Mr. Johnston saw and referred to in his site report, but did not see the three fish mouth fissures referred to in item b).

At Tab 1 is Mr. Harding's report to Emmons & Mitchell dated January 17, 2000, where in paragraph four thereof he notes that he found the existing membrane to be in good condition and the felt flashings also to be in good condition, though the felt flashing only extended to the outside of the inside masonry. He had a few sections of the membrane pull away from the deck while he removed the insulation that was bonded to it, but he notes in paragraph four that that would be expected.

With respect to the adhering of the insulation to the bitumen, it was Mr. Harding's evidence that the practice in 1985 went both ways, either bonded or not bonded to the bitumen. He noted that the industry was in a state of flux.

Mr. Harding noted that his work involved raising the parapet by the height of one block, but the net effect was not to raise the parapet above the roof level in that extra insulation was laid on the roof.

The cost of the work was $62,000 for the roofing, $32,000 for the mechanical work and masonry, and a $4,500 extra for the mechanical work for upgraded aluminium facer waterproofing.

In his report, Mr. Harding notes that there were some nuisance leaks at roof projections on the old roof but he notes that these were maintenance problems relating to roof fits and pitch pockets.

On the last page of his report he lists a number of items of his work under the contract which could have been done more cheaply.

Cross-examination by Mr. Garbutt

Mr. Harding has served as a director of the Ontario Industrial Roofing Contractors Association (OIRCA). He explained that the purpose of the Association is to keep members abreast with respect to changes in roofing standards. He was a member of OIRCA in 1985 and in fact served as a director from 1985 to 1989. On the issue of insulation, bonded or not bonded to the bitumen, it was his evidence that in 1985 the industry was in a state of transition and the application went both ways.

Cross-examination by Mr. Mueller

Mr. Harding had a foreman on the job, Darcy Timins and a field supervisor, Keith McGinn. Mr. Timins is still with Mr. Harding. Mr. McGinn has retired. Mr. Timins was on the job throughout; Mr. McGinn visited the site from time to time. He had a number of jobs to supervise. Mr. Harding was on the site six or seven times over the duration of the work and when there may have stayed one to one and a half hours.

Prior to 1985 and after he started up his company in 1981, he installed “a lot of” irma roofs.

It was Mr. Harding’s evidence that he did not think that Dow had ever said it was inappropriate to adhere the insulation to the bitumen.

Mr. Harding was referred to Tab 33, Mr. Johnston’s site report for November 2, 1998 and point number three where Mr. Johnston notes that some areas of the original membrane were exposed after removal of the lower insulation layer, which demonstrated embedment in asphalt flood coat as anticipated – “both partial and complete adhesion was noted.” Mr. Johnston also noted in point three that the adhered insulation had to be pried free and in some locations pulled sections of the asphalt and felt with it. Panels sheered and broke irregularly in most locations in removal. Mr. Harding confirmed that he saw some of what Mr. Johnston notes in point three, and conceded that he was not necessarily present when the insulation was being removed, but only saw the roof after it had been stripped of insulation. He was not therefore aware that almost all of the insulation adhered to the bitumen.

On cross-examination, Mr. Harding confirmed with Mr. Mueller that with respect to Tab 33 and Mr. Johnston's observations with respect to the roof, he, Mr. Harding did not see the craters referred to in item c), point four, or d), nor did he note the asphalt ridges pointed out by Mr. Johnston in item e).

In view of his comment with respect to Dow never indicating that adherence of the insulation to the bitumen was not appropriate, Mr. Harding was referred to Exhibit 55, Tab K, the Dow material, where under the heading 'Insulation' it states quite clearly that if a "bituminous membrane is used the final flood coat must be allowed to cool before applying the insulation loose. Some single ply membranes and coal tar pitch require a slip sheet. It is the roofer's responsibility to ensure that bonding of the insulation does not occur." Mr. Harding acknowledged that was what the Dow material stipulated.

Mr. Harding acknowledged that he has never seen the specifications with respect to the original roof. He acknowledged that the sloping of a roof is intended to avoid ponding and to facilitate drainage. He was referred to Exhibit 1, Tab 1, Division 7 relating to the roof, and particularly section 2 which states, for inverted roofs, slope structures to drains at least 5m/100 mm gradient. Mr. Harding was able to confirm that as a roofer he would have interpreted this to mean a variation of 100 millimetres every five metres, and acknowledged that Mr. Johnston wanted a sloped roof in the work that he, Mr. Harding was doing.

At Tab 12 of Exhibit 115A is a directive from Mr. Johnston to all bidders. Item 3 of that directive makes it clear that Mr. Johnston wanted a 2% slope on the roof.

At Tab 29 of Exhibit 115A is Mr. Kendall's letter to Mr. Johnston dated November 13, 1998 wherein Mr. Kendall confirms to Mr. Johnston that he went over the requirements of the design with Mr. Harding's crew, and confirmed that the finished metal would terminate two inches down the face of the brick, and that the modified bituminous sheets would terminate one and a half inches on to the brickwork. Mr. Harding acknowledged that that was part of his job, that is to extend the bituminous sheets fully across the block and brick wythes and down the brick wall one and a half inches. He confirmed as well that what he found prior to his work was that the sheets went across the blocks but not to the bricks.

With respect to Tab 1 of Exhibit 115, his report to E&M of January 17, 2000, where he notes sections of the membrane pulling away when the insulation was removed, Mr. Harding noted that even if one puts the insulation down on cool asphalt or cool bitumen and then puts the ballast on top of the insulation, there will be some degree of bonding between the insulation and the asphalt, but much more of course if the bitumen is hot.

Mr. Harding conceded that if there was serious leaking at the bottom of the parapet and a reverse slope to the parapet, he as a roofer would suggest remedial work involving a drain and proper sloping.

At the top of page two of Tab 1, Mr. Harding notes that at the perimeter edges a masonry block was installed over the existing masonry and all was anchored into the concrete deck. The face brick was anchored to this with a masonry anchor. Rob Santin would be more knowledgeable on this procedure (Rob Santin was his subcontractor on this job and apparently is related to Primo Santin). In any event, this was remedial work performed by Mr. Harding.

Mr. Harding confirmed that the report at Tab 1 of Exhibit 115 arose out of Mr. Mitchell telephoning Mr. Harding and asking him for a report as to what he did on the roof. Mr. Harding sent over the letter at Tab 1 and eventually sent over whatever he had in his files. Before doing so, Mr. Harding did not consider asking the plaintiffs' permission. He acknowledged that Mr. Johnston called him before this trial and they "played telephone tag" but never did speak. He assumed that the nature of Mr. Johnston's comments would have been critical of him. Since the report of January, 2000 Mr. Harding has had revenue from jobs given by E&M to the extent of possibly $100,000. Mr. Harding was not aware of any litigation until he got this job in 1998 and was not aware that E&M was involved in this litigation until he was in the latter part of his job. Nothing was paid by E&M for the report, and Mr. Harding was not looking for any payback.

Re-examination

In seeking the report from Mr. Harding, Mr. Mitchell did not promise any future work, make any threats, or offer to pay any money.

In the two years since the plaintiffs knew that Mr. Harding was going to be called on behalf of Emmons & Mitchell, they had no contact with him.

There is no date on Tab K of Exhibit 55.

Witness – Chris Roney (recalled)

Further cross-examination by Mr. Mueller

Exhibit 95 contains the calculations prepared by Mr. Roney relating to both his April 24, 2002 and his May 14, 2002 report. Mr. Roney explained once again that in both his April 24 and May 14 calculations, he used the 1984 Code to determine the forces that the building would have to resist. For the April 24 report to analyze the effect of the loads on the walls, he used the 1994 Canadian Masonry Standard. For the May 14 calculations to analyze the effect of the loads on the walls, he used M-78 which is referred to in the 1984 Code. So throughout, determining the forces that the building must resist he used the 1984 Code. For the April 24 analysis to determine the effect of those loads he used the 1994 Masonry Standard and for the May 14 analysis with respect to the effect of those loads he used M-78, the 1984 Code.

At page 13 of Exhibit 95, before the various calculations start, there is a notation by Mr. Roney relating to a conference call between 9:30 and 11:30. Mr. Mueller expressed great interest in this conference call and it has already been touched upon in Mr. Mueller's previous cross-examination. Arising out of that cross-examination, Mr. Mueller asked for Mr. Roney's time dockets which he brought with him on this re-cross and we filed as Exhibit 117.

The calculations on page 13 are those that were done for the April 24 report, which noted an overstress at the top of the building of some 39%, later in the May 14 report reduced to 6% using the old Code. Mr. Mueller went through Exhibit 117 with Mr. Roney and while there are some time frames referred to in his records which would encompass the 9:30 to 11:30 conference call both on April 22 and April 23, Mr. Roney could still not say with any certainty that the conference call noted related to this litigation. He was able to confirm that any conversations that he had with respect to this litigation were with either Mr. Nelson or Ms. Gambir, and so that if the notation on page 13 of his calculations relates to this file, then the telephone call would have to have been with one of them.

Witness – James Sinclair

Examination-in-Chief

James Sinclair is 70 years of age. He was trained in Scotland as a carpenter. After coming to Canada in the mid-1950's he worked in heavy construction. He came to Ontario in 1978 and eventually was employed by Emmons & Mitchell in 1984, where he worked until his retirement. He was the construction supervisor on the 275 Queen Street project. He initially thought that the building would consist of a basement and two levels but eventually it was changed to a basement and four levels with provision for the addition of two more levels some time in the future.

Don Emmons was the partner in charge of this particular job and as construction supervisor Mr. Sinclair reported to Mr. Emmons, to Mr. Vinkle and to Mr. Garbutt. He appreciated that Mr. Garbutt was a civil engineer and that there was no architect as such on the job which fact surprised Mr. Sinclair as it was the first project on which he had worked without an architect. Mr. Sinclair was on the job site from June of 1985 to May of 1986. He described it as a long drawn out job because it was being done during the winter.

E&M had a trailer on site. Mr. Sinclair testified that he did not have the contract drawings in the trailer but he did have the specifications and some "sparse" drawings.

Mr. Sinclair was on the site every day except on the weekends. He was not responsible for any other jobs during this project. On this job, as was his practice, he kept a complete and detailed diary which he threw out when he retired 10 years ago.

With respect to the attendances of the owners, Mr. Sinclair observed that Dr. Lawless came every day to see the project as his office was located but a short distance away. Dr. Reed attended perhaps a couple of times a week. They tended to ask a lot of questions. Mr. Sinclair characterized his working relationship with them as very stressed. They had a little knowledge with respect to construction but did not have full knowledge of the industry. They would say they wanted this and that, and at times Mr. Sinclair would write their requests down and take them to Mr. Emmons, Mr. Vinkle or Mr. Garbutt. He had a good working relationship with Mr. Garbutt.

Mr. Sinclair started his evidence by referring to the bricks and referred to them often during the course of his testimony. He complained about the use of John Price bricks on this project. He said they were not like the usual module bricks of today. There were about five or six forms used to make them and as a result each brick could be slightly different and they tended to be very porous.

Mr. Sinclair noted that the digging of the foundation commenced without there being a soil report, which apparently Dr. Lawless said was not needed.

Referring to the steel, Mr. Sinclair testified that it was erected plumb and in fact while it was being erected by Christmas Steel, the steel contractor, he, Mr. Sinclair was the individual on the transit making sure that it was plumb. He kept his eye on it and required adjustments to be made from time to time. He explained that the steel went up in two stages, firstly the first and second floors were erected and erected plumb. The decks were then put on these first two floors and the cement poured, then the next two floors of steel were erected with guide wires from the first two floors.

Referring to the shelf angle on the beams, Mr. Sinclair explained that when the brick came up to the brackets on the east and west sides of the building, there was a problem with coursing. The brick course would not fit under the plates. There was no such problem on the north and south sides where the brick coursing came to the underside of the plate. The beams on the east and west sides had been placed higher in order to accommodate the height of the steel deck which was sitting on floor joists which in turn were sitting on four inch shoes. It was Mr. Sinclair's evidence that with respect to this problem, he spoke to Mr. Garbutt who came over and gave permission to him to have Christmas Steel cut the shelf angles off the beams on the east and west sides. He is not sure that there were any shelf plates on the north end of the east side where the stairwell was located, but if they were there they too were cut off.

It was Mr. Sinclair's evidence that he would never have touched the shelf plates without Mr. Garbutt's permission and that he has no qualifications to work with the steel.

With respect to the bricks, Mr. Sinclair testified that they were selected by the owners, Drs. Lawless and Reed. Again he noted that the John Price bricks were difficult to work with because they came in different shapes. He recalls going over to the A&P and the plaintiffs showing him there the type of soldier courses that they wanted. A&P had used a standard brick and he suggested that they use a standard brick but they wanted the John Price brick. He explained to them the problem about the John Price brick. As well, the plaintiffs wanted a cove joint between the bricks, that is a half round joint. Mr. Sinclair recommended against a cove joint because the John Price brick was very porous and the water could lay in the joint. He recommended instead a flatter joint but they chose the cove joint. In his evidence he explained as well that the standard brick has three holes through its height, which makes for a "nice solid mass of concrete". The John Price brick however simply has a 'frog' in the bottom side of it, that is an indentation and in Mr. Sinclair's view that did not make as good a joint. Notwithstanding Mr. Sinclair's comments, their minds were made up and they chose the John Price brick.

With respect to the control joints, Mr. Sinclair noted that Primo Santin wanted a control joint at each end of the south wall, but the plaintiffs did not like the appearance of those joints and did not want them there. As a result, a single control joint down the centre of the south wall was settled upon.

It was Mr. Sinclair's testimony that what was supposed to be constructed was a solid wall and that it was a solid wall with a finger width of void between the brick wythe and the block wythe resulting from the fact that the brick layers had to hold on to the top of the brick to lay it.

Getting back to the control joints, Mr. Sinclair explained that they were needed because the masonry wall absorbs the heat of the sun at different rates than the underlying steel. The steel expands more than the masonry and can therefore fracture the masonry and cause cracks. He and Primo talked to the plaintiffs and eventually they settled on a control joint down the middle of the south wall.

It was at this point in Mr. Sinclair's evidence that Mr. Mueller rose to complain about the fact that Mr. Sinclair had given evidence with respect to the cutting of the shelf plates and the fact that he had sought and obtained Mr. Garbutt's authority to do so. At his examination for discovery on July 8, 1999 Mr. Mitchell in referring to the shelf plates and their being cut off gave the following answers:

> Q 331: And they were removed and nobody, be it Emmons or Mitchell or Abe Dick, went to the owners or the engineers to ask for permission to remove that?
> A. It's not in writing no.
> Q 332: Was it done orally?
> A. I can't confirm that.
> Q 333: If you do later take the position, I want to know exactly what was said, in other words, the consequences of it. You know, you go to the doctor and he says I am going to give you this pill, there's side effects, maybe, so I mean the consequences, namely that there be no support now from the angle iron, as I call it, on the east and west side and indeed this is now going to be what, five to 10 feet higher than the Building Code permitted.
> A. Correct.
> Q… without support. Do you understand?
> A. Correct. Yes.
>
> "U" Mr. Minnema (then counsel for defendant Emmons & Mitchell): If we get any information on that we will report it to you.
> "U" Mr. O'Donnell (then the plaintiffs' counsel): That's right but if there was some discussion you are alleging I want to know the particulars of it including the consequences.
> Mr. Minnema: We would give you the particulars of it regardless of whether or not the consequences were discussed I guess. Is that not what you are asking?
> Mr. O'Donnell: What I am saying is, I want the whole conversation as to what – if you are going to lead evidence as trial, I want the whole thing.
> Mr. Minnema: Most definitely.

In the face of those answers and undertakings located beginning at page 50 of Mr. Mitchell's examination for discovery, Ms. Gambir, Mr. Nelson's junior, on May 9, 2002 a few days before the start of this trial, wrote to Mr. Mueller and said as follows:

> Further to telephone discussions between our offices, please find the answers to the undertakings given on the examination for discovery of our client Mr. Mitchell.
> Undertaking #2: it is our position that we were instructed by Mr. Don Garbutt to cut the shelf angles.

This letter was filed as Exhibit D on Mr. Mueller's motion. In addition, during the cross-examination of Dr. Reed, apparently, although I have no actual recollection of it now, Mr. Nelson indicated to Dr. Reed that Mr. Sinclair would testify that he had obtained permission from Mr. Garbutt to cut the shelf plates. Mr. Nelson was to in due course bring this particular item of cross-examination to my attention which I assume he will do at some point in time.

Rule 31.09(1) provides that where a party has been examined for discovery and the party subsequently discovers that the answer to a question on the examination is incomplete, then the party shall forthwith provide the information in writing to every other party. Rule 31.09(3) provides that where a party has failed to comply with subrule (1) and the information subsequently discovered is favourable to the party's case, the party may not introduce the information at the trial except with leave of the trial judge.

In the final analysis I reserved on Mr. Mueller's motion and invited counsel to make their submissions on the admissibility of Mr. Sinclair's evidence with respect to the cutting of the plates during the course of their written submissions. It was agreed that Mr. Mueller and Mr. Garbutt could cross-examine Mr. Sinclair on the point without waiving their rights.

With respect to the windows, Mr. Sinclair testified that he had a conversation with Mr. Santin as the first floor went up. Mr. Sinclair initially understood that there would be a lintel across the top of the window opening between the bricks and that the window would go to the lintel. Then the stone surrounds came up. Mr. Santin expressed the thought that the stone surrounds would not work at the top of the window. The stone surround extended out past the surface of the brick to create a ledge and the lintel was perhaps a ¼ inch above the top of the stone surround which provided an entry point for rain which could collect on the ledge if it was not caulked properly. Mr. Sinclair enquired from Mr. Garbutt where the stone surrounds came from and Mr. Garbutt advised him that the owners wanted them. The stone surrounds were in fact installed and the opening caulked but Mr. Sinclair expressed concern about the caulking eventually perishing.

With respect to the ties, it was Mr. Sinclair's evidence that the strap ties were bent from the block down to the brick. In his view the only problem with respect to ties was that the roof level where for a height of about 18 inches around the building there were no ties.

It was Mr. Sinclair's evidence that in addition to the strap ties, ladder ties were inserted into the block joints every three courses. These ladder ties sat on top of the block and another block sat on top of the ladder tie. [This left the ladder tie completely enclosed in the block wythe, contrary to actual observations made in the test holes.]

The project was eventually completed in July or August of 1986.

That completed Mr. Sinclair's examination-in-chief, save for his comments during the showing once again of Dr. Lawless's video, Exhibit 7, taken on the construction project during 1985 and 1986. Very little of any significance came out of the viewing of the video. The following comments by Mr. Sinclair as he watched the video should be noted:

1) The video shows how the beams on the east and west wall had been raised up to accommodate the joist shoes of four inches. Mr. Sinclair noted that this was a "drafting error", but this comment was never followed up on during the course of cross.
2) Mr. Sinclair noted in the video where the initial soldier course was not acceptable. He noted that the original soldier course was out by some 3/8 inch and the next course up as well was 3/8 inch out. He noted "that's what you get with John Price brick." He noted as well that 12 to 14 feet of soldier course had to be taken out because it was not properly installed. He noted that the mason who had installed it had never worked with John Price brick before. After the original coursing was taken out and replaced there were no more complaints.
3) During the course of the video, Mr. Sinclair again demonstrated how the brick is laid, holding it at the top with a finger over the side, thus requiring a finger distance between

the brick wythe and the block wythe. In his view the mortar that extruded out from the bottom of the brick at the joint made it a solid wall. It should be noted that on the video one can see the void between the brick wythe and the block wythe.

4) At one point the video shows strap ties bent down from the block to the brick. At the point video taped, there appear to be almost two ties to a block. The video also shows that the block abuts the flange of the upright column but does not go into the flange. The video also shows that it appears that the bricks are laid from the outside and the blocks from the inside of the building.

Cross-examination by Mr. Mueller

Mr. Sinclair conceded that he had never read through the General Conditions which form a part of the Contract at Exhibit 1, Tab 1, and that he was not generally familiar with the General Conditions as published by the Canadian Construction Documents Committee (CCDC). He was not aware that the definition of 'consultant' under the General Conditions included both an architect or engineer. He did however understand and appreciate during the course of this work that Mr. Garbutt was taking the place of the architect and performing the same role as an architect. He understood the nature of a change order and new that Mr. Garbutt was issuing change orders.

> Mr. Sinclair acknowledged that his "Bible" consisted of the drawings and specifications. When Mr. Mueller asked him what drawings he had, Mr. Sinclair indicated he had drawings for the basement, the first floor, the second floor and the third floor as they pertained to the main part of the building, but with respect to the stairway he had drawings for the first and second floors only.

Mr. Sinclair was very specific that at no time did he have the three small drawings which are at Exhibit 56, Tab C.

Mr. Mueller referred Mr. Sinclair to the Contract and particularly Article A-2 at Exhibit 1, Tab 2 which lists the contract documents. Although Mr. Sinclair testified that he had gone over that list, he had never enquired with respect to the "details" that are referred to in the first part of the list, “Specifications and Details, dated June 27, 1985 as prepared by N.D. Garbutt, PEng.” Mr. Sinclair never asked what those details were.

With respect to the specifications at Tab 1 of Exhibit 1, which are dated June 27, 1985, Mr. Sinclair acknowledged that he had a set of the specifications in his trailer. He never did read through them but if a problem arose he looked at the specifications with respect to that problem. He never did compare the specifications to the plans unless with respect to a particular problem. He appreciated that the specifications over-ruled the drawings. Mr. Sinclair testified that in general he erected the building in accordance with the drawings and that if there was any problem that he ran into then he went to the engineer. He seemed not to be aware of General Condition 25.5 which required the contractor to review the contract documents and report to the consultant any error, inconsistency or omission he may discover.

With respect to his diaries, Mr. Sinclair explained that no one ever asked him for them prior to his throwing them out. He never asked Emmons & Mitchell if the company wanted them before he threw them out in 1992. By 1992 he was not aware that the plaintiffs and Emmons & Mitchell were talking about the problems existing in the building. He was first asked for the diaries within the last couple of years and was only contacted with respect to problems with the building within the last two years. He conceded that after 15 years it was rather hard to bring the job back. He had had many jobs over the years, however most jobs were different and he does not think that he is mixed up about this one.

With respect to the steel being out of plumb, Mr. Mueller brought to Mr. Sinclair’s attention the fact that Mr. Santin testified that at the top and at other places there was quite a gap between the block and the brick and that he blamed this gap on the steel being out of plumb. Mr. Mueller brought to Mr. Sinclair’s attention Mr. Santin’s evidence that he had discussed the problem with Mr. Sinclair and that Mr. Sinclair had said that there was nothing that could be done about it at that point. Mr. Sinclair did not agree that he and Mr. Santin had ever had such a discussion, and noted that if Mr. Santin had noticed that the steel was out of plumb he could have come to him and Mr. Sinclair could have had it fixed after getting hold of Mr. Garbutt. In Mr. Sinclair’s view, the block could have been moved to accommodate any lack of plumbness in the steel. In any event, Mr. Sinclair has no recollection of ever seeing a gap between the two wythes of one and a half inches at the top of the building. However he has not actual memory of ever looking down between the two wythes to see how far they were apart.

Mr. Mueller brought to Mr. Sinclair's attention that Mr. Roney had in his investigations noted voids between the brick wythe and the block wythe from ¾ inch to 1 ½ inches in width. Mr. Mueller enquired of Mr. Sinclair as to whether he had made the same observations. Mr. Sinclair noted that one finger's width is ¾ inch and if that is doubled up it would approximate 1 ½ inches. He agreed generally that there were measurements of that sort, perhaps ¾ inch to one inch between the wythes, but this was of no concern to him because of the ties that were being installed. He felt that the width of the gap was within tolerance. Mr. Mueller referred Mr. Sinclair to the specifications Division 4, section 2.1.1 which called for maintaining a continuous 19 millimetre air space behind the brick throughout the project. Mr. Sinclair acknowledged that aside from the mortar fins he knew that the opening was extending from one to two fingers in width. He never did discuss with Mr. Garbutt what Mr. Garbutt wanted.

Mr. Sinclair acknowledged having built cavity walls, and he acknowledged that the void between the wythes in a cavity wall is some 50 to 75 millimetres in width. Mr. Sinclair acknowledged having built veneer walls with brick in front of block but noted that they were different from this particular wall. It was his evidence that here he did not think that he was building a veneer wall. He understood a veneer wall to consist of total masonry, consisting of a block wall and a brick wall set out one to two inches from the block wall and tied in to the block wall with adjustable ties, the purpose of the adjustable ties being to accommodate a difference in coursing.

With respect to this particular job, Mr. Sinclair testified that adjustable ties were never mentioned. Mr. Garbutt saw the type of ties that were going into the wall and said nothing. In particular they discussed the type of tie being used right at the bottom of the wall, and there was no mention of any other type of tie. The tie being installed there was a strap tie. The only discussion he ever had with Mr. Santin about ties was when he told Mr. Santin to be sure to get lots of strap ties installed. Mr. Mueller brought to Mr. Sinclair's attention Mr. Santin's evidence that Mr. Sinclair told him to use ladder ties between the two courses when they corresponded, and where they did not correspond to use strap ties. Mr. Sinclair had no recollection of such a conversation and testified that the conversation could not have taken place because if it had, he would have gone to Mr. Garbutt to discuss it which he did not. At no time did he ever see ladder ties extending from the block to the brick wythe.

Mr. Mueller pointed out to Mr. Sinclair section 2.2.6 of Division 4, Masonry, which refers to truss type ties, sized 20 millimetres to 30 millimetres narrower than wall or partition, which he suggested was consistent with what Mr. Sinclair was saying about inserting the ladder ties completely into the block wythe, and Mr. Sinclair agreed.

Mr. Sinclair acknowledged that he knew the concept of a cavity wall as opposed to a veneer wall, and that he realized that a gap between the wythes was common to both types of construction. Simply put, he understood a cavity wall to be the same as a veneer wall but that the cavity wall had a wider gap between the wythes. He acknowledged as well that one had to use adjustable ties for both cavity wall and a veneer wall. Mr. Mueller asked Mr. Sinclair how he thought that he was building a solid wall when there could be up to two fingers' width of a gap between the two wythes. Mr. Sinclair explained that during the course of the laying of the brick wythe, the mortar fins extended from the brick to the block; although that did not produce a totally solid wall, it did constitute a solid wall as he always knew it. He acknowledged that he has been involved in the construction of a solid wall that was actually filled with mortar. He explained how in such a wall the block wall is parged and the brick and mortar comes right tight to it.

Mr. Sinclair acknowledged that in our case it never occurred to him to actually fill the void between the block and brick wythes, and he never knew that Mr. Garbutt intended the collar joint to be filled.

With respect to the shelf plates, Mr. Sinclair explained that he arranged with the steel contractor, Christmas Steel to cut the plates off the beams on the east and west walls. He first spoke to Don Garbutt about it. The plates were cut while Christmas Steel was on the job. He seems to recall the plates being cut on the east side at the first floor level to begin with. He is not aware whether the west side plates were cut at the same time. He explained that the wall did not go up in separate panels from bottom to top, rather the walls went up floor by floor on all four sides. Again he explained that Christmas Steel cut the plates while they were on the job site and were never called back on a special occasion to cut. Mr. Sinclair was unable to say who paid for the cutting off of the shelf plates. He knows that Christmas Steel never came to him and asked him to sign off. It was not a great deal of work and he suggests that it could have been a "freebie" as was his plumbing of the steel for Christmas Steel.

With respect to change orders, Mr. Sinclair appreciated that the formal way that they were achieved was by way of a revision notice, a change notice and a purchase order, all in writing. He explained however that he and Mr. Garbutt had a good working relationship and sometimes did not follow the formal procedures. Although not specifically familiar with the General Conditions, he was generally aware that changes in the work had to be processed in accordance with the provisions of G.C. 11 and 12.

With respect to the cutting of the shelf plates, Mr. Sinclair appreciated generally that the purpose of paper is to prevent disputes but often work is proceeded with on the basis of a gentleman's agreement. With respect to the plates, he knew he had no paper but they were being pressed by the owners to complete and they wanted to take advantage of a couple of warm days. He assumed that Mr. Garbutt would take care of the paper work. He noted as well that things were being changed so much that they were behind in their paper work, although he conceded that the number of changes in this job were not unusual for a job of this type.

Again with respect to the cutting of the plates, Mr. Sinclair conceded that if there is a point in issue, then one goes to the specifications. Then Mr. Mueller presented him with Division 5, section 1.10 of the specifications which provides "obtain written permission of engineer prior to field cutting or altering of structural members." Mr. Sinclair conceded that this specification covers the point.

Still with respect to the plates, Mr. Sinclair acknowledged that he was aware that the plates were required for compliance with the Building Code. He understood as well that Mr. Garbutt would have filed with the Municipality drawings showing the plates in place. Mr. Mueller asked Mr. Sinclair how he thought that Mr. Garbutt would permit the cutting off of the plates if they were required by the Code, and he really had no answer to that question, other than to say that one does not play with structural steel. He went on to assure Mr. Mueller on a stack of Bibles that he spoke to Mr. Garbutt and that he would not have touched the plates without Mr. Garbutt's permission. It was his evidence that the discussion with Mr. Garbutt took place in the field while they were looking at the steel. The discussion took place when, as the walls came up, he saw that the beam was in the wrong place.

It was his evidence that the first time he heard any discussion of the plates was when he was at the back of the courtroom at the beginning of this trial. He recalls Dr. Lawless testifying that he took a piece of the plate.

Mr. Sinclair acknowledged that it was the responsibility of the steel fabricator to prepare shop drawings for the steel and that in the usual course the steel fabricator designs from instructions received from the structural engineer. In this respect, section 1.3 of Division 5 provides, "submit shop drawings, clearly indicate framing plans, bearing and anchorage details, framed openings, accessories, schedule of materials, camber and loadings, welded and fasteners." Mr. Sinclair explained that in the ordinary course the steel fabricator does his shop drawings. They go to the general contractor for review and the general contractor sends on to the engineer for approval. Mr. Sinclair acknowledged that General Condition 34.3 provides that prior to submission to the consultant, the contractor shall review all shop drawings.

Still with the plates, Mr. Sinclair acknowledged that the problem might have been easy to fix in the field but no alternatives were discussed for keeping the plates.

With respect to the control joints and Mr. Sinclair's evidence to the effect that Mr. Santin wanted control joints at the east and west ends of the south face, Mr. Mueller brought to Mr. Sinclair's attention that Dr. Reed in his evidence testified that he was not aware of the control joint issue until the single joint had already been put in place on the south face. Mr. Sinclair said that could not have been the case in that Mr. Santin started the control joints at both corners right from the beginning. They were up a height of approximately three feet when Dr. Reed saw them and said he did not want them there. This was long before the work was being covered with a tarp. As a result of Dr. Reed's request they were taken out, although Primo was not happy with it.

Mr. Sinclair acknowledged that Division 4 section 2.2.2 requires that control joints be provided as dictated by structural design, and that means that one must go to the engineer. But in this case, testified Mr. Sinclair the owners dictated and he did not go to Mr. Garbutt on the point.

With respect to the windows and more particularly the stone around at the top, Mr. Mueller brought to Mr. Sinclair's attention Mr. Santin's evidence that he told Mr. Sinclair that he guaranteed the building would leak forever. In response, Mr. Sinclair acknowledged that both he and Mr. Santin were very concerned but he does not recall the details of the discussion. In any event, it was Mr. Sinclair's evidence here that he spoke to Mr. Garbutt who told him that the owners wanted the stone surround.

There then ensued a discussion between Mr. Mueller and Mr. Sinclair as to whether or not the drawings show the stone surrounds projecting out from the surface of the brick. Mr. Sinclair maintained that initially he thought the stones were to be flush with the brick and then the design was changed to cause them to project out from the surface of the brick, thus producing an overhang where water might collect and enter the building. Suffice it to say, from a review of Exhibit 8, the pre-contract drawings and Exhibit 9, the contract drawings, they do show the stone surround projecting out from the face of the brick, although some drawings in both sets do not show the projection because they were not intended to address the projection.

With respect to the issue as to who ordered the stone surrounds which was discussed during the course of Mr. Santin's evidence, Mr. Sinclair knew that there had been discussions between Emmons & Mitchell and Abe Dick as to who would supply the stone surrounds, but he does not know what the result of the discussion was.

With respect to the ties, Mr. Mueller brought to Mr. Sinclair's attention Mr. Santin's evidence that after discussions with Mr. Sinclair, Mr. Santin at alternate courses bent down the ties from block to brick and then bent down from brick to block. Mr. Sinclair denied the discussion. He did acknowledge that he did not discuss with Mr. Garbutt how to install the ties. Mr. Sinclair knew that Mr. Garbutt wanted strap ties. Mr. Garbutt saw them at the bottom of the wall and never discussed them again.

With respect to Mr. Sinclair's evidence that the block did not go into the flange of the beam, Mr. Mueller brought to Mr. Sinclair's attention Mr. Roney's evidence that in test hole number two, the beam was infilled between the flanges (see page 10 of Exhibit 87), and that on the east wall a similar observation was made. Mr. Sinclair had no memory of this being the case.

Mr. Mueller pointed to Mr. Sinclair's evidence in chief to the effect that for 18 inches from the top of the spandral beam at the roof level of the wall, there were no ties. Mr. Mueller brought to Mr. Sinclair's attention that in Exhibit 87, test hole number two, there were no ties for some 53 inches below the beam. That was not in accord with Mr. Sinclair's recollection.

With respect to the joints between the courses of brick where Mr. Sinclair says the owners wanted a cove joint, Mr. Sinclair conceded that he did not discuss with Mr. Garbutt the fact that the masons were applying a cove joint, notwithstanding that he did not approve of it.

Notwithstanding Mr. Santin's evidence that he had one drawing only, it was Mr. Sinclair's evidence that Abe Dick's foreman, Ron McCoy had access to all of the drawings in the E&M trailer, and did in fact look at them from time to time. Mr. Sinclair was not aware whether Mr. McCoy actually had his own full set of plans and specifications.

With respect to the roof, Mr. Sinclair acknowledged that the specifications called for a slope. He is also aware that the roof itself was kept level because it would eventually constitute the floor of the fifth floor to be added. He conceded in the circumstances the intelligent thing to do would have been to build a slope into the roofing materials themselves. For the most part he left the roof to the roofer.

He did however see the roof going on and he was strong in his evidence that the insulation was not put onto hot bitumen. The roof went on in the cold weather. The roof was sealed and it was not until the next day that the insulation was put on. He is not aware of any change order in existence with respect to the slope.

With respect to the roof specifications at Division 7, Mr. Sinclair conceded that he was not familiar with those specifications in that he pretty much left the roofing to the roofing contractor. Mr. Mueller persisted and showed him Exhibit 55, Tab H, photographs 3 and 5 which show the gap between the block and the brick at the top of the parapet, and Mr. Sinclair saw nothing unusual about this in that the flashing, if installed properly, completely covers the opening.

With respect to weepers and flashings, Mr. Sinclair was not aware that the specifications called for weepers and flashings. It was his evidence that a solid wall does not have weepers. With respect to weepers and flashings, he never looked at the specifications. Initially, his evidence here was that if he had looked and if he had seen that the specifications called for weepers and flashings, then he would have had them installed. He then changed his evidence, and testified that if he had seen the specifications calling for weepers and flashers then he would have checked with Mr. Garbutt. In any event, he never did look at the specifications in this regard and never did check with Mr. Garbutt.

With respect to Mr. Garbutt's attendances at the site, Mr. Sinclair recalls Mr. Garbutt coming once or twice a week. He recalls one particular period when he had difficulty getting in touch with Mr. Garbutt, and Mr. Garbutt was not attending the site. He was phoning Mr. Garbutt and

Mr. Garbutt was not returning his calls. He can't remember how long this period of absenteeism was but recalls it occurred some time around the middle of the construction project. It was more than a day or two but he can't remember how long it went on. He remembers that during the period he had a problem with the stairs from the second to the third and fourth floors and that he needed Mr. Garbutt's design input, and that was one of the things that he was calling him about.

Otherwise Mr. Sinclair testified that Mr. Garbutt was very attentive when he made his site visits.

Mr. Mueller then discussed with Mr. Sinclair his knowledge of the joint or lack thereof between the top of the block wall and the bottom of the spandral beam. Mr. Sinclair could not recall what had been done there and could not remember whether it was empty or filled. Mr. Mueller brought to Mr. Sinclair's attention Mr. Santin's evidence that Mr. Sinclair instructed him to fill the joint with mortar. Mr. Sinclair responded that he would not have given that instruction, and that common practice was to leave that space open.

Mr. Mueller asked Mr. Sinclair whether any clip angles had been installed between the top of the block wall and the underside of the beam, and he could not remember. Mr. Sinclair did comment however that usually clip angles extended from the vertical column to the sides of the block wall, and he can't recall generally having seen clips at the top of the wall.

Mr. Mueller discussed with Mr. Sinclair the joint between the top of the brick course and the underside of the shelf plate at each of the floors on the north and south sides. Mr. Sinclair's first thought was that that was usually where the copper flashing went. Mr. Mueller then referred Mr. Sinclair to the specifications, Division 4, section 2.2.1 which says, "provide 6 millimetre continuous expansion joint caulked continuously between steel support angles and brick panels each floor." Once Mr. Sinclair understood that this meant between the top of the brick and the underside of the shelf plate, Mr. Mueller disclosed to him that Mr. Santin testified that Mr. Sinclair had told Mr. Santin to fill that joint with mortar. Mr. Sinclair denied that to be the case. It was his evidence that there was no way that one would fill that joint with mortar because of the movement of the shelf plate. Again he referred to the heat factors with the steel expanding at a different rate than the bricks. It was his evidence that he would never have told Mr. Santin to fill this joint with mortar. In that case, the expansion of the steel would bring the brick with it. He never noticed on the site that those joints were filled with mortar. He never discussed it with Mr. Garbutt.

Getting back to flashings and weepers, Mr. Sinclair again confirmed that he never read the specifications. He knew that there were no weepers and he does not have a distinct memory about flashings. He thought that perhaps he had seen some flashings. He conceded that his memory might be a little eroded.

With respect to caulking, although Mr. Sinclair was not familiar with the actual specification, Division 7, section 6.1, he did understand that all exterior joints between different materials would need to be caulked as required by the specification. It would have to be caulking between the metal window frame and the stone surround and also between the stone surround and the brick. With respect to Tabs 39 and 41, invoices from Emmons & Mitchell which purport to charge $1,500 for the caulking of the windows, it was Mr. Sinclair's evidence that this would refer to the caulking between the window frames and the stone surrounds, and that the glass contractor, Fort Glass would have been responsible for the caulking from the frame to the stone surround. In his view, it would have been the responsibility of E&M to caulk the joint between the stone surround and the brick wall.

With respect to Exhibit 2, Tab 64, Mr. Garbutt's letter to Mr. Emmons dated May 10, 1993 raising questions about the collar, the brick and the ties, and Mr. Emmons's response thereto, although Mr. Sinclair testified that he would have disagreed with Mr. Garbutt's comment that "there is no such thing as a ½ inch air space", it was his evidence that at no time did he discuss Mr. Garbutt's enquiries with Mr. Emmons, and he was not the source of the information contained in Mr. Emmons's note. With respect to Mr. Garbutt's comment in the letter of May 10 that, "if the collar joint has not been filled, and the wall subsequently not constructed as a cavity or veneer wall (flashing, weep holes etc.), serious problems result," Mr. Sinclair acknowledged that this comment was at complete variance with what he thought he was building. As far as the answers which are noted in Mr. Emmons's note at Tab 65 relating to the cavity and the type of bricks and the type of ties, Mr. Sinclair conceded that he did not think there was anyone other than he, Mr. Garbutt or Mr. Santin who could have provided the answers.

With respect to as-built drawings, it was Mr. Sinclair's evidence that he kept a set on a special rack in the trailer, and that he transferred all changes to his set of as-built drawings. At the conclusion of the project he turned over the as-built drawings to the Emmons & Mitchell office. He considered the as-built drawings and his diaries as part of his Bible. The cutting off of the plates on the east and west walls would have been shown on the as-built drawings. Mr. Mueller brought to Mr. Sinclair's attention Mr. Garbutt's evidence that he requested as-built drawings but never received them, and Mr. Sinclair could offer no reason why he would not have received them. The practice was that he, Mr. Sinclair would turn over his as-built drawings as produced to the Emmons & Mitchell office, and in due course the owners would either get the original as-built drawings or copies thereof. He has no explanation as to why the owners never got as-built drawings.

With respect to the mock-up that is required by Division 4, section 9 of the specifications relating to masonry, Mr. Sinclair testified that he was generally aware of the requirement for a mock-up. On this job he had been told by Mr. Emmons that there had been a separate mock-up wall built, perhaps at Primo Santin's yard, that the parties had looked at the mock-up wall and that was it. It had been accepted. Mr. Mueller brought to Mr. Sinclair's attention Mr. Garbutt's evidence that the mock-up wall was the 12 to 15 foot length of real wall that was first constructed, and Mr. Sinclair again explained that he had been told by Mr. Emmons that there was an actual mock-up wall built, distinct from the 12 to 15 feet of real wall that first went up.

Cross-examination by Mr. Garbutt

Mr. Sinclair acknowledged that he never got copies of the sub-contracts, and that they had been kept in the Emmons & Mitchell office. Mr. Garbutt asked Mr. Sinclair how he could possibly do the work without the sub-contracts, and Mr. Sinclair explained that the paper work on this project was very thin.

Mr. Sinclair recalled that he had the steel drawings. He recalled that Christmas Steel brought the steel onto the site and starting putting it up. With respect to the brick, he is unaware of the name in which it was delivered, as is the case with the surrounds. Again, he went even further and said that the paper work on the job was a disgrace.

With respect to the stone surrounds, although he knew that the masonry contract, Exhibit 107 included stone surrounds as part of the contract, he did not know the configuration of the stone surrounds until they came on site. He agreed that usually the masonry contractor makes a drawing of the surrounds, but he never saw such drawings in this case. He believes Mr. Santin would have seen the drawings.

Not only did his as-built drawings go back to the E&M office, but also all the shop drawings. They were all put into the same box for delivery to the E&M office.

Mr. Sinclair agreed with Mr. Garbutt's comment that Mr. Sinclair had not been given a full deck of cards to deal with. In this respect, Mr. Sinclair admitted that he was so sick of it at one time that he wanted to quit, but he had never quit a job before. He described it as a "haphazard way of doing business", and that a lot of things happened right from the beginning. He testified that everything was very difficult in the field.

Re-examination by Mr. Nelson

Mr. Sinclair confirmed once again that he prepared as-built drawings as the job went along, that he turned those as-built drawings over to E&M, and that it was E&M's practice to turn them over to the architect or engineer who in turn would turn over the originals or a copy to the owners.

With respect to the caulking, again Mr. Sinclair confirmed that the window frame to stone surround caulking was the responsibility of the window contractor, and that the caulking from the stone surround to the masonry would constitute “different materials” under the specifications, and would be the general contractor’s responsibility.

Witness – Richard Quirouette

Examination-in-Chief by Mr. Nelson

After examination-in-chief by Mr. Nelson and cross-examination by Mr. Mueller with respect to Mr. Quirouette's qualifications, I found the witness qualified to give the court expert opinion evidence in the field of building envelope technology. Although not a licensed architect, Mr. Quirouette graduated from Carleton University with his Bachelor of Architecture in 1973. He thereafter worked from 1973 to 1987 with the National Research Council in the field of building envelopes. In 1987 he joined a private consulting firm and in 1993 started his own consulting firm. Over a period of some 29 years he has conducted over 250 cases relating to envelope failure, 90-95% of which have been related to moisture problems and 75% having been related to masonry construction. He has investigated many buildings constructed in the 1980's. About one-third of his work at the present time is for lawyers. He has given evidence in court on two previous occasions. He does not do the ordinary work of an architect, e.g. design of building, preparation of contract documents, on-going site inspections. He describes himself more or less as a 'trouble shooter'. Mr. Quirouette's c.v. was filed as Exhibit 119.

On behalf of Mr. Nelson, Mr. Quirouette prepared two reports, one dated April 29, 2002 (Exhibit 120) and a supplementary report dated May 10, 2002 (Exhibit 121). After delivering his first report, Mr. Quirouette received a copy of Dr. Reed's video which he reviewed and then issued his supplementary report.

Article 2.3 of Exhibit 120 refers to Mr. Quirouette's first site visit to the building on February 5, 2002 during which he walked around the building's perimeter and took photographs. At that time there was very little evidence that he could observe with respect to the alleged moisture problems with the exception of some light efflorescence. As well, photograph number 4 attached to his report shows staining underneath a course of brickwork that extends out from the vertical surfaces of the brick walls.

Mr. Quirouette attended at the site on a second occasion on March 16, 2002 and assisted Mr. Roney with respect to his test openings numbers 1 and 2. He and Mr. Roney had permission to make 4 test openings and to use the endoscope through drilled openings. Two of the openings were made on the 16th when Mr. Quirouette was on site. Because of weather conditions the last two test openings had to be postponed and Mr. Quirouette was not on site at the time nor did he observe first hand test openings 3 and 4.

Photographs 7 and 8 are photo's of test opening number 1 at the southwest corner of the building at the first floor level. Mr. Quirouette was present at this opening and noted that it took a long time because the brick and mortar were of good quality and in good condition. Mr. Quirouette noted that the east and west walls were constructed from the foundation to the parapets without intermediate supports, while the brick cladding of the south and north walls were constructed on shelf plates at each floor level. At this test hole, Mr. Quirouette believes that he saw ladder ties as he looked down between the wythes toward ground level, but did not observe any when looking up toward the second floor level. Photograph 8 shows the location of ties below the first floor level as seen through the scope. Photograph 9 shows a strap tie in test opening number 1. The tie appeared to be in good condition. Photograph 10 shows the shelf plate in test opening number 1. The plate showed no signs of water accumulation nor was there any indication of water accumulation anywhere in test hole number 1. Mr. Quirouette did observe, as noted in his report, that certain head and bed joints in the brick work of the corners near the top of the building were found to be cracked.

Photographs 13 and 14 attached to his report show the test opening number 2 on the north wall at the fourth floor level. The opening was to the left of the middle window on the fourth floor as shown circled in photograph 4. Again in test hole number 2 the brickwork and mortar were found to be solid and of good quality, however, Mr. Quirouette observed that the mortar was missing brick ties from the shelf plate up to the parapet and from the shelf plate down to the window. He conceded that this will require repair and noted that a heli tie had already been put in place as shown in photograph 14 through the brick and into the wood.

In his evidence Mr. Quirouette pointed out that he was not present at the time that test openings 3 and 4 were made. He was however provided with photo's of those test holes by Mr. Roney and it appeared to him from the photo's that the bricks and mortar in those test holes were in good condition as well.

Mr. Quirouette noted that during his first site visit and walk around he observed one control joint only on the south side of the building and it is shown in photograph 2 as a discernible white line.

With respect to the cracking in the mortar joints that he observed at the corners on the top floor, one of these is shown in photograph 15 which shows a significant step crack through several courses of the brick. Mr. Quirouette conceded that there were such cracks on all four corners of the building and explained that the cracks resulted from differential stresses between the north and south sides of the building on the one hand and the east and west on the other.

In article number 3 of Exhibit 120 is a list of the documents reviewed by Mr. Quirouette during the course of his investigation. It is noted that he reviewed both the 1980 National Building Code of Canada and the 1985 National Building Code of Canada. He explained that the 1980 National Building Code would have been adopted by Ontario to produce the 1983 Ontario Building Code which was applicable to this building.

In paragraph 4.1 of his report, Mr. Quirouette analyzes and discusses the wall that was actually constructed in this building. He explains the distinction between a mass wall and a cavity wall. He points out that the mass wall is the oldest form of masonry construction. By the mid-1900's the evolution of the mass wall comprised one course of four-inch brick with one course of four, six or eight inch concrete block with the joint in between (collar joint) filled with mortar. As building construction advanced into the second half of the 20th Century, the mass wall became impractical to construct because of its imperfect resistance to rain penetration, its poor thermal resistance, but mostly because of its weight and increasing construction cost. With a cavity wall, the masonry wall is provided with a modest cavity between the brick and block masonry to form a drainage path. The cavity wall includes flashing material to divert cavity water to the outside and also includes openings in the masonry veneer (weep holes). Mr. Quirouette points out in his report that while the cavity wall approach provides an improvement in rain penetration control over the mass wall, it does have a few inconveniences. Unlike the mass wall, when a rain or melt water leak appears on the inside of the building, it is difficult to locate the outdoor leak entrance location as it may have originated quite some distance from its appearance location indoor. Further, it is more difficult to repair than a mass wall. The cavity wall cannot always be repaired by mortar repointing or the application of a sealant to outside openings. In some cases, the cladding system must be dismantled so that the leakage path from the cladding surface to the inside of the building may be traced and repaired.

It was Mr. Quirouette's opinion that the wall as constructed was neither a mass wall nor a cavity wall. He saw it to be a hybrid wall with more features of a mass wall than a cavity wall. He noted that the design drawings do not clearly indicate a void between the block wall and the brick wall. The specifications on the other hand do provide for a 19 millimetre void between the two wythes. The test holes show that there was a lot of mortar in the cavity and that all ties observed had mortar on them, rendering them rigid.

In article 4.2 of his report Mr. Quirouette points out that he could find no records or surveys or exact moisture problem locations, date and time of occurrence or documented descriptions of the duration and severity of the moisture problems. From the plaintiffs' reports he was able to determine two principal moisture entry locations. The first was over the windows of the south wall near the east corner of the building with exact leak locations unknown. Again he found no particulars of the leak or any reference to the severity of the leak problem except to mention that on one occasion when the roof drain plugged up it caused major flooding of the interior through façade windows, stair lobby, window recesses and ultimately through the east stair well roof hatch where major flooding occurred (according to the Johnson report). Mr. Quirouette determined a second moisture entry problem on the northwest corner of the building in the lobby stair tower of the fourth floor, which he understood came from a roof leak as noted in the Johnson report.

It should be noted that in noting his complaints with respect to the lack of particularity relating to the leakage problems, Mr. Quirouette conceded that he did not have at the time of his first report production of Dr. Reed's videos.

In article 4.3 Mr. Quirouette continues his complaints that the plaintiffs have not actually investigated or revealed the exact causes of the alleged moisture problems in the exterior walls. Mr. Quirouette is very critical of Mr. Johnson's water spray test. It was his evidence that the placement of the hose and spray pattern of the test nozzle were determined to be between five and ten times as severe as the standard test method and was comparable, according to Mr. Quirouette to placing the water hose directly in the wall cavity. Mr. Quirouette concludes that the locations of the rain penetration if they exist at all would be opened brick head or bed joints near the top of the building and at the corners of the building near the top floor. Again, it should be noted that he comes to this conclusion without having had the advantage of seeing Dr. Reed's videos. He notes in his report that the cracked head and bed joints are caused by the absence or omission of adequate expansion control joints in the brick cladding system. As well he notes that the cutting of the plates on the east and west walls gave rise to numerous consequences including cracked head and bed joints at the upper level caused by differential expansion. He notes that when the plates were cut the designers should have authorized a change order to include a horizontal support below the 11 metre height and vertical expansion control joints at the corners of the building. Mr. Quirouette concludes at article 4.3 of his report by expressing the opinion that the alleged moisture problems were caused by blocked roof drains, a minor roof leak at the northwest corner of the building (which was subsequently repaired), the absence or omission of adequate expansion control joints (which caused cracking of brick head and bed joints at specific areas of the brick cladding), a poor parapet detail and the absence (omission) of window design details.

Referring to the roof, Mr. Quirouette points out that the design drawings give a clear indication that what is required is an irma type roof as opposed to a standard built up roof. In the irma type roof the insulation goes over the waterproofing, whereas with the standard built up roof the waterproofing goes over the insulation. Drawing number 4 of Exhibit 9 specifically refers to an irma roof system whereas the Specifications, Division 7 sections 2 through 6 very clearly describe a standard built up roof. Mr. Quirouette pointed out that the irma roof is more expensive and is a better roof than the standard built up.

With respect to the drawings, Mr. Quirouette notes that they fail to indicate the location of expansion/control joints, nor did they show any window detailing, flashings or weep holes. The drawings did show some parapet detail on a very small scale which rendered the detail hard to read. Mr. Quirouette notes that the Specifications called for flashings and weep holes but in fact none were installed. It is his opinion that the builder complied substantially with the drawings and built a hybrid wall with more features of a mass wall than a cavity wall. He noted that in his experience, builders use drawings to construct a building other than in circumstances where ambiguity arises where the builder will review the specifications for additional information. It was Mr. Quirouette's opinion that the architectural drawings were clear to the builder and that there were no contradictions in the drawings that signaled further inquiry or clarifications.

In article 4.6 of his report, Mr. Quirouette expresses his understanding that the irma roof leaked in only one location which was eventually repaired. In his view, apart from this one location the roof had nothing to do with the leaks into the cavity. In his view, there was no need to replace the roof in 1998 and it is his opinion that the roof could have provided another 15 years service with proper maintenance.

In article 4.7 of his report, Mr. Quirouette addresses the scope of the builder's responsibilities which is first and foremost the obligation to construct as per the design documents and to respect the agreed-upon schedule and budget. In addition, the builder is required to comply with all building regulations. The builder is not responsible for the design of the building. In this case it was not the responsibility of the builder to design the roof system, the roof slopes, the parapets, the brick and block infill wall system, the window details or the number and location of any expansion/control joints. In Mr. Quirouette's view the builder complied reasonably with the design drawings. At the same time the builder was in non-compliance with some of the specifications.

Mr. Quirouette explained that based on his review of the reports and his observations of the test holes numbers 1 and 2 he arrived at certain conclusions as set out in article 5 of Exhibit 120. He pointed out that these conclusions were reached before he had an opportunity to see Dr. Reed's video, Exhibit 7. His conclusions without reference to the Reed video, were as follows:

1) the plaintiffs failed to record or provide their expert architect Johnson a reasonable account of the moisture entry problems by location, date, duration and time;

2) there were two categories of alleged water penetration problems. The first category involved alleged drain water penetration through the exterior brick masonry walls. However Mr. Quirouette's findings noted the most serious water penetration problem occurred when the roof drains plugged up;

3) The second category of leak involved the roof system. A roof leak occurred in a stairwell at the northwest corner of the building resulting from a rupture in the vertical membrane flashing of the parapet in the area of the leak;

4) the drawings did not show any indications of expansion control joints, cavity flashings, brick weep holes or soft joint details. From the drawings, Mr. Quirouette concluded that Mr. Garbutt designed the building to be constructed with an irma roof and a mass wall type;

5) alternatively, the specifications indicated the roof was to be a standard built up roof and a 19 millimetre cavity wall with flashings and weep holes. The specifications then indicated to Mr. Quirouette a traditional roof system and a masonry cavity wall system. Mr. Garbutt then produced construction documents that were contradictory in concepts and incomplete in details;

6) builders construct buildings from the drawings. Specifications are used primarily to clarify ambiguous conditions;

7) the building construction complies essentially with the drawings but is at odds with certain project specifications;

8) the roof leaked at the northwest corner of the building. The leak was repaired. Subsequently the entire roof system was removed and replaced. There was no justification for removal and replacement;

9) the water test undertaken by Mr. Johnson did not comply with any known standard test methods for testing rain penetration control. It was far too severe to represent even a severe rain storm;

10) with the exception of one location on the south wall, there are no expansion/control joints in the brick cladding. As a result, there has been cracking in the head and bed joints evident at the corners of the building between the fourth floor and the roof. This is a design error;

11) Mr. Quirouette's investigation did not reveal the exact leak locations of water leaks in the exterior wall. He did understand that the roof felts did not extend over the face of the brick cladding and that the brick cavity may have been exposed to wind-driven rain. This is a design error;

12) it is not necessary to replace the masonry cladding to restore the rain penetration control of the exterior wall system;

13) the brick cladding of the Reed-Lawless office building can be prepared to provide the necessary water penetration control. The repairs would include minor structural repairs (see Roney Engineering Structural Report), installation of expansion/control joints at corners and other locations, repointing of mortar joints at selected areas; and window head flashing repairs;

14) The builder did not cause the alleged water problems to the building, with the possible exception of a roof membrane deficiency at one location in the northwest corner of the building which was not sufficient justification for the roof replacement. The builder is not responsible for the absence of a sloped-roof system or low parapets.

In his report, Exhibit 120, Mr. Quirouette made certain recommendations in article 6 thereof. Again, these recommendations were made without Mr. Quirouette having seen the Reed video, Exhibit 7. In due course, Mr. Quirouette saw the video and delivered a supplementary report dated May 10, 2002, Exhibit 121. When Mr. Quirouette viewed the video he was mindful of the four major causes of moisture problems in buildings as set out in article 1.3 of his report dated April 23, 2002, Exhibit 120. In Mr. Quirouette's opinion, the four major causes are rain, wind-driven rain, snow and ice melt, and condensation. In his *viva voce* evidence Mr. Quirouette testified that the stone surround was a grand place for condensation to develop. Interestingly, if one goes to article 6.5 of this report where the window stone surrounds are discussed, there is no mention of a condensation problem but according to Mr. Quirouette's evidence, once he viewed the Reed video he determined that the interior damage around the windows on the second floor was all due to condensation problems.

We then viewed portions of Dr. Reed's video as selected by Mr. Quirouette.

Mr. Quirouette referred to selected portions of the video showing the damage to the drywall over the top and along both sides of the window. He noted that this damage is seen literally on all windows of the second floor, on all sides of the building and in his view the area of damage is along the line of the stone surround where condensation accumulates as a result of interior humid air meeting the relatively cold window surround. He notes that there is no damage to the window sill and speculated that the stone surround either did not reach the area of the window sill or was somehow insulated from it. The fact of the matter is however that the window sill is constructed of formica and not drywall.

Portions of the video showing the damage to the drywall surrounding the windows on the second floor appear to have been taken in March of 2002.

Mr. Quirouette then selected a portion of the Reed video which was taken in April of 2000 while it was raining. It depicts the fourth floor and shows a constant drip of water coming from between the window frame and the drywall, down on to the sill on a north side window. The video shows the outside of the window covered with rain drops but Mr. Quirouette testified that this was not wind-driven rain (how he could tell that is beyond me) but rather would have been rain building up on a ledge and eventually entering the building.

Having seen the video, Mr. Quirouette then prepared a supplementary report dated May 10, 2002, Exhibit 121.

Mr. Quirouette in his evidence critiqued Mr. Johnson's sketch of the presumed spandrel assembly on the south façade which is found at Tab I of Mr. Johnson's folio number 3, Exhibit 56. Mr. Quirouette notes in his report that if the sketch produced is reasonably accurate then the path of the water entry shown is incorrect. Mr. Quirouette notes, again assuming that the sketch is reasonably accurate, that the window head and exterior wall junction was incorrectly designed. The sketch shows the window stone surround below the brick support angle and Mr. Quirouette notes this is unusual. Usually, the steel angle or steel lintel is below the stone surround. Mr. Quirouette notes that the sketch is probably not accurate in that it shows the stone surround up tight against the bottom of the steel lintel. In his opinion, if that was so then the stone surround would crack. He is of the view that there is in fact a space between the bottom of the steel lintel and the top of the stone surround. As well, he feels that the sketch is not accurate with respect to the top of the window being up tight against bottom of the stone surround. In his view there must definitely between a space between the top of the window and the underside of the stone. He put into evidence his sketch, Exhibit 122, where on the right hand side thereof he depicts what he thinks is likely the actual design of the window head detail which incorporates a space between the top of the stone surround and the bottom of the steel lintel, and a space between the top of the window and the bottom of the stone surround.

Mr. Quirouette next entered a sketch as Exhibit 123 which demonstrates his critique of Mr. Johnson's spray test on the south side of the building. Mr. Quirouette asked the rhetorical question, if water was passing through the brick façade and dripping down the inside of the brick wythe, how was it getting past the shelf plates. In his view, the shelf plates acted as a flashing and because there was no accompanying weep hole, the water would accumulate at the shelf plate and eventually produce spalling and efflorescence on the brick. However, he observed no such evidence of accumulation.

Mr. Quirouette then tendered three drawings, Exhibits 124, 125 and 126, which demonstrated his theories with respect to condensation, snow and ice accumulation, melting and rain penetration.

With respect to condensation, he explained that in his view humid air can get past the insulation in the wall and then pass up the void past cold surfaces where it condenses and either falls back down the cavity in warm weather or in cold weather frosts up and then runs back down when it thaws.

With respect to snow and ice accumulation, he noted that this could occur on the ledge of the stone surround and eventually melt and leak back into the void through the space between the stone surround and the steel lintel if not properly caulked.

With respect to rain penetration, he theorized that water hits the wall, accumulates on the surface of the wall, runs down the wall and then accumulates at the top of the stone surround and comes in between the steel lintel and the top of the stone surround, again, assuming that it is not properly sealed.

It should be noted that essentially none of this is detailed in Mr. Quirouette's report, Exhibit 121.

Mr. Quirouette confirmed that he is of the opinion that the subject building is in need of repairs. He confirmed that all of the recommendations set out beginning at page 20 of Exhibit 120 are still his recommendations after seeing the video, with the addition of one further recommendation with respect to the windows. The recommendations set out in Exhibit 120 are as follows:

1. shelf plates on the east and west walls are not required according to Mr. Roney;
2. the brick cladding requires expansion control joints to be cut in each corner in line with the ends of the plates from the foundation to the parapet. As well, additional control joints should be cut. The estimated cost of these joints would be about $10,000;
3. additional brick ties are required below the parapet and as well on either side of the vertical expansion control joints when cut;
4. where the mortar has cracked between the bricks at the top of the walls, it should be re-caulked. The estimated cost of re-pointing would be about $5,000;

5. had it not already been done, the brick cavity at the parapet would require sealing with a suitable membrane flashing extending from the face of the brick over the top and joining with the roof membrane under the flashing. This has now been done and should not have cost any more than approximately $10,000;

6. the original recommendation with respect to the stone surrounds which do not have any slopes to shed rain or melt water away from the building was to clean out the mortar joint between the stone surround and the steel lintel, place a backer rope and silicone sealant therein to prevent rain or melt water penetrating to the inside. The estimated cost was about $15,000. It was Mr. Quirouette's evidence that following viewing the video, he concluded that re-sealing would be insufficient and would not cure the condensation problem. He now has an additional recommendation. He notes in his report, Exhibit 121 that assuming the Johnson sketch to be accurate, the shelf angles supporting the brick above the window is in large measure as good as any flashing material with the exception that it does not project past the face of the brick. The steel lintel which is just above the window leak locations is also the exact location of a possible minor water penetration path not identified by others, according to Mr. Quirouette. He notes that any water migrating down the back side of the brick may have escaped to the outside on the surface of the shelf angle through the mortar by draining or wicking. However, he says, because of the outside seal between the stone and brick above the drainage wicking path is blocked and water may be forced to drain under the angle and into the window head on the inside of the building. For these reasons, his new recommendation is that the window stone surround be removed and the connections of the window stone surround and interior gypsum support be redesigned to provide a suitable thermal break and a better method of rain and/or melt water penetration control. He notes that this may cost something in the order of $1,000 per window or $65,000;

7. the soft joints under the shelf plate between the top of the brick and the shelf plate are not required;

8. the owners should undertake pressure balancing of the ventilation system to minimize air pressure differences at the brick façade, particularly during rain storms in order to reduce rain water penetration at parapets, window and minor brick joint openings.

Mr. Quirouette then gave us a wrap up of his final conclusions:

1. there are definitely moisture damages existing in the building and water penetration into the building;
2. a portion of the moisture damage is due to condensation;
3. ice and snow entry is another source of entry and damage;
4. the most prevalent source of entry and damage is rain and wind-driven rain;
5. the Johnson spray test was not properly done and Mr. Quirouette does not agree with the results;
6. the builder complied with the drawings, the masonry is well constructed. Except for a roof leak as a result of a tear of the roof membrane and a lack of ties at the top of the walls, the builder is not responsible for leakage of water into the building;
7. the most significant problem with this building is water entry at the window heads;
8. all the problems are easily and inexpensively repaired and the work required to remedy has been exaggerated by the plaintiffs' experts;
9. Mr. Quirouette does not believe that the masonry needs to be replaced.

Filed as Exhibit 127 was Mr. Quirouette's list of final conclusions.

Cross-examination by Mr. Mueller

Referring to his report of April 23, 2002, Exhibit 120, Mr. Quirouette confirmed that his first site visit was February 5, 2002 at which time he walked the perimeter of the building but did not go in.

His next site visit was on March 16, 2002, when he saw only two of the eventual four openings. Mr. Quirouette confirmed that neither test hole number 1 nor 2 permitted him to see the inside detail of the window opening. He confirmed that he has never seen into the window cavity as such.

He confirmed that it has been snowing in Kingston this week, the week of January 6, 2003 and that he did attend to see the building this week. Mr. Mueller suggested to him that one cannot observe any snow on the parapet at the present time and Mr. Quirouette conceded that he was not in a position to dispute that. With respect to possible accumulation of snow or ice on the top of the stone surrounds, Mr. Quirouette was not in a position to comment because he did not look on the occasion of his recent visit to the building.

Mr. Quirouette was referred to photographs 1, 3 and 4 of Appendix F in his own report, Exhibit 120, taken at the time of his walk around, and he confirmed that the photographs do not show any accumulation of snow or ice on the parapet.

Although he has never been on the roof to see the parapet, he was aware that originally it was designed with a flat top which was not good and that it was eventually changed so that the top sloped into the roof. It was suggested to him by Mr. Mueller that it would be fair to say that after the change, the reasonable expectation would be that any snow or ice melt would go to the roof, rather than to the outside of the building. Mr. Quirouette said this was not necessarily so, and described a situation where there might be a melt on the inner side of the top of the parapet causing an ice dam which would cause the melt to run toward the outside of the building.

Mr. Quirouette was referred to his own photograph 5 in Appendix F which shows some light accumulation of snow on the top of the window surround. On further questioning it was determined that this particular window was below ground and that the top of the window was at approximate foot level.

At Mr. Quirouette's photograph 15, one can see the south face of the building at the west corner, near the top of the wall. Cracking is evident on the south wall and Mr. Quirouette confirmed his observation at the time that the cracking was also in place on the adjacent west wall.

Mr. Quirouette was then referred to Mr. Garbutt's document, Exhibit 74, and particularly Tab 25 which is a photograph which shows a caulked crack at the southeast corner which Mr. Quirouette was aware of. He was referred as well to Tab 26, a photograph which depicts a caulked crack at the northeast corner which Mr. Quirouette was aware of. He confirmed that to his knowledge there were cracks at all of the corners.

Mr. Mueller brought to Mr. Quirouette's attention evidence that on the north wall there was a crack that had propagated from an original eight feet to approximately 20 feet at the present time. Mr. Quirouette agreed that this evidence would indicate continuing stresses on the wall causing propagation and agreed the propagation would be caused by two factors; the lack of control joints around the corners and the differential stresses caused by the fact that the north and south walls have shelf plates while the east and west walls have none. Mr. Quirouette confirmed that it was one of his recommendations that there be four vertical control joints placed from the parapet to the foundation at each end of each of the north and south walls and that in addition there be three more vertical control joints placed on each of the two walls, approximately 25 to 30 feet between each other.

Mr. Quirouette confirmed that he saw two test openings, one at the first floor level at the southwest corner (photograph 8 attached to his report) and one on the north face at the upper level. In addition to the actual test opening at the first floor level, some nine holes were drilled adjacent to the test opening and the scope was inserted to view the void between the two wythes. No such investigation was done on the north face at the upper level. Mr. Quirouette confirmed that his investigation indicated that this was not a solid wall, although he observed a fair amount of mortar in the cavity at the southwest corner. However, he saw no extrusions from the mortar beds that extended from one wythe to the other. What he did see was mortar droppings that had solidified on top of ties. Most of the nine ties that he saw with the scope were splattered with concrete.

With respect to the water test performed by Mr. Johnson, Mr. Quirouette confirmed that Appendix D to his report which is titled "Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Curtain Walls and Doors by Uniform or Cyclic Static Air Pressure Difference" does not specifically say that water testing should start at the lower level and work to the upper level. Rather it is simply a matter of practice. As well, Mr. Quirouette confirmed that ours is not a 'curtain wall'.

With respect to the cracks that Mr. Quirouette observed in the head and bed joints of the brick wythe which he said in his evidence in chief would have permitted some rain penetration, Mr. Quirouette confirmed that he took photographs of these cracks and also looked at them through his binoculars. He confirmed that he never looked at the wall with a microscope or a magnifying glass.

Mr. Quirouette confirmed that he is not an engineer or an architect, nor is he a roofing inspector by profession.

Mr. Quirouette confirmed that in his original report he was critical of the lack of record of rain penetration. Mr. Mueller presented him with Mr. Mueller's letter of January 2, 2002 to Emmons & Mitchell's then lawyer, setting out significant details of the leaks. Mr. Quirouette confirmed that he has seen that letter. Mr. Mueller then presented Mr. Quirouette with his letter of May 1, 2002 which again includes a description of the water problems in detail. Mr. Quirouette acknowledged that this letter was received by his office but he did not have an opportunity to read it until after his May 10, 2002 report. Both of Mr. Mueller's letters went in as Exhibits G and H, solely for the purpose of proving transmission of information.

In section 2.2 of his report, Exhibit 120, Mr. Quirouette notes his understanding of the history of the problem and that is that in the fall of 1986, not knowing exactly where the leaks occurred, E&M re-sealed the head stones at all windows of the building. Mr. Quirouette in his *viva voce* evidence confirmed that he understood that E&M were called in to deal with problem leaks and he confirmed that he would normally expect that at that time they would do some investigation to determine why the building was leaking. As well, in his report, Mr. Quirouette notes his understanding that in October of 1992 the plaintiffs engaged a contractor to reseal the windows, some of the brick joints and the roof parapet flashing. He confirmed in his *viva voce* evidence that this again referred to *inter alia* the top of the stone surround. He was not aware that E&M

had again been called in at this time to explain why there were leaks, but conceded that if that was so he would have expected E&M as a reasonable builder to investigate before spending money or advising others to do so.

Mr. Quirouette in his report, sections 4.2 and 5.2, makes reference to "one occasion when the roof drain plugged up and it 'caused major flooding of the interior on one occasion manifesting in major leakage through façade windows, stair/lobby window recesses and ultimately through the east stairwell roof hatch where major flooding occurred (from the Paul Johnston, Architect report, Folio 2, page 2)'". Mr. Quirouette in his *viva voce* evidence conceded that there may have been two such floodings through the roof hatch. He was not aware that Dr. Reed had investigated the floodings and found no plugging of the roof drain. He was aware that in 1994 when the first roof repairs were done, broken membranes were found at the base of the hatch area, similar to the membrane problem in the northwest corner. He was aware that the problem in the northwest corner was a very significant one, and indeed in his conclusion 5.3 of his report, he attributes the problem to the roof for which the builder is ultimately responsible.

At page 12 of his report, Mr. Quirouette refers to architect Johnson's speculation and Mr. Mueller asked him whether it was his intention to put doubt on Mr. Johnson's evidence as surmisal rather than reasoned conclusions. Mr. Quirouette responded that while Mr. Johnson made certain observations as did Drs. Reed and Lawless, he found most of the records of no value for the purpose of a technical analysis of moisture penetration. It was his view that if Mr. Johnson had the same information that he had, then Mr. Johnson must have been speculating.

Mr. Mueller then began to review with Quirouette exactly what he knew of what Mr. Johnson did or did not do.

Mr. Quirouette was aware that in October of 1994 Mr. Johnson opened an area above the window in a fourth floor storage room, and saw a pattern of rivulet staining on the vertical face of the outer lintel. To understand the configuration of the inner and outer lintel, see any of Mr. Quirouette's drawings, Exhibits 122 through 126. Simply put, over each of the window areas is placed a unit comprised of two steel angle irons, spot welded back to back with the front angle being somewhat lower than the back angle. The back angle holds the block portion of the wall across the window and the front angle holds the brick portion of the wall across the window. You will recall that Mr. Johnson's drawing shows the top of the stone surround to be up tight against the bottom of the front angle, and the top of the window to be up tight against the bottom of the stone surround. It is Mr. Quirouette's theory that there is a significant space in fact between the top of the stone surround and the bottom of the front angle, and the top of the window and the bottom of the stone surround.

In any event, Mr. Quirouette confirmed that he was aware that in October of 1994, Mr. Johnson observed the rivulet pattern on the inside vertical face of the outer lintel or angle iron.

As well, he was aware of the October, 1994 spray test by Johnson, when Mr. Johnson saw water coming through the bed joints and producing rivulets which came down the inner vertical face of the outer lintel.

Mr. Quirouette was aware that the inner face of the stone surround is recessed further out than the inner vertical face of the outer lintel.

Mr. Quirouette was referred to Mr. Johnson's Folio 3, Exhibit 56, Tab H which is the specific report of Mr. Johnson dated October 5, 1994 relating to the water test. Beginning at page 7, under the heading 'South Façade: Water Test', the report reads in part as follows:

> After establishing and recording the static condition of the wall after disassembly, a garden hose was set up over the parapet to spray the wall with a wide breadth spray. The nozzle was aimed downwards and slightly toward the wall for a slight angular incidence. It was retained in a fixed location centred above the window jamb (east) about nine inches from the wall. Care was taken to prevent water entering the open parapet and generally full spray started about 10 inches below the top of the brick. A sheet flow of water resulted on the surface of the brick. Although not intended to quantify scientifically any induced water penetration or absorption – empirically the following was noted:
>
> - after one minute and 15 seconds a single small trickle of water was noted on the inboard vertical face of the brick lintel about 20 inches east of the west

jamb of the window opening. No water was apparent on or above the horizontal leg of the block lintel adjacent [picture 24]

- one minute and 20 seconds later a similar trickle appeared on the same brick lintel face about six inches west of the east jamb [picture 22]
- both trickles emanated from the gap between the lintels

Mr. Quirouette indicated to Mr. Mueller that he understood Mr. Johnson's observations.

Mr. Quirouette was then referred to photograph 24 at Tab H, which is labeled, 'Inside view of back to back lintels during a water test' and which has the caption, "water rivulets emanate from between lintels, wetting of stone window surround proceeds ponding on soffit".

Mr. Mueller reminded Mr. Quirouette of his evidence in chief that Mr. Johnson when he took this photograph must have been looking at a downward angle and thus it did not disclose the space that he theorized exists between the top of the stone surround and the bottom of the forward or brick lintel. Looking at the photograph, one can see the block wythe sitting on the inside block lintel and the outside brick lintel sitting on top of the stone surround. There is no space evident between the top of the stone surround and the bottom of the forward lintel. The photographs appears to have been taken straight on, but Mr. Quirouette notwithstanding extensive questioning by Mr. Mueller would not concede the point. It was clear to me that Mr. Quirouette felt that to concede the fact that the photograph had been taken square on rather than at a downwards angle would be tantamount to admitting that there was in fact no discernible space between the top of the stone surround and the bottom of the outside steel lintel, which would have permitted the entry of water.

Referring to Mr. Quirouette's drawing, Exhibit 126, Mr. Quirouette confirmed that on the drawing he shows the space between the top of the stone surround and the bottom of the front angle iron to be some 15 millimetres, and conceded that that measurement would be slightly more than ¼ of the height of the stone surround. It was obvious to me that such a depth of space between the top of the stone surround and the bottom of the outside steel lintel would be equally obvious in photograph 24 if such a space existed, but again Mr. Quirouette was not prepared to concede the point.

Mr. Mueller pointed Mr. Quirouette to photographs 21 and 22 of Tab H, the same inside opening. Mr. Johnson in the caption of photograph 21 notes active oxidation and staining of the outside lintel, but Mr. Quirouette was unable to see such oxidation or staining. The caption on photograph 22 notes oxidation of the metal window channel at the top of the window as being noticeable and Mr. Quirouette was able to agree with that observation. Referring to Exhibit 124, that channel would lie directly under the insulation as marked by Mr. Mueller in red pen on Exhibit 124 and Mr. Quirouette conceded that some of the rusting in the channel could be accounted for by water dropping into the channel from above.

Again referring to Exhibit 124, Mr. Mueller asked Mr. Quirouette whether if in fact moisture came in through a brick joint above the lintel plates and if it could go between the plates, could surface tension take it along the bottom of the plate and permit it to drop down. Mr. Quirouette was prepared to answer 'yes' to that question assuming that the water was able to get between the two plates. In the face of that assumption he was prepared to concede that oxidation could be caused either by condensation or by rain getting between the plates.

At Tab K of Exhibit 56 is another report relating to an investigation conducted by Mr. Johnson on December 7, 8 and 9, 1994. This report relates in part to a further test hole where the area above the interior side of the window on the fourth floor was opened further to permit a greater view of the interior of the brick façade than had been previously visible because the concrete block had concealed Mr. Johnson's view in the investigation conducted two months previously. This opened area allowed Mr. Johnson to view the brick immediately below the spandral beam of the south wall structure and exposed approximately 16 inches more of the wall immediately above the window opening. The full concrete block inner wythe was removed for the full width of the window opening. Mr. Johnson in his report notes that the interior surface of the brick displayed obvious water flow markings where rain water had penetrated. Referring to a photograph which is at page 21 of the text of his report and which is the same photograph as photograph 44 at Tab K, Mr. Johnson in the body of his report at page 22 notes the following:

> Again in the picture at the right on page 21 (photograph 44, Tab K) an almost unfilled cap joint is evident at the underside of the spandrel beam and shelf plate where they meet the brick at the side of the opening and the discontinuance or stitched weld has obviously left water penetration openings. No flashings were discovered at any location of the interior block removal area. The complete bridging of the air space is obvious from the view into the void to the right of the sampling areas. In confirmation of our report of the early investigation no ties at all were found in the sampling anywhere above the fourth floor window.

Mr. Quirouette was able to agree that this was Mr. Johnson's conclusion but that it was not his conclusion as he was not there.

Going back to Tab H of Mr. Johnson's report, photograph 23 where Mr. Johnson notes rust droppings on disassembly lying on the window channel, Mr. Quirouette conceded that this could be as a result of rust dropping from above.

Going back to Tab K, the investigation by Mr. Johnson on December 7, 8, and 9, Mr. Mueller brought to Mr. Quirouette's attention Mr. Johnson's evidence that during the course of a natural rain storm he observed rain water coming through the bed joints and trickling down the old stains on the inside of the brick wythe and that this was occurring on either side of the window. To this Mr. Quirouette responded that he was aware that Mr. Johnson had testified to that extent but suggested that this could have been coming from the roof. [I must say that it is difficult to rely on Mr. Quirouette's evidence as an impartial expert. He is very much an advocate for the defendant builder and his evidence is tainted as a result thereof.]

Mr. Quirouette was asked by Mr. Mueller to describe what he saw to be the sequence of the installation of the windows. Mr. Quirouette described the following. The inside blocks would be taken up and a window opening created. The inside steel lintel would be put in place and then the blocks would be continued up on top of the lintel to finish under the spandrel beam. Then the brick courses would start and be taken up to the bottom of the window opening. The stone surround sill would be laid directly on top of the brick course with mortar, then the brick courses would be built up the sides, leaving space for the stone surround jambs. Once the bricks were to the top of the window opening, then the two side jamb stones would be pinned into the block and brick. The opening would be slightly larger than the window frame. Then the header of the stone surround would be placed on top of the jambs and pinned. The outside steel lintel would be put on and the bricks taken up. There would be a tolerance or space between the top of the header and the bottom of the steel lintel.

Throughout questioning by Mr. Mueller, Mr. Quirouette maintained that the tolerance or space would be between the top of the stone surround and the bottom of the steel lintel, as opposed to between the bottom of the sill and the top of the underlying brick course. He maintained this position notwithstanding construction photographs (photographs 36 and 44 from Exhibit 4) which appear to show the space at the bottom of the stone surround as opposed to the top.

Mr. Quirouette agreed generally with Mr. Mueller that if water somehow came through the bricks above the lintel plates and somehow ran down and past between the two lintel plates, then the surface tension would take it across a distance along the bottom of the brick lintel plate and it would eventually fall down onto the inside surface of the stone surround. As well, he agreed generally with the proposition that if water came through the brick wythe and started running down, it could be deflected out toward the centre of the void by mortar fins. As well, Mr. Quirouette agreed generally with the proposition that a dropped blob of mortar might conceivably block the space between the two vertical legs of the lintel plates to some extent.

Mr. Mueller pointed Mr. Quirouette to Mr. Johnson's Folio 3, Exhibit 56, page 14 thereof, where on this point Mr. Johnson commented as follows: "The rust marks and pattern were high enough on the angle leg to infer a flow from above and not simply through the horizontal joint between stone header and steel: equally there were no stains from the horizontal window to stone joint that was recaulked twice." Mr. Quirouette agreed that what Mr. Johnson was saying was that he found rivulet stains on the inner side of the outer lintel and none on the stone surround.

Mr. Mueller pointed Mr. Quirouette to the first full paragraph of Mr. Johnson's report at page 22 of Exhibit 56 where, *inter alia*, he said the following:

> It was recognizable that the pattern of water flow on the brick inherently matched the pattern of the visible and active rust on the steel angle lintels, particularly that which carried the front brick over the extent of the window opening… the top edge of both angles would contribute to an almost complete blockage in bridging on their own, even without mortar droppings.

Mr. Quirouette agreed that what Mr. Johnson was saying was that there was very little space available for water to get down between the two vertical faces of the inner and outer steel lintels.

Mr. Mueller pointed Mr. Quirouette to Tab H, photographs 22 and 24 of Mr. Johnson's report. Photograph 22 is of the test area over the window on the fourth floor looking east, and the caption reads, "during water test, rivulets and accumulation adjacent to window, block discolouration at jamb and oxidation of channel noticeable." Photograph 24 is an inside view of back to back lintels during the water test and the caption notes that "water rivulets emanate from between the lintels; wetting of stone window surround precedes ponding on soffit." Referring specifically to photograph 22, the water rivulets on the inside face of the stone surround clearly start in one case about a quarter of the way down the inside face and in the other case about half way down the face and both fan out toward the bottom. Mr. Mueller put it to Mr. Quirouette that what was seen in the photograph was consistent with water dropping onto the inside face of the stone surround and was inconsistent with the water having come along the top of the stone surround through the supposed space between the top of the stone surround and the bottom of the steel lintel, and over the edge of the stone surround. Notwithstanding that the obvious answer would have to be "yes, what we see is entirely inconsistent with the water having come over the top of the stone surround and down the inside face of the stone surround", Mr. Quirouette refused to agree with the proposition. He maintained that what is shown in photograph 22 is entirely consistent with water coming over the top of the stone surround from what he says must be a space between the stone surround and the bottom of the steel lintel, and proceeded to give an explanation of his position which frankly made no sense to me. Eventually, after much questioning he appeared to concede that if this occurred during the course of Mr. Johnson's water test, then yes, the water staining seen in photograph 22 could have come from either between the vertical legs of the two steel lintels, or along the top of the stone surround.

Mr. Mueller pointed Mr. Quirouette to photograph 24 which clearly shows water rivulets emanating from between the two lintels, and wetting the inside surface of the stone window surround. Mr. Quirouette agreed that the rivulet patterns line up on the steel lintel and the stone surround and conceded that surface tension would have brought the water across the bottom of the outside or brick steel lintel, and it would have dropped down on the stone surround.

With respect to the accumulation of snow and ice on the header ledge of the stone surround, Mr. Quirouette conceded that the only such accumulation he saw in February of 2002 was on a header of a window in the window well and he is aware that there was never any leakage complained of in the ground floor. As a matter of interest, he did go over during a snow storm last night, January 7, 2003 and viewed snow collecting on the parapet over the north wall, over a distance of approximately 20 feet.

At Tab K of Exhibit 56 is photograph 42 of the same opening which has had the block wythe fully removed at the lintel. It shows the inner lintel and a portion of the outer lintel which as we know is offset to some degree below the inner lintel. The photograph appears to have been taken straight on to the extent that the horizontal portion of the inner lintel is simply seen as a dark line in the photograph. Notwithstanding this, Mr. Quirouette was not prepared to concede that the photograph was taken entirely straight on. He did go so far as to agree that it was almost straight on but he could still discern a slight angle upwards. Mr. Mueller pointed out to Mr. Quirouette that the photograph shows no space at all between the bottom of the outer lintel and the stone surround and Mr. Quirouette said that this was simply an illusion.

In any event, rivulet markings are very apparent on the inside of the brick wythe in this photograph. One set of rivulets starts from the bottom of the 5th course of bricks up from the outer lintel. It runs down the 4th course and on to the 3rd course. This can be observed on the left side of the photograph. A similar set of rivulets appears on the right side of the photograph. Mr. Quirouette agreed that the rivulets are shown and also agreed that there is no rivulet on the 5th course of bricks which lies directly below the bottom of the shelf plate.

Photograph 44 shows the inner lintel before removal and photograph 43 shows the inner lintel removed, exposing the vertical face of the outer lintel. Mr. Quirouette agreed that photograph 44 shows mortar fins and conceded that with sufficient water flow the fins would cause water to be deflected into the centre of the cavity.

Mr. Quirouette conceded that photograph 43 shows a mixture of red primer and possibly some rust on the inside vertical face of the outside lintel. He conceded as well that the photograph shows no gap between the bottom of the lintel and the top of the stone surround. Further, he conceded that photograph 43 and photograph 45 show some mortar plugging of the space between the brick side of the lintel and the brick itself.

Mr. Mueller then turned to Mr. Quirouette's report, Exhibit 120 and confirmed with him that Mr. Quirouette blames the absence of control joints in the building on Mr. Garbutt's drawings which show none. Mr. Mueller drew Mr. Quirouette's attention to his Appendix C which are a portion of the Specifications. Mr. Quirouette conceded that Division 4, section 2.1.1 specifies "provide 6 millimetre continuous expansion joint caulked continuously between steel support angles and brick panels each floor." Mr. Mueller suggested this was a pretty specific description, but Mr. Quirouette initially suggested that the control joints could not be built with reference to this specification alone. He eventually conceded that indeed it was a pretty clear specification.

Mr. Mueller pointed Mr. Quirouette to Division 4, section 2.2.2 which states, "provide control joints as dictated by structural design" and Mr. Quirouette stated that in the face of that specification one would expect to find the control joints drawn on the plans. Mr. Mueller asked Mr. Quirouette whether in the absence of the joints on the plans, would not the builder have to say to the consultant, "where do you want them?" Mr. Quirouette conceded that in those circumstances the builder should ask the consultant and the consultant should advise.

At the top of page 13 of his report, Mr. Quirouette addresses the cutting of the shelf plates on the north and south walls. He reiterated that he could not fathom the builder having cut the plates without the permission of the engineer. This would be so in practice, it is dictated by the metal specifications and as well it is dictated by the Building Code. Mr. Quirouette finds it very strange that the builder might cut the plates without the engineer's permission. Mr. Quirouette conceded that it would be equally strange for the engineer to give permission if he initially required the plates in order to conform with the Building Code. As he put it, "there's something funny." Interestingly enough, when Mr. Mueller suggested that any such change would have shown up in the 'as-built' drawings, Mr. Quirouette testified that as far as he was concerned 'as-built' drawings are a 'myth'. This notwithstanding the evidence of Mr. Sinclair.

At the middle of page 13, Mr. Quirouette offers his opinion as to the cause of the alleged moisture problems, which as far as he is concerned were caused by blocked roof drains, a minor roof leak at the northwest corner of the building which was repaired, the absence of adequate expansion control joints, a poor parapet detail, and the absence of window design details. He made clear that in expressing that opinion he was not addressing any structural problems that may have caused or contributed to the leaks. He conceded that he was not qualified to do so.

With respect to the roof, Mr. Quirouette notes at the bottom of page 13, that in the process of roof design, the final drawings and specifications resulted in conflicting information. The drawings indicate an irma type roof system while the specifications indicated a standard built up roof system. He notes that the builder constructed the more expensive irma roof. It was his position in *viva voce* evidence that the specifications cannot be read as requiring an irma roof. Turning to the specifications, at Appendix C of his report, Division 7, Mr. Quirouette conceded that section 1 refers to CRCA roofing specifications which in turn call for a 2% slope on irma roofs. As well, he conceded that section 2 of the specification provides as follows, "for inverted roofs, slope structures to drains at least 5m/100mm gradient (2%)". He noted that the specification then goes on to specify a standard built up roof. He did concede that section 3.6 which addresses membrane flashings is equally consistent with a standard built up roof and an irma roof, as is section 5 relating to flashings. He noted that flashings are the same for both an irma roof and a standard built up roof. Mr. Mueller suggested to Mr. Quirouette that the builder needed help from the engineer to resolve the conflict. Mr. Quirouette responded that obviously the builder didn't think so because he went ahead and built an irma roof. Mr. Mueller suggested to Mr. Quirouette that everyone including the builder, the owners and the consultant, wanted an irma roof and the only issue with respect to that roof is the lack of slope. Mr. Quirouette testified that you have to have slope in every roof, whether it is an irma roof or a standard built up roof. Mr. Quirouette was aware that there was no slope built into this particular roof and in fact there was a reverse slope. He commented that there were complications because it was anticipated that another floor would be added. Mr. Quirouette went on to explain that the roofer here was faced with a flat roof as required by Mr. Garbutt for the anticipated 5th floor. Mr. Quirouette presumed then that the roofer would attempt to get the 2% slope in some other fashion and explained shortly that there are many ways of getting that slope in the roofing materials as there was in the mid-1980's.

With respect to the statement contained in Mr. Quirouette's report at the top of page 14, "the drawings for the most part indicated a mass wall, including a few detail drawings showing typical wall section details", Mr. Quirouette reiterated his evidence in chief and confirmed that he was not withdrawing from his position that this was in fact a hybrid wall which had more characteristics of a mass wall than a cavity wall. Mr. Mueller pointed Mr. Quirouette to the small detail drawing which shows a gap between the block and brick wythes where no hatch line has been drawn through, contrary to the hatch line that is drawn through the mortar joints of the bricks and the blocks. Mr. Quirouette explained that line hatchings are only used in same materials, however he did concede that the drawings were equivocal and that there are "question marks here". He conceded that this could have been constructed as a collar joint. He conceded that section 2.1.1 of the Specifications requires that a 19 millimetre air space be maintained throughout, behind the brick wythe and in this respect Mr. Quirouette conceded that the Specifications clearly call for a cavity wall, although it is a "pretty small gap for a veneer". Mr. Quirouette agreed with Mr. Mueller that Mr. Garbutt left the plans and specifications unclear and that the builder in this situation needed clarification. He went on to explain that the onus is on the consultant to provide clear instructions as to how to build in accordance with the plans and specifications, and that if a builder finds a contradiction or an error in the contract documents, then he should go to the consultant for clarification. Mr. Quirouette was unable to say what is called for if the builder does not find the error or contradiction.

He then went on to make one of the few admissions that he was prepared to make during cross-examination by Mr. Mueller. He said “to be honest here a contractor would have realized that something was out of whack and should have gone to question the consultant.” He said this was the case both with respect to the roof and the wall.

Referring to section 4.5 of his report, Mr. Quirouette testified that the major contradictions between the drawings and the specifications with respect to the roof and the wall are almost bizarre. In his report he expressed the view that the drawings themselves did not contain any ambiguities or contradictions that signaled further inquiry or clarifications. Obviously in his *viva voce* evidence at cross-examination he withdrew from that position and conceded that there was some lack of clarity in the drawings. As well, in his report he noted that the construction was reviewed by Mr. Garbutt who could have alerted the builder at any time during the construction that the brick cladding and block infill were not complying with the intended documents. In his *viva voce* evidence Mr. Quirouette added that when he sees both the plans and the specifications with respect to this building he is really puzzled how the building went up without discussion between Mr. Garbutt and the builder. As I understood his evidence, he said this was particularly true of the removal of the plates on the north and south walls and conceded that it would be fair to say that appropriate and reasonable review by the consultant would have disclosed the absence of the plates.

At the bottom of page 15 of his report, Mr. Quirouette notes that the designated professional, Mr. Garbutt approved the construction of the building during the construction process by the omission of any instructions to the contrary. In this respect, Mr. Quirouette conceded to Mr. Mueller that it is an industry standard without reference to specifications that if the consultant “messes up” the ultimate responsibility remains with the builder.

With respect to the missing soft joints (caulking) between the bottom of the shelf plates and the top of the brick course on the north and south sides, this constituted a failure on the part of the contractor to follow specifications and also a failure on the part of Mr. Garbutt to catch the mistake.

Mr. Quirouette conceded that the same thing could be said with respect to the absence of weepers and flashers which were called for in the specifications but not illustrated on the drawings.

Mr. Quirouette further conceded that the same thing could be said for the absence of ties in the top 60” of the wall and the absence of the slope on the roof.

In section 4.6 of his report relating to the roof, Mr. Quirouette expressed the opinion that based on the report by the roof contractor, Amherst Roofing and certain photographs that he looked at, that the existing roof following repairs to the parapets could have provided another 15 years of service with minimal maintenance. On cross-examination Mr. Quirouette conceded that he had never been on the roof and indeed has never seen the roof as it was being repaired or as the old roof was being removed and the new installed.

At section 5.1.1 of his report at page 18, Mr. Quirouette said the following:

> Our investigation did not reveal the exact leak locations of water leaks in the exterior wall either. However, we understand that the roof felts did not extend over the face of the brick cladding and that the brick cavity may have been exposed to wind-driven rain. This is a relatively minor problem which is easily corrected. This is an error or omission of the design documents.

In his *viva voce* evidence, Mr. Quirouette suggested that the metal flashing should have been taken over the parapet and down the brick a certain distance, but suggested that at that time in 1985 it was not required to take the felt completely across the top of the parapet. Mr. Mueller directed him to the small detail drawing which is in Appendix B of his report, and suggested that it showed the felt going completely across the top of the parapet. Mr. Quirouette conceded that looking at the drawing one could not tell what was required, and clarification was required.

Mr. Mueller then drew a cross-section of the parapet and pointed out some seven areas of concern, including working from the outside in, the overhang of the metal cladding down the face of the brick which apparently was not as specified, the absence of slope at the top of the parapet, the absence of felt completely across the top of the parapet, the absence of an interlock of the metal cladding on the inside face of the parapet, and the failure to continue the four roof felts up the cant and up a distance on the inside surface of the parapet. After some discussion with respect to the drawings, Mr. Quirouette agreed that the detail drawing in his Appendix B showed some details that were not in fact built and some matters that were not clear and required clarification. As well, there were some matters of good practice that were required and not done and the single layer at the cant was contrary to specifications. Mr. Quirouette was of the view that a reasonable review by the consultant should have caught the fact that only one felt membrane was going up the cant and the inside surface of the parapet, and that indeed all of the deficiencies relating to the parapet should have been seen by the builder's job supervisor and the consultant.

With respect to the replacement of the brick veneer, although in his report, section 5.12 and 5.14 Mr. Quirouette expresses the view that it is not necessary to replace the brick cladding to restore the rain penetration control of the exterior wall system, he conceded to Mr. Mueller on cross-examination that he expresses this opinion without getting into the issue of stability in relation to wind loads. He conceded that it was beyond his competence to comment on whether it would be necessary to take off the outer cladding in order to get at the inner block wythe.

Mr. Quirouette confirmed that there are no weepers and flashers incorporated into his remedial system because in his view this hybrid wall can be repaired without weepers and flashers, notwithstanding they are required by the Specifications. Mr. Quirouette went on to explain that if weepers and flashers are installed, then the building requires an air barrier, otherwise the weepers provide water penetration from outside due to air pressure. He explained that in effect that in order to get weepers and flashers the building needs other things or the situation is made worse. His solution is to not put in the weepers and flashers while Mr. Johnson's solution is to put in the weepers and flashers and do the other things that are required.

At section 5.14 of his report, Mr. Quirouette sets out a number of problems which in his opinion his client, Emmons & Mitchell Construction Ltd., did not cause. Again, he conceded that in this section he does not refer to any structural problems that may relate to leaking.

In recommendation 6.4, Mr. Quirouette talks of re-pointing the brick masonry cracks and he concedes that in this section he is not referring to microscopic cracks. As he put it, “if you don’t see them you don’t re-point them.”

In section 6.6 of his report, Mr. Quirouette confirms that in his opinion it is customary to install a soft joint between the underside of a shelf angle or plate, and the top of the brick masonry below.

Entered as Exhibit 130 was Mr. Mueller’s drawing of the parapet detail which he and Mr. Quirouette discussed at some length during the course of cross-examination.

Cross-examination by Mr. Garbutt

Mr. Quirouette confirmed that the limited number of strap ties that he was able to see appeared to be in good condition. Their life span depending on conditions can run from five years to 50 years. He was not qualified to say whether the strap ties are structurally adequate in mass or close to mass walls, but did comment that architecturally they are not acceptable today.

Mr. Quirouette confirmed his previous evidence that test openings 1 and 2 were difficult to achieve in that the bricks were extremely well bonded to each other.

In terms of construction procedures, Mr. Quirouette confirmed that the building is usually constructed form the drawings with the site supervisor (Sinclair) answering to the construction manager (Emmons). The supervisor should be familiar with the work of each sub-trade and shop drawings should go from the sub-trades to the supervisor to be taken up with the consultant. A lot of the specifications are incorporated in the shop drawings that the sub-trade submits. Mr.

Quirouette confirmed that the drawings referred to in his reports include both shop drawings and design drawings, however there is no suggestion here that there were any shop drawings available. Mr. Quirouette observed that he did specifically ask for window shop drawings and he was told that there were none.

When Mr. Garbutt pointed out to Mr. Quirouette that Mr. Sinclair had said he kept a journal, noted all changes to drawings and kept shop drawings, Mr. Quirouette commented that he had not requested Mr. Sinclair's journal. He did confirm for Mr. Garbutt that all those documents are the property of the general contractor. As well, Mr. Quirouette confirmed that Mr. Emmons, the contract manager is ultimately responsible to ensure conformance to the General Conditions.

Mr. Quirouette confirmed that the construction of the windows was the same throughout the building but the damage that he observed at the second floor was different from that at the fourth. At the second floor he saw damage due to condensation, at the fourth floor he saw damage due to penetration of rain from the exterior and from condensation.

Section 3.0 of his report, Exhibit 120 lists the documents that Mr. Quirouette reviewed in preparing his report. Mr. Quirouette was aware that each floor had its own mechanical system. He was aware that Dr. Reed's video showed the fourth floor occupied and the second floor vacant. He agreed that since the installation of the windows is the same throughout and the damage different at different floors, that it follows that window design is not the sole cause of the damage. He agreed that condensation could be managed by management of interior atmosphere. However, here he did not review the ventilation systems for each of the floors, although in article 6.7 of his report he recommends that to minimize air pressure differences at the brick façade, particularly during rain storms, he recommends that the owners undertake pressure balancing of the ventilation system which should reduce rain water penetrations at parapets, windows and minor brick joint openings.

Re-examination by Mr. Nelson

On January 8, the day before this re-examination evidence, Mr. Quirouette had attended at the site and taken photographs of the building during the snow storm. The photographs were filed as Exhibit 131 and show some snow accumulation on the parapet and the window headers on the north side of the building, and in one isolated case on the west side of the building.

With respect to Mr. Johnson's evidence relating to the actual leaks following the rivulet patterns during the rain storm on December 9, 1994, Mr. Nelson brought to Mr. Quirouette's attention the transcript of Mr. Johnson's evidence relating to that event, Exhibit 62 Mr. Johnson's report dated April 30, 2002 which comments in part:

> If Mr. Quirouette had been present as we were during the documented sleet and rain storm of December 9, 1994, and watched every water mark exposed in the wall disassembly recorded in photograph 42 (Appendix 3K) glisten with the rivulet flow of penetrating rain, he would not need the rigours of an ASTM test… (bottom of page 13)

and Mr. Johnson's report dated January 2002 which is Folio 3, Exhibit 56 at Tab K thereof, which is the actual report from December 7, 8 and 9, 1994. Mr. Nelson pointed out that at page 4 thereof where the particular hole at the fourth floor window transom area is described, there is no mention of this penetration occurrence. Mr. Quirouette commented that he would have expected the observation with respect to the results of flowing water during the rainstorm to be part of that report.

Witness – Robert Drysdale

Dr. Drysdale was re-called by Mr. Mueller as per the agreement of the parties, to respond to Mr. Roney's report dated October 21, 2002 (Exhibit 89) which in turn responded to Dr. Drysdale's report, Exhibit 32, and the calculations that Dr. Drysdale had prepared and filed as Exhibit 31. Dr. Drysdale had prepared a report containing a response to Mr. Roney's report of October 21, 2002 (Exhibit 89). That report was dated November 18, 2002 and was initially marked as Exhibit C, in order to permit Mr. Mueller to cross-examine Mr. Roney thereon. On this re-call of Dr. Drysdale, Exhibit C became Exhibit 132.

Because Mr. Mueller cross-examined Mr. Roney at some length with respect to Dr. Drysdale's report dated November 18, 2002, Exhibit 132, we are all fairly familiar with it at this point in time.

Dr. Drysdale began his evidence by referring to the cover page of his report, Exhibit 132, and pointing out that Mr. Roney performed a further computer analysis when he determined that at some point prior to construction the wall height had been changed from 2.4 metres to 2.6 metres. Dr. Drysdale notes that it is well known to structural engineers that greater wall heights produce more bending moment due to wind pressure, resulting in larger tensile stresses (cracking stresses). The increase is proportional to the square of the ratio of the 2.6 metres versus 2.4 metres wall height, which works out to be 1.17, that is 17% more bending with the corresponding 17% increase in tensile stress. He notes that therefore, notwithstanding all of these analyzed cases for the 2.4 metre block wall height showed that the brick and block walls were under-designed, the increase to 2.6 metre block wall height makes the situation 17% worse.

Continuing his response to Mr. Roney's report, Exhibit 89, Dr. Drysdale explained that today's Code must be used for <u>any</u> retrofit. In any event, it doesn't really matter which Code is used. Dr. Drysdale points out in the second paragraph on page 2 of Exhibit 132, the following:

> Despite the rhetoric there is really no disagreement between Mr. Roney and myself regarding which masonry code should be legally used for what purpose. However there is disagreement on the relative impact of using one versus the

> other. The new 1994 edition took advantage of research, much of which was done at McMaster University, to allow unreinforced masonry walls to carry about 30% more load where compression controlled the capacity. When tension controls the capacity, such as caused by bending due to wind pressure, this newer Code was calibrated to produce as nearly as possible the same capacities as in 1978, but for block the values are a little lower. For evaluation of the potential for brick cracking, it makes little difference which Code is used.

In Dr. Drysdale's opinion, Mr. Roney's analysis does not represent reality in that it satisfies none of the conditions of his bullet number 9 in Exhibit 32. In bullet number 9 of Exhibit 32, Dr. Drysdale points out that the wind pressure is distributed between the brick and block wythes of masonry according to calculated EI values representing section stiffness, only if,

a) neither wythe cracks (cracking of the brick was reported here);

b) the wythes have identical support conditions and are of the same height (neither of these conditions is met);

c) ties connecting the wythes are sufficiently stiff to force the wythes to deflect equally (very flexible ties were used);

d) additional bending due to eccentric axial loads is not applied to one or both wythes (the eccentric axial load is applied in indeterminate and inconsistent ways to both wythes);

e) additional stresses are not introduced by restraining deformation due to moisture and temperature changes and in-plane forces caused by the infill resisting lateral loads (all of these stresses do exist)

In Exhibit 132 Dr. Drysdale notes that even if it had been found that calculated tensile stresses were not high enough to cause concern, remedial ties would still be required over the entire building. However, what his calculations show is that introduction of remedial ties is not sufficient and that other strengthening or rebuilding measures are required because stresses are greater than the tensile capacity of the masonry. Therefore rebuilding of the concrete block wall using a larger size block or other strengthening measures is required to ensure that sufficient resistance to wind pressure is provided.

That is really the nut of Dr. Drysdale's position. In terms of making the block wall more solid, it would be more economic to install steel posts on either side of all windows in order to avoid removal of the block wall in its entirely. It is because the wall is weakened by the windows that the other area between the windows must be strengthened. At Exhibit 2, Tab 114, the estimate of Detra Buildings Inc. for remedial work, two alternatives are set out with respect to remedial work, A and B. It was Dr. Drysdale's evidence that alternate B which is the more expensive of the two is required in order to accommodate the posts.

In his report, Exhibit 89, Mr. Roney criticized Dr. Drysdale for choosing to omit the effect of axial compression on the wall. Dr. Drysdale notes in his report and noted in his *viva voce* evidence that not only are axial (vertical) loads difficult to predict, but it is certain that except for self weight, no axial load exists in some cases. For example, no applied vertical load exists on the concrete masonry wall under the steel beam near the columns. At this location, the steel beam does not deflect and cannot close the gap between the bottom of the beam and the top of the block wall caused by shrinkage of the concrete masonry. For the brick wythe, where the shelf plates supporting it were left intact, load will not be transferred from the frame into the brick. Where no shelf plate remains, vertical load will be limited to the weight of the veneer above. In other cases, near mid-span where the steel beam has deflected onto the concrete block wall, uncontrolled transfer of axial (vertical) force is a very dangerous condition. Dr. Drysdale explained that the I-beam has little resistance to twisting and as the beam twists, it bears on the edge of the wall as opposed to the centre of the wall so that the transfer of any axial force is in fact a negative feature.

At page 3 of his report, Exhibit 132, Dr. Drysdale points out the following. "Mr. Roney states that he has attempted to understand the actual mechanism of load transfer from the beams onto the wall and that he has modeled the effects in a conservative fashion. He has not provided any information on what he found and how he related this to a structural model. Basically, he has asked the reader to believe that he knew what he was doing; unfortunately the evidence is that he committed several major errors which completely invalidate his analysis. Dr. Drysdale considers the errors to be as follows:

1) Mr. Roney's method of distribution (sharing) the load between the brick and block wythes, even to be approximately true for walls of equal height and identical support conditions (neither of which is true here) requires that the walls be forced to deflect equally. Very stiff ties such as the helix type used in Dr. Drysdale's analysis are needed to satisfy this requirement. When Mr. Roney had the walls opened he should have seen that the existing ties, whether bent or not, do not satisfy this condition. Nevertheless, he relied on rigid ties for his analysis.

2) The benefit of axial compression by applying the vertical load from the frame near the centre of the wall is a grossly unconservative assumption. Mortaring in of the horizontal joint between the top of the block wall and the underside of the steel beam is very difficult to accomplish, even if it had been specified. As found by Mr. Johnson where this was attempted, filling the gap was poorly done and typically only on one side of the block wall. Therefore, even when attempted, uneven bearing would exist. In fact, the mason usually works from one side of the wall so that if mortar is pressed into this joint, it is mainly on the near face. However, even if the full mortaring was achieved, the natural rotation of the beam caused by floor loading, results in the beam bearing along one face of the wall. In this circumstance, the bending caused by the eccentric vertical axial compression is a <u>negative</u> factor, as the larger bending caused by the eccentricity creates a net increase in tension which adds to the tensile stresses caused by bending due to wind pressure. As such, any axial compression transferred from the steel frame to the concrete block wall is an adverse effect.

3) Mr. Roney, despite his stated attempt to model the wall in a logical and conservative fashion, neglected to account for the extremely large effects of window openings which weaken the wall.

Dr. Drysdale concludes his analysis by stating,

> Mr. Roney's calculations do not stand the test of even the most cursory scrutiny. They are wrong.

Mr. Roney in his report, Exhibit 89 states that given the stress levels predicted by the Drysdale analysis, it is inconceivable that after 16 years in service, the cracks would still be invisible. Dr. Drysdale in his report, Exhibit 132, item g, points out the following:

1) Visibility of cracks. Masonry is very stiff and very brittle in tension. Because of this, with a 2.6 metre vertical span from the floor to the underside of the steel beams, a one to two millimetre deflection is large. Naturally, the size of cracks associated with such a small deflection are microscopic (that is, cannot be seen with the naked eye). Therefore cracking can and does exist without being visible. Use of a magnifying glass can help. Dr. Drysdale does not personally use the term "invisible crack", except in response to a question, because if you use enough magnification, it will be visible. However other evidence of cracking that is not readily visible can be taken from more profuse passage of water through some mortar joints compared to others. When the bond is broken between the mortar and the brick, there is much less resistance to flow of water. Dr. Drysdale notes that in this regard Mr. Johnson reported observation of water coming through some mortar joints during a rainstorm. Dr. Drysdale has made the same observations, both in the field and in his laboratory.

2) With respect to the extent of cracking, Dr. Drysdale notes that some cracking is readily visible from the sidewalk. However extensive cracking is not visible. From his analysis, under design wind loads cracking would be expected to occur near the bottom of the steel beam because of the redundancy caused by running the veneer an extra .6 metres (compared to the block wall) past this point. [Note that the block wall spans vertically 2.6 metres from the top of the floor to the undersigned of the steel beams, whereas the brick wythe spans the full 3.2 metres in order to completely cover the face of the building.] Additionally, cracking would be expected near mid-height of the wall because of the high bending and the weakening effects of openings. Unless very high loading had been experienced, thereby causing large deformations, such cracks would be difficult to see. However, says Dr. Drysdale, hopefully in most locations such cracks do not yet exist because critical wind pressures have not been experienced by this building. Existence of widespread visible cracking would be clear evidence that failure of the walls

has already partially occurred. Retrofit is needed before a sufficiently strong wind occurs to cause the failure.

3) Likelihood of cracking. Dr. Drysdale points out that the likelihood of cracking and subsequent failure is linked to the likelihood of having the building experience the design wind pressure. In this regard, the design code accounts for probability of various wind pressures occurring, combined with the probability of variation in masonry strength, to ensure that overall probability of failure is in the order of approximately one in a million. While the flaws in the design and construction of this building have dramatically increased the probability of failure, it is fortunate that the critical wind pressure has not yet been experienced by the building. Dr. Drysdale points out that Mr. Roney takes comfort in the fact that the building has survived for 16 years. Dr. Drysdale says that that is like taking comfort from pulling the trigger on a six shooter three times without firing, as evidence that the next three times will also be safe.

Dr. Drysdale closes his comments on these points by saying that neither Mr. Roney, the owners nor anyone else has the right to accept a significantly increased probability of failure where lives may be endangered. He notes that Mr. Roney is on very dangerous ground when he uses the obvious current survival as proof of adequate capacity.

It is of interest to note that Dr. Drysdale in his many years' experience has seen building walls fail after having stood for very long periods of time, 25 years or more.

In his report, Dr. Drysdale points out that in Mr. Roney's item 3 of Exhibit 89, Mr. Roney omitted to mention that the internal pressure coefficient minus 0.3 only applies to buildings with uniformly distributed openings which is not representative of this building where windows are more concentrated along two faces. Correct application of the 1980 NBC gives the same plus/minus 0.7 coefficient used in Dr. Drysdale's analysis.

In his report, as well Dr. Drysdale comments that ties may be either in tension or compression, so that any mortar droppings are unlikely to be able to benefit both conditions. In Dr. Drysdale's opinion, reliance on random mortar droppings to provide sufficient strength and stiffness of ties is wildly optimistic. It is Dr. Drysdale's view that Mr. Roney's assertion in this respect helps justify the fact that his analysis is wrong.

Cross-examination by Mr. Nelson

Discussing the visibility of cracks and particularly item G at the bottom of page 3 of Exhibit 132, Dr. Drysdale confirmed that cracking results from tension and that the passage of water can be evidence of cracks. Wind-driven rain can penetrate through microscopic cracks and Mr. Johnson saw that in his field tests. The wind-driven rain penetrates cracks caused by stress, stress which in turn is caused by structural problems.

Mr. Nelson then referred Dr. Drysdale to his book, Masonry Structures – Behaviour and Design, published in 1999. At page 814 of Chapter 18, the book, which is co-authored with two others, states the following:

> Visible cracks can be as small as .1 millimetre or as wide as 25 millimetres, and can be classified on this basis. They can also be classified based on being active (live and working) or non-active (dormant). It is important to determine whether a crack is active because this may change the method of repair. Cracks smaller than .1 millimetre wide are not considered likely to increase water penetration due to wind-driven rain.

This portion of the book was marked as Exhibit 133. The reference for the last sentence in the above-noted quote is a technical publication authored by Clayford T. Grimm in 1988 (Exhibit 134) where at page 2 thereof Grimm says, "Cracks smaller in width than .1 millimetre are insignificant to water permeance because that is the least width through which wind-driven rain will enter." Dr. Drysdale confirmed that that reference is fairly consistent with the contents of his book but not exactly the same.

Entered as Exhibit 135 was page 819 of Dr. Drysdale's book which deals with field tests. Dr. Drysdale explained that in his book he described many types of field tests which took up some 10 pages in all and conceded that he had not done any field tests on this building.

Filed as Exhibit 136 was page 828 of Dr. Drysdale's book which deals with laboratory tests. Again Dr. Drysdale confirmed that he conducted no lab tests with respect to this particular building. Both Exhibits 135 and 136 speak to the usefulness of field tests and laboratory tests.

In general Dr. Drysdale explained that he and his co-authors included that portion of Mr. Grimm's article in order to present a well-rounded picture and in order to present "all sides" in the book. As well, he explained that he was not retained to perform laboratory tests or field tests.

With respect to Dr. Drysdale's comment on page 5 of Exhibit 132, to the effect that neither Mr. Roney, the owners, nor anyone else has the right to accept a significantly increased probability of failure where lives may be endangered, Dr. Drysdale was aware of the requirement to report dangerous buildings. He realizes that the public is using the building and he is presently unaware whether in fact the building has been reported.

Cross-examination by Mr. Garbutt

Dr. Drysdale confirmed for Mr. Garbutt that for his analysis he assumed a 19 millimetre air space between the wythes and nothing other than the ties in the void. He confirmed as well that his conclusion, *inter alia* is that the back-up wall by any code is not strong enough to transfer the load and needs reinforcement.

Re-examination by Mr. Mueller

Mr. Mueller reviewed with Dr. Drysdale the excerpt from his book, Exhibit 133 and the portion thereof that Mr. Nelson discussed with him in cross-examination. Mr. Mueller asked Dr. Drysdale how this comment with respect to cracks smaller than .1 millimetre wide related to his opinion here at trial that the evidence of Mr. Johnson with respect to penetration of water through the bed joints is indicative of non-visible cracks. Dr. Drysdale explained that he and his co-authors tried in the book to reference other material and in doing so he changed Mr. Grimm's statement which was unqualified, namely "Cracks smaller in width than .1 millimetre are insignificant to water permeance…" to, in his book, "Cracks smaller than .1 millimetre wide <u>are not considered likely</u> to increase water penetration…".

In any event, Dr. Drysdale's evidence on re-examination was that he was sticking by what he had to say in Exhibit 132.

As well, on re-examination he explained that he was not engaged to carry out an investigation of the building but rather to critique the other experts. He confirmed as well that the lack of field or lab tests did not impact on his conclusions and he is relying on Mr. Johnson's observations with respect to the envelope.

With respect to the obligation to report a dangerous building to the authorities and to the Architectural Society, Dr. Drysdale commented that Mr. Nelson had brought up a point that very much bothered him. He is aware of the obligation to report and indeed last August he reported a building where the consultant had not followed up with a dangerous situation. He explained here that the on-the-job consultant is Mr. Johnson and it is his responsibility to report what he, Dr. Drysdale considers a dangerous situation, and if Mr. Johnson does not report then Dr. Drysdale considers that it will be up to him, Dr. Drysdale to make the report to the authorities (the City) and to the Architectural Association. In his view it has to be done in the near future, that is someone will have to do it in the near future.

Witness – Antonio Pascoal

Examination-in-Chief by Mr. Nelson

Mr. Pascoal is a professional engineer and the President of AJV Masonry Limited. His curriculum vitae is filed as Exhibit 137 and it is impressive. He received his engineering degree from Queen's University in 1984 and became a licensed engineer in 1987. He founded AJV Masonry in 1988. It is a company that specializes in the installation, restoration and rehabilitation of masonry units. He is presently the Vice-President of the Canadian Masonry Contractors Association and will become the President in due course. He and his company have restored approximately 3 dozen significant buildings and have been involved in the restoration of existing buildings built in the 1980's.

Mr. Pascoal has prepared two reports for Emmons & Mitchell. One, Exhibit 138 is dated June 28, 1999 and the other, more a quote than a report, dated April 26, 2002 is marked as Exhibit 139.

Mr. Pascoal was initially retained by E&M to review the project and assist Mr. Roney in making some internal openings. He did so, made observations with respect to the building, and reviewed the reports, plans and specifications. After preparing Exhibit 138 Mr. Pascoal last year was asked to assist Mr. Roney in opening some external holes in the building. He was eventually given the reports of Mr. Roney and Mr. Quirouette and then prepared his quote which is Exhibit 139 dated April 26, 2002.

In his report Exhibit 138, Mr. Pascoal notes that the air space between the two wythes varied in thickness but generally was at least the thickness of an average mason's finger (19 millimetres) and often greater at some locations to accommodate building tolerances and the structural steel, concrete flooring etc. It appeared to Mr. Pascoal that the masons were going up with the block wall first in some instances and up with the brick wall first in other instances, and as a result the space varied.

In his report he notes that he observed mortar fins in the air space which for him were normal in this construction time and practice. He noted that the brick was irregular and hard to work with and often had to be pressed in order to keep the mason's line, thus resulting in a lot of fins.

Mr. Pascoal observed no evidence of masonry flashings, weep vents or holes, masonry air or vapour barriers or control expansion or soft joints.

In his report he notes that the exterior wall has not been provided with effective arrangements of wall flashings, weep holes, soft joints etc. and the rain water absorbed by the clay brick becomes trapped in the building systems. He notes that some water is absorbed by the concrete block back-up, some slowly evaporates, but some finds its way into the interior of the building causing damage and discomfort. He notes that this is most evident above the windows.

In his report he notes that the masonry construction is in general conformance with the original plans and specifications except that there is no 6 millimetre continuous expansion joint caulked continuously between the steel support angles and the brick panels at each floor, and that there are no brick weepers at all bearings and over horizontal obstructions. As well he notes that the shelf angle has been cut on the east and west elevations.

In his report, Mr. Pascoal notes that historically in the transition from using traditional mass wall masonry, the masons relied on the mass or volume capacity of the thick masonry wall to absorb rain water and then evaporate back out. Such examples are walls made of three, four or five wythes of brick or brick and stone mortared together. He notes that seldom did the water ever penetrate to damage the interior finishes. However, masonry was never and is still not considered a waterproof building material. He notes that with the introduction of engineered masonry or thin walled masonry or veneer applications such as those introduced after the 1960's but widespread into the 1970's, brick walls were reduced to one wythe with some sort of back up material. He notes that it was in this transition from thick to thin walls that building envelope sciences were just emerging. For example, for many years into the 1980's, most designers were unaware that clay brick veneers tended to expand with time while concrete block back up tended to shrink with time, making the solid collar joint an eventual wall system potential problem.

Mr. Pascoal notes that the walls in this building are not showing signs of distress such as visible bricks falling, brick cracking and mortar deterioration beyond normal maintenance expectations for a 15-year old building. He notes that there are signs of efflorescence and mortar deterioration at grade levels which are "perhaps also effected by de-icing salts".

In his report, Mr. Pascoal states that AJV Masonry has been involved with several remedial projects of buildings fairly similar to ours and in some cases identical shortcomings and masonry design and construction such as this one. His proposed remedial work is based on his opinion that it is not necessary to take off the brick wythe to effect repairs.

In his report, when discussing the masonry construction, Mr. Pascoal notes that there is some documentation that leads him to think that there was some confusion between using an air space (as constructed) versus a solid collar joint (where the air space would be mortar filled thus making a composite wall). He notes that had a solid joint been used, there should have also been notes about interior mortar parging of the back up block as well as a "tar like building paper" to prevent water migration to interior finishes.

Mr. Pascoal proposes remedial work as follows:

1. Waterproof membrane flashings to be introduced at all shelf angle locations, at all windows and door locations, and foundation perimeters;
2. PVC grade weep fence to be introduced over new flashings plus all locations just below shelf angles at the top of each floor to increase air circulation in the cavity;
3. new support angles at acceptable intervals be installed along the east and west elevations at each missing location;
4. new soft joints be saw cut and caulked horizontally at shelf angle locations;
5. new expansion joints be saw cut and caulked vertically at required strategic but aesthetically acceptable locations;

6. new helifix stainless steel skewer type brick ties be installed where any potential concern of inadequate anchorage such as at parapet locations and where remedial work requires saw cutting and brick replacement (for the purposes of a quotation he will allow 500 individual ties at $10 each for credit or addition);
7. detergent cleaning and soft bristle scrubbing of any efflorescence staining on the brick wall.

In Exhibit 139 Mr. Pascoal presents a quote of $90,000 for all of the above-noted work, exclusive of GST. With respect to his quote for the remedial masonry wall ties and the vertical control joints, he depends on Mr. Roney's report and with respect to the remaining items – re-pointing, re-sealing parapet cap, re-sealing window surrounds, etc. – he relies on Mr. Quirouette's report.

Cross-examination by Mr. Mueller

Mr. Pascoal confirmed that his observation with respect to the width of air space was based on two or three holes that were cut from inside the building.

With respect to the absence of flashings and weep holes, Mr. Pascoal explained that he was aware from the construction documents that there was a need for weepers but was not clear on whether or not there was a need for flashers. [How could there be one without the other.]

During the course of cross-examination, Mr. Mueller discussed with Mr. Pascoal the fact that he observed signs of efflorescence and mortar deterioration at grade levels. During the course of the discussion Mr. Pascoal commented that the transformer opening at the northwest corner of the building showed the steel lintel to be in excellent condition and there was no spalling or efflorescence. As well Mr. Pascoal confirmed that, in addition to the efflorescence and mortar deterioration at grade levels that he observed, he also observed efflorescence higher up on the walls, particularly on the "cold" walls that do not have the heat of the building behind them.

There was also some efflorescence on the “hot” walls, particularly around the windows which told him that there was a collection of moisture and dampness around the window surrounds.

With respect to his comment at page 3, item 2, Mr. Pascoal confirmed that had a solid joint been intended between the two wythes, then standard practice at the time would be to parge the interior of the block wall as well as to install a “tar-like” building paper on the inside of the block wall.

Mr. Mueller reviewed with Mr. Pascoal the exchange between Mr. Garbutt and Mr. Emmons (Exhibit 2, Tab 77 and Tab 65) relating to the design of the wall. Mr. Pascoal seems to recall having seen similar documents.

Finally, in cross Mr. Pascoal conceded that in his report Exhibit 138, he had not put his mind to whether or not the wall was sufficiently strong to withstand wind forces.

Re-examination by Mr. Nelson

On re-examination, Mr. Nelson introduced a photograph of the transformer vault that was put in as Exhibit 145 and Mr. Pascoal commented that there was no sign of weepers or flashings or soft joints, and at the same time no sign of distress or water damage around the opening. The photograph had been taken yesterday, January 8, 2003.

Interestingly, with respect to Mr. Pascoal’s evidence on cross-examination that he was not sure about the requirement for flashings, Mr. Nelson put the specifications to him that clearly call for flashings, and Mr. Pascoal responded that he did not have this particular set of specifications before him when he was doing the job.

Read-ins from the Examination for Discovery by the Defendant Emmons & Mitchell pursuant to the Rules

Exhibit 142 is the list of questions and answers taken to be read-in from the examination for discovery of Dr. Lawless and the examination for discovery of Mr. Garbutt as part of the defendant company's case.

In addition, by agreement between the parties, I am to read questions and answers 81 and 95 and 96 on the understanding that those three answers are <u>not</u> to form part of the defendant's case.

In addition, Mr. Mueller by letter dated January 2, 2002 (marked as Exhibit G), answered several undertakings. Again by agreement between the parties, where any of the read-ins of the defendant reference an undertaking, then I should read the answer to that undertaking, again without it forming part of the defendant's case.

Dr. Lawless
The read-ins from Discovery of Dr. Lawless disclosed the following.

Dr. Lawless and Dr. Reed discussed initially a two-story building and then discussed incorporating the potential to go a couple of more floors. Eventually they decided on four floors with the potential of building two more.

During the construction, Dr. Lawless went to the site almost every day. He saw Mr. Garbutt there approximately once a week.

With respect to when he first encountered water leakage into the building, Dr. Lawless did not know whether it was the fall of 1986 or into 1987. All he knew was that one would get a heavy rain storm or a period of wet weather which would produce dripping out of the inside of the drywall at the top of the ceiling, or at the top of the windows. If there was a long dry spell and then a bit of rain there was no problem, but if there were a series of wet days and wind, then there would be the dripping around the windows. It also dripped at the inside edge of the window well. The south side was the worst and the fourth floor was always the worst. Dr. Lawless was on the third floor and he had at times cans and pails and buckets sitting in the window wells. There was some dripping on the second floor but it was not as bad. It was a decreasing thing as one went down. Sometimes there was dripping on the north side but not very often. There was also some dripping on the east and west sides but the south side was always the worst. The window sill was replaced once on the fourth floor and the drywall around the windows was repaired several times. Fairly early on, in 1987 or 1988 they called Emmons & Mitchell and possibly Mr. Garbutt and advised of the problem of the leaking windows. An inspection was done Emmons and possibly Garbutt said the windows needed to be caulked and that there was no caulking on them. The caulking was done and paid for and it seemed to make a difference they thought. Whether it was just dry weather that they were experiencing, Dr. Lawless couldn't recall. Dr. Lawless was asked how long it seemed to make a difference, and he responded that it really never ceased. It got marginally better and then went back to the previous state. They went through a second caulking session somewhere in 1990 to 1992, because they were still having the problem and they kept asking why is the water coming in. After the first caulking, sustained rain would still cause leakage on the south side. It was getting worse prior to the 1992 caulking and accordingly the doctors sought advice on what to do.

The decision to completely replace the roof in 1998 was made after getting roofer Kendall's advice, who advised that it would be cheaper and better to just redo the whole roof rather than to repair the insufficiencies. The new roof would have slope and would have the membrane extended over the top of the parapet. Dr. Lawless and Dr. Reed hoped that this would have an effect on the water leakage that was coming in around the windows, but Mr. Johnson told them that it would not. Redoing the roof would at least cancel out the possibility of it being a roof problem.

The roof repair in 1994 did help in some areas but not the windows. It helped in the stairway and then in 1998 the whole roof was done. In the final result the placement of the new roof did not cure the leakage problems around the windows.

The cost of repairing the roof is part of the claim on a pro-rated basis. Amherst Roofing did the work. At discovery, it appears that the plaintiffs' position was that they were not seeking 100% indemnification for the roof. Their position was that the roof should have lasted for 20 years, so that to the extent that they got some years out of it, it had to be discounted. In other words, if the roof should have lasted 20 years and one gets 10 years out of it, then one recovers ½ of the cost of the brand new roof. The calculation given at discovery was "we got 8 years out of the first part and 12 years out of the second part. I think that was our calculation. So 8/20ths or two-fifths."

Mr. Garbutt

The read-ins from Mr. Garbutt's examination for discovery disclose the following.

Mr. Garbutt needed Tom Roberts to get a building permit. Roberts had to prepare sections for the building and enough wall detail to satisfy the building inspector. The wall detail would not have been accepted by the building inspector unless it was by an architect as opposed to an engineer. Accordingly, Roberts designed the wall including the block and the brick. Mr. Garbutt intended it to be a solid masonry wall and what Mr. Roberts did was different in that there was a ¾" gap between the wythes. When Roberts sent that to Mr. Garbutt, Mr. Garbutt accepted it.

Mr. Garbutt attended the site once or twice a week when the brick and block were going on. Mechanical work was going on at the same time. Mr. Garbutt saw the brick ties being installed.

There was no written contract between Mr. Garbutt and the plaintiffs but there is no question but that Mr. Garbutt was the engineer on the project and that he would design the building and supervise the work. Without getting into the extent to which Mr. Garbutt would supervise the work, he agreed that was the deal with the plaintiffs.

Mr. Garbutt was paid by percentage. He believes the rate was 6% which was something under scale at the time.